A Theory of Economic Slack

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Carloads, caravans, homeless and hungry; twenty thousand and fifty thousand and a hundred thousand and two hundred thousand. They streamed over the mountains, hungry and restless—restless as ants, scurrying to find work to do—to lift, to push, to pull, to pick, to cut—anything, any burden to bear, for food. The kids are hungry. We got no place to live. . . . In the camps the word would come whispering, There's work at Shafter. And the cars would be loaded in the night, the highways crowded—a gold rush for work. At Shafter the people would pile up, five times too many to do the work.

John Steinbeck, *The Grapes of Wrath*, 1939

Preface

This book is based for the most part on research that I conducted during graduate school at the University of California–Berkeley under the supervision of George Akerlof and Yuriy Gorodnichenko (2006–2010), and for fifteen years after that in collaboration with Emmanuel Saez (2010–2025).

In this preface, I explain what motivated the research, and provide the key references on which the book is based. The purpose is not to describe results, but to explain how the theory presented in the book emerged from a sequence of research questions, themselves motivated by macroeconomic puzzles and policy debates, and how it built on previous advances. Readers who prefer to begin with the case for putting slack back at the center of business cycle research may skip ahead to chapter 1.

Why do job seekers queue in recessions?

Early on in graduate school I read about Okun's misery index—the sum of the unemployment rate and inflation rate, designed as a simple measure of welfare in the US economy.[1] A low index means that the economy is doing well; a high index that it is doing poorly. The misery index posits that inflation and unemployment are the two main diseases affecting developed economies. The discovery of the misery index prompted me to start working on inflation. I particularly wanted to understand why seemingly tiny changes in the price of a baguette in France outraged customers so much (not least my mother). But that project proved too challenging, so I turned to the other component of the misery index: unemployment.

A first book that George advised me to read to start thinking about unemployment and business cycles was *Prices & Quantities* by Arthur Okun (1981). In it, Okun sketches a new macroeconomic framework in which labor and product markets are organized around implicit contracts between sellers and buyers. Although the slackish business cycle model

[1] See Asher, Defina, and Thanawala (1993) for a history of the misery index.

developed in the book is quite different from the framework that Okun had in mind, the book draws in numerous places on evidence that Okun collected, observations that he made, and references that he provided. Generally, the book shares Okun's conviction that Walrasian theory is inappropriate to describe labor and product markets, and that prices and wages are determined by norms that are somewhat insulated from cyclical forces.

George also suggested that I read *Equilibrium Unemployment Theory* by Christopher Pissarides (2000)—the leading textbook on unemployment theory. The textbook introduces the Diamond-Mortensen-Pissarides (DMP) model of the labor market, which was developed by Pissarides together with Peter Diamond and Dale Mortensen in the 1970s and 1980s. For their work, they received the Nobel Prize in Economics in 2010.[2]

An idea in the textbook that stuck with me is that it is possible, and in fact quite fruitful, to summarize the complex matching process in the labor market by a matching function—without modeling the matching process explicitly.[3] The idea plays a central role in this book, as trade in all markets is mediated by a well-behaved matching function. The function gives the number of trades that actually take place when a certain number of buyers and sellers are looking for each other on a market.

Reading the textbook, however, I was struck that the DMP model does not allow for the possibility—which seemed relevant in real life—that sometimes the economy just does not have enough jobs for all workers. This is an old, very Keynesian idea. Keynes had seen what was happening at the time of the Great Depression; he had seen the queues of workers waiting to apply for jobs, or to be hired, in front of job bureaus and factory gates. The photos collected in this preface, taken across the United States by Dorothea Lange, illustrate what Keynes (1936) had seen and wrote about in the *General Theory*: that in bad times there are just not enough jobs for everyone at the going wage, that jobs are rationed.

By ignoring job rationing, I felt, the DMP model cannot provide a good description of what happens in the labor market. In bad times, in the real world, workers queue for jobs. In that situation, a factory manager can just pick any worker that they want immediately; it is costless for firms to hire; matching becomes irrelevant. And yet, workers visibly queue for jobs. So it seems that something else happens in this situation: jobs are lacking.[4]

Thus, I became convinced that to describe the labor market better, we must allow for job rationing. My PhD thesis therefore developed a labor market model that features not only frictional unemployment—as in the DMP model—but also rationing unemployment—

[2]Albrecht (2011) reviews the early work by Diamond, Mortensen, and Pissarides, and the literature that built on it. Diamond (2011), Mortensen (2011), and Pissarides (2011) provide their own perspective on their research and the literature in their Nobel lectures.

[3]This idea goes back to work by Pissarides (1979, 1984a, 1985, 1986). Before that, people usually modeled specific trading processes that resulted in trading probabilities between 0 and 1, often from the urn-ball model that is commonly used in probability.

[4]In the 1970s and 1980s, the literature on efficiency wages built models producing job rationing. But the models fell out of favor, maybe because they could not easily be integrated into the fast-growing DMP framework. For surveys of the efficiency-wage literature, see Akerlof (1984), Yellen (1984), and Katz (1986).

A. Howard Street, known as “Skid Row,” the district of the unemployed (San Francisco, CA, 1937)

B. Unemployed men sitting on the sunny side of the San Francisco Public Library (San Francisco, CA, 1937)

C. Part of the daily lineup outside the State Employment Service Office (Memphis, TN, 1938)

D. Typical scene reflecting large population of unemployed in desperate need of work and looking for jobs (Camden, NJ, 1935)

Job rationing during the Great Depression in the United States

Sources. A: Dorothea Lange, available at `https://www.loc.gov/pictures/item/2017769707/`. B: Dorothea Lange, available at `https://www.loc.gov/pictures/item/2017769687/`. C: Dorothea Lange, available at `https://lccn.loc.gov/2017770574`. D: Franklin D. Roosevelt Library Public Domain Photographs, available at `https://catalog.archives.gov/id/195658`.

as in the *General Theory*. Technically, the issue is that the DMP model's labor demand is degenerate: it pins down a fixed level of labor market tightness, irrespective of the level of employment, instead of a proper downward-sloping demand for labor. I resolved this by replacing the standard linear production function, which has constant returns to labor, with a concave production function, which has diminishing returns to labor. This simple change delivers a downward-sloping labor demand curve, and with it, job rationing. In the model, firms may not want to create enough jobs for all the workers who want to work. This core idea was then published in the *American Economic Review* in 2012 (Michaillat 2012), and a streamlined version of the AER labor market model is presented in chapter 10.

Wages posed a related challenge. Shimer (2005) showed that with wages set by Nash bargaining, wages are too flexible: they absorb almost all of a productivity shock, leaving too little of it to generate sizable fluctuations in unemployment. Resolving this puzzle turned out to be straightforward, because Hall (2005b) had already shown that a rigid wage—one that does not fully track productivity—remains compatible with bilateral efficiency in the DMP model, and generates realistic labor market fluctuations. Blanchard and Gali (2010) went further, showing that the wage need not be completely fixed: it can instead be a rigid function of productivity and still generate realistic fluctuations. With a rigid wage, unemployment fluctuated, and a surprising result appeared: the recessionary tide does not lift all boats. By that I mean that in recessions, when unemployment goes up, rationing unemployment also goes up—jobs are scarcer—but frictional unemployment goes down.

An important idea in the paper—one that carries through to the book—is that the slackish model readily accommodates realistic, somewhat rigid price and wage norms, and that such rigidity generates realistic business cycle fluctuations. But how does one build realistic price and wage norms? In the real world, almost no prices are set at auction, as the Walrasian model assumes; instead, most sellers and buyers are engaged in long-term relationships in which norms of fairness matter a great deal. In a paper with Erik Eyster and Kristof Madarasz that stemmed from the baguette project, we documented these norms of fairness in the product market and modeled pricing under fairness concerns (Eyster, Madarasz, and Michaillat 2021). Earlier, Akerlof (1980, 1982) and Akerlof and Yellen (1990) had documented these norms of fairness in the labor market and modeled wage-setting under fairness concerns. The book does not use the pricing and wage-setting models developed in those papers, but it draws on many of the facts documented there to build realistic price and wage norms, especially in chapters 5 and 6.

Towards the end of my PhD thesis, I stumbled upon an interesting property of the model: it is highly state dependent, in that it operates very differently in booms and slumps. As a result, some policies are more effective when the economy is slack, while others are more effective when the economy is tight. For instance, the fiscal multiplier is larger when unemployment is high, thus explaining Yuriy and Alan Auerbach's discovery that fiscal multipliers are higher during recessions (Auerbach and Gorodnichenko 2012). This result was published in the *American Economic Journal: Macroeconomics* in 2014 (Michaillat 2014). The state dependence is present throughout the book in various contexts, but especially in chapters 11 and 13, where we obtain slack-dependent policy multipliers.

Should unemployment insurance be extended in recessions?

I went on the job market in 2009–2010, in the aftermath of the Great Recession. One of the big policy questions at the time was whether it was warranted to increase the generosity of unemployment insurance. Emmanuel attended a mock job talk that I gave in December 2009, thought that the model that I had developed in my thesis could be helpful to think about the question, and reached out to see if I wanted to collaborate with him and Camille Landais, who was doing a postdoc at Berkeley at the time, on a project to determine how unemployment insurance should be adjusted over the business cycle.

The model was promising because public economics could only take into account supply-side factors—how unemployment insurance affects job search—but did not have the tools to include in the analysis demand-side factors—the number of workers that firms want to hire. The key idea was that if firms did not want to hire more workers, then searching more for jobs would only result in a giant rat race, but not additional employment, which would severely reduce the purported welfare cost of higher unemployment insurance and make extending unemployment insurance in bad times more desirable.

So in the summer of 2010, Camille, Emmanuel, and I started working on the following question: how should the generosity of unemployment insurance respond to unemployment fluctuations? We found that in the presence of job rationing, optimal unemployment insurance is more generous than the conventional Baily (1978)-Chetty (2006) level when the labor market is inefficiently slack and less generous when the labor market is inefficiently tight. The implication was that unemployment insurance should become more generous in bad times—as it is in practice in the United States. The project was ultimately published in 2018, in a pair of papers in the *American Economic Journal: Economic Policy* (Landais, Michaillat, and Saez 2018b,a). Unemployment insurance is discussed in chapter 11.

The AEJ papers aimed to express the formula for optimal unemployment insurance in sufficient statistics. The idea was to express optimal policy in terms of statistics that apply across a range of models and that can be directly measured.[5] Sufficient statistics make the policy tradeoffs transparent and rooted in empirical evidence, but they had not been used to study the market-level effects of policy—those were typically studied in clunky structural models that kept the policy tradeoffs and mechanisms obscure. The AEJ papers

[5] See Chetty (2009) and Kleven (2021) for surveys of the sufficient-statistic approach. Emmanuel's job-market paper played a central role in the shift towards sufficient statistics in public economics (Saez 2001).

brought sufficient statistics to the macroeconomic design of policy; the book follows this methodology, especially in part V.

How are aggregate demand and labor demand connected?

After our project on unemployment insurance, Emmanuel and I decided to keep working together with the framework. A natural question that arose is why labor demand is sometimes high and sometimes low. In the model from my thesis, which we then used in our unemployment insurance project, fluctuations in labor productivity lead to fluctuations in labor demand. This is the same as in the DMP model. But in reality, labor productivity is unlikely to be the main driver of business cycles. Surely fluctuations in aggregate demand, stemming from changes in monetary policy or animal spirits, influence labor demand. The question was therefore how to connect aggregate demand to labor demand.

The answer, it turned out, was slack in the product market. Unemployment is the most studied form of slack but slack also exists in the product market: firms are always looking for customers to buy their products and services.[6] Using the architecture developed by Barro and Grossman (1971), we built a model in which both product and labor markets are organized around matching functions and therefore display slack. In the model, when prices and wages are rigid, aggregate demand influences unemployment. For example, after an increase in aggregate demand, firms find more customers for their products, which reduces the idle time of their employees and makes labor more profitable. This increases firms' labor demand, which reduces unemployment.

The model therefore features frictional unemployment as well as classical unemployment, caused by excessive real wages, and Keynesian unemployment, caused by insufficient product demand—thus bridging the DMP and Old Keynesian theories of unemployment.[7] This work was published in the *Quarterly Journal of Economics* in 2015 (Michaillat and Saez 2015) and is covered in chapter 15.

While our QJE paper offers a simple model that is useful to understand why slack exists and how slack on different markets interact, it is static so cannot be used to think about interest rates, which inherently involve dynamic saving decisions. And interest

[6] Early contributions to the literature on product market slack include Diamond (1982a, 1984), Walsh (1986), and Diamond and Yellin (1990). Diamond and Fudenberg (1989) and Boldrin, Kiyotaki, and Wright (1993) find that the model with product market slack exhibits periodic orbits; thus, models with slack might produce endogenous business cycles. Around the time of the Great Recession, several research teams started developing new models with product market slack, maybe inspired by Hall (2007), who had applied the DMP model to the product market to explain price rigidity and generate macroeconomic fluctuations. This crop of models includes Arseneau and Chugh (2007), Lehmann and Van der Linden (2010), Matha and Pierrard (2011), Gourio and Rudanko (2014), Petrosky-Nadeau and Wasmer (2015), and Bai, Rios-Rull, and Storesletten (2026).

[7] See Benassy (1993) for an overview of the Old Keynesian model. The model was initially known as the General Disequilibrium model (Barro and Grossman 1971), but it is now referred to as the Old Keynesian model (Barro 2025)—using the same terminology as Tobin (1992, 1993) to contrast with the New Keynesian model.

rates are the tool that central banks use to stabilize the economy. So, Emmanuel and I next developed a dynamic version of the model with interest rates. In this dynamic model, monetary policy sets interest rates, which determine aggregate demand, and through it unemployment. Hence, monetary policy can be used to stabilize the economy. This work was published in the *Oxford Economic Papers* in 2022 (Michaillat and Saez 2022). The model is covered in chapter 13.

What happens during long zero-lower-bound episodes?

One of the challenges that we faced in developing the OEP paper was how to build a nondegenerate aggregate demand curve. By nondegenerate I mean an aggregate demand curve that slopes downward, just like the old IS curve, and that can be used to study a monetary model at the zero lower bound or away from it. The standard aggregate demand curve, arising from a standard Euler equation, fails these requirements, because it is horizontal: it only tolerates a real interest rate equal to the time discount factor.

We discovered that it was possible to obtain such an aggregate demand curve by assuming that people's wealth directly enters their utility function. The justification for the assumption is that wealth is a marker of social status, and people value high social status. The assumption of wealth in the utility function produces a well-behaved aggregate demand curve, so we use it throughout part IV. The assumption had been used sporadically in macroeconomics after Kurz (1968) had first introduced it, especially to understand saving and investment behavior, and relatedly, long-run growth.[8] But it had not been used in business cycle research, so we could not publish the OEP paper until 2022, although it was written in 2014.

What we did manage to publish was a related paper showing that the wealth-in-the-utility assumption resolved all the anomalies of the New Keynesian model at the zero lower bound. That paper appeared in the *Review of Economics and Statistics* in 2021 (Michaillat and Saez 2021b). In chapter 13 we use the approach developed in the REStat paper to analyze forward guidance in the slackish business cycle model.

How large should stimulus packages be?

When monetary policy is constrained by the zero lower bound—as during the Great Depression and the Great Recession—the Federal Reserve is not able to stabilize the unemployment rate to its efficient level. In that case, monetary policy is ineffective, but fiscal policy can help stabilize unemployment. This is why the US government implemented a large stimulus package during the Great Recession. But it was unclear at the time how

[8] See for instance Zou (1994), Bakshi and Chen (1996), Carroll (2000), Piketty (2011), and Kumhof, Ranciere, and Winant (2015).

large that stimulus package should be, or even if a stimulus package was desirable at all, because there was no framework to answer the question.

The policy debate motivated Emmanuel and me to study optimal stimulus spending when there is unemployment, and in particular unemployment is inefficiently high. Public economics studied the optimal provision of public goods in normal times, but not in bad times (Samuelson 1954, 1955). Macroeconomics focused on the size of fiscal multipliers in bad times, but not on the optimal amount of public spending in bad times. Using a sufficient-statistic approach again, we characterized how optimal public spending deviates from the Samuelson rule when the unemployment gap is positive. We found that the size of the optimal stimulus package depends on several sufficient statistics, including the unemployment gap and the fiscal multiplier. These results were published in the *Review of Economic Studies* in 2019 (Michaillat and Saez 2019), and are covered in chapter 17.

How wide is the unemployment gap?

The AEJ and REStud papers showed that optimal unemployment insurance and optimal public spending depended critically on the unemployment gap. The results were not easily applicable, however, because we did not know what the efficient unemployment rate, u^*, was, so we did not know what the unemployment gap, $u - u^*$, was.

So our next project was to estimate the efficient unemployment rate and unemployment gap. We found that the efficient unemployment rate depended on a few sufficient statistics, which we measured in the United States, to find that the US unemployment gap is countercyclical—low (even negative) in good times and high (sharply positive) in bad times. This finding confirmed that unemployment insurance and public spending should be adjusted over the business cycle. This project was published in the *Journal of Public Economics Plus* in 2021 (Michaillat and Saez 2021a). The results are covered in chapter 9.

In the course of our work on the JPubE paper, Emmanuel and I uncovered that in the United States, given the values taken by the sufficient statistics, our formula for the efficient unemployment rate could be simplified to $u^* = \sqrt{uv}$. The value u^* also measures the full-employment rate of unemployment (FERU), because legal texts define full employment as the amount of employment maximizing social welfare.[9] These results were published in the *Brookings Papers on Economic Activity* in 2024 (Michaillat and Saez 2024b), and are presented in chapter 12.

[9] The full-employment mandate for the US federal government and central bank is defined by the Employment Act of 1946, the Federal Reserve Reform Act of 1977, and the Full Employment and Balanced Growth Act of 1978 (US Congress 1946, 1977; Congress 1978).

Are inflation and slack linked?

Much of the work in the book was motivated by the vast amount of slack that appeared in the US economy after the Great Recession. The aftermath of the pandemic, by contrast, led to historical tightness in the US labor market. These tight conditions led to new questions.

The first question was whether the excessive tightness on the US labor market could explain the inflation surge that occurred at the same time. Until then Emmanuel and I had assumed a fixed-inflation norm in our models because inflation had remained exceedingly stable in the decades prior to the pandemic. But the pandemic inflation pushed us to relax this assumption and study how slack and inflation could be connected.

We therefore developed a new theory of the Phillips (1958) curve connecting inflation to the unemployment gap (Michaillat and Saez 2024a). The Phillips curve emerged from directed search, as in Moen (1997), combined with a quadratic price-adjustment cost, as in Rotemberg (1982). The price-adjustment cost was required because without it, the Phillips curve would be degenerate: vertical, keeping unemployment at a fixed level, instead of downward-sloping. The slackish Phillips curve ensures that the divine coincidence described by Blanchard and Gali (2007) appears: inflation is on target when the economy is at full employment—in line with the evidence unearthed by Benigno and Eggertsson (2023). This work is the basis for chapter 14.

Does immigration affect slack?

A second question that arose after the pandemic is whether other policies besides cutting aggregate demand could help alleviate labor market tightness—thus helping firms fill vacant jobs and possibly dampening inflation. Naturally, another policy that could help was to expand labor supply via immigration. I therefore started to explore the effects of migration in a slackish labor market model, and confirmed that immigration reduces labor market tightness (Michaillat 2023). This work is discussed in chapter 11.

Has a new recession started?

Finally, after peaking in 2022, the US labor market cooled fairly rapidly in 2023–2025. This naturally led to the question: Has a new recession started? In the JPubE and BPEA papers, the combination of vacancy and unemployment data allowed us to compute the FERU and unemployment gap in real time. But it turns out that these data are useful to detect recessions in real time too—because recessions feature not only an increase in unemployment but also a decline in job vacancies as the economy moves down the Beveridge (1944) curve. By combining data on unemployment and job vacancies, Emmanuel and I constructed a recession rule (dubbed Michez rule by the *Financial Times*) that detects recessions faster and more robustly than the popular Sahm (2019) rule. It is even possible to do better than the Sahm and Michez rules by optimizing how the recession rules are constructed—by optimally filtering labor market data and selecting the detection threshold. The Michez rule was published in the *Oxford Bulletin of Economics and Statistics* in 2025 (Michaillat and Saez 2025); the optimal rules were described in Michaillat (2025); both papers are the foundation for chapter 18.

What is new in this book?

Together, the papers described above form the foundation of the book. Readers familiar with them will recognize various elements of the book. However, the book aims to be more than a collection of papers. It provides a unified and systematic treatment of the theory of slack, with consistent notation, concepts, and methodology. It also provides substantial evidence about slack—coming from diverse sources and spanning nearly one hundred years—to motivate and discipline the theory. Finally, the book places the model in context by clarifying its similarities and differences with leading business cycle models.

Many ingredients in the book's theory are not new: the matching function comes from the DMP literature; price and wage rigidities are ubiquitous in the Keynesian literature; wealth in the utility function has appeared before; sufficient statistics are commonplace in public economics; and so on. The novelty lies in how the book curates and combines these ingredients to offer a slack-centered theory of how economies operate and fluctuate, and to provide slack-based tools to monitor and respond to the cyclical inefficiencies that markets inevitably generate.

Terminology

Before we start, we define the core terms used throughout the book. Some of the terms are not used commonly in economics, some are often misconstrued, and some are used in the book in a way that is atypical. Either way, it is useful to clarify the meaning of these terms upfront.

Slack terms

- The term *slack* describes situations in which goods or services are available for sale but trades do not occur. Sometimes the goods are already produced; sometimes they are available to be produced at negligible marginal cost. In both cases, trades that would generate value for sellers and buyers do not materialize.
- The *slack rate,* denoted by u, is the share of goods and services that remain unsold.
- The *efficient slack rate,* denoted by u^*, is the slack rate that maximizes social welfare by minimizing the social costs from slack—both unused resources and resources devoted to matching.
- The *slack gap,* denoted by $u - u^*$, is the gap between the actual slack rate and the efficient slack rate. It measures departure from social efficiency.
- The term *slackish* describes a class of markets where slack is the central mechanism that equalizes supply and demand. Slackish does not mean the market is always too slack; rather, it refers to the structural mechanism through which the market operates. It means that market tightness is the key mediating variable, whether the market is inefficiently slack or inefficiently tight.

Model terms

- The term *equilibrium* is used to denote time-invariant points of differential equations or dynamical systems (instead of *steady state*). An equilibrium point has $\dot{x}(t) = 0$. This follows usage in dynamical systems and conveys the idea of a balanced state, which may be reached quickly.
- The term *solution* is used to denote the allocation that satisfies all the equations of a model (instead of *equilibrium*). This describes more transparently what the allocation is.
- The *matching function* refers to the aggregate function that mediates trades in slackish markets. With s sellers and b buyers looking to trade, the matching function $m(s, b)$ gives the number of trades that take place. It summarizes the random, costly, imperfect process of finding trading partners without modeling that process in detail.
- The term *tightness* refers to the central equilibrating variable in slackish markets. Tightness is denoted by θ and defined as the ratio of the two arguments in the matching function: the ratio b/s between the number of buyers and the number of sellers. Since trading probabilities are determined by tightness, tightness summarizes the state of the market.
- The *matching elasticity* refers to the elasticity of the matching function with respect to the number of sellers, holding the number of buyers fixed. The matching elasticity is a structural property of the matching function, denoted by η, and generally endogenous to tightness.
- The *matching wedge* denotes the gap between goods purchased and goods consumed caused by matching costs. Buyers must devote part of their expenditure to matching costs, so the amount spent on goods and services that are actually consumed is less than the total amount spent. If the wedge is τ, consuming one unit requires purchasing $1 + \tau$ units. Because matching probabilities depend on tightness, the wedge is endogenous to tightness. In the labor market, where matching costs take the form of recruiting costs, we refer to the matching wedge as the *recruiting wedge*.
- The *market supply* denotes the amount of goods actually sold given market tightness and market price. It is the effective supply; the *market capacity* is the notional market supply: the amount of goods that sellers place on the market and would like to sell given market tightness and market price. Because each good sells only with probability less than one, effective supply falls short of notional supply.
- The *market demand* denotes the amount of goods that buyers actually purchase given market price and market tightness. That is the effective demand. The notional market demand is the amount that buyers would like to consume. Because consuming one

good requires purchasing more than one, effective demand exceeds notional demand. Buyers purchase goods from sellers through *visits*. Visits are the demand-side input into the market's matching function, just as goods for sale are the supply-side input.

Labor market terms

- The term *unemployed workers* describes workers who are available and willing to work but unable to find a job. This definition is consistent with that used by the US Bureau of Labor Statistics (BLS). Based on the definition, the concept of voluntary unemployment is an oxymoron. Being unemployed means wanting a job but not finding one: it is by definition involuntary. Someone who does not have a job and does not want one (which is presumably what is meant with voluntary unemployment) is *out of the labor force*—not unemployed. Unemployed workers represent slack on the labor market.
- The *unemployment rate* is the number of unemployed workers as a share of the labor force.
- The term *job vacancies* denotes positions that firms report as open and for which they are recruiting. This definition is consistent with that used by the BLS. Vacancies measure recruiting effort; they are not a measure of unmet labor demand. Some vacancies never result in hiring, because the job-filling probability is always less than 1. Job vacancies correspond to visits on the labor market.
- The *vacancy rate* is the number of job vacancies as a share of the labor force.
- The term *full employment* describes a labor market that operates efficiently. Accordingly, the full-employment rate of unemployment (FERU) is the socially efficient rate of unemployment. Full employment therefore does not mean zero unemployment and does not automatically imply stable prices.
- The *unemployment gap* denotes the distance of labor market slack from social efficiency. It is computed by comparing the actual unemployment rate to the FERU.

Product market terms

- The term *idle capacity* describes the productive capacity of a firm that is available to be sold but cannot find customers and remains unsold. Idle capacity represents slack on the product market.
- The *idleness rate* is the share of productive capacity that remains idle.
- The term *idle labor* describes workers employed by firms but unproductive because their services are not bought by customers.

Curves

- The *Beveridge curve* is the negative relationship between the slack rate and the visit rate. In the labor market specifically, it is the negative relationship between the unemployment rate and the vacancy rate. In dynamic slackish models, it appears once market flows are balanced and the slack rate reaches equilibrium.
- The *Okun curve* is the relationship between the logarithm of output and slack. The slopes of Okun curves are useful to identify the shocks that drive business cycles. *Okun's law* is the empirical observation that the labor-market Okun curve is downward sloping.
- The *labor-market Okun curve* (which is the standard Okun curve) is the relationship between log output and unemployment (labor-market slack).
- The *product-market Okun curve* is the relationship between log output and idle capacity (product-market slack).
- The term *Phillips curve* denotes the negative relationship between inflation and slack. The slackish Phillips curve links inflation to the slack gap, unlike the New Keynesian Phillips curve, which links inflation to an output gap or marginal costs.

Policy terms

- The *inflation-slack coincidence* denotes the property that inflation is on target whenever the slack rate is efficient (instead of *divine coincidence*).
- The *product-labor coincidence* denotes the property that product-market slack is efficient whenever labor-market slack is efficient.
- The term *sufficient statistics* refers to statistics used in formulas describing efficient allocations and optimal policies. They have two defining properties, which distinguish them from structural parameters. First, they should be valid across a broad range of models. Second, they should be estimable directly from data.

Math terms

- The terms *positive* and *negative* are used for $x \geq 0$ and $x \leq 0$ (instead of *nonnegative* and *nonpositive*). The terms *strictly positive* and *strictly negative* are used for $x > 0$ and $x < 0$. This follows French usage and avoids the cognitive load from "non-" constructions.
- The terms *increasing* and *decreasing* are used for weakly monotone functions (instead of *nondecreasing* and *nonincreasing*). The terms *strictly increasing* and *strictly decreasing* are used for strictly monotone functions. This again avoids the cognitive load from "non-" constructions.

PART I.

Introduction

CHAPTER 1.

Introducing slack into business cycle research

In 1929, the Great Depression started in the United States. The event spurred a wave of research on business cycles and stabilization policy. Beginning with Keynes, economists naturally concentrated on the depression's most devastating aspect: mass unemployment. A quarter of the labor force was thrown into unemployment, which had terrible consequences—as Steinbeck (1939) described in the *Grapes of Wrath*.

Fifty years later, when the devastation of the depression had faded into memory, modern macroeconomics moved in a different direction: away from unemployment to focus on fluctuations in prices and quantities. But of course, unemployment surges remain a central feature of recessions. They return every once in a while, even when we have forgotten about them.

So this book returns the focus to economic slack—a concept that encompasses unemployment as well as other unsold and idle resources. There are many different forms of slack: people who cannot find a job and remain unemployed; machines left idle in a factory; employed workers left idle on the job; restaurant tables, airplane seats, hotel rooms, houses and other buildings that remain vacant; durable goods that cannot be sold and depreciate; nondurable goods that cannot be sold and perish.

The book starts by reviewing essential but little-known facts about slack. It then presents a theory to describe why slack emerges and why it fluctuates. Finally, it uses the theory to explain how monetary and fiscal policy should respond to surges and drops in slack. The book addresses a number of questions that policymakers recurrently face in downturns: How much should interest rates fall when unemployment rises? How large should stimulus spending be? And how early should we start worrying about an upcoming recession?

1.1. Slack in Keynes's General Theory

Perhaps even more than other disciplines, macroeconomics is the product of its time. Trying to make sense of the Great Depression and offer advice to avoid such future calamities, Keynes (1936) wrote the *General Theory of Employment, Interest and Money*. The *General Theory* offers a fairly balanced treatment of unemployment and inflation. It discusses inflation and unemployment to a significant extent, with slightly more coverage afforded to unemployment. In the book, inflation and deflation appear on 9 pages; unemployment is mentioned on 32 pages (table 1.1).

Although the *General Theory* does not cover forms of slack beyond unemployment, we know that slack was present in other forms during the Great Depression too: not only was it almost impossible for jobseekers to find jobs, but it was also almost impossible for shops to find customers and for factories to find buyers. As a result, many shops were empty, many factories were idle, and these businesses eventually had to lay their workers off and shut down. As DeLong (2022, p. 212) wrote about the Great Depression:

> Workers were idle because firms would not hire them to work their machines; firms would not hire workers to work their machines because they saw no market for goods; and there was no market for goods because idle workers had no incomes to spend.

This book presents a theoretical framework which shows us that these two phenomena are intrinsically linked. Slack in the product market—the fact that firms cannot find customers because aggregate demand is low—generates slack in the labor market—the fact that workers cannot find jobs because labor demand is low. A key insight of the book is that slack in the product market generates slack in the labor market. Plainly, if firms cannot sell their production, they will not hire workers.

In fact, it is not just that the phenomena of product market slack and labor market slack are linked. It is that they can be modeled entirely symmetrically. The same theory therefore applies to both types of slack, and both markets can be stacked to form a business cycle model with slack.

1.2. Absence of slack in New Keynesian texts

While Keynes was writing during a mass-slack event, by the 1970s, the economic environment had changed, leading to drastic changes in macroeconomic theory. Following the fast economic growth of the US economy in the 1950s and 1960s, and the spike in inflation

TABLE 1.1. Pages mentioning unemployment and inflation in the *General Theory* and New Keynesian texts, according to their indexes

	Keynes (1936)	Gali (2008)	Woodford (2003)
Unemployment	5–9, 13, 15, 16, 22, 109, 121, 128, 190, 235, 251, 274, 275, 278, 289, 334, 335, 346–349, 358, 359, 362–364, 381, 382	188	—
Inflation and related terms	119, 202, 207, 281, 291, 301, 303, 331, 332	1, 4, 11, 21–22, 31–32, 34, 37–38, 47, 49–52, 54–56, 60, 77, 79, 95–99, 103, 105, 107–108, 116, 120, 125–127, 130, 133–135, 137–139, 148, 155–156, 161–164, 168–176, 179, 189	3–5, 13–14, 17, 19–21, 39, 43–44, 46, 49, 90–101, 116–122, 126–138, 159–160, 176–177, 188, 203–205, 212–213, 215, 236, 248, 251, 262, 267, 272, 276–286, 289–295, 301–305, 317, 341, 347, 360–362, 381–382, 396–405, 408–409, 413–416, 418, 437, 440–441, 445–446, 460, 462–463, 467–485, 493, 499–501, 522–527, 544, 559–565, 576–582, 590–604, 615, 619–623, 639–641, 694–695, 714–715, 724–727
Number of unemployment pages	32	1	0
Number of inflation pages	9	60	195
Unemployment share	78%	2%	0%
Inflation share	22%	98%	100%

Besides the term "inflation," Keynes (1936) also features "deflation." In Gali (2008), the terms related to inflation include "inflation dynamics" and "inflation targeting." In Woodford (2003), the terms related to inflation include "inflation inertia," "inflation rates," "inflation-targeting central banks," "inflation-targeting rules," and "deflation." The unemployment and inflation shares are the shares of pages mentioning unemployment and inflation in the total number of pages mentioning either unemployment or inflation.

in the 1970s, the Old Keynesian model—which had formalized Keynes's ideas—was rapidly replaced by the Real Business Cycle model.

The Real Business Cycle model was built around Walrasian markets so it had no slack. Because it emphasized efficiency and monetary neutrality, it was not very useful for designing macroeconomic policies, and it was itself replaced by the New Keynesian model in the 1990s.

The New Keynesian model adopted the architecture of the Real Business Cycle model but swapped perfectly competitive markets for monopolistically competitive markets, which allowed researchers to introduce price rigidity and thus break efficiency and monetary neutrality. While the monopolistically competitive model produced rich price dynamics, it did not have room for unemployment or slack either, so the research squarely focused on inflation.

All in all, during the transition from Keynesian macroeconomics to New Keynesian macroeconomics, inflation was retained as a topic of interest but unemployment was lost. This omission of unemployment and slack more generally, and the concentration on inflation appears clearly in New Keynesian textbooks. The leading New Keynesian texts are *Interest and Prices: Foundations of a Theory of Monetary Policy* by Woodford (2003) and *Monetary Policy, Inflation, and the Business Cycle* by Gali (2008). *Interest and Prices* discusses neither slack nor unemployment. *Monetary Policy, Inflation, and the Business Cycle* does not discuss slack, but unemployment is mentioned on one page at the end of the book, when possible extensions to the New Keynesian model are discussed.

Inflation, by contrast, is ubiquitous in both books (table 1.1). The term "inflation" appears on 97 pages in *Interest and Prices*. Additional related terms, such as "inflation inertia," "inflation-targeting central banks," and "inflation-targeting rules," appear on a further 98 pages. In total, there are 195 pages that cover inflation. In *Monetary Policy, Inflation, and the Business Cycle*, the term "inflation" appears on 52 pages. Related terms such as "inflation dynamics" and "inflation targeting" appear on 8 additional pages. In total 60 pages discuss inflation in that book.

1.3. The cost of ignoring slack: the case of the Great Recession

It turns out that bringing slack back into business cycle research could matter quite a lot. To see this, we return to the start of the Great Recession in early 2008. The unemployment rate was 5.0%.

This was actually below what the New Keynesian literature stipulated. The unemployment target that comes out of the modern literature is the non-accelerating-inflation rate of unemployment (NAIRU): the unemployment rate that keeps inflation stable. Keeping inflation stable is not the same thing as keeping the labor market at full employment, and it is not what the full-employment mandate asks for. But because New Keynesian macroeconomics focuses on prices, the NAIRU has naturally become the preeminent unemployment target.[1] Moreover, the New Keynesian model does not natively feature unemployment, so to estimate the NAIRU, unemployment must be introduced somewhat artificially.[2] In any case, the NAIRU was at 5.5% at the time (figure 1.1).

The unemployment rate skyrocketed as the Great Recession unfolded, reaching 9.9% by 2009. The NAIRU also rose to 6.6%, so according to the New Keynesian paradigm, the unemployment rate was 3.3pp too high at the peak of the recession. Furthermore, estimates of the NAIRU are quite uncertain, so it would not be unreasonable to believe that the unemployment rate was only 2.0pp too high at the worst of the Great Recession (Crump et al. 2024, figure 2).

By contrast, this book develops a new theory of business cycles that places economic slack at the heart of the analysis. One of the key concepts emerging from the theory is the full-employment rate of unemployment (FERU). The FERU marks both full employment and social efficiency, so it is the appropriate target for policymakers.

As we will see in the book, the FERU differs markedly from the NAIRU. The FERU was below 4.0% during the Great Recession (2008–2009), and below 4.5% at any point during the long recovery from the recession (2010–2015). This means that the unemployment rate was 5.9pp above the FERU in 2009, at the peak of the Great Recession (figure 1.1).[3]

The picture painted by the FERU is therefore much more dire than what the NAIRU suggests. By ignoring slack, the New Keynesian paradigm sets a target for unemployment that is much too lenient: 2.2pp too high in 2008, 2.8pp too high in 2009, and 2.4pp too high in 2010 (figure 1.1). It is not until 2015 that the gap between NAIRU and FERU vanished.

How many jobs would have been saved by targeting the FERU instead of the NAIRU, assuming that policymakers would have tolerated the same unemployment gap under either target? A lot, it turns out. On average between 2008 and 2015, the NAIRU was 1.5pp

[1]According to the Council of Economic Advisers (2024, p. 24), "modern economics has generally defined full employment by citing the theoretical concept of the lowest unemployment rate consistent with stable inflation, which is referred to as u^*, ... the non-accelerating inflationary rate of unemployment (termed NAIRU)."

[2]As Crump et al. (2019, p. 166) write: "Although the New Keynesian model—which has become a popular framework for monetary policy analysis and the core structure in many monetary DSGE models—features a forward-looking Phillips curve, it is silent on u^*. Instead of relating inflation to the unemployment gap, it typically relates inflation to the output gap or real marginal costs." An option is to assume that the unemployment rate is the nonparticipation rate, so that fluctuations in unemployment are just negative fluctuations in employment (Hazell et al. 2022, p. 1317). Another option is to assume that the unemployment is the additional amount of labor that would be employed if labor unions did not charge markups; in that case, fluctuations in unemployment reflect fluctuations in the wage markup (Gali 2011, figure 1).

[3]Translating full employment into a number has long been a challenge for policymakers. The 1978 Full Employment and Balanced Growth Act did set an unemployment target of 3% within 5 years of the law, and 4% for young workers (Congress 1978). But these numbers were never taken seriously, in part because policymakers understood that the full-employment unemployment rate could not be a fixed number (DeLong 1996). When Congress asked Alan Greenspan in 1975 what the target for full employment should be, he quipped: "I do not think we should set a target" (Duboff 1977, p. 13).

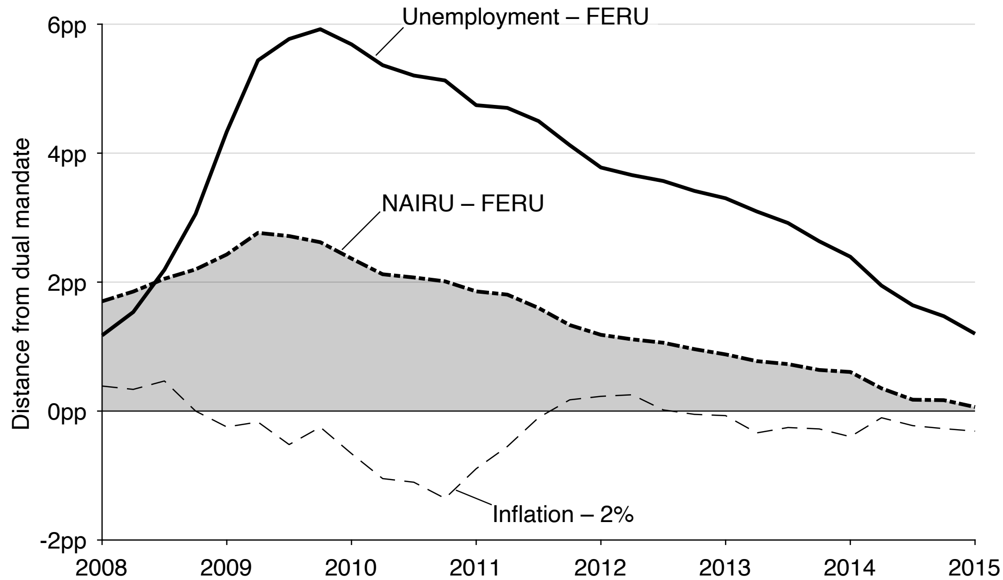


FIGURE 1.1. Costs of ignoring slack during the Great Recession in the United States

The unemployment rate is the quarterly average of the monthly, seasonally adjusted series produced by the Bureau of Labor Statistics (BLS 2025y). The inflation rate is the quarterly average of the monthly, seasonally adjusted, percentage change from a year ago of the consumer price index for all urban consumers, less food and energy, which is produced by the BLS (2025e). The non-accelerating-inflation rate of unemployment (NAIRU) is constructed by Crump et al. (2024, figure 2). The unemployment rates are reported as departures from the full-employment rate of unemployment (FERU), which is described in chapter 12. The inflation rate is reported as departure from the inflation target of 2%.

above the FERU, which translates into 16.3 million worker-years of unemployment.[4] So, by neglecting slack, the New Keynesian paradigm produces unemployment targets that are materially more permissive, allowing for an additional 16 million worker-years of unemployment during and after the Great Recession. The focus on inflation embodied in the NAIRU is especially odd given that inflation remained 0.3pp below its target of 2% on average between 2008 and 2015 (figure 1.1).

Additionally, the FERU formula developed in this book would have helped dismiss a hypothesis that was popular at the time of the Great Recession and hindered policy response. A substantial number of academic economists and policymakers argued that mismatch was a key cause of high unemployment and therefore that there was not much for the Federal Reserve to do—since it did not have the tool to tackle mismatch.[5] One prominent advocate of that view was Narayana Kocherlakota (2010), who was President of the Federal Reserve Bank of Minneapolis at the time. He argued that the Beveridge curve

[4]Each quarter, we compute the gap between NAIRU and FERU and multiply it by the number of workers in the labor force. We then sum these values over the 32 quarters between 2008 and 2015. Dividing by 4 gives 16.3 million worker-years.

[5]See Rothstein (2012) for an overview of the debate and related empirical evidence.

broke down at the onset of the Great Recession, so that there was not much that the Fed could do about the elevated level of unemployment:

> The relationship [between unemployment and job openings] has completely shattered.... Firms have jobs, but can't find appropriate workers. The workers want to work, but can't find appropriate jobs.... Whatever the source, though, it is hard to see how the Fed can do much to cure this problem.... Of course, the key question is: How much of the current unemployment rate is really due to mismatch, as opposed to conditions that the Fed can readily ameliorate? The answer seems to be a lot.... Most of the existing unemployment represents mismatch that is not readily amenable to monetary policy.

In essence, Kocherlakota argued that the shift in the Beveridge curve had drastically raised the unemployment rate the Fed should target. On this view, despite the high level of unemployment, the unemployment gap facing the Fed was not very large.[6]

As we will see in the book, the Beveridge curve is indeed central to understanding the workings of the labor market, and is at the heart of our FERU formula. In turn, the FERU formula allows us to quantify how much the Fed's full-employment target increased during the Great Recession. It shows that mismatch only had a small effect on the FERU: yes, the Beveridge curve shifted outward; but no, the rise in the FERU did not account for most of the increase in unemployment. The FERU only increased from 3.8% at the beginning of 2008 to 4.3% in 2010 and 4.5% in 2012. Since the FERU did not increase much, but the unemployment rate increased drastically, the unemployment gap was substantial, peaking at 5.9pp at the end of 2009 (figure 1.1). This means that 5.9pp of unemployment reflected deficient demand and was amenable to monetary policy, and to fiscal policy as well. However, due to the confusion created by the shift in the Beveridge curve and the lack of theory to translate that shift into a new full-employment target, the policy response remained muted, contributing to the slow recovery from the recession.

1.4. Rebalancing business cycle research

Policymakers in many countries have a dual legal mandate: keeping prices stable and maintaining the economy at full employment. Yet the New Keynesian model only speaks about price stability; it says nothing about full employment. The balanced treatment of unemployment and inflation that was a hallmark of Keynesian thought disappeared in the New Keynesian literature, as the focus turned solely to inflation. So the book develops

[6] Kocherlakota later softened his views. In an interview at the end of 2012, he explains: "I should be clear—it's not that I used to think structural factors were the sole source of elevated unemployment, and now I think they don't matter. It's more nuanced—the evolution of the data has led me to put less weight on structural factors than I did earlier" (Kocherlakota 2013).

a new business cycle model that provides a more balanced treatment of inflation and unemployment—one that accounts for fluctuations not only in prices, but also in slack, and that can help us think both about price stability and full employment.

Still, why should slack be taken into account to provide an accurate depiction of business cycles and relevant policy advice? The main reasons are that slack varies far more than prices over the business cycle, and slack carries substantial welfare costs. We begin the book in earnest in chapter 2 by making the case that slack is not only prevalent on almost all markets, but also highly countercyclical (low in good times and elevated in bad times). The chapter also argues that economic slack represents a waste of productive resources, and that in addition to its wastefulness, unemployment generates other, large welfare costs to society. Accordingly, a good model of business cycles should feature slack, and good economic policy should avoid periods of elevated slack.

The next question is: How does the book build a business cycle model with slack? The central assumption in the model is that all markets are organized around a matching function that gives the number of trades achieved when a certain number of buyers and sellers look for each other. Chapter 3 introduces the matching function and its properties, and explains why it naturally generates slack.

After chapter 3, the rest of part II develops the theoretical heart of the book: a model of a slackish market. Indeed, to understand why slack matters so much at the macroeconomic level, we must first understand how it operates at the level of individual markets. So, the book starts at the market level: it develops a model of markets with supply, demand, prices, but also slack. It begins in chapter 4 with the most basic slackish market model.

In the slackish market model, prices are set through norms. Because selling is difficult, sellers are glad when they can find a buyer. What this means is that there is a range of prices at which sellers would be willing to sell: the one they had in mind when they placed the good for sale, higher prices of course, but also lower prices that eat up some of their surplus. The same is true for buyers: it is not always easy to find exactly the right good, and buyers are happy when they can purchase what they were looking for, so they would be willing to pay a range of prices. Thus, many price norms can arise in the model. At the same time, the price norm has critical implications for the behavior of the model. Therefore, in chapter 5, we examine how prices are set in the real world to isolate the empirical properties of price norms. Then, in chapter 6, we insert a realistic, somewhat rigid price norm in the slackish market model and explore its properties. After that, we explore several extensions of the basic model: how market capacity is determined (chapter 7), and how the model operates when buyers and sellers enter into long-term customer relationships once they have matched (chapter 8).

We then define market efficiency, which provides a welfare benchmark against which to measure market outcomes (chapter 9). The difficulty in matching buyers and sellers—

illustrated by the presence of slack—implies that in reality we are very far from the auction market envisioned by Leon Walras. Many prices might prevail in markets: not just the efficient, market-clearing price that Walras focused on. The implication is that markets generally operate inefficiently. They are sometimes too slack, sometimes too tight, and efficient only in a knife-edge case.

Next, part III applies the slackish market model to the labor market. This is an important application because unemployment is the most socially costly form of slack. There are also good data on labor market slack, which allows us to calibrate the model and assess its quantitative behavior. In chapter 10, we build a slackish labor market model and show that it offers a unified theory of unemployment, merging job rationing and frictional unemployment. In chapter 11, we study the impact of labor market policies in the slackish labor market model: public employment, migration, and unemployment insurance. Last, in chapter 12, we compute the efficient unemployment rate and unemployment gap in the United States over the past century, to assess how far from social efficiency—and therefore full employment—the US labor market has been. We find that the US labor market has been generally inefficiently slack, and especially in recessions.

Having established that the unemployment gap observed in the US labor market is generally positive and sharply countercyclical, we must explain which macroeconomic forces generate such fluctuations and how macroeconomic policies—especially monetary policy—might stabilize them. This is what parts IV and V aim to do.

Part IV scales up the model from a slackish market to a slackish economy. We start with a basic one-market economy in chapter 13 to understand the core properties of slackish business cycles. With this basic model, we see that aggregate demand shocks can generate countercyclical fluctuations in the unemployment gap, and that monetary and fiscal policy can counteract such fluctuations. In chapter 14, we introduce directed search into the business cycle model to produce a Phillips curve that links inflation to slack. In chapter 15, we extend the model to a more realistic economy with both labor and product markets. The two-market economy allows us to understand how slack percolates from the product market to the labor market.

With a complete business cycle model in hand, we turn in part V to the policy questions posed by the unemployment-gap evidence: How should monetary and fiscal policy be conducted when the economy is not at full employment? We derive formulas for the optimal conduct of monetary policy (chapter 16) and fiscal policy (chapter 17) based on the unemployment gap. The part closes by showing how slack data can be used to detect recessions early and therefore trigger early policy responses (chapter 18).

Part VI concludes by summarizing the main arguments and implications of the slackish framework (chapter 19). It then outlines how the theory developed in the book could be fruitfully applied to countries besides the United States, particularly developing countries, where slack is widespread (chapter 20). It closes by situating the book's slackish business cycle model within the macroeconomic landscape—drawing connections to the Old Keynesian, Real Business Cycle, and New Keynesian models (chapter 21).

So, who is this book for? I have written it for a reader with undergraduate mathematical training and an interest in the social sciences—roughly an advanced undergraduate or a graduate student in economics or another quantitative social science. That reader should be able to go from beginning to end, absorb a coherent picture of how the economy operates and fluctuates over the business cycle, and later apply the theory. Mathematical training will help follow all algebraic arguments; curiosity beyond mathematics will help absorb various points that draw on psychology, sociology, public health, and economic history.

Although some sections of the book are necessarily mathematical, I hope the ideas developed here will speak to general readers interested in how modern economies function and sometimes falter. To make the data and findings accessible, the book relies throughout on theoretical diagrams and empirical plots, so that all arguments can be understood visually. Modern macroeconomics has abandoned models that can be represented and analyzed diagrammatically; but a theory that can be understood, manipulated, and extended through visual reasoning might have certain advantages over a theory that can only be understood through numerical solution of large dynamical systems.

It is true that this book diverges conceptually and methodologically from the traditional macroeconomic approach. Professional macroeconomists might find the material unfamiliar and difficult to engage with. When I presented the concept of FERU at a policy conference in Washington, DC, in 2023, Christina Romer criticized our effort to "divorce the concept of full employment from stable prices" on the grounds that "a concept of full employment that isn't consistent with stable inflation is not a sensible goal for policy".[7] Romer is one of the most accomplished scholars of the Great Depression; an expert on business cycles; and a leading authority on monetary and stabilization policy.[8] Furthermore, she understands how policy decisions are made in the real world: she chaired the Council of Economic Advisers at the time of the Great Recession, hired by the White House to guide the US economy out of the crisis. Her response reflected a common view—full employment should be defined through price stability—and revealed how inflation-centric current policy thinking is. Nevertheless, I hope that the book might entice macroeconomists to make a bit more room for slack in business cycle research.

[7]The quote is from the discussion that follows our paper on the FERU (Michaillat and Saez 2024b, p. 422).

[8]See for instance Romer (1993), Romer (1999), and Romer and Romer (2023). Her expertise is widely recognized: she was codirector of the NBER Monetary Economics Program for two decades and is part of the NBER Business Cycle Dating Committee.

CHAPTER 2.

Welfare cost, prevalence, and cyclicality of slack

Before we start building our slackish model of business cycles, it is critical to look at the data to get a good sense of what we want to model. As we will discuss in chapter 21, one of the three qualities of a good model according to Kuhn (1957) is to be descriptive—that is, to describe well the phenomena of interest. We therefore need to look at data from the real world to assess what the welfare cost of slack might be, how prevalent it is, and how it behaves over time. Throughout the book, we will incorporate these facts into the model—use them to justify model assumptions, calibrate model parameters, and discipline model mechanisms.

Everyone knows that during severe recessions such as the Great Depression or the Great Recession, a vast number of workers become unemployed. So, to understand these events, we must have a model that features unemployment in bad times. What we will come to realize is that unemployment and slack are much more prevalent than one might think. It is not true that slack only occurs in the labor market during deep downturns: slack is a pervasive feature of modern economies, present in most markets at all times and in varying amounts. Therefore, a good macroeconomic model should feature slack in all markets, as well as fluctuations in slack.

The evidence assembled in this chapter draws on data from the United States. I make this choice for two reasons. First, the models in this book are designed to describe the US economy, which is the benchmark economy in macroeconomics, and of course the largest global economy. International data are not directly applicable to the US economy because market-level and macroeconomic outcomes are shaped by country-specific regulations, institutions, and customs.

Second, the United States is the most demanding testing ground for a theory of slack. The United States is widely seen as the paragon of competition and free markets: the economy that should come closest to the Walrasian ideal in which slack does not exist. Yet, as we will see throughout this chapter, most US markets are actually riddled with slack—idle or unemployed workers, unsold goods, empty seats. If slack is pervasive even in the economy least hospitable to it, it is bound to be at least as pervasive elsewhere. In that way, the US focus sets the highest possible bar for the evidence.

Focusing on one country brings discipline to the data collection and model construction, but it certainly does not imply that the theory developed in this book does not apply to other countries. Chapter 20 shows that many patterns of slack observed in the US economy are also found, often more visibly, in other developed and developing economies. Slack therefore appears to be a universal feature of market economies.

2.1. Definition of economic slack

The concept of economic slack describes situations in which goods or services are available for sale but trades do not occur. When the goods or services are already produced, their costs of production are sunk, so the trades would typically generate a positive surplus. Sometimes the goods and services are not yet produced, but they are available to be produced at minimal marginal costs, so the main idea behind the concept of slack remains: trades that would generate value for sellers and buyers do not materialize.

There is no room for slack in the Walrasian model. In that model, anyone can sell anything at the market price. It is never the case that somebody would like to sell something at the market price but cannot find a buyer. Moreover, the marginal seller and buyer on a Walrasian market are indifferent between trading or not trading. Their marginal cost or marginal value for the good equals the market price, so it does not matter to them whether they trade or not. By contrast, in a world with slack, sellers are positively happy when they are able to trade, and they are distinctly hurt when they cannot trade. The same is true for buyers: they are positively happy when they are able to find the good that they are looking for.

The most famous example of slack is unemployment in the labor market. There are workers who are available and willing to work, and there are firms who are looking for workers, but the job seekers are unable to find jobs. As a result, the workers remain unproductive and their labor services are wasted.

But we also find slack in many other markets, not only in the labor market. For instance, in the market for services, we very often see slack. A hairdresser might be idle if no customers have come to the hair salon to get a haircut. The hairdresser is at work since the salon must remain open to attract customers; but they are idle at work until a customer arrives. We see this too in restaurants and cafes, where not all tables are always full and there are sometimes more empty seats than at other times. Depending on the number of customers sitting in their cafes and restaurant, waiters and cooks are more or less busy. Slack is also observed in hotels and airplanes, which are not always full, with hotel rooms and airplane seats sometimes remaining empty.

Slack still exists in other markets, for example in the market for goods. There, some goods are produced but cannot be sold and are thrown away: for instance food that nobody buys and that expires. But it is not just perishable goods. We also find slack in durable goods, such as clothes and appliances that remain unsold. Although durable goods do not go bad immediately, they depreciate over time, so stores would later have to sell them at a lower price and, eventually, discard them.

We can also see slack within firms. People employed by firms do not work at full capacity at all times; they are more or less utilized depending on the demands on their time. This sort of slack is apparent in all types of jobs. Manufacturing workers might be forced to work shorter hours if production lines are temporarily shut down due to low demand, and sometimes they might be forced to work overtime. In consulting firms, consultants are more or less busy. When business is booming, consultants work long hours on projects. But sometimes, when business is slower, consultants stay home because there is not enough business available for them to work. So slack also affects people who are employed within firms when firms do not have business.

2.2. Welfare cost of slack

Before building a theory of economic slack, we establish why having slack in a society is so problematic. In the first place, slack imposes a welfare cost because it represents a waste of productive resources. Labor that is available to produce value is left unemployed, and goods and services that are available to be consumed are left unsold. Additionally, the idleness associated with unemployment creates psychological hardship. A final welfare cost arises from the matching efforts exerted by buyers in a slackish world. Recruiting in the labor market and shopping in the product market absorb time and labor which could otherwise be used to produce valuable goods and services.

2.2.1. Welfare cost of labor market slack

Labor market slack takes the form of unemployed or underemployed labor. Mechanically, this imposes a welfare cost because it is a waste of productive resources. Labor that is available to generate value once incorporated in firms' production process is left unused.

In economics, we usually assume that leisure is valuable—this explains why people do not work all the time. Additional leisure time could in principle alleviate the welfare cost of unemployment. But, as Bakke (1969, p. 16) concludes after observing and interviewing unemployed workers for several years in New Haven during the Great Depression, "unemployment is not leisure." Bakke (1969, table 1) shows that unemployed workers engage in much fewer recreational activities than employed workers, despite having much more time on their hands. Among 24 recreational activities, they do strictly less of 18 activities, and strictly more of only 5 activities: sitting around home, sleeping during daytime, chatting and gossiping at corner stores and on the street, walking, and gardening and home repairs. So, of the activities that unemployed workers do more often than employed workers, 4 are various forms of being idle. As Bakke (1969, p. 16) notes, "Several hours in every day have been transferred to the leisure account. But the transfer is made under circumstances that cancel many of the opportunities usually related to leisure." One reason that he isolates is that workers who lose their jobs also lose their place in society, and their social network, which cuts them off from many forms of recreation. A second reason is that it is difficult for workers to change their routine and take part in new recreational activities. As Bakke (1969, p. 17) writes: "The plain fact is that these [activities] are not a normal part of their life; they have little or no experience with them; they have no normal incentive to make use of them. "

In fact, being unemployed is an extremely traumatic experience—among the worst events that can happen to an adult. That is why, beyond the wastefulness that it engenders, unemployment generates large additional welfare costs. People who are unemployed suffer from lower mental health and lower physical health than people who are employed. Robinson (1949, p. 11) for instance observed that "The most striking aspect of unemployment is the suffering of the unemployed and their families—the loss of health and morale that follows loss of income and occupation." At this point, the detrimental effects of unemployment on mental, physical, and public health are well documented.[1]

Critically, unemployment carries substantial costs even when controlling for income and personal characteristics. In the US General Social Survey, Di Tella, MacCulloch, and Oswald (2003) find that the nonpecuniary cost from becoming unemployed can be as profound as the emotional distress experienced when going through a divorce or dropping from the top income quartile to the bottom. In the same survey, Blanchflower and Oswald (2004, p. 1373) confirm that the nonpecuniary toll of unemployment can be substantial, akin to the emotional impact of losing $60,000 in annual income. Using two large, daily US surveys, Helliwell and Huang (2014) confirm that for workers who have fallen into unemployment, the nonpecuniary costs are several times as large as the pecuniary costs. They also find that an increase in local unemployment reduces the well-being of workers

[1]See for example Dooley, Fielding, and Levi (1996), Platt and Hawton (2000), Frey and Stutzer (2002), McKee-Ryan et al. (2005), Wanberg (2012), and Brand (2015).

who are still employed much more than what the reduction in local income would predict.

Where do the nonpecuniary costs of unemployment come from? There are various sources, which became clearly visible as early as the Great Depression. Eisenberg and Lazarsfeld (1938) review findings from clinicians who worked with unemployed workers, in the field, during the Great Depression. They isolate several causes for the elevated anxiety and widespread depression among jobless workers:

> Unemployment represents a personal threat to an individual's economic security; fear plays a large role;... the individual's prestige is lost in his own eyes, and as he imagines, in the eyes of his fellow men. He develops feelings of inferiority, loses his self-confidence, and in general, loses his morale.

What's fascinating is that all these negative facets of unemployment have been confirmed since. Bakke (1969) finds that unemployment strains family and other social relations. To illustrate the strain on personal relationships, and the social stigma associated with joblessness, Bakke (1969, p. 12) quotes a carpenter who was unemployed during the Great Depression:

> Every time I go to the home of one of my friends, they will say, 'Well, how about it? Do you have a job yet?' and constantly I'm being reminded of the fact that I am out of work. I hate to go even to see my relatives because I know that they're thinking 'So-and-so has a job; why can't Jim get one?' Perhaps this is only all in my own mind, but it keeps me from enjoying these relationships with them just the same.

Goldsmith and Darity (1992) provide a trove of evidence from psychology showing that an important reason why joblessness diminishes well-being is that it creates a sense of helplessness: that one's life is no longer under their control. Goldsmith, Veum, and Darity (1996) offer evidence that job loss and unemployment lead to a decline in an individual's self-esteem and sense of self-worth. In addition, Krueger and Mueller (2011) find that the process of unsuccessfully searching for a job significantly reduces unemployed workers' life satisfaction.

In fact, Jahoda (1981) emphasizes numerous important benefits of work—which are lost during unemployment. A lot of people care about their job and derive a lot of meaning from their employment. Their personal status and identity come from their jobs. Other benefits from work encompass a structured daily routine, regular interactions and shared experiences and goals with individuals beyond the immediate family, and the engagement in regular activities. Collectively, the loss of these benefits contributes to the psychological burdens associated with unemployment.

The nonpecuniary burdens from joblessness can have dramatic effects. Using administrative data for Pennsylvania workers, Sullivan and von Wachter (2009) find that high-tenure men who lose their jobs in mass layoffs face a 50%–100% higher mortality rate in the year after job loss, and a 10%–15% higher annual mortality hazard even twenty years later—enough to shorten life expectancy by 1.0–1.5 years for a worker laid off at age 40. Lost earnings and health insurance may play a role in the long run, as they deplete investment in health, but the sharp jump in mortality immediately after job loss does seem to point to the acute stress, social stigma, and loss of control that accompany unemployment.

As a final step, we assess the contribution of unemployed labor to social welfare. Unemployed workers divide their time across three activities: looking for a job; producing goods and services at home; and remaining idle—maybe waiting for callbacks from potential employers. How do these three activities contribute to social welfare? Jobseeking is required to find employment but—just like recruiting—does not contribute to welfare. Home production adds to welfare, while the psychological cost of idleness depletes welfare. Overall, it appears that the value of job seekers' home production, net of the psychological cost of idleness, is negligible. Using administrative data from the US military, Borgschulte and Martorell (2018) study how servicemembers choose between reenlisting and leaving the military. The choices allow them to estimate the value of home production plus public benefits received during unemployment minus the psychological cost of idleness. Subtracting the value of public benefits from these estimates, Michaillat and Saez (2021a, p. 11) find that the value of home production minus the psychological cost of idleness could be as low as 3% of the value of market production. Given its minimal value, we take the welfare contribution of unemployed labor to be zero.[2]

2.2.2. Welfare cost of product market slack

All goods, nondurable and durable, depreciate when they are not sold, which depletes welfare. The most costly such form of slack might be unsold food—both because food is the most necessary of goods, and because food perishes quickly.

Services that are not sold are obviously immediately lost. So the idle capacity in service firms imposes a direct welfare cost, since this capacity was readily available to provide value to customers.

Another welfare cost of product market slack takes the form of advertising. Because firms find it difficult to sell their production, they divert resources to advertising and selling it, in the hope of improving sales (Hall 2012). The OES indicates that 11% of US workers are in occupations devoted to sales and advertising, including retail salespersons, sales representatives, and advertising agents (Gourio and Rudanko 2014). Arkolakis (2010,

[2]Okun (1981, pp. 272–274) notes that unemployed workers have time available "for seeking other jobs" and "for the variety of pursuits that may be described as home activity". Then, to assess the welfare costs of recessions and slack, he weighs the value of job seekers' home activity against "the social stigma and family strains associated with unemployment." He agrees with us that the value of home activity net of the psychological cost of unemployment is negligible.

Appendix A) reports that 4%–8% of US GDP is devoted to marketing, with advertising alone constituting 2%–3% of GDP. Relatedly, based on a survey of 4,000 firms run by APQC, Fernandez-Villaverde et al. (2025) report that 7.5% of firms' revenue is devoted to attracting customers. From a welfare perspective, the activities of sales, marketing, and advertising are wasteful: they are required for sellers to find buyers, but do not directly produce welfare for consumers.

2.2.3. Welfare cost of recruiting

Employed workers must spend some of their time recruiting new hires for their firms, so they are unable to spend their entire time contributing to social welfare. Recruiting takes work: designing and advertising job vacancies, screening and interviewing candidates, and negotiating contracts.

Unfortunately, so far, there is only limited evidence on the number of recruiters in firms or the number of man-hours devoted to recruiting by firms. There are two sources of information about the amount of labor devoted to recruiting in the United States.

The first source of information about the amount of labor devoted to recruiting in the United States is from the National Employer Survey, which was conducted by the Census Bureau in 1997 (Villena Roldan 2010). The survey asked thousands of establishments across industries about their recruiting practices (Cappelli 2001). On average, establishments report that they devote 3.2% of labor costs to recruiting, which implies that servicing a job vacancy requires 0.92 workers at any point in time (Michaillat and Saez 2021a, p. 11).

A second source is the survey conducted by the consulting firm Bersin and Associates in 2011 (Gavazza, Mongey, and Violante 2018). The survey asked over 400 firms with more than 100 employees about their spending on all recruiting activities. Firms reported that recruiting one worker costs 93% of a monthly wage, which implies that it takes 1.16 workers to service a vacancy at any point in time (Michaillat and Saez 2024b, p. 332).

Both surveys show that in the United States, it takes about 1 full-time worker to service a job vacancy. Thus, the number of vacancies is a good measure of the number of recruiters in the economy: the number of workers diverted from producing and allocated to recruiting can be measured by the number of vacancies.

2.2.4. Welfare cost of shopping

A first welfare cost of shopping comes from the sheer amount of time spent by consumers shopping. Time-use data recorded by the BLS through the American Time Use Survey (ATUS) show that on an average day between 2003 and 2012, US consumers devote 42 minutes to shopping, including 18 minutes traveling to shops, 8 minutes buying groceries, gas, and food, 15 minutes researching and buying consumer goods and services, and 1 minute waiting at shops (Petrosky-Nadeau, Wasmer, and Zeng 2016, table 1).

A second welfare cost of shopping arises from the vast amount of goods that are returned to retailers. In 2020 in the United States, $428 billion of merchandise was returned to sellers—10% of retail sales—because the goods did not fulfill the needs of the buyers; online retail alone accounted for $102 billion in returns, or 18% of online sales (Janssen and Williams 2024, p. 387). These returns have substantial welfare costs, including extra shipping but especially the depreciation of the returned good. Indeed, returned goods can only rarely be sold as new; they are often sold at a discount or even discarded (Reagan 2019).

Shoppers also face stockouts, which are visibly costly: buyers are generally upset when they drive to a shop, expecting to buy a product, but are unable to find it (Taylor and Fawcett 2001). In a field experiment conducted in a mail-order catalog, Anderson, Fitzsimons, and Simester (2006) find that customers subjected to a stockout reduce their spending on other items in the order, and reduce long-run spending on the catalog's products. As Okun (1981, p. 276) notes, shortages and stockouts push buyers to "react in ways that involve costs—ordering further in advance, making commitments that sacrifice flexibility, and the like." Some of the goods that are bought in advance for fear of stockouts are eventually not needed and wasted.

And some markets are worse than others for shoppers. In the US housing market, on average, prospective buyers spend 8.2 months searching for a home and visit 10.0 homes before buying (Genesove and Han 2012, table 1). In addition to time spent, people hire real-estate agents to find them a home. In 2019, the commissions paid to these agents across the US economy amounted to 1.2% of total personal consumption expenditures (Bianchi, McKay, and Mehrotra 2024, p. 25). These 1.2% spent on brokers represent a welfare cost, since they do not deliver utility directly—just like the amount spent by firms on recruiters.

Finally, firms also devote labor to buying goods and services. As such labor is diverted from producing goods and services, it imposes a welfare cost. In data recorded by the OES between 2003 and 2021, firms allocate 1.4% of employment to ordering, buying, purchasing, and procurement (Fernandez-Villaverde et al. 2025, p. 2508).

2.3. Prevalence of slack in the labor market

The welfare cost of slack is severe, but severity alone does not tell us how often slack arises, or how widespread it typically is. Several of the examples above—particularly the observations of depression-era clinicians and the New Haven interviews—are drawn from the most extreme instance of slack in modern US history. But slack is not confined to episodes of that kind. As we will see in the rest of the chapter, slack, on both the labor market and the product market, is a constant presence in the US economy, whose magnitude fluctuates with the business cycle without ever disappearing.

In this section, we start by documenting the prevalence of the most well-known form of slack: the unemployment of workers in the labor market.

2.3.1. Definition of unemployment

Before looking at the data, we review how unemployment is defined in the United States.

The definition of unemployment involves several criteria developed by the Bureau of Labor Statistics (BLS 2023). First, the BLS isolates the part of the population that makes up the working-age civilian noninstitutional population. This group excludes individuals in the armed forces, individuals institutionalized in mental or penal institutions, and children below 16. The resulting group consists of all potential workers residing in the United States—including foreigners.

The working-age population is then divided into two categories. The smaller category is people who are out of the labor force. These people do not have a job and are currently not available or interested in working: people who are retired, studying, or parenting at home.

The larger category is the labor force—everyone who is either working or available and willing to work. People who have a job are counted as employed. Formally, to be employed, you must have worked at least 1 hour in the past week as a paid employee or for your own business. People who are temporarily absent from work, on vacation or because of illness, are also counted as employed.

People who do not have a job are counted as unemployed. From a statistical perspective, people are classified as unemployed when they meet 3 criteria: not currently working; being available to work; and wanting to work. The BLS gauges desire to work by asking respondents if they have been actively searching for a job in the past 4 weeks. Actively searching for a job means using a method that has the potential to result in a job offer, such as contacting an employer, sitting for a job interview, submitting a job application, or using a placement agency. People who are temporarily laid off and expecting to be recalled to their job are also counted as unemployed.

2.3.2. Measurement of unemployment

How is unemployment measured in practice? Unemployment is currently measured through the Current Population Survey (CPS), which is a household survey conducted by the BLS. This survey has been running since 1940: it was initially administered by the Census Bureau, and it has been administered by the BLS since 1948. Accordingly, we have a continuous measure of unemployment since January 1948. So, between January 1948 and December 2024, we compute the unemployment rate in the usual way: we take the number of job seekers measured by the BLS (2025x) from CPS, and divide it by the labor force measured by the BLS (2025c) from the CPS. This is the standard, official measure of unemployment rate, labelled U3 by the BLS. Because the unemployment rate measures the share of workers who would like to work but cannot find a job, it is measured as a share of the labor force, not of the entire population.

An issue is that, because the CPS data only start in 1948, we cannot see what happened during the Great Depression—the worst recession on record. Fortunately, historians have collected data from different sources that provide a picture of what was happening at that time. Between January 1929 and December 1947, we measure unemployment from the unemployment rate constructed by Petrosky-Nadeau and Zhang (2021). They extrapolate Weir (1992)'s annual unemployment series to a monthly series using various monthly unemployment rates compiled by the NBER.[3]

The resulting series is graphed in figure 2.1: it depicts the unemployment rate for the United States between January 1929 and December 2024. The shaded areas in the graph correspond to the official recession dates in the United States, which are identified by the Business Cycle Dating Committee of the National Bureau of Economic Research (NBER 2023). The NBER identifies these recessions by looking holistically at numerous macroeconomic variables. Between 1929 and 2024, the NBER identifies 15 recessions.

2.3.3. Prevalence of unemployment

Over the past century, there is always some unemployment in the United States (figure 2.1).

The unemployment rate is countercyclical, rising sharply at the onset of all recessions. During the pandemic recession, the unemployment rate reached its highest level since the end of World War 2: it peaked at 13.0% (2020:Q2). During the Great Recession, the unemployment rate rose to 9.9% (2009:Q4). During the Volcker twin recessions, the unemployment rate rose to 10.7% (1982:Q4). During the first-oil-crisis recession, the unemployment rate reached 8.9% (1975:Q2). The Great Depression was on another scale: the unemployment rate reached 25.3% (1932:Q3). This amount of joblessness is something that we have not seen since—even during the pandemic recession or Great Recession.

US recessions over the past century have had different sources, but in each one of them the unemployment rate rose. For instance, the recession in 2020 was caused by the coronavirus pandemic. The Great Recession in 2007–2009 coincided with the global

[3] There is no monthly measure of unemployment between January 1890 and March 1929. Instead, the monthly unemployment fluctuations are inferred from the spread between the yields of bonds of different quality. Given these limitations, we only begin our analysis in 1929. To be able to start at the beginning of 1929, we use the monthly values produced by Petrosky-Nadeau and Zhang (2021) for January–March 1929, although they are not directly computed from unemployment data. An advantage of using Weir's unemployment series instead of older series is that it counts relief workers, employed by the various New Deal programs, as employed instead of unemployed. This is consistent with the modern definition of unemployment, according to which people with any type of job—relief or not—are counted as employed. By contrast, older series counted these workers as unemployed, adding a handful of percentage points to the unemployment rate (Darby 1976).

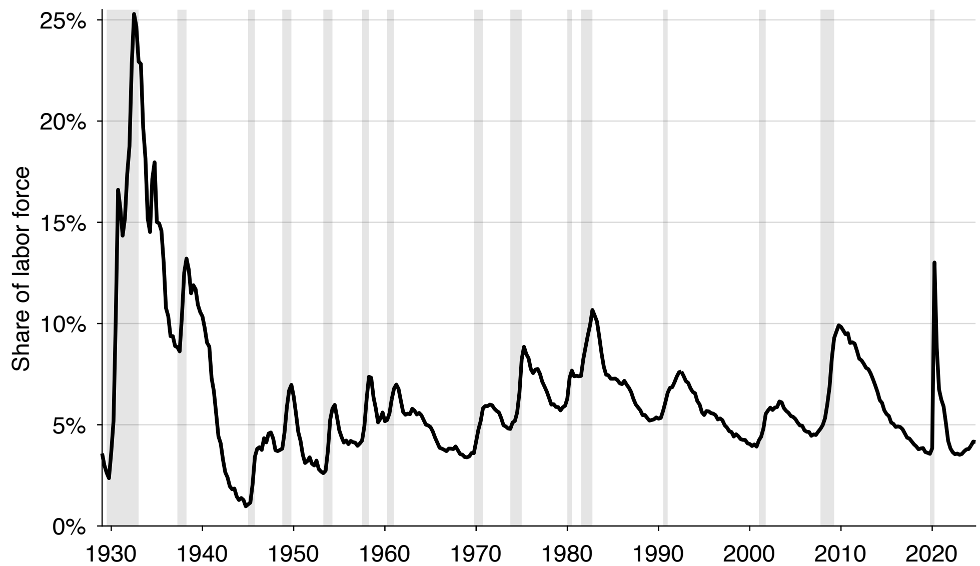


FIGURE 2.1. Unemployment in the United States, 1929–2024

The unemployment rate is a quarterly series computed from monthly, seasonally adjusted data produced by Petrosky-Nadeau and Zhang (2021) and the BLS (2025x,c). Shaded areas indicate recessions dated by the NBER (2023).

financial crisis. The recession in 2001 followed the burst of the dot-com bubble. The recession in 1990–1991 followed the Iraqi invasion of Kuwait and associated oil price shock. The twin recessions in 1980–1982 are associated with the tight monetary policy imposed by Volcker to fight inflation. The recession in 1973–1975 followed the first oil crisis. The biggest recession on record, the Great Depression in 1929–1933, was triggered by the October 1929 stock market crash, followed by a collapse of the banking system. In all these recessions the unemployment rate spiked.

Although there is more unemployment during recessions, unemployment exists at all times. The lowest unemployment rate on record is 1.0%, reached at the end of World War 2 (1944:Q4). This is low but still strictly positive: indeed, the unemployment rate has never reached zero. Such low unemployment was reached because of the war effort: it reduced unemployment through a combination of large-scale military production and large-scale enlistment of potential workers in the armed forces, which shrank the labor force. The World War 2 experience is typical: wartime generally leads to low unemployment. The unemployment rate fell to 2.6% at the end of the Korean War (1953:Q2), and it dropped to 3.4% in the middle of the Vietnam War (1969:Q1).

Overall, between 1929 and 2024, there are always some unemployed workers in the US economy. The unemployment rate fluctuates between 1.0% and 25.3% and averages 6.4% over the period. Over the more stable part of the period, 1948–2019, the unemployment rate averages 5.7%—lower but still very much positive.

2.3.4. Additional measures of unemployment

The concept of unemployment is meant to measure slack in the labor market. So in theory unemployment should comprise anyone who is available and willing to work but unable to find a job. The empirical challenge is to determine who wants to work. Reasonably, the BLS decided that somebody who wants to work is bound to search for a job, so the BLS asks people whether they have been searching for a job to determine whether they are unemployed. However, people search with different intensity and methods, maybe reflecting different willingness to work. The standard unemployment rate (U3) only counts as unemployed people who have been actively searching in the past 4 weeks (searching using a method that has the potential to result in a job offer). So there are workers who have been searching for a job in the past year but not in the past month who are not counted as unemployed. The job seekers might have stopped searching in recent weeks not because they do not want a job but because they could not find any appropriate jobs and lost hope. Similarly, there are people who have been searching for a job passively instead of actively who are not counted as unemployed. These job seekers have been using methods that could not result in a job offer unless additional steps are taken, for instance simply looking at job postings. These people may not have contacted firms because they did not see a good match, or because they were in training. To include all these job seekers into unemployment as well, the BLS developed additional measures of unemployment to count such workers.

The unemployment concept U4 includes everyone in the standard unemployment concept U3, plus discouraged workers. These are people who are available to start a job, have been actively searching for a job in the past 12 months, but have not been actively searching in the past 4 weeks because they became discouraged about their job prospects. When asked why they did not look for work during the last 4 weeks, discouraged workers respond for instance that "There are no jobs available," or "They have been unable to find work in the past." They are not counted in U3 because they did not actively search for work in the last 4 weeks.

The unemployment concept U5 includes everyone in U4 plus marginally attached workers. These are people who are available to start a job, have been actively searching for a job in the past 12 months, but have not been searching in the past 4 weeks for other reasons than discouragement about their job prospects. When asked why they did not look for work during the last 4 weeks, these workers respond for instance that they could not search because of family responsibilities, childcare, or illness. These additional workers are not classified in U3 because they did not actively search for work in the last 4 weeks; they are not classified in U4 because they were not discouraged about their job prospects.

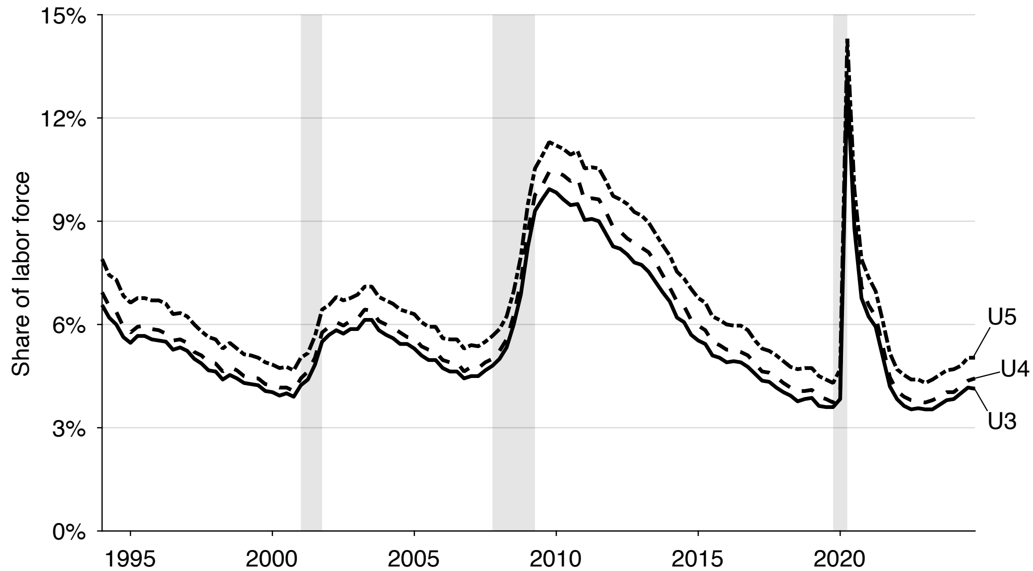


FIGURE 2.2. Alternative measures of unemployment in the United States, 1994–2024

The unemployment rates are quarterly averages of monthly, seasonally adjusted series computed by the BLS (2025y,v,w). Shaded areas indicate recessions dated by the NBER (2023).

The main takeaway is that the U4 and U5 rates are very close to the standard, U3 rate (figure 2.2). Between 1994 (when the U4 and U5 concepts were introduced) and 2024, the U3 rate averages 5.6%, the U4 rate averages 5.9%, and the U5 rate averages 6.6%. So on average the discouraged workers make up only 0.3% of the labor force, and the marginally attached workers make up just 1% of the labor force. Furthermore, these slivers of the labor force do not vary much over the business cycle. The distance between the U3 and U4 rates is always less than 0.8pp and the distance between the U3 and U5 rates is always less than 1.5pp.

A final form of unemployment is unemployed hours—originating from employees who work part time, although they were available and willing to work full time. Statistically, working part time means working less than 35 hours weekly, while working full time means working 35 hours or more. If a worker wants to work full time but can only find a part-time job, some of the hours that they want to supply are unemployed. However, such workers (part of the U6 concept of unemployment) are only a small sliver of the labor force. In 2024:Q4, only 0.7% of the labor force were working part time because they could not find a full-time job (BLS 2025g,c). And for each of these workers, only a part of their supply of 35 hours was unemployed.

2.4. Prevalence of slack in the product market

We have seen that slack is prevalent in the labor market. But slack is not limited to the labor market; it is also prevalent in the product market. Indeed, producers are never able to sell all the goods and services that they could potentially produce to customers.

2.4.1. Prevalence of unsold goods

Slack takes several forms in the product market. The most basic form is unsold goods, such as unsold food. Unsold food by farmers and manufacturers, and in grocery stores and restaurants, generates a vast amount of food waste in the United States (ReFED 2025). In 2023, 31% of the US food capacity (in tons) is surplus: either unsold, or unused by businesses, or uneaten at home. The value of the food unsold by farms, manufacturers, restaurants, and grocery stores represents 5% of the value sold.

It is not only food that US firms have difficulties selling: any good is difficult to sell. When firms produce nonperishable goods that they cannot sell, they generate unsold inventory. On average in the United States, in the manufacturing, wholesale, and retail sectors, only 51% of goods available for sale are sold in a given quarter (den Haan and Sun 2024, table 1). Moreover, the good-selling rate is highly procyclical: it is easier to sell goods in booms than in slumps (den Haan and Sun 2024, figure 1). As firms find it difficult to sell produced goods in slumps, the unsold goods pile up in inventory, so inventories swell up in slumps, and conversely shrink in booms (Bils and Kahn 2000).

A type of goods that is especially specialized, complex, and unique are capital goods, so they take significant time and effort to sell and to buy.[4] As a result, the market for capital goods exhibits notable slack: the unemployment rate of capital goods in the United States averages 11% and is sharply countercyclical (Ottonello 2026).

Relatedly, slack is well documented in the housing market: in the United States, hundreds of thousands of houses are for sale but not sold in any month (Realtor.com 2025a). Another manifestation of the slackishness of the US housing market is that it takes time to sell a house: the median time that a house stays on the market is always above 1 month, and sometimes as high as 3 months (Realtor.com 2025b).

2.4.2. Prevalence of idle capacity

Another form of slack in the product market is idle capacity in firms: capacity that is available and ready to produce goods and services, but remains unused because of a lack of demand. It is a type of slack that is less commonly discussed but still very prevalent, as we see from data collected by the Institute for Supply Management (ISM 2025). The

[4] See Gavazza (2011) for commercial airplanes and Ramey and Shapiro (2001) for aerospace equipment.

ISM keeps track of the "operating rate", both for manufacturing and nonmanufacturing industries. The operating rate indicates the actual production level of firms as a fraction of the maximum production level that these firms could have achieved given their current capital and labor. From this operating rate, we can create a rate of idleness, which is just 1 – operating rate. The idleness rate is the share of productive capacity that is idle because of a lack of demand: the amount of goods or services not sellable and therefore not produced, as a fraction of the total capacity of a firm given current capital and labor.

In the United States, we find that some capacity is idle at all times (figure 2.3). Between 1990 and 2024, the rate of idleness in the manufacturing sector averages 17.3%. And between 2000 and 2024, the rate of idleness in nonmanufacturing sectors averages 13.9%. This means that usually, given their capital and labor, firms could sell about 15% more than what they actually sell, if they had the demand for it.

The rate of idleness is the product-market equivalent of the rate of unemployment. Perhaps surprisingly, the rates of idleness in the manufacturing and nonmanufacturing sectors are much higher than the rate of unemployment. There are several ways to see this. The average rates of idleness are higher than the average rate of unemployment between 1990 and 2024, which is only 5.7%. The manufacturing rate of idleness is always above 10% and the nonmanufacturing rate of idleness is always above 8%. Both lower bounds on idleness correspond to already significant amounts of unemployment. Last, we can compare figure 2.1 to figure 2.3: we can see that at any time the unemployment rate is below the rates of idleness. The data show us that more labor is idle within firms than outside them: product market slack is a critical phenomenon.

Just like the rate of unemployment, the rates of idleness are sharply countercyclical, climbing in recessions. Idleness in the manufacturing sector peaked at 33.0%, during the Great Recession (2009:Q2). Manufacturing idleness was also elevated during the other recessions in the data: the idleness rate reached 20.6% after the Iraq War recession (1991:Q2), 22.5% after the dot-com bubble recession (2001:Q4), and 24.1% after the pandemic recession (2020:Q2). Slack in the nonmanufacturing sectors peaked at 26.7%, during the pandemic recession (2020:Q2). Nonmanufacturing idleness also reached 19.9% after the Great Recession (2009:Q2) and 16.9% after the dot-com bubble recession (2001:Q4). We see that idleness is a much bigger problem during recessions. We also see that the Great Recession was relatively worse for manufacturing sectors, while the pandemic recession was relatively worse for nonmanufacturing sectors.

2.4.3. When product market slack bleeds into the labor market

We have just seen that people employed by firms are idle some of the time at work because firms do not have enough business. In that case product market slack materializes as idleness at work. If workers are sent home by firms and their hours reduced, then product

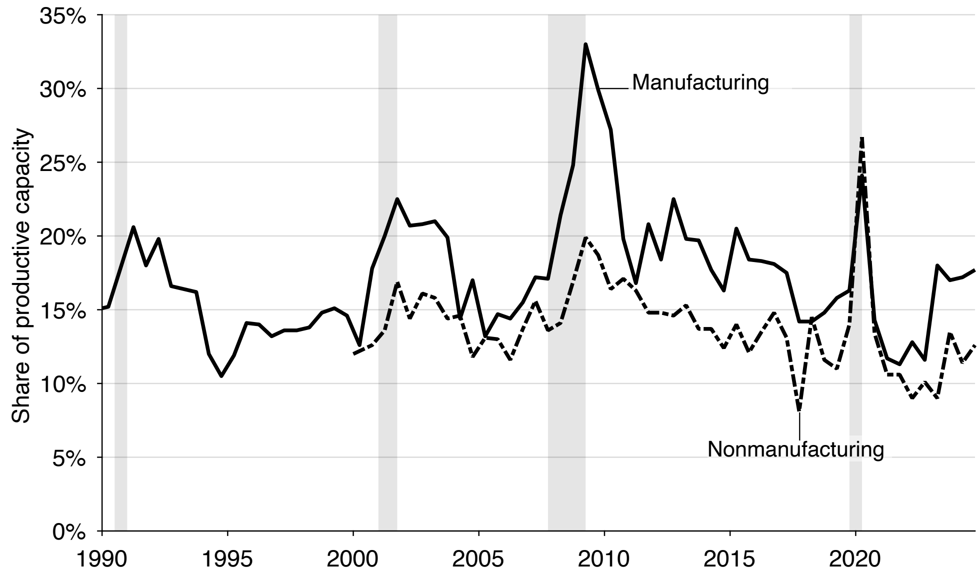


FIGURE 2.3. Idle capacity in the United States, 1990–2024

The idleness rates are quarterly series constructed by interpolating the semiannual operating rates measured by the ISM (2025). For nonmanufacturing sectors, the operating rate is only available after 2000. Shaded areas indicate recessions dated by the NBER (2023).

market slack materializes in another way: as full-time workers being forced to work part time because their employers do not have sufficiently many customers. These workers are counted as employed, but they are flagged as involuntary part-time workers in the CPS by the (BLS 2025h). Their usual hours of work are above 35 hours a week but they are now forced to work less than 35 hours because of unfavorable business conditions or low seasonal demand.

Again, this is a form of product market slack, since the issue is that the workers' labor services cannot be sold to customers—not that workers cannot sell their labor to firms. These workers are another component of the U6 unemployment concept measured by the (BLS 2023), but they are best interpreted as product market slack.

We see that many people are working part-time because of slack in the product market (figure 2.4). On average between 1956 and 2024, 2.1% of the labor force is working less than full time because of product market slack; that share is always above 1%. Just like the slack rate in figure 2.3, that share of workers is sharply countercyclical: much higher in bad times than in good times. During the Great Recession, the share spiked to 4.4% (2009:Q2); during the pandemic recession, it spiked to 5.8% (2020:Q2).

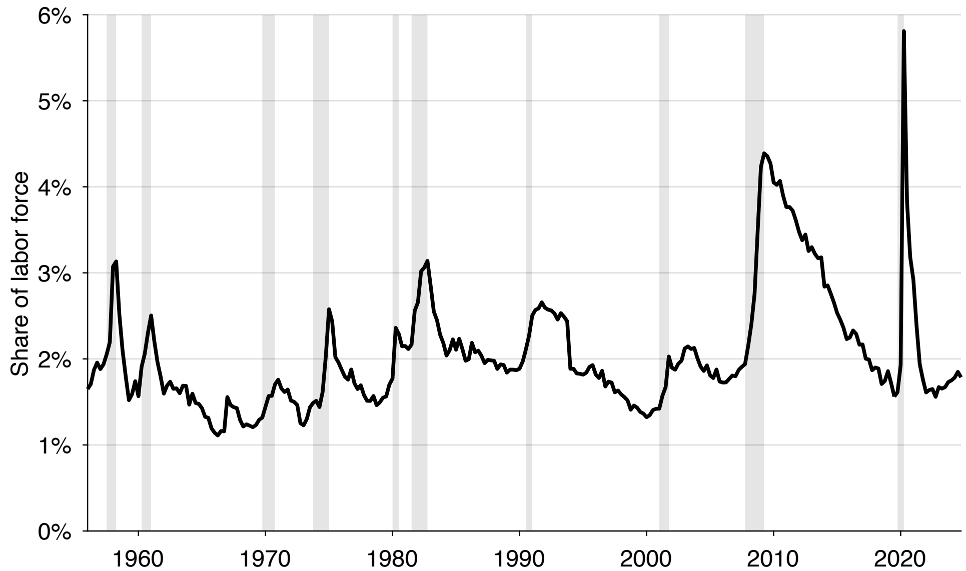


FIGURE 2.4. Workers forced to reduce hours because of product market slack in the United States, 1956–2024

The share of workers forced to reduce hours is a quarterly series computed from monthly, seasonally adjusted data produced by the BLS (2025h,c). Shaded areas indicate recessions dated by the NBER (2023).

2.5. Coexistence of sellers and buyers

We have observed slack in the labor market in the form of unemployed workers and slack in the product market in the form of idle capacity. To explain these patterns, we could simply use a nonclearing Walrasian model—a Walrasian model with a non-market-clearing price.[5] With an above-market-clearing price in the product market, there would be excess supply of goods and services, and we could explain why there are firms that have goods and services that are not sold. With an above-market-clearing wage in the labor market, there would be excess supply of labor, and we could explain why there are workers who want to work but are not employed.

However, the world is more complicated than that because just as there are sellers who are unable to sell their goods or services, we see buyers who are unable to purchase goods or services. The coexistence of sellers and buyers is inconsistent with the nonclearing Walrasian model. In a situation of excess supply, such a model can explain the existence of slack. But in excess supply, buyers are on their demand curves, so they are able to purchase the amount of goods or services that they desire at the prevailing price. In other words, all buyers can successfully and seamlessly purchase the amount of goods and services that they desire.

[5] Benassy (1993) offers a wonderful survey of that literature.

We now review evidence of this conundrum. We start with the labor market, where the presence of job vacancies has been well documented. We then turn to the product market, where the evidence is less readily available. In the book, to explain the coexistence of sellers and buyers that are unable to trade, we will organize each market around a matching function. The matching function will mediate all trades on the market, capturing the difficulties that sellers and buyers alike face in finding each other.

2.5.1. Coexistence of job seekers and vacant jobs

In the labor market, we always have unemployed workers trying to find jobs, as we have seen, but we also always have firms trying to fill vacant jobs. The presence of vacant jobs is the concrete manifestation that firms are looking to hire labor.

How can we assess whether firms want to hire workers at any point in time? Well, we have to look at the number of job vacancies that firms report. Job vacancies are measured through the Job Openings and Labor Turnover Survey (JOLTS), which is a firm survey conducted by the BLS since January 2001.[6] The definition that the BLS (2024a) adopted for job vacancy parallels the definition of an unemployed worker. There are three conditions. First, a specific position exists, and there is work available for that position. Second, the job must start within one month. And third, there is active recruiting for that position for workers outside the establishment. So each vacant position is associated with recruiting within firms, which ensures that firms are actively looking for new workers. Hence, between January 2001 and December 2024, we compute the vacancy rate from the number of job openings measured by the BLS (2025i) from the JOLTS. We divide the number of job openings by the size of the labor force, which is measured by the BLS (2025c) from the CPS, to express the number of vacant jobs as a share of the labor force.

There are no governmental measures of job vacancies in the United States before 2001, but other organizations have been collecting data on vacant jobs before that, and researchers have transformed these data into vacancy series. Between January 1951 and December 2000, we use the vacancy rate produced by Barnichon (2010) from the Conference Board's help-wanted advertising index. The Conference Board index aggregates help-wanted advertising in major metropolitan US newspapers. Given the shift from print advertising to online advertising with the advent of the internet, the Conference Board index needs to be adjusted in the 1990s, which Barnichon (2010) has done. Between January 1929 and December 1950, we use the vacancy rate produced by Petrosky-Nadeau

[6] To best align CPS and JOLTS data, we shift forward by one month the number of job openings from JOLTS. For instance, we would assign to October 2024 the number of job openings that the BLS assigns to September 2024. The motivation for this shift is that the number of job openings from the JOLTS refers to the last business day of the month (Monday 30 September 2024), while the labor force from the CPS refers to the Sunday–Saturday week including the 12th of the month (Sunday 6 October 2024 to Saturday 12 October 2024). So the number of job openings refers to a day that is closer to next month's CPS week than to the current month's CPS week. With this shift, the JOLTS conveniently starts in January 2001 instead of December 2000.

and Zhang (2021) from MetLife's help-wanted index. This index aggregates help-wanted advertisements from newspapers across major US cities, and reliably proxies for job vacancies (Zagorsky 1998).[7]

A first natural question is whether help-wanted indices proxy well for job vacancies. Comparing them to local data on job vacancies and other available evidence, (Abraham 1987) and (Zagorsky 1998) find that both the Conference Board index and the MetLife index proxy well for job vacancies. A second natural question is how the number of job vacancies can be recovered from an index? This is done by scaling the indexes to eventually match the number of job vacancies at the beginning of the JOLTS: the Conference Board index is scaled to align with the JOLTS vacancy rate in January 2001, and the MetLife index is scaled to align with the scaled Conference Board index in January 1951. Scaling effectively translates the indexes into vacancy rates.

Next, we splice the three vacancy series to create a continuous vacancy rate covering January 1929–December 2024 (figure 2.5). The series is procyclical, rising sharply in expansions. In the recovery from the coronavirus pandemic, the vacancy rate reached its highest level on record: it peaked at 7.2% (2022:Q2). Just before the Great Recession, the vacancy rate reached 3.1% (2007:Q2). Just before the burst of the dot-com bubble, the vacancy rate peaked at 4.1% (2000:Q1). Before the Volcker recessions, the vacancy rate was even higher, spiking at 5.1% (1979:Q2). Job vacancies are generally elevated in wartime: the vacancy rate reached 6.7% at the end of World War 2 (1945:Q1), 4.3% during the Korean War (1953:Q1), and 5.1% during the Vietnam War (1969:Q1).

While it is true that job vacancies exist in large numbers during expansions, vacant jobs exist at all times, which tells us that firms are always looking for some workers, just like some workers are always looking for jobs. Even in recessions, the vacancy rate remains positive. The lowest vacancy rate on record is 0.7%, reached in the depth of the Great Depression (1933:Q1). This is a very low number, but still positive. During the Great Recession, the vacancy rate dwindled to 1.5% (2009:Q3). The vacancy rate bottomed at 2.2% after the dot-com bubble recession (2003:Q2) and at 2.3% after the Volcker twin recessions (1982:Q4). The vacancy rate was also extremely low in the recessions that followed the Great Depression: 0.8% after the 1937–1938 recession (1938:Q2) and 1.6% after the 1948–1949 recession (1949:Q4).

Overall, between 1929 and 2024, the vacancy rate averages 3.2%. The vacancy rate reached its highest level on record, 7.2%, during the recovery from the pandemic and its lowest level on record, 0.7%, during the Great Depression. The bottom line is that, just as there always are unemployed workers, there always are vacant jobs.

[7] Petrosky-Nadeau and Zhang (2021) produce a vacancy series that starts in 1919, but we start the vacancy series in 1929 to align it with the unemployment data from figure 2.1.

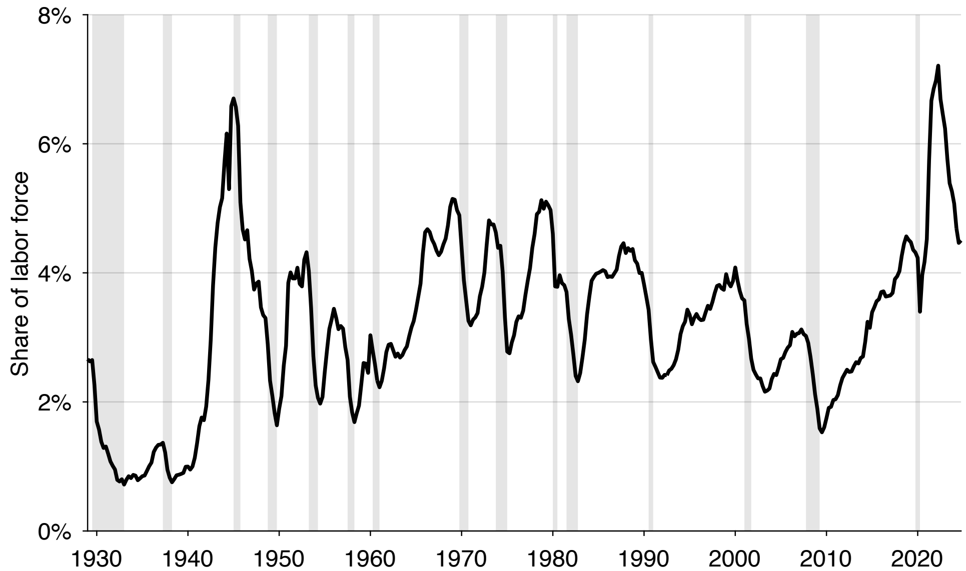


FIGURE 2.5. Job vacancies in the United States, 1929–2024

The vacancy rate is a quarterly series computed from monthly, seasonally adjusted data produced by Petrosky-Nadeau and Zhang (2021), Barnichon (2010), and the BLS (2025i,c). Shaded areas indicate recessions dated by the NBER (2023).

2.5.2. Beveridge curve

We have seen in figures 2.1 and 2.5 that the number of job seekers is countercyclical while the number of vacant jobs is procyclical. This implies that the unemployment and vacancy rates are negatively related. This negative relationship between unemployment and vacancy rates corresponds to the Beveridge (1944) curve: a key empirical relationship that guides our modeling and policy design throughout the book.

The Beveridge curve appears clearly on a scatter plot with unemployment rate on the horizontal axis and vacancy rate on the vertical axis (figure 2.6). Here to illustrate, we only use recent data, between 2000 and 2019. We stop before the pandemic to simplify the picture. The Beveridge curve says that when the economy is in a slump, there are a lot of job seekers and few vacant jobs. Good examples of such situations are 2003:Q2, when the unemployment rate reached 6.2% and the vacancy rate fell to 2.2%, and 2009:Q4, when the unemployment rate peaked at 9.9% and the vacancy rate plummeted to 1.6%. Conversely, when the economy is in a boom, there are few job seekers and many vacant jobs. Such a situation occurred in 2019:Q4, when the unemployment rate dipped to 3.8% and the vacancy rate reached 4.3%.

Although the Beveridge curve was first observed in Great Britain (Dow and Dicks-Mireaux 1958), it holds remarkably well in the United States (Blanchard and Diamond

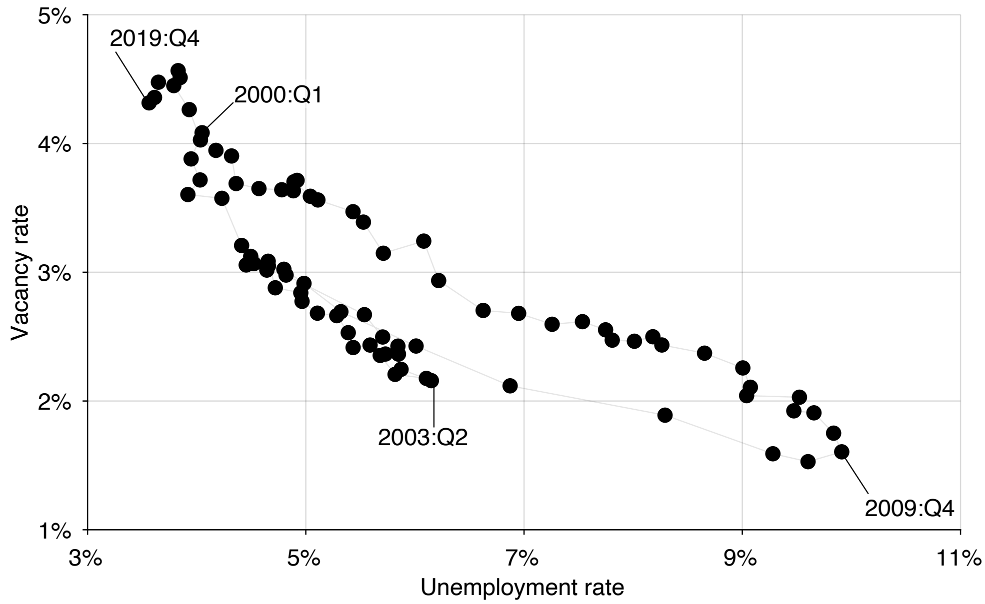


FIGURE 2.6. Beveridge curve in the United States in recent decades, 2000–2019

The unemployment rate comes from figure 2.1 while the vacancy rate comes from figure 2.5.

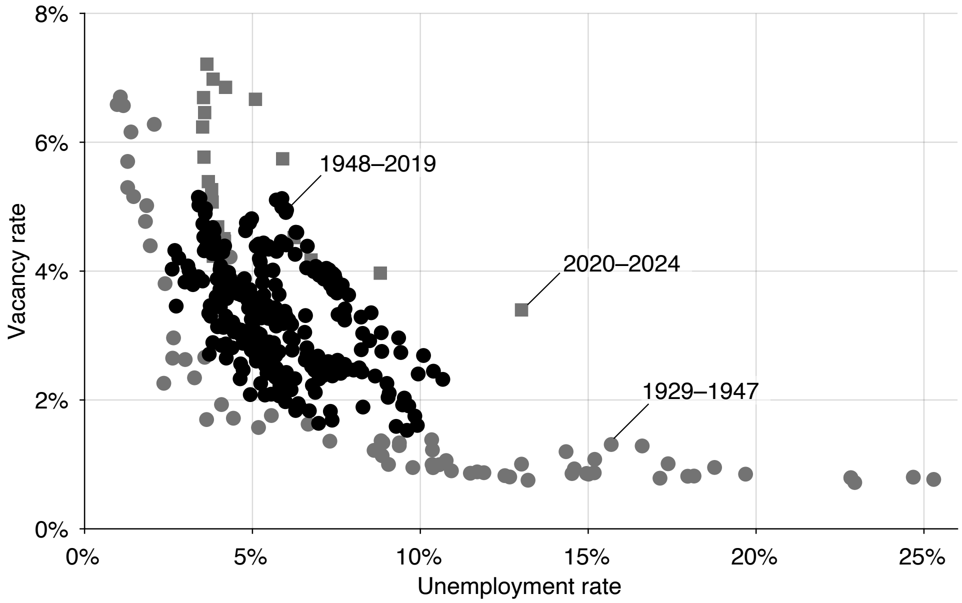
1989; Elsby, Michaels, and Ratner 2015).

To illustrate, figure 2.7 displays the Beveridge curve for the entire period of our data, 1929–2024. The figure offers two displays: a regular display (figure 2.7A) and a display on a log scale (figure 2.7B), which is useful to show that the Beveridge curve is isoelastic—that there is a linear relationship between log vacancy and log unemployment.

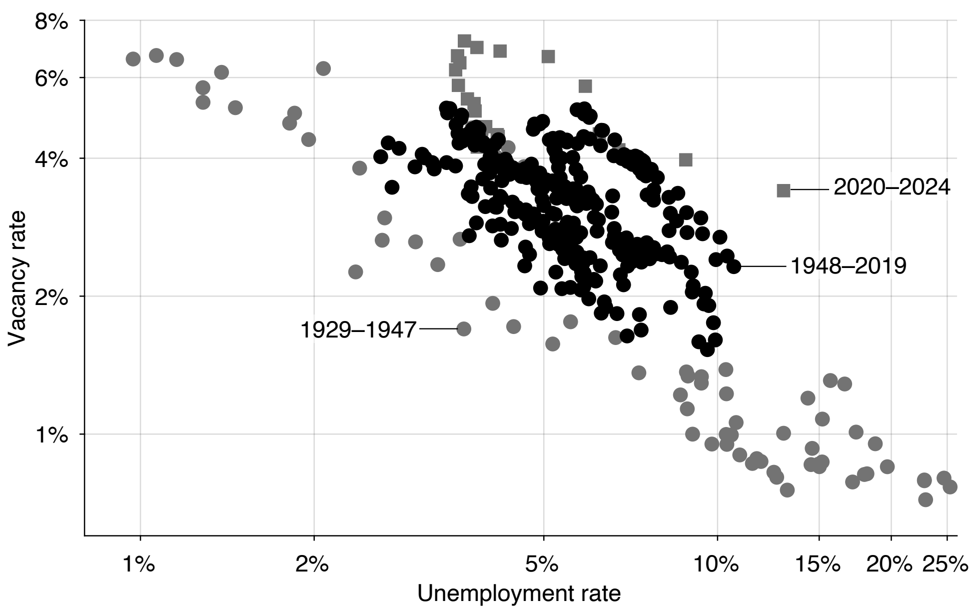
The figure distinguishes three subperiods. The initial period, 1929–1947, was extremely volatile, not least because it experienced two dramatic events, the Great Depression and World War 2. Data available for the period are also the noisiest since neither unemployment nor job vacancies were measured by the government. Around the middle of the Great Depression and at the end of World War 2, the Beveridge curve is subject to two large, outward shifts. These shifts are hard to discern on this figure but will be seen clearly in chapter 12.

In the next, postwar period, 1948–2019, the Beveridge curve is well-behaved. The US Beveridge curve is stable for long periods of time, before shifting inward or outward after a decade or more (Michaillat and Saez 2021a, figure 5). The structural breaks in the curve are of modest size compared to those that occurred in the 1929–1947 period. The economy is always tightly on the Beveridge curve.

Finally, the last subperiod corresponds to the coronavirus pandemic and its aftermath, 2020–2024. The pandemic shattered the US labor market and dramatically shifted the Beveridge curve outward. The Beveridge curve slowly recovered from the pandemic shift



A. Regular scale



B. Logarithmic scale

FIGURE 2.7. Beveridge curve in the United States, 1929–2024

The unemployment rate comes from figure 2.1 while the vacancy rate comes from figure 2.5.

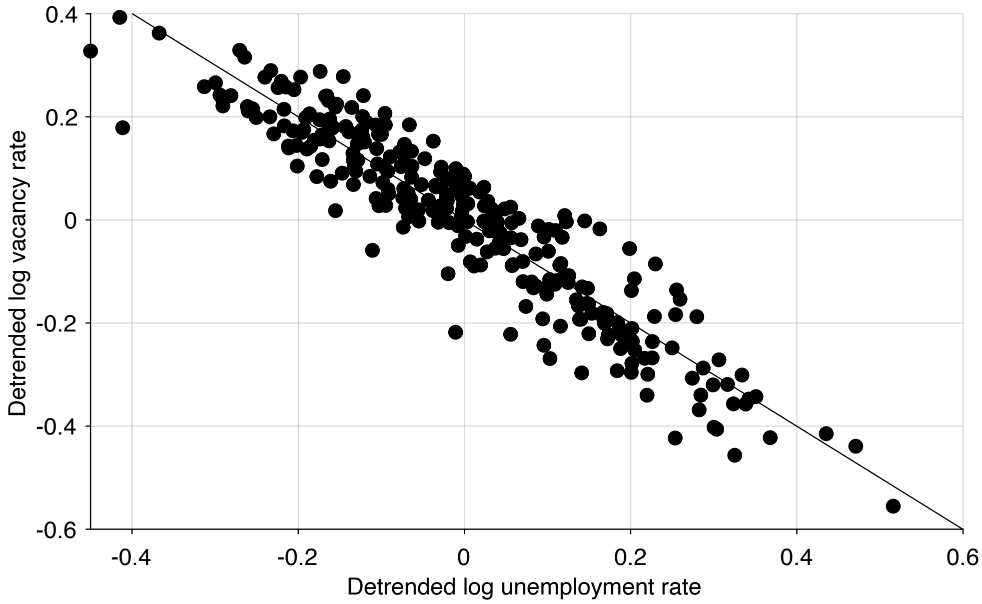


FIGURE 2.8. The US Beveridge curve is a rectangular hyperbola

The unemployment rate comes from figure 2.1 while the vacancy rate comes from figure 2.5. The figure plots the logarithm of the vacancy rate against the logarithm of the unemployment rate, both detrended using an HP filter with smoothing parameter 10,000. The data are for 1948–2019.

over the next few years and it only came back close to its pre-pandemic position in 2024.

In fact, unemployment and vacancy rates appear not only to be negatively related, but to be the inverse of each other. So doubling the unemployment rate cuts the vacancy rate in half, and conversely, doubling the vacancy rate cuts the unemployment rate in half. Mathematically, the property that the unemployment rate and vacancy rate are inversely related implies that the Beveridge curve is a rectangular hyperbola:

$$vu = A,$$

where $A \in (0, 1)$ is a constant. The rectangular hyperbola can be rewritten $v = A/u$, so we can see that the elasticity of the vacancy rate with respect to the unemployment rate along the Beveridge curve is -1.

It is possible to establish that the Beveridge curve is close to a rectangular hyperbola formally. This can be done by estimating the elasticity of the vacancy rate with respect to the unemployment rate along each of its branch. An elasticity of -1 corresponds to a rectangular hyperbola. It turns out that during the postwar period, the elasticity of the Beveridge curve on each branch remains between -0.84 and -1.02, so never far from -1 (Michaillat and Saez 2021a, figure 6).

It is possible to make the case for a hyperbolic Beveridge curve without introducing structural breaks and looking to branches in isolation. The idea is to remove a very slow-moving trend from the unemployment and vacancy rates to eliminate the decennial movements of the Beveridge curve, and to then estimate the elasticity of the detrended vacancy rate with respect to the detrended unemployment rate. To be precise, we remove from log unemployment rate a slow-moving trend obtained from a Hodrick-Prescott (HP) filter with smoothing parameter 10,000.[8] We do the same to the log vacancy rate. We then regress detrended log vacancy rate on detrended log unemployment rate. We focus on the 1948–2019 period because the large shifts in the Beveridge curve before and after that period are difficult to handle. We find that the slope is exactly −1.00 with a good fit, $R^2 = 0.87$ (figure 2.8). Thus, with detrended variables, the US Beveridge curve is a perfect rectangular hyperbola over seven decades, 1948–2019.

2.5.3. Coexistence of product sellers and buyers

In the product market, we have seen that sellers are continuously trying to sell goods and services; at the same time, buyers are continuously trying to purchase them.

Every day, US consumers spend a significant amount of time trying to buy goods and services: researching their shopping, shopping online and in person, driving to shops, and waiting in line at shops.

A first reason why buying takes time is that it is difficult to find the appropriate good that fits one's needs. This difficulty manifests itself in the large number of goods that are returned after purchase because they did not meet the buyer's needs—as noted above, 10% of all retail sales are eventually returned to sellers.

Another reason why shopping takes time is that goods are not always available in shops. Using the microdata underlying the Consumer Price Index, Bils (2016, table 1) finds that temporary stockouts are quite common: the average stockout rate is 4.6% across 180 categories of goods between 1988 and 2009. Taylor and Fawcett (2001, figure 1) surveyed the availability of 40 items across 20 US stores for 4 weeks. They find that for advertised items, stockouts occur on 12%–16.5% of the visits; for regular items, stockouts occur on 6.1%–7.6% of the visits. Stockouts are also prevalent for online retailers. Jing and Lewis (2011, table 1) find that 25.4% of orders placed to an online retailer were imperfectly filled—some of the items that were ordered could not be shipped to the customer. The value of the items in stockout represents 9.7% of the total value of the orders. Thus, overall, a notable share of the goods that shoppers seek are temporarily unavailable.

The activity of purchasing goods and services is not limited to households. Firms are also constantly trying to purchase goods and services from other firms. In fact, the amount of labor devoted by firms to source goods and services might not be very different

[8] Appendix C provides an introduction to the HP filter and explains the choice of the smoothing parameter. We use the HP filter with that smoothing parameter throughout the book, whenever detrending is required.

from the amount of labor devoted to recruiting workers. According to the Occupational Employment Survey (OES) run by the BLS, on average between 1997 and 2012, 560,600 workers were employed in buying, purchasing, and procurement occupations while 543,200 workers were employed in occupations involving recruitment, placement, screening, and interviewing (Michaillat and Saez 2015, figure 2B).

2.6. Prevalence of long-term relationships

The difficulties in matching sellers and buyers that create slack also incentivize sellers and buyers to form long-term relationships once they have found each other. Indeed, once buyer and seller are locked into a long-term relationship, they can trade seamlessly, without having to find new trading partners each time. In this section we show that long-term employment relationships are prevalent in the labor market, and long-term customer relationships are prevalent in the product market, providing indirect evidence of underlying matching difficulties.

2.6.1. Long-term employment relationships

US firms typically employ workers for long periods of time. Kalleberg (2001, table 8.2) finds that in 1997, 71% of men and 62% of women are engaged in standard employment, in which the work is full time and the relationship is assumed to continue for a substantial period or indefinitely. In addition, 8% of men and 22% of women are engaged in regular part-time employment, which is also expected to continue for a substantial period or indefinitely. So overall 79% of employed men and 84% of employed women are in long-term employment relationships. Workers who are not in long-term employment are for instance independent contractors (although these workers might be in long-term relationships with their customers) or workers in temporary-help agencies. These numbers have not changed much between the 1960s and 1990s (Kalleberg 2001, table 8.3). Between the 1990s and 2010s, the number of workers in alternative work arrangements increased by a few percentage points, driven partly by the emergence of the gig economy (Katz and Krueger 2019, table 2).

To assess how long employment relationships generally last, we turn to the job-separation rate. The rate indicates the number of worker-firm relationships that end in a given month, as a share of the total number of existing relationships. It is computed by the (BLS 2025u) from JOLTS data since 2001. As earlier, we omit data from the coronavirus pandemic and its aftermath because they are not representative of the typical behavior of the US labor market (the pandemic led to a dramatic, unprecedented increase in job separations).

Overall, we see that the job-separation rate is not that large, which tells us that people keep their jobs for quite a long time (figure 2.9A). Between 2001 and 2019, the monthly job-separation rate averages 3.6%. This means that every month, 3.6% of workers are willingly or unwillingly leaving their job. Assuming that job separations are a Poisson process, we infer that the average duration of a job is 1/0.036 = 28 months, or about 2.3 years.

A Poisson process is memoryless, which means that irrespective of how long a worker has been with the firm, conditional on being employed today, they can expect to be there for about 2 more years. In the US labor market, however, memorylessness does not hold. Initially, when a worker starts a new job, the separation risk is quite high. Then, as a worker's tenure with the firm increases, the separation risk falls drastically. As a result, workers who have stayed a while with a firm are likely to stay much longer still—they are likely to spend their career at the firm.

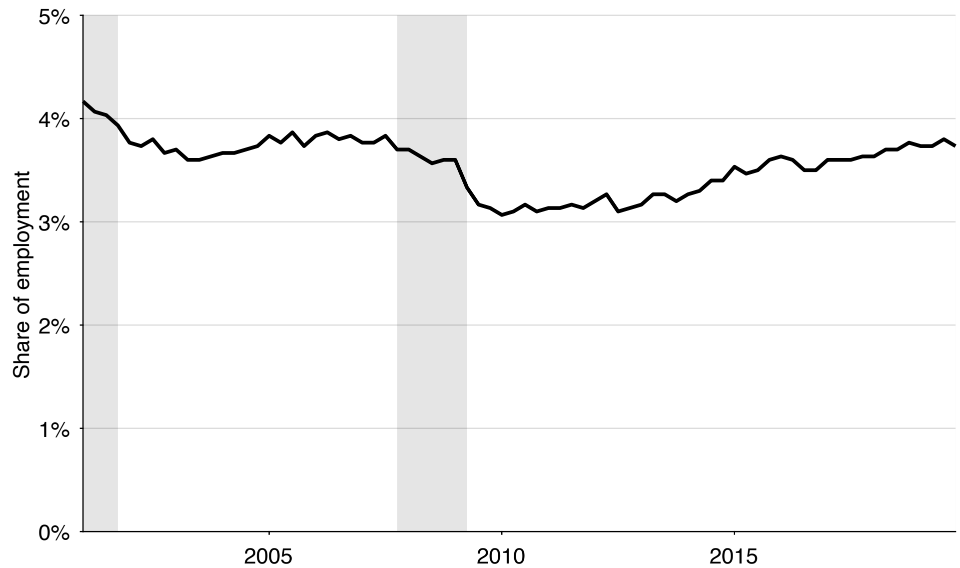
Indeed, Hall (1982) found that in the US in the 1960s and 1970s, the average worker kept their job for about 8 years, and 25% of workers held jobs that would last at least 20 years. Hall (1982, p. 724) explains that "The jobs held by middle-aged men are remarkably stable. . . . Monthly separation rates, which are of the order of 3% for workers in general, are about 0.25% for middle-aged men with at least ten years on the job." Moreover, the duration of US jobs has remained remarkably stable since the 1980s (Molloy, Smith, and Wozniak 2024, figures 1 and 2).

An interesting aspect of the job-separation rate is that it is stable over the business cycle. People typically assume that waves of job separations occur at the onset of each recession, but in recent decades, that has not really been the case: if anything the job-separation rate dropped during the course of the dot-com recession and Great Recession. How can that be? Do firms not lay workers off at the onset of recessions? It used to be true that the layoff rate would spike at the onset of recessions (Akerlof, Rose, and Yellen 1988, figure 2). But between 2001 and 2019, the layoff rate increases only slightly during recessions (figure 2.9B). Moreover, the rise in layoffs in recessions is offset by a drop in quits (see figure 2.9B and Akerlof, Rose, and Yellen 1988, figure 1). Overall, because quits (initiated by workers) are procyclical while layoffs (initiated by firms) are countercyclical, total job separations are close to acyclical.

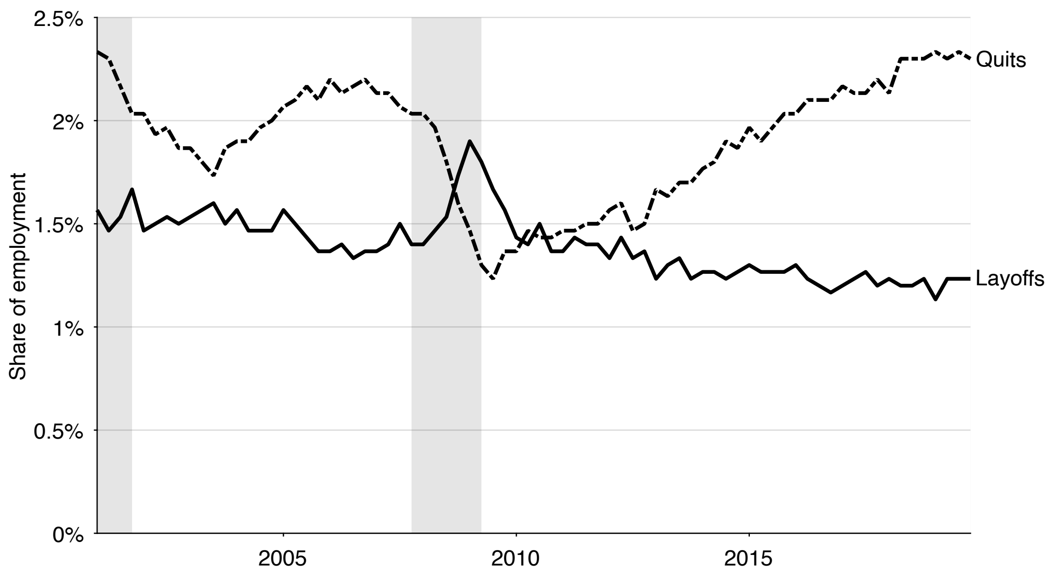
2.6.2. Long-term customer relationships

Although this is maybe less well known, long-term customer relationships are extremely common in the product market. Blinder et al. (1998, Table 4.12) report that on average, 85% of sales by US firms go to long-term customers—either other businesses or consumers.

Indeed, long-term customer relationships are exceedingly common in business-to-business transactions. A broad share of business-to-business transactions are conducted under explicit contracts that link firms over time. Among US firms trading intermediate



A. All job separations



B. Quits and layoffs

FIGURE 2.9. Monthly job separations in the United States, 2001–2019

The job-separation rates are quarterly averages of monthly, seasonally adjusted series computed by the BLS (2025u,m,r). Shaded areas indicate recessions dated by the NBER (2023).

goods and services, one-third of all transactions are conducted under contract (Goldberg and Hellerstein 2011). And more than 80% of US imports occur through long-term relationships (Monarch and Schmidt-Eisenlohr 2023, table 1). Fernandez-Villaverde et al. (2025, pp. 2509–2510) confirm that most relationships between firms and their suppliers in the United States are long term: the average customer-supplier relationship has been ongoing for 3.5 years.

Long-term customer relationships between consumers and sellers are also common. Dube, Hitsch, and Rossi (2010) documents that consumers have a higher probability of choosing products that they have purchased in the past—thus remaining loyal to the product and brand that they patronized in the past. But, as Okun (1981, pp. 150–151) observes, personal relationships with sellers are even more important than relationships with brands:

> The model of customer-supplier attachments applies most obviously to suppliers of services that are not easily evaluated by shopping. The resulting list is long and important in terms of the consumer's budget, including medical practitioners like the physician, dentist, and veterinarian; personal services like those of the barber, beautician, and dry cleaner; repair services of the automobile mechanic, house painter, plumber, and electrician; financial service suppliers like the lawyer, accountant, insurance agent, and branch banker. The considerations extend to goods as well as services . . . the consumer wants a reliable optician, pharmacist, antique dealer, butcher, and clothier.

2.7. Cyclicality of slack

This book is about business cycles. In it we will see that slack not only exists but also fluctuates. And fluctuations in slack are a central part of business cycles.

But what are business cycles? Here is a standard definition of business cycles, proposed by Burns and Mitchell (1946, p. 3):

> A cycle consists of expansions occurring at about the same time in many economic activities, followed by similarly general recessions, contractions, and revivals which merge into the expansion phase of the next cycle; this sequence of changes is recurrent but not periodic; in duration business cycles vary from more than one year to ten or twelve years; they are not divisible into shorter cycles of similar character with amplitudes approximating their own.

The typical measures of activity used to describe business cycles include industrial production, output, and consumption. They all move in tandem, deviating from trend by a few percent, up or down, at each expansion and recession (Stock and Watson 1999).

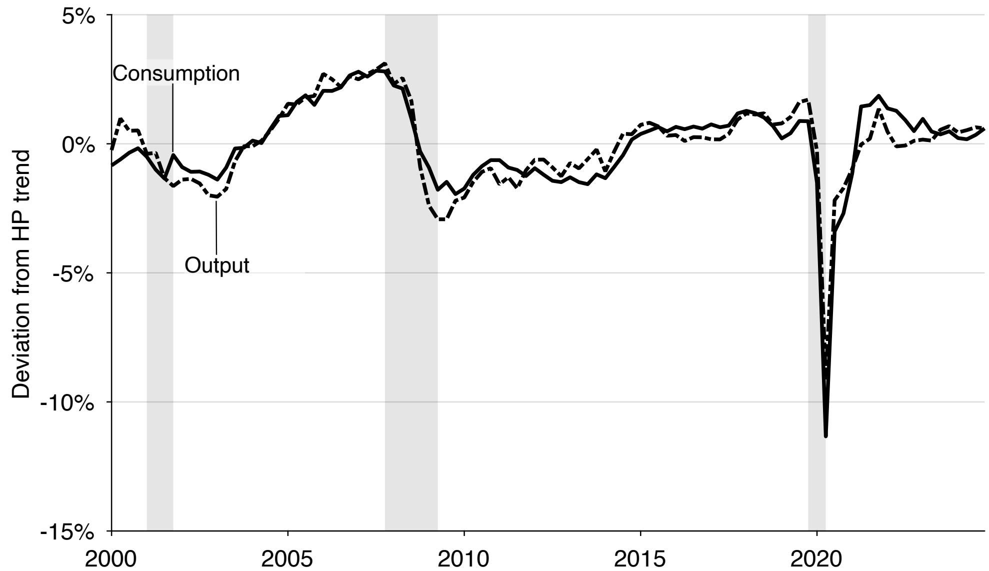


FIGURE 2.10. Fluctuations in consumption and output in the United States, 2000–2024

Output is real gross domestic product measured by the BEA (2025b). Consumption is real personal consumption expenditure measured by the BEA (2025c). The original quarterly, seasonally adjusted series are detrended by applying to their logarithms a HP filter with smoothing parameter of 10,000. Shaded areas indicate recessions dated by the NBER (2023).

The business cycle fluctuations to be explained are presented in figure 2.10. We recognize the typical pattern: output and consumption are above trend in expansion, and below trend in recessions. In the 21st century, output and consumption move essentially in tandem (this was not necessarily the case before). Between 2000 and 2024, the correlation between detrended output and consumption is 0.92.

Figure 2.10 addresses a possible concern about the approach taken in the book. Output is the sum of private and public consumption, private and public investment, and net exports. The models in the book abstract from imports and exports, and from capital and investment. While one might worry that a model abstracting from trade and investment could not explain output fluctuations, the figure shows that private consumption moves so closely with output that our focus is well-suited to the task.

The goal of business cycle research is to develop a model that can explain the fluctuations described by Stock and Watson (1999). The pervasiveness and cyclicality of slack suggest that business cycles may reflect fluctuations in slack in addition to fluctuations in productive capacity. Indeed, by definition, slack is the share of the economy's productive capacity that is either not used or not sold, so

$$\text{output} = (1 - \text{slack}) \times \text{productive capacity}. \tag{2.1}$$

This simple slack identity shows that there are two possible and nonexclusive sources of output fluctuations: fluctuations in productive capacity and fluctuations in slack.

There are three inputs into productive capacity. First is the capital stock. Second is the labor force: the share of the working-age civilian population that is available and willing to work. Third is technology: the blueprint to the production process, which shows how to combine capital and labor effectively. These inputs are combined through a production process to generate the productive capacity. We will see in the next sections that all these inputs, and therefore productive capacity, are broadly acyclical. This will leave fluctuations in slack as the main explanation for business cycle fluctuations.

2.7.1. Capital is broadly acyclical

How much does the capital stock vary over time and, in particular, over the business cycle? This is a fairly easy question to answer: we just need to look at capital stock at constant national prices for the United States graphed from 1950 to 2019 (University of Groningen and University of California–Davis 2023). It is clear that the capital stock is growing over time, but notice that the growth of the capital stock is extremely smooth—it is essentially unaffected by recessions or by the business cycle in general. This is not a new finding. If we go back to the first chapter of the main textbook of the Real Business Cycle literature, *Frontiers of Business Cycle Research*, one of the findings is that capital stock fluctuates much less than output and is largely uncorrelated with output (Cooley and Prescott 1995). Thus, we conclude that the US capital stock is acyclical.

2.7.2. Labor force is broadly acyclical

Next, how much does labor supply vary over the business cycle? The labor force is the labor supply. It is composed of the members of the working-age civilian population who want to work. Mathematically, the labor force is the product of the working-age civilian population and the labor force participation rate.

The working-age civilian population, of course, is growing at a very stable rate and not subject to cyclical fluctuations (BLS 2025q). What we need to look at is the share of the civilian population that wants to work: the labor force participation rate. This is the main decision that people make in the real world: whether to participate in the labor force or not.

It turns out that the participation rate does not really vary over the business cycle (BLS 2025j). The participation rate is molded by medium-run fluctuations that do not have much to do with the business cycle. Until the 1970s, the rate was fairly stable with some small fluctuations. There was then a big increase in the participation rate between the mid 1960s and the mid 1990s as women entered the labor force in large numbers (BLS

2025l). After the mid 1990s, there was a plateau followed by a decline in the participation rate from the 2000s. The participation rate for men has been declining since the 1950s, but this decline was more than compensated until the 1990s by women entering the labor force (BLS 2025k).

The medium-run forces might make it difficult to ascertain that the labor force participation rate is truly acyclical—although by looking separately at the participation rates for men and women, it is quite clear that they are acyclical (BLS 2025k,l). In any case, to be completely sure, researchers have explored the cyclicality of the participation rate with statistical techniques. Using US data covering 1946–1954, Rees (1957, p. 32) does not find evidence of the discouraged-jobseeker theory. In US data covering 1960–2006, Shimer (2009, p. 294) finds that the labor force participation rate is acyclical. Similarly, using US data spanning 1976–2009, Rogerson and Shimer (2011, pp. 624–625) find that over the business cycle, the labor force participation rate is nearly constant. In US data for 1976–2016, Cairo, Fujita, and Morales-Jimenez (2022, figure 1C) find that in the short-to-medium run, the labor force participation rate does not respond to productivity shocks (the typical shocks in the business cycle literature).

It is true that labor force participation dropped around the Great Recession (Erceg and Levin 2014), but the decline was primarily caused by population aging and other trends that preceded the recession (Aaronson et al. 2014; Krueger 2017). The only recession that led to a big change in the participation rate was the pandemic recession. When the pandemic started, there was a large drop in the participation rate, from 63% down to 60%. This drop was something we had never seen before. Workers decided to drop out of the labor force in large numbers because it became dangerous to work, and school closures imposed new childcare obligations. The participation rate recovered once working became safe again and schools reopened.

2.7.3. Technology is broadly acyclical

Technology, which is a blueprint to effectively use capital and labor, is likely to evolve even more smoothly than the capital stock. Think about a restaurant: it is clearly easier to buy an additional microwave oven than design new technology for the microwave. Technology is therefore something that we take as fixed over the business cycle.

Although it is difficult to say for sure, technology is likely acyclical. The invention process is slow and random. The diffusion process is slow and random. The depreciation process—the loss of know-how—is also slow and random. We can look at some proxies for technology to convince ourselves. The number of transistors per microprocessor grows steadily, without cyclical fluctuations (Rupp 2023). The same is true of the number of operations that can be carried out per second by the fastest existing supercomputers (Dongarra, Luszczek, and Petitet 2024).

It is true that technology is typically measured to be procyclical. But this is because technology is measured as a residual, whose fluctuations might be driven by slack (section 2.7.5).

2.7.4. Slack is sharply countercyclical

We saw previously that slack on the labor and product markets was countercyclical: higher in bad times than in good times. We now examine whether such fluctuations could possibly explain fluctuations in output—which are the most common marker of business cycles. We focus on the last 25 years of data, from 2000:Q1 to 2024:Q4, both because slack data are only fully available since 2000, and because the link between slack and output became tighter in the 21st century.

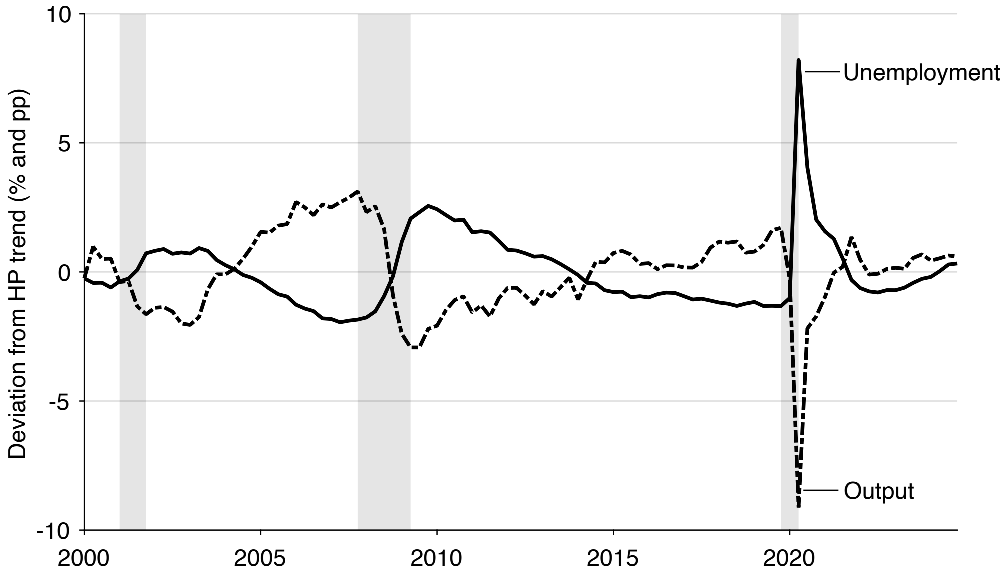
We first look at the connection between output and labor market slack. In figure 2.11A we superimpose cyclical fluctuations in output with those in unemployment. It appears that these fluctuations in unemployment could easily explain those in output since they are synchronized with them and of comparable amplitude. The correlation between the two series is almost −1, at −0.90.

In figure 2.11B we recast the results from figure 2.11A in the form of an Okun (1963) curve. The Okun curve relates short-run movements in output and unemployment. In the United States, between 1948 and 2013, output and unemployment rate are negatively correlated in the short run: an increase in the unemployment rate by 1pp is associated with a reduction in output by about 2% (Ball, Leigh, and Loungani 2017). Between 2000 and 2024, the connection is instead one-for-one: a 1pp increase in unemployment is associated with a reduction in output by about 1%.

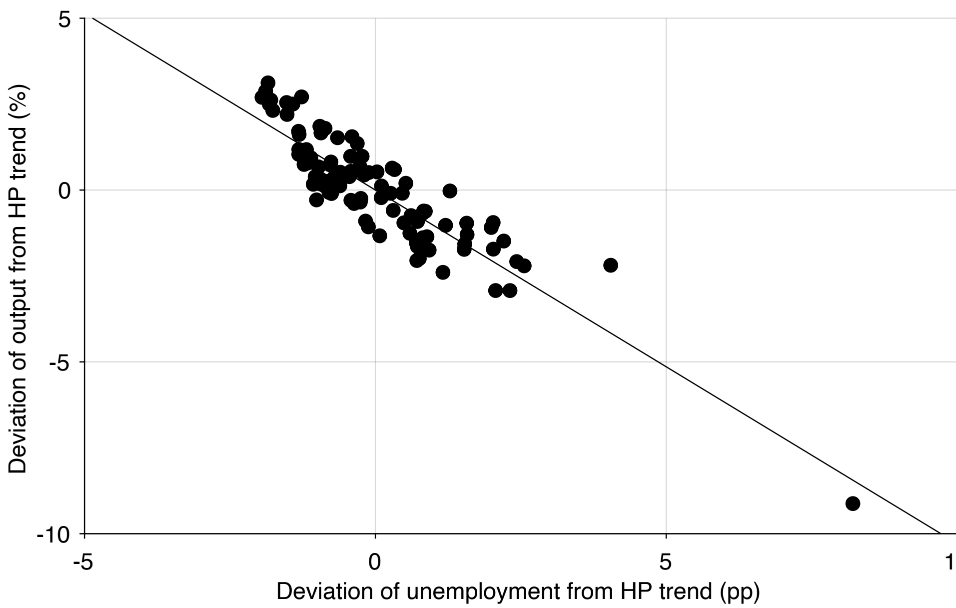
We also examine the connection between fluctuations in product market slack and output, by plotting product-market Okun curves. Fluctuations in product market slack also contribute to fluctuations in output, but because these slack data are noisier, the connection is not as tight as in the case of unemployment.

Figure 2.12 correlates short-run fluctuations in output with those in nonmanufacturing idleness. The fluctuations are quite negatively correlated, with a correlation coefficient of −0.72. So when there is more idle capacity in the nonmanufacturing sectors, output falls below trend. For the Okun curve relating output and nonmanufacturing idleness, the slope is −0.49: an increase in nonmanufacturing idleness by 1pp is associated with a reduction in output by roughly 0.5%.

The link between output and idleness is weaker in the case of manufacturing (figure 2.13). Output does fall below trend when manufacturing idleness rises, but the correlation between the two series is only −0.65. This is not so surprising, given that manufacturing now only employs less than 10% of the US workforce, and produces less than 15% of US GDP (Baily and Bosworth 2014, figure 1). For the Okun curve linking output



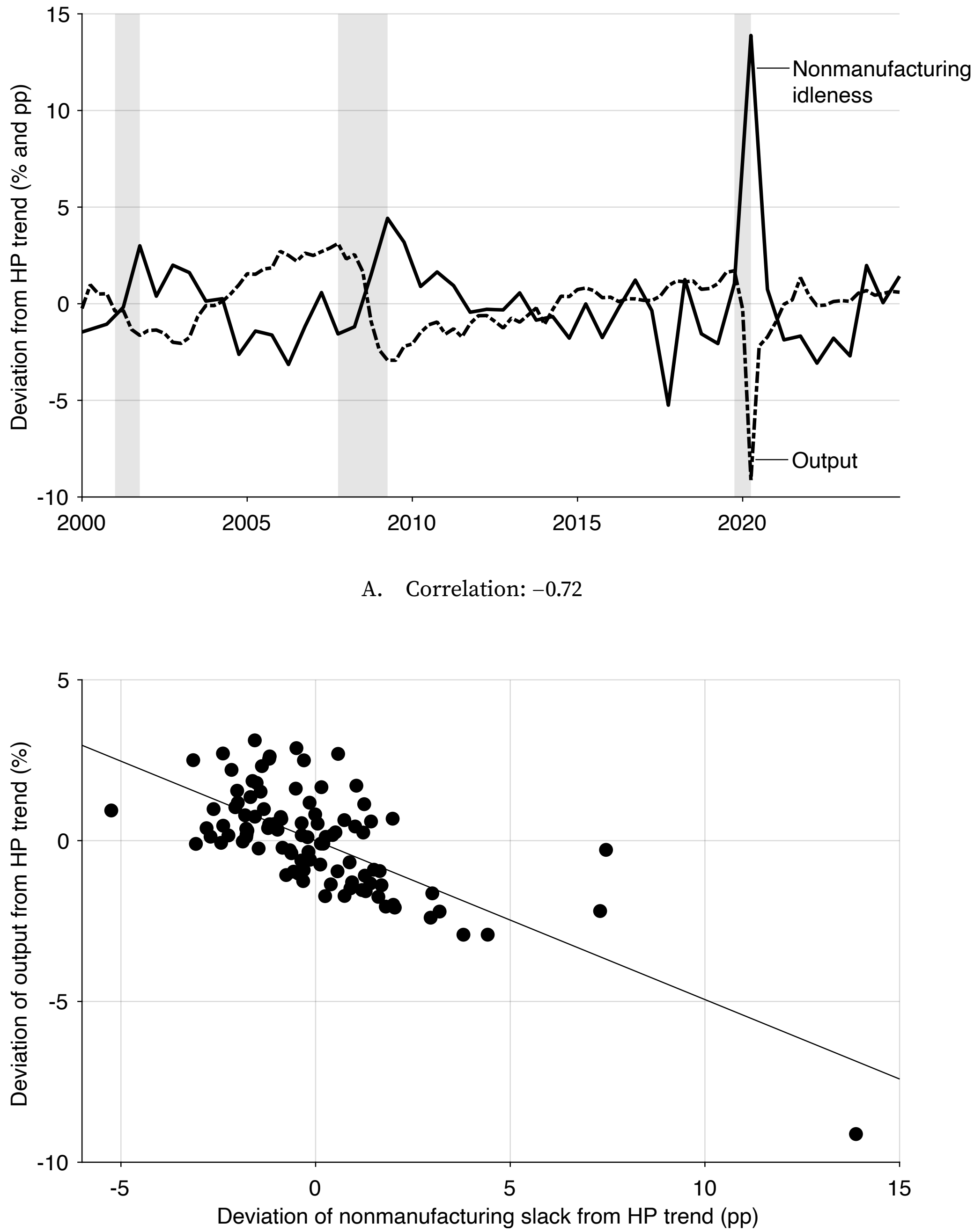
A. Correlation: −0.90



B. Estimated slope: −1.03

FIGURE 2.11. Okun curve with unemployment in the United States, 2000–2024

Detrended output comes from figure 2.10. Unemployment comes from figure 2.1. Unemployment is detrended by applying a HP filter with smoothing parameter of 10,000. Shaded areas indicate recessions dated by the NBER (2023).

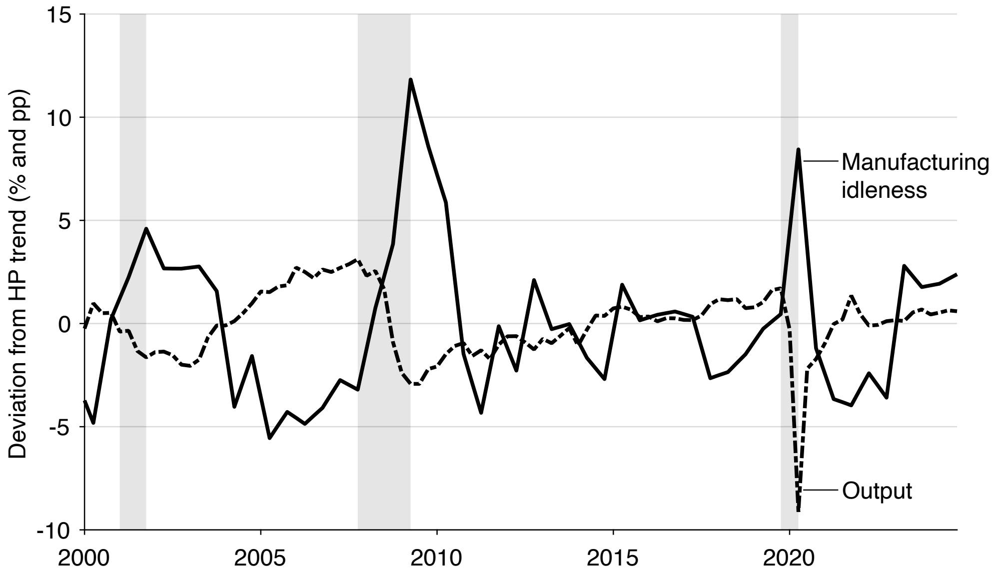


A. Correlation: −0.72

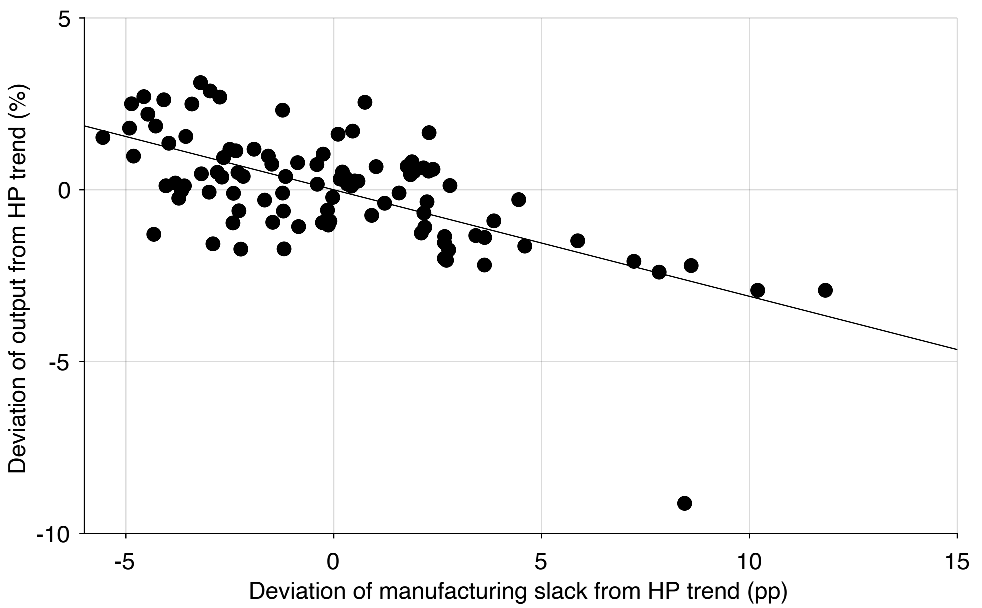
B. Estimated slope: −0.49

FIGURE 2.12. Okun curve with nonmanufacturing idleness in the United States, 2000–2024

Detrended output comes from figure 2.10. Nonmanufacturing idleness comes from figure 2.3. Idleness is detrended by applying a HP filter with smoothing parameter of 10,000. Shaded areas indicate recessions dated by the NBER (2023).



A. Correlation: −0.65



B. Estimated slope: −0.31

FIGURE 2.13. Okun curve with manufacturing idleness in the United States, 2000–2024

Detrended output comes from figure 2.10. Manufacturing idleness comes from figure 2.3. Idleness is detrended by applying a HP filter with smoothing parameter of 10,000. Shaded areas indicate recessions dated by the NBER (2023).

to manufacturing idleness, the slope is −0.31: an increase in the rate of manufacturing idleness by 1pp is associated with a reduction in output by about 0.3%.

2.7.5. The illusion of procyclical technology

Furthermore, the framework suggests that procyclicality in measured technology, which is a cornerstone of modern business cycle models, may be an illusion created by unmeasured variations in slack. The evidence of the procyclicality of technology comes from measuring technology as a Solow residual. In the Solow procedure, output, capital, and labor are measured, but technology is inferred. The fluctuations in output that cannot be explained by fluctuations in capital and labor are residually assigned to fluctuations in technology.

In our slackish model, however, there can be fluctuations in output even if capital and labor are fixed, because the rate of utilization of capital and labor varies over time, as shown in equation (2.1). Thus, fluctuations in technology, measured as a residual, could very well be fluctuations in slack. If we are in a world with slack and perform the Solow procedure to measure technology, variations in utilization will be interpreted as variations in technology. For instance, if customers stop eating out in bad times, then restaurant workers will be idle more of the time, so measured technology—the amount of meals sold per worker—will decrease, although the restaurant technology has not changed. Fewer meals are sold not because the ovens stopped working but because fewer people showed up to dine in the restaurant.

Basu, Fernald, and Kimball (2006) propose a more sophisticated Solow procedure that attempts to remove the fluctuations in labor and capital utilization that pollute measured technology. Their purified measure of technology is much less variable than standard measured technology—as anticipated. In fact, accounting for variable utilization has such an impact that measured technology turns countercyclical—and can therefore in no way be a major driver of business cycles.[9]

2.8. Lack of progress in stabilizing slack

Before we enter the main part of the book, it is important to recognize the scope for policy: despite major progress in other areas, very little progress has been made in stabilizing slack in general, and unemployment in particular.

Politicians and policymakers traditionally worry about three issues: output and its growth, inflation, and unemployment. The US economy has experienced a tremendous

[9] The countercyclicality of measured technology might be because good measures of slack are lacking. Basu, Fernald, and Kimball (2006) do observe hours per worker, which is one form of product market slack, as discussed in section 2.4.3. But they do not observe idle labor or idle capital in firms. Instead, they must impute them by assuming that underutilization of labor and capital is voluntary, to minimize costs. In a world with slack, underutilization is involuntary, due to insufficient demand. Reproducing their procedure with direct measures of idle labor and capital might well yield acyclical technology.

amount of growth in the last decades of the 19th century and during the 20th century. Over the entire period, per-capita output has grown at an average rate of around 2% per year Jones (2002, figure 1). With such growth and finite human needs, it seems that the amount of output available for consumption is sufficient in most places. Consumption is not distributed equally across people, of course, because US income and wealth remain unequal.[10] But the total amount of consumption available seems sufficient.

Another common macroeconomic woe is inflation. The United States has struggled with inflation in the past: it was for instance a big issue in the 1970s. And inflation imposes noticeable welfare costs, primarily because people are angered by inflation.[11] Since the 1990s, however, US inflation has been stable around low levels. For more than a quarter century, between 1994 and 2020, core inflation remained between 0.7% and 2.6%, despite tumultuous events that included the boom and bust caused by the dot-com bubble and the Great Recession (BEA 2025a). By contrast, in the previous quarter century, between 1969 and 1993, core inflation varied between 2.5% and 10.1%. So the Federal Reserve has made tremendous progress in taming inflation. Obviously, US inflation reached excessive levels in the aftermath of the coronavirus pandemic, but it subsided to about 3% in 2024.

Unlike other macroeconomic problems, little progress has been made about unemployment. The problem of unemployment manifests itself in several ways. First, over time, the average level of unemployment has not noticeably fallen (figure 2.1). Moreover, the peaks of unemployment continue to reach high levels. Of course, unemployment has never reached again the levels from the Great Depression—but the pandemic recession saw the highest peak in unemployment since 1940, and the Great Recession the third highest. Last, for certain socioeconomic groups, unemployment remains stubbornly high (Cairo and Cajner 2018, figure 1).

This lack of progress is likely due to the absence of slack in modern business cycle models: slack has been forgotten in the Real Business Cycle and New Keynesian models. That is why slack is the focus of this book. With a framework centered on slack, hopefully, policy will improve and fluctuations in unemployment and other forms of slack will subside, just like fluctuations in inflation have.

2.9. Summary

In this chapter we first asked why economic slack matters. Slack wastes productive resources, since goods and services that could generate value remain unsold. Moreover, in the labor market, unemployment imposes severe nonpecuniary costs, because many people derive identity, status, and daily structure from work. And the presence of slack is

[10] See for instance Piketty and Saez (2003), Saez and Zucman (2016), and Piketty, Saez, and Zucman (2018).
[11] See for instance Okun (1981, pp. 278–289), Shiller (1997), and Stantcheva (2024).

associated with matching efforts—such as recruiting and shopping—that absorb resources which could otherwise be used to produce goods and services.

We then examined how prevalent slack is. We found that slack is a pervasive feature of the US economy, not just a phenomenon of recessions. In the labor market, unemployment is always present, averaging about 6% since 1929 and spiking during recessions. In the product market, slack is even more prevalent: on average, firms operate with about 15% of capacity that is idle. Because slack is so prevalent, the Walrasian model is not well-suited to describe real-world markets.

An empirical puzzle is the coexistence of sellers and buyers who cannot find each other. In the labor market, this is captured by the Beveridge curve, which shows a negative relationship between unemployed workers and vacant jobs. In the product market, we also always have buyers trying to purchase goods and services, together with goods and services for sale. This coexistence of buyers and sellers on markets points to a fundamental matching problem in the economy. Because of this coexistence of sellers and buyers, we cannot use the nonclearing Walrasian model and must develop a new theory of slack.

Last, the chapter argued that business cycles correspond to fluctuations in slack, not in productive capacity. Inputs like capital stock, labor force, and technology are largely acyclical. In contrast, labor and product market slack is sharply countercyclical, and fluctuations in slack are of the same magnitude as output fluctuations. Despite the welfare cost of slack, however, little progress has been made in stabilizing it, which motivates the theory developed in the rest of the book.

PART II.

Slackish markets

CHAPTER 3.

Matching function

In slackish markets, all trades are mediated by a matching function. These markets therefore differ from Walrasian markets, where all trades are mediated via an auction, and have sharply different properties.

In this chapter we introduce the matching function, which is the key tool we use to model slackish markets and study slack and unemployment. The matching function summarizes the complex process through which buyers find sellers: for instance, how workers searching for jobs meet firms searching for employees, and how firms searching for customers meet consumers searching for goods and services.[1] Pissarides (2000, pp. 3–4) explains the role of the matching function wonderfully in the case of the labor market, but this book's argument is that his description in fact applies to the vast majority of markets. Slightly generalizing his words:

> Trade in [any] market is a nontrivial economic activity because of the existence of heterogeneities . . . and information imperfections. If all [buyers and sellers] were identical to each other, and if there was perfect information about their [attributes], trade would be trivial. But without homogeneity on either side of the market and with costly acquisition of information, [buyers and sellers] find it necessary to spend resources to find [desirable] trades. . . . The matching function gives the outcome of the investment of resources by [buyers and sellers] in the trading process as a function of inputs. It is a modeling device that captures the implications of the costly trading process without the need

[1]See Petrongolo and Pissarides (2001, pp. 425–427) for a detailed history of the matching function.

to make ... the features that give rise to it explicit.

After having introduced a generic matching function and discussed its properties, we will cover several specific matching functions.

3.1. Why do we need a matching function?

At any point in time, in almost any market, we have buyers and sellers who want to trade, but are not able to trade immediately. In the labor market, there are workers who want to sell their labor and firms who want to buy labor and yet, even though these two groups coexist, it takes time for workers to find a job and for firms to fill a vacant job. On the housing market, families trying to sell their houses coexist with families looking for a house to buy—but it takes time for families to sell their homes and for other families to find a new home. On the car market, there are cars that sit in inventory and people who are looking to buy cars. It takes time for people to find a car, and it also takes time for the cars in inventory to be sold. On the service market, there may be babysitters who are looking for work, and at the same time, there are families looking to hire—it still takes time for them to find the right babysitter. There are restaurant tables that are empty and hungry people looking for a good restaurant to eat at.

The bottom line is that in almost all markets, buyers who want to buy and sellers who want to sell always coexist—it takes time for the transaction to occur. To model such markets, we cannot use a Walrasian market. In the Walrasian market, one can always buy and sell any quantity of a good at the market price, immediately and without constraints. So a Walrasian market ignores the fact that it takes time and effort to buy and sell in most real-world markets.

To model this complex trading process, we introduce a tool called the matching function. The matching function is an aggregate function that captures and summarizes, at the macro level, all the complexities of trading that happen at the micro level. It is very similar to the production function, which summarizes at the aggregate level all the production that happens at the micro level. It is a simple, well-behaved function that depends on only a few aggregate variables.

Most trading is quite complicated—it is hard for buyers and sellers to get together to execute a trade that both sides desire. For instance, in the labor market, a worker with specific skills would look for a job with specific requirements, whether it be a specific industry, specific location, or specific working conditions or responsibilities. Similarly, a firm advertising a vacant job would have specific needs and would be looking for a worker with the right skill set, the right experience, the right qualifications and character.

This complexity is not limited to the labor market. It would also occur when firms trade with other firms. When firms look for suppliers for a specific part in a product that they are building or for a specific tool, they have a long list of technical requirements because that part is unique to their product or the tool is unique to their manufacturing process. They also have additional constraints on the quality of the part or tool, its reliability, the supplier's location and how it operates. The production capacity of the supplier and its existing commitments also matter, as they determine whether the supplier can produce the part or tool rapidly.

And the complexity of trading is not confined to the labor market or to sophisticated goods and services traded by firms. Even for seemingly simple goods and services purchased in everyday life, trading is often complex because buyers have heterogeneous needs and sellers offer differentiated products. Consider something as simple as a haircut. Hairdressers specialize: for women, men, children; for budget or high-end styles. Hairdressers have more or less experience, are more or less skilled, and are more or less talkative. Hair salons have different numbers of chairs and existing customers so they are more or less busy. And customers' preferences differ. Hours of operation and locations may or may not fit a customer's schedule and daily routine. As a result, it takes time and effort for customers to obtain a suitable haircut, and conversely, for hairdressers to attract and retain customers.

This complexity exists in most markets, so the matching function can be applied to almost all markets in which trading is complex. The only exceptions are auction markets, where the goods that are traded are available in large quantity and are homogeneous. Then, as all goods are the same, finding the right seller is not relevant. Goods traded on such auction markets include some securities—especially stocks—and some commodities—silver, gold, wheat, or cotton (Okun 1981, p. 134). These markets are well represented by a Walrasian model; in fact they are exactly the sort of markets that inspired Walras (Walker 1990, p. 653). But very few markets operate that way; in practice, most goods and services offered for sale are different, require shopping by buyers, and take time to sell for sellers. Even in the financial world that inspired Walras, many markets are better modeled using a matching function than a Walrasian apparatus: these include over-the-counter markets for securities—including corporate and government bonds, derivatives, and some stocks—but also credit markets.[2]

3.2. Definition of the matching function and market tightness

The matching function is an aggregate function that summarizes all the micro-level complexities of the trading process. Consider a market for a specific good with $s > 0$ sellers and $b > 0$ buyers, open for one period. Each seller places one good for sale, and each buyer aims to purchase one good. The matching function $m(s, b)$ gives the number of trades in

[2] See Wasmer and Weil (2004) and (Hugonnier, Lester, and Weill 2025).

the period.

Before going further, we introduce a new, fundamental variable: the market tightness, defined by

$$\theta = \frac{b}{s}.$$

On any market, the tightness is the ratio between the number of buyers and the number of sellers—the two inputs into the matching function.

Market tightness plays a crucial role in the book because it summarizes the state of any market. This is because, under the assumptions that we make, the trading probabilities given by the matching function only depend on market tightness.

Finally, we introduce the matching elasticity to characterize the shape of the matching function. The matching elasticity is the elasticity of the matching function with respect to the number of sellers:

$$\eta = \frac{s}{m(s, b)} \cdot \frac{\partial m(s, b)}{\partial s}. \tag{3.1}$$

The matching elasticity gives the percentage change in the number of trades when the number of sellers increases by 1%. Under our assumptions, the matching elasticity is a function of market tightness, $\eta(\theta)$.

3.3. Assumptions about the matching function

We make a few assumptions about the matching function, to reflect natural properties of markets and ensure that the resulting market model is well behaved. Researchers often assume specific functional forms for their matching functions—typically a Cobb-Douglas function. But most models continue to behave in a similar fashion as long as the matching function satisfies the assumptions listed here. Throughout the book, we stick to these assumptions.

First, the matching function is assumed to be twice continuously differentiable. This ensures that the matching function is smooth enough.

Second, naturally, the matching function is assumed to be strictly positive.

Third, the matching function is assumed to be zero without buyers or sellers: $m(s, b) \to 0$ when $s \to 0$ or $b \to 0$. This is also natural, as people are needed on both sides of the market for trades to occur.

Fourth, the matching function is assumed to be strictly increasing in each argument, s and b. This means that if there are more sellers in the market, or more buyers in the market, there are more trades. This is natural too: if more goods are available, chances are that a greater number of buyers find an appropriate good to buy; conversely, if there are more buyers on the market, chances are that a greater number of sellers find a buyer for their good.

Fifth, the matching function is assumed to have constant returns to scale:

$$m(zs, zb) = zm(s, b) \tag{3.2}$$

for any $z \geq 0$. This assumption critically simplifies the analysis, because it guarantees that the trading probabilities can be expressed as functions of market tightness. Because of constant returns to scale, the elasticity of the matching function with respect to the number of buyers is directly connected to the matching elasticity:

$$\frac{b}{m(s, b)} \cdot \frac{\partial m(s, b)}{\partial b} = 1 - \eta. \tag{3.3}$$

This result is a direct application of Euler's theorem for homogeneous functions (result A.2).

Sixth, the matching function is assumed to be strictly concave in at least one argument, s or b. Because of constant returns to scale, this assumption implies three other properties of the matching function (appendix D). First, the matching function is strictly concave in each argument ($\partial^2 m/\partial s^2 < 0$ and $\partial^2 m/\partial b^2 < 0$). Second, sellers and buyers are Edgeworth-Pareto complements in the matching function ($\partial^2 m/\partial b \partial s > 0$). Third, the matching function is jointly concave in both arguments. However, the joint concavity is weak and not strict because the matching function's Hessian is rank one. The sixth assumption implies that the matching function exhibits diminishing marginal returns to the numbers of sellers and to the number of buyers, which seems natural.

Seventh, the matching elasticity $\eta(\theta)$ is assumed to be increasing in tightness θ. The assumption captures the natural idea that as the market becomes tighter, and sellers are in smaller number relative to buyers, additional sellers contribute proportionally more to total trades.

A final, eighth assumption is that the number of trades that occur within the period considered is less than the numbers of sellers and buyers: $m(s, b) \leq \min(s, b)$. Indeed, there cannot be more goods sold than sellers, and more goods bought than buyers. This assumption is specific to discrete-time models: in a continuous-time model, the matching function gives the flow of trades occurring at any point in time, so there is no restriction on the level of $m(s, b)$.

3.4. Trading probabilities

The matching function $m(s, b)$ tells us that, during a certain period, not all s sellers are able to sell their good, and not all b buyers are able to buy a good. Therefore, we need to figure out the probability that a seller is able to sell,

$$f = \frac{m(s,b)}{s},$$

and the probability that a buyer is able to buy,

$$q = \frac{m(s,b)}{b}.$$

Our assumptions about the matching function have clear implications for how these trading probabilities behave.

3.4.1. Selling probability

We first look at the selling probability:

$$f = \frac{m(s,b)}{s} = m\left(\frac{s}{s}, \frac{b}{s}\right) = m\left(1, \frac{b}{s}\right),$$

since the matching function has constant returns to scale. We can re-express our selling probability as a function of market tightness:

$$f(\theta) = m(1,\theta). \tag{3.4}$$

Since the matching function is strictly increasing in its second argument, we infer that the selling probability is strictly increasing in tightness. The tighter the market, the more likely you are to sell. This makes sense because the tighter a market is, the more buyers there are for each seller. And since the matching function is strictly concave in its second argument, we infer that the selling probability is strictly concave in tightness. This means that a higher tightness leads to a higher selling probability, but with diminishing returns.

We can also see that when tightness is 0, the selling probability must be 0. This is because, in the matching function, when any of the two arguments is 0, there is no trade. This is intuitive because if there are no buyers, there is no chance of selling anything, so the probability of selling is 0.

Lastly, we know that the selling probability is always between 0 and 1 because the matching function is always less than the minimum of its two arguments. Since $m(s,b) \leq \min(s,b)$, then $m(s,b) \leq s$ and $f(\theta) \leq 1$.

3.4.2. Buying probability

Now, we shift our attention to the buying probability:

$$q = \frac{m(s, b)}{b} = m\left(\frac{s}{b}, \frac{b}{b}\right) = m\left(\frac{s}{b}, 1\right),$$

since the matching function has constant returns to scale. Using market tightness, we can rewrite the probability as:

$$q(\theta) = m\left(\frac{1}{\theta}, 1\right), \tag{3.5}$$

so that the buying probability only depends on tightness.

We can go over the properties of the buying probability in the same way. Because the matching function is strictly increasing in its first argument, the buying probability is strictly decreasing in tightness. This means that a buyer in a tight market is less likely to be able to buy the good they want, since there are very few sellers and a lot of buyers, increasing competition. This is true in any tight market: it is a good time to be a seller but a bad time to be a buyer.

Moreover, at the limit where tightness is infinite, the buying probability is 0. The reason is that at the limit, $1/\theta \to 0$, and $m(0, b) = 0$. This makes sense because when infinitely many buyers are competing to buy a good they want, the probability of buying that good is bound to go to 0.

Finally, we know that the buying probability is always between 0 and 1 because the matching function is always less than the minimum of its two arguments. Since $m(s, b) \leq \min(s, b)$, then $m(s, b) \leq b$ and $q(\theta) \leq 1$.

3.4.3. Relation between the trading probabilities

There is a simple relationship between the two trading probabilities. The number of trades is $m(s, b) = f(\theta)s = q(\theta)b$, so $f(\theta)/q(\theta) = b/s = \theta$. That is, for any matching function:

$$f(\theta) = \theta q(\theta). \tag{3.6}$$

From this result we also see that the elasticities of the trading probabilities with respect to tightness are necessarily related:

$$\epsilon_\theta^f = 1 + \epsilon_\theta^q.$$

From the expression (3.5) and result B.7, we see that the elasticity of the buying probability is directly related to the matching elasticity:

$$\epsilon_{\theta}^{q} = -\eta. \tag{3.7}$$

Thus, the elasticity of the selling probability is also related to the matching elasticity:

$$\epsilon_{\theta}^{f} = 1 - \eta. \tag{3.8}$$

Once again, these elasticities might or might not be constant, depending on the matching function.

3.5. Urn-ball matching function

We now study the urn-ball matching function. This matching function is useful because it has a simple microfoundation, which draws on the urn-ball model in probability theory.

3.5.1. Foundation

Several researchers proposed an urn-ball foundation to explain why trading probabilities might be less than 1 in a given period.

Butters (1977) used the urn-ball framework in the product market. In this model, firms advertise their products by dropping ads into customers' mailboxes. The probability to sell is less than 1 because several firms might drop their ad in the same mailbox, in which case the customer only buys from the cheapest firm. The probability to buy is also less than 1, because a customer might not receive any ad in their mailbox.

Hall (1979) used a similar setup in the labor market. In this model, firms simultaneously and randomly make job offers to job seekers. The probability to hire a job seeker is less than 1 because several firms might make a job offer to the same job seeker, who cannot accept more than one offer. The probability to find a job is also less than 1 because a job seeker might be unlucky and not receive any of the job offers.

Both of these situations correspond to an urn-ball setup. In Butters's example, the ads are the balls and the mailboxes are the urns. In Hall's example, the job offers are the balls and the job seekers are the urns.

In this section, we consider as an example the haircut market. There are b customers who are looking for a haircut, and s hairdressers. Each hairdresser can only accommodate one customer at a time, so if two customers pick the same hairdresser, only one can get a haircut. Because customers do not know where the others go, they don't know which hairdressers already serve a customer and which hairdressers are idle. Such coordination failure means that some customers cannot get a haircut and some hairdressers do not get customers.

3.5.2. Expression

We now derive the number of haircuts sold in a simple situation: $b > 0$ customers in need of a haircut each go to one of $s > 0$ open hair salons. The customers pick their hair salons simultaneously and randomly. That day, customers can only make one trip to the hair salon.

The probability that a specific customer goes to a specific hairdresser is $1/s$, and the probability that the customer does not go to the hairdresser is $1 - 1/s$. Accordingly, the probability that a specific hairdresser does not get a visit from any of the b customers is $(1 - 1/s)^b$. Conversely, the probability that a specific hairdresser gets at least one visit is $1 - (1 - 1/s)^b$. Since a hairdresser sells a haircut if at least one customer visits, this probability gives the probability to sell a haircut for a specific hairdresser, as well as the expected number of haircuts sold by the hairdresser (since the hairdresser sells either 0 or 1 haircut). Then, the expected number of haircuts sold by all hairdressers is just

$$m(s, b) = s\left[1 - \left(1 - \frac{1}{s}\right)^b\right].$$

This is the expected number of trades that occur on the haircut market. This matching function is a little cumbersome, but it can be greatly simplified when the number of hairdressers is large enough.

To simplify the above matching function, we use a linear approximation of the logarithm: $\ln(1 - x) \approx -x$ when x is small. This approximation allows us to rewrite the probability that a hairdresser gets no visit at all:

$$\left(1 - \frac{1}{s}\right)^b = \exp\left(b \cdot \ln\left(1 - \frac{1}{s}\right)\right) \approx \exp\left(b \cdot \frac{-1}{s}\right) = \exp\left(-\frac{b}{s}\right).$$

We can then rewrite our matching function, which is the expected number of haircuts sold:

$$m(s, b) = s\left[1 - \exp\left(-\frac{b}{s}\right)\right]. \tag{3.9}$$

This is the urn-ball matching function, which gives the number of trades for a given number of sellers, s, and buyers, b.

3.5.3. Properties

By looking at the matching function (3.9), we can verify that it satisfies all the assumptions listed in section 3.3.

First, we confirm that the matching function is smooth, strictly positive, and zero at the limit without sellers or buyers.

We easily see that the matching function has constant returns to scale:

$$m(zs, zb) = zs\left[1 - \exp\left(-\frac{zb}{zs}\right)\right] = zm(s, b).$$

We check that the matching function is strictly increasing in each argument. We immediately see from (3.9) that if the number of buyers goes up, the number of haircuts also increases. Formally, the partial derivative of the matching function with respect to the number of buyers is simply:

$$\frac{\partial m}{\partial b} = \exp\left(-\frac{b}{s}\right) > 0.$$

Showing that the number of haircuts increases with the number of hairdressers is a little bit trickier since s has two opposite influences on the matching function. We must calculate the derivative formally to check that it is positive. After simplifying, we get:

$$\frac{\partial m}{\partial s} = 1 - \left(1 + \frac{b}{s}\right)\exp\left(-\frac{b}{s}\right). \tag{3.10}$$

We know that $1 + x < \exp(x)$ for all $x > 0$ (result A.13). Thus, $(1 + x)\exp(-x) < 1$ for all $x > 0$, so we verify that $\partial m/\partial s > 0$. From this we conclude that the matching function is strictly increasing in s.

Moreover, the matching function is less than its two arguments. Since $\exp(-x) > 0$, it is clear that $m(s, b) \leq s$. Using again $\exp(x) \geq 1 + x$, we obtain

$$m(s, b) \leq s\left[1 - \left(1 - \frac{b}{s}\right)\right] = b.$$

So overall $m(s, b) \leq \min(s, b)$.

We verify that $\partial m/\partial b$ is strictly decreasing in b, which tells us the matching function is strictly concave in the number of buyers.

Next, we compute the matching elasticity of the urn-ball matching function to better understand the shape and behavior of the matching function. In particular, we show that the urn-ball matching function satisfies the assumption that the matching elasticity is increasing in tightness.

We start from the definition of the matching elasticity, given by (3.1). Using the partial derivative (3.10), we can write the matching elasticity as

$$\eta = \frac{s}{m(s, b)} \cdot \left[1 - \exp\left(\frac{-b}{s}\right)\right] - \frac{b}{m(s, b)} \cdot \exp\left(\frac{-b}{s}\right).$$

Given that the matching function $m(s, b)$ is given by (3.9), we express the matching elasticity as a function of the market tightness:

$$\eta(\theta) = 1 - \frac{\theta}{\exp(\theta) - 1}. \tag{3.11}$$

Hence, the matching elasticity is an increasing function of tightness, growing from 0 when tightness is 0, to $1 - 1/(e - 1)$ when tightness is 1, to 1 when tightness is infinite.

The properties of the matching elasticity appear once we write the exponential function using its Taylor series (result A.8), which then gives:

$$\frac{\exp(\theta) - 1}{\theta} = 1 + \sum_{n=1}^{\infty} \frac{\theta^n}{(n+1)!}.$$

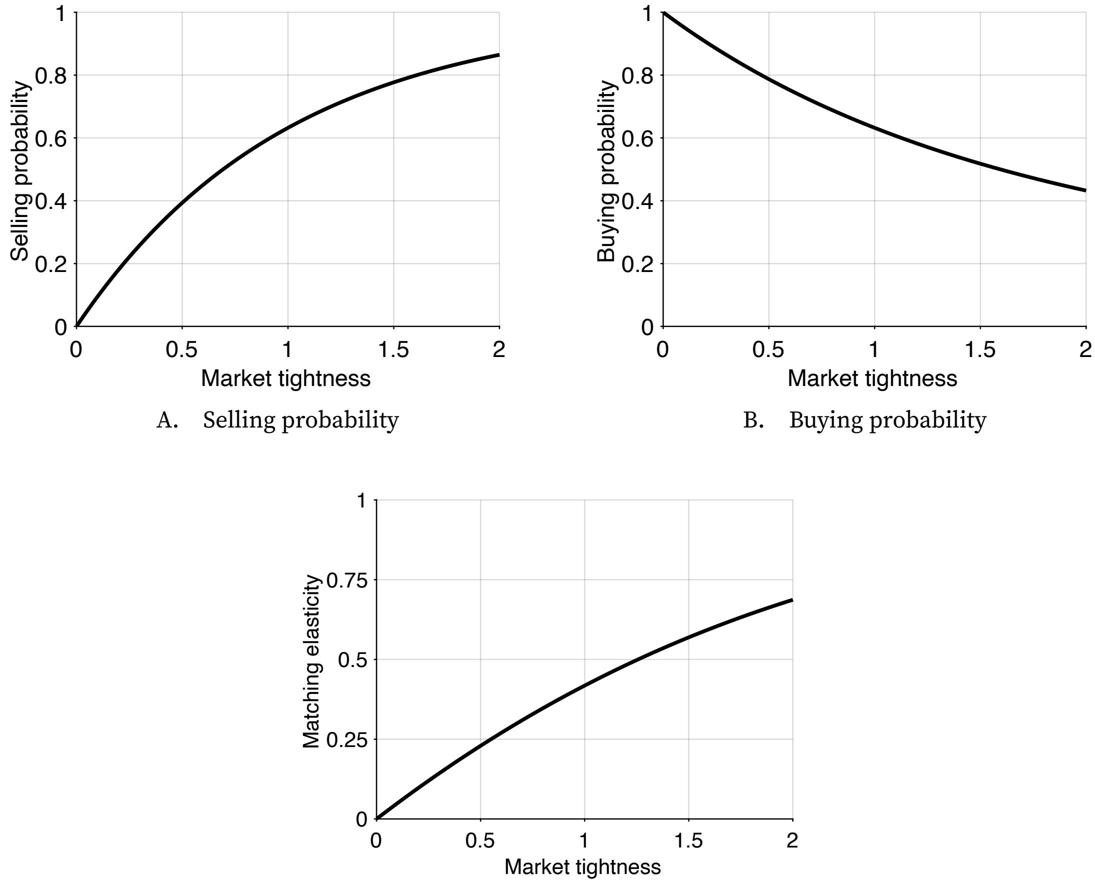
All the terms $\theta^n/(n+1)!$ in the sum are increasing in $\theta \geq 0$, so the function $[\exp(\theta) - 1]/\theta$ is increasing in θ, which tells us that $\eta(\theta)$ is increasing in θ. All the terms in the sum are 0 when $\theta = 0$, which tells us that $\eta(0) = 0$. And all the terms in the sum grow to ∞ when $\theta \to \infty$, which tells us that $\eta(\theta) \to 1$ as $\theta \to \infty$.

The behavior of the elasticity means that when there are no buyers on the market ($\theta = 0$), adding more sellers does not lead to any more trades. The intuition is that the market is already extremely congested with sellers, so more sellers do not bring more trade: only more buyers would generate more trades. On the other hand, when there are almost no sellers ($\theta \to \infty$), adding 1% more sellers generates 1% more trades, even with a fixed number of buyers. In that case the number of trades is almost solely determined by the number of sellers, so they grow proportionally.

Lastly, the selling and buying probabilities given by the urn-ball matching function and its matching elasticity are illustrated in figure 3.1.

3.6. Constant-elasticity-of-substitution matching function

Despite its simple microfoundation, the urn-ball matching function is not used often in theoretical work, because it is a little cumbersome. Instead, it is often preferable to use the constant-elasticity-of-substitution (CES) matching function, as it is much more tractable. While it does not have a microfoundation as simple as the urn-ball process, it is possible to obtain a CES matching function from a queuing process in which sellers call buyers to advertise their goods (Stevens 2007), and from a network in which sellers and buyers are connected (Angelis and Bramoulle 2026).



A. Selling probability

B. Buying probability

C. Matching elasticity

FIGURE 3.1. Properties of the urn-ball matching function

The urn-ball matching function is given by (3.9). The selling and buying probabilities are computed from the matching function with (3.4) and (3.5). The matching elasticity is given by (3.11).

3.6.1. Expression

The CES matching function takes the following form:

$$m(s,b) = \left(s^{-\sigma} + b^{-\sigma}\right)^{-1/\sigma}. \tag{3.12}$$

The parameter $\sigma > 0$ governs the elasticity of substitution between s and b.

3.6.2. Properties

We again check that the assumptions on the matching function listed in section 3.3 are satisfied here.

First, the CES matching function is smooth, strictly positive, strictly increasing in each argument, and zero at the limit without buyers or sellers.

The CES function has obviously constant returns to scale:

$$m(zs, zb) = \left[(zs)^{-\sigma} + (zb)^{-\sigma}\right]^{-1/\sigma} = zm(s, b).$$

Next, we verify that the CES matching function is less than the minimum of its two arguments. Since $b^{-\sigma} > 0$, then $s^{-\sigma} + b^{-\sigma} > s^{-\sigma}$. Hence,

$$m(s, b) = \left(s^{-\sigma} + b^{-\sigma}\right)^{-1/\sigma} < \left(s^{-\sigma}\right)^{-1/\sigma} = s.$$

Similarly, we find $m(s, b) < b$, which overall tells us that $m(s, b) \leq \min(s, b)$.

To verify that the function is concave in the number of sellers, we compute its derivatives. The partial derivative with respect to s is

$$\frac{\partial m}{\partial s} = \left[1 + \left(\frac{s}{b}\right)^{\sigma}\right]^{-\frac{1+\sigma}{\sigma}}.$$

The partial derivative is not only positive but also strictly decreasing in s. From this we infer that the CES matching function is strictly concave in the number of sellers.

Next, we verify that the CES matching elasticity is increasing in tightness. We compute the matching elasticity (3.1) using the results from appendix B:

$$\eta(\theta) = \frac{-1}{\sigma} \cdot \frac{s^{-\sigma}}{s^{-\sigma} + b^{-\sigma}} \cdot (-\sigma)$$

so that

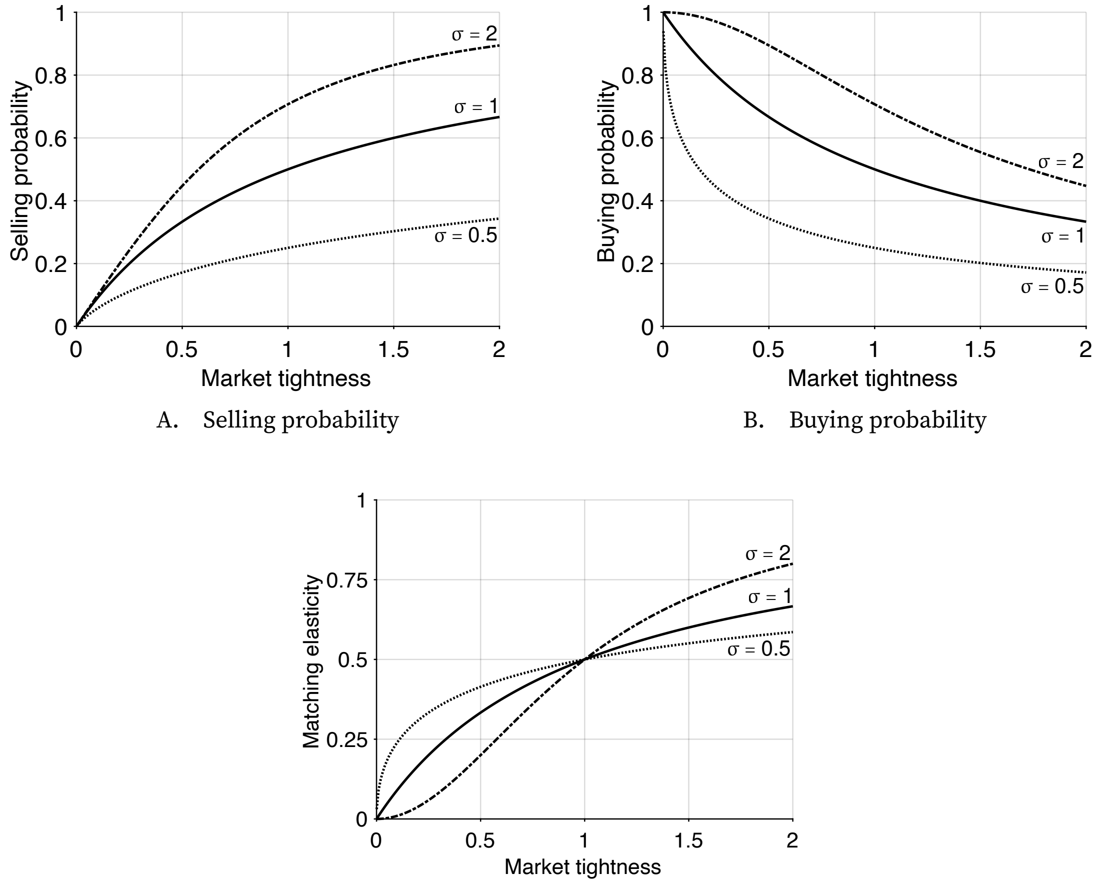
$$\eta(\theta) = \frac{1}{1 + \theta^{-\sigma}}. \tag{3.13}$$

We confirm that the matching elasticity is a strictly increasing function of tightness, growing from 0 when tightness is 0, to 1/2 when tightness is 1, to 1 when tightness is infinite. Just like in the case of the urn-ball matching function, the behavior of the matching elasticity implies that when there are no buyers on the market, adding more sellers does not lead to any more trades, while when there are almost no sellers, numbers of sellers and trades grow proportionally.

With the CES matching function, it is easy to see what happens when there are very few sellers and buyers. We can rewrite the matching function (3.12) as

$$m(s, b) = b \cdot [1 + \theta^{\sigma}]^{-1/\sigma},$$

where $\theta = b/s$ is the market tightness. When tightness is close to 0, θ^{σ} is negligible compared to 1, so the matching function approximates the number of buyers: $m(s, b) \approx b$. This means that when the number of buyers is small, the number of buyers gives the number



A. Selling probability

B. Buying probability

C. Matching elasticity

FIGURE 3.2. Properties of the CES matching function

The CES matching functions are given by (3.12) with $\sigma = 0.5$, $\sigma = 1$, and $\sigma = 2$. The selling and buying probabilities are computed from the matching functions with (3.4) and (3.5). The matching elasticity is given by (3.13).

of trades, irrespective of the number of sellers.

Since the matching function (3.12) is symmetric in b and s, we can also rewrite it as

$$m(s, b) = s \cdot \left[1 + \theta^{-\sigma}\right]^{-1/\sigma},$$

When tightness is very large, $\theta^{-\sigma}$ is negligible compared to 1, so the matching function approximates the number of sellers: $m(s, b) \approx s$. This means that when the number of sellers is small, the number of sellers gives the number of trades, irrespective of the number of buyers.

Finally, the selling and buying probabilities given by the CES matching function and its matching elasticity are illustrated in figure 3.2.

3.7. Cobb-Douglas matching function

Maybe the most popular of all matching functions is the Cobb-Douglas matching function. It is appealing because it is even easier to manipulate than the CES matching function, and it describes the US labor market well (Blanchard and Diamond 1989; Petrongolo and Pissarides 2001). It is possible to obtain a Cobb-Douglas matching function from the queuing process proposed by Stevens (2007), as long as matching costs are linear. The main drawback of the Cobb-Douglas matching function is that it does not guarantee that $m(s, b) \leq \min(s, b)$, so it must be truncated in discrete-time models to avoid trading probabilities greater than 1.

3.7.1. Expression

The Cobb-Douglas matching function takes the following form:

$$m(s, b) = \mu \cdot s^{\sigma} \cdot b^{1-\sigma}. \tag{3.14}$$

The parameter $\mu > 0$ governs the efficacy of the matching process. The parameter $\sigma \in (0, 1)$ governs the shape of the matching function.

3.7.2. Properties

We easily check that the Cobb-Douglas function satisfies the assumptions listed in section 3.3.

The Cobb-Douglas matching function is clearly smooth, strictly positive, strictly increasing in each argument, and zero at the limit without buyers or sellers. It also clearly exhibits constant returns to scale:

$$m(zs, zb) = \mu(zs)^{\sigma}(zb)^{1-\sigma} = zm(s, b).$$

And it is clearly strictly concave in the number of sellers and in the number of buyers.

A reason why the Cobb-Douglas matching function is particularly convenient is that its matching elasticity is constant: $\eta = \sigma$.

Relatedly, the trading probabilities that it produces are easy to deal with. We obtain them directly using (3.4) and (3.5):

$$f(\theta) = m(1, \theta) = \mu\theta^{1-\sigma}, \quad q(\theta) = m\left(\frac{1}{\theta}, 1\right) = \mu\theta^{-\sigma}. \tag{3.15}$$

In particular, the trading probabilities have constant elasticity with respect to tightness: $\epsilon_{\theta}^{f} = 1 - \sigma$ and $\epsilon_{\theta}^{q} = -\sigma$.

The Cobb-Douglas matching function has one main limitation: it is not always less than the minimum of its two arguments. In a continuous-time model, where the matching function describes the flow of trades at any point in time, this is not an issue. But in a discrete-time model, where the matching function describes the number of trades in a period, this is an issue because it might lead to trading probabilities that are greater than 1. To avoid trading probabilities that are greater than 1, the matching function must be truncated by imposing $m(s, b) = \min\left(\mu \cdot s^{\sigma} \cdot b^{1-\sigma}, s, b\right)$.[3] The truncation in turn creates difficulties in theoretical and numerical work (den Haan, Ramey, and Watson 2000). Hence, when the truncation is binding, it is often better to use another matching function.

The selling and buying probabilities given by the Cobb-Douglas matching function and its matching elasticity are illustrated in figure 3.3.

3.8. Empirical properties of the matching function

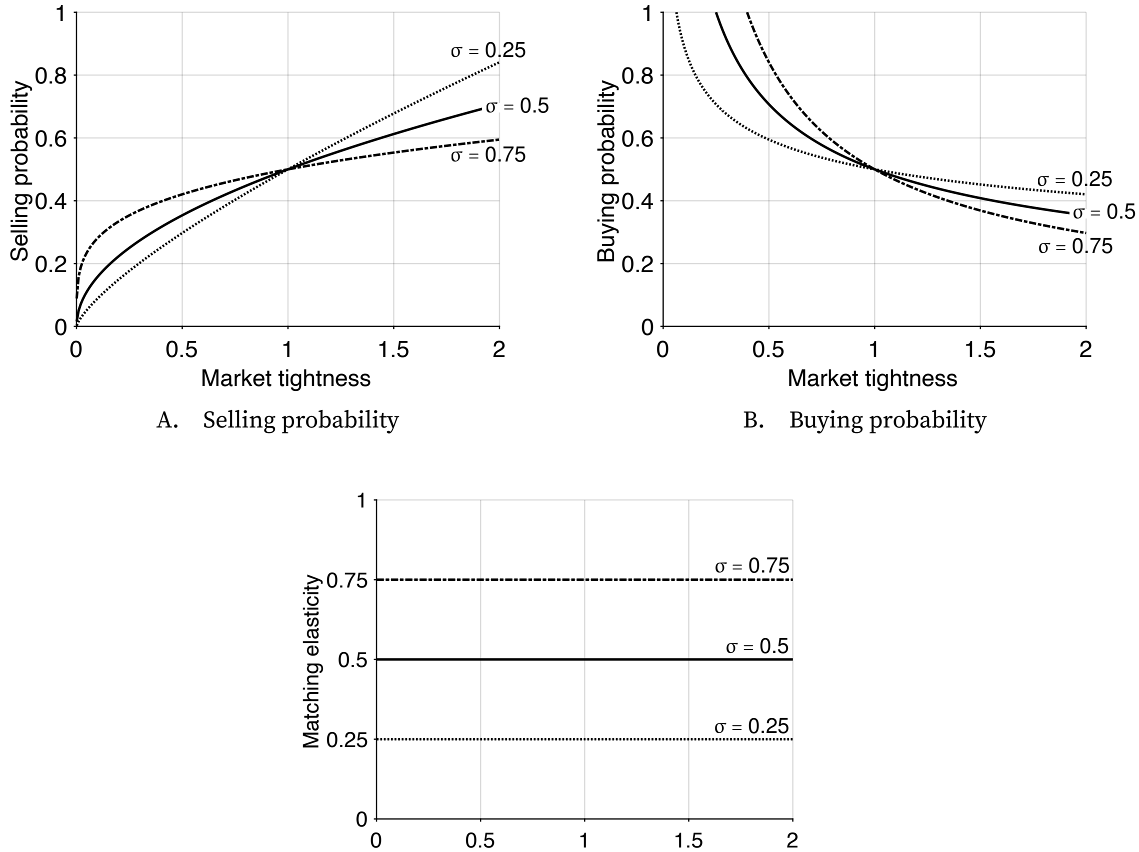
To conclude this chapter, we briefly review the empirical properties of the matching function to convince ourselves that our theoretical assumptions give us a matching function that describes the real world well. When matching functions were first developed, researchers examined the data to ensure that they built functions that were realistic (Blanchard and Diamond 1989; Petrongolo and Pissarides 2001). Here we review modern US data with the same goal.

Data on sellers and buyers are most readily available in the labor market, where we have good counts of who sells labor (job seekers) and who buys labor (vacant jobs). Thus we focus on the matching function for the labor market.

A central assumption is that the matching function has constant returns to scale. The main implication is that the market tightness determines the rates at which sellers and buyers trade. In the labor market, this means that labor market tightness determines the rate at which job seekers find jobs and the rate at which vacant jobs are filled. We only have data to compute the job-finding rate over a long period of time, so we concentrate on the link between job-finding rate and labor market tightness here. The data that we use start in 1948, which is when the BLS started to collect the data required to compute the job-finding rate (CPS data). Just like in figure 2.8, we stop the analysis in 2019 to concentrate on the 1948–2019 period, which is when the Beveridge curve and matching function are fairly stable.

For reference, we plot the US labor market tightness between 1948 and 2019. Labor market tightness is computed as the number of job vacancies per job seeker, or equiva-

[3] The expressions (3.15) show that $f(\theta)$ is larger than 1 when θ is high enough, and $q(\theta)$ is larger than 1 when θ is low enough. This is why the Cobb-Douglas matching function must be truncated in discrete-time models.



A. Selling probability

B. Buying probability

C. Matching elasticity

FIGURE 3.3. Properties of the Cobb-Douglas matching function

The Cobb-Douglas matching functions are given by (3.14) with $\mu = 0.5$ and $\sigma = 0.25$, $\sigma = 0.5$, $\sigma = 0.75$. The selling and buying probabilities are computed from the matching functions with (3.4) and (3.5). The matching elasticity is constant: $\eta = \sigma$.

lently, the ratio of vacancy rate to unemployment rate.[4] We see that labor market tightness is sharply procyclical. It averages 0.65 between 1948 and 2019. Over that period, tightness reached its lowest level, 0.16, during the Great Recession (2009:Q3) and its highest level, 1.60, during the Korean War (1953:Q1).

We start with the rate at which job seekers find jobs, and correlate this rate with labor market tightness. To compute the job-finding rate, we follow Shimer (2012). In month t, $u(t)$ workers are looking for jobs. Some of them find a job in the month, and some do not. Those who do not find a job remain unemployed, so $u(t) - u(t+1)$ seems to indicate the number of workers who have been able to find a job. There is a complication,

[4]In practice some vacancies are filled not by unemployed workers but by employed workers moving between jobs. The present empirical approach is valid as long as the number of vacancies captures well the demand for unemployed labor.

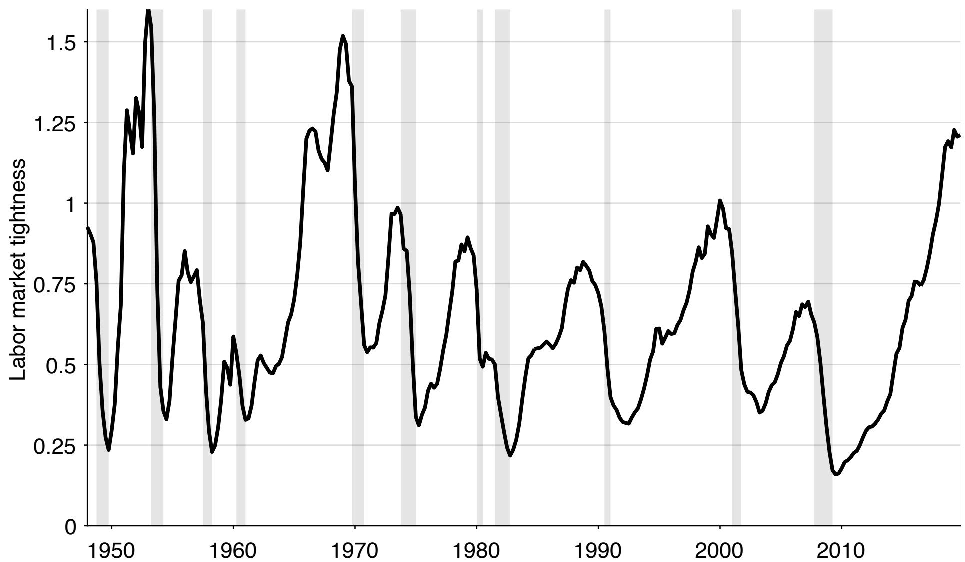


FIGURE 3.4. Labor market tightness in the United States, 1948–2019

Labor market tightness is the ratio of vacancy rate to unemployment rate. The unemployment rate comes from figure 2.1 while the vacancy rate comes from figure 2.5. Shaded areas indicate recessions dated by the NBER (2023).

however. Some workers who were employed lose their jobs in month t and join the pool of unemployed workers. We need to account for those when we compute the number of job seekers who found a job. We denote by $u^s(t+1)$ the number of workers who were previously employed and who have joined unemployment between months t and $t+1$. The number of workers who have found a job within month t is $u(t) - [u(t+1) - u^s(t+1)]$.[5] Dividing this number of job finders by the number of job seekers in month t, we obtain the probability of finding a job in month t:

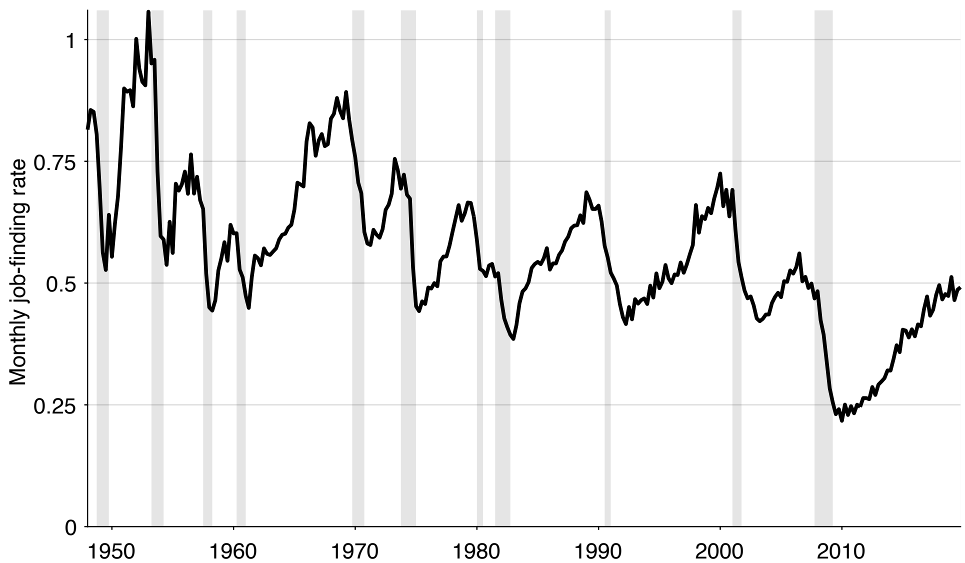
$$F(t) = 1 - \frac{u(t+1) - u^s(t+1)}{u(t)}.$$

To calculate $F(t)$, we measure $u(t)$ as the number of unemployed persons in month t computed by the BLS (2025x), and $u^s(t)$ as the number of persons who have been unemployed for less than 5 weeks in month t computed by the BLS (2025p).[6]

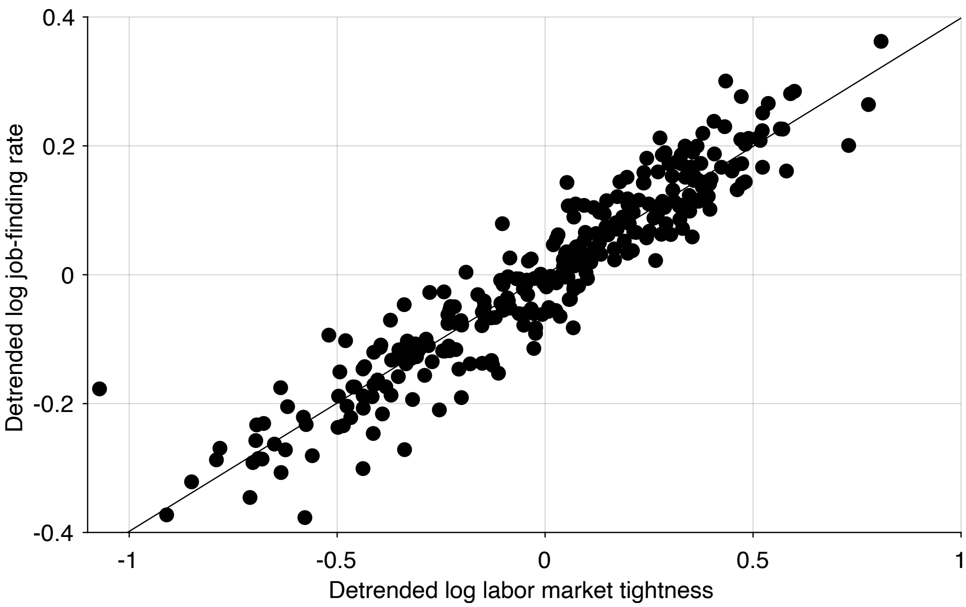
Assuming that unemployed workers find a job according to a Poisson process with monthly arrival rate $f(t)$, the probability that they have found a job within one month is

[5]This calculation assumes that workers only transition between employment and unemployment. In reality, some workers move in and out of the labor force. But the approach is valid as long as the numbers of job seekers who exit and enter the labor force are similar.

[6]Following Shimer (2012, appendix A), we multiply the series for $u^s(t)$ by 1.1 after January 1994 to correct for a change in the design of the CPS.



A. Monthly job-finding rate in the United States, 1948–2019



B. Detrended logarithm of tightness and job-finding rate, 1948–2019

FIGURE 3.5. The US job-finding rate is an isoelastic function of labor market tightness

The monthly job-finding rate is computed from (3.16). Labor market tightness comes from figure 3.4. In panel A, shaded areas indicate recessions dated by the NBER (2023). In panel B, the series are detrended by applying a HP filter with smoothing parameter of 10,000.

$F(t) = 1 - \exp(-f(t))$ (result A.4). Hence, we infer the job-finding rate from the job-finding probability:

$$f(t) = -\ln(1 - F(t)). \tag{3.16}$$

We thus obtain the monthly job-finding rate in the United States (figure 3.5A). Between 1948 and 2019, the job-finding rate is highly procyclical and averages 56% per month. This means that on average, a US job seeker takes $1/0.56 = 1.8$ months to find a job.

Our objective is to assess whether labor market tightness determines the job-finding rate. To uncover the relationship between tightness and job-finding rate, we plot the logarithm of the job-finding rate against the logarithm of tightness. To remove slow changes in the matching function, we detrend both series using an HP filter with smoothing parameter 10,000. We find that log job-finding rate is largely determined by log labor market tightness: the least-squares regression gives $R^2 = 0.89$, with a coefficient of 0.40 (figure 3.5B). This result indicates that the job-finding rate is well described as an isoelastic function of tightness, with an elasticity of 0.40.

Overall, we find that the job-finding rate is a stable function of tightness, which supports the assumption that the matching function has constant returns to scale. In addition, we have seen that the elasticity of the job-finding rate with respect to tightness appears constant, which suggests that the matching elasticity is constant. This finding suggests that a Cobb-Douglas matching function, which has a constant matching elasticity, describes the US labor market better than the urn-ball or CES matching functions, which have matching elasticities that respond noticeably to tightness (figures 3.1C and 3.2C). Furthermore, the matching elasticity that appears in the regression is $\eta = 0.60$, using (3.8).

The empirical findings in this section are in line with findings in the literature on the matching function. Early aggregate studies of the US labor market find that the matching function involves the stock of unemployed workers and job vacancies with constant returns to scale (Petrongolo and Pissarides 2001). The studies converge on a Cobb-Douglas form with a matching elasticity between 0.5 and 0.7. More recent studies obtain comparable results. Shimer (2005, table 2) estimates the matching elasticity at 0.72; Rogerson and Shimer (2011, p. 638) obtain an estimate of 0.58; and Borowczyk-Martins, Jolivet, and Postel-Vinay (2013, table 1) report a lower estimate of 0.30.

3.9. Summary

In this chapter we introduce the matching function as a key tool for modeling slackish markets, where trade is not instantaneous and requires time and effort from both buyers and sellers. Unlike Walrasian markets where trades are seamless, real-world markets are characterized by complexities that make it difficult for buyers and sellers to find each other. The matching function is an aggregate tool, similar to a production function, that captures the outcome of this complex trading process without modeling the micro-level details explicitly.

The chapter presents three specific types of matching functions: urn-ball, CES, and Cobb-Douglas, which each have their advantages and drawbacks, which we discussed. For instance, the CES function is useful in theoretical work, while the Cobb-Douglas function is especially parsimonious.

Finally, the chapter reviews empirical evidence on matching from the US labor market, which supports the use of a matching function with constant returns to scale. Furthermore, this evidence suggests a Cobb-Douglas matching function is a good approximation for the US labor market, because the matching elasticity appears roughly constant.

CHAPTER 4.

Slackish market model

This chapter presents the most basic model of a slackish market. In a slackish market, the market tightness, not the market price, adjusts to equilibrate supply and demand and to determine allocations and welfare. The market price is governed by a price norm, rather than by a market-clearing condition.

The model presented in this chapter is basic in many ways, and it will be extended in the next chapters. For example, this chapter's model is static—it lasts only one period. Of course, the real world is dynamic, so in chapter 8 we provide a dynamic extension of the model and show how we can move from a static to a dynamic slackish model. The dynamic extension allows us in particular to introduce long-term relationships.

4.1. Buyers and sellers

We assume that there are many sellers, each indexed by $i \in [0, 1]$. Each seller sells a fixed amount of goods $k_i > 0$. The aggregate amount of goods for sale on the market is $k = \int_0^1 k_i \, di$. The quantity $k > 0$ represents the market capacity.

We assume that there are many buyers as well, each indexed by $j \in [0, 1]$. Since the market is slackish and not Walrasian, it is difficult for buyers to find the goods they like. Thus, buyers need to visit various sellers to buy goods. The number of sellers that buyer j visits is denoted by v_j. The aggregate number of visits to sellers on the market is $v = \int_0^1 v_j \, dj$.

4.2. Matching function

First, we examine how sellers are able to actually find buyers for their goods. Based on the number of goods that are available, k, and the number of visits, v, we can determine the number of trades that are realized. Here, a trade is one good being bought at price $p > 0$. The number of trades, which is the output and sales in the model, is denoted by y.

Because we are working with a slackish market, the number of trades is determined by a matching function m. In this market, the demand-side input into the matching function is the aggregate number of visits to buy goods, v, rather than the mass of buyers (normalized to 1); the supply-side input is the aggregate number of goods for sale, k, rather than the mass of sellers (also normalized to 1). Hence, the number of trades is given by

$$y = m(k, v). \tag{4.1}$$

We assume that the matching function satisfies the generic properties listed in chapter 3. In particular, since we are in a static model, the number of trades cannot be more than the minimum of the number of visits and the number of goods available, so the matching function satisfies $y \leq \min(k, v)$.

4.3. Market tightness

As we saw in chapter 3, the trading probabilities are solely determined by the market tightness, θ, which here is the aggregate number of visits divided by the aggregate number of goods for sale:

$$\theta = \frac{v}{k}. \tag{4.2}$$

The selling probability is given by $f(\theta) = m(1, \theta)$. It satisfies $f(0) = 0$ and is increasing and concave in θ. These properties say that it is impossible to sell without visits and that it is easier to sell in a tighter market—when visits are in larger numbers relative to goods for sale.

The buying probability, which is the probability that a visit to a seller is successful, is given by $q(\theta) = m(1/\theta, 1)$. It is decreasing in θ, and $q(\theta) \to 0$ when $\theta \to \infty$. These properties say that a visit is less likely to be successful in a tighter market—as there is more competition for available goods—and that it becomes impossible to buy without goods for sale.

This is a general property of slackish markets: when the market is tighter, it is a better time to be a seller: goods are more likely to be sold. On the other hand, when the market is slacker, it is a good time to be a buyer: visits are more likely to be successful. Figures 4.1A and 4.1B display the selling and buying probabilities for an illustrative calibration.

4.4. Rate of slack

Because we are working with a slackish market, some goods always remain unsold. We define the rate of slack as the share of goods that remain unsold. It is easy to compute this slack rate and express it as a function of market tightness.

We previously saw that each good is sold with probability $f(\theta)$. For a seller with k_i goods for sale, omitting randomness at the seller level, the number of goods sold is $y_i = f(\theta)k_i$. The aggregate number of sales on the market is $y = \int_0^1 y_i \, di$, which can be expressed as

$$y = \int_0^1 f(\theta)k_i \, di = f(\theta) \int_0^1 k_i \, di = f(\theta)k. \tag{4.3}$$

Here we simply recover the matching function from the individual selling probabilities, since $f(\theta)k = m(1, \theta)k = m(k, \theta k) = m(k, v)$.

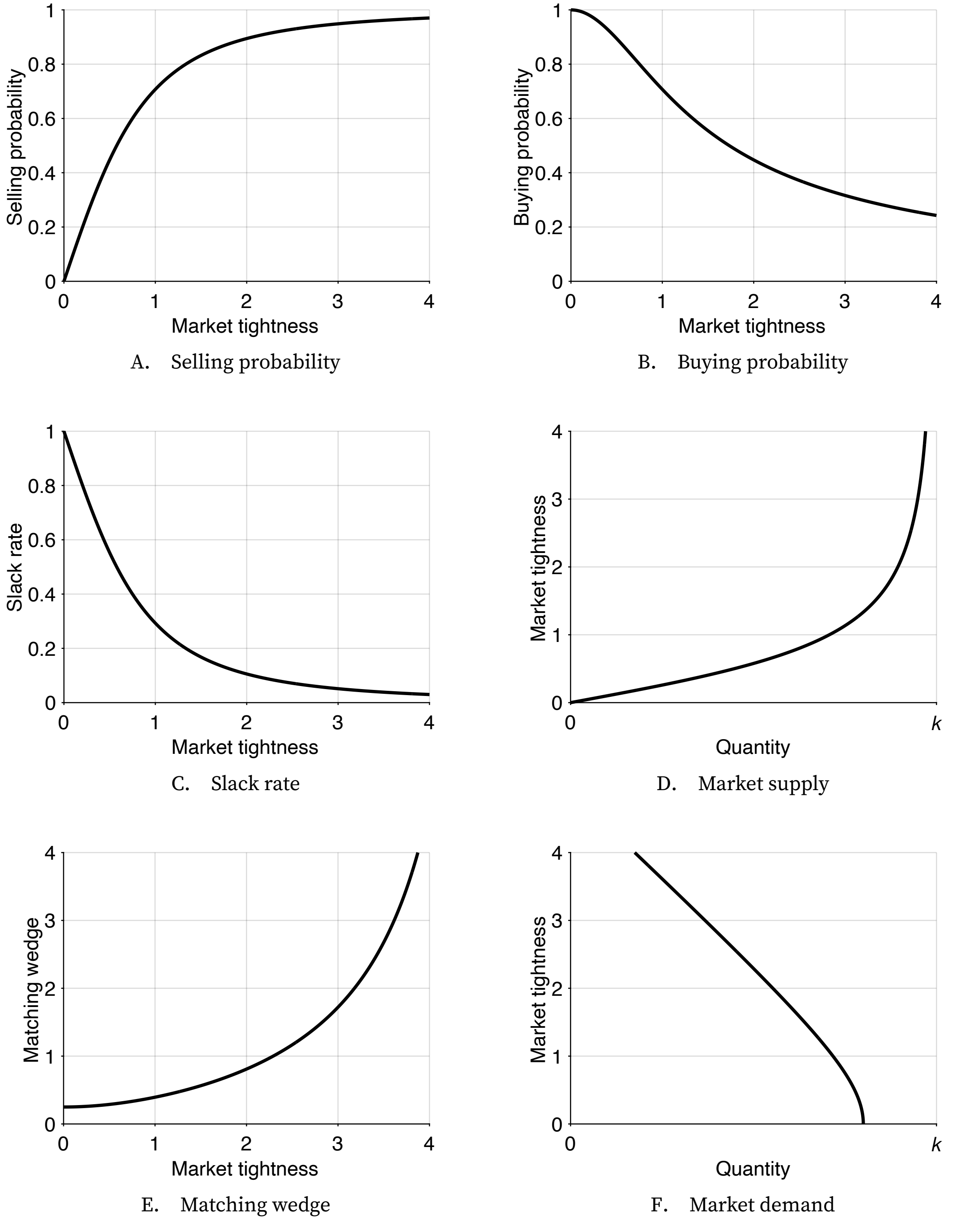
We define the rate of slack u in the market as the share of goods that remain unsold. In the aggregate, a share $f(\theta) \in [0, 1]$ of goods is sold, so the slack rate is a simple function of market tightness:

$$u(\theta) = 1 - f(\theta). \tag{4.4}$$

The properties of the slack rate follow from those of the selling probability $f(\theta)$. The slack rate is 1 when tightness is 0, because the selling probability is 0 when tightness is 0: when there are no buyers, nothing gets sold. The slack rate then decreases with tightness, because the selling probability increases with tightness. And since the selling probability is concave in tightness, the slack rate is convex in tightness.

The rate of slack can be interpreted in different ways depending on the type of goods sold on the market. If labor is sold on the market, then the rate of slack is just the rate of unemployment. If services are sold, then the rate of slack indicates the share of time that sellers are idle. If durable goods are sold, then the rate of slack gives the share of goods that are added to the unsold inventory. If perishable goods are sold, then the rate of slack indicates the share of goods that go to waste.

Figure 4.1C displays the rate of slack as a function of tightness to illustrate its properties. The figure also shows that the basic slackish model produces a Beveridge curve: a negative relationship between slack rate and visit rate. Indeed, in this static model, the visit rate is the number of visits per goods for sale, v/k, so the visit rate coincides with the market tightness. Therefore, the negative relationship between slack rate and market tightness corresponds to a Beveridge curve. In this static model the logic behind the Beveridge curve is trivial: a higher visit rate means a higher market tightness, which implies a higher selling probability, and thus fewer unsold goods and a lower slack rate.



A. Selling probability

B. Buying probability

C. Slack rate

D. Market supply

E. Matching wedge

F. Market demand

FIGURE 4.1. Numerical illustration of the slackish market model

The selling probability is given by (3.4). The buying probability is given by (3.5). The slack rate is given by (4.4). The market supply is given by (4.5). The matching wedge is given by (4.8). The market demand is given by (4.15). The matching function is CES, given by (3.12). Parameters are set to $\sigma = 2$, $\kappa = 0.2$, $k = 1$, $\alpha = 0.5$, $a = 2$, and $p = 1$.

4.5. Market supply

We now compute the market supply, which is the first key relation in analyzing how a slackish market behaves.

4.5.1. Expression of the market supply

A key distinction in this model is between the notional supply of goods and the effective supply. On the market, we have k goods that sellers would like to sell, which is the notional market supply. Here the notional supply is fixed, but it could depend on the market price and tightness, as we will see in chapter 7, where sellers determine the amount of goods that they bring on the market based on expected market conditions.

However, in the model, we use the effective market supply, which is the amount of goods actually sold given tightness, and which is less than the notional supply. The gap arises because the amount of goods that sellers are actually able to sell is less than the amount of goods that they put on the market, since, in our slackish market, each good is only sold with a probability that is less than one.[1]

By combining the sales equation (4.3) and slack equation (4.4), we obtain a simple expression for the market supply:

$$y^s(\theta) = [1 - u(\theta)]\, k, \tag{4.5}$$

where $u(\theta)$ is the slack rate and k is the market capacity. The market supply does not depend on the market price at all. To formalize the definition of market supply: it is just the amount of goods sold given the matching structure and the amount of goods supplied to the market by sellers. We have a gap between notional supply and effective supply that we use in our model, because not all goods supplied are traded: $u(\theta) > 0$ so $y^s(\theta) < k$.

4.5.2. Properties of the market supply

Let us now examine the properties of the market supply, given by (4.5). It is clear that the properties of the market supply follow from the properties of the slack rate, $u(\theta)$. When market tightness is 0, the slack rate is 1, so the supply is 0. Then, the supply is increasing in tightness, because the slack rate is decreasing in tightness. Finally, the supply is concave in tightness because the slack rate is convex in tightness.

We plot the market supply curve in a tightness-quantity plane to visualize its behavior (figure 4.1D). In Walrasian models, supply and demand curves are represented in price-

[1]The terminology of notional and effective supply and demand is borrowed from the Old Keynesian model, although the terms were used differently in that model: the notional supply and demand were Walrasian supply and demand, while the effective supply and demand took into account rationing in the other markets (Barro and Grossman 1971).

quantity planes. But in slackish models, tightness is the main equilibrating variable, so we conduct most of the analyses in tightness-quantity planes.

Because the function $y^s(\theta)$ is concave in θ, but the axes are inverted in the tightness-quantity plane (in the sense that tightness is on the y-axis instead of the x-axis, and quantities on the x-axis instead of the y-axis), then the supply curve appears convex.

4.6. Matching wedge

We have seen that buyers visit sellers to trade with them. These visits are required so buyers and sellers might match. Visiting sellers is costly: on each visit, a buyer incurs a matching cost $\kappa \in (0, q(0))$. The matching cost κ represents the number of goods spent on each visit. In the best-case scenario, a visit is successful with probability $q(0)$, so in expectation a visit yields $q(0)$ goods. If the matching cost is more than $q(0)$ to begin with, no buyer would ever try to buy anything. Thus, we impose $\kappa < q(0)$.

4.6.1. Expression of the matching wedge

The matching cost creates a wedge between the quantity of goods a buyer purchases and the quantity they actually consume. Indeed, when a buyer purchases goods, they use them for two purposes. First, they consume some of the goods, which provide utility. We denote buyer j's consumption by c_j. However, not all the goods are consumed—some goods are used for conducting visits and matching with a seller. This means that the number of goods consumed is less than the amount of goods purchased, which we denote y_j. Thus, there is a wedge between goods consumed and goods purchased. We need to assess how large this matching wedge is, and in doing so how consumption and purchases are linked.

Suppose buyer j conducts v_j visits with the aim of consuming c_j goods. Then the total number of goods purchased is $y_j = c_j + \kappa v_j$, as each visit requires κ goods. Where do the purchases come from? Each visit leads to a purchase only with probability $q(\theta)$. Thus, to obtain y_j purchases, a buyer needs to make $v_j = y_j/q(\theta)$ visits. Here we assume that there is no uncertainty at the buyer level: a fraction $q(\theta)$ of the visits provides a good. Hence, output and consumption are related by

$$y_j = c_j + \frac{\kappa y_j}{q(\theta)}. \tag{4.6}$$

Solving for y_j in the equation, we find that it is proportional to c_j:

$$y_j = [1 + \tau(\theta)]\, c_j, \tag{4.7}$$

where the matching wedge $\tau(\theta)$ is given by

$$\tau(\theta) = \frac{\kappa}{q(\theta) - \kappa}. \tag{4.8}$$

The matching wedge (4.8) is the gap between the amount of goods consumed and the amount of goods purchased. The gap is caused by the matching process: it arises due to the matching cost associated with each visit. Overall, to consume one good, a buyer must purchase $1 + \tau(\theta)$ goods. Reciprocally, if a buyer purchases one good, they only consume $1/[1 + \tau(\theta)]$ goods.

4.6.2. Properties of the matching wedge

We now discuss the properties of the matching wedge $\tau(\theta)$.

First, we restrict the analysis to $\theta \in (0, \bar{\theta})$ to ensure that the matching wedge is positive and finite, where the upper bound $\bar{\theta}$ is defined by

$$\bar{\theta} = q^{-1}(\kappa). \tag{4.9}$$

Since q is strictly decreasing from $(0, \infty)$ to $(0, q(0))$, the inverse q^{-1} is well defined on $(0, q(0))$. Hence $\bar{\theta}$ exists as long as $\kappa < q(0)$, which we have assumed earlier.[2] Moreover, as q is strictly decreasing, $q(\theta) > q(\bar{\theta}) = \kappa$ for any $\theta < \bar{\theta}$, and we verify that $\tau(\theta)$ is positive and finite for $\theta \in (0, \bar{\theta})$.

Next, we can see that the matching wedge is increasing in tightness, since it is the composite of two decreasing functions: $q(\theta)$ is decreasing in θ and $\kappa/(x - \kappa)$ is decreasing in x.

Moreover, $\tau(\theta) \to +\infty$ when $\theta \to \bar{\theta}$. The divergence $\tau(\theta) \to \infty$ reflects the fixed-point logic that appears in (4.6): matching resources must themselves be purchased through additional matching. As tightness approaches $\bar{\theta}$, the market is so tight that all goods purchased are devoted to matching: no goods are left for consumption.

We plot the matching wedge against tightness to visualize how it behaves (figure 4.1E). With the calibration used in the plot, the vertical asymptote of the wedge occurs at $\bar{\theta} = 4.9$.

4.6.3. Advantages of assessing the matching cost in terms of goods

We assume that the matching cost incurred by buyers is assessed in terms of goods. This means that to pay for the visits, buyers have to purchase an extra amount of goods that covers the visit costs. We make this choice for a few reasons, which we review here.

First, the assumption is tractable because then, we do not need to introduce an extra good to measure the matching cost. We can just focus on one market without thinking

[2] For the urn-ball and CES matching functions, $q(0) = 1$, so $\kappa < 1$ is sufficient.

about another market where that other good would be traded. The fact that the market is self-contained is helpful to think about welfare and efficiency.

Second, the assumption is portable: it allows us to think about any slackish markets, without having to specify the matching cost in different ways in different markets. In particular, it allows us to have isomorphic product and labor markets, which is especially convenient in the business-cycle model of chapter 15.

Third, the assumption is not unrealistic. In the labor market, a huge amount of labor is devoted by firms to recruiting workers—including headhunters hired by firms, human-resource workers within firms, and of course regular workers spending time reading resumes and interviewing future colleagues. In various markets for services, part of what is purchased is the service of a broker or agent who helps source the desired services. For instance, in the temporary staffing industry, the staffing agency is paid in part to recruit workers with the appropriate skills, experience, and availability. In the construction industry, the project manager is paid in part to find the appropriate construction workers required for the various tasks to complete the project (plumber, roofer, painter, landscaper, and so on). In the wedding industry, the wedding planner is paid in part to find the appropriate caterer, florist, band, dressmaker, and so on. It is maybe less clear how the assumption applies to markets for actual goods. However, in that case, we can introduce the matching cost in a different way: as the cost of a sample that the buyer procures to determine whether they want to purchase the good or not. This alternative representation is equivalent to the representation used in the model, and it applies to many goods markets (appendix E).

4.7. Market demand

The cornerstone of the demand side is the buyer's optimization problem: choosing how much to consume so as to maximize utility subject to a budget constraint. To build the market demand, we first need to specify buyers' utility function, then their budget constraint, before solving the maximization problem and deriving individual demands.

4.7.1. Buyer's utility function

We assume that buyers have utility over two things. First, buyer j has utility over consumption of the goods sold on the market, $c_j \geq 0$. Because we want to have a demand for goods, buyers must have the choice between spending on goods and spending on something else. If there is no choice, the concept of demand is not well-defined. We introduce money as the alternative that makes demand well-defined in our static environment. We assume that buyers derive utility from money balances, denoted b_j.

To further simplify the analysis, we assume that the buyer's utility function is quasi-linear. Specifically, we assume that buyer j has utility over consumption and money balances:

$$\mathcal{U}(c_j, b_j) = ac_j^{1-\alpha} + b_j, \tag{4.10}$$

where $a > 0$ governs the taste for goods relative to money and $\alpha \in (0, 1)$ governs the concavity of the utility over goods, which determines diminishing marginal utility of consumption.

4.7.2. Buyer's budget constraint

Buyer j maximizes their utility, but of course subject to a budget constraint. We assume that to start with, each buyer has an endowment of money B_j, which they use to hold money or buy goods. The budget simply says buyers can spend the endowment on goods or keep some of the endowment in money. In order to consume c_j goods, buyer j must purchase $[1 + \tau(\theta)]c_j$ goods at a price p. Hence, the expenditure to purchase goods is $p[1 + \tau(\theta)]c_j$. The buyer's budget constraint therefore is:

$$b_j + p\left[1 + \tau(\theta)\right]c_j = B_j. \tag{4.11}$$

The matching structure introduces a new term into the buyer's budget constraint: the matching wedge, $\tau(\theta)$. It appears because visits to sellers are costly. The wedge is specific to the slackish world; in the Walrasian world, anyone can buy any quantity at the given market price without any other cost, so there would be no matching wedge.

This new element modifies the buyer's behavior. Nevertheless, we can still treat the problem as we usually do: maximizing utility subject to a budget constraint. It's just that, in this case, the budget constraint is slightly modified.

4.7.3. Buyer's problem

We now need to solve the buyer's problem. This is a crucial element of the model because buyers are making the only behavioral decision here: how many goods to consume. Equivalently, buyers decide how many sellers to visit: once they have determined the number of visits, given the probability that a visit is successful, they also determine their consumption.

Buyer j's problem is to maximize their utility (4.10) given the budget constraint (4.11), taking aggregate variables as given: market price, p, and market tightness, θ. The easiest way to solve the problem is to express the buyer's money balances as a function of consumption, by reworking the budget constraint:

$$b_j = B_j - p\,[1 + \tau(\theta)]\,c_j.$$

Then, substituting these money balances into the buyer's utility function, the buyer's problem becomes $\max_{c_j \geq 0} \mathcal{U}(c_j, B_j - p[1 + \tau(\theta)]c_j)$, or

$$\max_{c_j \geq 0} a c_j^{1-\alpha} + B_j - p\,[1 + \tau(\theta)]\,c_j. \tag{4.12}$$

Note that we do not require money balances to be positive. Buyers may therefore spend more than their endowment and end up with negative balances—which we would interpret as borrowing instead of saving. We do this to avoid the complications that would arise if some buyers with low endowment were up against their borrowing constraint while other buyers with high endowment were not.

Here we assume that sellers and buyers take market tightness as given. This is quite a natural assumption because market tightness is the ratio of aggregate number of visits to aggregate number of goods for sale. As each market participant is small relative to the size of the market, it cannot influence tightness. We also assume that sellers and buyers take the market price as given; we will see when we discuss the formation of the market price in section 4.8 that this assumption is also reasonable.

Before we solve this maximization problem, let us briefly verify that it is well behaved. Since $\alpha > 0$, the first term is concave in c_j. The second term is linear in c_j, so concave in c_j. The objective function is the sum of two concave functions, so it is concave in c_j. In addition, the objective function's domain, $[0, \infty)$, is convex. Accordingly, we are solving a concave maximization problem. We can conveniently do this via first-order condition, because with this type of problems, the first-order condition is sufficient to find the global maximum (result A.20).

We therefore set the derivative of the objective function to 0 to determine the optimal c_j. The first-order condition directly yields

$$(1 - \alpha) a c_j^{-\alpha} = p\,[1 + \tau(\theta)]\,. \tag{4.13}$$

This condition says that at the optimum the buyer is indifferent between consuming one additional unit of good (which provides utility $(1 - \alpha)ac^{-\alpha}$) and holding an additional $p[1 + \tau(\theta)]$ units of money (which provides utility $p[1 + \tau(\theta)]$), which is the additional amount of money available to the buyer if they forego consumption.

Therefore, at the optimum, the buyer cannot improve their utility by consuming slightly more or slightly less: they are doing the best they can. This optimality condition is standard: it equalizes the marginal rate of substitution between consumption and money balances, $(\partial \mathcal{U}/\partial c)/(\partial \mathcal{U}/\partial b)$, with the price of consumption in terms of money. But, since money is the numeraire, and the marginal utility of money balances is 1, the condition simply equalizes the marginal utility from consumption, $\partial \mathcal{U}/\partial c$, with the price of consumption, $p[1+\tau(\theta)]$.

We now rewrite the first-order condition to get an expression for the buyer's desired consumption of goods as a function of the market price and tightness:

$$c_j = \left[\frac{(1-\alpha)a}{p} \cdot \frac{1}{1+\tau(\theta)}\right]^{1/\alpha}.$$

The consumption has intuitive properties. If the market price is higher, so goods are more expensive, the buyer consumes fewer goods. If the market tightness is higher, so goods are more difficult to find, the buyer consumes fewer goods. And finally if the buyer derives lower utility from goods (lower a), the buyer consumes fewer goods.

From the amount of goods consumed, c_j, it is easy to back out the rest of the buyer's behaviors. To consume c_j the buyer must purchase $y_j = [1+\tau(\theta)]c_j$ goods, and must visit $v_j = c_j/[q(\theta)-\kappa]$ sellers. The buyer spends $p[1+\tau(\theta)]c_j$ in total on goods and therefore holds $B_j - p[1+\tau(\theta)]c_j$ in money.

4.7.4. Expression of the market demand

We now aggregate the individual demands for consumption of goods to obtain the market demand for consumption: $c = \int_0^1 c_j \, dj$. We assumed that different buyers have different initial wealth. One might expect this to lead to a complex distribution of consumption choices. However, a crucial property of the quasilinear utility function (4.10) is that it eliminates wealth effects for the choice of c_j. Hence, all buyers choose the exact same amount c_j, regardless of whether they are rich or poor. This means we can aggregate easily. Since there is a mass 1 of buyers, the market demand for consumption $c_d(\theta, p)$ equals the individual demand that we have just derived:

$$c^d(\theta, p) = \left[\frac{(1-\alpha)a}{p} \cdot \frac{1}{1+\tau(\theta)}\right]^{1/\alpha}. \tag{4.14}$$

This equation gives the notional market demand. It indicates how much buyers would like to consume for a given market price and tightness.

However, in our model, we use an effective market demand, which is the amount of goods that buyers actually desire to buy for a given market price and tightness, which is more than the notional demand. The gap arises because the amount of goods that buyers are able to consume is less than the amount of goods that they buy, since, in a slackish market, some of the purchased goods are devoted to matching instead of consumption.

Conveniently, the concepts of market demand and supply are consistent—the supply and demand both measure goods that are traded. Hence, when we solve the model, we can use the condition that supply equals demand, as any good that is sold by a seller is a good that is bought by a buyer. By focusing on the number of trades, we have equality of supply and demand, even though the model is not Walrasian.

Formally, the market demand, $y^d(\theta, p)$, is the total amount of goods that all buyers purchase so as to maximize their utility given the market price, p and the market tightness, θ. To compute this market demand, we translate the amount of consumption desired, given by (4.14), into the amount of purchases desired, using the result that consuming one good requires purchasing $1 + \tau(\theta)$ goods. This means that

$$y^d(\theta, p) = [1 + \tau(\theta)]\, c^d(\theta, p),$$

which then gives the following expression:

$$y^d(\theta, p) = \left[\frac{(1-\alpha)a}{p}\right]^{1/\alpha} \cdot \frac{1}{[1+\tau(\theta)]^{1/\alpha - 1}}. \tag{4.15}$$

The market demand $y^d(\theta, p)$ is the number of goods demanded by buyers given market price and tightness.

4.7.5. Properties of the market demand

Let us turn to the properties of the market demand, given by (4.15). First, we see that it is decreasing in the market price. When the price level is higher, purchasing the good becomes less attractive compared to holding money, so demand is lower.

Similarly, the market demand is decreasing in market tightness. This property is visible in (4.15) because the matching wedge $\tau(\theta)$ is increasing in θ, while $1/\alpha > 1$. Intuitively, when the market is tighter, it is more difficult to buy goods: each visit is less likely to be successful, so buyers must visit more sellers per purchase, which pushes the matching wedge up. Hence, buyers have to devote a larger share of their purchase to matching instead of consumption, which makes consumption less appealing, and reduces demand.

Recall that $\bar{\theta}$ is the value of tightness at which $\tau(\bar{\theta}) = \infty$. We see that at this value, demand is 0. When tightness is 0, on the other hand, demand is given by

$$y^d(0, p) = \left[\frac{(1-\alpha)a}{p}\right]^{1/\alpha} [1+\tau(0)]^{1-1/\alpha}.$$

To wrap up our analysis of the market demand, we plot the demand curve in a simple graph with quantity on the x-axis and tightness on the y-axis (figure 4.1F). The graph depicts the properties of the market demand that we have just derived. This representation of quantity against tightness is specific to slackish markets. It differs from the Walrasian representation of quantity against price. In slackish markets, we plot quantity against tightness because tightness is the core variable that equilibrates the market.

4.8. Market price

Something that we haven't discussed yet is how the price at which goods are traded is determined. We address this important element now.

We are dealing with a slackish market where trades occur in a decentralized fashion, as described by the matching function. In a Walrasian market goods are sold at auction and the price is set by the auctioneer so as to equalize supply and demand. Here all trade occurs in individual buyer-seller pairs, who are free to negotiate the price as they wish. We cannot assume that a centralized auctioneer sets the price, as we do in a Walrasian market. We need to find a way to model how prices are formed in each of the buyer-seller pairs once they have met on the market.

We introduce a price norm that describes how prices are set in all these buyer-seller pairs. The price norm is very general and can take many forms. As we discuss in chapter 6, we may postulate a price that is a rigid function of the model parameters, or we may assume bargaining between buyers and sellers.

For the time being, we simply assume that prices in all trades are identical and are given by a generic price norm $p = p^n(\text{market variables})$. Since all market variables can be expressed as a function of the market tightness, we simplify the expression of the price norm to

$$p = p^n(\theta). \tag{4.16}$$

The goal should be to specify a price norm that reflects how prices are set in practice in the market studied—we want a price norm that is empirically valid. In chapter 5, we review evidence on how prices and wages are set, and in chapter 6, we specify a price norm that encodes these real-world properties.

Instead of assuming an auctioneer, as in Walrasian theory, our decentralized model gives us the freedom to make assumptions that are as realistic as possible about how prices are set in the real world. This is a vast improvement over the Walrasian model, because the assumption of perfectly competitive prices is probably the least realistic and therefore the most problematic assumption used in modern market models. Furthermore, the assumption of perfectly competitive prices has a raft of implications about welfare and policy, which are all erroneous if the pricing assumption itself is erroneous.

As in Walrasian theory we assume that sellers and buyers take prices as given. One might think that this is problematic because a buyer and a seller could bargain the transaction price once they have matched. However, the actual transaction price has no influence on anyone's decisions because the decisions are made before the match is realized. What matters is the price at which buyers and sellers expect to trade. This is what determines their decisions. The bargained price is only a transfer from buyer to seller: it has no effect on traded quantities, since bargaining would never lead to the destruction of a match. For instance if a price is a little lower than expected, the seller takes home a little less income, which means that they hold a little less money, while the buyer spends a little less, which means that they hold a little more money. But the market quantities—visits, sales, consumption, tightness, slack—would be unchanged.[3]

4.9. Solution of the model

Finally, we present a strategy to solve the model. The main step in solving the model is to determine the market tightness, which we do using a supply-equals-demand condition.

4.9.1. Structure of the solution

Now that we have introduced all the elements of the model and made all the assumptions we need, we are finally in a position to actually solve the model. Before we do so, let us first discuss the structure of the solution: the elements that we are looking for and what we can use to find them.

Formally, solving a model means figuring out the values of all the variables in the model as a function of the parameters of the model. Here, this is actually pretty simple. In our slackish market model, we have to find 6 aggregate variables using 6 independent equations, as shown in table 4.1. There are also individual variables in the model, such as consumption c_j, expenditure y_j, money balances b_j, and visits v_j by any individual buyer j, but these can easily be determined from the aggregate variables.

Now that we have the structure of our solution, the challenge is to find a simple way to compute all the variables in the model.

Before we do that, it is useful to realize that the description of the model in table 4.1 is not unique: it is one of many equivalent representations. Indeed, it is possible to reshuffle the independent equations into other equations, which offer alternative ways to solve the model. Different representations might be useful in different situations, depending on the question and property of interest.

The 6 equations form one valid representation of the model. We can algebraically combine and rearrange them to form a different but equivalent set of 6 independent

[3]We could generalize the model by allowing the transaction price to be the price norm plus some white noise. This would not change anything, so to keep things simple, we assume that actual prices are just the price norm, which buyers and sellers take as given.

TABLE 4.1. Structure of the slackish market model

Variable	Name	Condition	Interpretation	Reference
c	Consumption of goods	$c = c^d(\theta, p)$	Utility maximization	(4.14)
p	Price of goods	$p = p^n(\theta)$	Price norm	(4.16)
y	Sales of goods	$y = [1 + \tau(\theta)]c$	Matching wedge	(4.7)
v	Visits	$q(\theta)v = y$	Matching function	(4.1)
θ	Market tightness	$\theta = v/k$	Definition of tightness	(4.2)
u	Rate of slack	$u = 1 - f(\theta)$	Definition of slack	(4.4)

equations. The key is that any valid system yields the same solution. For instance, if we divide $y = q(\theta)v$ by $\theta = v/k$, we get $y/\theta = q(\theta)k$, or $y = \theta q(\theta)k$, which is just $y = f(\theta)k = y^s(\theta)$. This manipulation is telling us that we could determine sales from the market supply. However, then we would need to replace either the condition for v or θ, since otherwise the 3 equations for y, v, and θ would not be independent. Conceptually, if we did not replace the equations for v or θ, we would lose the information about the matching wedge and matching cost in the model solution—the information that used to be conveyed by the condition that $y = [1 + \tau(\theta)]c$.[4]

4.9.2. Strategy to solve the model

To solve the model, we need to figure out the values of the 6 variables that are determined by 6 independent equations (table 4.1). We follow a sequential strategy for solving the model: we first determine the market tightness, θ; then, we compute the values of the 5 remaining variables. We will compute the market tightness in the next section. Here, we describe how the 5 other variables can easily be determined once we know tightness.

First, we know via the price norm that $p = p^n(\theta)$ so we can easily compute p given θ. Similarly, the rate of slack is directly determined by tightness: $u = 1 - f(\theta)$.

For consumption, we simply use the notional market demand: $c = c^d(\theta, p)$. Since the price p is itself a function of tightness, we can determine c if we know θ.

Then, to compute sales, we simply use that $y = [1 + \tau(\theta)]c$. Since c can be determined from θ, we can compute y given θ.

Last, to calculate the number of visits, we have that $v = y/q(\theta)$. Again, since y is a function of tightness θ, we can compute v as soon as we have θ.

[4] For instance, we could determine the number of visits by $y = c + \kappa v$, or $v = (y - c)/\kappa$. That condition would capture properly how the matching cost enters into the model. And this condition contains the same information as the previous equations $y = [1 + \tau(\theta)]c$ and $q(\theta)v = y$, so no information would be lost by replacing the equations for y and v in table 4.1 by $y = y^s(\theta)$ and $v = (y - c)/\kappa$. Indeed, $y = [1 + \tau(\theta)]c$ implies $y = q(\theta)c/[q(\theta) - \kappa]$, which gives $y[q(\theta) - \kappa] = q(\theta)c$, then $q(\theta)(y - c) = \kappa y$, and finally $(y - c)/\kappa = y/q(\theta)$. But then $q(\theta)v = y$ tells us that $v = y/q(\theta)$, so combining both equations we learn that $v = (y - c)/\kappa$, which is just the new equation used to determine v.

We have shown that we can easily determine the values of the remaining 5 variables once we know market tightness. Tightness is the most critical variable and the next step is to figure out how to compute it.

4.9.3. Computing market tightness from demand and supply

We are now ready to compute the market tightness, θ. From it we can determine the values of all the other variables in the model.

First, we need to think about the key decisions in our model. The first is the buyers' purchasing decision given market tightness, which is characterized by the market demand:

$$y = y^d(\theta, p^n(\theta)). \tag{4.17}$$

The market demand gives the amount of output demanded by buyers.

The second important decision is buyers' shopping decision, where the number of visits is chosen such that buyers purchase the desired amount of goods given market tightness:

$$y = vq(\theta).$$

By definition of tightness, we know that

$$v = \theta k.$$

So, if we know the number of goods that are supplied to the market, k, and tightness, θ, we can calculate the aggregate number of visits. Then,

$$y = \theta q(\theta) k.$$

Recall that $\theta q(\theta) = f(\theta)$ and $f(\theta)k = y^s(\theta)$, so

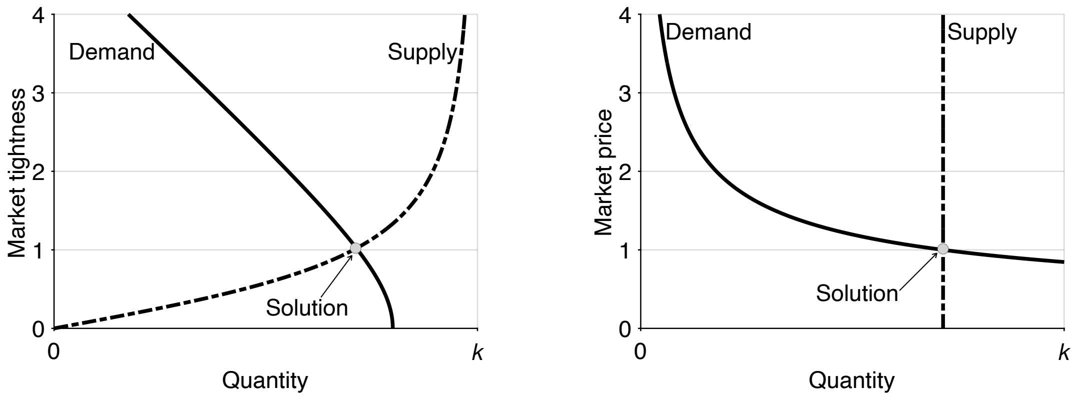
$$y = y^s(\theta), \tag{4.18}$$

which just says that output is given by the market supply too.

Thus, in our model, output is given by both market demand and supply (equations (4.17) and (4.18)). The solution of the model must therefore ensure that market demand equals market supply—otherwise how could output be equal to both market demand and market supply. Accordingly, market tightness must equalize market demand and supply:

$$y^d(\theta, p^n(\theta)) = y^s(\theta). \tag{4.19}$$

The value for market tightness that solves the model can be found at the intersection



A. Solution in tightness-quantity plane B. Solution in price-quantity plane

FIGURE 4.2. Numerical solution of the slackish market model

The market supply is given by (4.5). The market demand is given by (4.15). The matching function is CES, given by (3.12). Parameters are set to $\sigma = 2$, $\kappa = 0.2$, $k = 1$, $\alpha = 0.5$, and $a = 2$. In panel A, the price is set to $p = 1$ and tightness varies. In panel B, tightness is set to $\theta = 1.02$, which is obtained from the intersection of the supply and demand curves in panel A, and p varies.

of the market supply and demand curves. The market supply curve captures the number of trades, for any tightness, given the market capacity and the matching process. The market demand curve captures the number of trades, for any tightness, that maximize the utility of buyers given their budget constraints. Thus, at the solution, the matching process is respected, as well as the utility maximization process.

The supply-equals-demand condition (4.19) is the equivalent of the market-clearing condition of Walrasian theory. The Walrasian market-clearing condition imposes that at the market price, the quantity that buyers desire to buy equals the quantity that sellers desire to sell. This condition is required to ensure the consistency of the Walrasian model because sellers and buyers make their decisions expecting to be able buy and sell any quantity at the quoted price. Here, the supply-equals-demand condition imposes that at the market tightness, the quantity that buyers desire to buy equals the quantity traded through the matching process. The condition ensures the consistency of the slackish model: that buyers visit the appropriate number of shops given how much they desire to consume.

Just as a Walrasian market, a slackish market can be conveniently analyzed via supply and demand curves in a market diagram. Figure 4.2A plots the market supply and demand curves, assuming a simple price norm ($p = 1$). The key difference with a Walrasian market is that it is the market tightness, not the market price, that equilibrates supply and demand. That's why the supply and demand curves are plotted in a tightness-quantity plane, not in the Walrasian price-quantity plane. The tightness that solves the slackish model is found at the intersection of the two curves, just like the price that solves the Walrasian model is found at the intersection of the Walrasian supply and demand curves. In this numerical example, the solution is $\theta = 1.02$.

Of course we can also represent the supply-equals-demand condition in a Walrasian, price-quantity plane (figure 4.2B). In this graph we set the tightness to its solution value: $\theta = 1.02$. The supply curve is just vertical in this plane, because it does not depend on the price. The demand curve is downward sloping. The intersection between supply and demand curves occurs at the price norm, $p = 1$, because we set tightness to the value such that $y^s(\theta) = y^d(\theta, 1)$. Thus, when $p = 1$, supply equals demand. But this graph does not bring new information. It is just repeating what we saw in the tightness-quantity graph: that at $\theta = 1.02$ and $p = 1$, supply equals demand. Furthermore, unlike tightness, the price is not determined by market forces: it is given by the price norm. So the price-quantity graph is not meaningful in a slackish market.

4.9.4. Brief discussion of the solution

To conclude, let us discuss a few things about the solution of the model.

First, just like there are alternative ways to describe the structure of the model, as we saw in table 4.1, there are alternative ways to get the market tightness. An equivalent approach, followed by Michaillat and Saez (2015), is to formulate the market supply and demand in terms of consumption instead of output. Then the market demand $c^d(\theta, p)$, given by (4.14), indicates the number of goods that buyers want to consume. The market supply $c^s(\theta)$ indicates the number of goods traded given the matching process and the amount placed for sale by sellers. This alternative market supply is given by $c^s(\theta) = y^s(\theta)/[1 + \tau(\theta)]$. Condition (4.19) is equivalent to $y^d(\theta, p)/[1 + \tau(\theta)] = y^s(\theta)/[1 + \tau(\theta)]$ and so to $c^d(\theta, p) = c^s(\theta)$. The consumption-based solution makes it easier to think about welfare, but is otherwise slightly less natural, so we follow the output-based approach in this book.

Second, in many models with matching functions, the key decision by buyers is the effort that they put into searching for goods. For instance, in a DMP model, this effort is the vacancies posted by firms. In these search-and-matching models, the solution of the model is organized around the search effort. As we show in appendix F, it is possible to reframe the solution of the model in terms of the number of visits to sellers—which is search effort in this model. The two solutions are equivalent, but the visit-based approach is less convenient analytically.

Third, a natural question is about the realism of the solution concept. When we solve the model, we assume that buyers maximize their utility given market tightness. But how can buyers determine what market tightness is on the market? This is a typical challenge in markets in which buyers and sellers must figure out the values of market variables on which they base their decisions. It is quite a challenging task for buyers to consider the entire market around them and assess what the market tightness will be once all buyers and sellers enter the market.

We discuss these issues further in appendix G and propose a simple resolution. We assume that a statistical agency announces tightness in advance, and that buyers act based on the tightness that has been announced. Then, if the agency announces the tightness equalizing market supply and market demand, the prevailing tightness is the announced tightness. So the agency is correct, and the supply-equals-demand tightness prevails, just like in this chapter.

In fact, even if the statistical agency is unable to build market demand and supply, or unable to equalize both, the agency may find the tightness solving the model via a tatonnement process. Under certain conditions, the model behaves well enough that the tatonnement process converges to the model solution, at which point the tightness quoted by the agency is the same as the tightness realized on the market.

We have now assembled the basic machinery of a slackish market. The core equation of the model is the supply-equals-demand condition, given by (4.19), which determines market tightness. For the time being, we cannot solve the equation explicitly or perform comparative statics because the solution depends on the price norm, $p^n(\theta)$. In the next chapter, we will look at different assumptions for the price norm and solve what the market tightness is in each case and determine how the model responds to different shocks for each norm.

4.10. Model with fixed prices

To complete this chapter, we analyze a slackish market model in which the price norm gives a completely fixed price. The fixed-price model is not only simple but also quite useful because any price that's not fully flexible produces the same qualitative results as a fixed price—with of course different quantitative results—as we will show in chapter 6. Furthermore, the model with a fixed price is the most striking example of a slackish model: all adjustments to shocks occur through market tightness, and none through prices.

4.10.1. Solution of the model

We start by solving the slackish market model when prices are fixed. We assume that the price norm imposes a fixed price $p > 0$ for all goods.

To solve the model, we need to find the tightness θ that equalizes market demand and supply:

$$y^d(\theta, p) = y^s(\theta). \tag{4.20}$$

We first establish the existence and uniqueness of the solution by introducing the excess demand function:

$$z(\theta, p) = y^d(\theta, p) - y^s(\theta). \tag{4.21}$$

With the excess demand function, the solution of the model simply is

$$z(\theta, p) = 0.$$

We know that $y^d(\theta, p)$ is strictly decreasing from $y^d(0, p) > 0$ when $\theta = 0$ to 0 when $\theta = \bar{\theta}$. At the same time, $y^s(\theta)$ is strictly increasing in tightness, starting from 0 when $\theta = 0$. Moreover, both y^d and y^s are continuous in tightness.

Thus, we know that the excess demand function z is continuous in θ, and it declines from $z(0, p) = y^d(0, p) > 0$ to $z(\bar{\theta}, p) = -y^s(\bar{\theta}) < 0$ when θ increases from 0 to $\bar{\theta}$. Thus, by Bolzano's theorem, the equation $z(\theta, p) = 0$ has a solution on $(0, \bar{\theta})$ (result A.1). Moreover, since the excess demand function is strictly decreasing, we know that the solution is unique.

This argument is illustrated in figure 4.3B.

4.10.2. Graphical representation of the solution

To understand the mechanics of the model, and later derive comparative statics, it is very helpful to represent the solution of the model in a tightness-quantity plane. We know that the market demand $y^d(\theta, p)$ is strictly decreasing from $y^d(0, p) > 0$ when $\theta = 0$ to 0 when $\theta = \bar{\theta}$. At the same time, the market supply $y^s(\theta)$ is strictly increasing in tightness, starting from 0 when $\theta = 0$. The market demand curve and market supply curve are illustrated in figure 4.3A. We confirm that the curves $y^d(\theta, p)$ and $y^s(\theta)$ cross once for $\theta \in (0, \bar{\theta})$. The crossing point is the solution of the model.

Comparing figure 4.3A to figure 4.3B, we see that the supply-demand representation conveys more information than the excess-demand representation, which is why we use it throughout the book. The supply-demand representation provides not only the tightness θ that solves the model, but also the corresponding output y, and the amount of slack in the market, $k - y = u(\theta)k$. In particular, we see that in a slackish market, although supply equals demand, there always is some slack: there always are some goods that are available for sale but not sold.

4.10.3. Comparative statics

Now that we have solved the model, we can look at comparative statics: how the model responds to shocks to various parameters. We use our graphical supply-demand representation to visualize the behavior of the model in response to the shocks.

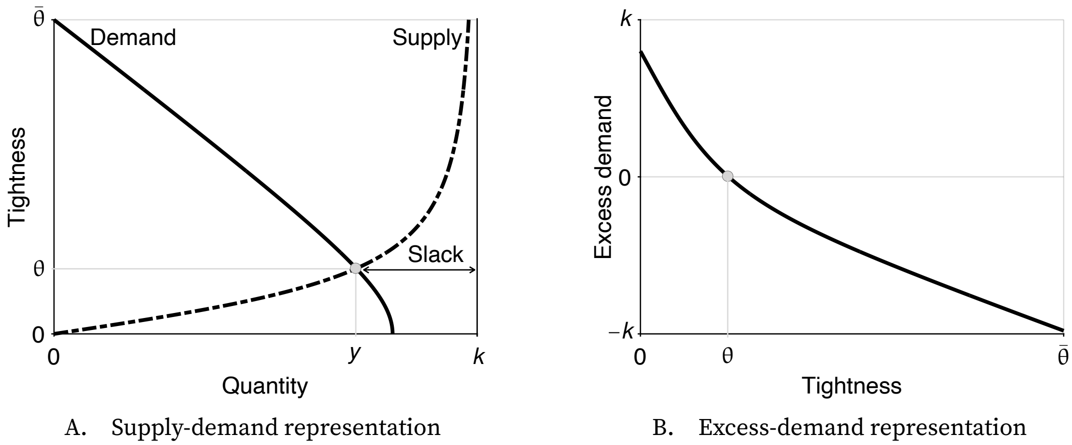


FIGURE 4.3. Solution of the slackish market model with fixed prices

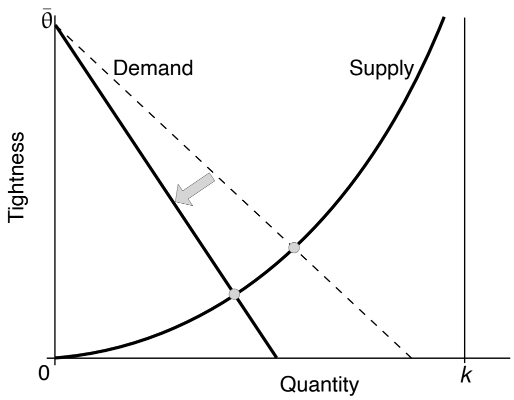
The market supply is given by (4.5). The market demand is given by (4.15). The excess demand is given by (4.21). The matching function is CES, given by (3.12). Parameters are set to $\sigma = 2$, $\kappa = 0.2$, $k = 1$, $\alpha = 0.5$, $a = 2$, and $p = 1$.

Let us start by looking at a negative demand shock: a decrease in preference for goods, a. We start with a high preference, and then reduce the parameter value to a low preference in order to see how the solution of the model changes.

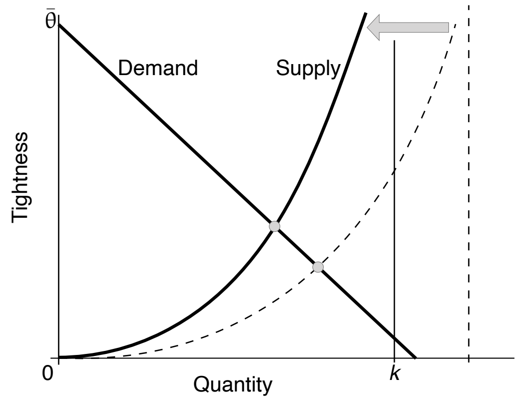
From figure 4.4A, we see that a negative demand shock brings the demand curve inward, which results in a lower value of both tightness θ and sales y. This was to be expected: as buyers value goods less, they aim to consume and therefore purchase fewer goods. They therefore visit fewer sellers, which reduces tightness and the amount of goods sold. The amount of slack in the market increases as it becomes harder for sellers to sell their goods. Note that at this stage we cannot say what happens to consumption. Although purchases decline, the matching wedge $\tau(\theta)$ also decreases, so a larger share of purchases is devoted to consumption. That is, buyers purchase fewer goods but devote a larger fraction of them to consumption. The reason is that with lower demand, visits are more likely to be successful, so fewer visits are required per purchase. Thus, the response of consumption depends on the specific situation, as we clarify in chapter 9.

Another way to be convinced that tightness must fall after a negative demand shock is to assume that tightness remains the same and realize that we reach a contradiction. Since tightness and price remain the same, but the parameter a is lower, we know that buyers want to purchase fewer goods. However, through the matching process, if tightness is the same, the number of goods sold remains the same. So here's the contradiction: if tightness remains the same, it is not possible that both buyers maximize utility and the matching process is satisfied. In other words, keeping the same tightness as before is not a solution of the model.

Next we turn to a negative supply shock: a decrease in the market capacity, k. We start



A. Negative demand shock



B. Negative supply shock

FIGURE 4.4. Comparative statics in the slackish market model with fixed prices

The market supply is given by (4.5). The market demand is given by (4.15). The price is fixed to p. A negative demand shock is a reduction in the preference for goods a. A negative supply shock is a reduction in market capacity k.

with a high capacity, and then change the parameter value to a lower capacity in order to see how the model solution changes. In figure 4.4B, we see that a decrease in capacity brings the supply curve inward. What happens to the solution of our model? The new value of sales y is lower than it was before, while the new value of tightness θ is higher. This makes sense because if fewer goods are supplied, through the matching process, fewer goods are bound to be purchased. And because there is more competition among buyers, tightness is bound to increase. Hence slack falls in the market, both in the sense that the share of goods that remain unsold, $u(\theta)$ is lower, and in the sense that the number of goods that remain unsold, $u(\theta)k$ is lower. Since tightness is higher, the matching wedge $\tau(\theta)$ also increases, so a larger share of fewer purchases is devoted to matching. In that case, we know that consumption decreases. What we do not know here is what happens to the number of visits. Tightness is higher so visits are less likely to be successful, which means mechanically that buyers would have to visit more sellers to keep the amount of purchases constant, but which also means that purchases become more complicated and thus less desirable. As we see in appendix F, visits may go up or down after a negative supply shock.

To summarize, both negative demand and supply shocks lead to lower sales; the key difference is that negative demand shocks lead to lower tightness while negative supply shocks lead to higher tightness. After a negative demand shock, it becomes harder to sell goods, while after a negative supply shock, the fewer goods that are for sale are sold more easily. The price does not respond at all to the shocks: it is tightness that absorbs all the shocks and re-equilibrates the market.

4.11. Summary

In this chapter we introduce the basic model of a slackish market: a market in which slack is the central mechanism for equalizing supply and demand. Unlike a Walrasian market, where the price adjusts to equilibrate supply and demand, in a slackish market, the market tightness—here the ratio of buyers' visits to goods for sale—is the variable that equilibrates the market.

The model is built around sellers with a fixed capacity of goods and buyers who must visit sellers to make purchases. The number of trades is determined by a matching function, which captures the complex, decentralized process of buyers finding sellers. Due to the properties of this function, the probability of selling and the probability of buying depend solely on the market tightness. A tighter market favors sellers but harms buyers. The rate of slack, which is the share of goods that remain unsold, is a decreasing function of tightness. The matching wedge, which represents the additional goods buyers must purchase to cover the costs of visiting sellers, is an increasing function of tightness. Importantly, the matching wedge creates a gap between what is purchased and what is consumed.

Buyers maximize their utility, which yields a downward-sloping market demand curve when quantities are plotted against tightness. The market supply is upward-sloping in a tightness-quantity plane because more tightness means a higher probability of selling goods.

The model is solved by finding the market tightness that equalizes market demand and supply. This tightness then determines all other market outcomes, including output, consumption, and the rate of slack. The price at which goods are traded is set by a price norm, which can take various forms.

In the simple case with fixed prices, tightness adjustments absorb all shocks. Negative demand shocks reduce tightness and then sales because of increased slack. By contrast, negative supply shocks increase tightness so they reduce sales despite lower slack.

CHAPTER 5.

Prevalence of price and wage rigidity

In a slackish market, there are infinitely many ways to split the bilateral surplus enjoyed by traders who have met, and therefore many possible ways to set prices. We assume that it is price norms that determine prices in the market.

Price norms are determined by both economic and social forces and can take many forms. Okun (1981, p. 105) explains this wonderfully in the case of wages—and the same applies to prices:

> Wage standards are conventional and habitual. They have their logic and take on a life of their own. They become entrenched because firms have incentives to get into step with other participants in the labor market. There may be more determinacy in the persistence of any reasonable convention than in its precise nature.

Because price norms can take many forms, before modeling them we must examine how prices behave in real-world markets to understand their properties. Accordingly, in this chapter, we review evidence on prices and wages in the United States. This review will show that prices and wages are somewhat rigid: they adjust slowly to shocks that disturb the market. Fairness appears to be a key reason behind price and wage rigidity. In chapter 6, we will model a realistic price norm and plug it into our slackish market model.

5.1. What are price norms?

A price norm is just how buyers and sellers set prices in a market. There is no such thing in a Walrasian market because there the price is set to equalize supply and demand. Sellers have no choice but to adhere to the market price. If a seller raises their price 1 cent above the market price, they will lose all customers. Buyers have no choice but to adhere to the market price. If a buyer asks for a discounted price 1 cent below the market price, the seller will just refuse to trade and sell their goods to other buyers. And of course there is no reason for sellers to lower their price below the market price or for buyers to offer a price above the market price since sellers and buyers can already trade the desired quantity at the market price.

But assuming a price norm is not as strange as it might seem at first. In the real world, there are also many ways in which prices are formed—a lot of it being cultural or institutional. In the service industry in the United States, for instance, prices are jointly determined by buyers and sellers. Sellers set a floor price, and then buyers add a tip on top, at their discretion in theory, but following well-established tipping norms in practice. A tremendous amount of goods and services are transacted that way, where cultural norms determine the price. Between 2008 and 2018 in the United States, tips declared on W-2 forms amounted to $30 billion per year (Basker, Foster, and Stinson 2024). Furthermore tips are widely underreported: only between 45% and 60% of tips are reported. If underreporting is 50%, the W-2 numbers imply yearly tips of $60 billion—offered entirely voluntarily by buyers after consumption.

Tips are an example of how culture influences prices. The minimum wage and other price controls show how institutions might influence prices. Huet-Vaughn and Piqueras (2023) provide fascinating evidence from a minimum wage reform in Iowa in 2015, which affected wages paid by firms for many years although the reform itself only lasted a bit more than a year. In 2015–2016 in a county in Iowa, the minimum wage was raised from $7.25 an hour to $10.10 an hour. The reform obviously raised the wages of minimum-wage workers. But the minimum-wage reform was abolished after 17 months by the state of Iowa, bringing the minimum wage in the county back down to $7.25. Despite the abolition, wages remained elevated for minimum-wage workers at least until 2022. Firms voluntarily chose to pay workers better for many years, influenced by a temporary policy intervention.

Price norms determine price levels, but as Okun (1981, pp. 151–154) observed, they also include various practices to build the confidence of customers that prices are as fair and as predictable as possible. For instance, firms might commit to fixed-time scheduling: to hold price constant for a fixed amount of time. They might for instance commit to adjust prices only once a year or once a season. Firms might also pre-notify customers in advance of any price change, and explain why an upcoming price increase is necessary.

5.2. Why do price norms exist?

A key property of slackish markets is that buyers and sellers are always happy to trade. Anytime a seller and a buyer find each other and there is an exchange, there is a bilateral surplus from the trade so that both buyer and seller take something away from it—they are strictly better off trading than not trading. This surplus offers room for price norms to emerge and exist.

This is true in the real world: sellers are usually happy when they can sell their goods while buyers are happy when they can buy what they were looking for. This is different from the Walrasian model, where the marginal buyer and seller are indifferent between buying or selling and not buying or not selling, because the price exactly equals the seller's marginal cost of production and the buyer's marginal utility.

Since the solution to such a bilateral-monopoly problem is indeterminate, there are many possible ways to split the bilateral surplus between seller and buyer, including norms.[1] Assuming a price norm is unusual in economics models, but the reality is that in a slackish model, there is no obvious economic mechanism that can determine the price. Any price that leaves seller and buyer with a positive individual surplus would be acceptable to both parties. There is no deviation from that price that generates a Pareto improvement. Of course, a seller would be better-off with a higher price, but a buyer would be worse-off. In this setting, market forces cannot erode prevailing price norms.

In fact, such theoretical indeterminacy might explain why there exist so many different cultural and institutional arrangements to determine prices—effectively norms. In some places, sellers simply set a fixed price. In other places, people normally bargain over the prices of goods. There are places where a price is posted but customers are expected to add a tip whose size is dictated by complex conventions. Sometimes, prices are regulated or directly set by governments.

The situation of bilateral monopoly arises because the pairing of a buyer and a seller generates a positive bilateral surplus. Intuitively, the reason is that it is difficult for sellers to sell their goods and for buyers to find a good, so there is a surplus when they meet and have the opportunity to trade. We now formally establish what the bilateral surplus is. We consider the ex-post surplus from trade, evaluated once the buyer has incurred the cost to visit the seller and the seller has incurred the cost to produce the good.[2] The outside option is to walk away after sinking those costs. To start, we compute the seller's and buyer's surpluses: what they take away from the trade of a good, compared with what they would take away if the trade did not occur.

[1] The indeterminacy of the solution to the bilateral-monopoly problem has been known since Edgeworth (1881). It is discussed by Howitt and McAfee (1987) and Hall (2005b) in the context of matching models.

[2] In this chapter production is exogenous and costless—each seller just brings k_i to the market—but in chapter 7 production will be endogenous and costly.

5.2.1. Seller surplus

We assume that just like buyers, sellers have a quasilinear utility function, so their marginal utility of money is 1. The price of a transaction is p. Then, for p units of money, the seller takes away p utils from the trade. If there was no trade, the seller would get nothing. The seller does not benefit from the good that they put for sale because they cannot consume their own production. The seller surplus therefore simply is:

$$\mathcal{S} = p.$$

Critically, the seller surplus is positive for any positive price, so the seller is always happy to trade.

5.2.2. Buyer surplus

Given that a buyer's utility is given by (4.10), and that we evaluate the surplus at the buyer's planned consumption level c, the marginal utility of one good is:

$$\frac{\partial \mathcal{U}}{\partial c} = (1 - \alpha)ac^{-\alpha}.$$

This is what the buyer takes away from the trade of an extra good at the planned consumption level. If the trade did not occur, however, the buyer would get to keep their p units of money, which provide p utils since the utility is quasilinear. Of course, the buyer does not recover the already-sunk matching expenditures that brought them here. Thus, the buyer surplus is

$$\mathcal{B} = (1 - \alpha)ac^{-\alpha} - p. \tag{5.1}$$

However, the buyer's consumption is not random. It is given by the solution to the buyer's optimization problem. It turns out that consumption is such that the buyer's surplus is always positive. The first-order condition from the buyer's problem, given by (4.13), tells us that the marginal utility from consumption, $(1 - \alpha)ac^{-\alpha}$, equals $[1 + \tau(\theta)]p$, where $\tau(\theta)$ is the matching wedge. Combining this result with the expression for the buyer surplus, we simply get:

$$\mathcal{B} = \tau(\theta)p. \tag{5.2}$$

Similarly to the seller surplus, the buyer surplus is positive. The buyer surplus has a simple interpretation: it is the monetary cost sunk to match with this seller, since consuming one good requires purchasing $\tau(\theta)$ goods for matching beforehand, each costing a price p. Another interpretation is that the buyer surplus is the monetary cost required to match with a different seller if the present trade fell through. Indeed, matching with a new seller to purchase one good would require to spend again on matching cost. The implication is that the buyer surplus is larger when matching costs are larger, because a larger matching cost implies a larger matching wedge, τ. In any case, given that the buyer decides to shop fully knowing what the price norm is, when they find a good, their surplus is always positive, so they are always happy to buy it.

5.2.3. Total surplus

Since the buyer and seller surpluses are both positive, we conclude that the total surplus from a trade, $\mathcal{T} = \mathcal{S} + \mathcal{B}$, is also positive. The total surplus admits two simple, equivalent expressions. The first expression involves the price of the good and is obtained by using (5.2):

$$\mathcal{T} = [1 + \tau(\theta)]\, p. \tag{5.3}$$

The second expression involves the marginal utility of consumption and is obtained by using (5.1):

$$\mathcal{T} = (1 - \alpha) a c^{-\alpha}. \tag{5.4}$$

Hence we see that the total surplus is simply the marginal utility of consumption for buyers. This is quite obvious in retrospect. Without trade, the good is not sold and therefore not consumed by anyone, so one unit of consumption is lost, which is valued at the buyer's marginal utility. In the slackish model, therefore, all trades that occur generate a positive total surplus, irrespective of the price norm.

In sum, any trade generates a positive surplus for both the buyer and the seller. This is a key distinction from Walrasian models where marginal traders are indifferent. The model also differs from monopolistic models, in which sellers enjoy a surplus but the marginal buyer is indifferent between buying and not buying. And the model differs from monopsonistic models, in which buyers enjoy a surplus but the marginal seller is indifferent between selling and not selling. With price norms giving all bargaining power to either seller or buyer, however, the slackish model can mimic a monopolistic or monopsonistic model. But it is much more general than these models: it can feature any price between the monopolistic and monopsonistic extremes—in general, both traders derive some surplus from any trade.

5.2.4. Bilateral efficiency

The theoretical reason why any price norm may prevail in reality is that any price norm respects bilateral efficiency. Price norms that do not respect bilateral efficiency would impose undeniable losses on traders and would therefore be too costly to be sustainable. It therefore seems that at the minimum, to be sustainable, a price norm must respect bilateral efficiency.

Bilateral efficiency is just pairwise rationality: a minimal amount of cooperation so that when two traders meet, they do the best they can for the pair as a whole. If a pair has the opportunity to trade for a positive total surplus, they do it. If bilateral efficiency was violated, pairs would forego opportunities to generate a positive joint surplus and would in effect lower their joint well-being. They would forego available improvements to their joint well-being. If there was minimal cooperation, pairs would not ignore these Pareto improvements and would take the opportunity to increase their total surplus. Hence, price norms under minimal cooperation should be bilaterally efficient.

To respect bilateral efficiency, a price must be at such level that when a trade generates a positive total surplus, buyer and seller are also left with positive individual surpluses. In that way, any time a trade creates some bilateral gains it also creates individual gains, which makes it acceptable to both trading parties. This ensures that any trade that creates bilateral gains is conducted. We have just established that for any price norm, all trades conducted on the market generate both individual and total surpluses. Therefore, any price norm respects bilateral efficiency.[3]

5.2.5. Barro critique

Starting with the *General Theory*, many models have used wage rigidity to generate unemployment. Barro (1977) offered a famous critique of such wage rigidity. He noted that wage rigidity generated unemployment by forcing firm to fire workers whose current marginal product of labor fell below the real wage. However, Barro noted that such firing violated bilateral efficiency: firms and workers could both be better off and generate a positive total surplus by renegotiating a lower wage and continuing to work together. Barro added that since workers are employed by firms for a long time, such renegotiations might be likely to occur. He concluded that generating unemployment from such rigidity was therefore unsatisfactory. The price and wage norms used in our slackish model respect

[3] A famous result due to Hall (2005b) is that in a DMP model, fixed wages are theoretically legitimate as long as they remain within a certain band. Why is the price norm not restricted to a price band here? The reason is that Hall works with a linear model, whereas our utility function satisfies the Inada condition: it has infinite marginal utility when consumption falls to 0. This ensures that any price is acceptable to buyers when their planned consumption is low enough. Given the buyers take the price norm as given when they decide to shop, they only plan to purchase goods that deliver at least as much utility as the price they pay for it, so the buyer surplus is always positive. Given the Inada condition, there is no restriction on the price norm: the price band becomes infinitely wide when planned consumption becomes arbitrarily low.

bilateral efficiency—all relationships generating a positive surplus are preserved—and are therefore immune to the Barro critique.

5.3. Evidence of price rigidity

In theory, price norms can take many forms. We therefore need to examine how prices are set in the real world to understand how price norms look in reality.

In this section, we argue that US prices are somewhat rigid. What do we mean when we say that prices are rigid? We mean that prices do not fully respond to underlying shocks. An extreme case of this is when prices do not move at all—in that case the price is fixed. This is the opposite of a flexible price, which fully responds to shocks and absorbs them.

5.3.1. Frequency of price changes

A typical piece of evidence in favor of price rigidity is that prices stay at the same level for a number of months. Nakamura and Steinsson (2013, table 1) report the frequency of price changes and duration for which prices stayed the same in the United States across many different products. The statistics come from the Consumer Price Index microdata, collected by the BLS between 1988 and 2005.

For regular prices, which exclude sales prices, the median durations are about 7–10 months, while the mean durations are a bit higher, around 9–12 months. The mean is bigger than the median because some products have prices that stay stable for a very long time. A classic example is the bottle of Coca-Cola, which remained priced at a nickel (5 cents) for more than 70 years, from 1886 until 1959 (Levy and Young 2004).

The noticeable difference between median and mean durations of price spells tells us that there is quite a lot of heterogeneity in the frequency of price changes across items. We see this clearly on the histogram reported by Nakamura and Steinsson (2013, figure 3). Many items have a frequency of regular-price changes that is very close to 0, meaning that prices are fixed over years. The biggest mass is for products that have a price-change frequency of 5% per month. These products' prices are fixed on average for 1/0.05 = 20 months, so about a year and a half. For most of the products, the probability of a price change is below 10% per month; these products' prices are fixed on average for at least 1/0.1 = 10 months, so close to a year. The rest of the products are scattered all over the frequency range. There is clearly a lot of heterogeneity in how prices change, which might be unsurprising, because both pricing norms and the variability of demand and production costs vary across products.

5.3.2. Prevalence of rigid prices

In slackish markets, goods are traded only once buyers and sellers have matched. The price is determined in a situation of bilateral monopoly, in which both buyer and seller derive a surplus from trading. Critically, the total surplus is determined by the buyers' valuation for the good (equation (5.4)). As a result, the central empirical question is how prices respond to shifts in how buyers value the good. Unfortunately, the frequency of price changes does not tell us how prices respond to shocks. If the demand for a good is not changing, there is no reason for its price to change. We therefore need to turn to different evidence.

In effect, we are interested in the passthrough of valuation shocks into prices. The idea would be to determine how much the price of a good changes if customers suddenly value it differently. A flexible price would adjust one-for-one, a rigid price would adjust some but less than one-for-one, and a fixed price would not respond.

Unfortunately, again, estimates of the passthrough of marginal valuation into prices are not readily available. But there is plenty of other evidence that firms are reluctant to adjust prices in response to demand shocks. Based on copious econometric evidence, Nordhaus (1974, pp. 183–184) reported that:

> Faced with temporary changes in demand, firms generally alter production and employment rather than price. Prices are based on long-run profitability and other managerial objectives and are not significantly adjusted to cyclical conditions.

Okun (1975, p. 362) observed in discussions with business people that "empirically, the typical standard of fairness involves cost-oriented pricing with a markup," implying that firms would not think about responding to a shift in demand. Blinder et al. (1998, p. 156) asked 121 US firms the following: "When the demand for your product is weak, are your customers willing to let you hold your prices, or do they insist on price reductions?" If the price norm was flexible, all customers would insist on price reductions when they value the product less, but that is not what happened in practice: only 29% of firms answered that customers insist on price reductions; 51% of firms answered that they can hold their prices; and the rest said that customers have mixed attitudes.

The total surplus from a trade can also be expressed as a function of market tightness (equation (5.3)). This is because market tightness determines the matching wedge—the amount of resources that buyers must devote to matching with a seller. When it is difficult to find sellers, then having found one is particularly valuable. If we take this perspective, then we might expect prices to respond directly to market tightness. But here too, survey evidence suggests that prices are quite rigid. Among 189 US firms interviewed by Blinder et al. (1998, p. 153), 64% agree with the idea that "firms have implicit understandings with their customers—who expect the firms not to take advantage of the situation by raising prices when the market is tight."

5.3.3. Possible reasons for price rigidity

We now briefly try to explain why prices might be rigid: why might sellers not fully respond to demand shocks? In a Walrasian market, the price goes up whenever demand goes up, so why are sellers reluctant to charge a higher price for their goods if buyers value them more?

The main reason for price rigidity is that people care about the fairness of prices and react negatively when they perceive prices as unfair. As a result, sellers avoid charging prices that customers might consider unfair. In a famous survey, Kahneman, Knetsch, and Thaler (1986, p. 729) document that buyers find it unfair for sellers to raise prices when the demand for their good is high. They describe the following situation:

> A hardware store has been selling snow shovels for $15. The morning after a large snowstorm, the store raises the price to $20.

Among 107 respondents, only 18% regard this pricing as acceptable, whereas 82% regard it as unfair.

We saw in chapter 4 that if demand for a good is higher but the market price does not respond, market tightness rises, making it more difficult for buyers to buy the good. One might wonder whether, in this situation, it would become acceptable for sellers to raise prices. Kahneman, Knetsch, and Thaler (1986, p. 732) actually inquire about this situation:

> A shortage has developed for a popular model of automobile, and customers must now wait two months for delivery. A dealer has been selling these cars at list price. Now the dealer prices this model at $200 above list price.

Among 130 respondents, only 29% regard this pricing as acceptable, whereas 71% regard it as unfair. So buyers also find it unfair for sellers to raise prices when the market for their good is tight.

We also saw in chapter 4 that market tightness rises if supply of a good is lower and the market price does not respond. One might wonder in this situation too whether it would become acceptable to raise prices. Kahneman, Knetsch, and Thaler (1986, p. 734) inquire about this question by describing the following vignette:

> A severe shortage of Red Delicious apples has developed in a community and none of the grocery stores or produce markets have any of this type of apple on their shelves. Other varieties of apples are plentiful in all of the stores. One grocer receives a single shipment of Red Delicious apples at the regular

> wholesale cost and raises the retail price of these Red Delicious apples by 25% over the regular price.

Among 102 respondents, only 37% view this pricing as acceptable, whereas 63% view it as unfair. This vignette confirms that buyers find it unfair for sellers to raise prices when market tightness is high.

Kahneman, Knetsch, and Thaler conducted their survey in Toronto and Vancouver, but subsequent studies have confirmed their results in the United States. For instance, Shiller, Boycko, and Korobov (1991) obtained similar results from a survey conducted in the greater New York area in 1990.

A natural question is: why should firms care about what is fair and what is not fair? This question has a simple answer: customers appear to reduce purchases when they feel unfairly treated (Piron and Fernandez 1995). More generally, unfair prices seem to make customers angry, which of course firms would like to avoid (Rotemberg 2009).

Moreover, firms do identify fairness as a major concern in price setting. Blinder et al. (1998, tables 5.1 and 5.2) finds that it is in part because of "implicit contracts" with customers that firms do not respond to shocks more: "firms tacitly agree to stabilize prices, perhaps out of fairness to customers." In fact, US firms report that the main reason why they do not change prices more often is that "it would antagonize or cause difficulties for our customers" Blinder (1994, table 4.5). And when firms must change prices, they devote significant resources to justify the price increase to customers and avoid antagonizing them, especially when the increase is large (Zbaracki et al. 2004).

5.4. Evidence of wage rigidity

In this section, we turn from prices to wages. We review evidence on how wages are set in the real world, both for workers in established relationships with firms and for new hires. The main takeaway is that wages are somewhat rigid in reality, which means that they respond in a muted fashion to shocks to labor productivity and labor market slack.

Studying wages separately from prices is important because wages are prices in the labor market model of chapter 10. Moreover, in our core business-cycle model of chapter 13, workers are self-employed, so the product and labor markets are blended, which means that prices are also wages. In that context, understanding wage rigidity is crucial to understand price rigidity. Finally, in the extension of the business-cycle model with separate product and labor markets, covered in chapter 15, both price and wage rigidity play a crucial role.

5.4.1. Frequency of wage changes for job stayers

Just as we did with prices, we begin by looking at the frequency of wage changes in the United States. Dickens et al. (2007, figure 1) report the distribution of percentage changes of wages received by workers who stayed in the same job in the United States in 1987. The data come from the Panel Study of Income Dynamics (PSID), a large-scale household survey run by the University of Michigan.

The first interesting observation is that there is a substantial spike at zero: more than 15% of workers do not have a wage change between 1986 and 1987; they have a wage freeze or a fixed nominal wage. This is surprising since the United States was experiencing inflation during that time. Therefore, real wages for these workers actually fell from one year to the next.

Another interesting point is that a significant mass of wage changes is around 4%. In that period, US inflation was around 4%, so this is evidence of real wages being fixed: nominal wages adjusted exactly by the amount of inflation.

The presence of nominal and real wage freezes is typical in the PSID and other household surveys. These freezes imply that for many workers, their nominal or real wage does not change from year to year: it remains fixed for several years. Kahn (1997) finds such patterns in the PSID between 1970 and 1988. Card and Hyslop (1997) find the same patterns in CPS microdata for 1979–1993.

Interestingly, there are a significant number of workers whose nominal wages decrease. Since Keynes (1936), macroeconomic models often assume that nominal wages are downwardly rigid—that nominal wages cannot be reduced. However, we do see nominal wage reductions in the data. Kahn (1997, table 1) reports that a nominal wage cut from one year to the next occurs 18% of the time. Card and Hyslop (1997, p. 75) find that 15%–20% of job stayers report nominal wage cuts in any year.

One possible issue with household surveys is measurement error, which might explain the nominal wage cuts that we see in the data (Akerlof, Dickens, and Perry 1996). Lebow, Saks, and Wilson (2003, figure 3) address this issue by using the BLS Employment Cost Index microdata for 1981–1999, which are based on establishment records and therefore less prone to measurement error. They find the same patterns: nominal and real wage freezes are common, but nominal wage cuts do occur. Elsby and Solon (2019, table 1) report an even more convincing piece of evidence on the frequency of downward nominal wage rigidity: in administrative data covering all workers from the US state of Washington, 2005–2015, between 20% and 33% of workers receive year-to-year nominal wage cuts, depending on the year. Finally, Grigsby, Hurst, and Yildirmaz (2021, pp. 430–431) use administrative data collected by a large payroll processing company (ADP), covering millions of US workers between 2008 and 2016. They report that on average the share of workers experiencing an annual cut in nominal base wage is only 2.5%, but this share grows to 6% during the Great Recession, and to 10% during the Great Recession in industries particularly badly affected by the downturn. The average share of workers experiencing an annual reduction in nominal hourly compensation (including bonus and overtime) is 17%.

We have seen that many job stayers keep the same wage from one year to the next. So what is the expected duration of wages in jobs? Barattieri, Basu, and Gottschalk (2014, table 6) look at the Survey of Income and Program Participation, which is administered by the US Census Bureau, and which they correct for measurement error. They find that between 1996 and 2000, the probability of a within-job nominal wage change is 15%–22% per quarter. The implied duration of the nominal wage is between 1/0.22 = 4.5 quarters and 1/0.15 = 6.7 quarters, so between one year and one year and a half. Grigsby, Hurst, and Yildirmaz (2021, table 3) report similar results: a quarterly probability of nominal base wage change of 19%, which implies a mean duration of wage contracts of 1/0.19 = 5.3 quarters.

Overall, as Altonji and Devereux (2000, p. 383) write:

> The data overwhelmingly rejects a model of flexible wage changes and provides some evidence against a model of perfect downward rigidity in favor of a more general model.

Based on the evidence, and unlike in Keynesian models, this book does not rely on downward nominal wage rigidity to generate unemployment and unemployment fluctuations. Instead, we will use real wages that are somewhat rigid both upward and downward.

5.4.2. Frequency of wage changes for new hires

We have seen that within a firm-worker relationship, wages are not moving freely. However, when we consider dynamic labor market models, the key decision that the firm makes is whether they hire a worker or not, and what matters to them at the time of making that decision is how they expect to pay the worker during the entire duration of their relationship. Thus, when a worker is hired, the fact that their wages are going to be fixed over time is not sufficient information for the firm. What the firm needs to know is the total wage bill expected over the entire relationship. And what affects the firm's decision to post vacancies and hire workers is whether the overall wage payments are rigid or not.

This is not to say that the wages of existing workers are irrelevant, but they are only one piece of the puzzle. We need to know what happens to the wages of new hires. Once we combine these two bits of information, we will have the full picture of the wage that is paid during the entire relationship. And we will then build a model around that.

To know what happens to wages of new hires, we need to look at wages that are attached to vacancies that are posted. This will tell us what happens when a new worker comes into a firm. Hazell and Taska (2025) do exactly this. They use data from a business analytics company that collects vacancies of many online job portals and company websites (Burning Glass). The data show how the wage associated with each job vacancy in a given firm and location varies between 2010 and 2020 in the US labor market. They find significant rigidities in wages that are posted with vacancies. They report the quarterly probability of nominal posted wage change for one job within a specific firm in a specific location is only 17%, and the expected duration for which the wage remained at the same level is 5.4 quarters—about a year and a half (Hazell and Taska 2025, table 3). This is in line with what we saw earlier for job stayers: many workers do not get a change in nominal wage from year to year.

When wages do change, Hazell and Taska (2025, table 3) also look at whether they go up or down. They find that there is a 4% chance of nominal wages decreasing, and a 12% chance of them increasing. This means that 4/16 = 25% of wage changes are decreases, and 75% are increases. It is not surprising that increases are more common, especially as the economy was recovering from the Great Recession: technological progress, inflation, and the tightening of the economy all pushed wages up. The 25% of nominal wage decreases is also in line with the fraction of job stayers experiencing cuts in nominal wages.

Last, Hazell and Taska (2025, figure 3) look at the distribution of wage changes. It is only for non-zero wage growth, meaning it eliminates the mass of vacancies for which wages do not change from quarter to quarter. There are some nominal wage decreases, and many more wage increases. A large share of the nominal wage increases amount to 3%, and occur every six quarters. The peak of the distribution therefore implies that the typical posted wage grows at 2% per year, which is roughly the level of inflation in the United States during that time. This shows that fixed real wages are common for job vacancies.

5.4.3. Prevalence of rigid wages

We have seen that nominal and real wages move sluggishly, both in existing employment relationships and new employment relationships. However, when we talk about wage rigidity, we are thinking about how wages respond to shocks. Real wages are rigid if they do not respond fully to a change in productivity, and flexible if they respond one-for-one to a change in productivity.

In fact, wage rigidity to productivity shocks is isomorphic to price rigidity to demand shocks. In a slackish market, wages and prices are not set to clear markets but to split the bilateral surplus generated by a match. A change in labor productivity shifts the marginal value of a match to the firm, just as a change in preferences shifts the marginal value of a match to the buyer in the goods market. In both cases, the price paid in the relationship would adjust one-for-one to the underlying shock under flexible prices or wages, and less than one-for-one under rigid prices or wages.

Haefke, Sonntag, and van Rens (2013) look at how the real wage paid to US workers changes when there is a change in that worker's productivity, combining several BLS datasets. They look at what happens to wages in both new and existing employment relationships, painting a complete picture of what happens to real wages over the entire relationship.

Over the 1984–2006 period, the elasticity of real wages to productivity is 0.24–0.37 for all workers and 0.79–0.83 for new hires (Haefke, Sonntag, and van Rens 2013, table 4). However, data are available for a longer period, 1979–2006. Over that longer period, wages are more rigid: the elasticity of real wages to productivity is 0.18–0.20 for all workers and 0.30–0.49 for new hires (Haefke, Sonntag, and van Rens 2013, table 9). The elasticities of real wages with respect to productivity below 1 are clear evidence of wage rigidity. At the same time, wages of job stayers and new hires respond to changes in productivity, and therefore real wages are not completely fixed.

Overall, the elasticity of new hires' real wages with respect to productivity is in the 0.3–0.8 range. A reasonable estimate for the elasticity is 0.5, as it is in the middle of the range and corresponds to the most compelling estimate produced by Haefke, Sonntag, and van Rens (2013, table 9): the elasticity of hourly wages over the complete data sample. Haefke, Sonntag, and van Rens (2013, p. 898) also find that wages in ongoing contracts are close to a random walk, meaning that in expectation the wage in an entire employment relationship is determined by the wage at the beginning of the relationship. The elasticity of the present value of wages with respect to labor productivity is therefore close to the elasticity of the wages of newly hired workers: about 0.5.

As we discussed in the case of prices, the total surplus from a trade can also be expressed as a function of market tightness, because market tightness determines the amount of resources that buyers must devote to matching with a seller. In the case of the labor market, this means that the total surplus from matching is lower when the labor market is slack and recruiting is easy; one wonders in this situation whether it would become acceptable to lower wages. But here too, evidence suggests that wages are quite rigid. A wave of in-migration into a labor market raises the labor force, which reduces market tightness if wages do not fall, following the mechanism described in chapter 4. As the surplus from firm-worker matches is lower, wages might fall.

However, migration does not generally seem to significantly affect local wages. Card (1990) examines the impact of international immigration on wages in a famous natural experiment: the arrival of Cuban immigrants from the Mariel boatlift on the Miami labor market in the 1980s. The well-known findings by (Card 1990, p. 255) are that "the Mariel immigration had essentially no effect on the wages . . . of non-Cuban workers in the Miami labor market " and "perhaps even more surprising, the Mariel immigration had no strong effect on the wages of other Cubans." The lack of response of wages is especially surprising because the wave of immigration was substantial: Miami's labor force increased by 7% due to the Mariel boatlift.

Boustan, Fishback, and Kantor (2010) reach a similar conclusion by studying the effect of internal migration in a quasi experiment that occurred during the Great Depression in the United States. They use variation in the generosity of New Deal programs and extreme weather events to instrument for migrant flows to and from US cities. They find that in-migration had little effect on the hourly earnings of local workers.

Maybe even more surprisingly, policies that directly affect the surplus from worker-firm matches do not seem to influence wages. Take unemployment insurance: unemployment benefits provide income to job seekers, which improves their outside option, and according to bargaining theory should raise wages. But that is not what we observe: in the United States, unemployment benefits do not seem to affect wages. Marinescu (2017) explores data on job vacancies and job applications from a major job board (`CareerBuilder.com`). She looks at the extensions of unemployment insurance implemented in the United States between 2008 and 2011, which raised benefit duration from 26 weeks up to 99 weeks. She finds that the benefit extensions did not affect the level or distribution of wages advertised by firms (Marinescu 2017, table 6). Johnston and Mas (2018) study in administrative data the effects of a natural experiment: an unexpected 16-week reduction in benefit duration in Missouri in 2011. They find that re-employment earnings do not change after a cut in the duration of UI benefits Johnston and Mas (2018, p. 2501).

5.4.4. Possible reasons for wage rigidity

We have seen that wages are quite rigid and do not adjust like flexible wages would. We now briefly explain why that is.

To understand wage rigidity, it helps to look at the history of the US labor market. In the early 20th century, American firms shifted from a spot-market system—where workers were hired and paid on a daily basis—to a system of internal labor markets that produced rigid wages (Jacoby 1984). Internal labor markets were more bureaucratic, rule-based, and rigid than spot markets. Wages were tied to job descriptions rather than to how productive an individual worker happened to be on a given day or how bad the labor market was that month. The motivation for the organization of firms around internal labor markets was to provide a fair, equitable remuneration to workers—to improve their job satisfaction and eventually their productivity and loyalty to the firm.

In fact, this change in how US firms pay workers might have started with industrialization at the end of the 19th century. Hanes (1993, p. 751) reports that wages became more rigid at that time as workers started to organize and did not hesitate to strike: "Having learned that cutting nominal wages in a downturn might provoke a strike, firms avoided doing so even though none was bound by an enforceable wage contract."

Internal labor markets might explain why, during the Great Depression, job seekers offered to work at wages much below pre-depression wages for their skill and experience in their occupation, and yet firms chose to pay existing and new workers wages that were much closer to pre-depression wages (Simon 2001). Today, internal labor markets are prevalent and they render wages insensitive to marginal productivity or labor market slack (Doeringer and Piore 1971). Having access to 20 years of personnel data from a large US firm, Baker, Gibbs, and Holmstrom (1994, p. 923) find evidence of an internal labor market: the firm "seems to be shielding its employees from some of the market-induced variations in marginal product."

Internal labor markets also impose an internal fairness constraint: new hires and incumbent workers ought to be treated equitably. Hence, these institutions ensure that wages and wage rigidity are comparable for new and incumbent workers in similar jobs, as Grigsby, Hurst, and Yildirmaz (2021, p. 464) find.

Managers themselves have now completely internalized the ideas that led to the adoption of internal labor markets. They are convinced that cutting pay in bad times is a bad idea because it is seen as unfair by workers. Detailed interviews of compensation managers by Bewley (1999) in hundreds of US firms show that they avoid wage cuts at all costs, even in recessions, because they think wage cuts will antagonize workers and damage morale. Low morale in turn has a range of negative consequences for firms: workers get demotivated or angry, they stop trying as hard, turnover goes up, hiring gets harder, and eventually profitability drops. These findings are confirmed by other interviews and surveys of US firms (Blinder and Choi 1990; Campbell and Kamlani 1997).

And experimental evidence supports managers' views. In 1914, Ford decided to double pay for workers by introducing a five-dollar-a-day minimum wage. Raff and Summers (1987, tables 7 and 8) report that workers' attachment to the firm increased significantly: the yearly turnover rate fell from 370% in 1913 down to 16% in 1915, and the daily absenteeism rate fell from 10% in 1913 down to 2.5% in 1915. Simultaneously, the factory's productivity and profitability increased noticeably. While Ford's motivations for the wage increase are unclear, he was very pleased with the results, as Raff and Summers (1987, p. S59) recount:

> There was . . . no charity in any way involved. . . . We wanted to pay these wages so that the business would be on a lasting foundation. We were building for the future. A low wage business is always insecure. . . . The payment of five dollars a day for an eight hour day was one of the finest cost cutting moves we ever made.

More recent evidence is consistent with what happened at the Ford factory more than a century ago. In a natural experiment involving New Jersey police departments, Mas

(2006) finds that when police officers are paid less than what they think is a fair wage, their effort drops—leading to lower arrest rates, lower sentence length, and more numerous crime reports. And in numerous laboratory experiments, researchers have found that people do not just work for money; they very much care about fairness, and when they feel shortchanged by firms, performance suffers (Fehr, Goette, and Zehnder 2009). By running experiments in firms, organizational psychologists have also found that fair pay leads to better morale and loyalty to the firm (Bewley 2005).

Labor market institutions can play a role, too, in explaining wage rigidity. During the Great Depression, for example, the National Industrial Recovery Act is often blamed for keeping wages higher than they otherwise would have been, even while the economy was collapsing (Temin 1990). Another institution that plays an important role in explaining wage rigidity in more recent times, at least for low-wage workers, is the minimum wage. Indeed, the minimum wage compresses the bottom of the wage distribution, especially for female workers, which necessarily generates rigidities (DiNardo, Fortin, and Lemieux 1996, figure 1). Unions might also be a source of wage rigidity, as they raise and stabilize wages of their members. As Freeman and Medoff (1984, p. 54) explain:

> The wages of union workers tend to be less sensitive to business cycle ups and downs than the wages of nonunion workers; this is true in part . . . because unions are less likely to accept real wage reductions in recessions unless the senior employees' jobs are imperiled. By contrast, nonunion firms are more likely to alter wages in the short run, although nonunion wages are also not extremely sensitive to the business cycle.

However, the impact of unions on overall wages may be more limited today since union coverage in the United States has shrunk to 10% of the workforce as of 2023 (Jaeger, Naidu, and Schoefer 2025, p. 253).

Overall, internal labor markets, managerial norms, and labor market institutions all form barriers that prevent wages from adjusting flexibly to product market or labor market conditions. Firms do not tie wages tightly to productivity and treat new and existing workers alike within their internal labor market, and managers believe cuts are poison for morale.

5.5. Summary

The evidence from US markets reveals that prices and wages exhibit substantial rigidity. This rigidity stems primarily from fairness concerns. Customers find price increases due to a higher product demand or a tighter product market to be unfair. Similarly, workers find wage cuts due to lower labor demand or a slacker labor market to be unfair. Accordingly, producers keep prices stable to maintain customers' goodwill, just as firms keep wages stable to preserve worker morale. In the slackish model, the price norm ought to feature such rigidity to be realistic.

Although the evidence presented in this chapter goes against the Walrasian orthodoxy, it is in a way completely pedestrian. It has been known for a long time that the assumption that price and wage are perfectly flexible is unrealistic and that prices and wages are rigid in reality. Hicks (1946, p. 265) for instance observed:

> This assumption [that prices are perfectly flexible] must now be dropped, for it is of course highly unrealistic. In most communities there are a large number of prices which, for one reason or another, are fairly insensitive to economic forces. . . . This rigidity may be due to legislative control, or to monopolistic action . . . It may be due to lingering notions of a "just price". The most important class of prices subject to such rigidities are wage-rates; they are affected by rigidity from all three causes. They are particularly likely to be affected by ethical notions, since the wage-contract is very much a personal contract, and will only proceed smoothly if it is regarded as "fair" by both parties.

Besides making it obvious that prices and wages are somewhat rigid, Hicks (1946, p. 266) anticipates one of the main themes of this book—that we should look at unemployment alongside other forms of slack:

> Mr. Keynes goes so far as to make the rigidity of wage-rates the corner-stone of his system. . . . It seems to me that the more fundamental sociological implications are brought out better if we treat rigid wage-rates as merely one sort of rigid prices. It is hard to exaggerate the immediate practical importance of the unemployment of labour, but its bearing on the nature of capitalism comes out better if we look at it alongside the unemployment (and even the misemployment) of other things.

CHAPTER 6.

Rigid price norm

In this chapter, we model a realistic price norm and plug it into our slackish market model. Building on the evidence presented in chapter 5, the price norm imposes a price that is neither fixed nor flexible but somewhat rigid. We then study how this rigidity shapes the properties of the model—especially the response to shocks.

6.1. Model with rigid prices

In this section we go back to our slackish market model and assume a price norm that encodes how prices move in the real world: they do not fully respond to underlying shocks.

6.1.1. Expression of the rigid price norm

We will now look at what happens when we have a rigid price: a price that moves in the direction of the flexible price, but less than it. To impose this, we specify the following rigid price norm:

$$p^n = \rho \cdot \left(a \cdot k^{-\alpha}\right)^{1-\gamma}, \tag{6.1}$$

where $\gamma \in (0, 1]$ determines the amount of rigidity of the price norm and $\rho > 0$ determines the level of the price norm. At $\gamma = 1$, the price is fixed and $p^n = \rho$. At $\gamma = 0$, the price would be flexible. We do not consider $\gamma = 0$ below because we focus on somewhat-rigid prices.

6.1.2. Solution of the model

Next, we solve the model with rigid prices. The tightness that solves the model is given by the supply-equals-demand condition, (4.19), where the price norm is (6.1). Because the price norm is just a function of the parameters, and not of tightness, we can show that the model admits a unique solution, just like we did in the case with a fixed price, and as illustrated in figure 4.3. Indeed, the argument that we presented was valid for any price, so it is valid in particular if the price takes the value given by (6.1).

The solution of the model also continues to be represented graphically by figure 4.3A. What is different is the response to shocks, which we examine next.

6.1.3. Comparative statics

Let us look at a negative demand shock with rigid prices, represented by a reduction in the demand parameter a. We know that for a given price, the market demand is reduced when a falls. But the price also drops with a, which tends to push market demand back up. To determine the combined effect of a, we incorporate the rigid price (6.1) into the market demand (4.15):

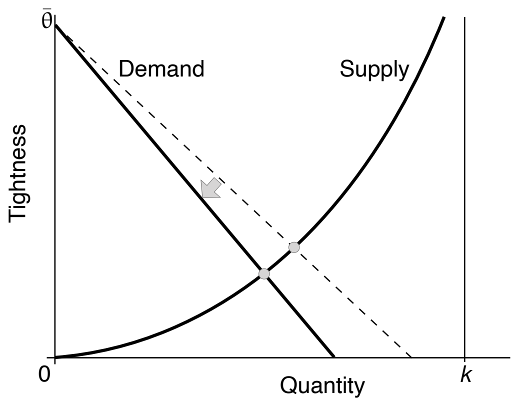
$$y^d(\theta) = \left[\frac{(1-\alpha)a^\gamma}{\rho}\right]^{1/\alpha} \cdot \frac{k^{1-\gamma}}{[1+\tau(\theta)]^{1/\alpha-1}}. \tag{6.2}$$

This is a similar equation to the one with fixed prices, except that the parameters appear in a slightly different fashion. In particular, the demand parameter a enters with an exponent γ. Since γ is strictly positive, because of price rigidity, the demand parameter has the exact same effect as with fixed prices, except that the impact of shocks to a will be attenuated by the exponent γ.

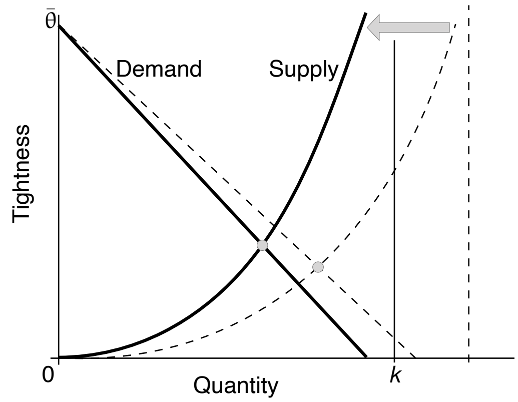
So after a negative demand shock, the market demand curve moves inward while the market supply curve remains unmoved, as shown in figure 6.1A. Hence, the comparative statics under demand shocks are the same as with a fixed price. The only difference is quantitative: the size of the effects is slightly attenuated, especially when prices are becoming more flexible (γ closer to 0). Formally, the elasticities of all the variables with respect to a shrink by a factor $\gamma \leq 1$.

Next, let us look at a negative supply shock with rigid prices, represented by a reduction in the market capacity k. The market supply is reduced when k falls. In addition, prices rise, since k appears in the denominator of the price norm (6.1). This increase captures the fact that goods are scarcer so more valuable. That rise in prices will depress the market demand, as illustrated in figure 6.1B.

To perform the comparative statics, we have to determine what happens to tightness, so we have to determine which of the inward movements dominates: the reduction in



A. Negative demand shock



B. Negative supply shock

FIGURE 6.1. Comparative statics in the slackish market model with rigid prices

The market supply is given by (4.5). The market demand is given by (6.2). The price norm is given by (6.1). A negative demand shock is a reduction in the preference for goods a. A negative supply shock is a reduction in market capacity k.

supply (pushing tightness up) or the reduction in demand (pushing tightness down). To do that, we rewrite the supply-equals-demand condition with the expression (6.2) and separate the terms that depend on tightness from the terms that depend on the supply parameter:

$$\left[\frac{(1-\alpha)a^{\gamma}}{\rho}\right]^{1/\alpha} k^{-\gamma} = [1-u(\theta)]\,[1+\tau(\theta)]^{1/\alpha-1}. \tag{6.3}$$

Now, when the capacity k drops, the left-hand side rises, since $-\gamma < 0$. So the right-hand side must rise as well, to maintain the equality. We know that both $1-u(\theta)$ and $1+\tau(\theta)$ are increasing functions of θ, and $1/\alpha > 1$, so the right-hand side is an increasing function of θ. This tells us that θ must rise when k falls. So just as with fixed prices, tightness goes up after a negative supply shock. As with demand shocks, the response of tightness is attenuated when prices are less rigid (lower γ). All the other variables respond as in the case with fixed prices, except of course the market price which rises here instead of remaining fixed.

6.2. Comparing fixed, rigid, and flexible prices

Overall, we have seen in the previous section that all the comparative statics remain the same if the market price is not completely fixed but simply rigid (table 6.1). It is only when prices are flexible that comparative statics completely change.

This can be seen in (6.3): for any $\gamma \in (0,1]$, including the case of fixed prices ($\gamma = 1$), tightness responds the same to demand and supply shocks. But for flexible prices ($\gamma = 0$), the terms a^{γ} and $k^{-\gamma}$ drop out, so tightness does not respond to either demand or supply shocks. In that case the shocks are entirely absorbed by prices, so tightness does not change.

An implication is that demand shocks are entirely neutral under flexible prices. Because tightness remains the same, the other quantities (output, consumption, rate of slack, visits) also remain the same.

Additionally, supply shocks do not affect tightness under flexible prices, although they do affect some quantities. Because tightness does not change, the rate of slack $u(\theta)$ does not change. However, because capacity has decreased, output $y = [1 - u(\theta)]k$ decreases, consumption $c = y/[1 + \tau(\theta)]$ decreases, and visits $v = y/q(\theta)$ decrease too.

Why is the response so different under flexible prices, compared to rigid prices? Consider a negative demand shock, represented by a fall in demand parameter a. Under a flexible price, the price norm moves proportionally with a, leaving the market demand curve unchanged and therefore leaving tightness unchanged. This is because the flexible price decreases enough to exactly offset the fall in marginal utility and keep the market demand curve in the same position. We can see this in the expression for the flexible price, by setting $\gamma = 0$ in the price norm (6.1). The demand parameter a is in the numerator, so the price drops in proportion to a after the shock. Hence the price drops as much as the marginal utility of consumption, which neutralizes that drop and brings back the market demand to its original position. After incorporating the drop in price, the market demand is unchanged, so tightness is unchanged.

A similar logic explains why flexible prices also neutralize shocks to market capacity. Consider a negative supply shock, represented by a fall in market capacity k. For a given tightness, lower capacity reduces output and consumption. Under flexible prices, this reduction in consumption raises the marginal utility of consumption. Because the flexible price is proportional to marginal utility, the price increases exactly enough to reflect the increased scarcity of goods. This increase in the price shifts the market demand curve inward just enough to offset the inward shift of the market supply curve. The two effects cancel out, leaving market tightness unchanged. Formally, this can be seen by setting $\gamma = 0$ in the price norm (6.1), which implies that the price rises proportionally when k falls. Under flexible prices, supply shocks are therefore fully absorbed by prices, so tightness does not respond.

6.3. A possible source of price flexibility: bargaining

The price norm (6.1) introduces a flexible benchmark, which moves in step with demand and supply parameters so as to absorb supply and demand shocks. A natural question is: How could this flexible benchmark arise in the market? In this section, we show that flexible prices appear when sellers and buyers bargain over prices.

TABLE 6.1. Comparative statics in the slackish market model for various price norms

	Sales y	Tightness θ	Price p	Slack rate u	Consumption c	Visits v
A. Fixed price norm: $\gamma = 1$						
Decrease in demand a	–	–	0	+	?	–
Decrease in supply k	–	+	0	–	–	?
B. Rigid price norm: $\gamma \in (0, 1)$						
Decrease in demand a	–	–	–	+	?	–
Decrease in supply k	–	+	+	–	–	?
C. Flexible price norm: $\gamma = 0$						
Decrease in demand a	0	0	–	0	0	0
Decrease in supply k	–	0	+	0	–	–

The comparative statics are obtained from equation (6.3) and are illustrated in figures 4.4 and 6.1. A variable's increase is denoted by "+", a decrease by "-", no response by "0", and an indeterminate response by "?".

6.3.1. Outcome of the bargaining process

Following Diamond (1982b), we assume that the outcome of the bargaining process is that buyer and seller share the surplus of their trade. This means that the seller surplus $\mathcal{S}$ amounts to a share χ of the total surplus $\mathcal{T}$ and the buyer surplus $\mathcal{B}$ amounts to a share $1 - \chi$ of the total surplus, where $\chi \in (0, 1)$ is the bargaining power of the seller.

In section 5.2, we established that for a trade at price p, the seller's surplus is just the price, $\mathcal{S} = p$, and the total surplus is just the marginal utility from consumption, $\mathcal{T} = (1 - \alpha)ac^{-\alpha}$. Since the surplus-sharing solution requires that $\mathcal{S} = \chi \cdot \mathcal{T}$, the bargained price is

$$p = \chi \cdot (1 - \alpha)ac^{-\alpha}. \tag{6.4}$$

That is, through bargaining, sellers keep a share χ of the trade surplus, which is just the marginal utility from the good enjoyed by buyers. Buyers are left with a fraction $1 - \chi$ of the surplus. They enjoy the full marginal utility by consuming the good but must pay a share χ of the marginal utility to sellers; so they are left with $1 - \chi$ of the marginal utility in the end. If sellers have all the bargaining power ($\chi = 1$), the price is just the marginal utility. If buyers have all the bargaining power ($\chi = 0$), the price is 0.

In chapter 4, we noted that all price norms reduce to functions of market tightness. This is true here too. In the model, consumption and output are related by $c = y/[1 + \tau(\theta)]$, and output and tightness are related by $y = [1 - u(\theta)]k$, so consumption can be written a function of tightness too:

$$c = \frac{1 - u(\theta)}{1 + \tau(\theta)} \cdot k. \tag{6.5}$$

By plugging this expression into (6.4), we write the bargained price as a function of tightness:

$$p = \chi(1-\alpha)ak^{-\alpha}\left[\frac{1-u(\theta)}{1+\tau(\theta)}\right]^{-\alpha}. \tag{6.6}$$

6.3.2. Solution with bargained prices

Solving the model with bargaining is a bit different because the market demand takes a very special form once it is combined with the bargained price. When we derived the market demand in chapter 4, we obtained it by equalizing the marginal utility of consumption to the marginal cost of consumption: $(1-\alpha)ac^{-\alpha} = [1+\tau(\theta)]p$. And we have just seen that the bargained price satisfies $p = \chi(1-\alpha)ac^{-\alpha}$. Combining both, we see that once we incorporate the bargained price into the market demand, the market demand becomes degenerate. It pins down a unique tightness irrespective of quantities traded: $1 = [1+\tau(\theta)]\chi$. This condition on tightness can be rewritten as a condition on the matching wedge:

$$\tau(\theta) = \frac{1-\chi}{\chi}. \tag{6.7}$$

In the tightness-quantity diagram, the market demand is therefore horizontal:

$$\theta = \tau^{-1}\left(\frac{1-\chi}{\chi}\right), \tag{6.8}$$

where τ^{-1} is the inverse of the matching wedge. The inverse is well defined on $(0,\infty)$ because τ is strictly increasing on $(0,\bar{\theta}) \to (0,\infty)$. The inverse itself is strictly increasing.

6.3.3. Flexibility of bargained prices

From the expression (6.8) for market tightness, we can express the bargained price (6.6) as a function of the parameters of the model. The bargained price is given by

$$p = (1-\alpha)\chi^{1-\alpha}\left[1-u\left(\tau^{-1}\left(\frac{1-\chi}{\chi}\right)\right)\right]^{-\alpha} ak^{-\alpha}. \tag{6.9}$$

We simplified the expression using the fact that when (6.8) holds, $1+\tau(\theta) = 1/\chi$.

We now see that if we set our price norm (6.1) with $\gamma = 0$ and

$$\rho = (1-\alpha)\chi^{1-\alpha}\left[1-u\left(\tau^{-1}\left(\frac{1-\chi}{\chi}\right)\right)\right]^{-\alpha},$$

we obtain the bargained price described by (6.9).

Hence, price bargaining provide an explanation for price movements in the direction of flexibility. Bargained prices are too flexible to be realistic: they prevent the rate of slack from changing, which is not consistent with the real world, where the amount of slack in markets varies considerably over the business cycle (chapter 2). But with a small amount of bargaining in markets, prices might not be completely fixed but somewhat flexible towards the bargained price. Thus, price bargaining might provide a foundation for the price norm (6.1).

6.4. Quantifying the response of the model to shocks

Comparative statics tell us in which directions the model variables respond to demand and supply shocks. But it is also possible to determine the amplitude of the response of the variables. The key step is to compute the elasticity of market tightness with respect to the parameters at the source of the shocks (a and k). Given that tightness determines all other variables in the model, we can quantify the response of any variable from the response of tightness. From the elasticity of tightness, we will see in particular that the response of the model to shocks is determined critically by the rigidity of prices. Throughout the section, we use the elasticity results introduced in appendix B.

6.4.1. Response to demand shocks

We start with the elasticity of tightness with respect to the demand parameter a. This elasticity tells us how much the model responds to demand shocks. We start from the supply-equals-demand equation, which implicitly gives tightness: $y^d(\theta, a) = y^s(\theta)$, where the market demand is given by (6.2) and the market supply takes the usual form, (4.5). Using implicit differentiation with elasticities, we get

$$\epsilon_a^d + \epsilon_\theta^d \cdot \epsilon_a^\theta = \epsilon_\theta^s \cdot \epsilon_a^\theta,$$

where ϵ_a^d is the partial elasticity of market demand y^d with respect to the demand parameter a, ϵ_θ^d is the partial elasticity of market demand y^d with respect to tightness θ, ϵ_θ^s is the elasticity of market supply y^s with respect to tightness θ, and ϵ_a^θ is the elasticity of tightness θ with respect to the demand parameter a—which is the elasticity that we are aiming to compute.

Refactoring this condition, we isolate the elasticity of tightness with respect to market demand:

$$\epsilon_a^\theta = \frac{\epsilon_a^d}{\epsilon_\theta^s - \epsilon_\theta^d}. \tag{6.10}$$

This expression shows that the response of tightness to a demand shock depends of course on the response of demand to the shock (ϵ_a^d), and critically on the relative slopes of the demand and supply curves ($\epsilon_\theta^s - \epsilon_\theta^d$).

Next, we compute the individual elasticities involved in (6.10). Using the elasticities (3.7) and (3.8), we get

$$\epsilon_a^d = \frac{\gamma}{\alpha}, \quad \epsilon_\theta^d = -\frac{1-\alpha}{\alpha} \cdot \epsilon_\theta^{1+\tau}, \quad \epsilon_\theta^s = 1 - \eta(\theta), \tag{6.11}$$

where $\epsilon_\theta^{1+\tau}$ is the elasticity of $1+\tau$ with respect to tightness θ. Given (3.7) and (4.8), we infer that the elasticity of $1+\tau(\theta) = q(\theta)/[q(\theta)-\kappa]$ with respect to θ is

$$\epsilon_\theta^{1+\tau} = [-\eta(\theta)] - [-\eta(\theta)] \cdot \frac{q(\theta)}{q(\theta)-\kappa} = \eta(\theta)\tau(\theta). \tag{6.12}$$

Hence, the elasticity of the market demand with respect to market tightness is

$$\epsilon_\theta^d = -\frac{1-\alpha}{\alpha} \cdot \eta(\theta)\tau(\theta). \tag{6.13}$$

Plugging in (6.10) the values of the elasticities, we obtain:

$$\epsilon_a^\theta = \frac{\gamma}{[1-\eta(\theta)]\,\alpha + (1-\alpha)\eta(\theta)\tau(\theta)}. \tag{6.14}$$

This expression highlights several properties of the response of the market to demand shocks. First, tightness goes up when demand is stronger ($\epsilon_a^\theta > 0$). Second, if prices are flexible ($\gamma = 0$), then tightness does not respond to demand shocks ($\epsilon_a^\theta = 0$), which means that the market does not respond to demand shocks—in line with what we saw when we looked at bargained prices (table 6.1). Third, the more rigid prices are (higher γ), the stronger is the response of tightness, and thus of the other variables, to demand shocks.

6.4.2. Response to supply shocks

Next we move to the elasticity of tightness with respect to the market capacity k, which determines the market supply. This elasticity tells us how much the model responds to supply shocks. We start again from the supply-equals-demand equation, which implicitly gives tightness: $y^d(\theta, k) = y^s(\theta, k)$, where the market demand is given by (6.2) and the market supply takes the usual form, (4.5). Using implicit differentiation with elasticities, we get

$$\epsilon_\theta^d \cdot \epsilon_k^\theta + \epsilon_k^d = \epsilon_\theta^s \cdot \epsilon_k^\theta + \epsilon_k^s,$$

where ϵ_k^s is the partial elasticity of market supply y^s with respect to the market capacity k and ϵ_k^θ is the elasticity of tightness θ with respect to the market capacity k—which is the elasticity that we are aiming to compute here. Besides the elasticity above, we also have

$$\epsilon_k^d = 1 - \gamma, \quad \epsilon_k^s = 1.$$

Combining these results, we obtain first that

$$\epsilon_k^\theta = \frac{\epsilon_k^d - \epsilon_k^s}{\epsilon_\theta^s - \epsilon_\theta^d}, \tag{6.15}$$

which shows that the response of tightness to a supply shock depends on the response of demand and supply to the shock (ϵ_k^s and ϵ_k^d), and critically on the slopes of the demand and supply curves ($\epsilon_\theta^s - \epsilon_\theta^d$). Next, we can plug in this abstract expression the various values of the elasticities and we obtain:

$$\epsilon_k^\theta = \frac{-\gamma}{1 - \eta + \frac{1-\alpha}{\alpha} \cdot \eta\tau(\theta)}. \tag{6.16}$$

From (6.16) we see several properties of the response of the market to supply shocks. First, tightness goes down when supply is higher ($\epsilon_k^\theta < 0$). Second, if prices are flexible ($\gamma = 0$), tightness does not respond to supply shocks ($\epsilon_k^\theta = 0$). Third, the response of tightness to supply shocks is stronger when prices are more rigid (higher γ).

6.4.3. State dependence

A critical property of the slackish model emerges from the elasticity formulas: the model exhibits state dependence. This means that the model behaves differently when the market is slack or tight. To display such state dependence, we now compute the response of output to demand and supply shocks.

We begin by picking a relationship that relates output to tightness. A convenient choice is the market supply: $y = y^s(\theta, k)$. From this relationship, we then obtain $\epsilon_a^y = \epsilon_\theta^s \cdot \epsilon_a^\theta$ so

$$\epsilon_a^y = \frac{\gamma}{\alpha + \frac{\eta(\theta)}{1-\eta(\theta)} \cdot (1-\alpha)\tau(\theta)}.$$

So the response of output to demand shocks has broadly the same properties as the response of tightness.

In addition, we see here that the response of output to demand shocks is stronger in a slacker market. Indeed, the matching wedge $\tau(\theta)$ is highly procyclical. Moreover, by the assumption from chapter 3 that the matching elasticity $\eta(\theta)$ is increasing in tightness, the ratio $\eta(\theta)/[1-\eta(\theta)]$ is also increasing in tightness. Combined with the strict increase of $\tau(\theta)$, this implies that the term $\eta(\theta)\tau(\theta)/[1-\eta(\theta)]$ is strictly increasing in tightness. Thus, since all other terms in the elasticity are constant, the elasticity ϵ_a^y is decreasing in tightness. This means that when the market is slack, the elasticity is high: output responds sharply to changes in market demand. By contrast, when the market is tight, the elasticity is low: output does not respond much to changes in market demand.

For supply shocks, we have $\epsilon_k^y = \epsilon_k^s + \epsilon_\theta^s \cdot \epsilon_k^\theta$ so

$$\epsilon_k^y = 1 - \frac{\gamma}{1 + \frac{\eta(\theta)}{1-\eta(\theta)} \cdot \frac{1-\alpha}{\alpha} \cdot \tau(\theta)}.$$

Since $\gamma \leq 1$, the elasticity is strictly positive ($\epsilon_k^y > 0$). So unlike tightness, which falls after an increase in supply, output rises after an increase in supply. Furthermore, output rises more when prices are less rigid. With a flexible price ($\gamma = 0$), output moves one-for-one with capacity ($\epsilon_k^y = 1$); when prices are rigid, the move is less than one-for-one ($\epsilon_k^y < 1$).

We also see that the response of output to supply shocks is stronger in a tighter market. Following the same logic as above, we know that the term $\eta(\theta)\tau(\theta)/[1-\eta(\theta)]$ is strictly increasing in tightness. This result implies that the elasticity ϵ_k^y is increasing in tightness. This means that when the market is slack, the elasticity is low: output does not respond much to changes in market capacity. By contrast, when the market is tight, the elasticity is high: output responds significantly to changes in market capacity.

We have just established that the slackish model is state-dependent: the response of output to shocks depends on the state of the market. The reason behind state dependence is the nonlinearity of the slackish model. Because of the curvature of the market demand and supply curves, shocks have systematically different effects when tightness is high or low. We will come back to this result in the dynamic model, where state dependence is exacerbated by a stronger nonlinearity (chapter 8).

6.5. Summary

In this chapter we incorporate price rigidity into our slackish market model. Prices respond partially to shocks with some rigidity parameter $\gamma \in (0, 1]$, so adjustment to shocks is between the extremes of fixed and flexible prices. Both prices and tightness adjust, with the elasticity of tightness to demand and supply shocks determined by γ. This price rigidity explains why slack varies over the business cycle—a central feature of real markets that models with flexible prices cannot capture.

A key finding is that all the qualitative properties of the model (the comparative statics) remain the same if the market price is not completely fixed but simply rigid. It is only when the market price is completely flexible that comparative statics change. This result would not be surprising to Old Keynesian scholars. Tobin (1992, p. 394) was for instance fully aware that all but completely flexible prices produce the same results:

> Price flexibility is not a yes-or-no circumstance. Consider instead a spectrum

> of the degree of price flexibility, from complete flexibility at one extreme to complete rigidity at the other.... Who owns the middle ground? We Keynesians do, despite common beliefs to the contrary. Keynes and Keynesian economists did not assume complete rigidity, nor did they need to.... It is not true that only an arbitrary and gratuitous assumption of complete rigidity, converting nominal demand shocks into real demand shocks, brings into play ... demand-determining processes ... Any degree of stickiness that prevents complete instantaneous price adjustment has the same qualitative implications.

Finally, because of the curvature of the market supply, the slackish model is state dependent. Output responds sharply to demand shocks in slack conditions but little in tight conditions. Conversely, output responds little to supply shocks in slack conditions but more in tight conditions.

CHAPTER 7.

Endogenous capacity

So far we have assumed that the market capacity—number of goods sold on the market—is given exogenously. The motivation is that supply decisions do not adjust much in the short run. It takes time to systematically supply more or fewer goods. Nevertheless, to understand completely how slackish markets work, it is good to examine how the model works when the capacity is endogenous. This is what we do in this chapter.

7.1. Market supply with endogenous capacity

The cornerstone of the supply side is the seller's optimization problem: choosing how much to sell so as to maximize utility. We set up and solve the seller's problem, and then derive the properties of the market supply.

7.1.1. Seller's utility function

Once again there is a mass 1 of sellers. Each seller $i \in [0, 1]$ can either enter the market to sell k goods, or remain outside of the market and not sell anything. The seller must decide before the market opens to enter or not. If they enter, they place k goods for sale, and sell a fraction of them. If they do not enter, they do not place anything for sale, but they enjoy utility ξi^{ϕ}. The parameter $\xi > 0$ governs the utility from nonparticipation relative to money. The parameter $\phi > 0$ ensures that the utility from nonparticipation is increasing in i. Sellers with high i enjoy nonparticipation—leisure or recreation—very much. Sellers with low i enjoy working instead of leisure. In addition all sellers have linear utility over money balances.

A seller's only decision is whether to enter the market or not. The number of sellers who enter the market is $h \in [0,1]$. If person i refuses to enter, she receives no income but enjoys leisure utility ξi^{ϕ}. If a seller decides to enter, she receives utility from her expected income but no utility from leisure. A fraction $1 - u(\theta)$ of all goods are sold at price p, where $u(\theta)$ is the slack rate. So any seller's expected income is simply $[1 - u(\theta)]pk$.

7.1.2. Market tightness

Importantly, because the number of goods for sale is $\int_0^h k\,di = hk$, the matching function says that the number of trades on this market is

$$y = m(hk, v).$$

Hence, the market tightness in the case with endogenous capacity is defined as

$$\theta = \frac{v}{hk}. \tag{7.1}$$

7.1.3. Seller's problem

The seller's problem is simple: the seller must simply decide to participate or not. The h sellers with sufficiently low utility from nonparticipation enter the market. The remaining $1-h$ sellers, who have sufficiently high utility from nonparticipation relative to the expected market income, abstain from entering.

Formally, people opt to participate when $\xi i^{\phi} \leq [1 - u(\theta)]pk$, and they decline to participate when $\xi i^{\phi} > [1 - u(\theta)]pk$. Accordingly, the capacity of the market is implicitly defined by

$$\xi h^{\phi} = [1 - u(\theta)]pk.$$

This equation says that the marginal market participant ($i = h$) is indifferent between participating and not, because her nonparticipation utility (ξh^{ϕ}) equals the expected income ($[1 - u(\theta)]pk$). Reshuffling the equation, we express the market participation rate as a function of market tightness and market price:

$$h = \left[\frac{[1 - u(\theta)]pk}{\xi}\right]^{1/\phi}. \tag{7.2}$$

The participation rate is increasing in market tightness and market price because both raise sellers' expected income.

We see in passing that the elasticity $1/\phi$ determines the strength of the participation response to market tightness. Using (7.2), we infer that the elasticity of the participation rate with respect to market tightness is

$$\epsilon_\theta^h = \frac{1-\eta}{\phi},$$

where $1-\eta$ is the elasticity of $1-u(\theta) = f(\theta)$ with respect to θ (equation (3.8)). If the elasticity $1/\phi$ is small, then the participation rate does not respond much to variations in tightness. In the case of the labor market, the elasticity $1/\phi$ is akin to a Frisch elasticity of labor supply. We know that US workers exhibit a low Frisch elasticity (Chetty et al. 2012). Thus, the US labor force participation rate should not respond much to fluctuations in labor market tightness—explaining why any cyclicality in the labor force participation rate is hard to discern (chapter 2).

To illustrate, we plot the participation rate h as a function of tightness (figure 7.1A). To ensure that $h \leq 1$ for all θ, we assume that the parameter ξ is large enough. We want $h \leq 1$ even when $u(\theta) = 0$, which requires $\xi \geq pk$.

7.1.4. Expression of the market supply

On the market, we have h sellers, each selling k goods. The notional supply kh is not fixed any more: it depends on the market price and tightness via equation (7.2).

Once again, we use the effective market supply, which is the amount of goods actually sold given tightness, and which is less than the notional supply because sellers are unable to sell all the goods that they put for sale. The effective supply is $[1-u(\theta)]kh < kh$. From equation (7.2), it is pretty simple to express the market supply:

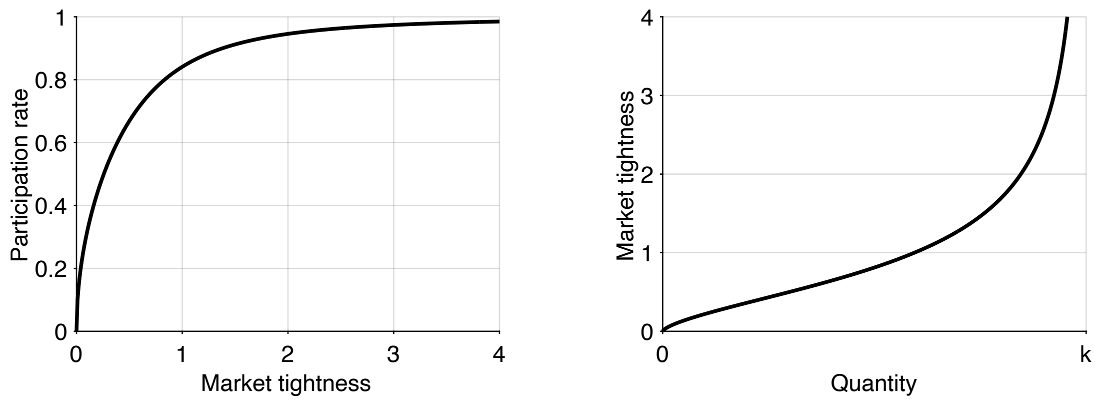
$$y^s(\theta, p) = [[1-u(\theta)]k]^{1+1/\phi}\left(\frac{p}{\xi}\right)^{1/\phi}. \tag{7.3}$$

Notice that when $\phi \to \infty$, the market supply with endogenous capacity reduces to the market supply with fixed capacity, $y^s(\theta) = [1-u(\theta)]k$. This is because in this case, the utility of leisure ξi^ϕ goes to 0 for all sellers $i < 1$, so all sellers enter the market ($h = 1$), and the market capacity is fixed at k.

7.1.5. Properties of the market supply

Let us now look at the properties of the market supply, $y^s(\theta, p)$. First, we see that it is increasing in the market price p. When the price is higher, entering the market becomes more attractive, so supply is higher.

We also see that when market tightness θ is 0, the market supply is 0, because $1-u(0) = f(0) = 0$. Then, the market supply is increasing in market tightness. This is visible in (7.3) because the slack rate $u(\theta)$ is decreasing in tightness. When the market is tighter, more goods are sold, which has a mechanical effect on supply, and which also provides



A. Participation rate

B. Market supply

FIGURE 7.1. Numerical illustration of the slackish market model with endogenous capacity

The market participation rate is given by (7.2). The market supply is given by (7.3). The matching function is CES, given by (3.12). Parameters are set to $\sigma = 2$, $k = 1$, $p = 1$, $\xi = 1$, and $\phi = 2$.

an incentive for sellers to enter the market by raising the expected income. Hence, when tightness is higher, more sellers offer goods for sale, and a larger fraction of these goods are sold, raising supply.

To wrap up our analysis of the market supply, we plot the supply curve in a tightness-quantity graph (figure 7.1B). The graph depicts the properties of the market supply that we have just derived.

7.2. Model with fixed prices

We now solve the model with endogenous capacity in the simple case with fixed prices. We assume that the price for all goods is fixed, so the price norm simply imposes that the price of all goods is fixed to $p > 0$.

7.2.1. Solution with fixed prices

To solve the model, we need to find the tightness θ that equalizes market demand and supply. With endogenous capacity, the condition is

$$y^d(\theta, p) = y^s(\theta, p), \tag{7.4}$$

where the market demand is the same as in the basic model, given by (4.15), and the market supply is specific to the model with endogenous capacity, given by (7.3).

The first step is to ensure that the solution indeed exists and is unique. We know from chapter 4 that $y^d(\theta, p)$ is decreasing from $y^d(0, p) > 0$ when $\theta = 0$ to 0 when $\theta = \bar{\theta}$. At the same time, we have seen in this chapter that $y^s(\theta, p)$ is increasing in tightness, starting from 0 when $\theta = 0$. Moreover, both y^d and y^s are continuous in tightness. Thus, we can be sure that equation (7.4) has a unique solution, since we can be sure that the curves $y^d(\theta, p)$ and $y^s(\theta, p)$ cross once for $\theta \in (0, \bar{\theta})$. The crossing point is the solution of the model. This argument is the same really as in the model with exogenous capacity, which was illustrated in figure 4.3A. The only graphical difference is that the slope of the market supply is different.

7.2.2. Comparative statics with fixed prices

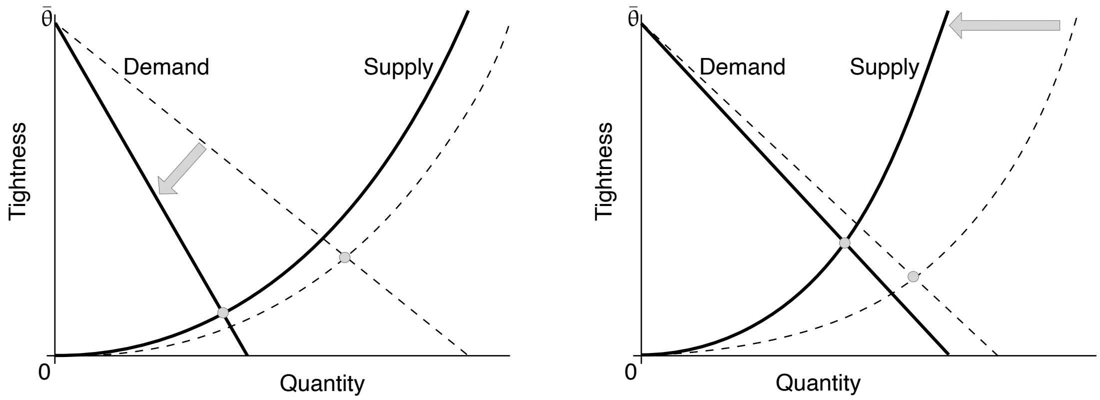
Now that we have solved the model, we can look at comparative statics, starting with a negative demand shock: a decrease in preference for goods, a. From figure 4.4A, which continues to apply here for qualitative thinking, we see that a negative demand shock brings the demand curve inward, resulting in a lower value of both tightness θ and aggregate sales y. This was to be expected: as buyers value goods less, they visit fewer sellers, which reduces tightness and the amount of goods sold. The slack rate in the market, $u(\theta)$, increases as it becomes harder for sellers to sell their goods with the lower tightness.

The rise in the slack rate lowers participation in the market, as shown by (7.2). Sellers realize that the market is slacker, which means they are less likely to sell their goods, and that they expect a lower income. As a result, entering the market is not as profitable, so more sellers stay home.

Overall sales fall both because fewer goods are offered for sale and a smaller fraction of these goods are sold. The quantity of goods that are unsold, $u(\theta)hk$, may go up or down depending on the relative response of the participation rate and the slack rate.

Next we turn to a negative supply shock. With endogenous capacity, several parameters can change and create such a supply shock, with slightly different implications. A negative supply shock could be a drop in the capacity k of each seller, or an increase in the outside option ξ of sellers. As (7.3) shows, both a lower k and a higher ξ lead to a lower market supply. From figure 4.4B, which continues to apply here too for qualitative analysis, we realize that a negative supply shock brings the supply curve inward, which results in a higher tightness θ but lower sales y. The logic is that fewer goods are supplied, so through the matching process, fewer goods are bound to be purchased. And because there is more competition among buyers for the fewer available goods, tightness is higher. Because tightness is higher, the slack rate falls in the market, as well as the number of goods that remain unsold (since fewer goods are placed for sale, and a smaller share of those are unsold).

What happens to the participation rate after the negative supply shock? The logic depends on the source of the shock. Let's consider first an increase in the utility from nonparticipation, ξ. Here the crux is that output and participation rate are related by



A. Negative demand shock B. Negative supply shock

FIGURE 7.2. Comparative statics in the slackish market model with endogenous capacity and rigid prices

The market supply is given by (7.3). The market demand is given by (4.15). The price norm is given by (7.5). A negative demand shock is a reduction in the preference for goods a. A negative supply shock is a reduction in market capacity k or an increase in seller outside option ξ.

$y = [1 - u(\theta)]kh$, so $h = y/[[1 - u(\theta)]k]$. We know that when the outside option ξ is higher, output y is lower and slack rate $u(\theta)$ is lower, so clearly the participation rate h is lower. The reason is that, despite a higher tightness and lower slack rate, participation has become less attractive after the increase in outside option. The increase in tightness compensates a little bit the increase in the outside option, but not completely.

Next, let's look at a drop in individual capacity k. We see from (7.3) that the market supply is determined by $[1 - u(\theta)]k$. Since output falls after the negative supply shock, and output is on the supply curve, it must be that $[1 - u(\theta)]k$ falls—despite the increase in θ. From (7.2), we infer that the participation rate falls when $[1 - u(\theta)]k$ falls. Overall, the participation rate falls when individual capacity falls. The reason is that, despite a higher tightness and lower slack rate, participation has become less attractive after the drop in individual productivity. The increase in tightness compensates a little bit the lower productivity, but not completely.

7.3. Model with rigid prices

Just like in the case with fixed capacity, it is possible to generalize all the results when prices are rigid instead of fixed. This is the most general form of a slackish market: with endogenous demand and capacity, as well as rigid prices that respond partially to demand and supply disturbances. Because of the generality, the analysis is a little bit more cumbersome. But the qualitative results are the same as with endogenous capacity and fixed prices, which are themselves the same as in the model with fixed capacity and fixed prices, so intuitions carry over naturally. The quantitative results (the elasticities of tightness with respect to various parameters) are more complex but are at the end of the day natural extensions of the elasticities with exogenous capacity and fixed prices.

7.3.1. Expression of the rigid price norm

With endogenous capacity, the rigid price must be slightly adjusted to capture the factors that influence market supply. We specify the rigid price norm as follows:

$$p^n = \left[a^{\frac{\phi}{\alpha+\phi}}\, \xi^{\frac{\alpha}{\alpha+\phi}}\, k^{-\frac{(1+\phi)\alpha}{\alpha+\phi}}\right]^{1-\gamma} \cdot \rho, \tag{7.5}$$

where $\gamma \in (0,1]$ determines the amount of rigidity of the price norm and $\rho > 0$ determines the level of the price norm. At $\gamma = 1$, the price is fixed and $p^n = \rho$. Notice that when $\phi \to \infty$, the price norm reduces to the price norm (6.1), which we used when capacity was exogenous.

It's natural to wonder where the rigid price norm comes from. Why does it take this specific form? It is built that way to capture all the forces that affect market tightness (which itself determines all other variables in the model). To build it, simply rewrite the supply-equals-demand condition and separate all the parameters on the left-hand side and all the terms that depend on tightness on the right-hand side. Using (4.15) and (7.3), we obtain the following equation:

$$(1-\alpha)^{1/\alpha} \cdot \frac{a^{1/\alpha}\xi^{1/\phi}k^{-\frac{1+\phi}{\phi}}}{p^{1/\alpha+1/\phi}} = [1-u(\theta)]^{\frac{1+\phi}{\phi}}\,[1+\tau(\theta)]^{1/\alpha-1}. \tag{7.6}$$

We then build the rigid price norm so the term $p^{1/\alpha+1/\phi}$ moves with $a^{1/\alpha}\xi^{1/\phi}k^{-\frac{1+\phi}{\phi}}$ but in a subdued fashion.

7.3.2. Solution with rigid prices

Because the price norm (7.5) is just a function of the parameters, and not of tightness, we can show that the model admits a unique solution, just like we did in the case with a fixed price. Indeed, the argument that we presented was valid for any price, so it is valid in particular if the price takes the value given by (7.5).

7.3.3. Comparative statics with rigid prices

We can now consider a negative demand shock: a reduction in the preference for goods a. It leads to a reduction in market demand, as seen in (4.15), but also to a drop in price, as shown in (7.5). However, the response of the price norm to the demand shock is less than the shock, so even when we take into account the price reduction, the market demand falls. This can be seen because in the market demand, a appears only in the term a/p. Given the expression of the price norm (7.5), the term featuring a is proportional to

$$a^{1-\frac{(1-\gamma)\phi}{\alpha+\phi}}.$$

Since $1-\gamma \in (0,1)$ and $\phi/(\alpha+\phi) \in (0,1)$, the exponent on a is strictly positive, even with the price response included. Thus, the market demand moves together with a, just like in the case with fixed prices. The drop in a also leads to a drop in market supply because of the reduction in price, which makes participation in the market less attractive.

We need to look at market supply and demand jointly to see what happens to tightness and output. To do that, we rewrite (7.6) by using the complete expression for the price norm, given by (7.5):

$$(1-\alpha)^{1/\alpha} \cdot \frac{\left[a^{1/\alpha}\xi^{1/\phi}k^{-\frac{1+\phi}{\phi}}\right]^{\gamma}}{\rho^{1/\alpha+1/\phi}} = \left[1-u(\theta)\right]^{\frac{1+\phi}{\phi}}\left[1+\tau(\theta)\right]^{1/\alpha-1}. \tag{7.7}$$

Overall, we see that when demand a drops, the left-hand side drops, since $\gamma > 0$. So the right-hand side must drop as well, to maintain the equality. We know that both $1-u(\theta)$ and $1+\tau(\theta)$ are increasing functions of θ, and $1/\alpha > 1$, so the right-hand side is an increasing function of θ. This tells us that θ must fall when a falls. This comparative static is illustrated in figure 7.2A.

In fact, it is straightforward to compute the elasticity of tightness with respect to a from (7.7). It is even faster to go that way than to start from the supply-equal-demand condition, as we did in chapter 6, because the parameters are all arranged together, and the terms with tightness are all grouped together. We see that the elasticity of the left-hand side with respect to a is simply

$$\epsilon_a^l = \frac{\gamma}{\alpha}.$$

We use (6.12) to obtain the elasticity of $1+\tau(\theta)$ with respect to θ, and (3.8) to obtain the elasticity of $1-u(\theta) = f(\theta)$ with respect to θ. From those, we see that the elasticity of the right-hand side with respect to tightness is

$$\epsilon_\theta^r = \frac{1+\phi}{\phi}(1-\eta) + \frac{1-\alpha}{\alpha}\eta\tau(\theta). \tag{7.8}$$

Now, implicit differentiation of (7.7) gives $\epsilon_a^l = \epsilon_\theta^r \epsilon_a^\theta$, so that

$$\epsilon_a^\theta = \frac{\gamma/\alpha}{\frac{1+\phi}{\phi}(1-\eta) + \frac{1-\alpha}{\alpha}\eta\tau(\theta)} = \frac{\gamma}{(1+1/\phi)(1-\eta)\alpha + (1-\alpha)\eta\tau(\theta)}.$$

We confirm that the elasticity is strictly positive, so tightness drops when demand falls. At the same time, we see that the response is stronger when prices are more rigid (high γ). Indeed, if prices are more rigid, they cannot absorb the demand shock much, so tightness must absorb more of it. We also see that the elasticity is larger when the supply elasticity $1/\phi$ is lower. This is because when the supply is less elastic, it responds less to the price reduction caused by the lower demand, so fewer sellers are discouraged from entering the market, which means that competition among sellers will be more drastic. In other words, tightness will fall more. And in fact we recognize that when $1/\phi \to 0$, the elasticity reduces to the elasticity with fixed capacity, given by (6.14).

We can repeat the same analysis with a negative supply shock: say a drop in seller capacity k. (We could similarly look at an increase in seller outside option ξ.) The reduced capacity leads to a reduction in market supply, naturally, despite the increase in price which attracts some new sellers into the market. This can be seen because in the market supply (7.3), k appears only in the term $k^{\frac{1+\phi}{\phi}} p^{\frac{1}{\phi}}$. Given the expression of the price norm (7.5), we realize that the term featuring k is proportional to

$$k^{\frac{1+\phi}{\phi}\left[1-(1-\gamma)\frac{\alpha}{\phi+\alpha}\right]}.$$

Since $1-\gamma \in (0,1)$ and $\alpha/(\alpha+\phi) \in (0,1)$, the exponent on k is strictly positive, even with the price response included, so the market supply moves together with capacity (just like with fixed prices). The drop in capacity also leads to a drop in market demand because of the higher price.

We need to look at (7.7), which captures jointly market supply and demand, to see what happens to tightness and output. We see that when capacity k drops, the left-hand side rises, since $-(1+\phi)\gamma/\phi < 0$. So the right-hand side must rise as well, to maintain the equality. The right-hand side is increasing in tightness θ, so tightness must rise when capacity falls. This comparative static is illustrated in figure 7.2B.

We can again without much effort compute the elasticity of tightness with respect to capacity from (7.7). The elasticity of the left-hand side with respect to capacity is

$$\epsilon_k^l = -\frac{(1+\phi)\gamma}{\phi}.$$

Now, implicit differentiation of (7.7) gives $\epsilon_k^l = \epsilon_\theta^r \epsilon_k^\theta$, where ϵ_θ^r is given by (7.8), so that

$$\epsilon_k^\theta = -\frac{\gamma}{(1-\eta) + \frac{1-\alpha}{\alpha} \cdot \frac{\phi}{1+\phi}\eta\tau(\theta)}.$$

We confirm that the elasticity is strictly negative, so tightness is higher when capacity is reduced. Again, we see that the response of tightness is stronger (the amplitude $\left|\epsilon_k^\theta\right|$ of the elasticity is higher) when prices are more rigid (higher γ). The logic is the same as with the demand shock: if prices are more rigid, they cannot absorb the supply shock much, so tightness must absorb more of it. We also see that the amplitude of the elasticity is larger when the supply elasticity $1/\phi$ is higher. (A higher supply elasticity $1/\phi$ corresponds to a lower ϕ, so a lower term $\phi/(1+\phi)$ in the denominator, making the absolute value of the elasticity $|\epsilon_k^\theta|$ larger.) Intuitively, when the supply is less elastic, it responds less to the price increase caused by the lower supply, so fewer sellers are encouraged to enter the market—which would have alleviated the drop in seller capacity but here does not happen much. As a result, without much new entry, tightness will rise more. Again we realize here that when $1/\phi \to 0$, the elasticity reduces to the elasticity with fixed capacity, given by (6.16).

7.4. Summary

This chapter extends the slackish market model by making market capacity endogenous rather than exogenous. We allow potential sellers to decide whether to enter the market or not, based on their expected income from sales, which they compare against their utility from nonparticipation. The resulting market supply now depends on both the participation rate and the selling probability, making it increasing in both market tightness and price.

We analyze the model under both fixed and rigid prices and find that the comparative-static results remain consistent with those under exogenous capacity. Negative demand shocks reduce both tightness and sales while decreasing market participation, as sellers find entry less attractive due to lower expected selling probabilities. Negative supply shocks (whether from reduced individual capacity or increased nonparticipation utility) increase tightness but reduce sales and participation rates. Furthermore, when prices are more rigid, tightness absorbs more of the shocks. The same is true when supply elasticity is lower: tightness responds more because fewer sellers adjust their participation in response to price changes.

You may have noticed that here we did not look at the case with bargained prices. That is because bargained prices are exactly the same with endogenous capacity. Combined with the market demand that is unchanged, the bargained price determines tightness, as (6.8) shows. Hence demand shocks have no effect on the model, even with endogenous capacity, and supply shocks have no effect on tightness but affect output, and here the participation rate, in a straightforward way.

CHAPTER 8.

Customer relationships

This chapter presents a dynamic version of the slackish market model, in which buyers and sellers enter into long-term customer relationships once they have matched, as most buyers and sellers do in the real world. The customer relationships break down every once in a while for random reasons. For simplicity, we focus on the equilibrium of the dynamic model, where market flows are balanced: as many customer relationships are created as destroyed at any point in time. This is a good approximation to study the day-to-day behavior of markets as long as the convergence to the equilibrium is rapid—which is the case at least in the US labor market.

8.1. Trading with customer relationships

In this dynamic world, goods are traded through customer relationships. Because it is difficult for sellers to find buyers and buyers to find sellers, once a buyer and seller have found one another, they enter a customer relationship. Indeed, if the match is good, there is no need to look for another trading partner. Each customer relationship delivers one good per unit time to the buyer.

However, customer relationships do not last forever. At any point in time, relationships might be broken for exogenous reasons. We denote by $\lambda > 0$ the rate at which separations occur. Once there is a separation, buyers have to visit sellers to find a new supplier, and sellers have to display goods in their shops to entice buyers and find a new customer.

8.1.1. Matching with customer relationships

We assume that time, $t \geq 0$, is continuous. This allows us to carry out a continuous-time analysis, which is simpler than a discrete-time analysis.

We denote by k the number of goods for sale per unit time, and by $y(t)$ the number of goods sold to customers per unit time. Since all sales occur through customer relationships, $y(t)$ is also the number of ongoing buyer–seller relationships. Therefore $k - y(t)$ is the number of goods unsold and available for sale to visiting buyers per unit time.

In a dynamic world it becomes important to distinguish between visit level and visit rate, and between slack level and slack rate. The visit level is the number of visits per unit time, which we denote $V(t)$. The visit rate is the number of visits as a share of the number of goods for sale k; we denote it by $v(t) = V(t)/k$. Similarly the slack level is the number of goods unsold per unit time, which we denote $U(t) = k - y(t)$. The slack rate is the number of unsold goods as a share of the number of goods for sale:

$$u(t) = \frac{k - y(t)}{k} = 1 - \frac{y(t)}{k}. \tag{8.1}$$

How does matching work for those sellers and buyers that are not in customer relationships? As usual, the number of new matches is given by a matching function, m, that takes as arguments the number of goods for sale that are not already committed to buyers, $U(t)$, and the number of buyers' visits to sellers' shops, $V(t)$. Accordingly, the number of new matches per unit time is $m(U(t), V(t))$.

The market tightness is as usual the ratio of the two arguments in the matching function:

$$\theta(t) = \frac{V(t)}{U(t)}. \tag{8.2}$$

The market tightness determines the rate at which each uncommitted good is sold to a new customer, $f(\theta(t))$, and the rate at which each visit leads to a purchase, $q(\theta(t))$. The rates are given by the usual expressions, (3.4) and (3.5), and have the usual properties.

8.1.2. Law of motion of customer relationships

Next, we write down the law of motion for prevailing customer relationships. The law of motion also tells us how sales vary over time in this dynamic environment, since one good is sold per unit time in each relationship. The law of motion is

$$\dot{y}(t) = f(\theta(t))\,[k - y(t)] - \lambda y(t).$$

In words, the change in the number of relationships ($\dot{y}(t)$) is equal to the number of new customer relationships ($f(\theta(t))[k - y(t)]$) minus the number of existing customer relationships that broke down ($\lambda y(t)$).

From these dynamics, we infer the law of motion for the rate of slack, $u(t)$. The rate of slack is the share of goods that are available for sale but unsold, as shown by (8.1). Accordingly, the law of motion of the slack rate follows from the law of motion for the number of customer relationships:

$$\dot{u}(t) = -\frac{\dot{y}(t)}{k} = \lambda\frac{y(t)}{k} - f(\theta(t))(1 - \frac{y(t)}{k}).$$

This gives the law of motion for the slack rate:

$$\dot{u}(t) = \lambda[1 - u(t)] - f(\theta(t))u(t). \tag{8.3}$$

The law of motion says that the slack rate increases when customer relationships are severed ($\lambda[1 - u(t)]$) and decreases when new customer relationships are formed ($f(\theta(t))u(t)$).

8.2. Beveridge curve

Now that we have the differential equation describing the dynamics of the slack rate, we solve it to determine what the slack rate converges to, and how quickly it does so. We find that a Beveridge curve—a negative relationship between visits and slack—naturally arises as the equilibrium point of the law of motion of slack.

8.2.1. Building a Beveridge curve from balanced flows

For the analysis here, we consider that market tightness θ and thus the selling rate $f(\theta)$ are constant. Given those, we study the dynamics of the slack rate.

First, let us determine the equilibrium point of the differential equation (8.3). By definition, the equilibrium is the slack rate u such that $\dot{u} = 0$. Thus, the equilibrium slack rate satisfies

$$-[\lambda + f(\theta)]u + \lambda = 0,$$

which implies that the equilibrium slack rate is solely determined by the selling rate $f(\theta)$ and separation rate λ:

$$u = \frac{\lambda}{\lambda + f(\theta)}. \tag{8.4}$$

In fact, the equilibrium point is decreasing in the selling rate and therefore market tightness. The intuition of course is that when goods are sold more rapidly, fewer goods remain unsold at any point in time, so there is less slack. The equilibrium point is increasing in the separation rate, on the other hand (result A.9). Here the intuition is that when customer relationships break down more often, more goods are unattached and therefore unsold at any point in time, so there is more slack.

Furthermore, the relationship between slack rate and tightness given by (8.4) is a Beveridge curve: it describes a negative relationship between slack rate and visit rate. This can be seen by introducing visits into the equation. Visits are implicitly present because market tightness is just the ratio of visit rate to slack rate: $\theta = v/u$. We start by rewriting our equation (8.4):

$$\lambda u + f(\theta)u = \lambda.$$

The selling rate is given by $f(\theta) = m(1, \theta)$, as we saw with equation (3.4). Since the matching function has constant returns to scale, $f(\theta)u = m(u, \theta u) = m(u, v)$. Hence, the equilibrium condition for the slack rate links the visit and slack rates by

$$m(u, v) = \lambda(1 - u). \tag{8.5}$$

The equilibrium condition is called a balanced-flow condition and we see clearly why here: the slack rate is in equilibrium when the number of new customer relationships, given by the matching function $m(u, v)$, equals the number of customer relationships that separate, which itself is given by the share of goods in relationships, $1 - u$, times the rate at which those separate, λ. So, slack is constant over time when the numbers of relationships that form and separate are equal.

How can we see that the balanced-flow equation (8.5) is a Beveridge curve? Well, it suffices to differentiate it implicitly (result A.7). The equation implicitly defines the visit rate as a function $v(u)$ of the slack rate, and implicit differentiation gives

$$\frac{\partial m}{\partial u} + \frac{\partial m}{\partial v} \cdot v'(u) = -\lambda. \tag{8.6}$$

Reshuffling terms to isolate the derivative $v'(u)$, we get:

$$v'(u) = -\frac{\lambda + \partial m/\partial u}{\partial m/\partial v}. \tag{8.7}$$

Since the matching function is increasing in both arguments, we have $\partial m/\partial u > 0$ and $\partial m/\partial v > 0$, which tells us that $v'(u) < 0$. The balanced-flow equation therefore generates a Beveridge curve: it defines the visit rate as a decreasing function of the slack rate. The main reason is that when there is more slack, more goods are available to be sold, so given the properties of the matching function, fewer visits are necessary to generate the required number of new matches. A secondary reason is that when there is more slack, there are fewer established customer relationships, so fewer separations for a given separation rate. Fewer new matches are then required to offset these separations.

8.2.2. Convergence to the Beveridge curve

We have seen that a Beveridge curve appears at the point where the market flows—creation and destruction of customer relationships—are balanced. The next question is: How long does it take for the market to converge to the Beveridge curve? That is, how long does it take for the slack rate $u(t)$ to converge to the equilibrium point $u(\theta)$, defined by (8.4)?

We can rewrite the differential equation as

$$\dot{u}(t) + (\lambda + f)u(t) = \lambda.$$

This is an autonomous first-order linear differential equation, which is easily solvable (result A.27). The solution is

$$[u(t) - u(\theta)] = [u(0) - u(\theta)]\exp(-(\lambda + f)t).$$

The interpretation of the solution is that the gap between the slack rate $u(t)$ and the equilibrium point $u(\theta)$ shrinks at a rate of $\lambda + f$. So if the selling rate f and separation rate λ are large enough, convergence to the equilibrium—to the Beveridge curve—is rapid.

The speed of the convergence to the Beveridge curve depends on the market. In some markets selling might be faster or there might be more turnover in relationships, so convergence might be faster. In other markets, selling might take more time or relationships might be more stable, so convergence might be slower.

In this book, we take the perspective that convergence to the Beveridge curve is fast enough that we can ignore it. The motivation for the assumption comes from the US labor market. There, λ averages 3.6% per month (section 2.6.1), f averages 56% per month (section 3.8), so $\lambda + f$ is about 59.6% per month. This means that the gap between the unemployment rate and the Beveridge curve shrinks very quickly. The half-life $t_{1/2}$ of this exponential decay is the time it takes to halve the initial distance between the unemployment rate and the Beveridge curve: $[u(t_{1/2}) - u(\theta)] = (1/2) \times [u(0) - u(\theta)]$. Here the half-life satisfies:

$$t_{1/2} = \frac{\ln(2)}{\lambda + f} = \frac{0.693}{0.596} = 1.16.$$

This means that it will take only 1.16 months to halve the initial distance between the unemployment rate and the Beveridge curve. So convergence to the Beveridge curve is extremely fast. After one quarter, only $(1/2)^{3/1.16} = 17\%$ of the initial gap is left.

We also make the assumption because it greatly simplifies the analysis of the dynamic model, since it replaces a differential equation (equation (8.3)) by a static relationship (equation (8.4)). Thus, we can simplify our model by getting rid of the dynamics of slack.[1]

The assumption that at the time scale of the model market flows are always balanced simply neglects the short period of time during which flows are unbalanced, before the market converges to its balanced state. This is very much like the assumption that people are always rational in economic models—which neglects the learning period before people converge to a rational behavior. It is also akin to the assumption that people always play the Nash equilibrium in game theory—which neglects the learning or coordination period that it takes to reach such equilibrium.

In any dynamic model, we therefore assume that market flows are balanced at all times so the slack rate is just a function of the market tightness:

$$u(\theta) = \frac{\lambda}{\lambda + f(\theta)}. \tag{8.8}$$

The qualitative properties of the equilibrium slack rate are the same as the properties of the slack rate in the static model: when market tightness is 0, the slack rate is 1; then the slack rate is a decreasing and convex function of tightness. These properties follow from the properties of the selling rate, $f(\theta)$, which is 0 when tightness is 0 and is increasing and concave in tightness. To establish convexity, we also use the facts that the function $\lambda/(\lambda+x)$ is decreasing and convex in $x \geq 0$ (result A.9), and that the composite of a function that is convex and decreasing with a concave function is convex (result A.17).

Figure 8.1A displays the equilibrium slack rate as a function of tightness to illustrate its properties in this dynamic model.

8.2.3. Why can slack never disappear?

The expression for the slack rate (8.8) explains why there is always some slack in the market: in a dynamic world, slack can never completely disappear. Mathematically, the reason is that since the separation rate λ is strictly positive and the selling rate $f(\theta)$ is finite, the equilibrium slack rate $u(\theta)$ is bound to be strictly positive.

Intuitively, the reason is that at any moment some customer relationships are bound to be terminated—because the circumstances of some buyers and sellers constantly change so that they must quit their relationships. The goods that were traded in these relationships become available again, but it takes some time for sellers to find new customers. As a result, the goods remain unsold for some time: there is some slack on the market.

In the context of the labor market, it has been known for a long time that labor-market flows impose a minimum level of unemployment: unemployment can never completely

[1] Just as we do here, it is common in the DMP literature to assume that the Beveridge curve holds at all times. See for instance Pissarides (1986), Pissarides (2009), Hall (2005b), Hall (2005a), and Elsby, Michaels, and Solon (2009).

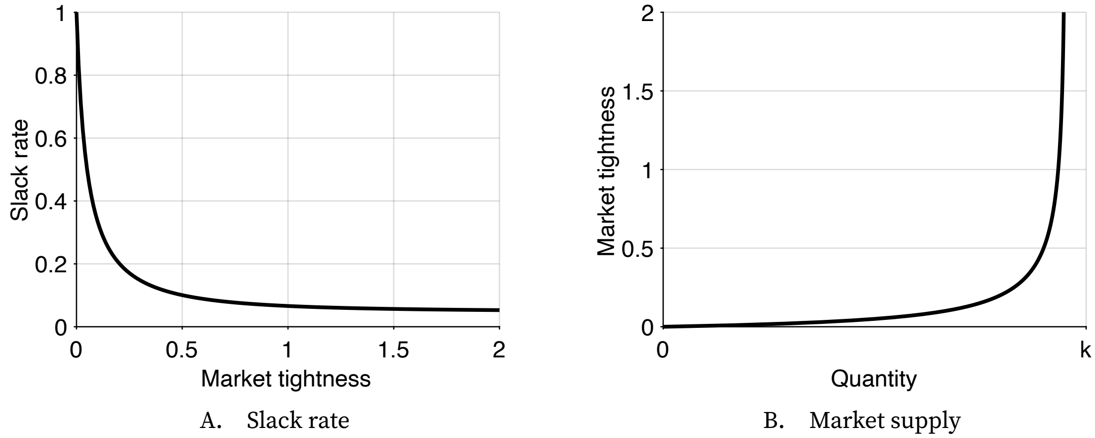


FIGURE 8.1. Numerical illustration of the dynamic slackish market model in equilibrium

The slack rate is given by (8.8). The market supply is given by (8.9). The matching function is CES, given by (3.12). Parameters are set to $\sigma = 2$, $\kappa = 4$, $k = 1$, $\alpha = 0.5$, $a = 2$, $p = 1$, and $\lambda = 5\%$.

disappear. For instance, Robinson (1946, pp. 169–170) made the following observation:

> In a changing world there are always bound to be, at any moment, some workers who have left one job and have not yet found another.... Changes in occupation for personal reasons will always be going on. So long as such shifts in employment are taking place there is always likely to be some unemployment even when the general demand for labor is very high.

8.3. Equilibrium market supply

Now that we have computed the equilibrium slack rate, it is easy to compute the market supply. Indeed, output $y(t)$, slack rate $u(t)$, and market capacity k are related by

$$y(t) = [1 - u(t)]\, k,$$

because by definition the slack rate is the share of capacity that is unsold at any time (which is also formalized by (8.1)).

But in equilibrium, the slack rate is related to tightness by (8.8), so that output is a function of tightness too. This function of tightness gives the amount of output sold through the matching process when flows are balanced. It is the equilibrium market supply in this dynamic world, and it is given by

$$y^{s}(\theta) = [1 - u(\theta)]\, k, \tag{8.9}$$

where $u(\theta)$ is the equilibrium slack rate given by (8.8), and k is the market capacity.

The qualitative properties of the equilibrium market supply are the same as the properties in the static model: when market tightness is 0, the supply is 0; then the supply is increasing and concave in tightness. These properties follow from the properties of the equilibrium slack rate, which is 1 when tightness is 0 and is decreasing and convex in tightness.

We plot again the market supply curve in a tightness-quantity plane to visualize its behavior in this dynamic model (figure 8.1B). Compared to the market supply curve in a static model (figure 4.1D), this market supply curve is much more curved—although both supply curves are obtained with the same CES matching function, under the same calibration.

We can see why the market supply curve is more curved in the dynamic world by comparing the expressions for the two supply curves. Once the expressions (4.4) and (8.8) for the slack rates are substituted into the market supplies, the supplies in the static and dynamic models are given by

$$y^s(\theta) = f(\theta)k \quad \text{and} \quad y^s(\theta) = \frac{f(\theta)}{\lambda + f(\theta)}k. \tag{8.10}$$

Hence, in a dynamic model, the concave function $f(\theta)$ is composed with a concave and increasing function, $x/(x+\lambda)$, which makes the market supply $y^s(\theta)$ a more concave function of tightness.

Because of this sharp curvature of the market supply, the slackish model is sharply state dependent. This state dependence has a range of implications, because it implies that the model behaves very differently when tightness is high and when tightness is low. Section 8.8 describes the first manifestations of this state dependence by contrasting the effects of supply and demand shocks in tight and slack markets.

8.4. Equilibrium matching wedge

Now switching to the demand side of the model, our first step is to compute the matching wedge: the share of consumption that must be devoted to matching on a slackish market. In the dynamic model, the matching wedge has a slightly different expression. Here we maintain the assumption that the market is in equilibrium, so the market remains the same at any point in time. We therefore omit the time index to keep notation simpler.

Consider a buyer j who conducts V_j visits to consume c_j goods. Because the buyer must also purchase some goods for matching, their purchases are larger than their consumption: $y_j > c_j$. Since we assume that market flows are balanced, the number $y_j$of customer relationships through which the buyer purchases the y_j goods remains stable over time, which requires that the number of new relationships created at any point in time equals the number of customer relationships that are destroyed at that point. The number of relationships that are destroyed is λy_j, as λ is the separation rate, while the number of relationships that are created is $q(\theta)V_j$, as $q(\theta)$ is the buying rate. Thus, to maintain y_j relationships with sellers, a buyer must visit

$$V_j = \frac{\lambda y_j}{q(\theta)} \tag{8.11}$$

shops. Given that each visit requires κ goods, output and consumption are related by

$$y_j = c_j + \frac{\kappa\lambda y_j}{q(\theta)}. \tag{8.12}$$

The equation relating purchases and consumption in this dynamic world (equation (8.12)) is the same as the equation in the static world (equation (4.6)), except that the term κ in the static world becomes $\kappa\lambda$ in the dynamic world. Therefore, the matching wedge relating purchases to consumption admits the same expression as in the static model once κ is replaced by $\kappa\lambda$. Thus, $y_j = [1 + \tau(\theta)]c_j$ where

$$\tau(\theta) = \frac{\kappa\lambda}{q(\theta) - \kappa\lambda}. \tag{8.13}$$

The matching wedge has the same properties as in the static model except that κ must be replaced by $\kappa\lambda$. For instance, the constraint on the matching cost becomes $\kappa\lambda \in (0, q(0))$, and the upper bound $\bar{\theta}$ on tightness becomes $\bar{\theta} = q^{-1}(\kappa\lambda)$.

8.5. Equilibrium market demand

In a dynamic world, buyers ought to have a utility function that integrates consumption at all times, discounting future consumption appropriately. Then each buyer would maximize this present-discounted utility subject to their intertemporal budget constraint, which allows buyers to use money to smooth consumption over time. However, the dynamics along the convergent path to the equilibrium, where the buyer's consumption is constant over time, are not relevant to the present market model. To abstract from them, we simply assume that agents' time discount rate equals 0. This simplifies the analysis by allowing us to directly focus on the equilibrium of the buyer's problem, rather than having to solve an optimal control problem.[2]

Under the assumption of zero discount rate, the buyer's problem is just a static problem:

[2] Pissarides (1984b) and Hosios (1990) make the same assumption in their analysis of labor market efficiency: they set discount rate to 0 to focus on the equilibrium of the model and abstract from transition dynamics. Another equivalent assumption would be that all money not spent (wealth) is taxed at the end of each period by the government, and transfers of money are made at the beginning of each period. Then buyers would only face static—not intertemporal—constraints so they would simply maximize their flow utility at any point in time, irrespective of their discount rates.

TABLE 8.1. Structure of the dynamic slackish market model in equilibrium

Variable	Name	Condition	Interpretation	Reference
c	Consumption of goods	$c = c^d(\theta, p)$	Utility maximization	(4.14)
p	Price of goods	$p = p^n(\theta)$	Price norm	(4.16)
y	Sales of goods	$y = [1 + \tau(\theta)]c$	Matching wedge	(4.7)
v	Visit rate	$v = \lambda y/[q(\theta)k]$	Balanced flows	(8.11)
θ	Market tightness	$\theta = v/u$	Definition of tightness	(8.2)
u	Slack rate	$u = \lambda/[f(\theta) + \lambda]$	Beveridge curve	(8.8)

the buyer simply maximizes the instantaneous utility subject to the equilibrium budget constraint, without any intertemporal considerations. Once we move to a model of the entire economy (part IV), we will of course allow for saving and intertemporal dynamics—but these are not the object of the present analysis.

Accordingly, we simply assume that each buyer maximizes utility (4.10) subject to the budget constraint (4.11), where the matching wedge is given by (8.13). This is the same problem as in the static model, so the market demand remains the same, given by (4.15), with the same properties.

8.6. Solution of the model

Now that we have introduced all the elements of the dynamic model, we are in a position to solve it. The structure of the model remains the same as in the basic model, but with a few adjustments, as described in table 8.1.

The approach to find the solution of the dynamic model is the same as in the static case: first, determine the market tightness, θ; then, compute the values of the 5 remaining variables (c, p, y, u, v).

Just as in the static model, tightness is determined from a supply-equals-demand condition in the dynamic model. Indeed, the amount of output demanded by buyers imposes that $y = y^d(\theta, p^n(\theta))$, where the equilibrium market demand is given by (4.15). At the same time, output is related to tightness via the equilibrium market supply, $y = y^s(\theta)$, where the equilibrium market supply is given by (8.9). The market tightness that solves the model equalizes equilibrium market demand and supply: $y^d(\theta, p^n(\theta)) = y^s(\theta)$. The main difference here with the static model is that the market supply is substantially different. Nevertheless, graphically, the solution of the model is found at the intersection of the market supply and demand curves. The equilibrium market supply curve captures the number of trades, for any tightness, given market capacity and balanced flows. The equilibrium market demand curve captures the number of trades, for any tightness, that maximize the per-period utility of buyers given their budget constraints.

8.7. Model with rigid prices

In this section, we solve the model under rigid prices (which includes the special case of fixed prices) and perform comparative statics with respect to demand and supply shocks. The analysis is brief as most of the results are the same as in the static model.

8.7.1. Solution with rigid prices

In this dynamic world the rigid prices remain given by (6.1), and the fixed price is a special case of the rigid price with $\gamma = 1$. The expression for the rigid price does not need to be amended in this dynamic model because the structures of the market demand and supply are close enough to those in the static model.

The model behavior is determined by the supply-equals-demand condition, which pins down market tightness. The market demand comes from (6.2), and the market supply comes from (8.9), so the supply-equals-demand condition takes the following form here:

$$\left[\frac{(1-\alpha)a^{\gamma}}{\rho}\right]^{1/\alpha} k^{-\gamma} = [1-u(\theta)]\,[1+\tau(\theta)]^{1/\alpha-1}.$$

Formally, this condition is exactly the same as condition (6.3) in the static case. However, two small changes appear in the background. First, in the factor $1+\tau(\theta)$, the term κ is replaced by the term $\kappa\lambda$, which incorporates the separation of relationships. This change does not modify the properties of the model, since none of the properties rely on the specific value of the matching cost κ. Second, in the factor $1-u(\theta)$, the term $f(\theta)$ is replaced by $f(\theta)/[\lambda+f(\theta)]$, which reflects the flows on the market. This change does not affect the qualitative properties of the model, because both functions $f(\theta)$ and $f(\theta)/[\lambda+f(\theta)]$ have the same qualitative properties: they are 0 when $\theta = 0$ and are increasing and concave in θ. Hence, through the same logic as in the static model, the model admits a unique solution (figure 4.3).

8.7.2. Comparative statics with rigid prices

We can quickly run through the comparative statics of the dynamic model, using our analysis of the static model in chapters 4 and 6.

After a negative demand shock (a reduction in the demand parameter a), the market demand curve moves inward while the market supply curve remains unmoved, as shown in figure 6.1A, so tightness and sales fall.

After a negative supply shock (a reduction in the market capacity k), the market supply shifts inward. Because prices increase a little when goods are scarcer—except if they are completely fixed—the market demand is depressed too, as illustrated in figure 6.1B.

Overall, however, whether prices are fixed or only rigid, tightness increases and sales decrease.

Despite the formal similarities between the static and dynamic models, the market adjusts to shocks slightly differently, and maybe not entirely as expected, in the dynamic world. Consider for instance a negative demand shock: suddenly buyers do not value the goods offered on the market as much. We have seen that market tightness drops so the slack rate rises: more goods remain unsold. One might imagine that more goods are unsold because more buyers terminate their relationships with sellers. But in the model this is not what happens: the reason why more goods remain unsold is that buyers reduce the number of new relationships they form. It becomes harder for sellers to find buyers for their goods, so the slack rate goes up. Existing buyer-seller relationships continue to operate as before, but their number dwindles by natural attrition. And because fewer new relationships are created at any point in time, there is more slack on the market: the stock of unsold goods balloons.

8.8. State dependence

One aspect of the model that is exacerbated in the dynamic environment is the curvature of the market supply curve. The market supply curve is much more curved—much more nonlinear—in the dynamic model, as can be seen by comparing figures 4.1D and 8.1B. This mathematical property—a highly curved market supply—has important economic implications. It means that the model behaves very differently when the market is slack or tight: the model is very state dependent. In this section we formalize this state dependence.

8.8.1. Curvature of the market supply

The entire approach followed in chapter 6 to compute the response of tightness to demand and supply shocks remains valid. The one key difference is that the market supply has a different curvature, so the elasticity of market supply with respect to tightness is different.

We compute the elasticity of the market supply with respect to tightness using the expression (8.10). We know that the elasticity of $f(\theta)$ with respect to θ is $1-\eta(\theta)$, where $\eta(\theta)$ is the matching elasticity (equation (3.8)). Then, using the results on elasticities from appendix B, we obtain the following elasticity of market supply with respect to tightness:

$$\epsilon^{s}_{\theta} = [1-\eta(\theta)] - \frac{f(\theta)}{f(\theta)+\lambda}[1-\eta(\theta)] = [1-\eta(\theta)]\, u(\theta). \tag{8.14}$$

The market supply is more curved in this dynamic environment because of the factor $u(\theta)$, which was not there in the static environment (see equation (6.11)). When the slack rate is $u(\theta)=1$, the elasticity is just $1-\eta(\theta)$, as in the static case. But as the slack rate falls, the elasticity drops, which means that the supply is less and less responsive to tightness. The elasticity converges to 0 when the slack rate goes to 0: at that point the market supply is perfectly vertical in a tightness-quantity plane.

8.8.2. Response to demand shocks

The sharp state dependence of the dynamic slackish model is clearly visible once we look at the response of sales to shocks. Let us start by looking at the response of sales to a demand shock. The response is given by the elasticity $\epsilon_a^y = \epsilon_\theta^s \cdot \epsilon_a^\theta$ where ϵ_a^θ is given by (6.10). This gives

$$\epsilon_a^y = \frac{\epsilon_a^d}{1 - \epsilon_\theta^d / \epsilon_\theta^s}.$$

We plug into this equation the appropriate values for the elasticities: $\epsilon_a^d = \gamma/\alpha$, the value of ϵ_θ^d given by (6.13), and the value of ϵ_θ^s given by (8.14). We get

$$\epsilon_a^y = \frac{\gamma}{\alpha + (1 - \alpha) \cdot \frac{\eta(\theta)}{1-\eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)}}. \tag{8.15}$$

We see that demand shocks have larger effects on output when the market is slack. The matching wedge $\tau(\theta)$ is strictly increasing in tightness, and the slack rate $u(\theta)$ is strictly decreasing in tightness, so the ratio $\tau(\theta)/u(\theta)$ is strictly increasing in tightness. Furthermore, by the assumption in chapter 3 that the matching elasticity $\eta(\theta)$ is increasing in tightness, the ratio $\eta(\theta)/[1 - \eta(\theta)]$ is also increasing in tightness. Hence, the product of the ratios $\eta(\theta)/[1-\eta(\theta)]$ and $\tau(\theta)/u(\theta)$ is strictly increasing in tightness. This implies that, since all other terms in the elasticity are constant, the elasticity ϵ_a^y is strictly decreasing in tightness. This means that when the market is slack, the elasticity is high: output responds sharply to changes in market demand. By contrast, when the market is tight, the elasticity is low: output does not respond much to changes in market demand. Furthermore, the cyclicality of the elasticity ϵ_a^y is exacerbated in the dynamic world because of the presence of the countercyclical slack rate $u(\theta)$ in (8.15). The term $u(\theta)$ captures the extra curvature of the market supply in a dynamic world.

The result that demand shocks have larger effects when the market is slack is intuitive. Consider an increase in demand. If tightness remained the same, that increase in demand would directly translate into an increase in output. In figures 8.2A and 8.2B, that increase corresponds to the distance A–B. However, to generate the additional matches required to trade the additional output, more visits are required. These additional visits raise market tightness, which make consumption less attractive—as there is more competition among buyers for the goods for sale. The increase in tightness therefore somewhat reduces output demanded: the reduction corresponds to the movement B–C along the market demand.

In a slack market, depicted in figure 8.2A, visits are highly successful, so few extra visits are required to generate the additional trades. Hence, tightness does not increase much, so the reduction B–C is small, and the overall increase in output, A–C, is large. This means that demand disturbances have large effects on output in slack markets.

In a tight market, depicted in figure 8.2B, things are different. Then, visits are much less likely to be successful, so many more visits are required to generate the additional trades. Tightness increases significantly, so the reduction B–C is large, which means that the overall increase in output, A–C, is small. The implication is that demand disturbances have small effects on output in tight markets.

To illustrate the state dependence of the dynamic slackish model, we plot the demand elasticity of output, ϵ_a^y (figure 8.3A). We see that when tightness increases from 0 to 2, the demand elasticity of output drops sharply, from 2 to 0. The numerical values of the elasticity would depend on the calibration of the model, and would change for instance if the price rigidity γ was calibrated differently. The goal of the numerical exercise is only to illustrate that the slackish model's state dependence is strikingly strong.

8.8.3. Response to supply shocks

We obtain opposite results with supply shocks: supply shocks have larger effects on output when the market is tight.

The response of output to supply shocks is given by the elasticity $\epsilon_k^y = \epsilon_k^s + \epsilon_\theta^s \cdot \epsilon_k^\theta$ where ϵ_k^θ is given by (6.15). This gives

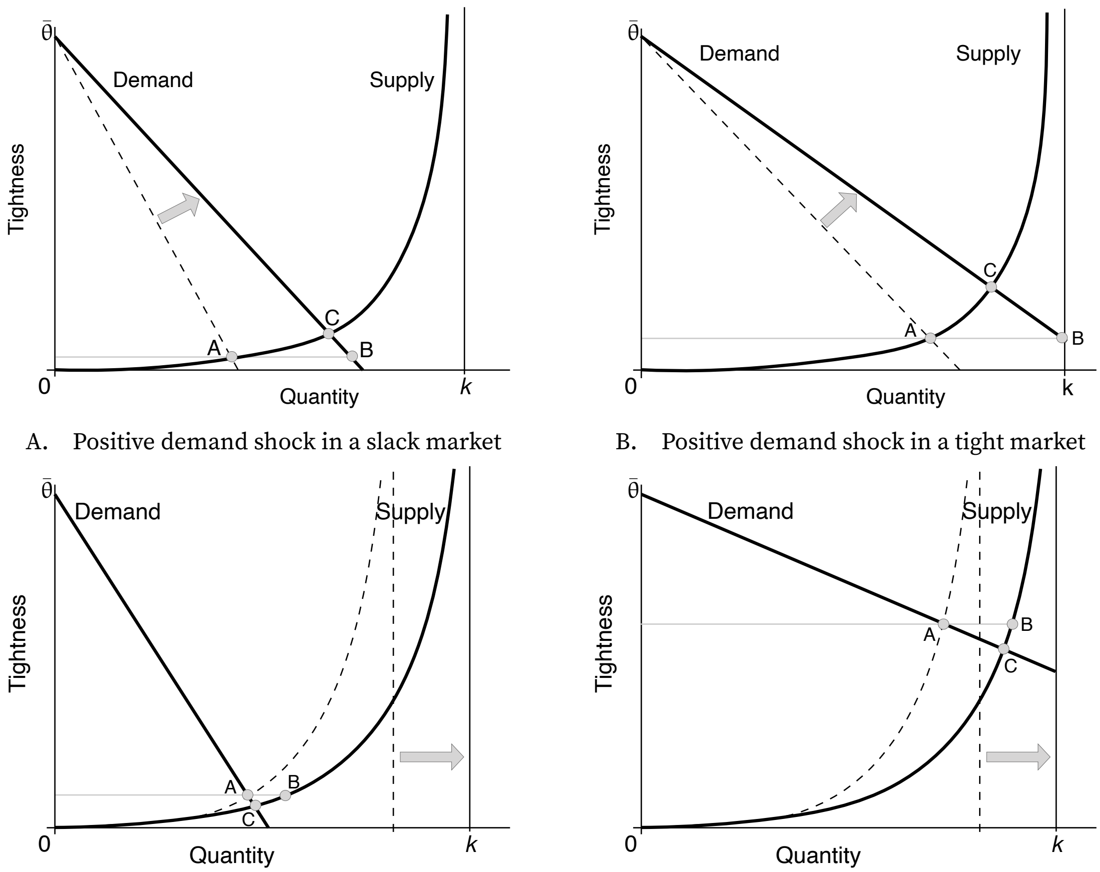
$$\epsilon_k^y = \epsilon_k^s + \frac{\epsilon_k^d - \epsilon_k^s}{1 - \epsilon_\theta^d / \epsilon_\theta^s}.$$

We plug into this equation the appropriate values for the elasticities: $\epsilon_k^s = 1$, $\epsilon_k^d = 1 - \gamma$, the value of ϵ_θ^d given by (6.13), and the value of ϵ_θ^s given by (8.14). We get

$$\epsilon_k^y = 1 - \frac{\gamma}{1 + \frac{1-\alpha}{\alpha} \cdot \frac{\eta(\theta)}{1-\eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)}}. \tag{8.16}$$

Following the same logic as above, we see that the response of output to supply shocks is stronger in a tighter market. We know that the product of the terms $\eta(\theta)/[1-\eta(\theta)]$ and $\tau(\theta)/u(\theta)$ is strictly increasing in tightness. Hence, the elasticity ϵ_k^y is increasing in tightness. This means that when the market is tight, the elasticity is high: output responds significantly to changes in market capacity. By contrast, when the market is slack, the elasticity is low: output does not respond much to changes in market capacity.

To understand the result, consider an increase in market capacity. If tightness remained the same, that increase in supply would directly translate into an increase in



A. Positive demand shock in a slack market

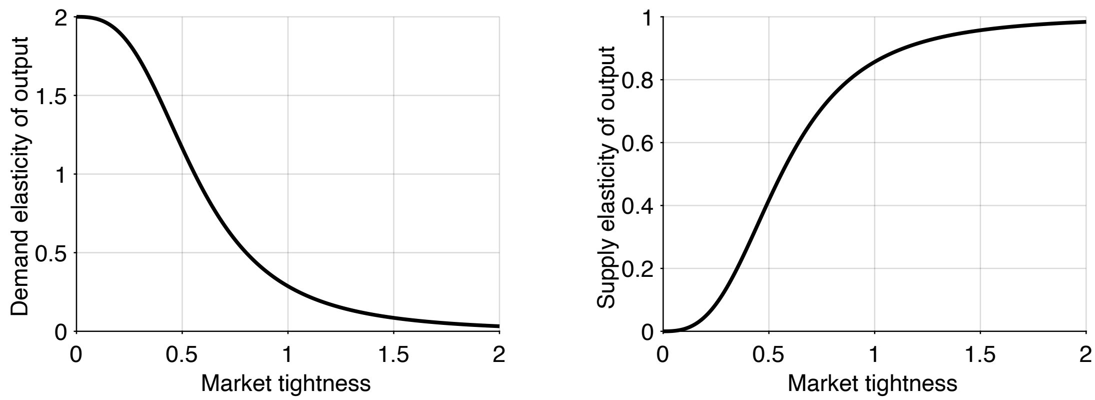
B. Positive demand shock in a tight market

C. Positive supply shock in a slack market

D. Positive supply shock in a tight market

FIGURE 8.2. State dependence in the dynamic slackish market model

The market supply is given by (8.9). The market demand is given by (6.2). The price norm is given by (6.1). A negative demand shock is a reduction in the preference for goods *a*. A negative supply shock is a reduction in market capacity *k*.



A. Elasticity of output to demand

B. Elasticity of output to supply

FIGURE 8.3. Numerical illustration of state dependence in the dynamic slackish market model

The elasticity of output to demand is given by (8.15). The elasticity of output to supply is given by (8.16). The matching function is CES, given by (3.12). Parameters are set to $\sigma = 2$, $\kappa = 4$, $k = 1$, $\alpha = 0.5$, $a = 2$, $p = 1$, $\gamma = 1$, and $\lambda = 5\%$.

output. However, because the market demand has not changed, buyers would not be willing to absorb this output at the current tightness. In figures 8.2C and 8.2D, that gap between demand and supply corresponds to the distance A–B. Buyers thus start reducing their visits to reduce their purchases, which lowers tightness. This reduction in tightness has two implications: Buyers are willing to purchase more goods than before (a movement down the demand curve from A), and fewer matches are made so fewer goods are traded (a movement down the supply curve from B). Overall, the gap between demand and supply shrinks until the market reaches C, where demand and supply are equalized again. The overall increase in output from the increase in supply is the distance A–C, which is less than A–B.

In a tight market, depicted in figure 8.2D, the supply curve is relatively steep and the demand curve is relatively flat, so the new output level is much closer to point B than point A. This means that the increase in output, A–C, is large: supply disturbances have large effects on output in tight markets.

In a slack market, depicted in figure 8.2C, things are different. The supply curve is relatively flat and the demand curve is relatively steep, so the new output level is much closer to point A than point B. This means that the increase in output, A–C, is small: supply disturbances have small effects on output in slack markets.

To illustrate further the state dependence of the dynamic slackish model, we plot the supply elasticity of output, ϵ^y_k (figure 8.3B). We see that when tightness increases from 0 to 2, the supply elasticity of output rises sharply, from 0 to 2. Once again, the numerical exercise illustrates that the slackish model is strongly state dependent.

8.9. Flexibility of bargained prices in the dynamic model

In the static model, we showed that prices obtained by bargaining were flexible, in the sense that they could be written as the price norm (6.1) with $\gamma = 0$ (see section 6.3). We therefore argued that bargaining might push prices toward the flexible benchmark. To conclude this chapter, we show that the same is true in the dynamic model. Determining the bargained price is more complicated in a dynamic world, but the main insight remains: the bargained price can again be written as the price norm (6.1) with $\gamma = 0$.

8.9.1. Expression of bargained prices

As in the static model, we assume that after bargaining, seller and buyer agree to share the trade surplus: the seller keeps a fraction χ of the surplus and the buyer gets a fraction $1 - \chi$ of the surplus, where $\chi \in (0, 1)$ is the bargaining power of the seller. Let $\mathcal{S}$, $\mathcal{B}$, and $\mathcal{T}$ be the seller, buyer, and total surplus. The surplus-sharing outcome requires that $\mathcal{S} = \chi \cdot \mathcal{T}$ and $\mathcal{B} = (1 - \chi) \cdot \mathcal{T}$. If we eliminate the total surplus from these equations, we see that the buyer surplus and seller surplus are related by

$$\mathcal{B} = \frac{1 - \chi}{\chi} \cdot \mathcal{S}. \tag{8.17}$$

The buyer surplus may be bigger or smaller than the seller surplus, depending on the bargaining power. Let's now compute these surpluses to determine the bargained price p.

What is the surplus accrued by a buyer from establishing an extra buyer-seller relationship? The value from consuming an extra good is the marginal utility of consumption, $(1 - \alpha)ac^{-\alpha}$. At the same time, the buyer must pay the seller the price, p, which reduces their quasilinear utility by p utils. Hence, the buyer accrues $(1 - \alpha)ac^{-\alpha} - p$ utils per unit time from the trade. How long can the buyer expect the relationship to last? Since relationships are dissolved at a rate λ, following a Poisson process, the expected duration of the seller-buyer match is $1/\lambda$. By contrast, if the negotiations with the seller break down and the trade does not go through, the buyer gets nothing. Thus, the expected surplus from trading with the seller is

$$\mathcal{B} = \frac{(1 - \alpha)ac^{-\alpha} - p}{\lambda}.$$

We turn next to the seller surplus. In the trade, the seller receive the price, p. If the negotiations break down, the seller does not receive anything for the good until they can find a new buyer for it. From the seller's perspective, trading with the buyer has value only until the relationship breaks down or the seller finds a new trade partner. As soon as either event occurs, the surplus from trading vanishes. Now, existing trade relationships break down at a rate λ, while new trade relationships are formed at a rate $f(\theta)$. Here we have two Poisson events, one occurring at rate λ and the other at rate $f(\theta)$, and we are looking for the rate at which the first of the two events happens. A well-known result from statistics is that the first of these two events happens at a rate of $\lambda + f(\theta)$. Thus, the advantage of starting in a trade relationship compared to starting without a trade relationship vanishes according to a Poisson process with rate $\lambda + f(\theta)$. Hence, the seller enjoys a surplus for an expected duration of $1/[\lambda + f(\theta)]$. As a result, the expected surplus from trading with a buyer is

$$\mathcal{S} = \frac{p}{\lambda + f(\theta)}.$$

We can now calculate the price given by the bargaining process. The price is such that (8.17) holds. Plugging in the expressions for buyer and seller surplus derived above we get:

$$\frac{(1-\alpha)ac^{-\alpha} - p}{\lambda} = \frac{1-\chi}{\chi} \cdot \frac{p}{\lambda + f(\theta)}.$$

Hence, the bargained price is set at a markdown below the marginal utility of consumption:

$$p = \frac{(1-\alpha)ac^{-\alpha}}{1 + \frac{1-\chi}{\chi} \cdot u(\theta)}. \tag{8.18}$$

If the seller bargaining power is 0, the price falls to 0. If the seller bargaining power is 1, the markdown vanishes and the price rises to the marginal utility of consumption. Interestingly, the markdown increases with the slack rate. If the slack rate was 0, the markdown would vanish and the price would reach the marginal utility of consumption. When the slack rate reaches 1, the markdown reaches $1/\chi$, which implies that the price is just a fraction χ of the marginal utility of consumption, just as in the static case. This property appears because sellers' outside option is worse when the slack rate is higher: if they do not agree to the price offer, they are forced to remain idle for a longer time. Sellers must therefore settle for lower prices when the slack rate is higher—which in effect allows buyers to impose a higher markdown.

8.9.2. Solution with bargaining

Plugging the price (8.18) into the market demand equation (4.13), we see that the market demand is degenerate under price bargaining:

$$\frac{\tau(\theta)}{u(\theta)} = \frac{1-\chi}{\chi}. \tag{8.19}$$

Just as in the static case, the demand curve under bargaining pins down a unique tightness, irrespective of quantities. The function τ/u is strictly increasing on $(0, \bar{\theta}) \to (0, \infty)$ so its inverse $(\tau/u)^{-1}$ is well defined on $(0, \infty)$, and strictly increasing. In the tightness-quantity diagram, the market demand is therefore horizontal at

$$\theta = (\tau/u)^{-1}\left(\frac{1-\chi}{\chi}\right). \tag{8.20}$$

8.9.3. Flexibility of bargained prices

We can see that bargained prices are flexible by rearranging the results above. In the price expression (8.18), we express consumption as a function of tightness as in (6.5): $c(\theta) = [1 - u(\theta)]k/[1 + \tau(\theta)]$. We then set tightness to the value given by (8.20). These steps yield

$$p = (1-\alpha)\left[1 + \frac{1-\chi}{\chi} \cdot u\left((\tau/u)^{-1}\left(\frac{1-\chi}{\chi}\right)\right)\right]^{\alpha-1}\left[1 - u\left((\tau/u)^{-1}\left(\frac{1-\chi}{\chi}\right)\right)\right]^{-\alpha} \cdot a \cdot k^{-\alpha},$$

after using (8.19) to replace $1 + \tau(\theta)$ by $1 + \frac{1-\chi}{\chi}u(\theta)$. Hence, the bargained price takes the form of the price norm (6.1) with $\gamma = 0$ and

$$\omega = (1-\alpha)\left[1 + \frac{1-\chi}{\chi} \cdot u\left((\tau/u)^{-1}\left(\frac{1-\chi}{\chi}\right)\right)\right]^{\alpha-1}\left[1 - u\left((\tau/u)^{-1}\left(\frac{1-\chi}{\chi}\right)\right)\right]^{-\alpha}.$$

Hence, we verify that the bargained price can be written as a flexible price norm in the dynamic model, just as in the static model.

8.10. Summary

This chapter presents a dynamic version of the slackish market model, in which trade is conducted through customer relationships. Overall, the structure of the dynamic slackish model is almost identical to that of the static slackish model. The main difference is that there are flows on the market—new customer relationships being created and destroyed. In equilibrium, however, these flows are balanced, and the model has a similar structure to the static model. The dynamic model can be analyzed with supply-and-demand diagrams, just like the static model.

The Beveridge curve emerges naturally in the dynamic model at the equilibrium: it links visits to slack when the number of new customer relationships formed equals the number of relationships that separate. Evidence from the US labor market suggests convergence to the Beveridge curve is rapid, justifying the assumption that markets operate in equilibrium. This assumption considerably simplifies the analysis by replacing differential equations with static equations.

In a dynamic world, the market supply is a more curved function of tightness than in the static model, which generates strong state dependence in the model's responses to shocks. Demand shocks have larger effects when the market is slack, while supply shocks have large effects when the market is tight. This state dependence implies that the same disturbance has very different consequences depending on the prevailing state of the market, which will become important in later chapters when we analyze the effectiveness of government policies.

CHAPTER 9.

Market efficiency and inefficiency

We have now built a slackish model of markets, and studied its positive properties: what price, slack, and output prevail in the market, and how these variables respond to supply and demand shocks. In this chapter we turn to the normative properties of the model: What is the socially efficient market allocation? Can we expect the market to operate efficiently? If not, how far from efficiency might the market be?

What do we mean by efficient? The efficient allocation is the allocation that maximizes the social welfare generated by the market. Knowing markets' efficient allocations is critical for designing the best possible stabilization policies, as we will see in part V.

9.1. Overview of the sufficient-statistic approach

Before we begin we must decide which model to use for the welfare analysis. We have introduced a range of slackish models: a static model with exogenous capacity; a static model with endogenous capacity; a dynamic model with exogenous capacity, which itself could be extended with endogenous capacity. We have also seen that these models can be built around a number of matching functions—as long as they satisfy a few assumptions—and a wide array of price norms. Which one should we pick?

It turns out that we do not need to answer this question, because we adopt the sufficient-statistic approach to welfare analysis. We simply solve the problem of a market planner who is trying to maximize welfare in a more general framework. Typically, being more general adds complexity. Here, however, the generalization cleans things up and highlights the mechanisms and forces at play better.

Traditionally, welfare analysis is structural. This approach comes with two limitations that the sufficient-statistic approach resolves. First, the structural approach assumes the entire structure of the market and studies welfare and policy within this structure. The limitation is that the market structure is assumed religiously but largely determines the policy conclusions. The policy insights are in many ways baked into the initial assumptions about model structure. Second, the structural approach requires calibrating all the parameters in the model even though many of these parameters are generally not observable in the real world.

With the sufficient-statistic approach, we do not assume the entire structure of the market. Instead, we make a minimal set of assumptions on the welfare function and market structure that allow for welfare analysis. This is beneficial because, unlike the structural approach, it allows the analysis to be applied to a range of models: any model that satisfies the minimal set of assumptions. Additionally, from a practical side, since we are making fewer assumptions, it is more likely that our analysis is valid in the real world.

At the end of the chapter, we will apply the general analysis to each of the models that we have introduced in the book, and we will map the sufficient-statistic results into structural results.

9.2. Market planner's problem

We consider a market planner who aims to allocate the goods supplied to the market efficiently. The market planner allocates some of the goods for consumption, some for matching, and leaves some of them unsold. The planner's objective is to maximize market welfare. What should the planner do?

9.2.1. Market slack

We assume that there is slack on the market, so a share $u \in (0, 1)$ of all goods available on the market are unsold. The rest of the goods are sold to buyers.

9.2.2. Beveridge curve

We also assume that the market features a Beveridge curve. This means that the visit rate is a strictly decreasing and convex function $v(u)$ of the slack rate. This is the sole structural assumption that we make in the welfare analysis. We do not need to assume a matching function or price norm; we only need the reduced-form Beveridge curve.

Why do visits enter the planner's problem? Because visits require resources, measured by the matching cost, $\kappa > 0$. The matching cost is the number of goods that have to be devoted to any one visit.

The Beveridge curve says that as the planner reduces the slack rate, the visit rate increases. So as the number of unsold goods falls, more goods must be allocated to visits and matching buyers and sellers. This is the central tradeoff that the planner must resolve: how to balance unsold goods against goods allocated to matching.

9.2.3. Market welfare

Market welfare measures the well-being generated by market activity. In our slackish market model, market welfare is not very complicated. It starts from the market capacity, which is k goods.

First, not all goods are sold. The slack rate is u, so only $(1-u)k$ goods are sold to buyers. The uk goods that remain unsold have no social value.

However, not all $(1-u)k$ sold goods are consumed. Some goods are instead devoted to matching and, as they are not consumed, they do not produce direct social value. Thus, we must net out the amount of goods devoted to matching: we need to subtract $\kappa v k$ goods from the $(1-u)k$ goods sold to buyers. (Since v is the visit rate, vk is the number of visits in the market.)

Overall, $(1-u)k - \kappa v k$ goods are consumed through market purchases, so market welfare is

$$[1 - u - \kappa v]\, k. \tag{9.1}$$

9.2.4. Planner's problem

We are now in a position to state the market planner's problem. We begin by plugging the Beveridge curve into market welfare, to write market welfare as a function of the slack rate:

$$\mathcal{W}(u) = [1 - u - \kappa v(u)]\, k. \tag{9.2}$$

The market planner chooses a slack rate $u \in (0, 1)$ to maximize market welfare $\mathcal{W}(u)$. The slack rate u^* that maximizes market welfare is the efficient slack rate. The efficient slack rate is the best slack rate from a social perspective.

9.3. Solution to the planner's problem

We now determine the efficient slack rate, u^*. Ideally the planner would prefer to have no unsold goods and no goods allocated to matching, but that is not feasible in a slackish market. The Beveridge curve imposes that some goods are always unsold, and some visits are always required so some buyers and sellers match.

9.3.1. Analytical solution

We start by analytically determining the efficient slack rate, u^*. Because the Beveridge curve $v(u)$ is strictly convex, we see in (9.2) that the welfare function $\mathcal{W}(u)$ is strictly concave. Indeed, $-v(u)\kappa k$ is the opposite of a strictly convex function and thus strictly concave, and adding the linear term $[1-u]\,k$ preserves strict concavity (result A.16). Because the welfare function is strictly concave, the first-order condition is sufficient to find the unique maximum of the welfare function (result A.20).

To find the efficient slack rate, we simply set the derivative of the welfare function to zero: $\mathcal{W}'(u) = 0$. The efficient slack rate satisfies

$$0 = -\left[1 + \kappa v'(u)\right] k.$$

This means that the efficient slack rate is implicitly defined by

$$v'(u) = -\frac{1}{\kappa}. \tag{9.3}$$

Efficiency requires the slope of the Beveridge curve equals $-1/\kappa$. There is a tradeoff between too much slack and too little slack, both of which are undesirable. The tradeoff is modulated by the Beveridge curve. The efficient slack rate guarantees that there is some slack to keep resources devoted to matching low, but not too much to keep the number of unsold goods low.

Formally, formula (9.3) says that when the market operates efficiently, the welfare costs and benefits from moving one good from sold to unsold are equalized. When one good is not sold anymore, consumption drops by 1 good. Having one more good unsold also means having $-v'(u) > 0$ fewer visits. Each visit reduces consumption by the matching cost, κ, so consumption increases by $-v'(u)\kappa > 0$ goods. When costs and benefits are equalized, we have $1 = -v'(u)\kappa$, which is equivalent to (9.3).

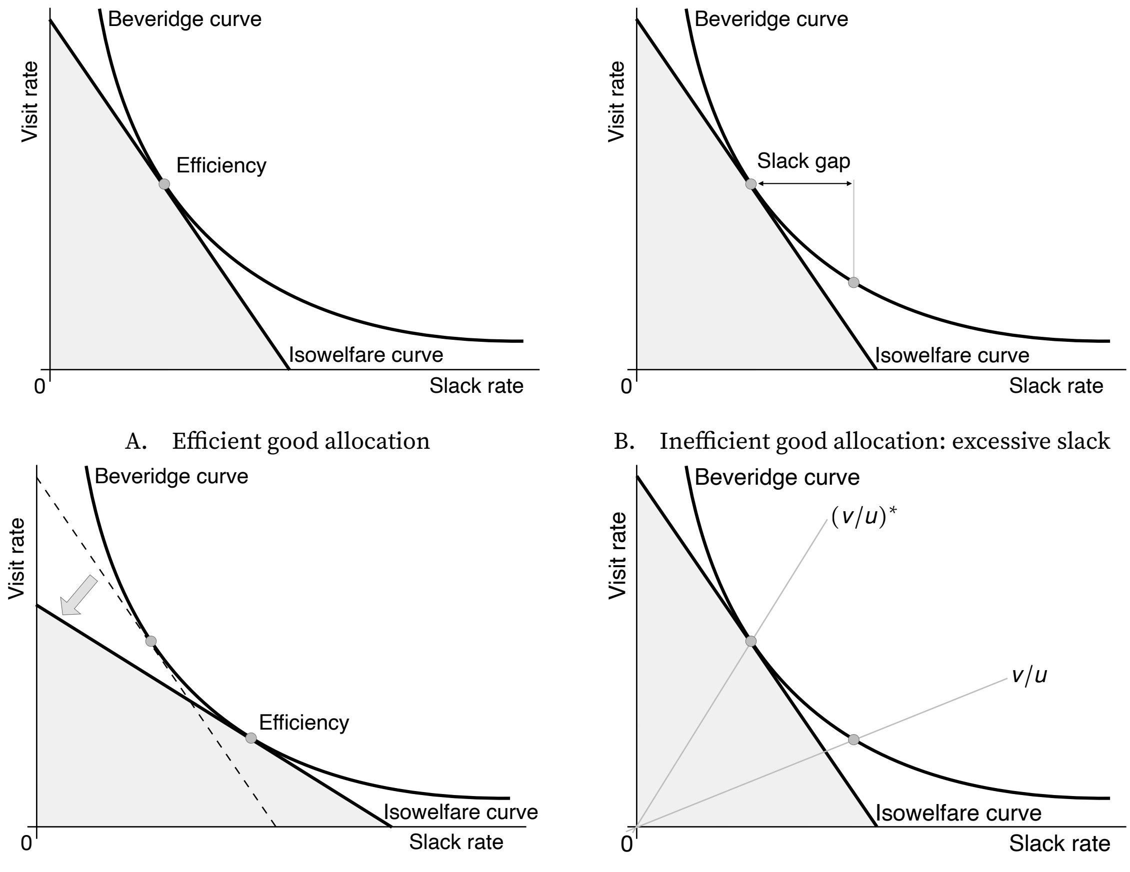
9.3.2. Graphical solution

Next, we graphically determine the efficient slack rate (figure 9.1A). To do that, we introduce isowelfare curves:

$$v = A - \frac{1}{\kappa} \cdot u, \tag{9.4}$$

where $A > 0$ governs the welfare level along the curve. On any isowelfare curve, the visit rate and slack rate are such that social welfare does not change.

Suppose that the planner could place the market on any isowelfare curve. Which one would they choose? Well, the planner would want to be on an isowelfare curve as close



A. Efficient good allocation

B. Inefficient good allocation: excessive slack

C. Comparative statics: higher matching cost

D. Efficient and inefficient v-u ratios

FIGURE 9.1. Efficiency and inefficiency in slackish market model

The isowelfare line is given by (9.4). The shaded area indicates all good allocations with higher welfare than the isowelfare line.

to the origin as possible. Points closer to the origin are characterized by fewer visits and fewer unsold goods. Visits and unsold goods are costly, so the points on isowelfare curves that lie closer to the origin have higher welfare than the points that are further out.

However, the isowelfare curves that are deep inside, close to the origin, do not cross the Beveridge curve: they are not feasible. The market planner must respect the Beveridge curve so must choose a point that is also on the Beveridge curve. Accordingly, the optimal choice for the planner is to pick the isowelfare curve that is as far in as possible while still touching the Beveridge curve: the isowelfare curve that is tangent to the Beveridge curve. The tangent isowelfare curve has the highest welfare while respecting the feasibility constraint represented by the Beveridge curve.

Then, the efficient allocation is the point of tangency between the Beveridge curve and isowelfare curve. Since the slope of the Beveridge curve is $v'(u)$ and the slope of the isowelfare curve is $-1/\kappa$, we recover the efficiency condition (9.3) with our graphical approach.

At the point of tangency, the market operates efficiently; at other points on the Beveridge curve, it operates inefficiently. At any point below the efficiency point, for instance, slack is inefficiently high (figure 9.1B). This situation generates a positive slack gap, $u - u^* > 0$. As the market becomes tighter and tighter, at some point it crosses efficiency. If the market keeps tightening, the slack gap becomes negative, that is, $u - u^* < 0$. In this case, the slack rate is below the efficient slack rate.

9.3.3. Comparative statics

From equation (9.3), we see what happens if the matching cost changes. Given that the Beveridge curve $v(u)$ is convex, $v'(u)$ is increasing in u. If the matching cost κ goes up, then $-1/\kappa$ falls, meaning that the Beveridge curve must be flatter at efficiency. As a result, the efficient slack rate u^* increases. If the matching cost goes up, that makes the concern about too many goods devoted to matching more important, which tilts the balance in the tradeoff towards allowing a bit more slack.

We can also perform the comparative statics on the efficiency diagram (figure 9.1C). If the matching cost κ goes up, the isowelfare curve (9.4) becomes flatter. A flatter line means that the new tangency point occurs further out. Thus, the new efficient slack rate is now higher.

9.4. Sufficient-statistic formula for the efficient v-u ratio

There are two potential issues with the efficiency formula (9.3) that make it hard to implement in practice. First, the formula defines the efficient slack rate or visit rate only implicitly. If the formula holds, then the slack and visit rates are efficient, but their values are not given. Second, both in theory and in practice, the Beveridge curve is convex, so the slope of the Beveridge curve constantly changes with the tightness of the market. Measuring a slope in the data is hard enough: measuring the slope at any point in time to know if the condition is satisfied would be so hard as to be impractical.

To alleviate these issues, we reformulate the efficiency condition so that it involves model variables explicitly and so that it only involves stable statistics (unlike the Beveridge slope). We are looking for a formula in which we can plug market statistics that give us the efficient level of a market variable directly. Computing such formula is actually simple.

9.4.1. V-u ratio

Our first step in improving the efficiency formula is to introduce the v-u ratio, v/u. This is the ratio of the visit rate to the slack rate, which is also the ratio of the number of visits to the number of unsold goods. Furthermore, in the dynamic model of chapter 8, the v-u ratio coincides with the market tightness, $v/u = \theta$.

In general the market tightness is the ratio of the number of visits to the number of goods for sale in the matching function. In static models the number of goods for sale is the market capacity, while in dynamic models the number of goods for sale is the number of goods currently unsold, which is why the v-u ratio corresponds to the market tightness only in dynamic models. In any case, the rest of the section is about the v-u ratio.

9.4.2. Beveridge elasticity

Our second step is to introduce the Beveridge elasticity, which we denote β. The Beveridge elasticity is defined by

$$\beta = -\frac{u}{v} \cdot v'(u).$$

Hence the Beveridge elasticity is the elasticity of the Beveridge curve, normalized to be positive by multiplying it by -1. (Recall that the Beveridge curve is downward sloping so $v'(u) < 0$.)

9.4.3. Sufficient-statistic formula

To obtain a sufficient-statistic formula, we rework the efficiency condition (9.3). We start by multiplying both sides of (9.3) by $-(v/u)(u/v) = -1$. We obtain:

$$\frac{v}{u} \cdot \left[-\frac{u}{v} \cdot v'(u)\right] = \frac{1}{\kappa}.$$

The left-hand side is just the v-u ratio times the Beveridge elasticity, β. Dividing both sides by β, we therefore get:

$$\left(\frac{v}{u}\right)^{*} = \frac{1}{\beta\kappa}. \tag{9.5}$$

Equation (9.5) is our sufficient-statistic formula for market efficiency. It says that the efficient v-u ratio is determined by two sufficient statistics. The first sufficient statistic is the matching cost, κ. The key comparative static is that if the matching cost increases, then $(v/u)^{*}$ falls and u^{*} increases. If you have a higher matching cost, it is much more costly to visit shops, so the planner prefers to leave more goods unsold so fewer goods are devoted to matching and in the end more goods are consumed.

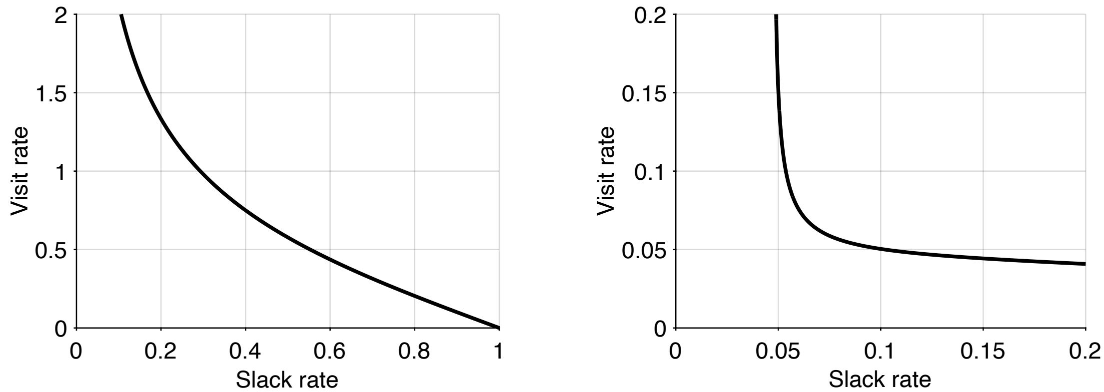
The second sufficient statistic is the Beveridge elasticity, β. This statistic matters because the market planner is trying to solve the tradeoff between visits and slack. When the elasticity of the Beveridge curve is high, it is costly to cut slack a bit, in the sense that a small reduction in slack requires many visits. Conversely, by increasing just a bit the slack rate, you can reduce the number of visits considerably. In other words, slack is more favorable when the Beveridge curve is steeper: a higher Beveridge elasticity leads to a lower $(v/u)^{*}$, so a slacker market.

This formula for market efficiency is quite simple given that it only involves two statistics. If a government wants to know the efficient v-u ratio in any market, they just need to know the Beveridge elasticity and matching cost in that market to determine the efficient market allocation.

The efficient v-u ratio is depicted in figure 9.1D, together with a v-u ratio that is inefficient. In the diagram, the v-u ratio is represented by the slope of any ray through the origin. The efficient slack and visit rates are then given by the intersection of the Beveridge curve with the ray whose slope is given by $(v/u)^{*}$.

9.5. Applying the formula to the static market

Formula (9.5) expresses the efficient v-u ratio as a function of two sufficient statistics. It is natural to ask how the efficiency condition would look in specific models. The sufficient statistics do not necessarily correspond to parameters from the models, so it is unclear how efficiency might look. In the rest of the chapter we show how the formula applies to all the models presented so far in the book. In this section we apply the formula to the basic static model of chapter 4 .



A. Static model

B. Dynamic model

FIGURE 9.2. Beveridge curve in slackish market models

In the static model, the Beveridge curve is given by (9.6). In the dynamic model, the Beveridge curve is given by (8.5). The matching function is CES, given by (3.12). Parameters are set to $\sigma = 2$ and $\lambda = 5\%$.

9.5.1. Verifying the sufficient-statistic assumptions

We need to check that the static model satisfies all the assumptions made in the sufficient-statistic analysis.

First, welfare is solely determined by the consumption of buyers. Assuming that all buyers are the same to avoid distributional issues, welfare is solely determined by aggregate consumption. This in turn is just the number of goods for sale k minus unsold goods uk minus goods used for matching $\kappa v k$. Hence, welfare is proportional to $1 - u - \kappa v$, just as in the sufficient-statistic analysis.

Second, the model features a Beveridge curve. The slack rate is given by $u = 1 - f(\theta)$. But in the static model the market tightness θ is just equal to the visit rate v, so the equation relates visit rate and slack rate:

$$u = 1 - f(v). \tag{9.6}$$

Given that the selling probability f is strictly increasing and concave, we learn that the slack rate u is a strictly decreasing and convex function of the visit rate v. Equivalently, the visit rate v is a strictly decreasing and convex function of the slack rate u (result A.18). Thus, the model admits a Beveridge curve—illustrated in figure 9.2A.

9.5.2. Applying the formula

All the sufficient-statistic conditions are satisfied, which tells us that formula (9.5) holds. We now transform the sufficient-statistic formula into a structural formula, which links the efficient market tightness θ^* to parameters and functions of the static model.

The main step is to compute the Beveridge elasticity in the static model. Equation (9.6) implies that the elasticity of u with respect to v is

$$\epsilon_v^u = -\frac{f(\theta)}{1-f(\theta)} \cdot [1-\eta(\theta)],$$

where $\eta(\theta)$ is the matching elasticity, so $1-\eta(\theta)$ is the elasticity of $f(\theta)$ with respect to θ. The Beveridge elasticity is defined by $\beta = -\epsilon_u^v = -1/\epsilon_v^u$. Therefore, the Beveridge elasticity relates to the market tightness as follows:

$$\beta(\theta) = \frac{1}{1-\eta(\theta)} \cdot \frac{u(\theta)}{1-u(\theta)}. \tag{9.7}$$

Given the matching elasticity $\eta(\theta)$ and slack rate $u(\theta)$ are determined by the matching function, the Beveridge elasticity is solely determined by the matching function.

We then rapidly apply formula (9.5) to the static model. Since the visit rate equals market tightness in that model, the formula imposes

$$\theta^* \beta(\theta^*)\kappa = u(\theta^*),$$

where the Beveridge elasticity $\beta(\theta)$ is given by (9.7) and the slack rate $u(\theta)$ by (4.4). As both $\beta(\theta)$ and $u(\theta)$ are solely determined by the matching function in this model, we have obtained a structural efficiency condition that links the efficient tightness θ^* to the matching function and the matching cost κ.

9.5.3. Hosios condition

By looking closely, we see an interesting property of formula (9.5.2): it implies the famous Hosios (1990) condition. The Hosios condition says that in a market organized around a matching function, and with bargained prices, efficiency is achieved when the seller's bargaining power equals the matching elasticity:

$$\chi = \eta.$$

It turns out that this is an implication of formula (9.5.2).

In the static model under bargaining, we saw with equation (6.7) that the matching wedge satisfies $\tau(\theta) = (1-\chi)/\chi$.

Formula (9.5.2) can be rewritten in a comparable way. Using the expression for the Beveridge elasticity, given by (9.7), we rewrite the efficiency condition (9.5.2) as

$$\kappa\theta^* = \left[1-\eta(\theta^*)\right]\left[1-u(\theta^*)\right].$$

Then, using the expressions of the matching wedge and slack rate, given by (4.8) and (4.4), we know that

$$\frac{\tau(\theta)}{1+\tau(\theta)} \cdot [1 - u(\theta)] = \kappa\theta.$$

Combining these last two equations, we infer that the efficient tightness given by (9.5.2) also satisfies

$$\tau(\theta^*) = \frac{1-\eta(\theta^*)}{\eta(\theta^*)}. \tag{9.8}$$

Hence, for the market to operate efficiently under bargaining, it must be that $\tau(\theta) = (1-\chi)/\chi = [1-\eta(\theta^*)]/\eta(\theta^*)$, which requires that $\chi = \eta(\theta^*)$—which is just what the Hosios condition requires.

In section 6.3, we saw that the price obtained through bargaining equals a fraction χ of the marginal utility of consumption. From this we infer that the static slackish market model operates efficiently if the market price equals a share $\eta(\theta)$ of the marginal utility of consumption. Equivalently, the static market operates efficiently if the market price allocates to the seller a fraction $\eta(\theta)$ of the marginal utility of consumption enjoyed by the buyer.

9.6. Applying the formula to the market with endogenous capacity

The sufficient-statistic analysis did not mention endogenous capacity, so it is natural to worry that the formula might not apply with endogenous market participation. In this section, we show that this worry is not warranted: by an envelope argument, the sufficient-statistic formula continues to apply even when market participation is endogenous, as in the model of chapter 7.

9.6.1. Validity of the formula with endogenous capacity

Is formula (9.5) still valid when sellers enter endogenously? We now turn to the welfare function in this generalized framework. Note first that we do not need to take the utility from money into account since the stock of money is fixed and money just changes hands between buyers and sellers, who value it similarly. So we only need to take into account utility from consumption and disutility from participation.

Market welfare is the sum of individual utilities for all buyers and sellers. All buyers are the same by assumption (to avoid distributional issues) so their utility is just the utility from aggregate consumption: $\mathcal{B}(c(u)) = ac(u)^{1-\alpha}$. Aggregate consumption c is a function of the unemployment rate u given by

$$c(u) = [1 - u - \kappa v(u)]\, kh(u),$$

where $\nu(u)$ is the Beveridge curve. Aggregate consumption is obtained by subtracting matching costs $\kappa\nu kh(u)$ from output $(1-u)kh(u)$.

All sellers have different utility from nonparticipation, so the aggregate utility is obtained by integrating the individual utilities of all potential sellers who remain outside of the market: $\mathcal{S}(h(u)) = \int_{h(u)}^{1} \xi i^{\phi}\, di$, where the participation rate $h(u)$ comes from (7.2).

Overall, then, market welfare is a function of the slack rate: $\mathcal{W}(u) = \mathcal{B}(c(u)) + \mathcal{S}(h(u))$. The planner cannot do anything about matching on the market, so it takes the Beveridge curve as given. It cannot force people to enter the market or leave it, so it must respect people's voluntary entry or exit of the market, described by (7.2). What the planner can do is influence prices, for instance through taxes or subsidies or price caps, or purchase goods directly on the market. Through these actions it can determine the slack rate u.

The planner chooses the slack rate u to maximize market welfare $\mathcal{W}(u)$. The first-order condition for the maximization problem is $0 = \mathcal{W}'(u)$, which becomes:

$$0 = \mathcal{B}'(c) \cdot \left\{\left[-1 - \kappa\nu'(u)\right] kh(u) + \left[1 - u - \kappa\nu(u)\right] kh'(u)\right\} + \mathcal{S}'(h)h'(u).$$

Rearranging terms, we get

$$0 = \mathcal{B}'(c)kh(u) \cdot \left[-1 - \kappa\nu'(u)\right] + \frac{h'(u)}{h(u)} \cdot \left[\mathcal{B}'(c)c(u) + \mathcal{S}'(h)h(u)\right].$$

We now need to rework the second term in the first-order condition. Equation (4.13) tells us that

$$\mathcal{B}'(c) = a(1-\alpha)c^{-\alpha} = p\left[1 + \tau(\theta)\right] \quad \text{so} \quad \mathcal{B}'(c)c = p\left[1 + \tau(\theta)\right]c = py.$$

By Leibniz's integral rule, we see that $\mathcal{S}'(h) = -\xi h^{\phi}$ (result A.6). Then, using equation (7.2), we learn that

$$\mathcal{S}'(h) = -p(1-u)k \quad \text{so} \quad \mathcal{S}'(h)h = -p(1-u)kh = -py.$$

Hence, we have just established that

$$\mathcal{B}'(c)c(u) + \mathcal{S}'(h)h(u) = 0.$$

This result is key, and it hinges on an envelope-theorem logic. Because market participants behave optimally, they enter until the marginal cost from entering (foregone leisure) is just equal to the marginal benefit from entering, which is determined by the price of goods. But the price of goods also determines the marginal utility of consumption given that buyers behave optimally. All in all, when the planner changes the slack rate, it does affect participation, but this change in participation has no effect on welfare because participation has been chosen optimally by sellers.

Critically, we have just learned that the second term in the first-order equation is just 0. Therefore, the first-order condition reduces to $v'(u) = -1/\kappa$, just as in the baseline case with fixed participation. We therefore recover the result that the efficient v-u ratio is given by $(v/u)^* = 1/(\beta\kappa)$.

In sum, endogenizing market participation does not change the welfare analysis. This result stems from an envelope-theorem logic that is classic in public economics. Even if the market planner alters market participation by changing the slack rate, welfare is unaffected because the sellers who move in or out of the market are indifferent between participating or not. Indeed, if sellers strictly preferred participation to nonparticipation, they would move into the market. Conversely, if they strictly preferred nonparticipation, they would move out of the market.

9.6.2. Applying the formula

In the static market with endogenous capacity, the welfare function is different because it also includes the welfare of sellers, who decide whether to enter the market or not. What makes this model all the more intriguing is that the planner takes into account the fact that it can affect the number of sellers in the market by picking the tightness of the market, as equation (7.2) shows. Despite these differences, we have seen that the v-u ratio remains given by the sufficient-statistic formula (9.5) in this model.

How does the sufficient-statistic formula (9.5) look in the model with endogenous capacity? Just like in the model with exogenous capacity, the Beveridge curve is obtained from the definition of the slack rate—$u = 1 - f(v)$—so the sufficient-statistic formula translates to the structural formula (9.5.2).

9.7. Applying the formula to the dynamic market

Finally, we apply the sufficient-statistic formula to the dynamic model of chapter 8. Although we did not mention any dynamics in the welfare analysis, the formula remains valid here because the dynamic model features a Beveridge curve too.

9.7.1. Verifying the sufficient-statistic assumptions

We now check that our dynamic model satisfies all the assumptions made in this chapter's welfare analysis. First, welfare is solely determined by the consumption of buyers, just as in the static case.

Second, the model features a Beveridge curve, given implicitly by (8.5). In chapter 8 we established that the Beveridge curve $v(u)$ is downward sloping but we did not check its convexity, which is key for the welfare analysis. We can do that here. We need to differentiate (8.5) twice. The first implicit differentiation gives (8.6). We now implicitly differentiate that equation again with respect to u and obtain

$$0 = \frac{\partial^2 m}{\partial u^2} + 2 \cdot \frac{\partial^2 m}{\partial u \partial v} \cdot v'(u) + \frac{\partial^2 m}{\partial v^2} \cdot \left[v'(u)\right]^2 + \frac{\partial m}{\partial v} \cdot v''(u).$$

Reshuffling the terms to isolate the second derivative of the Beveridge curve, we obtain:

$$v''(u) = \frac{-\frac{\partial^2 m}{\partial u^2} - 2 \cdot \frac{\partial^2 m}{\partial u \partial v} \cdot v'(u) - \frac{\partial^2 m}{\partial v^2} \cdot \left[v'(u)\right]^2}{\frac{\partial m}{\partial v}}.$$

However, the cross-partial derivative of the matching function is linked to the own-partial derivatives (result D.2):

$$\frac{\partial^2 m}{\partial b \partial s} = \sqrt{\left(-\frac{\partial^2 m}{\partial b^2}\right) \cdot \left(-\frac{\partial^2 m}{\partial s^2}\right)}.$$

Hence, we simplify the expression for the curvature of the Beveridge curve:

$$v''(u) = \frac{\left[\sqrt{-\frac{\partial^2 m}{\partial u^2}} - v'(u) \cdot \sqrt{-\frac{\partial^2 m}{\partial v^2}}\right]^2}{\frac{\partial m}{\partial v}}.$$

Using this expression, we proceed to show that $v''(u) > 0$. Since the matching function is strictly increasing in both arguments, the denominator is strictly positive: $\partial m/\partial v > 0$. Furthermore, we have assumed that the matching function is strictly concave in its second argument, $\partial^2 m/\partial v^2 < 0$, and we have shown in result D.1 that the matching function is strictly concave in its first argument, $\partial^2 m/\partial u^2 < 0$. We have also shown with (8.7) that the Beveridge curve is strictly decreasing: $v'(u) < 0$. This means that all the elements in the square are strictly positive: $\sqrt{-\partial^2 m/\partial v^2} > 0$, $-v'(u) > 0$, and $\sqrt{-\partial^2 m/\partial u^2} > 0$. Hence the square is strictly positive, and so is the whole expression: $v''(u) > 0$. Thus, the Beveridge curve in the dynamic model is strictly convex.

As an illustration, figure 9.2B displays the Beveridge curve obtained in the dynamic model under our usual calibration. Just like the market supply, the Beveridge curve is much more convex in the dynamic model than in the static model.

9.7.2. Applying the formula

All the sufficient-statistic conditions are satisfied, so formula (9.5) holds. We now transform the sufficient-statistic formula into a structural formula, which links the efficient market tightness θ^* to parameters and functions of the dynamic model.

The key step is to compute the Beveridge elasticity in the dynamic model. From the balanced-flow equation, $m(u, v) = \lambda(1 - u)$, which implicitly defines the Beveridge curve, we computed the slope of the Beveridge curve, and found that it was given by equation (8.7). We now rework this equation to compute the Beveridge elasticity. We first multiply both sides of equation (8.7) by $-u/v$:

$$-\frac{u}{v}v'(u) = \frac{\lambda \cdot \frac{u}{m(u,v)} + \frac{u}{m(u,v)} \cdot \frac{\partial m}{\partial u}}{\frac{v}{m(u,v)} \cdot \frac{\partial m}{\partial v}}.$$

We divided numerator and denominator of the right-hand side fraction by the number of matches $m(u, v)$ to make the elasticities of the matching function appear.

We now simplify this expression. The left-hand side is just the Beveridge elasticity β, by definition. The second term in the right-hand numerator is the matching elasticity, η. From the balanced-flow equation, $m(u, v)/\lambda = (1 - u)$, so the first term in the right-hand numerator boils down to $u/(1 - u)$. The right-hand denominator is the elasticity of the matching function with respect to the number of visits, which is $1 - \eta$, as we established in (3.3). All in all, we obtain the following expression for the Beveridge elasticity:

$$\beta(\theta) = \frac{1}{1 - \eta(\theta)} \left[\eta(\theta) + \frac{u(\theta)}{1 - u(\theta)}\right]. \tag{9.9}$$

Just as in the static case, the Beveridge elasticity is a function of tightness that is closely related to the matching elasticity, $\eta(\theta)$. In this dynamic case, the Beveridge elasticity is determined by the matching function and the separation rate λ.

We see that in the dynamic model, the Beveridge elasticity is likely to be stable even if slack varies over time. Indeed, the slack rate is generally small, so in (9.9) the term $u/(1 - u)$ is much smaller than η, which keeps the elasticity stable. This is different from what happens in the static model, where the Beveridge elasticity varies strongly with slack (equation (9.7)). In the dynamic model, formula (9.5) is therefore more usable than the original formula (9.3) as the statistics that it involves (β, κ) are stable over time and do not need to be constantly re-estimated.

In the dynamic model, the v-u ratio is the market tightness: $\theta = v/u$. Thus formula (9.5) simply gives:

$$\theta^* \beta(\theta^*)\kappa = 1, \tag{9.10}$$

where the Beveridge elasticity is given by (9.9). This is the structural condition for efficiency in the dynamic model. It links the efficient tightness θ^* to the matching function, the separation rate λ, and matching cost κ.

9.7.3. Hosios condition

Here too, just like in the static case, formula (9.10) implies the Hosios (1990) condition.

In the dynamic model under bargaining, we saw with equation (8.19) that the market tightness satisfies $\tau(\theta)/u(\theta) = (1-\chi)/\chi$, where χ is the seller's bargaining power.

Formula (9.10) can be rewritten in a comparable way. Given the expression of the Beveridge elasticity in (9.9), the formula gives:

$$\kappa\theta^* \left[(1-u(\theta^*))\eta + u(\theta^*)\right] = \left[1-\eta(\theta^*)\right]\left[1-u(\theta^*)\right].$$

From the expressions for the matching wedge and slack rate, given by (8.13) and (8.8), we know that

$$\frac{\tau(\theta)}{1+\tau(\theta)} \cdot \frac{1-u(\theta)}{u(\theta)} = \kappa\theta.$$

Hence, the efficiency condition above becomes

$$\frac{\tau(\theta^*)}{u(\theta^*)}\left[(1-\eta(\theta^*))u(\theta^*) + \eta(\theta^*)\right] = \left[1-\eta(\theta^*)\right]\left[1+\tau(\theta^*)\right].$$

Subtracting $[1-\eta(\theta^*)]\tau(\theta^*)$ from both sides, and dividing both sides by $\eta(\theta^*)$, we finally find that the efficient tightness θ^* is implicitly defined by

$$\frac{\tau(\theta^*)}{u(\theta^*)} = \frac{1-\eta(\theta^*)}{\eta(\theta^*)}. \tag{9.11}$$

This alternative specification of the efficient tightness in a dynamic slackish model is more cumbersome than (9.10), and it has a less direct link to the Beveridge curve and a less direct interpretation, but it is useful here and in other situations in the next chapters.

Just as in the static model, for the market to operate efficiently under bargaining, it must be that $\tau/u = (1-\chi)/\chi = [1-\eta(\theta^*)]/\eta(\theta^*)$, which again requires that $\chi = \eta(\theta^*)$—which is what the Hosios condition requires.

In (8.18), we saw that in the dynamic model the bargained price is set at a markdown $1+[(1-\chi)/\chi]u$ below the marginal utility of consumption. From this we infer that the dynamic slackish market model operates efficiently if the market price is set at a markdown

$$1 + \frac{1-\eta(\theta^*)}{\eta(\theta^*)} \cdot u(\theta^*)$$

below the marginal utility of consumption.

9.8. Modulating the social cost of slack

So far we have assumed that unsold goods are pure waste: they have no social value or cost besides their wastefulness.

This is a valid assumption for services that are not rendered or goods that are perishable. But if, for instance, unsold goods can be donated, these goods would have some social value. In the United States in 2023, retailers donate 12% of donatable unsold food to food banks (ReFED 2025). On the other hand, if unsold goods are sent to landfills where they produce noxious gases while decomposing, these goods impose a social cost. In the United States in 2023, about 25 million tons of surplus food end up in landfill, emitting a sizable amount of methane (ReFED 2025).

Similarly, the assumption seems appropriate for employed workers who are idle. But if an unemployed worker engages in home production, their time might have social value. If on the other hand the time spent unemployed is painful psychologically, their time generates a social cost.

To allow for the possibility that unsold goods have some social value or cost beyond the waste they impose, we introduce a statistic $\zeta < 1$, which captures the social value of unsold goods compared to sold goods. The case $\zeta = 0$, which we have considered so far, is when unsold goods are pure waste. The case $\zeta < 0$ is when unsold goods generate a cost on top of being wasteful. And the case $\zeta \in (0, 1)$ is when unsold goods have some social value that alleviates the wastefulness of unsold goods.

In this extension, the market welfare has an additional element. Since each of the uk unsold goods have a social value ζ relative to sold goods, an amount ζuk must be added to market welfare (9.1). In total, market welfare becomes

$$\mathcal{W}(u) = [1 - \kappa v(u) - (1 - \zeta)u]\, k.$$

Given that the welfare cost of a good failing to sell is reduced by $1 - \zeta$, the first-order condition (9.3) generalizes to

$$v'(u) = -\frac{1 - \zeta}{\kappa}. \tag{9.12}$$

If slack has some social value ($\zeta > 0$), the isowelfare curve is flatter in the efficiency diagram, so the tangency point occurs further out, at a higher efficient slack rate (just like in figure 9.1C). Along the Beveridge curve, the efficient slack rate u^* increases. Intuitively, when the cost of having unsold goods is a bit less, the efficient allocation of goods tolerates a bit more slack. Conversely, if slack imposes a social cost beyond its wastefulness ($\zeta < 0$), the isowelfare curve is steeper, so the efficient slack rate is lower.

Finally, the sufficient-statistic formula (9.5) generalizes to

$$\left(\frac{v}{u}\right)^{*} = \frac{1-\zeta}{\beta\kappa}. \tag{9.13}$$

In this generalized formula, a third sufficient statistic appears: the social value of unsold goods, ζ. The efficient v-u ratio responds to changes in the social value of unsold goods just like to changes in the matching cost κ: when the matching cost increases or when the social value increases, the efficient v-u ratio decreases, so the efficient slack rate increases.

9.9. Are markets efficient?

The last question we ask in this chapter is: Can we expect markets to operate efficiently? One of the key theorems of Walrasian theory is that markets operate efficiently. But slackish markets are different: there is absolutely no guarantee that a slackish market operates efficiently.

From a theoretical perspective, the key reason why Walrasian markets are efficient while slackish markets are not is that prices are set differently. In a slackish market, there exists a price norm that guarantees efficiency, but this price norm might not be adopted in the real world. Under all other price norms, efficiency is not guaranteed. In effect, there is no guarantee that the price norm that prevails in the real world leads to efficiency.

In any slackish market, the market tightness is given by the supply-equals-demand condition. When we solve the model, we use the price given by the price norm, plug it into the market demand, and determine the tightness that equalizes supply and demand.

Here we use the condition in the other way: we set tightness to its efficient level θ^*. Then we use the supply-equals-demand condition to find the efficient price p^*, defined by

$$y^d(\theta^*, p^*) = y^s(\theta^*).$$

We know that the market demand is strictly decreasing in the price, so that equation defines a unique efficient price p^*. If the price norm happens to set the price to p^*, the market will operate efficiently. But there is no guarantee that this occurs in reality, as the formation of norms is a complex process that is not geared toward efficiency.

Furthermore, by using the functional forms for market supply and demand, we can obtain an expression for the efficient price. For concreteness, let's find the expression for the efficient price in the basic model of chapter 4. Supply and demand curves are given by (4.15) and (4.5). Hence, the supply-equals-demand condition requires:

$$\left[\frac{(1-\alpha)a}{p^*}\right]^{1/\alpha}\left[1+\tau(\theta^*)\right]^{1-1/\alpha} = \left[1-u(\theta^*)\right]k.$$

From (9.8), we know that $1+\tau(\theta^*) = 1/\eta(\theta^*)$. By reshuffling terms, we obtain the following expression for the efficient price:

$$p^* = (1-\alpha)\left[1-u(\theta^*)\right]^{-\alpha}\eta(\theta^*)^{1-\alpha}\cdot ak^{-\alpha},$$

where the efficient tightness θ^* is equivalently defined by (9.5.2) or (9.8).

An interesting aspect of the efficient price is that it is a flexible case of the price norm (6.1) introduced in chapter 6. Indeed, we obtain the price p^* from the price norm by setting the price-rigidity parameter to $\gamma = 0$ and the price-level parameter to

$$\rho = (1-\alpha)\left[1-u(\theta^*)\right]^{-\alpha}\eta(\theta^*)^{1-\alpha},$$

where the efficient tightness θ^* is independent of the parameters a and k, as it is defined by (9.5.2).

We argued earlier in the book that bargaining was a possible mechanism pushing prices towards flexibility. We see here that efficiency is another one. Any social or economic mechanism pushing markets towards efficiency makes prices more flexible. In chapter 14 we will introduce such a market mechanism: Moen (1997)'s directed search.

Because market inefficiency is generic, government interventions might be needed to bring markets and the economy closer to efficiency—as we discuss in part V. Here there is no guarantee that Smith's invisible hand maintains markets at efficiency. The reason is that with slack, the hand is unable to influence prices and keep them at their efficient level. Hence, there is generally too much or too little slack on any market.

9.10. Summary

In this chapter we define efficiency as the level of slack that maximizes market welfare. Rather than specifying a complete structural model, we adopt a sufficient-statistic approach: We analyze efficiency in any market that admits a Beveridge curve, and express the results in terms of observable statistics.

We find that the efficient slack rate equates the slope of the Beveridge curve to the inverse of the matching cost. From this, we obtained a sufficient-statistic formula that links the efficient v-u ratio to the matching cost κ and Beveridge elasticity β: $(v/u)^* = 1/(\beta\kappa)$.

Finally, we argue that slackish markets are generically inefficient, since no market mechanism ensures that prices adjust to their efficient level. In practice, price norms might generate either excessive or insufficient slack, justifying policy interventions that move markets toward their efficient operating points.

PART III.

Application to the labor market

CHAPTER 10.

Slackish labor market model

We now apply the slackish market model to the labor market. We devote an entire part to the labor market because from a social perspective, unemployment is the most costly form of slack, so precisely understanding it is important. We will see that the slackish labor market model offers a unified theory of unemployment—blending frictional unemployment and job rationing. The slackish model is therefore appropriate to describe both good times and bad times and to capture the influence on unemployment of both labor demand and labor supply factors and policies.

In this chapter, we start by presenting the slackish labor market model. We describe the origins of unemployment and the sources of its fluctuations. We also use the richness of US labor market data to calibrate the slackish labor market model and compare its quantitative behavior with real-world patterns.

10.1. Slackish model of the labor market

In this section we apply the slackish market model of chapter 8 to the labor market. The mapping between the generic market model and the labor market model is described in table 10.1. Most of the notation remains the same, but some notation is adjusted to respect conventions. We do not repeat the derivations here: we only summarize the main equations and results, and we provide some interpretation.

TABLE 10.1. Application of the slackish market model to the labor market

Generic market			Labor market		
Symbol	Name	Definition	Symbol	Name	Definition
k	Market capacity	–	h	Labor force	–
V	Visits	–	V	Job vacancies	–
U	Unsold goods	–	U	Job seekers	–
v	Visit rate	$v = V/k$	v	Vacancy rate	$v = V/h$
u	Slack rate	$u = U/k$	u	Unemployment rate	$u = U/h$
$m(u, v)$	Matching function	–	$m(u, v)$	Matching function	–
θ	Market tightness	$\theta = v/u$	θ	Labor market tightness	$\theta = v/u$
f	Selling rate	(3.4)	f	Job-finding rate	(3.4)
q	Buying rate	(3.5)	q	Job-filling rate	(3.5)
κ	Matching cost	–	κ	Recruiting cost	–
τ	Matching wedge	(8.13)	τ	Recruiting wedge	(10.4)
y	Output	–	l	Employment	–
y^s	Market supply	(8.9)	l^s	Labor supply	(10.1)
y^d	Market demand	(4.15)	l^d	Labor demand	(10.7)
c	Consumption	–	n	Producers	–
$\mathcal{U}$	Utility function	(4.10)	$\mathcal{Y}$	Production function	(10.5)
a	Demand shifter	–	a	Labor productivity	–
α	Diminishing marginal utility of consumption	–	α	Diminishing marginal productivity of labor	–
λ	Buyer-seller separation rate	–	λ	Job-separation rate	–
p	Market price	–	w	Real wage	–
p^n	Price norm	(6.1)	w^n	Wage norm	(10.10)
ρ	Price norm level	–	ω	Wage norm level	–
γ	Price rigidity	–	γ	Wage rigidity	–

10.1.1. Matching function and labor market tightness

The labor market is composed of a labor force of size $h > 0$ and a mass 1 of firms, each indexed by $j \in [0, 1]$.

The labor force comprises l employed workers and $U = h - l$ unemployed workers.

Each firm j posts V_j job vacancies to hire job seekers. The reason why firms must constantly recruit new workers is that they constantly lose existing workers—who retire, quit, or are dismissed. Formally, each firm faces a job-separation rate $\lambda > 0$. The aggregate number of vacancies is $V = \int_0^1 V_j \, dj$.

From the unemployment level U and vacancy level V, we also define the unemployment rate $u = U/h$ and the vacancy rate $v = V/h$.

The rate at which job vacancies and job seekers match is given by a matching function $m(U, V)$ that satisfies the assumptions laid out in chapter 3.

The labor market tightness is the ratio of the two arguments in the matching function: $\theta = V/U = v/u$.

The labor market tightness determines the rate at which job seekers find jobs, $f(\theta) = m(1, \theta)$, and the rate at which job vacancies are filled, $q(\theta) = m(1/\theta, 1)$.

10.1.2. Equilibrium labor supply

The equilibrium labor supply is the employment level when labor market flows are balanced:

$$l^s(\theta) = [1 - u(\theta)]\, h, \tag{10.1}$$

where the equilibrium unemployment rate is given by

$$u(\theta) = \frac{\lambda}{\lambda + f(\theta)}. \tag{10.2}$$

In the labor market, the number of job seekers who remain unemployed and that of workers who are employed are determined by how quickly job seekers find jobs and how long employed workers hold on to their jobs. With a fixed job-separation rate λ, these numbers are determined by the job-finding rate $f(\theta)$ and therefore labor market tightness θ. From any initial division of the labor force between employed and unemployed workers and a given tightness, the unemployment rate converges to the equilibrium unemployment rate given by (10.2) and the employment level converges to the equilibrium labor supply given by (10.1).

As we saw in chapter 8, the convergence to the equilibrium unemployment rate is rapid: in the US labor market, the convergence process has a half-life of about 1 month. As a result, in practice, the actual and equilibrium unemployment rates are indistinguishable

(Michaillat and Saez 2021a, figure A3). This is why we abstract from the law of motion of the unemployment rate (given by (8.3)) and consider the unemployment rate as a function of tightness (given by (10.2)).

When labor-market flows are balanced, at any point in time, equally many people lose their job and find a job; equally many employment relationships are created and dissolved. Thus, employment and unemployment remain constant over time, so we can work with static supply and demand relationships instead of differential equations, greatly simplifying the analysis.

10.1.3. Equilibrium labor demand

Because it takes time to fill vacancies and each vacancy requires the attention of a recruiter, firms must allocate a share of their workforce to recruiting. Not all workers can be employed at producing goods and services that can be sold to customers. Instead, each firm j employs two types of workers: n_j producers and some recruiters. Because each job vacancy requires $\kappa > 0$ recruiters, the total number of recruiters in firm j is κV_j and the total number of workers is $l_j = n_j + \kappa V_j$. Because of the recruiters, a wedge appears between the number of employees in the firm and the number of producers:

$$l_j = [1 + \tau(\theta)]\, n_j, \tag{10.3}$$

where $\tau(\theta)$ is the recruiting wedge, given by

$$\tau(\theta) = \frac{\kappa\lambda}{q(\theta) - \kappa\lambda}. \tag{10.4}$$

The recruiting wedge corresponds to the ratio between the number of recruiters in the firm, $\tau(\theta) n_j$, and the number of producers, n_j. The wedge is an equilibrium object because it is computed under the assumption that labor market flows are balanced.

Next, each firm j produces goods or services using the concave production function

$$y(n_j) = a \cdot n_j^{1-\alpha}, \tag{10.5}$$

where a governs labor productivity, n_j is the number of producers in the firm, and $\alpha \in (0, 1)$ governs the concavity of the production function, which determines diminishing marginal product of labor. Recruiters are busy recruiting, so only producers participate in the production process. This explains why the production function takes the number of producers n_j and not the number of employees l_j as an argument.

Why might the production function exhibit diminishing instead of constant marginal product of labor? A basic reason is that in the short run, capital and land (and other natural resources) are fixed. For instance, if production is given by a Cobb-Douglas function of labor and capital, but capital is fixed in the short run, then the short-run production function takes the form that we use here.

Each firm chooses the size of their workforce to maximize the flow of real profits:

$$\mathcal{P}_j = \mathcal{Y}(n_j) - wl_j = a \cdot n_j^{1-\alpha} - w\left[1 + \tau(\theta)\right] n_j, \tag{10.6}$$

where $w > 0$ is the real wage paid by the firm to all its workers. The firm's real profits are solely determined by the firm's number of producers, n_j, so the firm chooses that number to maximize profits.

In principle, because the model is dynamic, firms should maximize the discounted sum of flow real profits, subject to the law of motion of employment. But because firms operate in a balanced-flow environment, they can pick the number of producers that they employ at any point in time by posting the appropriate number of vacancies. So firms are not subject to any intertemporal constraints, and maximizing the discounted sum of flow real profits is equivalent to maximizing the flow of real profits at any point in time.

The nice thing about the firm's profit function (10.6) is that it is isomorphic to the buyer's objective function (4.12) that we derived in the generic market model, up to a constant. The constant is the endowment of money balances (B_j), which appears in the buyer's problem but not in the firm's problem. Obviously, the constant has no effect on the first-order condition and demand curve. The market demand (4.15) can therefore be applied to the labor market, giving us the following labor demand:

$$l^d(\theta, w) = \left[\frac{(1-\alpha)a}{w}\right]^{1/\alpha} \cdot \frac{1}{\left[1 + \tau(\theta)\right]^{1/\alpha - 1}}. \tag{10.7}$$

The labor demand gives the firm's profit-maximizing employment level for any labor market tightness θ, real wage w, and productivity a. The labor demand reflects the property that when profits are maximized, the marginal product of labor equals the marginal cost of labor, which includes both real wage and recruiting costs:

$$(1-\alpha)an_j^{-\alpha} = w\left[1 + \tau(\theta)\right]. \tag{10.8}$$

The first-order condition of the firm problem (10.8) allows us to derive the labor share in the model, which is quite helpful. The first-order condition implies that firm j's real wage bill is

$$wl_j = w[1 + \tau(\theta)]n_j = (1-\alpha)an_j^{1-\alpha} = (1-\alpha)\mathcal{Y}(n_j). \tag{10.9}$$

Accordingly, for all firms, the wage bill is a share $1 - \alpha$ of their output, which implies that the economy's labor share—the aggregate wage bill as a share of aggregate output—is just

$1 - \alpha$. And since firms' profits are their output minus their wage bill, firms' profits amount to a share α of their output, and the economy's profit share is α. These profits represent broadly operating profits, calculated before incorporating payments to capital and land.

10.1.4. Wage norm

Just as in chapter 6, we assume a rigid wage norm. This means that the real wage depends on labor productivity a and on the size of the labor force h, but the response to shocks to a and h is muted.

Accordingly, we assume that the real wage norm depends on labor productivity and the labor force, just as in the price norm (6.1):

$$w^n = \omega \cdot \left(a \cdot h^{-\alpha}\right)^{1-\gamma}. \tag{10.10}$$

The parameter $\gamma \in (0,1]$ determines the amount of wage rigidity. At $\gamma = 1$, the real wage is fixed at $w^n = \omega$. At $\gamma = 0$, the real wage would be flexible (we do not consider this case here because we focus on somewhat-rigid wages). Finally, the parameter $\omega > 0$ determines the level of the wage norm.

Once we incorporate the wage norm (10.10), the labor demand takes the form of the market demand (6.2):

$$l^d(\theta) = \left[\frac{(1-\alpha)a^{\gamma}}{\omega}\right]^{1/\alpha} \cdot \frac{h^{1-\gamma}}{[1+\tau(\theta)]^{1/\alpha-1}}. \tag{10.11}$$

10.1.5. Solution of the model

The labor market tightness that solves the model is given by a supply-equals-demand condition:

$$l^d(\theta) = l^s(\theta) \tag{10.12}$$

where the labor demand incorporates the wage norm and is given by (10.11), and the labor supply is given by (10.1).

In the slackish labor market model, the wage is given by the wage norm w^n, so it is tightness that equalizes labor demand and labor supply. As we showed in chapters 4 and 6, the supply-equals-demand equation admits a unique solution θ, and all the other variables in the model can be computed from tightness.

The solution of the model is represented graphically in figure 10.1 in a tightness-employment plane. The solution is at the intersection of the labor demand and supply curves. The intersection directly gives labor market tightness but also employment and

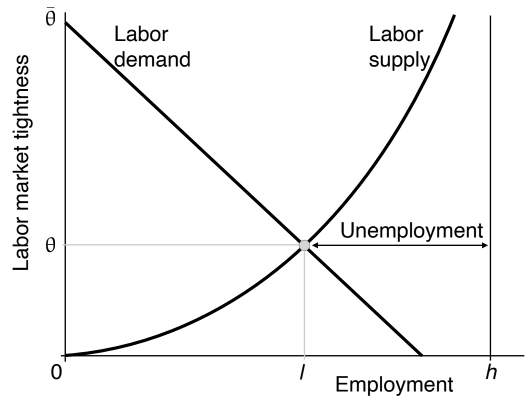


FIGURE 10.1. Solution of the slackish labor market model

The labor supply curve is given by (10.1). The labor demand curve is given by (10.11). The solution of the model is at the intersection of the labor demand and supply curves.

unemployment in the model.

10.2. Calibration of the model

In this section, we calibrate the model to the US labor market. As much as possible, we use evidence from the 1948–2019 period. During this period, the labor market operated in a consistent fashion. We omit earlier and later evidence because the monumental disruptions imposed by the Great Depression, World War 2, and the coronavirus pandemic are not representative of the typical behavior of the US labor market.

The calibrated values of the model parameters are summarized in table 10.2. Throughout the rest of the chapter, we will compare the quantitative behavior of the calibrated model with patterns observed in the US labor market. For completeness, the table also summarizes the calibrated values of the main model variables.

The calibration proceeds in two steps. We calibrate as many parameters as possible by directly measuring their values in US data. Then, parameters for which we do not have direct measurements are calibrated to match select, important facts about the US labor market and economy—such as the average unemployment rate and elasticity of the Beveridge curve.

10.2.1. Baseline labor productivity and labor force

In the labor market model, labor demand shocks are represented by variations in labor productivity, a, and labor supply shocks are represented by variations in labor force, h. Nevertheless, for the calibration, we need to set the two parameters to baseline values. We will then explore how the model responds to changes in these values.

TABLE 10.2. Calibrated values of the parameters and variables of the slackish labor market model

Value	Description	Source	Reference
		A. Normalized parameters	
$a = 1$	Labor productivity	Normalization	Baseline
$h = 1$	Labor force	Normalization	Baseline
		B. Parameters calibrated from direct measurement	
$\kappa = 1$	Recruiting cost	Borgschulte and Martorell (2018)	Section 2.2.3
$\gamma = 0.5$	Wage rigidity	Haefke, Sonntag, and van Rens (2013)	Section 5.4.3
$\lambda = 3.6\%$	Monthly job-separation rate	BLS (2025u)	Section 2.6.1
		C. Parameters calibrated to match target	
$\alpha = 0.35$	Diminishing marginal returns to labor	Labor share = 65%	Autor et al. (2020)
$\sigma = 0.47$	Matching elasticity	Beveridge elasticity = 1	Figure 2.8
$\mu = 0.75$	Monthly matching efficacy	Unemployment rate = 5.7%	Equation (10.13)
$\omega = 0.65$	Wage level	Labor market tightness = 0.65	Equation (10.14)
		D. Variables calibrated from historical evidence	
$\theta = 0.65$	Labor market tightness	Average over 1948–2019	Figure 3.4
$u = 5.7\%$	Unemployment rate	Average over 1948–2019	Figure 2.1
		E. Other variables	
$v = 3.7\%$	Vacancy rate	$v = \theta \cdot u$	
$f(\theta) = 0.60$	Monthly job-finding rate	$f(\theta) = \mu\theta^{1-\sigma}$	
$q(\theta) = 0.92$	Monthly job-filling rate	$q(\theta) = \mu\theta^{-\sigma}$	
$\tau(\theta) = 4.1\%$	Recruiting wedge	$\tau(\theta) = \kappa\lambda/[q(\theta) - \kappa\lambda]$	
$l = 0.943$	Number of employees	$l = (1-u)h$	
$n = 0.906$	Number of producers	$n = l/[1+\tau(\theta)]$	
$y = 0.938$	Output	$y = an^{1-\alpha}$	
$w = 0.65$	Real wage	$w = \omega a^{1-\gamma}$	

As usual, we normalize the baseline productivity level to $a = 1$. We also normalize the baseline labor force to $h = 1$, which conveniently equalizes the employment, unemployment, and vacancy rates and levels at the baseline.

10.2.2. Recruiting cost

We argued in section 2.2.3 that it takes about 1 full-time worker to service a job vacancy. Accordingly, we set the recruiting cost to $\kappa = 1$.

10.2.3. Job-separation rate

We found in section 2.6.1 that between 2001 and 2019, the monthly job-separation rate averages 3.6%. Hence we set $\lambda = 3.6\%$.

10.2.4. Concavity of the production function

Accordingly, we set the concavity parameter in the production function (10.5) to $\alpha = 0.35$, which produces a labor share of $1 - \alpha = 65\%$. This value corresponds to the typical, historical US labor share, before it started falling in the 1990s (Autor et al. 2020, figure 1).

10.2.5. Matching function

We showed in section 3.8 that a Cobb-Douglas matching function describes the US labor market well, so we use this functional form: $m(U, V) = \mu U^{\sigma} V^{1-\sigma}$.

Next, we set the parameter σ in the matching function to obtain a Beveridge elasticity of 1, as observed in the US (section 2.5.2). In a dynamic slackish model like here, and with a Cobb-Douglas matching function, the Beveridge elasticity β is directly linked to σ, as (9.9) shows:

$$\beta = \frac{1}{1-\sigma}\left(\sigma + \frac{u}{1-u}\right) \quad \text{so} \quad \sigma = \frac{1}{1+\beta}\left(\beta - \frac{u}{1-u}\right).$$

Compared to (9.9), we replace the matching elasticity η by the parameter σ since both are the same when the matching function is Cobb-Douglas.

We target a Beveridge elasticity of $\beta = 1$. We set the unemployment rate to its average value between 1948 and 2019: $u = 5.7\%$ (section 2.3.3). Plugging these numbers in the expression above, we obtain a matching elasticity of $\sigma = 0.47$.[1]

[1] We could also calibrate σ from the elasticity of the job-finding rate with respect to labor market tightness (chapter 3). This alternative approach yields $\sigma = 0.6$. We prefer to target the Beveridge elasticity because it is a key sufficient statistic for the efficient unemployment rate, so targeting it keeps the model's welfare properties realistic. The two strategies do not yield the same value of σ because the model cannot simultaneously match both elasticities (the model does not perfectly describe the data). Relatedly, the Cobb-Douglas matching function produces an isoelastic job-finding rate, just as in US data (figure 3.5B), but not an exactly isoelastic Beveridge curve, as US data also suggest (figure 2.8). So the model is not able to produce both an isoelastic job-finding rate and an isoelastic Beveridge curve. With a penalized Cobb-Douglas matching function, the

Finally, we calibrate the parameter μ in the matching function so the unemployment rate takes its average value when labor market tightness is at its average value. Unemployment rate and labor market tightness are related by (10.2). Using the expression for the matching function, this gives $u = \lambda/(\lambda + \mu\theta^{1-\sigma})$ so

$$\mu = \lambda\theta^{\sigma-1} \cdot \frac{1-u}{u}. \tag{10.13}$$

Between 1948 and 2019, the unemployment rate averages 5.7% (section 2.3.3), while labor market tightness averages 0.65 (section 3.8). Setting unemployment and tightness to their average levels, and setting $\lambda = 3.6\%$ and $\sigma = 0.47$, we obtain $\mu = 0.75$.

10.2.6. Wage norm

We first calibrate the wage-rigidity parameter in the wage norm (10.10). We showed in section 5.4.3 that the elasticity of real wages with respect to labor productivity is about 0.5. Thus, we set the wage-rigidity parameter to $\gamma = 0.5$, which ensures that the elasticity of real wages with respect to productivity is $1 - \gamma = 1 - 0.5 = 0.5$.

Next, we calibrate the wage-level parameter ω in the wage norm so that the tightness takes its average value at the baseline. Through the labor demand (10.11), the parameter ω is related to various other parameters and to labor market tightness. At the tightness that solves the model, the labor demand just gives the employment level, which here is also the employment rate, which is $1 - u$, where u is the unemployment rate. Next, note that we can express the term $1 + \tau(\theta)$ as a function of the model parameters and tightness. Starting from (8.13) and using (3.15), we obtain

$$1 + \tau(\theta) = \frac{q(\theta)}{q(\theta) - \lambda\kappa} \quad \text{so} \quad \frac{1}{1+\tau(\theta)} = 1 - \frac{\lambda\kappa}{\mu} \cdot \theta^{\sigma}.$$

Combining these results with the labor demand (10.11), and setting $a = h = 1$, we obtain

$$\omega = (1-\alpha)(1-u)^{-\alpha}\left(1 - \frac{\kappa\lambda}{\mu} \cdot \theta^{\sigma}\right)^{1-\alpha}. \tag{10.14}$$

Setting $u = 5.7\%$, $\theta = 0.65$, $\alpha = 0.35$, $\kappa = 1$, $\lambda = 3.6\%$, $\mu = 0.75$ and $\sigma = 0.47$, the equation gives $\omega = 0.65$.

model can generate an exactly isoelastic Beveridge curve, but at the cost of a job-finding rate whose tightness elasticity is no longer constant and of somewhat more cumbersome algebra (Michaillat and Saez 2024a). We therefore stick with the traditional Cobb-Douglas matching function in the book.

TABLE 10.3. Comparative statics in the slackish labor market model

	Employment l	Tightness θ	Wage w	Unemployment rate u	Vacancy rate v
A. Negative labor demand shock					
Decrease in labor productivity a	–	–	–	+	–
B. Negative labor supply shock					
Decrease in labor force h	–	+	+	–	+

The comparative statics are obtained from equation (6.3) and are illustrated in figure 10.2. A variable's increase is denoted by "+" and a decrease by "-".

10.3. Fluctuations in unemployment

In this section we use our labor-market model to explain the fluctuations in unemployment and other variables observed on the US labor market.

10.3.1. Sources of fluctuations

The first question is: what is the main source of labor market fluctuations? The key insight to answer this question is that labor demand shocks produce a positive correlation between employment and labor market tightness, while labor supply shocks produce a negative correlation (figure 10.2). This insight can be used to distinguish demand and supply shocks on any slackish market, and especially in the labor market.

In the model, a labor demand shock is a shock that impacts worker productivity, a. If there is a decrease in productivity, the labor demand decreases for any level of tightness (equation (10.11)). Indeed, if workers suddenly become less productive, it is less profitable for firms to hire workers—as wages are rigid ($\gamma > 0$) and therefore do not decrease as much as productivity. In the labor market diagram of figure 10.2A, the labor demand curve shifts inward. At the new solution, the tightness is lower. The employment level decreases as well, as the market slides down along the labor supply curve.

A labor supply shock is a shock to the size of the labor force, h. After a decrease in labor force, the labor supply mechanically decreases for any level of tightness (equation (10.1)). The real wage increases slightly when the labor force decreases, as labor becomes scarcer (equation (10.10)). As a result, the labor demand is slightly depressed by a decrease in the labor force (equation (10.11)). In the labor market diagram of figure 10.2B, the labor supply curve shifts inward, and the labor demand moves inward slightly as well. As we established using equation (6.3), the shift in labor supply dominates, so at the new solution labor market tightness is higher. In contrast, the employment level decreases as both labor supply and demand are lower.

Table 10.3 summarizes the comparative statics. The main takeaway is that fluctuations

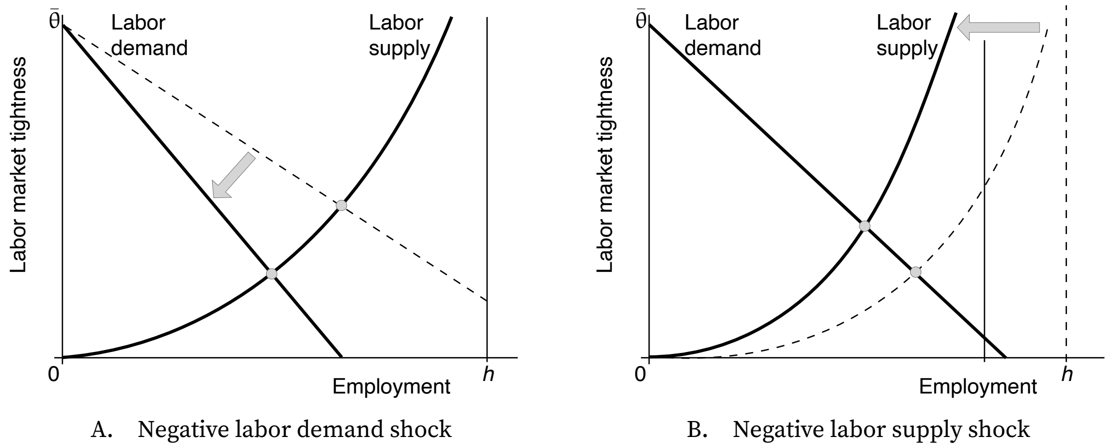


FIGURE 10.2. Comparative statics in the slackish labor market model

The labor supply is given by (10.1). The labor demand, including the wage norm, is given by (10.11). A negative labor demand shock is a reduction in labor productivity a. A negative labor supply shock is a reduction in labor force h.

in labor demand—which stem from labor productivity shocks—lead to positive comovements between employment level and labor market tightness. By contrast, fluctuations in labor supply—which stem from labor force shocks—lead to negative comovements between employment level and labor market tightness.

To determine the main source of labor market fluctuations, we therefore examine the correlation between employment and labor market tightness in the US labor market. Labor market tightness was constructed in section 3.8. Employment simply is the employment level computed by the BLS (2025f) from CPS data. Just like in figure 2.8, we stop the analysis in 2019 to concentrate on the 1948–2019 period so the analysis is not unduly influenced by the dislocation of the US labor market caused by the coronavirus pandemic.

We first look at the comovements of employment and labor market tightness by superimposing cyclical fluctuations in employment with those in tightness (figure 10.3A). It is clear that fluctuations in employment and tightness are positively correlated. The correlation between the two series is substantial, at 0.85. The two series appear especially strongly positively correlated after 1970—when they track each other very closely. Such positive correlation indicates that labor demand shocks, not labor supply shocks, are the main source of labor market fluctuations.

In figure 10.3B we recast the results from figure 10.3A in scatter form. We see again that employment and labor market tightness are positively correlated: an increase in tightness by 10% is associated with an increase in employment by about 0.3%. We also see that employment fluctuations are for the most part explained by tightness fluctuations: the least-squares regression has an R^2 of 0.89.

To conclude in the slackish labor market model, labor demand shocks produce the sort of comovements between employment and tightness observed in reality. Labor supply shocks would produce opposite comovements. In the rest of the chapter, we therefore use labor demand shocks to generate labor market fluctuations. The labor demand equation (10.11) shows that labor demand is increasing in labor productivity. So good times are represented by a high labor productivity and bad times by a low labor productivity. A higher productivity could reflect higher technology in good times, or more realistically, a higher utilization rate of employed labor—as displayed in figure 2.3.

It is unsurprising that we find that changes in the size of the labor force do not explain short-run labor market fluctuations. We saw in chapter 2 that the labor force participation rate is moving quite slowly over time, compared to the pace of business cycles, so changes in labor force were always unlikely to explain cyclical labor market fluctuations. Instead, changes in labor force participation are likely to have medium run effects in the labor market—as the labor force participation rate evolves in the medium run.

10.3.2. Amplitude of fluctuations

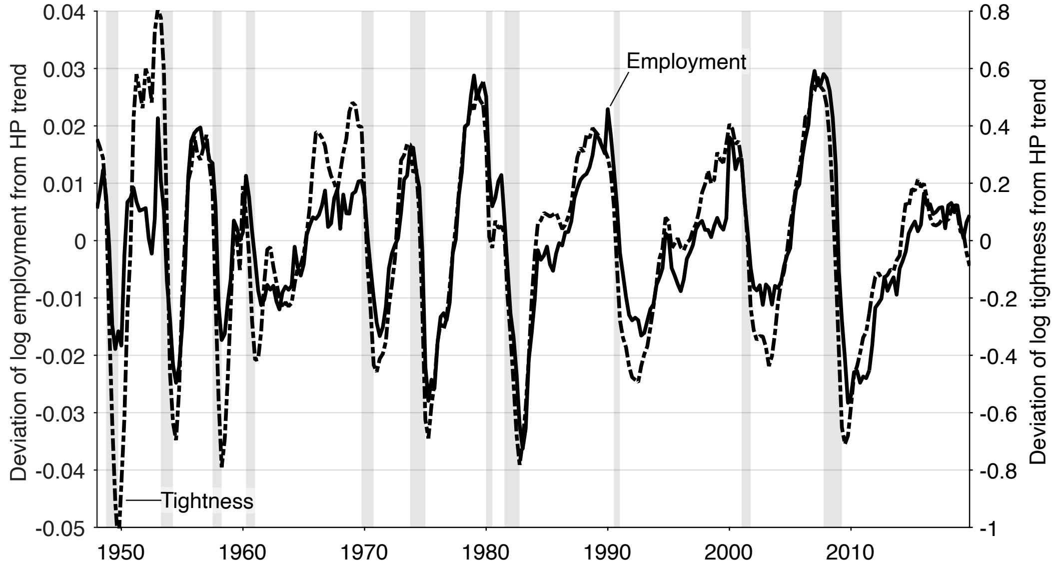
A more quantitative question arises next: can the model actually match the amplitude of the fluctuations we see in the data? That is, can the calibrated model generate an elasticity of labor market tightness with respect to labor productivity that we see in US data?

First, we need to measure labor productivity. We simply measure labor productivity a as a Solow residual: $\ln(a) = \ln(y) - (1-\alpha)\cdot\ln(n)$ where y is real output in the nonfarm business sector and n is employment in the nonfarm business sector, both constructed by the BLS (2025o,n).[2] Labor productivity incorporates both technology and labor utilization—this is by design since both affect labor demand. Here we treat labor utilization as exogenous; we will endogenize it in chapter 15.

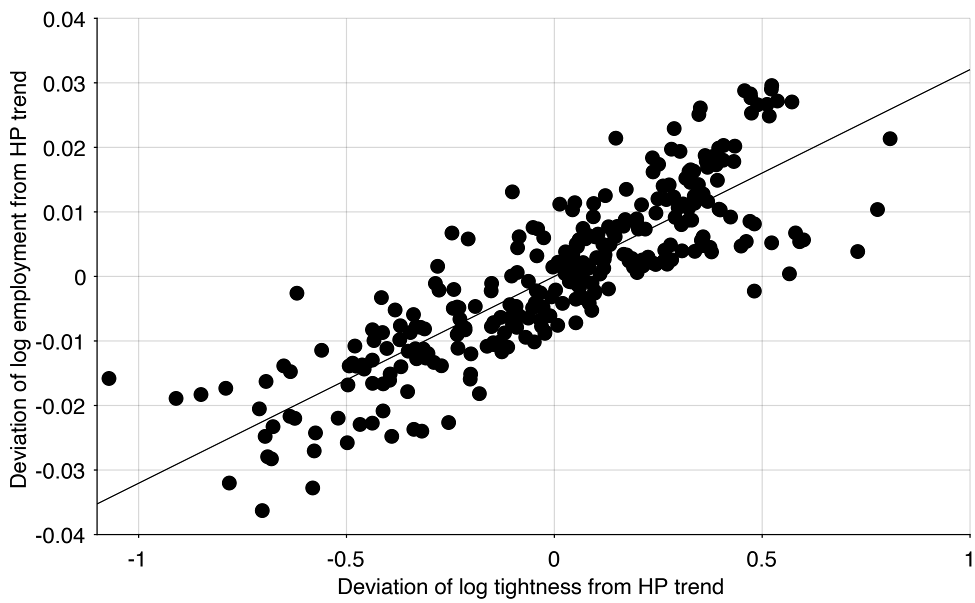
To estimate elasticity of labor market tightness with respect to labor productivity, we regress log tightness on log productivity, after detrending them with our HP filter (figure 10.4). We continue to focus on the 1948–2019 period. We see that labor market tightness rises when labor productivity rises, with an elasticity estimated at $\epsilon^{\theta}_{a} = 12.8$: an increase in productivity by 1% generates an increase in tightness by 12.8%.

We are interested in reaching the target for the elasticity of tightness with respect to productivity: $\epsilon^{\theta}_{a} = 12.8$. Can we generate such an elasticity for a plausible amount of wage rigidity? To answer this question, we exploit the elasticity of labor market tightness with respect to labor productivity, which is given by equation (6.14), but adjusted for the fact that in a dynamic model the elasticity of labor supply with respect to tightness is given by (8.14). Applying this equation here, with a Cobb-Douglas matching function, we get the

[2]According to the model, n should be the number of producers in the nonfarm business sector—the number of employees net of the number of recruiters. Because the number of recruiters is not available, we simply use the number of employees for n.



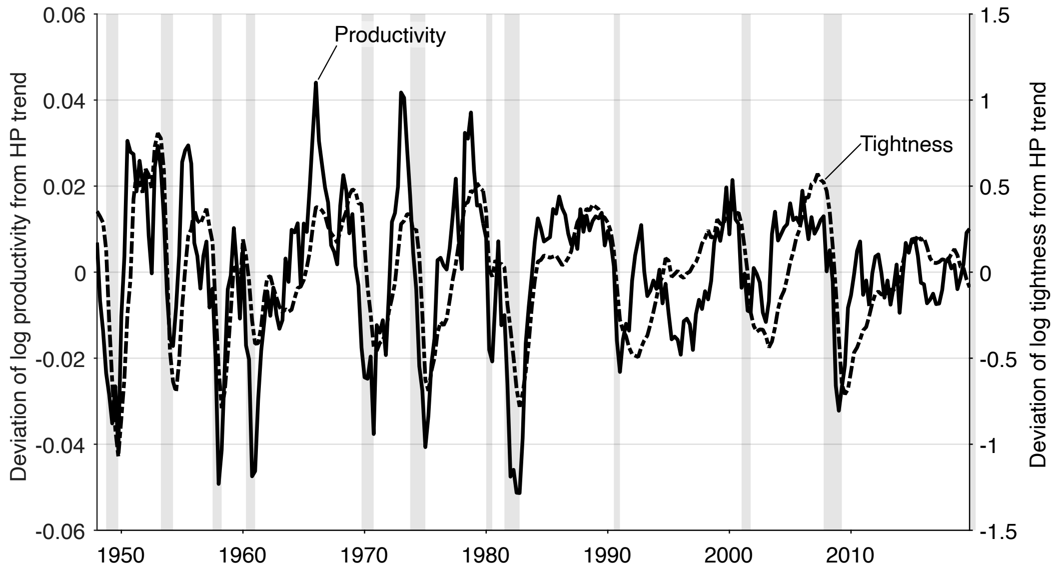
A. Correlation: 0.85



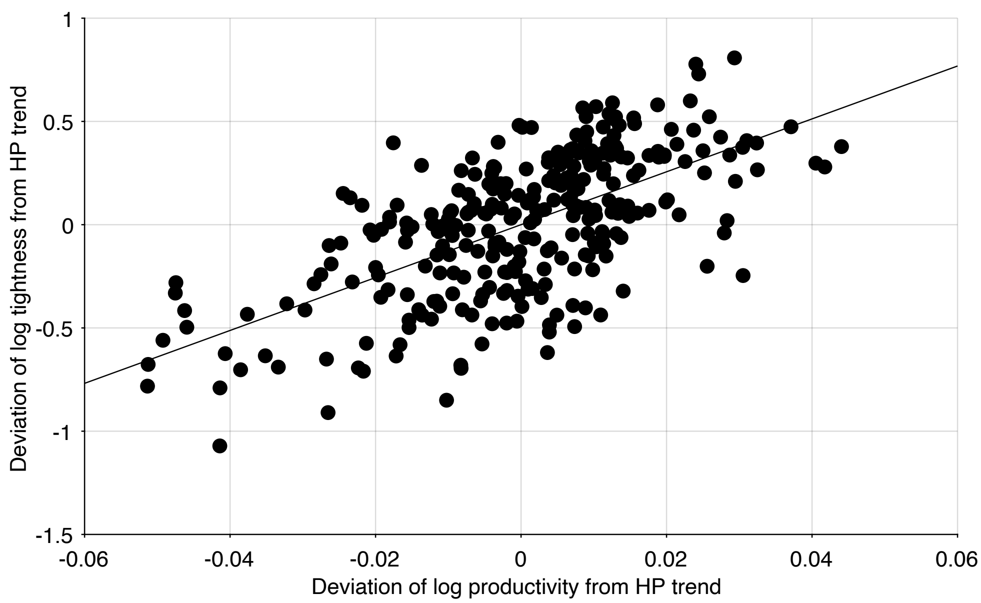
B. Estimated slope: 0.032

FIGURE 10.3. Correlation between employment and labor market tightness in the United States, 1948–2019

The employment level is the quarterly average of the monthly, seasonally adjusted series computed by the BLS (2025f). Labor market tightness comes from figure 3.4. Both series are detrended by applying a HP filter with smoothing parameter of 10,000. Shaded areas indicate recessions dated by the NBER (2023).



A. Correlation: 0.64



B. Estimated elasticity: 12.8

FIGURE 10.4. Response of labor market tightness to labor productivity in the United States, 1948–2019

Labor productivity is computed as the residual $\ln(a) = \ln(y) - (1 - \alpha) \cdot \ln(n)$ where y is quarterly, seasonally adjusted real output in the nonfarm business sector and n is quarterly, seasonally adjusted employment in the nonfarm business sector, both constructed by the BLS (2025o,n). Labor market tightness comes from figure 3.4. Both series are detrended by applying a HP filter with smoothing parameter of 10,000. Shaded areas indicate recessions dated by the NBER (2023).

elasticity of tightness with respect to productivity:

$$\epsilon_a^\theta = \frac{\gamma}{(1-\sigma)\alpha u + (1-\alpha)\sigma\tau}. \tag{10.15}$$

First, as we saw in chapter 6, if the wage is flexible ($\gamma = 0$), then there are no fluctuations in tightness with changes in productivity ($\epsilon_a^\theta = 0$). If the wage is flexible, it moves in proportion with productivity, so it absorbs any changes in productivity, which leaves firms no incentives to change the size of the firm when productivity varies. Thus, the only way to produce fluctuations in the model ($\epsilon_a^\theta > 0$) is if the wage is somewhat rigid ($\gamma > 0$).

We now calibrate the elasticity of tightness with respect to productivity by plugging in values for the parameters and variables of the model listed in table 10.2. Setting the production-function shape to $\alpha = 0.35$, the matching-function shape to $\sigma = 0.47$, the unemployment rate to $u = 5.7\%$, and the recruiting wedge to $\tau = 4.1\%$, we find $\epsilon_a^\theta = 43 \times \gamma$.

The last parameter that we need to calibrate is the wage-rigidity parameter γ, which is critical. We have set $\gamma = 0.5$, based on the rigidity of real wages of new hires observed by Haefke, Sonntag, and van Rens (2013). This gives us $\epsilon_a^\theta = 43 \times 0.5 = 22$: significantly higher than the elasticity of tightness with respect to productivity observed in the data.

Of course, we do not yet have a precise estimate of the wage-rigidity parameter. The best available evidence suggests that it is in the 0.2–0.7 range. How much of that range is rigid enough to produce the fluctuations in tightness observed in reality? To obtain an elasticity of at least 12.8, the wage rigidity must be at least $12.8/43 = 0.30$. Hence any wage rigidity in the 0.3–0.7 range is both realistic in itself and sufficient to amplify productivity shocks as much as in the data. Wage rigidity in the 0.2–0.3 range is slightly too low to amplify productivity shocks sufficiently. Overall, except at the most flexible end of the range, the small amount of wage rigidity observed in real wages for new hires is sufficient to amplify productivity shocks as much as observed in labor market data.

For completeness, we also compute the elasticities of the unemployment rate u, vacancy rate v, and employment level l with respect to productivity a. This is very easy to do in the model. First, using the Beveridge curve $v(u)$, we have $v(a) = v(u(a))$ so $\epsilon_a^v = (-\beta) \cdot \epsilon_a^u$, where $\beta = -\epsilon_u^v$ is the Beveridge elasticity. Second, by definition of tightness, $\theta(a) = v(a)/u(a)$ so that $\epsilon_a^\theta = \epsilon_a^v - \epsilon_a^u = -(1+\beta)\epsilon_a^u$. In other words, the Beveridge elasticity distributes the fluctuations in tightness between fluctuations in unemployment and fluctuations in vacancies:

$$\epsilon_a^u = \frac{-1}{1+\beta} \cdot \epsilon_a^\theta \quad \text{and} \quad \epsilon_a^v = \frac{\beta}{1+\beta} \cdot \epsilon_a^\theta.$$

Finally, the employment level is simply given by $l(a) = [1 - u(a)]h$ so $\epsilon_a^l = -[u/(1-u)]\epsilon_a^u$, which gives

$$\epsilon_a^l = \frac{1}{1+\beta} \cdot \frac{u}{1-u} \cdot \epsilon_a^\theta.$$

We have calibrated the model to match the empirical observation that the Beveridge curve is a rectangular hyperbola, so the Beveridge elasticity is just $\beta = 1$. Hence, in the calibrated model, $\epsilon_a^v = \epsilon_a^\theta/2 = 11$ and $\epsilon_a^u = -\epsilon_a^\theta/2 = -11$. This means that fluctuations in unemployment and vacancies contribute equally to fluctuations in labor market tightness. We have also calibrated the model so $u = 5.7\%$, which gives $u/(1-u) = 0.060$ and $\epsilon_a^l = \epsilon_a^\theta \times 0.03 = 0.66$. We notice also that under productivity shocks, the observed elasticity of employment with respect to tightness is $\epsilon_\theta^l = \epsilon_a^l/\epsilon_a^\theta = 0.030$, which is exceedingly close to the elasticity of 0.032 estimated in figure 10.3B.

Finally, figure 10.5 displays labor market tightness, unemployment rate, vacancy rate, and employment as a function of productivity to illustrate the quantitative properties of the calibrated model. The figure also shows the Beveridge curve: the negative relationship between unemployment rate and vacancy rate traced as productivity varies.

10.3.3. Shimer puzzle

By looking closely, we see that our calibration solves the famous Shimer (2005) puzzle.

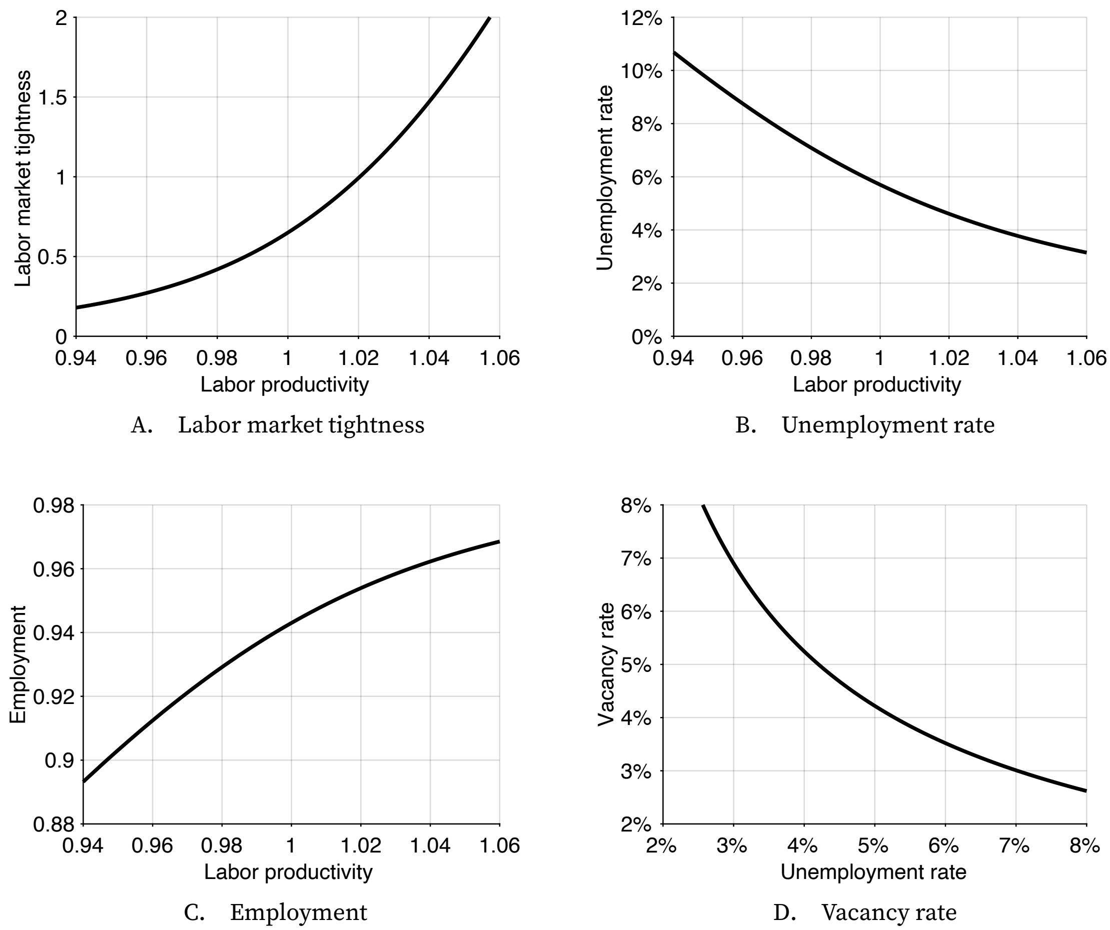
The Shimer puzzle says—broadly—that in a labor market model built around a matching function, and with bargained wages, productivity shocks are not amplified enough to produce realistic fluctuations in labor market tightness, unemployment, and vacancies.

We can see the Shimer puzzle in the present analysis, because the bargained wage is flexible, in the sense that it can be represented by the wage norm (10.10) with $\gamma = 0$. And as we have just seen in (10.15), if $\gamma = 0$, tightness does not respond to productivity shocks at all, so unemployment rate and vacancy rate do not respond either.

Are we entirely sure that the bargained wage is flexible? Well yes: we established that result in chapter 8 in the case of a generic dynamic market, but the result directly applies to this chapter's labor market. In fact, an interesting result from section 8.9 is that the bargained wage is set at a markdown below the marginal product of labor:

$$w = \frac{(1-\alpha)an^{-\alpha}}{1 + \frac{1-\chi}{\chi} \cdot u}. \tag{10.16}$$

If the worker bargaining power χ is 0, the markdown becomes infinite and the wage falls to 0. If the worker bargaining power is 1, the markdown vanishes and the wage rises to the marginal product of labor. Interestingly, the markdown increases with the unemployment rate. If the unemployment rate was 0, the markdown would vanish and the wage would reach the marginal product of labor. When the unemployment rate reaches 1, the markdown reaches $1/\chi$, which implies that the wage is just a fraction χ of the marginal



A. Labor market tightness

B. Unemployment rate

C. Employment

D. Vacancy rate

FIGURE 10.5. Response to productivity fluctuations in the slackish labor market model

Labor market tightness θ is given by (10.12). The unemployment rate u is given by (10.2). The employment level is given by $l = (1 - u)h$. The vacancy rate is given by $v = \theta u$. The calibration of the model is described in table 10.2.

product of labor. This property appears because workers' outside option is worse when the unemployment rate is higher: if they do not agree to the wage offer, they are forced to remain jobless for a longer time. Workers must therefore settle for lower wages when the unemployment rate is higher—which allows firms to impose a higher markdown.

Just as we did in the generic dynamic market, we plug the wage expression into the labor demand equation (10.8). We find that the labor demand is degenerate under wage bargaining, in the sense that it does not involve quantities:

$$\frac{\tau(\theta)}{u(\theta)} = \frac{1-\chi}{\chi}. \tag{10.17}$$

In the tightness-quantity diagram, the labor demand is therefore horizontal, pinning down a unique labor market tightness:

$$\theta = (\tau/u)^{-1}\left(\frac{1-\chi}{\chi}\right). \tag{10.18}$$

From this equation and the underlying structure of the matching wedge $\tau(\theta)$ and slack rate $u(\theta)$, we infer that labor market tightness is solely determined by the worker bargaining power χ, the recruiting cost κ, the job-separation rate λ, and the matching function. The Shimer puzzle is directly visible in the labor demand (10.18), as labor market tightness does not depend in any way on labor productivity a. This means that productivity shocks have no effect on tightness, and therefore no effect on unemployment, employment, or vacancies.

10.3.4. Mechanism behind fluctuations

A common idea is that unemployment goes up in recessions because firms lay off a large number of workers. However, this is not what happens in the model, and this is not what happens in the data.

In the model, the pool of unemployment grows in bad times because it takes a longer time for people to find a job, not because employed workers lose their jobs at a higher pace. Formally, when labor demand drops, labor market tightness falls, so the job-finding rate plummets, and job seekers get stuck in unemployment. Conversely, the pool of unemployment shrinks in good times because it is much quicker for unemployed workers to find a job, not because employed workers keep their jobs longer.

This is also what appears in the data. As we see in figures 3.5A and 2.9A, the job-finding rate is sharply procyclical, while the job-separation rate is acyclical. It is true that the layoff rate was countercyclical in the past (Akerlof, Rose, and Yellen 1988, figure 2). But it is less true now, and the increase in layoffs in recessions is offset by a simultaneous drop in quits, as figure 2.9B shows. As Hall (2005a, p. 397) eloquently puts it,

> In the modern US economy, recessions do not begin with a burst of layoffs. Unemployment rises because jobs are hard to find, not because an unusual number of people are thrown into unemployment.

More formally, Shimer (2012, p. 128) shows that the job-finding rate explains 77% of unemployment fluctuations over 1948–2010, while the job-separation rate explains only 23%. Over 1987–2010, the share explained by the job-finding rate rises to 90%, and the share explained by the job-separation rate declines to 10%.

10.4. Origins of unemployment

Although the slackish model is built around a matching function, not all unemployment is frictional in it. In fact, part of unemployment is due to job rationing, and part to matching frictions. Moreover, in bad times, when total unemployment is high, frictional unemployment is low: job rationing accounts for almost all of unemployment.

In that way the slackish model unifies two theories of unemployment that have coexisted for a long time but have always only been applied to different contexts. Before the Great Depression people were working on frictional unemployment, caused by matching and recruiting (Reder 1969). But the Great Depression and the *General Theory* killed that theory and replaced it with one explaining unemployment from a lack of jobs due to low labor demand combined with a wage floor (Okun 1981, figure 1-2). In the 1970s, researchers started asking: Why should nominal wages be sticky? This led to the development of efficiency-wage models, which aimed to complete the *General Theory* by providing a justification for the wage floor (Akerlof and Yellen 1986). However, the emergence of the DMP model put a stop to the efficiency-wage literature, and once again researchers started focusing solely on frictional unemployment.

The slackish framework allows us to blend both sources of unemployment, offering a unified way to study the labor market in good and bad times. In this section, we establish that the slackish model incorporates both rationing and frictional unemployment, and we study the cyclicality of the two components. We also show that the DMP model can be seen as a special case of the slackish model without rationing unemployment.

10.4.1. Why is a unified theory of unemployment beneficial?

A common view among New Keynesian macroeconomists is that a market-clearing model—or a model with frictional unemployment—is accurate enough to describe good times. Of course, they also realize that such a model does not work in bad times—market failures are just too prevalent. As Mankiw and Romer (1991) wrote at the onset of their text on *New Keynesian Economics*:

> Perhaps, the invisible hand guides the economy in normal times, but events like the Great Depression require different theories.

Thus, there is this idea that it is acceptable to have separate models to describe bad times and good times. But surely we can do better. A satisfactory model of the labor market ought to capture both good times and bad times. The labor market and its participants are the same in good times and bad, so should we not aim for a unified treatment of the labor market over the business cycle? Should we not be able to use a common theory to think about bad times, when there is high unemployment, and good times, when there is low unemployment? Scientifically, this seems like the right thing to do.

Practically, the coronavirus pandemic demonstrates why it is very helpful to have a labor market model that describes both good and bad times. Initially, when the pandemic started, there was a vast increase in unemployment, so we needed a model that could help us think about that. In the US, unemployment rose to almost 15%, which we had not seen since the Great Depression. However, the labor market recovered rapidly. Suddenly, we were in a very tight labor market, with a vast amount of vacancies posted, and low unemployment. It was important to have a model that could help us understand this situation.

So it is evident that the labor market can move quickly from a situation with a massive amount of slack to a situation that is massively tight. We need to be able to think about these two things simultaneously; this is what the model developed in this chapter aims to do. We do not want to have two separate models to think about good times and bad times, but rather one model that captures both. This is one of the motivations for the model: to offer a unified treatment of unemployment that applies at all times. The unified treatment captures both the lack of jobs that we observe in slumps as well as the matching frictions especially visible in booms.

10.4.2. Job rationing in the real world

Why is it a quality that the slackish model features job rationing—unlike the DMP model? It is because we know that when matching frictions vanish—when workers are desperate to work so recruiting for firms is costless—unemployment does not disappear. This means that some unemployment must be caused by a lack of jobs: it cannot all be due to matching frictions.

How do we know this? It is because whenever a significant recession occurs, people queue for jobs at factory gates, in front of firms, at job bureaus, and job fairs. Many people want to work so recruiting is essentially costless, and yet, there is still some unemployment—so unemployment cannot possibly all be frictional.

The photos in the preface illustrate this phenomenon. The same queues of workers in front of factory gates featured prominently in Charlie Chaplin's *Modern Times* as well. It is clear that queues were everywhere during the Great Depression in the United States—a clear manifestation of job rationing.

10.4.3. Job rationing in the model

We are interested in the following question: Can the slackish model still have some unemployment when the recruiting cost goes to zero? The key finding is that, if the labor demand is low enough, it is possible that some unemployment remains although the recruiting cost is zero.

We define rationing unemployment as the amount of unemployment that prevails when the recruiting cost is 0. Rationing unemployment comes from the fact that even when recruiting is free, firms do not want to hire all workers in the labor force. This means that there just are not enough jobs for everyone.

To see that rationing unemployment exists in the slackish model, and to study its properties, let's first compute the labor demand when the recruiting cost is set to $\kappa = 0$. In that case, the recruiting wedge $\tau(\theta)$ becomes 0, so the labor demand does not depend on tightness anymore. The labor demand reduces to the Walrasian labor demand, which equalizes marginal product of labor to the real wage:

$$l^d(w, \kappa = 0) = \left[\frac{(1-\alpha)a}{w}\right]^{1/\alpha}. \tag{10.19}$$

Then the rationing unemployment rate is defined as

$$u^r = \max\left(0, 1 - \frac{l^d(w, \kappa = 0)}{h}\right).$$

A natural question is whether rationing unemployment always exists in our slackish model. The answer is no. There are certain situations in which jobs are lacking and other situations in which jobs are plentiful. Rationing unemployment exists whenever $l^d(w) < h$, which can be rewritten

$$w > (1-\alpha)ah^{-\alpha}.$$

Thus, rationing unemployment exists whenever the real wage is higher than the marginal product of labor of the last worker in the labor force—who is the least productive worker in the labor force by diminishing marginal returns to labor under a concave production function. The logic is simple. If the wage the firm has to pay is above the marginal product of the least productive worker in the labor force, then no firm wants to hire that worker. This scenario leads to job rationing because some workers are less productive than what they are paid. Essentially, job rationing arises in any situation in which the wage is too high.

Are we sure that for some level of productivity there is some job rationing? The answer is yes. Given the wage norm assumed here, the labor demand in the absence of recruiting cost can be expressed as a function of model parameters:

$$l^d(a, h, \kappa = 0) = \left[\frac{(1-\alpha)a^\gamma}{\omega}\right]^{1/\alpha} \cdot h^{1-\gamma}. \tag{10.20}$$

Job rationing occurs whenever $l^d(a, h, \kappa = 0) < h$, which can be rewritten

$$a < \left(\frac{\omega}{1-\alpha}\right)^{1/\gamma} h^\alpha. \tag{10.21}$$

This gives us a threshold of productivity below which job rationing appears.

Finally, we represent rationing unemployment graphically (figure 10.6). With our Cobb-Douglas matching function, when $\theta \to 0$, $q(\theta) \to \infty$ so $\tau(\theta) \to 0$. From this, we learn that the labor demand when $\theta = 0$ is the same as the labor demand when the recruiting cost is 0. This means that the position of the vertical, zero-recruiting-cost labor demand corresponds to the x-intercept of the regular, positive-recruiting-cost labor demand with the x-axis. Hence, when the recruiting cost goes to zero, the labor demand curve rotates clockwise around the fixed x-intercept. And for our purpose, this means that rationing unemployment is measured by the distance between the x-intercept of the labor demand and the labor force.

There are several possible situations, depending on the value of labor productivity. If labor productivity is above the threshold described in (10.21), the x-intercept of the labor demand occurs farther to the right than the labor force h, so there is no rationing unemployment (figure 10.6A). If labor productivity is below the threshold, the x-intercept of the labor demand occurs within the labor force h, so there is some rationing unemployment (figure 10.6B). If labor productivity is even lower, the x-intercept is even closer to the origin so rationing unemployment is larger (figure 10.6C).

10.4.4. Frictional unemployment in the model

We now complete the decomposition of unemployment into two parts: rationing unemployment u^r and frictional unemployment u^f. Frictional unemployment is the extra unemployment that exists beyond that caused by the lack of jobs in the economy:

$$u^f = u - u^r.$$

We next express frictional unemployment as a function of the parameters and tightness. We focus on the situations in which productivity is low enough so that rationing unemployment prevails. Without rationing unemployment, all unemployment is fric-

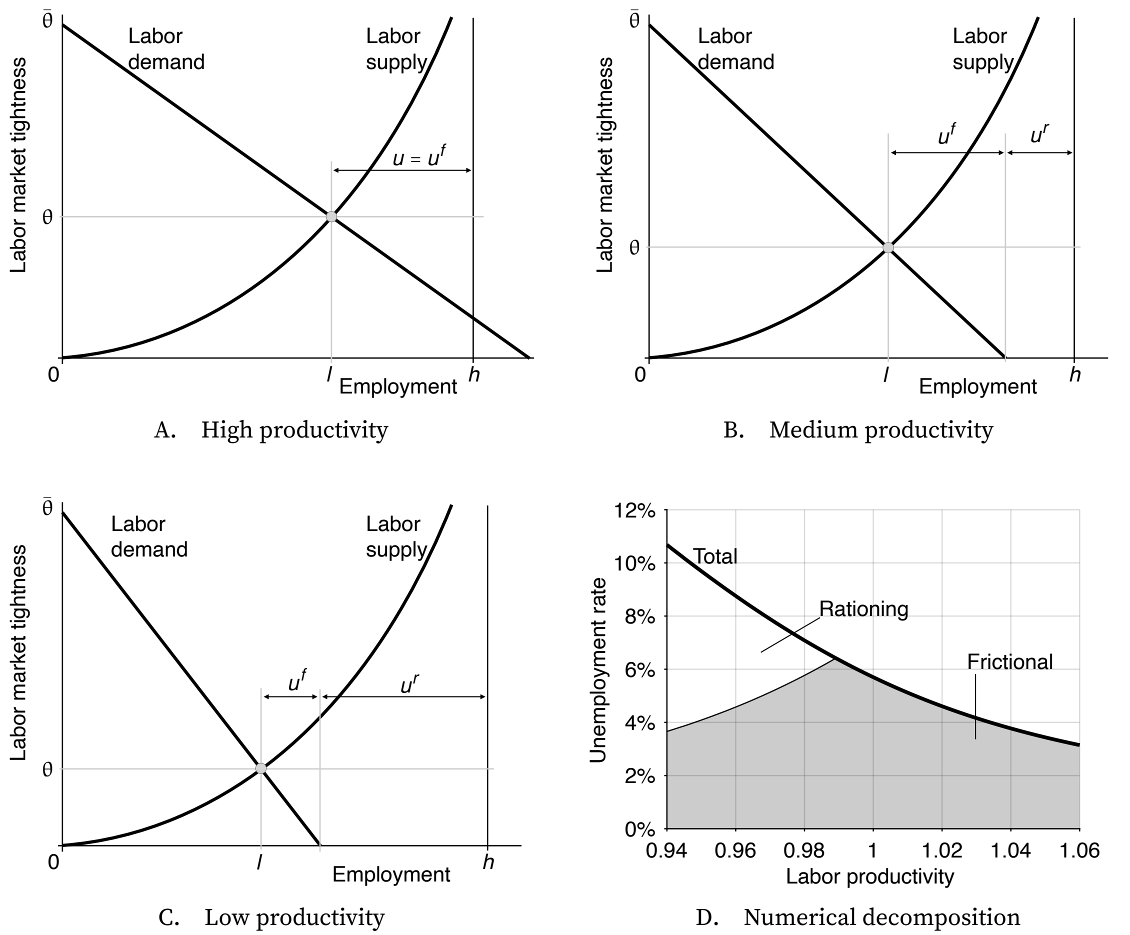


FIGURE 10.6. Rationing and frictional unemployment in the slackish labor market model

In panels A–C, the labor supply is given by (10.1) and the labor demand, including the wage norm, is given by (10.11). In panel D, the calibration of the model comes from table 10.2.

tional, so there is no special expression for frictional unemployment. With rationing unemployment, $u = 1 - l^d(\theta)/h$ and $u^r = 1 - l^d(\kappa = 0)/h$ so

$$u^f = \frac{l^d(\kappa = 0)}{h} - \frac{l^d(\theta)}{h}.$$

Using the expressions (10.11) and (10.20) for the labor demands, we find that frictional unemployment satisfies

$$u^f = \left[\frac{1-\alpha}{\omega}\right]^{1/\alpha} h^{-\gamma} a^{\gamma/\alpha} \left[1 - \frac{1}{(1+\tau(\theta))^{1/\alpha - 1}}\right]. \tag{10.22}$$

We graphically decompose unemployment in figure 10.6. In figure 10.6A, productivity is high, so all unemployment is frictional. In figures 10.6B and 10.6C, labor productivity is below the threshold (10.21), so there is some rationing unemployment. However, total unemployment is greater than this amount. Indeed, tightness is strictly positive so firms hire fewer workers than at the point where tightness is zero. Thus, there is more unemployment than when tightness is zero and this extra unemployment corresponds to frictional unemployment. Frictional unemployment corresponds to the horizontal distance covered as we move along the labor demand from the x-intercept to the supply-demand intersection.

10.4.5. Fluctuations in rationing and frictional unemployment

It is interesting to study how rationing and frictional unemployment vary when the labor market fluctuates. From this we will see how the sources of unemployment change as the state of the labor market evolves. We would presume that the tide lifts all boats: that when unemployment is high, all types of unemployment are high. It turns out that this is not what happens. In very good times, all unemployment is frictional, so of course total and frictional unemployment move together. But then a divergence occurs. As rationing unemployment becomes larger, overall unemployment grows, but frictional unemployment shrinks. So in bad times, rationing unemployment is large but frictional unemployment is low.

It is easy to establish these results from the expressions that we derived for frictional and rationing unemployment. We only consider situations where labor productivity is below the threshold in (10.21) so rationing unemployment exists. From expression (10.20), we see that as labor productivity a drops, the zero-recruiting-cost labor demand $l^d(\kappa = 0)$ shrinks, so rationing unemployment $1 - l^d(\kappa = 0)/h$ grows.

What happens to frictional unemployment at the same time? In expression (10.22) for frictional unemployment, we see that as labor productivity drops, the term a^γ drops. We have also seen earlier (see table 10.3) that when productivity drops, tightness drops, which means that the recruiting wedge $\tau(\theta)$ decreases, and implies that the term in square brackets in (10.22) decreases. Accordingly, frictional unemployment decreases when labor productivity falls, just as total unemployment increases.

We can also see these movements graphically (figure 10.6). As labor productivity drops, the labor demand curve shifts inward, so rationing unemployment and total unemployment grow. The economic mechanism is the following: in bad times, productivity falls, wages do not fall as much, so the profitability of workers falls and firms do not want to hire as many workers. This reduces the number of jobs in the economy, which boosts unemployment. Since labor demand is weaker, tightness falls.

What happens to frictional unemployment as the labor market cools? Tightness is much lower as the economy moves down the labor supply curve, so the amount of employment that has been reduced due to recruiting costs is very small. This leads to less frictional unemployment. The general intuition is that in bad times it is easy for firms to recruit workers since labor is readily available. The total increase in unemployment due to the negative productivity shock is dampened by the reduction in frictional unemployment.

Overall, in bad times there is high rationing unemployment and low frictional unemployment. A natural question is how big the fluctuations in rationing and frictional unemployment might be. Figure 10.6D shows what happens in the calibrated version of the slackish model. For any unemployment rate below 6.4% (which is slightly above the average unemployment rate of 5.7%), all unemployment is frictional. Once unemployment rises further, above 6.4%, job rationing appears, and frictional unemployment starts declining. When unemployment reaches 7%, rationing unemployment is 1.2% and frictional unemployment falls to 5.8%. When unemployment rises further to 8% and then 9%, rationing unemployment reaches 2.9% and then 4.6% and frictional unemployment declines to 5.1% and 4.4%. Finally, when unemployment reaches 10%, rationing unemployment reaches 6.0% and frictional unemployment declines to 4.0%.

10.4.6. Absence of job rationing in the DMP model

The DMP model is the workhorse model of unemployment in macroeconomics and labor. However, the idea of job rationing—the idea that irrespective of recruiting costs firms do not want to hire everyone because there is not enough demand for labor—is absent from the DMP model. This means that when the recruiting cost goes to zero, unemployment disappears. All unemployment is therefore frictional in that model. As we will see, this property colors how the DMP model operates and its policy implications.

The DMP model is a special case of the slackish labor market model with a linear production function and a bargained wage. So we can use the analysis above to show quickly that there is no job rationing in that model.

Let's first gather the labor demand and supply in the DMP model. The labor supply is the same as in the generic model, given by (10.1). The labor demand follows from the first-order condition (10.8) with $\alpha = 0$, so it is perfectly elastic:

$$[1 + \tau(\theta)]\, w = a. \tag{10.23}$$

Given that the labor demand is perfectly elastic, it is horizontal in our usual tightness-employment plane (figure 10.7A).

To completely specify the DMP model, we need to specify the wage norm. The wage norm imposes that wages are determined through Nash bargaining, so that the employment surplus is shared between worker and firm. Specifically, the worker keeps a share χ of the surplus and the firm keeps a share $1 - \chi$, where $\chi \in (0, 1)$ is the worker's bargaining power. Using equation (8.18), with $\alpha = 0$, we see that the wage in the DMP model is a markdown below productivity:

$$w = \frac{a}{1 + \frac{1-\chi}{\chi} \cdot u(\theta)}. \tag{10.24}$$

With wage bargaining, as we saw in (8.19), the labor demand equation becomes

$$\frac{\tau(\theta)}{u(\theta)} = \frac{1-\chi}{\chi}. \tag{10.25}$$

The labor demand pins down a unique labor market tightness, determined by the bargaining power χ, the matching function, as well as the parameters underlying the functions $\tau(\theta)$ and $u(\theta)$: job-separation rate λ and recruiting cost κ.

Since we perform an experiment where the recruiting cost κ goes to 0, it is good to spell out what the recruiting wedge $\tau(\theta)$ is, since it depends on the recruiting cost. Using the expressions (10.4) and (10.2) for the functions $\tau(\theta)$ and $u(\theta)$, we rewrite the labor demand as

$$\kappa \cdot \frac{f(\theta) + \lambda}{q(\theta) - \kappa\lambda} = \frac{1-\chi}{\chi}.$$

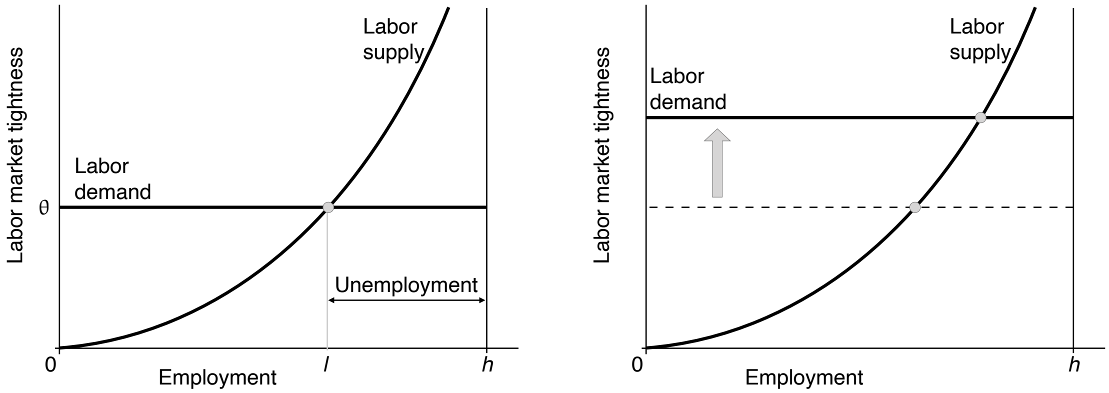
Multiplying both sides by $\chi[q(\theta) - \kappa\lambda]$, and collecting terms, we obtain

$$\lambda\kappa + \chi\kappa f(\theta) = (1 - \chi) q(\theta).$$

Finally, dividing both sides by $q(\theta)$ to keep all the terms with κ and θ together, and using again (3.6), we find that the DMP model's labor demand is

$$(1 - \chi) - \kappa \left[\chi\theta + \frac{\lambda}{q(\theta)}\right] = 0. \tag{10.26}$$

The left-hand side of the equation is continuous and strictly decreasing from $1 - \chi > 0$



A. Tightness-employment representation B. Reduction in recruiting cost

FIGURE 10.7. Solution of the DMP labor market model

The labor supply is given by (10.1) and the labor demand, including the wage norm, is given by (10.25).

to $-\infty$ when θ goes from 0 to ∞. Indeed, $q(\theta)$ is strictly decreasing in θ so $-\lambda/q(\theta)$ is also strictly decreasing. Furthermore, with the Cobb-Douglas matching function, when $\theta \to 0$, $q(\theta) \to \infty$ so $\lambda/q(\theta) \to 0$. Hence, the DMP model's labor demand defines a unique tightness.[3]

To conclude, we want to know what happens when the recruiting cost κ goes to 0. As $\kappa \to 0$, the term in square brackets in (10.26) goes to 0 for any finite θ, and the left-hand side goes to $1 - \chi$, so the equation is violated. The equation can only continue to hold if tightness becomes arbitrarily large. Hence, in the DMP model, $\theta \to \infty$ when $\kappa \to 0$. The economic intuition is that with bargaining, the wage is strictly below productivity, so if there are no recruiting costs, firms are willing to hire everyone and post an arbitrarily large number of vacancies. This drives tightness to infinity. The movement of the labor demand as the recruiting cost falls is depicted in figure 10.7B, showing how tightness rises as the recruiting cost dwindles.

What happens to employment and unemployment? Well, employment is determined by the labor supply, so it is the limit of the labor supply when $\theta \to \infty$. Given that the labor supply is given by (10.1), and that $f(\theta) \to \infty$ when $\theta \to \infty$ with the Cobb-Douglas matching function, the labor supply asymptotes to the labor force h as tightness becomes infinite. This means that everyone in the labor force has a job when tightness is infinite. Thus, when the recruiting cost vanishes, unemployment disappears. From this we conclude that there is no job rationing in the DMP model and all unemployment is frictional. The only reason there is unemployment in the DMP model is because there is a recruiting cost.

[3]The labor demand is the same as the job-creation curve in Pissarides (2000, equation (1.24)), once we set unemployment benefits to $z = 0$ and the interest rate to $r = 0$ in Pissarides' model.

10.4.7. Another definition of job rationing based on infinite job-search effort

We now introduce a parameter $e > 0$ for job-search effort by unemployed workers. In the previous specification we effectively set $e = 1$; here we ask what happens when e changes, and in particular when e goes to infinity—when people are really desperate to work, what happens to unemployment? We will see that the limit of the unemployment rate when job-search effort is infinite is just rationing unemployment.

To allow for job-search effort, the matching function becomes $m(e \cdot u, v)$: total search input on the worker side is $e \cdot u$, with u the number of job seekers, while v continues to capture recruiting effort on the firm side. Labor market tightness is the ratio of the two arguments, $\theta = v/(e \cdot u)$. By constant returns to scale, the job-filling rate is still $q(\theta)$ with the same properties as before. But the job-finding rate becomes $e \cdot f(\theta)$—so it depends on both tightness and effort. This is because $f(\theta)$ is the rate at which one unit of effort is successful, so $e \cdot f(\theta)$ is the rate at which e units of effort are successful. We can quickly formally see this, as the job-finding rate is

$$\frac{m(e \cdot u, v)}{u} = e \cdot \frac{m(e \cdot u, v)}{e \cdot u} = e \cdot m\left(1, \frac{v}{e \cdot u}\right) = e \cdot m(1, \theta) = e \cdot f(\theta).$$

Because the job-filling rate is unchanged, nothing on the firm side moves at given θ: labor demand is unaffected by e and still given by (10.11).

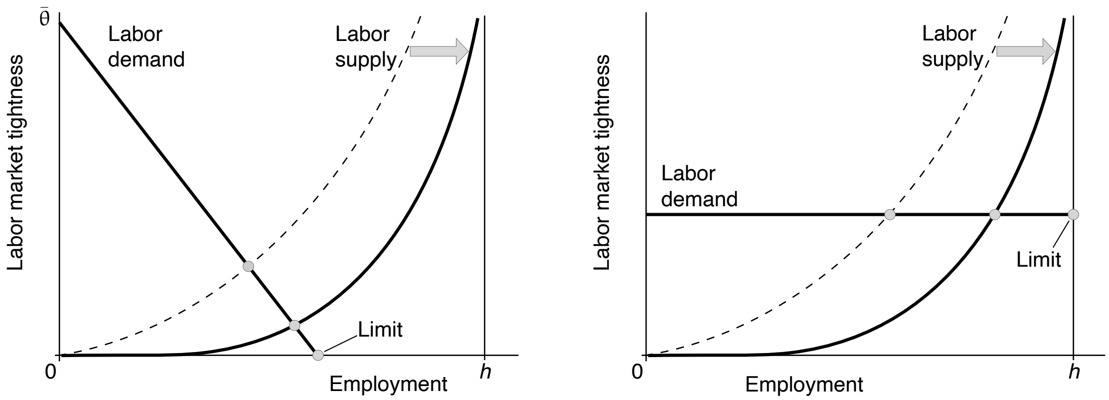
Only labor supply changes. Since the job-finding rate is altered from $f(\theta)$ to $e \cdot f(\theta)$, the labor supply becomes

$$l^s(\theta, e) = \frac{e \cdot f(\theta)}{\lambda + e \cdot f(\theta)} \cdot h. \tag{10.27}$$

The properties with respect to tightness θ are as before. For effort, labor supply is 0 when effort is 0, labor supply is strictly increasing with effort, and the limit of the labor supply is h when effort is infinite. So raising effort is like raising tightness—if people search more, more people have jobs.

The easiest way to see what happens in the model when the job-search effort goes to infinity is to use the tightness-employment diagram (figure 10.8). We know that the labor demand is unaffected by search effort, while the labor supply shifts out when the search effort increases. In the limit with infinite search effort, the labor supply curve becomes rectangular with a right angle at $l = h$. Using this, we can see what happens when search effort goes to infinity.

In a generic slackish model, when search effort goes to infinity, unemployment still does not disappear when jobs are rationed. The economy converges to the same point as in the limit of zero recruiting cost, to the x-intercept of the labor demand curve. The unemployment that remains at that stage is rationing unemployment. The condition for



A. Generic slackish model B. DMP model

FIGURE 10.8. Limit of the labor market models when job-search effort becomes infinite

The labor demand is given by (10.11) in panel A and by (10.25) in panel B. The labor supplies are given by (10.27), for a lower job-search effort (dashed line) and a higher job-search effort (solid line).

it to exist is unchanged: the intercept must lie below h, equivalently productivity must be low enough as in (10.21). Whether search effort goes to infinity or recruiting cost goes to 0, the same amount of rationing unemployment prevails.

Hence, even infinitely intense job search does not eliminate unemployment. This is because even if it is immensely easy for firms to find workers, firms do not want to hire these workers when their wage exceeds their marginal product. This is what happened during the Great Depression: people were desperate to work but unemployment did not vanish, and queues formed at factory gates.

In the DMP model, as job-search effort goes to infinity, unemployment disappears. As search effort rises, the labor supply curve shifts out and the labor market moves along the horizontal labor demand curve. When effort becomes infinite, all workers in the labor force find a job, and no unemployment remains. So we confirm that there is no job rationing in the DMP model. The infinite-effort thought experiment illustrates that the DMP model provides an excessively supply-centric view of the labor market. Because its labor demand is degenerate—it is perfectly elastic—the labor demand is never a hindrance to employment; the labor supply is the main issue.

10.5. Summary

In this chapter, we apply the slackish market model of chapter 8 to the labor market. We derive equilibrium labor supply from balanced flows in the labor market, equilibrium labor demand from profit maximization by firms, and close the model with a rigid wage norm. The solution of the model is found at the intersection of the labor supply and demand curves in a tightness-employment diagram. We then calibrate the model to the

US labor market over 1948–2019 and use the calibration to connect the model's quantitative properties to evidence on unemployment, vacancies, tightness, and the Beveridge curve.

We find that the main driver of labor market fluctuations in the United States is a labor demand shock moving labor productivity, not movements in labor force: employment and tightness comove positively, as the model predicts under productivity shocks but not under labor force shocks. A modest amount of wage rigidity is enough to amplify productivity shocks into realistic movements in tightness, unemployment, and vacancies. We also emphasize that fluctuations in the model—and in modern US data—work primarily through fluctuations in the job-finding rate rather than fluctuations in job separations.

Finally, we show that although the matching function plays a central role, not all unemployment is frictional in the slackish model. That is, unemployment would not disappear if recruiting costs vanished or unemployed workers searched infinitely hard for jobs. While all unemployment is frictional in good times, frictional unemployment falls as rationing unemployment grows in bad times. In the calibration, when total unemployment reaches 10%, rationing unemployment amounts to 6.0% while frictional unemployment is only 4.0%.

In that way, the slackish model departs sharply from the DMP model. Although both labor market models use a matching function, all unemployment is frictional in the DMP model—because its labor demand is degenerate. This difference has notable consequences. For instance, when unemployment rises in recessions, economists often advocate for policies that improve matching in the labor market, such as investment in placement agencies or harsher monitoring of workers' search effort. This is because all unemployment is frictional in the DMP model, so the natural reflex is to reduce matching frictions. However, when part of unemployment is caused by a lack of job, as in the slackish model, this advice falls flat. In the slackish model, frictional unemployment shrinks in slumps, so there is very little scope for policies that improve search and matching. In chapter 11, we will delve into policies that are able to alleviate unemployment in bad times in the presence of job rationing.

CHAPTER 11.

Policies in a slackish labor market

In the previous chapter we introduced the slackish labor market model. The model provides a unified theory of unemployment, involving both rationing unemployment—as in older models from the Keynesian tradition—and frictional unemployment—as in the modern DMP model.

As Diamond (2013, p. 37) observed, however, the decomposition of unemployment into rationing and frictional unemployment is here to clarify the mechanisms at play in the slackish model, but does not have direct policy implications:

> This division [between rationing and frictional unemployment] sharpens awareness of the implications of the critical new modeling assumptions employed in [Michaillat (2012)]. It comes from comparing two notions of equilibrium, with and without a recruiting cost, which is used to capture the role of frictions. . . . The model has the property that . . . the effect of unemployment benefits varies systematically with the tightness of the labor market . . . While a comparison with the hypothetical frictionless alternative is a way of highlighting the workings of this effect, it is the direct estimate of the change in equilibrium from a policy change that we are really interested in.

Diamond's observation is entirely correct. In this chapter, we turn from mechanisms to policy. We discuss policies used in the United States to tackle labor-market problems of three kinds: too much unemployment (job shortages), too many vacancies (labor shortages), and the need to support workers in bad times. We examine the direct effects of these policies on tightness, employment, and unemployment; and we assess how their effectiveness depends on the state of the labor market—tight or slack—which informs their costs and benefits.

We have seen, for instance in figure 10.6D, that rationing unemployment appears only when the labor market is slack enough; otherwise all unemployment is frictional. This dichotomy might make people think that the model's behavior is itself dichotomous—for instance that policies operate as in a DMP model when unemployment is frictional and differently when some unemployment comes from job rationing. But as we show in this chapter, this is not the case.

To illustrate how policies operate in the slackish labor market model, we consider three policies that operate in the labor market through three different channels: labor demand, labor supply, and the Beveridge curve. The three policies are also present on many real-world labor markets: public employment, which relieves job shortages; migration, which relieves labor shortages; and unemployment insurance, which supports workers when jobs are scarce. We analyze these policies using our tightness-employment diagram, and we find that the effects of policies change smoothly with market tightness, and are not affected by the presence or absence of rationing unemployment. In fact, the effects of the policies are governed by a single object in any state: the ratio of the labor supply and labor demand elasticities.

11.1. A labor-demand policy: public employment

In this section, we begin by describing the effects of a policy stimulating labor demand. Specifically, we look at the effect of public employment in the labor market. We characterize the public-employment multiplier, which is a typical statistic to describe the effects of this type of policy. The multiplier measures the number of workers added to the workforce when one worker is added to the public sector.

11.1.1. Modeling public employment

We make the public sector as similar to the private sector as possible to simplify the analysis. Essentially, we assume that the government behaves just like a private firm. If the government wants to hire a worker, it needs to post vacancies exactly like a private firm. The government also pays the same wage as private firms, and faces the same job separation rate. Because private and public jobs are the same, people indiscriminately apply for jobs in the public and private sectors.[1]

[1] These simplifying assumptions are not entirely accurate. In practice, in the United States, public jobs tend to differ from private jobs. A first difference concerns wages: during the New Deal, hourly wages were substantially lower in relief jobs than in private jobs (Neumann, Fishback, and Kantor 2010); and on average, public wages are higher and more rigid than private wages (Quadrini and Trigari 2007). A second difference concerns separation rates: public jobs have substantially lower separation rates than private jobs,

11.1.2. Solution of the model with public employment

The labor supply $l^s(\theta)$ is not affected by public employment: it remains given by (10.1). The only underlying difference is that the number of job vacancies used to compute labor market tightness now includes both private and public vacancies. Because we assumed that public and private jobs are the same, from workers' perspective, the fact that the government is an employer does not change anything. Public employment changes the number of vacancies, but for a given labor market tightness, whether the labor market is rich with private jobs or with public jobs does not matter.

The aggregate labor demand is composed of two elements: the labor demand from private firms and the labor demand from the government. We assume that the government employs a certain number of workers, g, so the public labor demand is simply g. The private labor demand $l^d(\theta)$ remains given by (10.11), because for a given labor market tightness, nothing has changed for private firms. Indeed, the presence of the government affects firms only through tightness. The aggregate labor demand amounts to $l^d(\theta) + g$.

Hence, with public employment, labor market tightness is determined by

$$l^d(\theta) + g = l^s(\theta). \tag{11.1}$$

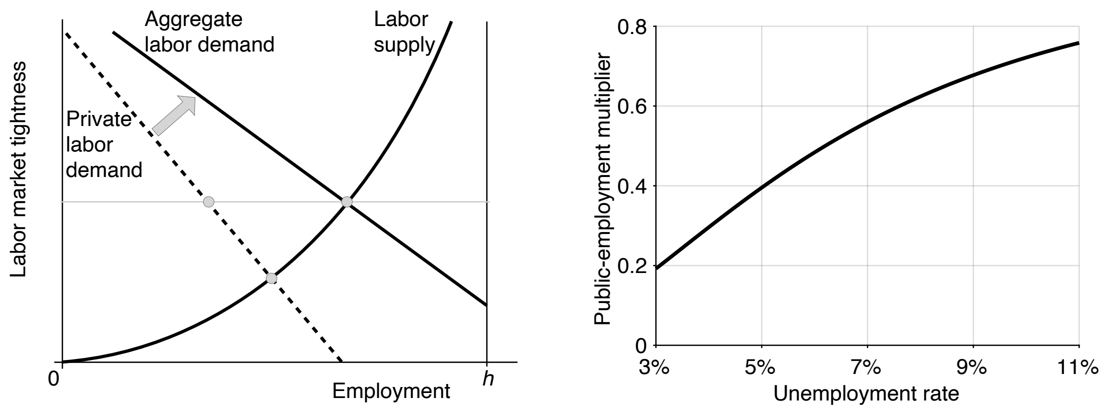
Through the supply-equals-demand condition, public employment affects labor market tightness, and then total employment, given by $l^s(\theta)$, private employment, given by $l^d(\theta)$, and the unemployment rate $u(\theta)$, which remains given by (10.2).

The easiest way to see the impact of public employment is to turn to our tightness-employment diagram (figure 11.1A). The private labor demand corresponds to the labor demand without public employment, and the labor supply is the same as without public employment. The aggregate labor demand is the private labor demand plus public employment, g. The solution of the model is at the intersection of the labor supply and aggregate labor demand.

We immediately see the impact of adding public jobs to the labor market. Adding public jobs boosts labor market tightness and aggregate employment. As a result, it reduces unemployment.

What happens to private employment with the introduction of public jobs? Since tightness rises, private employment, given the private labor demand curve $l^d(\theta)$, falls. Indeed, when tightness is higher, firms are less prone to hiring because it is harder to recruit and thus less profitable to hire. Essentially, as the government starts hiring workers, there are fewer unemployed workers to recruit and more vacancies. This added competition discourages private firms from recruiting: there is crowding out of private

and therefore last substantially longer (BLS 2025t). Of course, if jobs differ in the two sectors, job seekers might direct their search toward a specific sector.



A. Effects of public-job creation

B. Public-employment multiplier in good and bad times

FIGURE 11.1. Public employment in the slackish labor market model

In panel A, the labor supply is given by (10.1), the private labor demand is given by (10.11), and the aggregate labor demand is just the private labor demand plus public employment, g. In panel B, the public-employment multiplier is given by (11.2) under the calibration in table 10.2.

jobs by public jobs. As a result of crowding out, the increase in total employment is not as large as the initial increase in public employment.

What are the distributional effects of public employment? Workers benefit, since the employment rate improves and wages remain the same. Private firms, on the other hand, are hurt: firm profits fall after public jobs are created. Indeed, there are fewer workers in private firms, and a higher tightness, so fewer producers and hence less private output. But the firm profit share is just α, so firm profits are proportional to output: as output falls, so do firm profits. Overall, public employment is a policy that benefits workers but costs firm owners.

11.1.3. Public-employment multiplier

To quantify the effect of public employment on total employment, we compute the public-employment multiplier. The public-employment multiplier is the extra number of workers with a job when one extra worker is hired in the public sector:

$$\frac{dl}{dg}.$$

A multiplier of 1 means that public jobs do not crowd out private jobs at all. A multiplier of 0 means that public jobs crowd out private jobs one-for-one, so total employment does not increase with public employment.

For simplicity and transparency, we evaluate the public-employment multiplier at the point where public employment is $g = 0$. So we compute the effect of the first worker employed in a public job on total employment.

To compute the multiplier, we implicitly differentiate the supply-equals-demand condition (11.1), which defines tightness as an implicit function of public employment (result A.7). We obtain:

$$\frac{dl^d}{d\theta} \cdot \frac{d\theta}{dg} + 1 = \frac{dl^s}{d\theta} \cdot \frac{d\theta}{dg}.$$

This expression gives the change in tightness generated by a change in public employment:

$$\frac{d\theta}{dg} = \frac{1}{\frac{dl^s}{d\theta} - \frac{dl^d}{d\theta}}.$$

Since the change in total employment generated by a change in public employment is obtained by measuring the movement along the labor supply curve, we get

$$\frac{dl}{dg} = \frac{dl^s}{d\theta} \cdot \frac{d\theta}{dg} = \frac{\frac{dl^s}{d\theta}}{\frac{dl^s}{d\theta} - \frac{dl^d}{d\theta}} = \frac{1}{1 - \frac{dl^d/d\theta}{dl^s/d\theta}}.$$

Since we evaluate the multiplier at $g = 0$, we have $l^s(\theta) = l^d(\theta) = l$. So we can multiply the two derivatives in the denominator of the multiplier expression by θ/l and transform them into elasticities. Accordingly, the public-employment multiplier is given by

$$\frac{dl}{dg} = \frac{1}{1 - \epsilon^d_\theta/\epsilon^s_\theta}, \tag{11.2}$$

where ϵ^d_θ is the elasticity of private labor demand with respect to labor market tightness, given by (6.13), and ϵ^s_θ is the elasticity of labor supply with respect to labor market tightness, given by (8.14).

Hence, the properties of the public-employment multiplier are determined by the positive ratio between the tightness elasticity of labor demand and the tightness elasticity of labor supply. With our Cobb-Douglas matching function, the positive elasticity ratio is

$$-\frac{\epsilon^d_\theta}{\epsilon^s_\theta} = \frac{1-\alpha}{\alpha} \cdot \frac{\sigma}{1-\sigma} \cdot \frac{\tau(\theta)}{u(\theta)}. \tag{11.3}$$

As its name indicates, the positive elasticity ratio is strictly positive; it is also strictly increasing in tightness. As we discussed in chapter 8, the elasticity ratio determines the effects of demand and supply shocks in the slackish market model. In the slackish labor market model, it determines all the properties of the public-employment multiplier. This simply means that the public-employment multiplier is determined by the slope of private labor demand relative to labor supply.

From equations (11.2) and (11.3), we see that the public-employment multiplier is strictly positive, strictly less than 1, and strictly decreasing with tightness. Another interesting point is that when the labor market operates efficiently, the elasticity ratio is $-\epsilon_{\theta}^{d}/\epsilon_{\theta}^{s} = (1-\alpha)/\alpha$, which implies that the public-employment multiplier is just α—the production-function shape, equal to one minus the labor share, and calibrated to $\alpha = 0.35$. The reason is that, as (9.11) shows, when the labor market operates efficiently, $\tau/u = (1-\sigma)/\sigma$. (Here the matching function is Cobb-Douglas so the matching elasticity is just $\eta(\theta) = \sigma$.)

The properties of the public-employment multiplier mean that creating public jobs increases total employment and reduces unemployment, but it reduces private employment—as illustrated in figure 11.1A. Furthermore, when tightness is lower, the multiplier is higher: public employment crowds out private employment less and reduces unemployment more. When tightness is higher, the multiplier is lower: public employment crowds out private employment more and reduces unemployment less. Since in the United States tightness is high in booms and low in slumps, we infer that public employment is more effective at reducing unemployment in slumps and less effective in booms—because then it crowds out private employment more.

The geometric intuition for the slack dependence of the multiplier is that relative to the labor demand curve, the labor supply curve is much flatter in a slack market than a tight market, as we saw in figures 8.2A and 8.2B. Hence, when the aggregate labor demand shifts, it boosts mostly employment and not tightness in a slack market, but it boosts mostly tightness and not employment in a tight market.

The economic intuition for the slack dependence is the following. In a slack market, the pool of unemployed workers is large, so when the government steps in and starts hiring workers out of that pool, it does not make a big difference for private firms. Additionally, the government needs few vacancies to hire public workers because the matching process is congested by jobseekers, so private vacancies do not have to compete with many new public vacancies. Overall, crowding out is low, so the multiplier is closer to one.

In a tight market, it is the opposite. The pool of unemployed workers is small, so if the government starts hiring the few workers who are looking for a job, it becomes much more difficult for firms to hire workers. Furthermore, the government needs many vacancies to hire additional workers because the matching process is congested by vacancies, so private vacancies now face more competition. Because of public hiring, it becomes much harder for firms to find workers, so firms are forced to cut back on hiring. Crowding out is substantial, and the multiplier is closer to zero.

Numerically, the public-employment multiplier takes a value of 0.46 at the average unemployment rate of 5.7% and is sharply slack dependent (figure 11.1B). When the unemployment rate is 3%, the public-employment multiplier is quite low, at 0.19. When the unemployment rate reaches 11%, the multiplier quadruples, and reaches 0.76. This means that when the government creates 10 jobs in the public sector, 8.1 private jobs are lost when unemployment is 3%, 5.4 private jobs are lost when unemployment is 5.7%, and only 2.4 private jobs are lost when unemployment is 11%. Accordingly, there are only 1.9 fewer unemployed workers when the unemployment rate is 3%, 4.6 fewer unemployed workers when the unemployment rate is 5.7%, but 8 fewer unemployed workers when the unemployment rate is 11%. The slack dependence—a form of state dependence—is significant.

A final result is that the public-employment multiplier would be 0 in the DMP model. Mathematically, the DMP production function is linear, so $\alpha = 0$, which implies that the elasticity ratio $-\epsilon_{\theta}^{d}/\epsilon_{\theta}^{s}$ diverges to infinity, and the multiplier converges to 0. In the generic slackish model, the number of jobs in the private sector is somewhat limited, so creating jobs in the public sector decreases unemployment. But there is no such lack of jobs in the DMP model, so when the government hires workers, it has absolutely no effect on unemployment: it only crowds out private employment one-for-one, leaving the number of unemployed workers unchanged.

11.1.4. Public employment in the United States

The effects of public employment in the model appear to align well with the effects of public employment observed in the United States.

To fight unemployment during the Great Depression, the Roosevelt administration launched the New Deal to employ millions of people who were without jobs.[2] These workers were employed by the federal, state, and local governments as part of various programs, such as the Works Progress Administration, the Civil Works Administration, and the Federal Emergency Relief Administration. These workers were used, among other things, to build civil infrastructure across the country, including roads, schools, and dams.[3]

The crowding-out mechanism described in this chapter seems to have been at play at the time, and was certainly a concern for businessmen and policymakers. The government was concerned that the public jobs created by the New Deal might take away job applicants from firms, thus making it difficult to hire workers in the private sector. Neumann, Fishback, and Kantor (2010, p. 196) report that

> The government tried to ensure the private industry was not affected by focus-

[2] See for instance Fleck (1999) and (Fishback 2007, pp. 395–401).

[3] In fact, the photos displayed in the preface of this book were taken in 1935–1938 as part of an effort by the US government to assemble an “extensive pictorial record of American life between 1935 and 1944” (Library of Congress 2026). This effort was funded by the Resettlement Administration and then the Farm Security Administration—programs that spun off from the Federal Emergency Relief Administration (Fishback 2007, p. 403)

> ing work relief on building public works that had traditionally been the role of government. Moreover, work relief jobs were made less attractive by keeping work relief payments per hour worked well below private wages in most areas. Relief officials also encouraged workers to accept private employment when available. Despite these practices, private employers complained that WPA work relief made it more difficult for them to hire workers. . . . Wallis and Benjamin [find] that an additional relief job was associated with a reduction of about half of a private job.

In their study of relief work in the United States during the Great Depression, Benjamin and Matthews (1992, chart 8) confirm both predictions of the theory: crowding out of private jobs by public jobs, and stronger crowding out in tighter markets. They compute the number of private jobs crowded out for each public relief job between 1933 and 1939. In 1933, at the worst of the Great Depression, each public job only crowded out about 0.3 private jobs, implying a public-employment multiplier of 0.7. The number of jobs crowded out increased to 0.4 in 1934, 0.6 in 1935–1936, 0.8 in 1937–1938, and 0.9 in 1939. Accordingly, the public-employment multiplier fell from 0.7 to 0.1 as the US economy recovered from the depression. These multiplier fluctuations are not too far off from those presented in figure 11.1B.

11.2. A labor-supply policy: migration

Next, we describe the effects of a policy stimulating labor supply: migration. We characterize the migration multiplier, which, in broad terms, measures the number of workers added to the workforce when one migrant worker is added to the labor force.

11.2.1. Modeling migration

To simplify the analysis, we assume that migrant workers are indistinguishable from local workers: they have the same productivity, command the same wage, and are hired through the same channels. Because local and migrant workers are the same, firms indiscriminately hire both types of workers.[4]

We also adjust the wage norm for the study of migration. We showed in section 5.4.3 that changes in the size of the labor force through migration do not generate noticeable changes in wages. To capture this fact, we slightly adjust the wage norm (10.10) and make

[4]While these simplifying assumptions seem reasonable for internal migration, they might not be accurate for international migration. In the United States, for instance, immigrant workers earn less but find jobs faster than native workers (Albert 2021); moreover, immigrants and natives generally differ in terms of education and experience (Borjas 2003; Ottaviano and Peri 2012).

it unresponsive to the labor force:

$$w^n = \omega \cdot a^{1-\gamma}.$$

11.2.2. Solution of the model with migration

The labor demand $l^d(\theta, w)$ is not affected by migration: it remains given by (10.7). The only underlying difference is that the number of job seekers used to compute labor market tightness now includes both local and migrant workers. Because we assumed that local and migrant workers are the same, from firms' perspective, the presence of migrant workers does not change anything. Because we have adjusted the wage norm to study migration, the expression of the labor demand incorporating the wage norm changes from (10.11) to

$$l^d(\theta) = \left[\frac{(1-\alpha)a^{\gamma}}{\omega}\right]^{1/\alpha} \cdot \frac{1}{[1+\tau(\theta)]^{1/\alpha-1}}. \tag{11.4}$$

The aggregate labor supply is composed of two elements: the labor supply from local workers and the labor supply from migrant workers. We denote the extra number of workers in the labor force due to migration by $x \geq 0$. The local labor supply $l^s(\theta)$ remains given by (10.1), because for a given labor market tightness, everything remains the same for local workers. Since local and migrant workers find and lose jobs at the same rate, they all face the same unemployment rate $u(\theta)$ given by (10.2), so the aggregate labor supply amounts to $[1-u(\theta)](h+x)$, which can be rewritten as

$$l^s(\theta)\left(1+\frac{x}{h}\right), \tag{11.5}$$

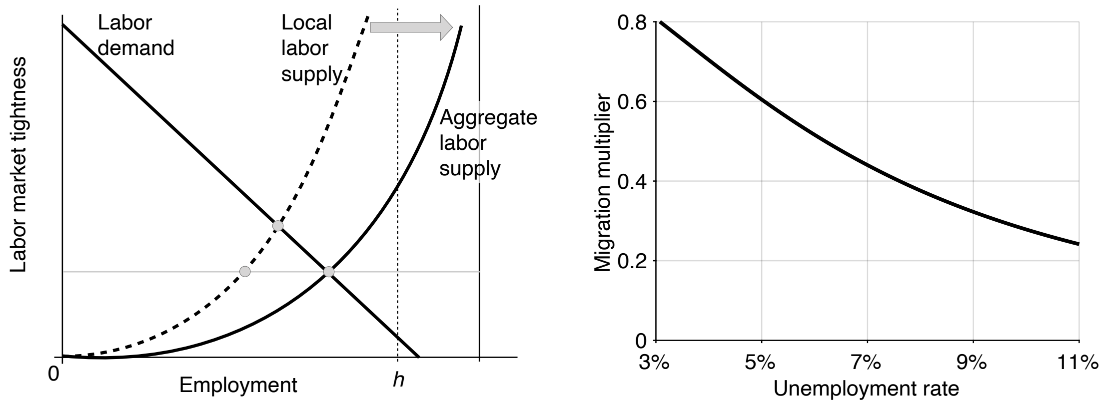
where x/h is the number of migrant workers in the labor market as a share of the local labor force.

Hence, with migration, labor market tightness is determined by

$$l^d(\theta) = l^s(\theta)\left(1+\frac{x}{h}\right). \tag{11.6}$$

Through the supply-equals-demand condition, migration affects labor market tightness, and then total employment, given by $l^d(\theta)$, local employment, given by $l^s(\theta)$, and the unemployment rate $u(\theta)$.

The easiest way to see the impact of migration is to turn to our tightness-employment diagram (figure 11.2A). The local labor supply corresponds to the labor supply without migration, and the labor demand is the same as without migration. The aggregate labor supply is the local labor supply plus the labor supply of migrants, $[1-u(\theta)]\cdot x$. The solution



A. Effects of in-migration

B. Migration multiplier in good and bad times

FIGURE 11.2. Migration in the slackish labor market model

In panel A, the labor demand is given by (11.4), the local labor supply is given by (10.1), and the aggregate labor supply is given by (11.5). In panel B, the migration multiplier is given by (11.7) under the calibration in table 10.2.

of the model is at the intersection of the labor demand and aggregate labor supply.

We immediately see the impact of migration in the labor market. The arrival of migrant workers depresses labor market tightness but stimulates aggregate employment. Because tightness falls, the unemployment rate—faced by both local and migrant workers—is higher. Since the unemployment rate increases and the size of the labor force increases, the total number of unemployed workers—some local and some migrant—increases sharply.

What exactly happens to local workers when migrant workers enter the labor market? Because tightness drops, local workers' job-finding rate drops. A lower job-finding rate means a higher unemployment rate, so migration increases the unemployment rate faced by local workers and decreases their employment rate—given by the local labor supply $l^s(\theta)$. The underlying reason is that after an influx of migrant workers into the labor market, there is the same number of jobs but more job seekers, so it becomes harder to find a job. As a result, a larger fraction of local workers remains unemployed, and a smaller fraction is employed.

What are the distributional effects of migration? Local workers are clearly hurt, since the unemployment rate increases and wages remain the same. Local firms, on the other hand, benefit: firm profits increase once migrant workers enter the labor market. Indeed, firms employ more workers, and tightness is lower, so firms have more producers and generate more output. As firm profits are proportional to output, profits increase. Thus, migration is a policy that benefits local firms but hurts local workers—just the opposite of what happens with public employment.

11.2.3. Migration multiplier

To quantify the effect of migration on total employment, we compute the migration multiplier. The migration multiplier is the percentage increase in employment when one percent of the labor force migrates into the labor market:

$$\frac{h}{l} \cdot \frac{dl}{dx}.$$

A multiplier of 1 means that migrant workers are seamlessly absorbed into the labor market, without affecting local workers. A multiplier of 0 means that employed migrants displace local workers one-for-one, so total employment does not increase with migration.

For simplicity and transparency, we evaluate the migration multiplier at the point without migration: $x = 0$. So we compute the effect of the first migrant worker entering the labor market.

To compute the multiplier, we implicitly differentiate the supply-equals-demand condition (11.6), which defines tightness as an implicit function of migration:

$$\frac{dl^d}{d\theta} \cdot \frac{d\theta}{dx} = \frac{dl^s}{d\theta} \cdot \frac{d\theta}{dx} \cdot \left(1 + \frac{x}{h}\right) + l^s(\theta) \cdot \frac{1}{h}.$$

This expression gives the change in tightness generated by a change in migration:

$$\frac{d\theta}{dx} = \frac{l^s(\theta)}{h} \cdot \frac{1}{\frac{dl^d}{d\theta} - \frac{dl^s}{d\theta} \cdot \left(1 + \frac{x}{h}\right)}.$$

Since the change in total employment generated by a change in migration is obtained by measuring the movement along the labor demand curve, we get

$$\frac{dl}{dx} = \frac{dl^d}{d\theta} \cdot \frac{d\theta}{dx} = \frac{l^s(\theta)}{h} \cdot \frac{\frac{dl^d}{d\theta}}{\frac{dl^d}{d\theta} - \frac{dl^s}{d\theta} \cdot \left(1 + \frac{x}{h}\right)} = \frac{l^s(\theta)}{h} \cdot \frac{1}{1 - \frac{dl^s/d\theta}{dl^d/d\theta} \cdot \left(1 + \frac{x}{h}\right)}.$$

Since we evaluate the multiplier at $x = 0$, we have $l^s(\theta) = l^d(\theta) = l$ and $x/h = 0$. So in particular we can multiply the two derivatives in the denominator by θ/l and transform them into elasticities. Accordingly, the migration multiplier is given by

$$\frac{h}{l} \cdot \frac{dl}{dx} = \frac{1}{1 - \epsilon_\theta^s/\epsilon_\theta^d}, \tag{11.7}$$

where ϵ_θ^d is the elasticity of labor demand with respect to labor market tightness, given by (6.13), and ϵ_θ^s is the elasticity of local labor supply with respect to labor market tightness, given by (8.14).

Hence, just like the public-employment multiplier, the properties of the migration multiplier are determined by the positive ratio between the tightness elasticity of labor demand and the tightness elasticity of labor supply, given by (11.3). This again means that the migration multiplier is determined by the slope of labor demand relative to local labor supply.

From equations (11.7) and (11.3), we see that the migration multiplier is strictly positive, strictly less than 1, and strictly increasing with tightness. Furthermore, when the labor market is operating efficiently, the migration multiplier is just $1 - \alpha$: it is equal to the labor share, so 0.65 in our calibration. The logic is the same as with the public-employment multiplier—it relies on efficiency condition (9.11).

These properties mean that the arrival of migrant workers increases total employment, but it reduces local employment and raises local unemployment—exactly as illustrated in figure 11.2A. Furthermore, when tightness is lower, the multiplier is lower: migrant workers displace local workers and increase local unemployment more. When tightness is higher, the multiplier is higher: migrant workers displace local workers and increase local unemployment less. Since in the United States tightness is high in booms and low in slumps, we infer that migration has more adverse effects on local workers in slumps and less adverse effects in booms.

The geometric intuition for the slack dependence of the multiplier is that relative to the labor demand curve, the local labor supply curve is much flatter in a slack market and much steeper in a tight market, as we saw in figures 8.2C and 8.2D. Hence, when the aggregate labor supply shifts, the resulting drop in tightness has a large adverse effect on local employment in a slack market, but a small adverse effect on local employment in a tight market.

The economic intuition for the slack dependence is the following. In a slack market, the pool of unemployed workers is large, so the matching process is congested by job seekers, and it is easy for firms to recruit workers. The arrival of migrant workers only adds to the job-seeker congestion, and does not make it noticeably easier for firms to hire—since they already have no issue recruiting. Since recruiting is not much easier, firms do not create many new jobs and the multiplier is low. This also means that most of the jobs that migrant workers get are obtained at the expense of local workers.

In a tight market, the opposite occurs. The pool of unemployed workers is small, and the matching process is congested by vacancies. The arrival of migrant workers relaxes the vacancy congestion and makes it much easier for firms to recruit. Facing much easier hiring, firms create many new jobs, so the multiplier is larger. As a result, migrant workers fill new jobs and do not displace local workers much.

Numerically, the migration multiplier takes a value of 0.54 at the average unemployment rate of 5.7% and is sharply slack dependent (figure 11.2B). When the unemployment rate is 3%, the migration multiplier is quite high, at 0.80. When the unemployment rate reaches 11%, the multiplier is much smaller, dropping to 0.24. This means that when a wave of migrants amounting to 10% of the labor force arrive in the labor market, broadly, the number of jobs increases by 8% when unemployment is 3%, by 5.4% when unemployment is 5.7%, and only 2.4% when unemployment is 11%. So, roughly, after the wave of migration, the unemployment rate increases by 2pp when the initial unemployment rate is 3%, by 4.6pp when the initial unemployment rate is 5.7%, and by 7.6pp when the initial unemployment rate is 11%. The slack dependence is again significant.

A final result is that the migration multiplier would be 1 in the DMP model. The mathematical argument is the same as in the case of the public-employment multiplier: the DMP production function is linear ($\alpha = 0$), so the ratio $\epsilon^s_\theta/\epsilon^d_\theta$ converges to 0 (as can be seen from (11.3)), and the multiplier converges to 1. In the generic slackish model, the number of jobs is somewhat limited, so the arrival of migrant workers puts upward pressure on the unemployment rate. But there is no such lack of jobs in the DMP model, so when migrant workers arrive, they are seamlessly absorbed: the unemployment rate remains unchanged, and employment grows in proportion to the number of arrivals. This could be seen in the tightness-employment diagram of figure 10.7A. The labor supply shifts outward with migration, moving the labor market along the horizontal labor demand: tightness does not change and the unemployment rate is unaffected.

11.2.4. Migration in the United States

The slackish labor market model predicts that an influx of new workers into a local labor market raises the unemployment rate among local workers. US evidence collected from domestic migration during the Great Depression and international migration from Cuba supports this prediction.

Boustan, Fishback, and Kantor (2010) study the effect of internal migration in the United States during the Great Depression. Just like native workers protest immigration from abroad, locals in areas where the depression was less severe protested against the arrival of migrants from other less fortunate regions. Locals accused newcomers of taking jobs away from them. For example, Californians tried to scare possible migrants away from the state, highlighting how slack the Californian labor market was. A billboard in Oklahoma carried the message "NO JOBS in California / If YOU are looking for work—KEEP OUT / 6 men for every job / No state relief available for non-residents" (Boustan, Fishback, and Kantor 2010, p. 720).

To assess the impact of internal migrants on local residents, Boustan, Fishback, and Kantor (2010) use variation in the generosity of New Deal programs and extreme weather events to instrument for migrant flows to and from US cities. They find that in-migration prompted some residents to move away from their city and others to lose weeks of work or access to relief jobs. They estimate that for every 100 arrivals, 21 locals were displaced from relief jobs. An additional 19 local workers were forced to shift from full-time to part-time work. And a further 19 residents moved out to other cities. They also find that, just as in-migration diminished the work opportunities of local workers, out-migration improved local opportunities symmetrically.

Perhaps surprisingly, the results from Card (1990)'s famous study on the impact of the Cuban immigrants from the Mariel Boatlift on the Miami labor market in the 1980s are not inconsistent with migration-induced unemployment. Indeed, (Card 1990, p. 251) reports that the unemployment rate for Cuban workers increased drastically after the boatlift:

> Unlike the situation for whites and blacks, there was a sizable increase in Cuban unemployment rates in Miami following the Mariel immigration. Cuban unemployment rates were roughly 3 percentage points higher during 1980-81 than would have been expected on the basis of earlier (and later) patterns.

The fact that the Cuban workers faced a higher unemployment rate is evidence that the labor market could not absorb all new arrivals, and that jobs were somewhat rationed. In fact, the boatlift raised the Cuban labor force in Miami by 20% (Card 1990, p. 246). So for each 100 new arrivals into Miami's Cuban labor force, $3/20 \times 100 = 15$ local Cubans were pushed into unemployment.

Relatedly, Anastasopoulos et al. (2021, figure 5) compute the response of labor-market tightness in Miami upon the arrival of Cuban immigrants from the Mariel Boatlift in the 1980s. Relative to the tightness in comparable cities, they find that Miami tightness fell by 40% after the Mariel Boatlift. This implies that it became much harder for local workers to find jobs after the boatlift.

11.2.5. Departure from the academic consensus on migration

This slackish model and empirical evidence presented in this section suggest that in-migration raises local unemployment. Yet, academic economists tend to reject the notion that in-migration negatively affects local workers (Raphael and Ronconi 2007). Federman, Harrington, and Krynski (2006, p. 302) observe for instance:

> One of the central questions in the debate over immigration policy is whether immigrants adversely affect labor market outcomes for natives. Some Americans believe they do, worrying that immigrants take jobs away from native workers. Most of the empirical evidence produced by economists, however, does not support these concerns.

The general perception by academic economists that immigration has no adverse effect on local workers might be an indirect consequence of not having models to think about the effect of migration on unemployment—neither the Walrasian model nor the DMP model allows immigration to raise unemployment. If such a phenomenon was present in existing models, researchers might have paid more attention to the empirical findings in existing studies. As Michaillat (2025, table 1) shows, experimental evidence from different countries and historical periods, involving both international and domestic migrations, paints a consistent pattern: an influx of new workers into a local labor market raises the unemployment rate among local workers.

Because existing models only allow migration to affect wages, the empirical literature has focused on wages, which do not appear to respond much to immigration. This has created a puzzle: "The depth of public concern over immigration is somewhat puzzling, given that most studies find only small economic impacts on the native population" (Card, Dustmann, and Preston 2012, p. 78). The adverse impact of immigration on unemployment is not emphasized at all in economic research—but it is often there, offering a resolution of the puzzle.

The absence of migration-induced unemployment in existing migration models severely limits the scope of empirical inquiry about migration. A wonderful example of such limitation appears in a seminal paper by Scheve and Slaughter (2001). The paper documents people's perceptions about immigration. Scheve and Slaughter (2001, footnote 8) candidly admit that they do not examine people's perceptions of job stealing because such perceptions would be inconsistent with the effects of immigration in standard models:

> The 1992 National Election Studies survey asked other questions about immigration that we do not analyze. For example, respondents were asked whether they think Asians or Hispanics 'take jobs away from people already here.' We do not focus on this question because its responses cannot clearly distinguish among our three competing economic models. All our models assume full employment, so no natives could have jobs 'taken away' by immigrants.

The response of respondents in the 1992 National Election Studies is presented in table 11.1. More than 80% of respondents are indeed worried that Hispanic and Asian immigrants take jobs away from them. It is quite possible that the existence of migration-induced unemployment has not received the attention it deserves just because standard models do not feature it.

The public anxieties about migration often focus on the idea that migrant workers "steal jobs" from local workers (as displayed in table 11.1). Given this section's results, these anxieties are unsurprising. In a slackish model, the job-finding rate and employment rate of local workers both fall when immigrants arrive. So local workers in the model might feel that immigrants steal their jobs: the fraction of local workers who have a job is lower, and the fraction who remain unemployed is higher. The reason is that immigrant workers are now employed in some of the available jobs, relegating local workers to unemployment.

TABLE 11.1. Popular perceptions about immigration

The growing number of these immigrants takes jobs away from people already here.	How likely is it? Extremely	Very	Somewhat	Not at all
Hispanics	20%	29%	38%	13%
Asians	19%	30%	37%	13%

Source. 1992 American National Election Studies survey. The responses to the question about Hispanic workers (question Q4c) can be accessed at `https://electionstudies.org/data-tools/anes-variable/variable.html?year=1992&variable=V926238`. The responses to the question about Asian workers (question Q5c) can be accessed at `https://electionstudies.org/data-tools/anes-variable/variable.html?year=1992&variable=V926241`.

Of course the labor market is not completely zero-sum: the number of jobs is not fixed, so displacement is not one-for-one. But displacement does exist, and is especially strong in bad times. In that way, the labor market is somewhat zero-sum, and it becomes more zero-sum in bad times, perhaps explaining why the pushback against immigration intensifies in depressed labor markets. Hoffman (1974) shows for instance that high unemployment and cut-throat competition for existing jobs during the Great Depression intensified anti-Mexican sentiment, leading to the pressured or forced repatriation of half a million Mexicans from the United States between 1929 and 1939. As Hoffman (1974, p. 2) explains, "With the onset of the depression, pressure mounted to remove aliens from the relief rolls and . . . from the jobs they were said to hold at the expense of American citizens."

11.3. A Beveridge-curve policy: unemployment insurance

Migration affects the labor supply but not the Beveridge curve, which is invariant to the size of the labor force. We finally look at a policy that affects the labor supply via the Beveridge curve: unemployment insurance.

11.3.1. Modeling unemployment insurance

Unemployment insurance (UI) improves the lot of unemployed workers by providing them with benefits when they have no labor income. However, UI has two possible adverse effects in the labor market.

First, UI might depress labor demand by raising wages. Indeed, UI elevates the outside option of job seekers, so if they bargain with firms, they might be able to obtain higher wages. However, we showed in section 5.4.3 that changes in UI have no observable effects on wages. We therefore omit this channel here and assume that wages are unaffected by UI, and remain given by (10.10).

Second, UI might depress labor supply by reducing job-search effort and therefore keeping workers in unemployment longer. This effect has been well documented (Krueger and Meyer 2002). For instance, using the Continuous Wage and Benefit History dataset, which contains administrative data from 12 US states, Katz and Meyer (1990, p. 45) find that a one-week increase in benefit duration increases the average duration of the unemployment spells of UI recipients by about 0.2 weeks. Accordingly, we assume that UI reduces the job-search effort e of unemployed workers.

We've been looking at policies that stimulate employment all along in this chapter. For consistency, here too we consider a policy change that boosts employment: a cut in UI benefits.

11.3.2. Solution of the model with unemployment insurance

The labor demand $l^d(\theta, w)$ is unaffected by UI, and wages are unaffected by UI, so the expression of the labor demand incorporating the wage norm remains (10.11). The labor supply now incorporates the job-search effort of unemployed workers, which enters the labor supply as shown in (10.27). Hence, with UI, labor market tightness is determined by

$$l^d(\theta) = l^s(\theta, e). \tag{11.8}$$

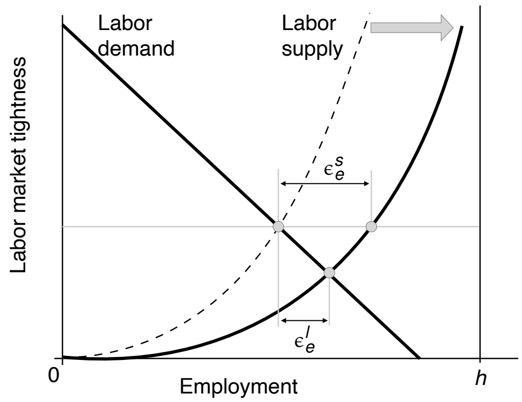
A UI cut boosts job-search effort e, and then through the supply-equals-demand condition, affects labor market tightness, total employment, given by $l^d(\theta)$, and the unemployment rate, $u = 1 - l^d(\theta)/h$.

The easiest way to see the impact of a cut in UI benefits is to turn to our tightness-employment diagram (figure 11.3A). The UI reduction stimulates labor supply without affecting the labor demand. Hence, it reduces labor market tightness and stimulates employment, as the labor market moves down the labor demand curve. Because employment rises, the unemployment rate is lower: unemployed workers search harder, so they find jobs faster, which shrinks the unemployment pool. However, because tightness falls, the increase in employment is not as large as the shift in labor supply: after shifting, the labor market moves down the labor supply to the new intersection with labor demand.

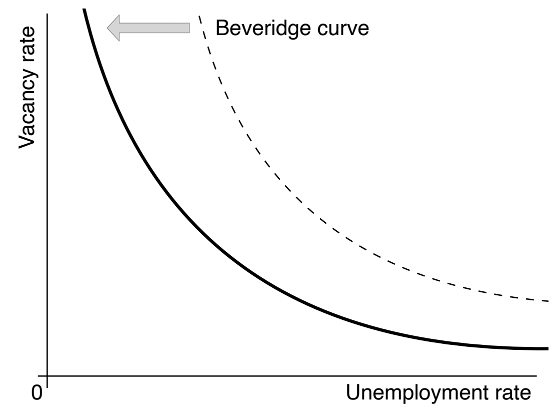
Although that's not visible in the tightness-employment diagram, a cut in UI also improves the location of the Beveridge curve, moving it inward in a vacancy-unemployment diagram (figure 11.3B). Indeed, with job-search effort e, the Beveridge curve is implicitly defined by the balanced-flow condition:

$$\lambda(1 - u) = m(e \cdot u, v). \tag{11.9}$$

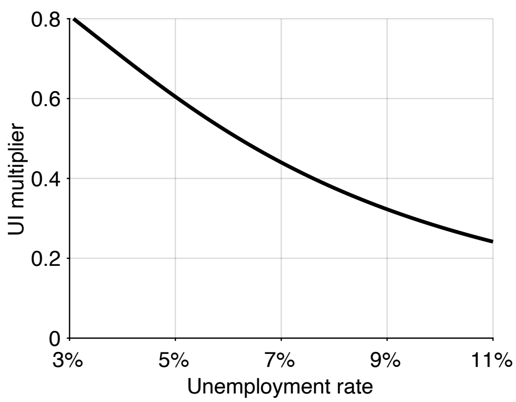
As the value of e increases, lower values of u and v are required to satisfy the condition.



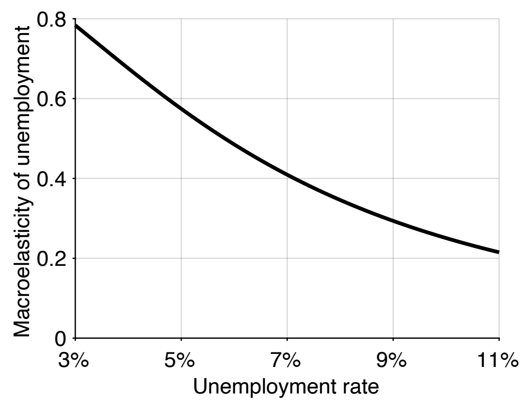
A. Effects of a UI reduction



B. Effects on the Beveridge curve



C. UI multiplier in good and bad times



D. Macroelasticity of unemployment in good and bad times

FIGURE 11.3. UI in the slackish labor market model

In panel A, the labor demand is given by (10.11), and the labor supply is given by (10.27). In panel B, the Beveridge curve is implicitly given by (11.9). In panel C, the UI multiplier is given by (11.10) under the calibration in table 10.2. In panel D, the macroelasticity of unemployment with respect to job-search effort is given by (11.11) under the calibration in table 10.2.

11.3.3. UI multiplier

To quantify the effect of UI on total employment, we compute the UI multiplier. The UI multiplier is the ratio of the elasticity of employment with respect to effort to the elasticity of labor supply with respect to effort:

$$\frac{\epsilon_e^l}{\epsilon_e^s}.$$

The UI multiplier has a more complicated definition than the public-employment and migration multipliers because UI operates in the labor market in a slightly more complicated way. A UI multiplier of 1 means that the increase in job-search effort directly translates into an increase in employment: the increase in employment is the same as the shift in labor supply generated by higher effort. A multiplier of 0 means that the increase in job-search effort has no effect on employment.

We can rewrite the UI multiplier in an informal way that might nonetheless be useful for intuition:

$$\frac{\epsilon_e^l}{\epsilon_e^s} = \frac{\frac{e}{l} \cdot \frac{dl}{de}}{\frac{e}{l^s} \cdot \frac{\partial l^s}{\partial e}} = \frac{dl}{dl^s|_\theta},$$

where we simplified the expression using the fact that we take elasticities around the point where $l^s(\theta, e) = l$. In that way, we see that the UI multiplier is the change in employment relative to the shift in labor supply triggered by high search effort.

For simplicity and transparency, we evaluate the UI multiplier at the point where effort is $e = 1$. So we compute the effect of the initial departure of job-search effort from its normal value.

To compute the UI multiplier, we implicitly differentiate the supply-equals-demand condition (11.8) with elasticities (result B.10):

$$\epsilon_\theta^d \cdot \epsilon_e^\theta = \epsilon_\theta^s \cdot \epsilon_e^\theta + \epsilon_e^s.$$

This expression gives the elasticity of tightness with respect to job-search effort:

$$\epsilon_e^\theta = \frac{\epsilon_e^s}{\epsilon_\theta^d - \epsilon_\theta^s}.$$

Since the change in employment generated by a change in effort is obtained by measuring the movement along the labor demand curve, we get

$$\epsilon_e^l = \epsilon_\theta^d \cdot \epsilon_e^\theta = \epsilon_e^s \cdot \frac{\epsilon_\theta^d}{\epsilon_\theta^d - \epsilon_\theta^s} = \epsilon_e^s \cdot \frac{1}{1 - \epsilon_\theta^s / \epsilon_\theta^d}.$$

Accordingly, the UI multiplier is given by

$$\frac{\epsilon_e^l}{\epsilon_e^s} = \frac{1}{1 - \epsilon_\theta^s/\epsilon_\theta^d}, \tag{11.10}$$

where ϵ_θ^d is the elasticity of labor demand with respect to labor market tightness, given by (6.13), and ϵ_θ^s is the elasticity of local labor supply with respect to labor market tightness, given by (8.14).

The UI multiplier is exactly the same as the migration multiplier, given by (11.7), so it has the same properties: the UI multiplier is strictly positive, strictly less than 1, and strictly increasing with tightness.

This means that a cut in UI, by stimulating search effort, increases total employment, but not as much as the shift in labor supply would suggest. The increase in employment is less than the supply shift because labor market tightness drops—as displayed in figure 11.3A. Furthermore, when tightness is lower, the multiplier is lower: a cut in UI stimulates employment and reduces unemployment less. Since the US labor market tightness is high in booms and low in slumps, we infer that a cut in UI is more potent to raise employment in booms and less potent in slumps.

The economic intuition for the slack dependence is the following. In a slack market, the pool of unemployed workers is already large, so the matching process is congested by job seekers, and it is easy for firms to recruit workers. Forcing jobless workers to search harder only adds to the job-seeker congestion, and does not make it noticeably easier for firms to hire—since they already have no issue recruiting. As recruiting is not noticeably easier, firms do not create many new jobs, so employment barely increases despite the higher effort: the multiplier is low. At that point the labor market operates almost like a rat race: by searching harder job seekers improve their position in the queues for jobs, but this improved position only comes at the expense of other job seekers. The number of jobs is almost fixed—almost independent of labor supply.

In a tight market, the opposite occurs. The pool of job seekers is small, and the matching process is congested by vacancies. The increase in job-search effort drastically relaxes the vacancy congestion and makes it much easier for firms to recruit. Facing much easier hiring, firms create many new jobs, so the multiplier is larger. Job seekers fill the new jobs and do not noticeably displace other job seekers in the queues for jobs.

The UI multiplier is the same as the migration multiplier, so it takes the same numerical values (figure 11.3C). Thus, the response of employment to a UI cut is 80% of the response of labor supply when the unemployment rate is 3%, 54% at the average unemployment rate of 5.7%, and only 24% when the unemployment rate reaches 11%. This means that as the unemployment rate increases, less and less of the labor-supply shift produced by the UI cut translates into an increase in employment and a reduction in unemployment.

Landais, Michaillat, and Saez (2018b,a) refer to ϵ_e^l as macroelasticity, including effects of individual search effort and labor market tightness, and ϵ_e^s as a microelasticity, including only effects of individual search effort. This section therefore shows that the macroelasticity is less than the microelasticity, especially in slack labor markets.

There is a macroelasticity that is closely related to the UI multiplier and that we can compute numerically to finalize the results: the macroelasticity of unemployment with respect to job-search effort, which captures the percentage decrease in unemployment when job-search effort increases by 1%:

$$-\epsilon_e^u = \frac{1-u}{1-\epsilon_\theta^s/\epsilon_\theta^d}. \tag{11.11}$$

The macroelasticity of unemployment is just $1-u$ times the UI multiplier, so it is quite close in value to it, and has the same properties: it is strictly positive, strictly less than 1, and strictly increasing with tightness.[5] This means that by raising job-search effort, a cut in UI reduces unemployment, and the reduction is especially strong in tight labor markets. In the calibration, the macroelasticity decreases from 0.78 when the unemployment rate is 3%, to 0.51 at the average unemployment rate of 5.7%, and down to only 0.21 when the unemployment rate reaches 11%. This means that a 10% increase in job-search effort reduces the unemployment rate by 7.8%, then 5.1%, and then only 2.1% as the unemployment rate grows from 3% to 11%.

A final result is that the UI multiplier would be 1 in the DMP model—just like the migration multiplier. The reason is again the same: there is no lack of jobs in the DMP model, so when unemployed workers search harder, they are seamlessly absorbed into new jobs: tightness remains unchanged, and employment grows as much as the labor supply shifts. This could be seen in the tightness-employment diagram of figure 10.7A. The labor supply shifts outward with a cut in UI, moving the labor market along the horizontal labor demand: tightness does not change and the employment increase is determined by the labor supply shift. In the DMP model, the macroelasticity of unemployment with respect to job-search effort, given by (11.11), is $-\epsilon_e^u = 1-u$, so it is always approximately equal to 1.

11.3.4. Unemployment insurance in the United States

The slackish labor market model predicts that reducing UI, by incentivizing job search, raises employment and reduces the unemployment rate. However, the effect on employment is less than the underlying shift in labor supply because labor market tightness

[5] We obtain the macroelasticity in three steps. First, given (10.27), we see that $\epsilon_e^s = 1 - l/h = u$. Second, since $u = 1 - l/h$, we know that $\epsilon_e^u = -\epsilon_e^l \cdot (1-u)/u$. Third, we conclude that $-\epsilon_e^u = (1-u)\cdot(\epsilon_e^l/\epsilon_e^s)$, which yields (11.11) given that $\epsilon_e^l/\epsilon_e^s$ is given by (11.10).

drops. US evidence collected during the Great Recession and before appears in line with the prediction.

The most direct evidence comes from the work by Marinescu (2017), who measures how labor market tightness responded to the UI extensions implemented in the United States between 2008 and 2011. As we discussed in section 10.4.7, with varying search effort, labor market tightness is the ratio of job vacancies to aggregate search effort, which is itself individual search effort times number of job seekers. Typically it's difficult to measure such a tightness because job-search effort is unobservable. But Marinescu is able to do it using data from the job board `CareerBuilder.com`, which contains information on job vacancies posted by firms and job applications sent by job seekers, which serves as a measure of job search effort. With an event-study design, she finds that the UI extensions reduced the number of job applications without affecting the number of vacancies. This means that an increase in UI raised labor market tightness, and that a decrease in UI would lower tightness, as the slackish model predicts.

Another way to measure the effect of UI on tightness is to examine the effect of changing UI on workers who do not receive UI. If there are spillovers between insured and uninsured job seekers, we would know that labor market tightness has changed. For instance, if uninsured job seekers exit unemployment faster when insured job seekers in their market receive more generous UI, we would learn that the increase in UI has raised tightness. Levine (1993) studies such spillovers in various US datasets and finds that when UI increases in certain labor markets, insured job seekers in those markets search less, and uninsured job seekers in the same markets find jobs faster. This implies that uninsured job seekers benefited from a higher job-finding rate, and therefore that labor market tightness increased after UI increased. Conversely, it means that tightness would decrease after UI decreased, as the slackish model says.

Finally, the slackish model predicts that the macroelasticity of unemployment with respect to job-search effort is smaller in slacker markets. This finding reflects this property of the model: when the labor market is slack, an increase in job-search effort cannot lower unemployment much as jobs are rationed. Toohey (2017) obtains evidence supporting this prediction. He exploits variations in job-search requirements in UI programs across US states and over time. He finds that when search requirements are more stringent, UI recipients search more and find jobs faster. But increasing search effort has a smaller effect on the unemployment rate in bad times than in good times, as the slackish model predicts.

11.4. Summary

In this chapter, we study the effects of common labor policies in the slackish labor market model of chapter 10: public employment, migration, and unemployment insurance. The model connects these policies although they operate in the labor market through different channels: labor demand, labor supply, and Beveridge curve. The aim is to understand how these policies might affect unemployment and tightness.

The main finding is that all these policies affect both tightness and unemployment, and are strongly state dependent. Because jobs are somewhat scarce, public employment does not completely crowd out private employment and so reduces unemployment. The reduction is especially large in bad times. For the same reason, in-migration is not completely absorbed in the labor market and so raises unemployment. The rise is especially large in bad times, when jobs are scarcer. Finally, the increase in unemployment caused by a rise in unemployment insurance is not as large as what the reduction in job-search effort would suggest. The increase in unemployment is especially small in bad times, when frictional unemployment is low. Formally, the effects of all the policies are governed by a single statistic: the ratio of the labor supply and labor demand elasticities.

None of these policy properties can be true without the others—they all arise from job rationing. This means that, within the confines of the model, it would not be logical to believe jointly (as a Democrat might) that government spending can stimulate employment and that local workers are unaffected by immigration. It would not be logical either to believe jointly (as a Republican might) that unemployment insurance has a substantial adverse effect on employment and that local workers are adversely affected by immigration.

CHAPTER 12.

Full employment and unemployment gap

We have seen that in a slackish market model there is no guarantee that the market is operating efficiently. This means that in theory, the labor market may be socially inefficient: the amount of unemployment might be inefficiently high or low. Is that the case in practice in the United States?

To answer this question, we compute the socially efficient rate of unemployment over the past century—which corresponds to the full-employment rate of unemployment (FERU). We then compare the FERU to the actual US unemployment rate to measure the unemployment gap: the distance of the US labor market from social efficiency. We will see that most of the time, the US labor market is inefficiently slack—operating below full employment.

12.1. Efficient labor market tightness

In this chapter, we produce a measure of the efficiency or inefficiency of the US labor market. To do that, we apply the efficiency analysis of chapter 9 to the labor market.

12.1.1. Social planner's problem

The social planner aims to organize production as efficiently as possible. This means that the planner must allocate as efficiently as possible across production, recruiting, and job seeking to maximize the amount of production in the labor market.

Our starting point is the labor market's productive capacity, which is given by the labor force: h workers.

There is slack in the labor market, so a share $u \in (0, 1)$ of all workers remain unemployed. The other $(1-u)h$ workers are employed by firms.

However, labor is used not only for production but also for recruiting via job vacancies. Recruiting takes work: designing and advertising job vacancies, screening and interviewing candidates, and negotiating contracts. Workers involved in recruiting are unable to spend their entire time contributing to social welfare.

As usual, we assume that the labor market features a Beveridge curve. The Beveridge curve links the vacancy rate v—the number of job vacancies as a share of the labor force—to the unemployment rate u. The function $v(u)$ is strictly decreasing and convex.

Moreover, filling job vacancies requires resources, measured by the recruiting cost $\kappa > 0$. The recruiting cost is the number of workers allocated to one job vacancy per unit time.

We see that not all employed workers are directly productive. Some workers are instead devoted to recruiting and, as they do not produce output, they do not contribute to social welfare. So we must subtract the $\kappa v h$ recruiters from the $(1-u)h$ workers employed by firms to obtain the number of producers: $(1-u)h - \kappa v h$.

We assume that social welfare is determined by production in the labor market and thus by the number of producers. Accordingly, social welfare is given by

$$\mathcal{M}(u) = \left[1 - u - \kappa v(u)\right] h.$$

The social planner's problem is to pick an unemployment rate u to maximize social welfare $\mathcal{M}(u)$.

12.1.2. Sufficient-statistic formula

The social planner's problem is exactly the same as the market planner's in chapter 9. Hence, the efficiency formula is the same, given by equation (9.5):

$$\left(\frac{v}{u}\right)^* = \frac{1}{\beta\kappa}, \tag{12.1}$$

where κ is the amount of labor devoted to recruiting per job vacancy, β is the elasticity of the Beveridge curve.

In the labor market model, the ratio v/u corresponds to the labor market tightness θ. Hence, the formula becomes

$$\theta^* = \frac{1}{\beta\kappa}. \tag{12.2}$$

12.1.3. Calibration of the sufficient statistics

We now calibrate the efficiency formula for the US labor market.

First, as we saw in figure 2.8, the labor-market Beveridge curve is a rectangular hyperbola, so we set the elasticity to $\beta = 1$.

Second, as we saw in chapter 2, it takes about 1 full-time worker to service a job vacancy, so $\kappa = 1$.

Since $\beta = 1$ and $\kappa = 1$, the efficient labor market tightness is therefore given by $\theta^* = 1$. Thus, the US labor market is inefficiently slack whenever labor market tightness is below 1, inefficiently tight whenever tightness is above 1, and efficient when tightness equals 1.

In section 9.8, we saw that the welfare analysis could be extended to include nonzero social value of unemployment. However, we noted in chapter 2 that beyond lost production, the idleness of labor generates significant psychological costs. We saw that the value of home production by unemployed labor minus the psychological cost of idleness is about zero, so the social value is about 0. This is why we keep things simple and do not insert a nonzero social value of unemployment here.

12.1.4. Interpretation of the efficiency condition

Social efficiency corresponds to a labor market tightness of 1, so it prevails when the unemployment and vacancy rates are equal ($u = v$). When they are not equal, the labor market is operating inefficiently. The labor market is inefficiently tight when there are more job vacancies than job seekers ($v > u$). In that case, increasing u and reducing v would increase social welfare. The labor market is inefficiently slack when there are more job seekers than job vacancies ($u > v$). Then, reducing u and increasing v would increase social welfare.

12.2. Equivalence between efficiency and full employment

Before applying the theory to US data, it is useful to note that the concept of efficiency studied in this chapter and the following ones is synonymous with the notion of full employment used in US laws. Accordingly, the unemployment gap computed in this chapter is a marker not only of the inefficiency of the labor market, but also of the distance from full employment.

12.2.1. Full employment in US law

In the United States, the federal government and its central bank are mandated to stabilize the labor market at full employment by the Employment Act of 1946, the Federal

Reserve Reform Act of 1977, and the Full Employment and Balanced Growth Act of 1978 (US Congress 1946, 1977; Congress 1978).[1]

12.2.2. Full employment in the model

The Employment Act and Full Employment and Balanced Growth Act clearly state that achieving full employment is a way to maximize economic well-being, so we translate full employment in the model as social efficiency—a state of affairs in which social welfare is maximized. The Employment Act states for instance that reaching full employment is designed "to foster…the general welfare" (US Congress 1946, p. 1)—so in theoretical words, to reach efficiency. This is also the spirit of the Full Employment and Balanced Growth Act. Congress (1978, p. 1888) was motivated to act by the fact that "the Nation has suffered substantial unemployment and underemployment, …over prolonged periods of time, imposing numerous economic and social costs on the Nation." The hope of the law is to ensure that in contrast, with appropriate policy, the country is not deprived of "the full utilization of labor…and the related increases in economic well-being that would occur under conditions of genuine full employment."

We therefore interpret full employment as the allocation of labor that maximizes social welfare. Using the result above, we can state that the labor market is at full employment whenever the unemployment and vacancy rates are equal ($u = v$).

12.2.3. Connection to Beveridge's criterion for full employment

Famously, Beveridge (1944, p. 18) defined full employment to mean that "unemployment is reduced to short intervals of standing by, with the certainty that very soon one will be wanted in one's old job again or will be wanted in a new job that is within one's powers." He then stated that "Full employment…means having always more vacant jobs than unemployed men, not slightly fewer jobs."

Beveridge's criterion has been used in academic research. Early on, Rees (1957, chart 5) applied Beveridge's criterion to the United States. He computed the ratio between number of vacancies and number of unemployed workers and examined under which conditions the US labor market reached full employment (a ratio above 1). More recently, Benigno and Eggertsson (2023) used Beveridge's criterion to explain the kink that they observed in the US Phillips curve.

Beveridge's criterion is well known in government circles as well. During a press conference in 2022, Jerome Powell, the chair of the Federal Reserve, was asked by jour-

[1]"Full employment" is sometimes referred to as "maximum employment." During the debate preceding the Employment Act, "maximum employment" was adopted as a less stringent goal than "full employment" (Duboff 1977, p. 6). In 1978, the Full Employment and Balanced Growth Act amended the Employment Act and replaced "maximum employment" by the more ambitious target of "full employment" (Weir 1987, p. 398).

nalist Howard Schneider which tightness the Fed might target, Powell (2022, pp. 12–13) responded: "So in terms of the vacancy-to-unemployment ratio, we don't have a goal in mind.... I think when we got to one-to-one in the, you know, in the late teens, we thought that was a pretty good number." A vacancy-to-unemployment ratio of 1 corresponds to Beveridge's criterion for full employment.

How did Beveridge come up with this criterion? The logic was simple: he interpreted job vacancies as unmet labor demand and accordingly defined full employment as the point where there would be enough labor demand for all workers, so the point where job vacancies exceed job seekers. Beveridge then explains that some workers are always unemployed "however high the demand for labor" because there always is some "frictional unemployment."

Our criterion for full employment shares some similarities but is more symmetric than Beveridge's criterion. Beveridge thought that the labor market could be either too slack, when $v < u$ or at full employment, when $v \geq u$. By contrast, we find that having more vacancies than job seekers is a sign of inefficient tightness, just like having more job seekers than vacancies is a sign of inefficient slack. We argue that full employment is achieved exactly when $v = u$, and that on either side of full employment, the labor market is operating inefficiently: either too slack, when $v < u$, or too tight, when $v > u$.

Although our full-employment criterion overlaps with Beveridge's, the logic is entirely different. We do not interpret vacancies as unmet labor demand, but we use vacancies as an indirect measure of the labor devoted to recruiting. In the United States, it takes roughly one recruiter to handle one vacancy, so the number of vacancies tracks the amount of labor devoted to recruiting. This empirical regularity is what we use in deriving the FERU. If we had a direct measure of man-hours devoted to recruiting, we would not even need vacancies to compute the FERU.

Indeed, in our theory of slack, job vacancies do not represent unmet labor demand: they represent recruiting effort—an effort to find new workers through the matching process. As we saw in chapter 10, if a firm wants to recruit one worker this month and knows that a vacancy is only filled with probability 1/2, the firm will post 2 different vacancies to hire one worker in expectation. And each vacancy will require time and effort from the firm's human-resource workers to be filled.

Our approach appears to be consistent with reality. ZipRecruiter's Julia Pollak explained in an interview with CBS News that firms routinely post several vacancies for each hire they are planning to make (Ivanova 2023):

> When you have fewer candidates per opening, you have to be more creative. The high openings figure does partly reflect recruiting intensity, and not actual roles and seats and slots.

12.3. Efficient, full-employment rate of unemployment

We have established that when the labor market operates efficiently—which corresponds to a state of full employment—labor-market tightness is 1. In this section, we construct the rate of unemployment at full employment—the FERU. Having the rate of unemployment when the labor market operates efficiently, at full employment, is useful because researchers and policymakers more commonly think about unemployment than about labor market tightness, and because the effects of stabilization policies on unemployment are better understood than those on labor market tightness.

To translate the efficient tightness into an efficient unemployment rate, we need to specify the Beveridge curve. As we discussed in chapter 2, since the Beveridge elasticity is 1, the Beveridge curve is a rectangular hyperbola:

$$vu = A,$$

where $A > 0$ governs the location of the Beveridge curve. At efficiency, the labor market tightness is 1, so $v^* = u^*$, which imposes

$$u^* = v^* = \sqrt{A}. \tag{12.3}$$

To derive the expression for the FERU, we start from equation (12.3) and substitute A out of it by using the Beveridge curve $A = uv$. We find that the FERU is the geometric average of the unemployment and vacancy rates:

$$u^* = \sqrt{uv}. \tag{12.4}$$

Since $uv = A > 0$, expression (12.4) implies that the FERU is strictly positive. Hence, full employment should not be interpreted as zero unemployment.

A first reason why full employment does not mean zero unemployment is that zero unemployment is infeasible. Indeed, the Beveridge curve prevents unemployment from ever reaching zero. Since vacancies require labor, the number of vacancies posted in the labor market is bounded above. Accordingly, the Beveridge curve $u = A/v$ implies that the unemployment rate is bounded below by some positive number.

The fact that labor market flows impose a minimum level of unemployment—and therefore that full employment cannot be zero unemployment—has been known for a long time. Beveridge (1944, p. 125) realized that "however great the unsatisfied demand for labor, there is an irreducible minimum of unemployment, a margin in the labor force required to make change and movement possible." As a result, "even under full employment, there will be some unemployment,. . . on each day some men able and willing to work will not be working." Robinson (1946, pp. 169–170) made the same observation: "In a changing world there are always bound to be, at any moment, some workers who have left one job and have not yet found another. . . . Changes in occupation for personal reasons will always be going on. So long as such shifts in employment are taking place there is always likely to be some unemployment even when the general demand for labor is very high."

A second reason why full employment does not mean zero unemployment is that zero unemployment is undesirable. Unemployment is clearly a waste of economic resources as people who would like to work are not able to be productive. Yet, reducing the unemployment rate to zero is not desirable because it would require diverting a vast amount of labor toward recruiting.

Equation (12.3) shows that full employment occurs when unemployment and vacancy rates are equal. The equation also shows that the location of the Beveridge curve, A, solely determines these rates at full employment. In a dynamic labor-market model the Beveridge curve's position is determined by the job-separation rate and the efficacy of the matching function (chapter 10). Any change in either parameter shifts the curve, affecting the FERU. However, which parameter causes the shift is irrelevant; only the shift itself matters for welfare and the FERU.

12.4. Inefficiency of the US labor market over the past century

In this section we investigate whether the US labor market has operated efficiently over the past century or not. We find that it has not.

12.4.1. Labor market tightness over the past century

We start by looking at the US labor market tightness over the past century. Over 1929–2024, labor market tightness averages 0.73. Tightness is extremely volatile before the end of World War 2. Tightness reached its most extreme values during that period: tightness plunged to 0.03 during the Great Depression and climbed all the way to 6.8 at the end of World War 2. Things are more quiet in the postwar period. Tightness peaked at 1.60 in 1953Q1, during the Korean War, and it bottomed at 0.16 in 2009Q3, during the Great Recession. Twice, the labor market reached full employment just before entering a recession. This happened before the 1973–1975 recession (tightness peaked at 0.99 in 1973Q3) and before the 2001 dot-com recession (tightness peaked at 1.01 in 2000Q1). In the aftermath of the coronavirus pandemic, the US labor market has become historically tight. In 2022Q2, tightness reached 1.98, a value which it had last reached in 1945.

A first finding is that over almost a century, tightness is generally inefficiently low, so the tightness gap is negative. Furthermore, this gap is exacerbated in recessions (figure 12.1). This means that the labor market does not generally operate efficiently. Instead, it is generally inefficiently slack, especially during recessions.

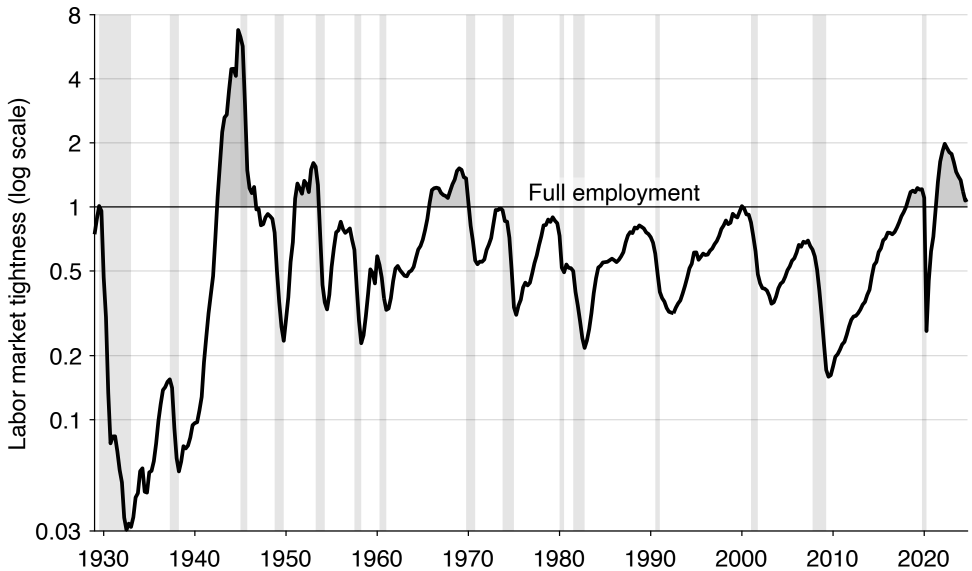


FIGURE 12.1. Labor market tightness in the United States, 1929–2024

Labor market tightness is v/u where the vacancy rate v comes from figure 2.5 and the unemployment rate u comes from figure 2.1. Shaded areas indicate recessions dated by the NBER (2023). The labor market is at full employment when tightness equals 1, inefficiently slack when tightness is below 1, and inefficiently tight when tightness exceeds 1.

The labor market is not always inefficiently slack, however. There are several episodes when it becomes inefficiently tight. And these episodes do not appear at random. Before 2018, the labor market had only been inefficiently tight during major wars—World War 2, the Korean War, and the Vietnam War. Keynes (1936, p. 322) doubted that a labor market could reach full employment in peacetime. He was essentially right: before 2018 the US labor market had only reached full employment in wartime.

Since 2018, the labor market has been inefficiently tight just before the coronavirus pandemic (2018Q3–2020Q1), and in the aftermath of the pandemic (2021Q3–2024Q4). The state of the labor market around the pandemic is therefore a rarity: it is the only peacetime episode during which it became inefficiently tight—the only episode of peacetime overheating.

12.4.2. FERU over the past century

Next, we combine the US unemployment and vacancy rates between 1929 and 2024 to compute the FERU. Given that the US labor market experienced extreme fluctuations during the entire period, especially in the first two decades, we plot the unemployment and vacancy rates, as well as labor market tightness and FERU, on logarithmic scales. Besides improving the readability of the figures, logarithmic scales have several advantages. First,

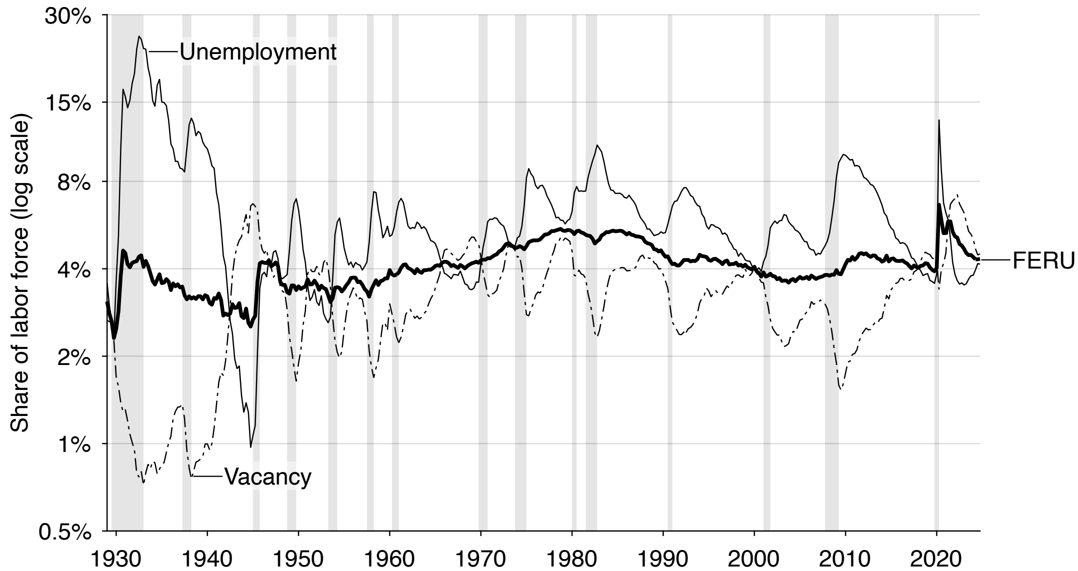


FIGURE 12.2. FERU in the United States, 1929–2024

The unemployment rate u comes from figure 2.1. The vacancy rate v comes from figure 2.5. The FERU is $u^* = \sqrt{uv}$, so on a logarithmic scale it is the midpoint between the unemployment and vacancy rates. Shaded areas indicate recessions dated by the NBER (2023).

the symmetry of the unemployment and vacancy movements on a logarithmic scale makes it clear that the Beveridge curve is a rectangular hyperbola. Second, the FERU is particularly easy to construct on a logarithmic scale. Indeed, the FERU is just the midpoint between the unemployment and vacancy rates: as $u^* = \sqrt{uv}$, then $\ln(u^*) = 0.5 \times \ln(u) + 0.5 \times \ln(v)$.

Over 1929–2024, the FERU averages 4.1% (figure 12.2). The FERU is quite stable over time, remaining between 2.3% and 6.7% over almost a century. Despite significant macroeconomic volatility during the prewar period, 1929–1947, the FERU is stable: it averages 3.4% and remains between 2.3% and 4.6%. The FERU is also stable during the postwar period, 1948–2019: it averages 4.2% and remains between 3.1% and 5.5%. The Beveridge curve shifts in and out during the postwar period (figure 2.7), but the shifts are not large enough to produce noteworthy changes in the FERU. Finally, the FERU temporarily rose above 6% during the pandemic, before falling back down below 5% in 2022 and below 4.5% in 2023.

The sharp increase of the FERU at the onset of the pandemic is unprecedented: the FERU increased by 2.7pp, from 4.0% in 2020Q1 to 6.7% in the following quarter. This sharp increase is explained by the gigantic outward shift of the Beveridge curve that took place in the spring of 2020 as the US economy was shut down by the coronavirus (figure 2.7). Indeed, the FERU is solely determined by the location of the Beveridge curve, so only an outward shift of the Beveridge curve can raise the FERU.

Two other notable increases in the FERU occurred during the two other major catastrophes of the century: the Great Depression and World War 2. At the onset of the Great Depression, the FERU increased by 2.3pp: from 2.3% in 1929Q4 to 4.6% one year later. At the end of World War 2, the FERU increased by 1.7pp: from 2.5% in 1944Q4 to 4.2% one year later. In both cases, the Beveridge curve also shifted outward in dramatic fashion, in response to the dislocation of the labor market that occurred when the US economy plunged into the depression and when veterans flocked back to the US labor market.

It is maybe suprising that the FERU has not declined noticeably in the past two or three decades, despite the advent of the internet and online job search and recruiting. One could have imagined that internet could have helped matching in the labor market—thus shifting the Beveridge curve inward and reducing the FERU. But in the 21st century, the FERU reached its lowest value in 2003:Q4, at 3.6%, when the internet was in its infancy. This macro finding is actually consistent with micro evidence, which finds that the internet and online job portals might not have much effect on the job market: the introduction of the internet or specific online job portals (such as Craigslist) did not have any effect on the local unemployment rate or on job-finding rates of jobseekers (Kuhn and Skuterud 2004; Kroft and Pope 2014). So it is unclear that the advent of the internet drastically improved labor-market matching.[2]

12.4.3. Unemployment gap over the past century

The inefficiency of the US labor market over the past century is clearly visible by computing the US unemployment gap, $u - u^*$ (figure 12.3).

Over 1929–2024, the unemployment gap averages +2.3pp. The unemployment gap was of course positive and very large during the Great Depression: the labor market was much too slack then. The unemployment gap reached +20.9pp in 1932Q3, its highest level on record. The labor market recovered only slowly from the depression. The labor market reached full employment in 1942Q3, a few quarters after the United States had entered World War 2. The unemployment gap kept falling during the war; it reached −1.6pp in 1945Q1, its lowest level on record.

In the postwar period, the unemployment gap peaked at +5.9pp in 2009Q4, during the Great Recession. At the end of the Volcker recession, in 1982Q4, the gap reached the slightly lower value of +5.7pp. The lowest value taken by the unemployment gap is −0.8pp, in 1969Q1, during the Vietnam War. During the Korean War, the unemployment gap was

[2] Looking at the academic job market, it does not seem that the amount of time and effort required to recruit colleagues is appreciably diminished by online job portals, and that the outcome of recruiting is appreciably improved by them. Most of the recruiting time is spent on reading applications and papers; interviewing candidates; flying out candidates; and debating with colleagues. These four tasks also determine the outcome of the recruiting process. They are mostly unaffected by the internet or online job boards.

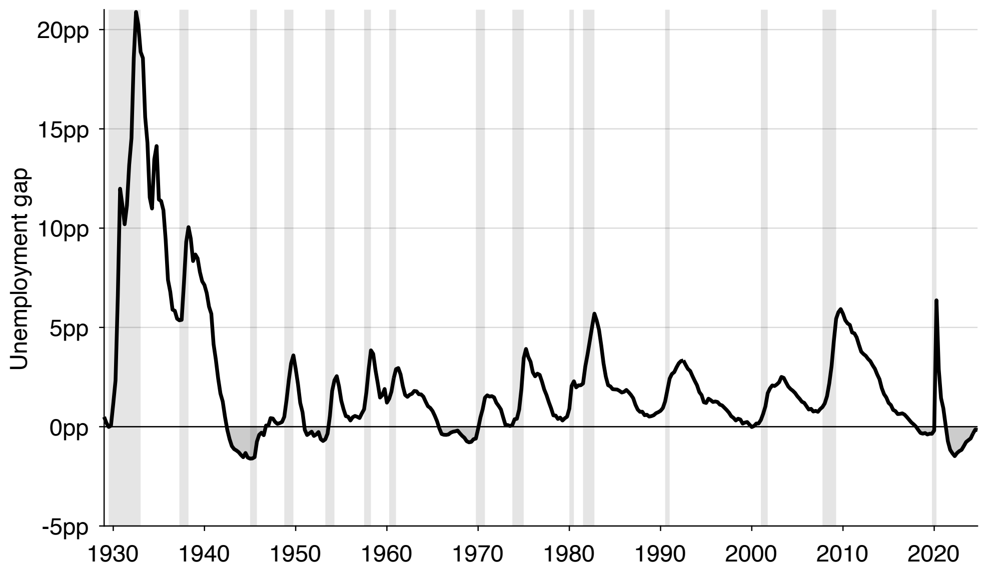


FIGURE 12.3. Unemployment gap in the United States, 1929–2024

The unemployment gap is $u - u^{*}$, where the unemployment rate u comes from figure 2.1 and the FERU u^{*} comes from figure 12.2. Shaded areas indicate recessions dated by the NBER (2023).

almost as low, reaching −0.7pp in 1953Q1.

While the unemployment gap averages 0 over the pandemic and post-pandemic period, the labor market experienced sharp departures from full employment. The unemployment gap was initially positive and large: the labor market was much too slack in the first year of the pandemic. The unemployment gap peaked at +6.3pp in 2020Q2, its highest level since 1945. But the labor market recovered quickly and reached full employment in the middle of 2021. The unemployment gap turned negative after that, reaching −1.5pp in 2022Q2, its lowest level since 1945. The gap then shrank to −0.2pp in 2024Q4. So during 2022–2024, the labor market was well beyond full employment.

Because the US labor market has generally been inefficient, there is a role for the government to stabilize the labor markets and attempt to bring unemployment to its efficient level. Reassuringly, this result is consistent with US law. If the labor market always operated efficiently, then there would be no need to pass laws that say that the government and the central bank have to cooperate to maintain the labor market at full employment. Yet this is exactly what the Employment Act of 1946, the Federal Reserve Reform Act of 1977, and the Full Employment and Balanced Growth Act of 1978 do.

12.5. Explaining fluctuations in the unemployment gap

We have just established that the unemployment gap is countercyclical: it rises in recessions and recedes in expansions (figure 12.3). Is the pattern consistent with fluctuations in the slackish labor market model of chapter 10?

We saw that in the slackish labor market model, labor demand shocks—in the form of shocks to labor productivity—produce fluctuations that resemble US labor market fluctuations. They produce a procyclical labor market tightness, countercyclical unemployment rate, procyclical vacancy rate, and procyclical employment level. Moreover, the fluctuations in tightness in response to fluctuations in labor productivity are commensurate to the fluctuations observed in the US labor market.

In this section, we show that these shocks can also generate a countercyclical unemployment gap in the slackish labor market model.

First, we must apply the sufficient-statistic formula (12.2) to the slackish labor market model. It is not as easy as it might seem because in the model, in general, the Beveridge elasticity β is not fixed but is a function of tightness. So as equation (9.10) shows, the efficient labor market tightness really is defined by $\theta^* \beta(\theta^*)\kappa = 1$, where the Beveridge elasticity $\beta(\theta)$ is given by equation (9.9). The Beveridge elasticity only depends on the matching function and job-separation rate λ.

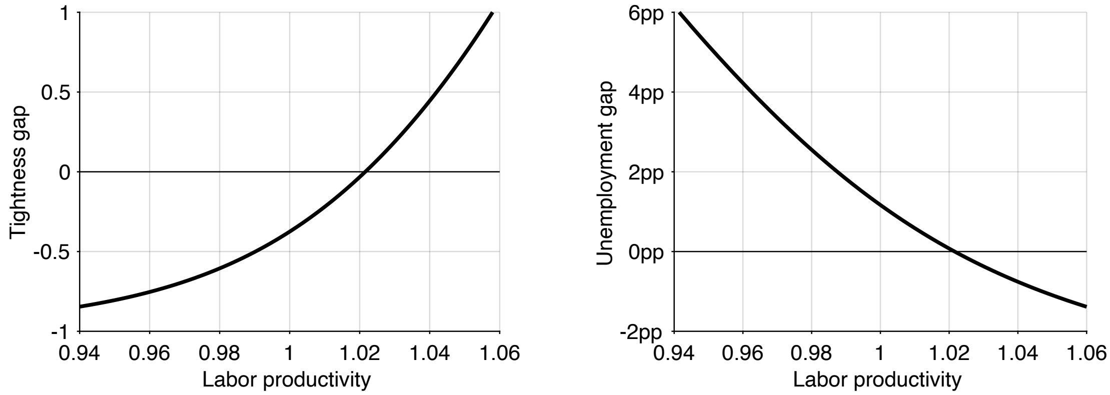
So there is a tightness θ^* that is efficient and that does not respond to labor productivity shocks—as these shocks do not appear anywhere in the structural equation $\theta^* \beta(\theta^*)\kappa = 1$. In the calibration of table 10.2, the efficient tightness is not exactly 1 but close to it: $\theta^* = 1.03$. It is not exactly 1 because at θ^*, the Beveridge elasticity is not exactly 1 but $\beta(\theta^*) = 0.98$.

In response to labor productivity shocks, the actual tightness θ responds sharply, while the efficient tightness θ^* remains at 1.03. Hence, the simulated tightness gap $\theta - \theta^*$ is sharply procyclical, moving with tightness (figure 12.4A).

The FERU is just the efficient unemployment rate, $u(\theta^*)$. In the calibrated model, $u^* = 4.5\%$. Just as the efficient tightness, the FERU does not respond to productivity shocks. Accordingly, the simulated unemployment gap $u - u^*$ is sharply countercyclical, moving with the unemployment rate (figure 12.4B). Moreover, the amplitude of the simulated unemployment gap matches what we observe in figure 12.3. The highest simulated value is +6.0pp, close to the unemployment gap observed during the Great Recession; and the lowest value is −1.4pp, close to the unemployment gap observed after the coronavirus pandemic.

12.6. Existing unemployment targets do not capture full employment

In this final section, we briefly review other series that US policymakers have used to measure full employment: the natural rate of unemployment (NRU) and the nonaccelerating-



A. Tightness gap B. Unemployment gap

FIGURE 12.4. Tightness gap and unemployment gap in the slackish labor market model

The tightness gap is $\theta - \theta^*$, where labor market tightness θ comes from figure 10.5A and the efficient tightness is $\theta^* = 1.03$. The unemployment gap is $u - u^*$, where the unemployment rate u comes from figure 10.5B and the FERU is $u^* = 4.5\%$. The calibration of the model is described in table 10.2.

inflation rate of unemployment (NAIRU).[3] We explain why this is an unsatisfactory choice.

Indeed, why should policymakers use the FERU if there are perfectly acceptable series such as the NAIRU or NRU to serve as full-employment targets? The main reason is that the NAIRU and the NRU have nothing to do with labor market efficiency, so they are not useful to measure full employment.

12.6.1. The NRU does not capture full employment

A first measure that the federal government, Federal Reserve, and other policymakers often use to measure full employment is the NRU. The NRU is a nebulous concept, but generally refers to some trend of the unemployment rate (Rogerson 1997). Specifically, policymakers often rely on the NRU series produced by the (CBO 2025).[4]

The CBO opened its doors in 1975 (US Congress 1974). It started producing estimates of the NRU shortly after that, which have become well known and widely used.[5] For example, when he was President of the Federal Reserve Bank of Boston, Rosengren (2014, p. 180) measured the departure of the Fed from its full-employment mandate by "the squared deviations of unemployment from an estimate of full employment utilizing the Congressional Budget Office assessment of the natural rate for each year."

[3] See Backhouse, Forder, and Laskaridis (2023) for a comprehensive history of the two concepts.

[4] The CBO rebranded the "natural rate of unemployment" as "noncyclical rate of unemployment" after 2021—leaving the acronym NRU unchanged and highlighting the fact that the natural rate is generally considered as the trend or noncyclical component of unemployment.

[5] See for instance this report produced by the CBO in January 1987: `https://www.cbo.gov/sites/default/files/100th-congress-1987-1988/reports/doc01b-entire_0.pdf`.

The CBO's NRU is a slow-moving trend of the unemployment rate computed by assuming that the labor market was at full employment in 2005, and then by incorporating changes in the demographic composition of the labor force over time.[6] Although the NRU conveys information about the demographic forces exerted in the labor market, without a theory of full employment, it is impossible to know whether the US labor market was at full employment in 2005 or not, and by induction, whether the NRU in any year measures full employment. More generally, in a slackish labor market, there is no guarantee that the trend of unemployment is efficient (chapter 9). Thus, the NRU cannot be a satisfactory measure of efficiency and thus full employment.

12.6.2. The NAIRU does not capture full employment either

Another measure of full employment that policymakers sometimes use is the NAIRU. The concept of NAIRU was proposed by Modigliani and Papademos (1975), although in a slightly modified form—they introduced NIRU for "noninflationary rate of unemployment". The acronym NAIRU then appeared in Baily (1976) and Baily and Tobin (1977). These early papers also offered measures of the NAIRU for the United States, that could be used as targets for monetary policy.

Today, the Federal Reserve and government continue to use the NAIRU as a full-employment target. For instance, the Council of Economic Advisers (2024, p. 24) describes the concept of full employment as follows: "Modern economics has generally defined full employment by citing the theoretical concept of the lowest unemployment rate consistent with stable inflation, which is referred to as u^*, ...the non-accelerating inflationary rate of unemployment (termed NAIRU)." The quote is particularly meaningful because the Council of Economic Advisers was created by the Employment Act to ensure that the government achieved its full-employment mandate.

Although the NAIRU might contain information relevant to the Fed's price-stability mandate, there is absolutely no guarantee that the unemployment rate coming out of the NAIRU estimation measures the efficient rate of unemployment. Because the NAIRU is the unemployment rate at which inflation remains stable, it is measured by estimating accelerationist Phillips curves.[7] These estimation procedures therefore rely on price dynamics. They do not measure the social costs and benefits from unemployment, so they cannot produce an estimate of the efficient unemployment rate, or equivalently of full employment.

A secondary issue with the NAIRU is that its estimates are very imprecise (Staiger, Stock, and Watson 1997; Laubach 2001). The slackish Phillips curve developed in chapter 14

[6] The construction of the CBO's NRU is described in Shackleton (2018, appendix B) and reviewed in Bok et al. (2023).

[7] For different methods to estimate the NAIRU, see Staiger, Stock, and Watson (1997), Gordon (1997), Laubach (2001), and Ball and Mankiw (2002).

might explain why. In that model, inflation might be stable at any level, for different unemployment rates: the Phillips curve is not accelerationist in the slackish model. If unemployment is at full employment, inflation is stable at the target; if unemployment is above the FERU, inflation is stable below the target; and if unemployment is below the FERU, inflation is stable above the target. So the concept of NAIRU is not well defined under a slackish Phillips curve. Ball and Mankiw (2002) argue that the NAIRU concept is implicit in any model in which monetary policy influences both inflation and unemployment. But it's not present in a slackish business cycle model in which monetary policy affects inflation and unemployment. If the world is slackish, the NAIRU is not well defined, explaining why it might be very difficult to estimate.

12.6.3. The NRU is higher than the FERU

Given that the NRU and NAIRU are defined entirely differently from the FERU, it is no surprise that their levels differ. But it is noteworthy that both NRU and NAIRU are generally markedly higher than the FERU.

Between 1949 and 2019, the CBO's NRU is always above the FERU (figure 12.5). During that period, the NRU averages 5.5%, which is 1.2pp above the average FERU. The fact that the NRU is above the FERU does not reflect a deep insight about the US labor market: it only results from the assumption made when constructing the NRU. The CBO constructs the NRU by assuming that the labor market was at full employment in 2005; indeed, the NRU and actual unemployment rate overlap at 5.0% in 2005:Q4. But the FERU was lower than that at the same time: in 2005:Q4, the FERU was 3.8%, so 1.2pp below the NRU. The gap between the NRU and FERU in 2005 permeates throughout the entire period.

While the NRU is always above the FERU between 1949 and 2019, the gap between the two series is not constant over the entire period—reflecting the different forces driving the FERU and NRU. In the 1950s, the NRU was often more than 2pp above the FERU. For instance, the NRU stood 2.3pp above the FERU in 1953 and 2.2pp above the FERU in 1957. The FERU hovered between 3.1% and 3.7% during that decade, while the NRU was between 5.3% and 5.5%. In the 1960s, the gap between the two series was still significant: the FERU stayed between 3.8% and 4.2% during that decade, while the NRU was between 5.5% and 5.9%.

Interestingly, the values of the FERU for the 1950s and 1960s are in line with estimates of full employment produced by economists at the time, while the values of the NRU are much above these contemporaneous estimates. Indeed, from its creation in 1946 to 1956, the Council of Economic Advisers (CEA) used an unemployment rate of 3% as a marker of full employment (Duboff 1977, p. 8). Then the CEA started raising their unemployment target. In 1962, the CEA wrote that an unemployment rate of 4% was "a reasonable and prudent full employment target for stabilization policy" (Duboff 1977, p. 10). Then, in 1969, Bakke (1969, p. 280) reported that "Since the [CEA] identified an unemployment rate of 4% with a condition of practically full employment, this figure served as a constant in the equation for computing the potential output." These values of 3% in the 1950s and 4% in the 1960s are perfectly consistent with our FERU, but much below the CBO's NRU.

The gap between NRU and FERU closed in the 1970s and 1980s as the Beveridge curve drifted outward, and reopened in the 1990s and 2000s, as the Beveridge curve shifted back inward. The gap closed again in the 2010s.

In the aftermath of the coronavirus pandemic, 2020–2024, the NRU was actually below the FERU. This is because the dramatic outward shift of the Beveridge curve during the pandemic raised the FERU but did not affect the NRU.

12.6.4. The NAIRU is also higher than the FERU

There is no standardized time series for the NAIRU. Here, we use the NAIRU computed by Crump et al. (2024) using state-of-the-art techniques. That series covers 1960–2023.

Just like the NRU, the NAIRU is significantly higher than the FERU. Between 1960 and 2019, the NAIRU is above the FERU in all but a few quarters (figure 12.5). The NAIRU averages 5.9% over 1960–2019. This is 1.5pp more than the average FERU over the same period.

The gap between the NAIRU and FERU is particularly large around 1980. In the decade between 1974 and 1983, the NAIRU is on average 3.1pp above the FERU. In fact, the NAIRU reaches 9.0% in 1975, 8.7% in 1980, and 8.7% again in 1982. These are incredibly high numbers for what is supposed to be a measure of full employment.

The gap between NAIRU and FERU shrank in the late 1980s and 1990s, but it reopened again at the time of the Great Recession. Between 2007 and 2011, the NAIRU was on average 2pp above the FERU. In 2009, the gap between the two series reached 2.8pp, as the NAIRU climbed to 6.6% in 2009. The two series became quite close in the late 2010s, but the NAIRU diverged from the FERU again after the pandemic. In 2022, the NAIRU reached 7.0%. In 2022–2023, the gap between NAIRU and FERU averaged 2.1pp.

12.7. Summary

This chapter argues that the US labor market has seldom operated at its socially efficient level of unemployment. Using the efficiency framework of slackish markets, and combining it with the available empirical evidence, we find that in the United States efficient labor market tightness equals 1: full employment corresponds to one vacancy per job seeker. Historically, tightness has been well below unity, indicating pervasive inefficiency and excessive slack, except during wartime and, more recently, during the post-pandemic expansion when the market became inefficiently tight.

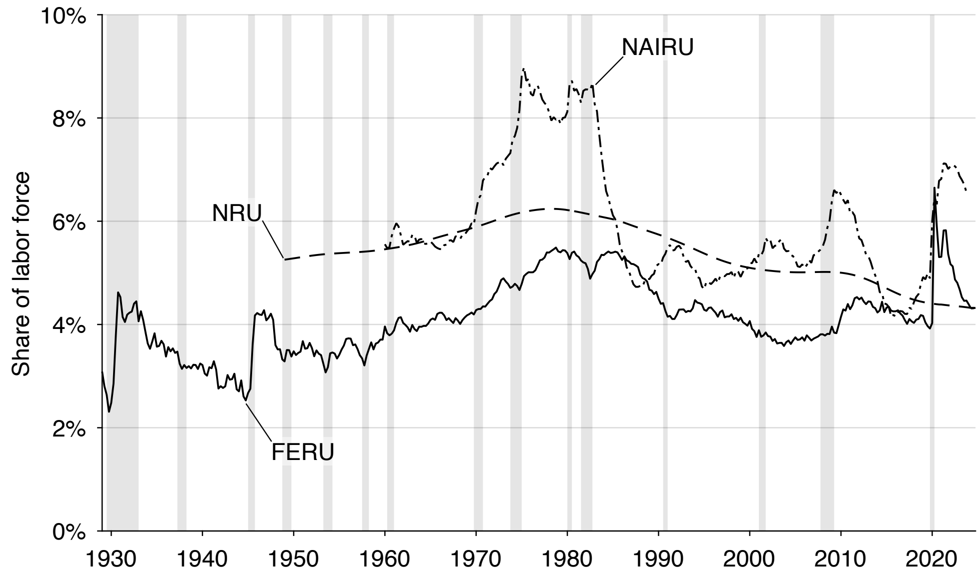


FIGURE 12.5. FERU, NRU, and NAIRU in the United States, 1929–2024

The FERU comes from figure 12.2. The NRU is constructed by the CBO (2025) for 1949–2024. The short-term NRU is constructed by the CBO (2021) for 1949–2020. The NAIRU is constructed by Crump et al. (2024, figure 2) for 1960–2023. Shaded areas indicate recessions dated by the NBER (2023).

We also show how the efficient tightness can be translated into an efficient unemployment rate, which we call the full-employment rate of unemployment (FERU). In our slackish model, the FERU simply is the geometric mean of the unemployment and vacancy rates. The FERU formalizes the concept of full employment as a state of social efficiency rather than as zero unemployment. The FERU remains fairly stable over time—around 4%—despite decennial shifts of the Beveridge curve.

Although the Federal Reserve has a mandate to maintain the US labor market at full employment, there is no agreed-upon measure of the FERU, which makes it difficult for them to design policy to achieve full employment, and for observers to assess their performance. One contribution of this chapter is to propose a simple, welfare-based formula for the FERU that can be implemented in real time: $u^* = \sqrt{uv}$.

12.8. Onto business cycles

Having measured inefficiency in one central market—the labor market—we now ask how economy-wide forces generate that inefficiency over time and how policy should stabilize it. To answer these questions, the remainder of the book takes an aggregate perspective and studies not just individual markets and their fluctuations but the entire economy and business cycles.

12.8.1. Which shocks produce fluctuations in the unemployment gap?

The historical analysis reveals that the US labor market has been inefficiently slack for most of the past century, especially in slumps. The only exceptions are wartime mobilizations and the post-pandemic period. We have also seen that in the slackish labor market model, labor demand shocks generate fluctuations in the unemployment gap similar to those we observe in the United States: the gap swells in recessions and recedes in expansions. But labor demand shocks are not the end of the story: why do firms sometimes want to hire more labor and sometimes less? The first question therefore is: In the economy, what are the shocks that trigger countercyclical fluctuations in the unemployment gap? We will answer this question in part IV.

12.8.2. Can policy mitigate fluctuations in the unemployment gap?

We have seen that the US labor market is generally inefficiently slack, and furthermore that the unemployment gap is countercyclical, so that the labor market drifts further from efficiency in recessions. Such fluctuations in the unemployment gap open the door to stabilization policies: monetary and fiscal policy might be able to counter the shocks that lead to these fluctuations and bring unemployment closer to its efficient level. But this possibility raises a second question: Can monetary policy and fiscal policy mitigate fluctuations in the unemployment gap? In chapter 11 we saw that public employment is able to boost labor demand and reduce unemployment, so there clearly is scope for some form of fiscal policy. But the main stabilization policy in the United States is monetary policy. So how does monetary policy affect unemployment and can it shrink the unemployment gap? We will also address these questions in part IV.

12.8.3. Should policy mitigate fluctuations in the unemployment gap?

Because the historical analysis reveals that the unemployment rate has often drifted above the FERU, one wonders whether the US government has failed its legal mandate to maintain full employment. Things are not so simple, however, because the government and Federal Reserve have a dual mandate: not only full employment but also price stability. One possibility is that maintaining price stability forced the government and Fed to accept an elevated unemployment rate. More generally, stabilization policies come with tradeoffs, which raises a third question: To optimize social welfare over the business cycle, how should monetary and fiscal policy be conducted? We will also address this question in part V.

PART IV.

Slackish business cycles

CHAPTER 13.

Slackish business cycle model

Chapter 12 showed that the unemployment gap in US data is generally positive and sharply countercyclical: it widens in recessions and narrows in expansions. The first task of part IV is to explain which aggregate shocks can generate these fluctuations. The second task is to clarify whether and how monetary policy—the main stabilization policy in the United States—can partially or completely undo such shocks so as to stabilize unemployment fluctuations. The final task is to assess whether fiscal policy can support monetary policy in stabilizing unemployment.

This chapter develops the simplest possible slackish business cycle model for that purpose. The model clarifies how aggregate demand and supply determine unemployment, and how monetary policy steers aggregate demand to stabilize unemployment in response to shocks. The model covers both normal times, when monetary policy operates in a conventional way by setting the nominal interest rate, and zero-lower-bound episodes, when monetary policy operates by forward guidance. The model also explains how fiscal policy in the form of public spending is able to stabilize unemployment.

13.1. Overview of the model

The model is organized around one slackish market in which self-employed workers sell their production to other households.

In the model, households produce services and not goods. This simplifies the model a lot because if a service is available and no customer purchases it, the service is just lost and wasted. Services cannot be stored, so we do not need to worry about inventory of unsold goods and the depreciation of unsold goods.

Furthermore, households produce services directly—there are no firms. Hence, we do not need to distinguish between a labor market and a product market: we only have the service market.

One might wonder how realistic it is to assume that households produce only services and not goods. It turns out that it is not such a bad assumption. In US data, there have always been more workers in service-producing industries than in goods-producing industries, but the gap between these two types of industries has widened significantly. From 1943 to today, the number of US workers in goods-producing industries has not changed much: 18.8 million workers in 1943, peaking at 25.0 million workers in 1979, and falling slightly to 21.7 million workers in 2024 (BLS 2025a). By contrast, the number of workers in service-providing industries has increased dramatically, from 23.8 million workers in 1943 to 136.3 million in 2024 (BLS 2025b). So the share of US workers in service-providing industries rose from 56% in 1943, to 72% in 1979, and to 86% in 2024. That is, of all workers, only 14% are producing goods in 2024, whereas 44% were producing goods in 1943. That is why modeling the US economy as a service economy seems like an accurate simplification.

We assume that households cannot produce services for themselves, so households also purchase and consume services produced by other households. The key idea is that a matching function mediates all the trades on the market for services, which produces slack.[1]

In the following chapters, we expand this baseline model to refine our understanding of unemployment fluctuations—both their causes and consequences. For instance, in chapter 15, we include separate product and labor markets to trace how slack propagates across markets.

13.2. Slackish market for services

The model is dynamic and organized around a single market for services. The structure of the market is similar to that of the dynamic slackish market model of chapter 8, and of the slackish labor market model of chapter 10.

We assume that there are many large households, each indexed by $j \in [0, 1]$. In each household, $h_j > 0$ workers are part of the labor force: either working or searching for a job. The aggregate number of workers in the labor force is $h = \int_0^1 h_j \, dj$. The labor force comprises l employed workers and $U = h - l$ unemployed workers.

Each worker in the labor force has the capacity to produce $k > 0$ services per unit time. Here the parameter k describes the productivity of labor. The economy's aggregate

[1]This is very much in the spirit of Diamond (1982a), who developed a model of the product market in which self-employed workers must trade because they cannot consume their own production.

capacity is kh services per unit time.

Services are sold through long-term worker-household relationships. Once workers have matched with a household, they become full-time employees of the household. They remain so until they separate, which occurs at rate $\lambda > 0$. The separations reflect retirements, dismissals, quits, and so on.

To find new employees, each household j posts V_j help-wanted ads. The reason why households must constantly hire help is that they are constantly losing existing workers. Filling a help-wanted ad requires the work of $\kappa > 0$ recruiters per unit time. The recruiters advertise the ads opened by their employer, read applications, and interview and select suitable candidates. The aggregate number of help-wanted ads is $V = \int_0^1 V_j \, dj$.

From the unemployment level U and vacancy level V, we also define the unemployment rate $u = U/h$ and the vacancy rate $v = V/h$.

The rate at which help-wanted ads and workers match is given by a matching function $m(U, V)$ that satisfies the assumptions laid out in chapter 3. The aggregate tightness is the ratio of the two arguments in the matching function: $\theta = V/U = v/u$.

The aggregate tightness determines the rate at which unemployed workers find jobs, $f(\theta) = m(1, \theta)$, and the rate at which help-wanted ads are filled, $q(\theta) = m(1/\theta, 1)$.

13.3. Equilibrium aggregate supply

The equilibrium aggregate supply is the output level when employment flows are balanced:

$$y^s(\theta) = [1 - u(\theta)]\, kh, \tag{13.1}$$

where the equilibrium unemployment rate is given by (8.8).

As we saw in chapter 8, the convergence to the equilibrium unemployment rate is rapid. Hence, the actual and equilibrium unemployment rates are indistinguishable: this is why we abstract from the law of motion of the unemployment rate (given by (8.3)) and consider the unemployment rate as a function of tightness (given by (8.8)).

13.4. Inflation rate and interest rates

Before studying aggregate demand, we describe the prices of services and assets in the model.

13.4.1. Inflation rate

At time t, services are sold at a unit price $p(t)$. As in any slackish model, we need to specify a price norm to determine $p(t)$. In a dynamic model, the price norm determines the rate of inflation, $\pi(t) = \dot{p}(t)/p(t)$.

We saw in chapter 5 that in reality market prices respond very sluggishly to changes in market conditions. This appears to be true in the aggregate too. Of course, prices always rise, but the pace at which they do so does not seem to be affected by economic conditions. In figure 1.1, we saw that US inflation barely fell when unemployment spiked during the Great Recession, and then barely rose when unemployment declined. More systematically, US inflation does not seem to respond much to shocks to monetary policy or to fluctuations in unemployment.[2]

Hence, we simply assume that the price of services grows at a fixed rate. This price norm could well be the result of constant communication from the central bank, which always tries to convince people that they will do anything to keep inflation at a fixed level—2% in the United States. Formally, we assume that prices grow at a constant inflation rate π:

$$p(t) = p(0)\exp(\pi t).$$

In chapter 14, we will move away from fixed inflation and introduce endogenous inflation and a Phillips curve to study how fluctuations in slack might affect inflation. For now, we study the model with fixed inflation.

13.4.2. Nominal interest rate

Next, we need to specify how nominal interest rates are set. We assume that the central bank follows an interest rate peg: they maintain the nominal interest rate $i(t)$ at some specific level i at all times in the absence of shocks. We also assume that nominal interest rates are subject to the zero-lower-bound constraint: $i(t) \geq 0$.

13.4.3. Real interest rate

The key price in our model is the real interest rate, because it is the price of government bonds relative to services:

$$r(t) = i(t) - \pi(t).$$

Because the inflation rate and nominal interest rate are fixed, the real interest rate is also fixed at $r = i - \pi$. Note that the real interest rate must be less than the time discount rate ($r < \delta$) because if not, households would not want to consume anything and aggregate demand would collapse.

[2] For evidence, see Christiano, Eichenbaum, and Evans (1999, figure 2), Stock and Watson (2010, figure 1), and Stock and Watson (2020, figure 1).

13.5. Preference for wealth

In this chapter and the rest of part IV, we make the unusual assumption that wealth enters the utility function. Thanks to this assumption, the aggregate demand is nondegenerate and the model is much more robust. How can we justify such an assumption?

The standard model assumes that people save to smooth consumption over time, but it has long been recognized that people enjoy accumulating wealth irrespective of future consumption. In fact, people accumulate wealth not just because they want to use it for future consumption, but also because they enjoy the social status and power that comes with wealth.

Describing the European upper class of the early 20th century, Keynes (1919, chapter 2) noted that:

> The duty of saving became nine-tenths of virtue and the growth of the cake the object of true religion.... Saving was for old age or for your children; but this was only in theory—the virtue of the cake was that it was never to be consumed, neither by you nor by your children after you.

Fisher (1930, p. 17), who invented the model of consumption and saving that we use today, added that:

> A man may include in the benefits of his wealth ... the social standing he thinks it gives him, or political power and influence, or the mere miserly sense of possession, or the satisfaction in the mere process of further accumulation.

A few decades later, describing the process of economic development across countries, and the importance of people's willingness to save, Lewis (1955, pp. 26–27) observed that:

> In most communities the attractions of asceticism are small when compared with the attractions of wealth, either as a means to power, or as a mark of superior social status.... Wealth is also desired as a means to power—whether it be the power of bribery, political power, power over employees, or other forms of power.... In every country of the world wealth wins respect and prestige, even though in some there is a time lag.

Keynes and Fisher and Lewis all observed that people enjoy accumulating wealth because it offers social status and power. Today, we have scientific evidence to back this claim: neuroscientists have actually confirmed that wealth itself provides utility, independently of the consumption it can buy. Camerer, Loewenstein, and Prelec (2005, p. 32) explain:

> Brain-scans conducted while people win or lose money suggest that money activates similar reward areas as do other 'primary reinforcers' like food and

> drugs, which implies that money confers direct utility, rather than simply being valued only for what it can buy.

A survey of 2000 Americans aged 62–75 conducted by the Employee Benefit Research Institute in 2020 and analyzed by Coy (2021) confirms these ideas. The survey first asks respondents what they plan to do over the course of their retirement with the wealth that they hold in their financial accounts. About 15% plan to grow their wealth, 1% plan to spend down none of their wealth, and another 35% plan to spend down a small portion of their wealth. So more than half of the respondents do not plan to spend down their wealth in retirement.

Asked why they do not plan to consume their wealth in old age, about 35% of respondents say that spending down is not necessary, and another 30% say that not spending makes them feel better. The survey explores that finding further and asks respondents whether they agree with the statement that "saving as much as I can makes me happy and fulfilled." About 65% of the respondents strongly or somewhat agree with the statement—a substantial validation of the wealth-in-the-utility assumption.

Since people do value holding wealth, it is reasonable to assume that it enters the utility function. Among all the reasons why people may value wealth, we focus on social status: we postulate that people enjoy wealth because it provides status in society. We therefore introduce relative wealth into the utility function. The assumption seems plausible. Adam Smith, Ricardo, John Rae, J.S. Mill, Marshall, Veblen, and Frank Knight all believed that people accumulate wealth to attain high social status (Steedman 1981). More recently, a broad literature has documented that people seek to achieve high social status, and that accumulating wealth is a prevalent pathway to do so.[3]

13.6. Equilibrium aggregate demand

We now turn to the aggregate demand in the model, which we derive by solving the consumption-saving problem of households.

13.6.1. Household's utility function

To study the aggregate demand side of the model, we introduce the utility function for households. Each household j values two things: services and relative real wealth.

The utility from consumption of services, $c_j(t) \geq 0$, is

$$c_j(t)^{1-\alpha},$$

[3] See for instance Frank (1985), Cole, Mailath, and Postlewaite (1995), Weiss and Fershtman (1998), Heffetz and Frank (2011), Fiske (2010), Anderson, Hildreth, and Howland (2015), Cheng and Tracy (2013), Ridgeway (2014), Mattan, Kubota, and Cloutier (2017).

where $\alpha \in (0, 1)$ governs the concavity of the utility over services, which determines diminishing marginal utility of consumption.

Each household holds $B_j(t) \geq 0$ government bonds, which is their sole source of wealth. The utility from relative real wealth is simply

$$\frac{1}{a} \cdot \frac{B_j(t) - B(t)}{p(t)},$$

where $B(t) = \int_0^1 B_j(t)\, dj$ is the average nominal wealth across households, so $B_j(t) - B(t)$ is the household's relative nominal wealth. The relative nominal wealth is divided by the price level $p(t)$ to compute the relative real wealth, $[B_j(t) - B(t)]/p(t)$. The parameter $a > 0$ governs the taste for consuming services relative to holding real wealth. For instance, a higher a means that households value consumption more relative to saving and accumulating wealth and building social status.

Then, the utility that household j maximizes is the discounted sum of flow utilities:

$$\int_0^\infty \exp(-\delta t) \left[c_j(t)^{1-\alpha} + \frac{1}{a} \cdot \frac{B_j(t) - B(t)}{p(t)} \right] dt. \tag{13.2}$$

13.6.2. Household's budget constraint

The next step to deriving the aggregate demand curve is to set up the budget constraint of the household. First, we need to figure out household j's income. There are two components to income: labor income and saving income. Labor income comes from the $[1 - u(\theta(t))]\, h_j$ people who are working in the household. These workers sell $k[1 - u(\theta(t))]h_j$ services per unit time. All services have a price $p(t)$, so labor income is

$$p(t)\, [1 - u(\theta(t))]\, kh_j.$$

Household j also holds $B_j(t)$ government bonds, which have a nominal interest rate $i(t)$. Hence the household's saving income is

$$i(t) B_j(t).$$

Next, we need to look at the sources of expenditure. First, to consume $c_j(t)$ services, household j must purchase more than $c_j(t)$ services. Indeed, some services are used only for recruiting, to fill help-wanted ads, and do not result in consumption. The amount of services paid for at any time is

$$y_j(t) = [1 + \tau(\theta(t))]\, c_j(t) \tag{13.3}$$

where $\tau(\theta)$ is the matching wedge, given by (8.13). A household spends $p(t)\left[1+\tau(\theta(t))\right]c_j(t)$ on services at time t: $c_j(t)$ of these services are consumed while $\tau(\theta(t))c_j(t)$ of these services are used for recruiting.

Additionally, to finance the interest payments on government bonds, the government levies a lump-sum tax $T(t)$ on each household.

Bringing together the household's income and expenditure, we build the household's nominal budget constraint:

$$\dot{B}_j(t) = iB_j(t) + p(t)\left[1 - u(\theta(t))\right]kh_j - p(t)\left[1+\tau(\theta(t))\right]c_j(t) - T(t), \tag{13.4}$$

where $\dot{B}_j(t)$ is the change in savings, or nominal wealth, at time t.

To express the budget constraint in real terms, we introduce the household's real stock of bonds,

$$b_j(t) = \frac{B_j(t)}{p(t)}.$$

A useful relationship is $\ln(b_j(t)) = \ln(B_j(t)) - \ln(p(t))$. Taking the derivative of this relationship with respect to time, we get

$$\frac{\dot{b}_j(t)}{b_j(t)} = \frac{\dot{B}_j(t)}{B_j(t)} - \frac{\dot{p}(t)}{p(t)} = \frac{\dot{B}_j(t)}{B_j(t)} - \pi.$$

Thus, the temporal change in real wealth is linked to the temporal change in nominal wealth by

$$\dot{b}_j(t) = \frac{\dot{B}_j(t)}{p(t)} - \pi b_j(t), \tag{13.5}$$

because $b_j(t)/B_j(t) = 1/p(t)$.

By combining the nominal budget constraint (13.4) with equation (13.5), we finally obtain the law of motion of real wealth:

$$\dot{b}_j(t) = rb_j(t) + \left[1 - u(\theta(t))\right]kh_j - \left[1+\tau(\theta(t))\right]c_j(t) - \frac{T(t)}{p(t)}. \tag{13.6}$$

13.6.3. Household's problem

We are now ready to solve household j's problem, which will bring us closer to deriving the model's aggregate demand. The household's problem is to choose a path for consumption $c_j(t) \geq 0$ and for real wealth $b_j(t) \geq 0$ to maximize utility (13.2) subject to the budget constraint (13.6). The household takes as given the paths of aggregate variables $\theta(t)$, $r(t)$, $b(t)$, and $T(t)/p(t)$, and the endowment of real wealth $b_j(0) \geq 0$. The household is also subject to a no-Ponzi condition that prevents them from running Ponzi schemes: $\lim_{t\to\infty} \exp(-rt) b_j(t) \geq 0$.

To solve this optimal control problem, we set up a Hamiltonian (result A.23):

$$\begin{aligned}\mathcal{H}(t, c_j(t), b_j(t)) = c_j(t)^{1-\alpha} + \frac{1}{a}\left[b_j(t) - b(t)\right] \\ + \psi_j(t)\left\{rb_j(t) + [1 - u(\theta(t))]\, kh_j - [1 + \tau(\theta(t))]\, c_j(t) - \frac{T(t)}{p(t)}\right\}.\end{aligned}$$

The variable $\psi_j(t)$ is the costate variable on household j's real budget constraint.

We now introduce the first-order conditions that characterize the household's optimal behavior (result A.24). First, $\partial\mathcal{H}/\partial c_j = 0$, which directly links consumption to the costate variable:

$$\psi_j(t) = \frac{1}{1 + \tau(\theta(t))} \cdot (1 - \alpha) c_j(t)^{-\alpha}. \tag{13.7}$$

From this equation we see the meaning of the costate variable on real wealth, $\psi_j(t)$: it gives the marginal utility derived from one unit of real wealth. Indeed, one unit of real wealth allows the household to purchase one service, which allows them to consume $1/(1+\tau)$ services. The marginal utility from consumption is $(1-\alpha)c_j(t)^{-\alpha}$, so the marginal utility from one unit of real wealth is $[1/(1+\tau)] \times (1-\alpha)c_j(t)^{-\alpha}$, which is just the costate variable.

The second first-order condition is $\partial\mathcal{H}/\partial b_j = \delta\psi_j(t) - \dot{\psi}_j(t)$, which requires

$$\frac{1}{a} + r\psi_j(t) = \delta\psi_j(t) - \dot{\psi}_j(t).$$

This in turn gives the differential equation describing how the costate variable associated with real wealth evolves when household j saves and spends optimally:

$$\dot{\psi}_j(t) = (\delta - r)\psi_j(t) - \frac{1}{a}. \tag{13.8}$$

In addition to these first-order conditions, optimality requires that the appropriate transversality condition is satisfied: $\lim_{t\to\infty} \exp(-\delta t)\psi_j(t) b_j(t) = 0$.

Here, any interior solution to the first-order conditions (13.7)–(13.8) is a global maximum of the household's problem. This is because the utility function (13.2) is concave in both consumption and real wealth, and the budget constraint (13.6) is linear in both consumption and real wealth (result A.25).

13.6.4. Euler equation

The differential equation (13.8) implicitly describes household j's consumption and saving over time: it is the Euler equation of the model. By solving it, we will obtain the model's aggregate demand. The Euler equation is an autonomous linear first-order differential equation, and to solve it, we need to understand its dynamical properties. First, its equilibrium point is

$$\psi(r) = \frac{1}{(\delta - r)a}. \tag{13.9}$$

The equilibrium value of the costate variable on real wealth has a simple interpretation. The Euler equation (13.8) shows that people's saving decision are determined by the time discount rate, which says how patient they are, the real interest rate, which determines financial returns on savings, plus marginal utility of real wealth, which determines the hedonic returns on wealth—indeed, with wealth in the utility function, the returns on wealth are not only financial but also hedonic. People keep consumption constant in equilibrium only if they are indifferent between consuming now or later, which requires the hedonic and financial returns on wealth to exactly offset time discounting.

To be precise: instead of consuming one unit of real wealth today, saving it to consume it later add utility $r\psi$ since the real wealth has grown in the future, removes utility $\delta\psi$ since future consumption is discounted, and add utility $1/a$, which holding real wealth today provides some utility directly. These effects must add up to zero in equilibrium: $\delta\psi = r\psi + 1/a$, which is equivalent to (13.9).

Second, since $\delta - r > 0$, the differential equation is a source. These properties appear on the phase line plotted in figure 13.1A.

The variable $\psi_j(t)$ is a costate variable, so it is not a predetermined variable at time t: it can jump at any time. If $\psi_j(t)$ jumps above the equilibrium point, it would start increasing over time and would diverge to ∞. This means that consumption would decrease and eventually become negative, violating one of the problem's feasibility constraints (see equation (13.7)). If $\psi_j(t)$ jumps below the equilibrium point, it would decrease over time, which means that consumption would increase over time and would diverge to ∞, eventually violating the resource constraint in the economy. Therefore, $\psi_j(t)$ must jump to the equilibrium point at $t = 0$—transition to the equilibrium point is immediate. At $t = 0$, $\psi_j(t)$ jumps to $\psi(r)$, so that at all times, $\psi_j(t) = \psi(r)$.

Therefore, using (13.7), we see that at all times t, household j's consumption is

$$c_j(t) = \left[\frac{(1-\alpha)(\delta - r)a}{1 + \tau(\theta)}\right]^{1/\alpha}, \tag{13.10}$$

where $c_j(t) > 0$. This consumption path is admissible and interior, and it satisfies all the

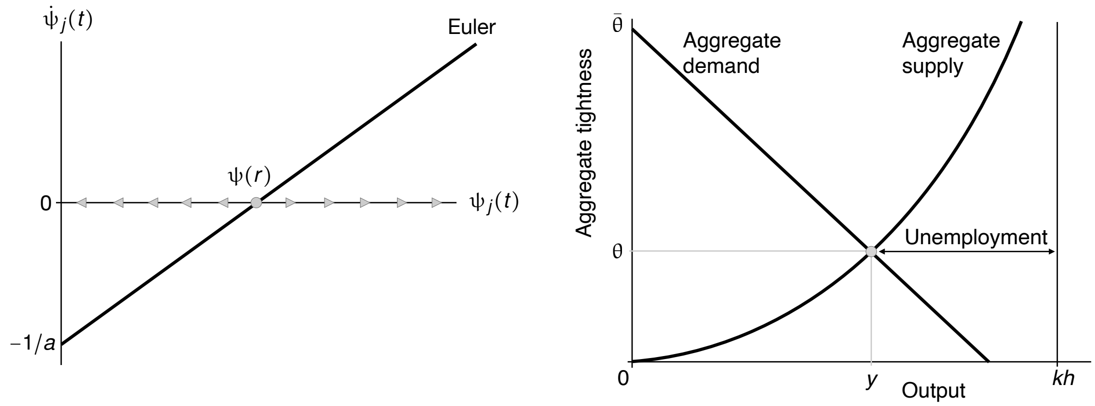


FIGURE 13.1. Solution of the slackish business cycle model

The variable $\psi_j(t)$ is the costate variable associated with the household j's real wealth. The Euler equation is given by equation (13.8). The equilibrium point $\psi(r)$ is given by equation (13.9). The aggregate supply curve is given by equation (13.1). The aggregate demand curve is given by equation (13.11).

first-order conditions, so it is a global maximum of the household's problem.

13.6.5. Recovering the standard Euler equation

The Euler equation (13.8) looks a bit different from the standard Euler equation in macroeconomic models—which might be worrisome. But there is no need to worry: it is easy to recover the standard form by eliminating the matching wedge and wealth in the utility function.

We remove the matching wedge by setting $\tau(\theta) = 0$ in (13.7) and (13.8), and we remove the utility of wealth by setting $1/a = 0$ in (13.8). The two equations become

$$(1-\alpha)c(t)^{-\alpha} = \psi(t), \qquad \frac{\dot{\psi}(t)}{\psi(t)} = \delta - r(t).$$

Next, we take the logarithm of the first equation and differentiate it with respect to time. We obtain

$$-\alpha \cdot \frac{\dot{c}(t)}{c(t)} = \frac{\dot{\psi}(t)}{\psi(t)}.$$

Combining these equations, we recover the standard Euler equation:

$$\frac{\dot{c}(t)}{c(t)} = \frac{1}{\alpha} \cdot [r(t) - \delta].$$

As usual, the equation says that the growth rate of consumption is determined by the gap between the real interest rate and the time discount rate.

We can also see why the standard Euler equation is degenerate. When $r(t) \neq \delta$, $\dot{c}(t) \neq 0$

for all $c(t) > 0$: the differential equation does not admit an interior equilibrium point. That is, there is no $c(t) > 0$ such that $\dot{c}(t) = 0$. If $r(t) = \delta$, then there is another strange situation: $\dot{c}(t) = 0$ for all $c(t) > 0$. So the household's Euler equation is not well behaved without wealth in the utility function. This ultimately means that the aggregate demand is degenerate without wealth in the utility function: it imposes that the real interest rate equals the time discount rate ($r = \delta$) but does not determine a consumption level.

13.6.6. Expression of the equilibrium aggregate demand

We have seen that although our model is dynamic, consumption jumps directly to the equilibrium point of the Euler equation at $t = 0$ and stays there. Thus, we can solve the business cycle model by supply-demand analysis, using tightness-quantity diagrams, with a very similar solution method to what we did for the market models.

The first step is to write down the aggregate demand curve, which gives us the aggregate amount of output that is demanded by households according to their Euler equations. The aggregate amount of consumption demanded is $c = \int_0^1 c_j \, dj$, where the households' demands for consumption c_j are given by (13.10). Then, the aggregate amount of output demanded is $y = [1 + \tau(\theta)]\, c$. This gives the equilibrium aggregate demand:

$$y^d(\theta, i) = \frac{[(1-\alpha)(\delta + \pi - i)a]^{1/\alpha}}{[1+\tau(\theta)]^{1/\alpha - 1}}. \tag{13.11}$$

We see that the aggregate demand is decreasing in aggregate tightness, from $y^d(0, i) > 0$ to $y^d(\bar{\theta}, i) = 0$ as tightness rises from $\theta = 0$ to $\theta = \bar{\theta}$. The aggregate demand is also decreasing in the nominal interest rate i and in fact in the real interest rate $r = i - \pi$.

Note that the aggregate demand is an equilibrium object because the labor flows are balanced at any point in time, so the number of workers who find a job in any household equals the number of workers who lose their job there.

13.7. Solution of the model

We now solve the slackish business cycle model. We distinguish normal times, when the nominal interest rate is positive, and zero-lower-bound episodes, when the nominal interest rate is at zero.

13.7.1. Solution in normal times

The structure of the solution to this chapter's business cycle model is virtually identical to the structure of the solution of the market model of chapter 4. In particular, the key step in solving the model is to find aggregate tightness. Then, from the value of tightness, we can infer the values of all the other variables in the model.

We know that aggregate tightness must equalize aggregate demand and aggregate supply:

$$y^d(\theta, i) = y^s(\theta). \tag{13.12}$$

The aggregate demand and aggregate supply functions have the same properties as the market demand and market supply in the fixed-price model of chapter 4, and therefore, we can use the same argument to show that equation (13.12) admits a unique solution, and therefore that the business cycle model admits a unique solution.

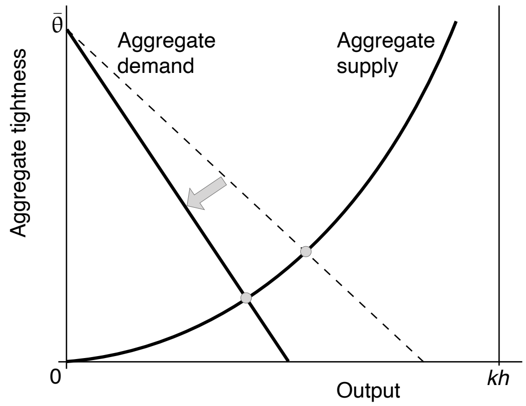
Accordingly, equation (13.12) implicitly defines tightness as a function of the nominal interest rate, $\theta(i)$. Then, from θ, we can back out all the other variables.

The solution of the model can be represented graphically just as in chapter 4. The tightness that solves the model is at the intersection of the aggregate demand and supply curves, as shown in figure 13.1B. The aggregate supply curve captures the capacity of the economy plus the matching process. The aggregate demand curve captures the fact that households maximize their utility given their budget constraint. At the intersection of the aggregate supply and aggregate demand curves, the matching process is respected (including hiring and separations), and households' utility maximization process is respected.

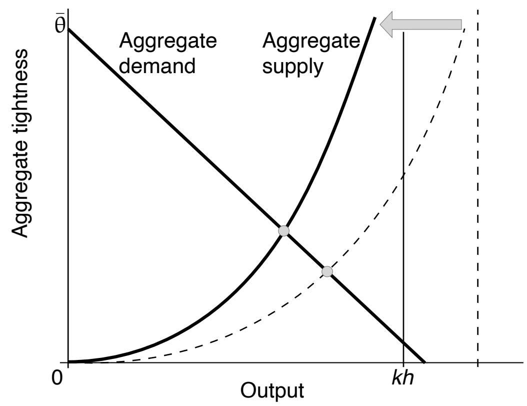
13.7.2. Solution at the zero lower bound

Our model behaves the same at the zero lower bound and away from it. In both situations, the aggregate demand and supply curves are identical. Indeed, in equations (13.1) and (13.11), nothing changes whether the economy is at the ZLB ($i = 0$) or not ($i > 0$). Hence, the model accommodates permanent zero-lower-bound episodes, and business-cycle shocks and policies have identical effects then, with the obvious difference that conventional monetary policy cannot stimulate the economy further by lowering the nominal interest rate at the zero lower bound.

This property of the slackish business cycle model is entirely different from what occurs in a New Keynesian model. The New Keynesian model predicts that when the zero lower bound lasts sufficiently long, output and inflation collapse to implausibly low levels, and policies such as public spending and forward guidance have extraordinarily large effects (Michaillat and Saez 2021b). More generally, in New Keynesian models, the zero lower bound is "a topsy-turvy world, in which many of the usual rules of macroeconomics are stood on their head" (Eggertsson and Krugman 2012, p. 1472). The zero lower bound is not topsy-turvy in slackish models.



A. Negative aggregate demand shock



B. Negative aggregate supply shock

FIGURE 13.2. Aggregate demand and supply shocks in the slackish business cycle model

The aggregate supply curve is given by equation (13.1). The aggregate demand curve is given by equation (13.11).

13.8. Effects of aggregate shocks

Now that we have characterized the solution in our dynamic model of slack, we use comparative statics to look at the effect of aggregate demand and supply shocks on aggregate tightness, output, and employment.

13.8.1. Qualitative effects of aggregate demand shocks

The most natural source of a negative aggregate demand shock is a decrease in the taste for consumption relative to saving, a.[4] After such a shock, households become more thrifty: they desire to save more and consume less, which depresses aggregate demand (equation (13.11)). Graphically, the shock triggers an inward rotation of the aggregate demand curve (figure 13.2A).

We see that as the aggregate demand curve rotates downward, both aggregate tightness θ and output y decrease. The unemployment rate, $u(\theta)$, increases due to a decrease in tightness, as shown by (8.8). A negative demand shock means people want to consume less, so there is less output produced and a lower tightness. There is more slack in the economy, as a larger share of the economy's productive capacity remains unsold and unused.

The efficiency analysis of chapter 9 continues to apply to this model, as the economy is organized around a single market that features a Beveridge curve with two sources of welfare cost: labor services that are unemployed and labor services that are devoted to recruiting instead of consumption. Hence, the efficient aggregate tightness θ^* is given by (9.10): $\theta^* \beta(\theta^*) \kappa = 1$, where $\beta(\theta)$ is the Beveridge elasticity, given by (9.9). The key is

[4]Another possible negative aggregate demand shock is a decrease in the time discount rate, δ.

TABLE 13.1. Comparative statics in the slackish business cycle model

	Tightness θ	Output y	Employment l	Unemployment u	Gap $u - u^*$
A. Aggregate demand shocks					
Decrease in taste for consumption a	–	–	–	+	+
B. Aggregate supply shocks					
Decrease in labor productivity k	+	–	+	–	–
Decrease in labor force h	+	–	–	–	–
C. Policy shocks					
Decrease in nominal interest rate i	+	+	+	–	–
Increase in public spending g	+	+	+	–	–

The comparative statics are obtained from figure 13.2A, figure 13.2B, figure 13.4A, and the adaptation of figure 11.1A to this model. A variable's increase is denoted by "+" and a decrease by "-".

that the efficient tightness is not affected by aggregate demand shocks, so the efficient unemployment rate $u^* = u(\theta^*)$ is not affected either. This implies that the unemployment gap $u - u^*$ moves together with the unemployment rate. In particular, the unemployment gap widens when the unemployment rate rises after a negative aggregate demand shock.

13.8.2. Quantitative effects of aggregate demand shocks

To quantify the effect of aggregate demand shocks on the unemployment rate, we derive the semielasticity of unemployment with respect to the taste for consumption, ε_a^u. The semielasticity is defined as follows:

$$\varepsilon_a^u = a \cdot \frac{du}{da},$$

and it is related to the elasticity of unemployment with respect to the taste for consumption by $\varepsilon_a^u = u \cdot \epsilon_a^u$. Semielasticities are typically easier to interpret than elasticities for rate variables that take small values close to zero (such as the unemployment rate or slack rate or interest rate). In terms of notation, we use a curly ε instead of the regular ϵ to denote semielasticities.

Using the analysis of demand shocks in slackish markets, it is straightforward to obtain the elasticity of tightness with respect to the taste for consumption. Formula (6.10) applies here too, so

$$\epsilon_a^\theta = \frac{\epsilon_a^d}{\epsilon_\theta^s - \epsilon_\theta^d},$$

where ϵ_a^d is the elasticity of aggregate demand with respect to the taste for consumption, $\epsilon_a^d = 1/\alpha$, ϵ_θ^d is the elasticity of aggregate demand with respect to tightness, given by

(6.13), and ϵ_θ^s is the elasticity of aggregate supply with respect to tightness, given by (8.14). Combining all these elements, we find that the elasticity is

$$\epsilon_a^\theta = \frac{1}{\alpha\left[1-\eta(\theta)\right]u(\theta)+(1-\alpha)\eta(\theta)\tau(\theta)}.$$

Next, we compute the elasticity of the unemployment rate with respect to tightness from equation (8.8) (using again results from appendix B):

$$\epsilon_\theta^u = -\frac{f(\theta)}{f(\theta)+\lambda}\left[1-\eta(\theta)\right] = -\left[1-\eta(\theta)\right]\left[1-u(\theta)\right]. \tag{13.13}$$

Accordingly, the elasticity of the unemployment rate with respect to the taste for consumption is

$$\epsilon_a^u = \epsilon_\theta^u \cdot \epsilon_a^\theta = -\frac{1-u(\theta)}{\alpha u(\theta)+(1-\alpha)\cdot\frac{\eta(\theta)}{1-\eta(\theta)}\cdot\tau(\theta)}.$$

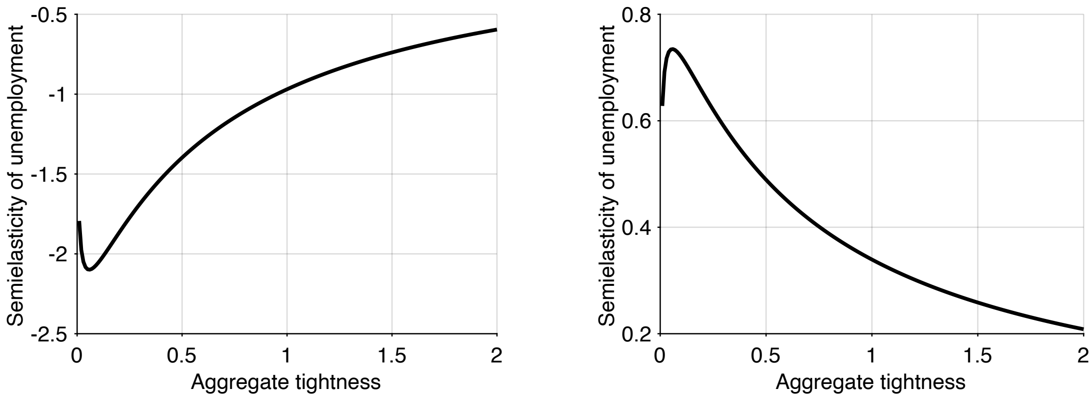
Last, the semielasticity of the unemployment rate with respect to the taste for consumption is

$$\varepsilon_a^u = u\cdot\epsilon_a^u = -\frac{1-u(\theta)}{\alpha+(1-\alpha)\cdot\frac{\eta(\theta)}{1-\eta(\theta)}\cdot\frac{\tau(\theta)}{u(\theta)}}. \tag{13.14}$$

The semielasticity ε_a^u gives the percentage-point increase in the unemployment rate when the taste for consumption rises by 1%. We see that it is negative, confirming that the unemployment rate rises when the taste for consumption drops (a negative aggregate demand shock). The effect is sharply dependent on tightness—so the slackish business cycle model is state-dependent.

It's not easy to establish the monotonicity of the semielasticity because both the numerator and denominator are increasing in tightness. But we can obtain a few results, and guess a few properties of the semielasticity too—which we will validate numerically. First, when the economy is operating efficiently, we learn from (9.11) that the semielasticity is just $-(1-u^*) \approx -1$. Because $u(\theta)$ is small compared to 1, while the two terms in the denominator are commensurate, it is quite likely that the change in the denominator term when θ changes dominates the change in $1-u(\theta)$ in the numerator, which would then imply that the semielasticity is decreasing in θ around efficiency.

Figure 13.3A validates this conjecture when the slackish business cycle model is calibrated just like the labor market of chapter 10 (table 10.2). Most of the parameters have similar interpretations in the two models, so it seems appropriate to use the calibration. The sole relevant parameter with a different interpretation is α: it determines the shape of the production function in the labor market model and the shape of the utility function in this business cycle model. Both functions are concave, and we keep the same value of



A. Semielasticity of unemployment to the taste for consumption

B. Semielasticity of unemployment to aggregate capacity

FIGURE 13.3. Effects of aggregate shocks on unemployment in the slackish business cycle model

The semielasticity of unemployment to the taste for consumption is given by (13.14). The semielasticity of unemployment to aggregate capacity is given by (13.15). The model is calibrated as in table 10.2.

α in the two models for transparency and simplicity.

We see that the semielasticity is around −1 when the economy is efficient (tightness around 1). The amplitude of the semielasticity is for the most part decreasing in tightness. The logic is the same as in figure 8.3A: when the economy is slack, changes in aggregate demand have large effects on quantities and thus the unemployment rate; conversely, when the economy is tight, changes in aggregate demand have small effects on quantities and thus the unemployment rate. The monotonicity breaks down only when tightness is very low and the unemployment rate very high, because then the movements in the numerator of the semielasticity, $1 - u(\theta)$, take over.

13.8.3. Qualitative effects of aggregate supply shocks

The next step is to look at the effects of aggregate supply shocks on the economy and unemployment. A negative aggregate supply shock could be caused by a decrease in either labor productivity, k, or labor force, h.

After a decrease in labor force or productivity, the aggregate capacity shrinks, so the aggregate supply shifts inward (figure 13.2B). As a result, aggregate tightness increases but output decreases. Note that unlike under aggregate demand shocks, tightness and output move in opposite directions for aggregate supply shocks—the usual pattern that we have been observing since chapter 4. The unemployment rate $u(\theta)$ decreases because tightness increases. So here, people want to consume a similar amount of services, but fewer services are offered in the economy, so the economy becomes tighter: there is less slack, and labor services are sold more easily.

Although shocks to productivity and labor force have similar effects on output and tightness, they have a different effect on the employment level. After a reduction in labor force, the number of employed workers l drops: fewer workers want to work, so fewer workers work. Formally: $l = y/k$ and y decreases, so l decreases. After a reduction in productivity, on the other hand, the number of employed workers rises: workers are less productive so more workers are required to provide demanded services. Formally, $l = (1-u)h$ and u decreases, so l actually increases here. In this model, the fact that output and employment move in opposite directions is specific to productivity shocks.

13.8.4. Quantitative effects of aggregate supply shocks

To quantify the effect of aggregate supply shocks on the unemployment rate, we derive the semielasticity of unemployment with respect to aggregate capacity, ε^u_{kh}. The semielasticity is defined as follows:

$$\varepsilon^u_{kh} = kh \cdot \frac{du}{dkh}.$$

It is related to the elasticity of unemployment with respect to aggregate capacity by $\varepsilon^u_{kh} = u \cdot \epsilon^u_{kh}$.

Using the analysis of supply shocks in slackish markets, it is straightforward to obtain the elasticity of tightness with respect to aggregate capacity. Formula (6.15) applies here too, so

$$\epsilon^\theta_{kh} = \frac{-\epsilon^s_{kh}}{\epsilon^s_\theta - \epsilon^d_\theta},$$

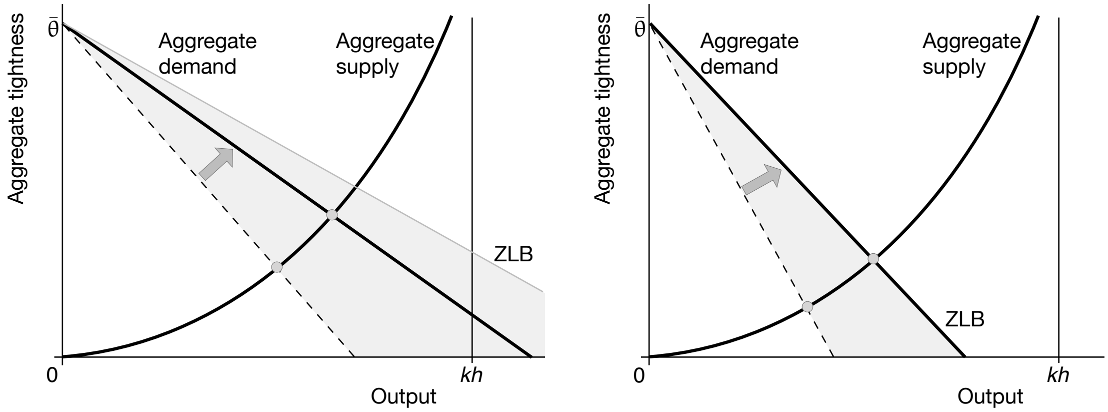
where ϵ^s_{kh} is the elasticity of aggregate supply with respect to aggregate capacity, $\epsilon^s_{kh} = 1$, ϵ^d_θ is the elasticity of aggregate demand with respect to tightness, given by (6.13), and ϵ^s_θ is the elasticity of aggregate supply with respect to tightness, given by (8.14). (We have also set $\epsilon^d_{kh} = 0$ in (6.15) since aggregate demand does not depend on aggregate capacity.) Combining these elements, we obtain

$$\epsilon^\theta_{kh} = \frac{-1}{[1-\eta(\theta)]\, u(\theta) + \frac{1-\alpha}{\alpha} \cdot \eta(\theta)\tau(\theta)}.$$

Finally, using the elasticity of unemployment with respect to tightness given by (13.13), we find that the semielasticity of the unemployment rate with respect to aggregate capacity is

$$\varepsilon^u_{kh} = u \cdot \epsilon^u_\theta \cdot \epsilon^\theta_{kh} = \frac{1-u(\theta)}{1 + \frac{1-\alpha}{\alpha} \cdot \frac{\eta(\theta)}{1-\eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)}}. \tag{13.15}$$

The semielasticity ε^u_{kh} gives the percentage-point increase in the unemployment rate when the aggregate capacity rises by 1%. We see that it is positive, confirming that the



A. Decrease in nominal interest rate B. Nominal interest rate at the zero lower bound

FIGURE 13.4. Expansionary monetary policy in the slackish business cycle model

The aggregate demand and supply curves are constructed in figure 13.1B. The gray cone indicates all the positions that the aggregate demand curve can reach when the nominal interest rate is reduced from its current level. The zero-lower-bound curve shows the position of the aggregate demand curve at the zero lower bound; it is the most outward position that the aggregate demand curve can reach; it is obtained from equation (13.11) when $i = 0$.

unemployment rate rises when aggregate capacity rises (a positive aggregate supply shock). The effect is sharply dependent on tightness. As with semielasticity (13.14), we see that when the economy is operating efficiently, the semielasticity is just $(1 - u^*)\alpha \approx \alpha$. We also see that the semielasticity is likely decreasing in θ around efficiency.

Figure 13.3B validates these results in the calibrated slackish business cycle model. The semielasticity is around α when the economy is efficient (tightness around 1). The semielasticity is for the most part decreasing in tightness. Why is that? We know that output responds strongly to supply shocks in tight markets and weakly in slack markets. But the response of output, $y = (1 - u)kh$, is the mechanical response to capacity kh, dampened by the response of the unemployment rate u. So, it follows that there is little dampening in tight markets but significant dampening in slack markets. This then means that the unemployment rate responds little in tight markets but significantly in slack markets—just as the figure shows.

13.9. Monetary policy

We now show how conventional monetary policy steers aggregate demand by modulating the nominal interest rate. In doing so, it affects aggregate tightness and thus the unemployment rate.

13.9.1. How does conventional monetary policy operate?

We start by describing how the model responds to conventional monetary policy. We consider an unexpected permanent decrease in the nominal interest rate i. As inflation π remains constant, the real interest rate $r = i - \pi$ falls. When the real rate falls, wealth has lower returns, so households desire to save less and consume more, which boosts aggregate demand (equation (13.11)). Conventional monetary policy therefore operates just like an aggregate demand shock.

Graphically, when the nominal interest rate drops, the aggregate demand curve rotates outward, so output and tightness increase (figure 13.4A). Since tightness rises, the unemployment rate decreases. And since the efficient tightness and unemployment rate are unaffected by monetary policy, the unemployment gap moves with the unemployment rate: it shrinks whenever the nominal interest rate is reduced.

13.9.2. The monetary multiplier

Next, we quantify the effect of conventional monetary policy on the unemployment rate. We compute and study the monetary multiplier,

$$\frac{du}{di}.$$

The monetary multiplier gives the percentage-point increase in the unemployment rate when the nominal interest rate increases by 1pp.

We follow the usual steps to compute the monetary multiplier. Formula (6.10) applies here too, so we get the elasticity of tightness with respect to the interest rate:

$$\epsilon_i^\theta = \frac{\epsilon_i^d}{\epsilon_\theta^s - \epsilon_\theta^d},$$

where ϵ_i^d is the elasticity of aggregate demand with respect to the interest rate, given by

$$\epsilon_i^d = -\frac{1}{\alpha} \cdot \frac{i}{\delta + \pi - i},$$

and ϵ_θ^d and ϵ_θ^s are the elasticities of aggregate demand and supply with respect to tightness, given by (6.13) and (8.14). Combining these elements, we find that the elasticity is

$$\epsilon_i^\theta = -\frac{i}{\delta + \pi - i} \cdot \frac{1}{\alpha\left[1 - \eta(\theta)\right] u(\theta) + (1 - \alpha)\eta(\theta)\tau(\theta)}.$$

Next, introducing the elasticity of unemployment with respect to tightness, given by

(13.13), we find that the monetary multiplier is

$$\frac{du}{di} = \frac{u}{i} \cdot \epsilon_{\theta}^{u} \cdot \epsilon_{i}^{\theta} = \frac{1 - u(\theta)}{\delta - r} \cdot \frac{1}{\alpha + (1-\alpha) \cdot \frac{\eta(\theta)}{1-\eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)}}. \tag{13.16}$$

We see that the monetary multiplier takes the same expression as the semielasticity of unemployment with respect to the taste for consumption, given by (13.14), up to the $-1/(\delta - r)$ factor. This means that the properties of the multiplier follow from those of the semielasticity. First, the monetary multiplier is positive, confirming that the unemployment rate rises when the nominal interest rate increases. Second, the monetary multiplier is also sharply slack-dependent. When the economy is operating efficiently, the multiplier is $(1 - u^*)/(\delta - r)$, and around efficiency, the multiplier is likely to decrease with tightness.

We plot the monetary multiplier in the calibrated model in figure 13.5. To do that, however, we also need to calibrate the real interest rate r and time discount rate δ. As of 2016, in the United States, the real interest rate consistent with full employment is about $r = 0.5\%$ (Williams 2017, table 1). We use that value for the real interest rate. Estimates of the time discount rate δ obtained in field and laboratory experiments and from real-world behavior are quite dispersed, but the majority of them point to discount rates well above prevailing real interest rates. Frederick, Loewenstein, and O'Donoghue (2002, table 1) and Andersen et al. (2014, table 3) survey a total of 54 studies.[5] For each of the 54 studies we compute the mean estimate, then take the median across studies. We obtain a median annual discount rate of $\delta = 36\%$. The estimates are very dispersed: the lower quartile is $\delta = 13\%$ and the upper quartile is $\delta = 175\%$. To capture the impatience observed in many studies, we also compute a mean annual discount rate, after excluding 9 studies whose estimates are implausibly high. We obtain a truncated mean annual discount rate of $\delta = 91\%$.[6] We adopt $\delta = 91\%$ as a baseline calibration.[7]

[5]Many of the experiments elicit discount rates using financial flows, not consumption flows. As the goal is to elicit the discount rate on consumption, this could be problematic (Cohen et al. 2020, section 4.2). Ideally, we would rely on experiments using time-dated consumption rewards instead of monetary rewards. Such experiments directly deliver estimates of the discount rate. Many such experiments have been conducted; a robust finding is that discount rates are systematically higher for consumption rewards than for monetary rewards (Cohen et al. 2020, section 3.1). Hence, the experimental estimates used here are, if anything, lower bounds on actual discount rates.

[6]Frederick, Loewenstein, and O'Donoghue (2002, table 1) contains 42 studies and Andersen et al. (2014, table 3) contains 16 studies. There are 4 studies that appear in both tables, so 54 distinct studies are included in total. When we compute the mean discount rate, we remove 9 studies in which respondents are so impatient that the behavior cannot be easily rationalized with an exponential discounting model—they imply immense annual discount rates, sometimes above $10^{10}\%$. The annual discount rate cutoff that we use to remove the 9 studies is 1,000%.

[7]The discount rate used here is higher than the typical calibration in macroeconomic models. That is because in models without wealth in the utility function, the discount rate must equal the average real interest rate, which is single digit. With wealth in the utility function, the discount rate is not tied to the interest rate and can be calibrated from experimental evidence. The finding by experimental studies find that annual discount rates are double digit may not be so surprising considering that annual interest rates on US credit card are typically around 20% (Rossman 2026), and that the penalty rates on these same cards typically reach

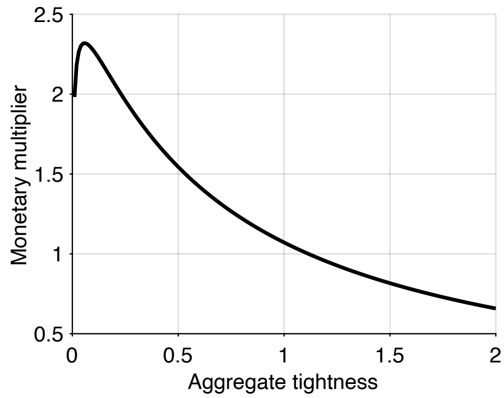


FIGURE 13.5. Monetary multiplier in the slackish business cycle model

The monetary multiplier is given by (13.16). The model is calibrated as in table 10.2, with in addition $\delta = 91\%$ and $r = 0.5\%$.

We see that the monetary multiplier is just above 1 when the economy is efficient (tightness around 1). The amplitude of the monetary multiplier is for the most part decreasing in tightness, from $du/di = 2.3$ when tightness is 0.1 down to $du/di = 0.7$ when tightness is 2. The logic is the same as in figure 8.3A: when the economy is slack, changes in aggregate demand have large effects on quantities and thus the unemployment rate; conversely, when the economy is tight, changes in aggregate demand have small effects on quantities and thus the unemployment rate.

13.10. Forward guidance at the zero lower bound

In this section, we move away from conventional monetary policy and instead study the effects of forward guidance around zero-lower-bound episodes. At the zero lower bound, the nominal interest rate is stuck at zero, so conventional monetary policy does not operate and cannot stimulate the economy. This is why the central bank must turn to forward guidance—to circumvent the zero-lower-bound constraint.

Forward guidance is a type of monetary policy that consists of announcing ahead of time future interest rates. The goal is to make promises that can stimulate the economy today, although the current interest rate is stuck at zero. Forward guidance became popular around the time of the Great Recession because policymakers had no choice—since the zero lower bound became binding in many countries.

30% (Irby 2026).

TABLE 13.2. Timeline of the zero-lower-bound episode and forward guidance

Taste for consumption a	Nominal interest rate i	Equilibrium costate variable ψ
A. Initial normal times: $t < 0$		
a_n	$i_n > 0$	$\psi_n = 1/[(\delta + \pi - i_n)a_n]$
B. ZLB episode: $0 < t < t_z$		
$a_z < a_n$	0	$\psi_z = 1/[(\delta + \pi)a_z] > \psi_n$
C. Forward guidance: $t_z < t < t_z + \Delta$		
a_n	0	$\psi_f = 1/[(\delta + \pi)a_n] < \psi_n$
D. Terminal normal times: $t_z + \Delta < t$		
a_n	$i_n > 0$	ψ_n

The parameter $t_z > 0$ gives the duration of the zero-lower-bound episode. The parameter $\Delta > 0$ gives the duration of forward guidance.

13.10.1. Timing of forward guidance

We consider a two-stage scenario where a forward-guidance episode follows a zero-lower-bound episode, as in Cochrane (2017). The complete scenario is presented in table 13.2.

Temporary zero-lower-bound episodes and forward guidance require a dynamic analysis. We therefore come back to the Euler equation (13.8). All households face the same Euler equation, so their costate variables are all the same: $\psi_j(t) = \psi(t)$. The costate variable $\psi(t)$ indicates how tight the finances of households are, and therefore how much they wish to consume, via equation (13.7).

Importantly, at any point in time, the costate variable determines the aggregate tightness $\theta(t)$. Indeed, from equations (13.7) and (13.3), we see that the costate variable ψ determines the location of the aggregate demand at any point in time:

$$y^d(\theta, \psi) = \left[\frac{1-\alpha}{\psi}\right]^{1/\alpha} \frac{1}{[1+\tau(\theta)]^{1/\alpha - 1}}. \tag{13.17}$$

This new expression of the aggregate demand curve is valid for any value of the costate variable, whether it is in equilibrium or not. In fact, when the costate variable is at its equilibrium value $\psi = 1/[(\delta - r)a]$, expression (13.17) reduces to the standard expression for the aggregate demand curve, given by (13.11). Conventional monetary policy moves aggregate demand by changing the interest rate and therefore the equilibrium costate variable; forward guidance moves aggregate demand by changing the costate variable off that equilibrium.

The aggregate demand in forward guidance, (13.17), has the same properties as the aggregate demand curve in normal times, (13.11), except that it depends on the costate variable. The aggregate demand in forward guidance y^d is strictly decreasing in the costate variable ψ.

Of course, aggregate demand must equal aggregate supply at any point in time, and this equality implicitly defines tightness as a function $\theta(\psi)$ of the costate variable:

$$y^d(\theta, \psi) = y^s(\theta), \tag{13.18}$$

where the aggregate demand curve is given by (13.17). The tightness that solves the model is at the intersection of the aggregate demand and supply curves, as shown in figure 13.1B. When the costate variable increases, the aggregate demand curve shifts inward, which lowers tightness along the aggregate supply curve. Conversely, when the costate variable decreases, the aggregate demand curve shifts upward, which raises tightness. Hence, we learn that tightness $\theta(\psi)$ is a strictly decreasing function of the costate variable at any point in time.

Hence, once we determine the dynamics of the costate variable, we can immediately trace the dynamics of aggregate tightness, and then of the unemployment rate and all the other variables in the model. Whenever the costate variable is high, aggregate demand is depressed, so the economy is slack, and conversely, whenever the costate variable is low, the economy is tight.

Before time 0, the economy is in normal times: the taste for consumption is at a high, normal level: $a = a_n$. The nominal interest rate is strictly positive, at $i_n > 0$, to keep the equilibrium costate variable ψ at a normal level:

$$\psi_n = \frac{1}{(\delta + \pi - i_n)a_n}. \tag{13.19}$$

This keeps the equilibrium tightness at the desired level θ_n, which is related to ψ_n via equation (13.18).

Between times 0 and t_z, there is a zero-lower-bound episode. The taste for consumption is low, depressing aggregate demand: $a = a_z < a_n$. To compensate, the nominal interest rate is lowered to $i = 0$. However, this is insufficient to keep tightness at the desired level: the equilibrium costate variable ψ is elevated, taking a value

$$\psi_z = \frac{1}{(\delta + \pi)a_z} > \frac{1}{(\delta + \pi - i_n)a_n} = \psi_n. \tag{13.20}$$

Accordingly, equilibrium tightness is depressed during the zero-lower-bound episode: $\theta_z < \theta_n$.

To alleviate the situation, the central bank makes a forward-guidance promise at time 0: that it will maintain the nominal interest rate at 0 for a duration Δ once the ZLB is over.

After time t_z, the taste for consumption returns to its normal, higher level: $a = a_n > a_z$. Between times t_z and $t_z + \Delta$, the central bank fulfills its forward-guidance promise and keeps the nominal interest rate at 0. As a result, the equilibrium costate variable ψ is low, taking a value

$$\psi_f = \frac{1}{(\delta + \pi)a_n} < \frac{1}{(\delta + \pi - i_n)a_n} = \psi_n. \tag{13.21}$$

After time $t_z + \Delta$, monetary policy returns to normal, $i = i_n$, the equilibrium costate variable is back to its desirable level, ψ_n, and the equilibrium tightness is back to its desirable level, θ_n.

13.10.2. How does the zero lower bound affect the economy?

To fix ideas, we start with a zero-lower-bound episode without forward guidance. We analyze the episode using the phase lines in figure 13.6A. We go backward in time. After time t_z, monetary policy maintains the economy at its desired location by setting the appropriate nominal interest rate. The costate variable $\psi(t)$ is maintained at its normal value ψ_n given by (13.19).

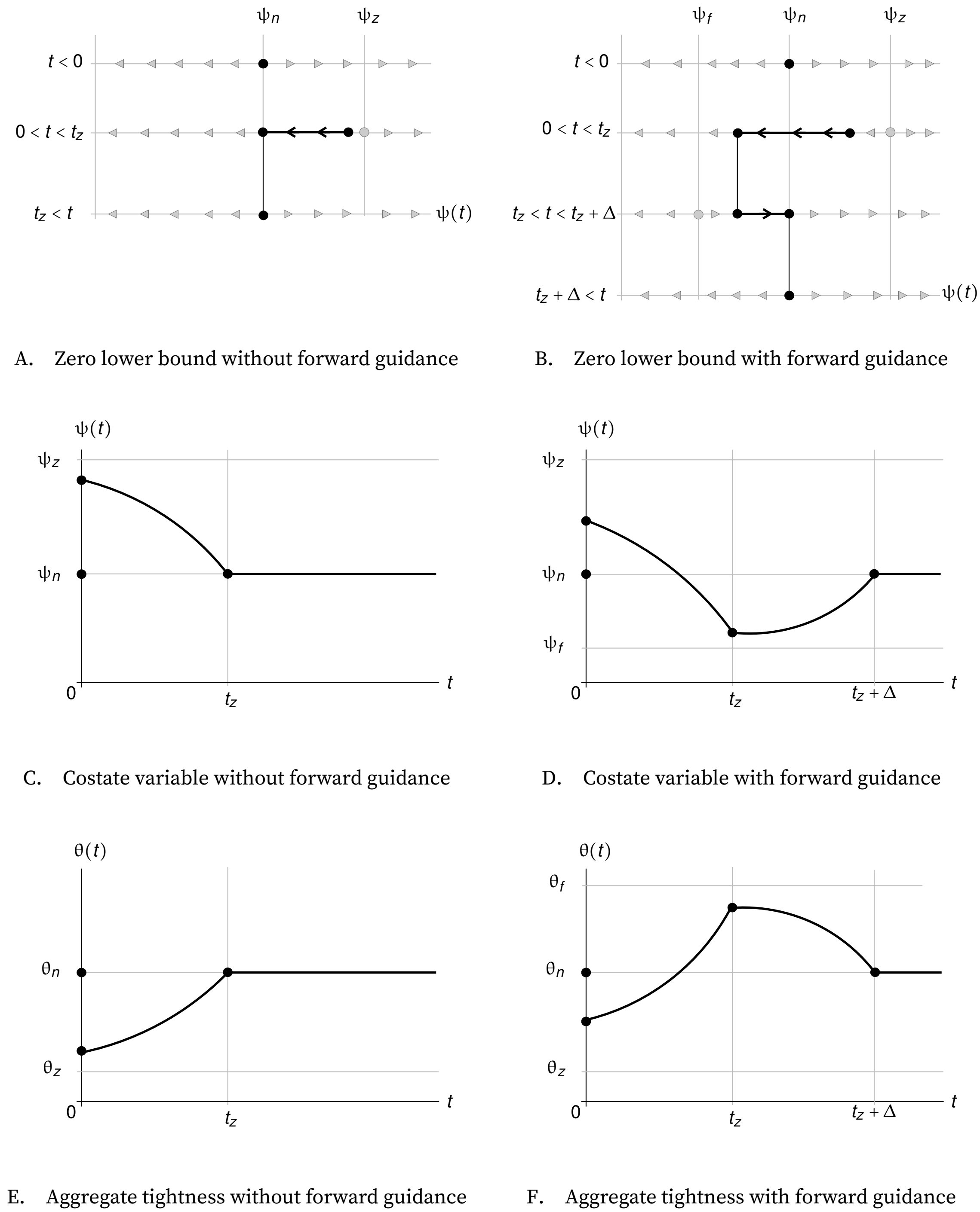
We look next at the zero-lower-bound episode, between times 0 and t_z. Since trajectories are continuous, the economy is at the same point at the end of the zero-lower-bound episode and after it. However, during the zero-lower-bound episode, the point $\psi = \psi_n$ is not an equilibrium: it is below the equilibrium point, so $\dot{\psi} < 0$ at that point. Hence, the costate variable decreases over time between $t = 0$ and $t = t_z$ to reach that point. From this we infer that at time 0, when the zero-lower-bound episode starts, the costate variable $\psi(0)$ must jump somewhere between the zero-lower-bound equilibrium ψ_z, given by (13.20), and the normal equilibrium ψ_n.

Note that the costate variable can only jump at time 0, because this is the only time when some new information is revealed. Without new information, the optimality of the behavior of households requires that the costate variable evolves continuously over time.

Overall, we see that the costate variable is above normal during the zero-lower-bound episode: it initially jumps to its highest point and then slowly decreases to its normal value (figure 13.6C). Since aggregate tightness is negatively related to the costate variable, as (13.18) shows, we infer that tightness jumps down at time 0, when the taste for consumption jumps down and the zero lower bound becomes binding. After that, the aggregate tightness slowly recovers to reach its normal level at time t_z (figure 13.6E).

13.10.3. How does forward guidance operate?

Now, how can forward guidance improve things at the zero lower bound? Conventional monetary policy improves things by reducing the nominal interest rate to 0 at time 0, as soon as the negative aggregate demand shock hits. But we assume that the decrease in the taste for consumption is so large that the reduction in interest rate is insufficient



A. Zero lower bound without forward guidance

B. Zero lower bound with forward guidance

C. Costate variable without forward guidance

D. Costate variable with forward guidance

E. Aggregate tightness without forward guidance

F. Aggregate tightness with forward guidance

FIGURE 13.6. Zero-lower-bound episodes without and with forward guidance

The timeline of the analysis is presented in table 13.2. Panels A and B display the phase lines of the Euler equation (13.8) at the different stages of the analysis. Panels C and D describe how the costate variable $\psi(t)$ is moving over time—as determined from the phase lines. Panels E and F describe how aggregate tightness $\theta(t)$ is moving over time—as inferred from (13.18).

to maintain the economy at the desired level: $\psi_z > \psi_n$. To supplement conventional monetary policy, the central bank therefore introduces forward guidance.

The economic logic of forward guidance is simple, even if the dynamics look intricate. During the zero-lower-bound episode, the central bank cannot cut the nominal interest rate any further. But it can promise that, once the taste for consumption has recovered, it will keep the interest rate at 0 for a while—below the rate that would normally be required to stabilize the economy. That promise engineers a temporary boom after the zero lower bound ends: with aggregate demand back to normal and rates still at 0, households find wealth less attractive relative to consumption, so they spend freely and tightness rises above its normal level.

Because households are forward looking, that future boom feeds into today's decisions. Anticipating a spell of unusually low interest rates after the zero lower bound ends, households expect low financial returns on wealth and therefore a lower marginal value of wealth in the future. Because accumulating wealth is less desirable in the future, working backward, it is also somewhat less desirable now, during the zero-lower-bound episode. As households feel less financially constrained now, aggregate demand shifts outward, tightness rises, and slack falls, even though the current interest rate remains stuck at 0. In short, forward guidance stimulates the present by committing to keep rates too low—and the return on wealth low too—in the near future.

We now formalize the analysis of the zero-lower-bound episode with forward guidance using the phase lines in figure 13.6B. We go backward in time. After time $t_z + \Delta$, monetary policy maintains the economy at its desired location by setting the appropriate nominal interest rate. The costate variable is maintained at the normal equilibrium ψ_n.

Between times t_z and $t_z + \Delta$, the economy is in forward guidance. Since trajectories are continuous, the economy is at the same point at the end of forward guidance and after it. However, during the forward-guidance episode, the point $\psi = \psi_n$ is not an equilibrium: it is above the equilibrium point, $\psi = \psi_f$, so $\dot{\psi} > 0$ at that point. Hence, the costate variable increases over time between $t = t_z$ and $t = t_z + \Delta$ to reach that point. This means that at the beginning of forward guidance, the costate variable must be below the normal equilibrium ψ_n. We also see that when the forward guidance lasts longer (higher Δ), the costate variable is pushed to lower levels at the beginning of forward guidance.

We then turn to the zero-lower-bound episode, between times 0 and t_z. At time t_z, the costate variable is lower than the normal equilibrium ψ_n and lower than the zero-lower-bound equilibrium ψ_z. This means that $\dot{\psi} < 0$ at that point. Hence, the costate variable decreases over time between $t = 0$ and $t = t_z$ to reach that point. This implies that at time 0, when the zero-lower-bound episode starts, the costate variable $\psi(0)$ must jump somewhere between the zero-lower-bound equilibrium ψ_z and the forward-guidance equilibrium ψ_f, which is given by (13.21). We also see that when the zero lower bound lasts longer (higher t_z), the costate variable is pushed to higher levels at time 0.

In sum, we see that the costate variable is below normal during the forward-guidance episode. During the zero lower bound, the costate is higher than during forward guidance, but it may be above or below normal, depending on the durations of the zero-lower-bound and forward-guidance episodes. If the zero-lower-bound episode is long enough, the costate variable initially jumps to its highest point, then slowly decreases, and finally increases back to its normal level (figure 13.6D). Since aggregate tightness is negatively related to the costate variable, tightness jumps down at time 0, when the zero lower bound becomes binding. After that, the aggregate tightness slowly increases until time t_z, before decreasing back down to reach its normal level at time $t_z + \Delta$ (figure 13.6F).

The boom engineered during forward guidance therefore improves the situation at the zero lower bound. Instead of reaching the normal equilibrium at time t_z, the economy reaches a point with above-normal tightness at that time, so at any time before t_z, tightness is higher than without forward guidance, so the economy is less slack. Forward guidance is more powerful if it lasts longer and if the zero lower bound is shorter.

13.10.4. Closed-form expressions for the costate variable

So far we have used the phase lines for a qualitative analysis of the zero lower bound and forward guidance. But, because the Euler equation (13.8) is an autonomous linear first-order differential equation, it is not difficult to obtain closed-form expressions for the costate variable. We need to find an expression for the zero-lower-bound episode and a separate expression for the forward-guidance episode. We go backward in time again. From these expressions, we can derive the dynamics of the aggregate tightness $\theta(t)$ using equation (13.18).

During forward guidance, the costate variable's dynamics are governed by

$$\dot{\psi}(t) - (\delta + \pi)\psi(t) = -\frac{1}{a_n},$$

with boundary condition $\psi(t_z + \Delta) = \psi_n$. Applying result A.27, we find that during forward guidance, when $t_z < t < t_z + \Delta$, the costate variable is given by

$$\psi(t) = \psi_f + \left(\psi_n - \psi_f\right) \exp(-(\delta + \pi)(t_z + \Delta - t)).$$

This expression actually tells us where the costate must be at the end of the zero-lower-bound episode, and at the start of forward guidance. When $t = t_z$, the costate variable equals

$$\psi(t_z) = \psi_f \cdot [1 - \exp(-(\delta + \pi)\Delta)] + \psi_n \cdot \exp(-(\delta + \pi)\Delta). \tag{13.22}$$

So the costate variable is between the forward-guidance and normal equilibrium points, ψ_f and ψ_n, and it is closer and closer to the forward-guidance equilibrium point, so lower and lower, as the forward guidance lasts longer (higher Δ).

During the zero lower bound, the costate variable's dynamics are governed by

$$\dot{\psi}(t) - (\delta + \pi)\psi(t) = -\frac{1}{a_z},$$

with boundary condition that $\psi(t_z)$ is given by (13.22). Thus, we find that during the zero lower bound, when $0 < t < t_z$, the costate variable is given by

$$\begin{aligned} \psi(t) &= \psi_z \cdot [1 - \exp(-(\delta + \pi)(t_z - t))] \\ &+ \Big(\psi_f [1 - \exp(-(\delta + \pi)\Delta)] + \psi_n \exp(-(\delta + \pi)\Delta)\Big) \exp(-(\delta + \pi)(t_z - t)). \end{aligned} \tag{13.23}$$

Here we see again that the costate variable is higher when the zero lower bound lasts longer (higher t_z) but lower when forward guidance lasts longer (higher Δ).

The logic behind the forward guidance explains the comparative statics obtained here. Longer forward guidance (higher Δ) deepens and lengthens the post-zero-lower-bound boom, so it pulls the costate variable ψ down more at the start of that boom and, working backward, throughout the zero-lower-bound episode: the policy is more powerful. A longer zero-lower-bound episode (higher t_z) does the opposite. The promised boom is farther away, so it has less influence on the value of wealth at the beginning of the episode. Households then remain closer to the depressed zero-lower-bound situation, and forward guidance does less to relieve slack today.

13.10.5. Forward-guidance puzzle

One possible reason why forward guidance became popular at the time of the Great Recession is that the New Keynesian model predicts that the policy might be extremely powerful.

Michaillat and Saez (2021b, proposition 4) provides two results that describe the potency of forward guidance in the New Keynesian model. Just as described in table 13.2, consider a zero-lower-bound episode of duration t_z followed by a forward-guidance episode of duration Δ. The first result is that there exists a threshold Δ^* such that a forward-guidance episode longer than Δ^* transforms a zero-lower-bound episode of any duration into a boom. In addition, when forward guidance is longer than Δ^*, a long-enough forward-guidance episode or a long-enough zero-lower-bound episode generates an arbitrarily large boom. These results emerge in the New Keynesian model because its aggregate demand is degenerate: without wealth in the utility function, the dynamical system representing the model is a saddle at the zero lower bound, which creates explosive dynamics.

However, in the data, future monetary policy has much more subdued effects on output and inflation (Del Negro, Giannoni, and Patterson 2023). The effects of future monetary policy on current output and inflation are implausibly strong in the New Keynesian model. The gap between the New Keynesian model's prediction and the empirical evidence is known as the forward-guidance puzzle.

The slackish model does not suffer from the forward-guidance puzzle, as forward guidance only has limited effects in the model—especially if the zero-lower-bound episode lasts a long time. The intuition follows from the discussion above. If the zero lower bound lasts long enough, the future boom and future financial comfort is too distant to matter, and forward guidance becomes ineffective—which is why the model does not generate a forward-guidance puzzle.

The closed-form expressions above allow us to establish these results formally. First, there exists a threshold t_z^* such that a zero-lower-bound episode longer than t_z^* prompts a slump—a costate variable above normal and thus tightness below normal—irrespective of the duration of forward guidance. That is, for $t_z > t_z^*$, and for any Δ, $\psi(0) > \psi_n$, which implies $\theta(0) < \theta_n$ through (13.18).

This can be seen using the expression (13.23) for the costate variable during the zero-lower-bound episode. At time 0, the costate variable jumps to

$$\psi(0) = \psi_z \cdot [1 - \exp(-(\delta + \pi)t_z)] + \psi(t_z) \exp(-(\delta + \pi)t_z),$$

where $\psi(t_z)$ is given by (13.22). The value $\psi(t_z)$ is a weighted average of ψ_f and ψ_n, so the lowest value that it can take, irrespective of the duration of forward guidance, is ψ_f (which is reached if the forward-guidance episode lasts an infinitely long time). Since $\psi(0)$ is itself a weighted average of $\psi(t_z)$ and ψ_n, a lower bound on $\psi(0)$ is

$$\psi_z \cdot [1 - \exp(-(\delta + \pi)t_z)] + \psi_f \exp(-(\delta + \pi)t_z). \tag{13.24}$$

That lower bound is a weighted average of ψ_f and ψ_z, and the weight on $\psi_z > \psi_f$ is larger when t_z is larger. Thus, when t_z is large enough, $\psi(0)$ is close enough to ψ_z and therefore above ψ_n. In fact the threshold t_z^* is such that

$$\psi_z \cdot \left[1 - \exp\left(-(\delta + \pi)t_z^*\right)\right] + \psi_f \exp\left(-(\delta + \pi)t_z^*\right) = \psi_n,$$

guaranteeing that any zero-lower-bound episode longer than t_z^* prompts an initial slump. Reshuffling the equation, we extract a closed-form expression for the threshold:

$$t_z^* = \frac{1}{\delta + \pi} \cdot \ln\left(\frac{\psi_z - \psi_f}{\psi_z - \psi_n}\right).$$

Any zero-lower-bound episode longer than t_z^* generates a slump, irrespective of the length of the ensuing forward-guidance episode.

Furthermore, the slump approaches the zero-lower-bound equilibrium as the zero-lower-bound duration approaches infinity, irrespective of forward guidance. We see this by looking at the lower bound on the costate variable given by (13.24). As $t_z \to \infty$, the lower bound converges to ψ_z, which means that $\psi(0)$ itself converges to ψ_z—as $\psi(0)$ is sandwiched between the lower bound and ψ_z. So when the zero lower bound lasts a very long time, there is nothing that forward guidance can do: it is entirely ineffective, and the economy is stuck in the vicinity of the zero-lower-bound equilibrium.

Finally, arbitrarily large booms are impossible in the slackish model. The costate variable always remains between ψ_f and ψ_z, as (13.23) shows, so the costate variable is bounded below by ψ_f, implying that tightness is bounded above by θ_f.

13.11. Fiscal policy

Monetary policy plays a central role for stabilization in the United States and most other developed economies, but it is not always sufficient. In particular, when the nominal interest rate is constrained at the zero lower bound and aggregate demand remains too weak, it is natural to ask whether fiscal policy can provide additional support. Such support is especially important if the zero lower bound is expected to last a long time—because then the effectiveness of forward guidance diminishes greatly.

Indeed, the fiscal mechanism developed in chapter 11 in the context of public employment carries over directly to public spending. To simplify, we assume that public spending g does not affect the demand for private goods and services $y^d(\theta, i)$. This is what happens for instance if the utility over public goods and services g and private goods and services c is separable:

$$(1 - \aleph) \cdot c^{1-\alpha} + \aleph \cdot g^{1-\alpha},$$

where $\aleph \in (0, 1)$ measures the value of public goods relative to private goods.

Then, if the government purchases an amount g of goods and services to stimulate the economy, aggregate demand is boosted and becomes

$$y^d(\theta, i) + g, \tag{13.25}$$

where $y^d(\theta, i)$ is the private demand given by (13.11). As public spending stimulates aggregate demand, it also stimulates aggregate tightness, given by the modified equation:

$$y^d(\theta, i) + g = y^s(\theta). \tag{13.26}$$

As tightness rises, the unemployment rate, still given by (8.8), falls.

Hence, in this business cycle model, public spending operates just like public employment in the labor market model of chapter 11. The effect of public spending can be illustrated in the tightness-output diagram. As shown in figure 13.7A, after an increase in public spending, the aggregate demand curve shifts outward and the economy moves up along the aggregate supply curve: aggregate tightness and output rise. As tightness rises, unemployment falls. However, as tightness rises, it becomes more difficult and expensive for households to procure services: households therefore reduce their purchases. Just like public employment crowds out private employment, public spending crowds out private spending (a movement up along the private demand curve). The effects of public spending are summarized in table 13.1.

In fact the multiplier computed in chapter 11 continues to apply in this setting. It remains the case, as in (11.2), that the public-spending multiplier is given by

$$\frac{dy}{dg} = \frac{1}{1 - \epsilon^d_\theta / \epsilon^s_\theta}, \tag{13.27}$$

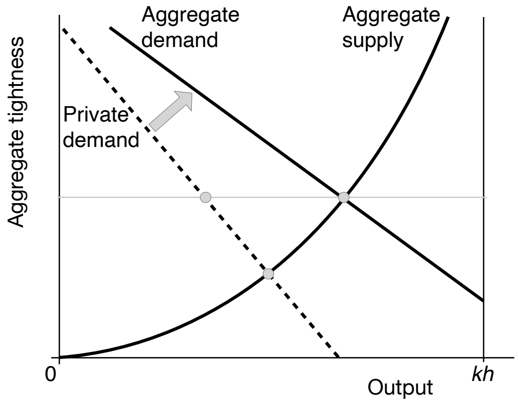
where ϵ^d_θ is the elasticity of private aggregate demand with respect to aggregate tightness, still given by (6.13), and ϵ^s_θ is the elasticity of aggregate supply with respect to aggregate tightness, still given by (8.14). Hence, the properties of the public-spending multiplier are determined by the positive elasticity ratio

$$-\frac{\epsilon^d_\theta}{\epsilon^s_\theta} = \frac{1-\alpha}{\alpha} \cdot \frac{\eta(\theta)}{1-\eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)}.$$

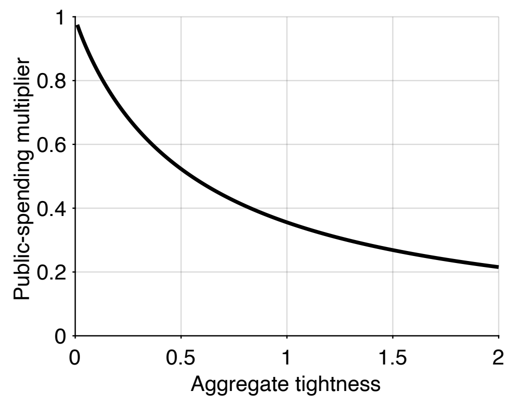
In chapter 8, we showed that the elasticity ratio determines the effects of demand and supply shocks in the slackish market model. Here, it determines all the properties of the public-spending multiplier.

Just like the public-employment multiplier, the public-spending multiplier is strictly positive, strictly less than 1, and strictly decreasing with tightness. The multiplier is α when tightness is efficient. The numerical values of the public-spending multiplier under the same calibration as above are presented in figure 13.7B. The fiscal multiplier is close to 1 when tightness is close to 0, it drops to 0.84 when tightness is 0.1, and then to 0.52 when tightness is 0.5. When the economy is efficient (tightness around 1), the multiplier is 0.35. The multiplier shrinks further as the economy becomes tighter: down to 0.27 when tightness is 1.5 and to 0.22 when tightness is 2.

Hence, purchasing public services increases output and reduces unemployment, but it diminishes purchases of private services by households. There is some crowding out of private services by public services: when the government purchases public services, it makes it harder for households to purchase private services because the government and households compete to hire the same workers.



A. Effect of public spending



B. Fiscal multiplier in good and bad times

FIGURE 13.7. Fiscal policy in the slackish business cycle model

The aggregate supply curve is given by equation (13.1). The aggregate demand curve is given by equation (13.11). The fiscal multiplier is given by (13.27). The model is calibrated as in table 10.2.

Furthermore, when tightness is lower, public spending crowds out private spending less and reduces unemployment more. So the public-spending multiplier is slack dependent, as public spending stimulates output more effectively in a slack economy than in a tight economy. This result is supported by several US studies, including Auerbach and Gorodnichenko (2012), Candelon and Lieb (2013), Fazzari, Morley, and Panovska (2015), and Ghassibe and Zanetti (2022). Relatedly, Brunet (2026) finds that employment multipliers were much lower during World War 2 than in later decades in the United States. This is consistent with our model, as the US labor market was the tightest in history during World War 2 (figure 12.1).

13.12. Summary

This chapter develops a slackish model of a service-based economy in which households both produce and consume services. Because all transactions must pass through a matching function, households cannot sell all of their productive capacity, giving rise to slack. The assumption that households derive utility not only from consumption but also from wealth ensures that aggregate demand is well defined. Aggregate tightness equilibrates aggregate demand and aggregate supply, and it in turn determines output, consumption, and unemployment.

The model delivers transparent comparative statics. Negative demand shocks reduce tightness and output, and raise unemployment. Negative supply shocks have distinct signatures: declines in labor force participation and declines in productivity both reduce output and raise tightness, but they have opposite implications for employment: lower productivity raises employment while lower participation decreases it. Finally, the chapter shows that the model continues to behave perfectly normally at the zero lower bound—although of course conventional monetary policy becomes ineffective then.

At the zero lower bound, forward guidance can stimulate the economy, especially when the zero-lower-bound episode is short. The model dynamics during forward guidance are determined by the Euler equation for the costate variable associated with households' wealth. The costate variable shifts the aggregate demand curve and therefore determines aggregate tightness at each date through the aggregate-demand-equals-aggregate-supply equation. This mapping allows us to study temporary zero-lower-bound episodes and forward guidance using transition paths for tightness.

Fiscal policy also stabilizes the economy in the model. Public spending stimulates aggregate demand, raising tightness and reducing unemployment. The public-spending multiplier—which measures the increase in output for each service purchased by the government—is positive, below one, and larger when the economy is slack.

In this chapter, we assumed that production occurs within households, which gives the business cycle model a labor-market flavor and allows slack to be interpreted as unemployment. An isomorphic version of the model has workers producing in their own shops and households visiting these shops to purchase services. Help-wanted ads then become visits, and slack becomes idle capacity, giving the model a product-market flavor. This malleability arises because the model has a single market that blends product and labor markets. The same abstraction will remain in chapter 14, which adds a Phillips curve to the model. It will be resolved in chapter 15, where we introduce separate product and labor markets.

CHAPTER 14.

Slackish Phillips curve

In chapter 13 we assumed that inflation was exogenously determined by price norms. However, the post-pandemic spike in inflation in the United States, which coincided with a very hot economy, revealed that inflation might arise when the economy is inefficiently tight. This situation had not occurred in the United States in the modern, post-Volcker era of monetary policy as the US economy had not been inefficiently tight since the end of the 1960s.

In this chapter, we insert a Phillips curve linking inflation to unemployment in the slackish business cycle model of chapter 13. The Phillips curve emerges from directed search: workers search for buyers offering the highest prices for their services, knowing of course that they will face more competition for high-paying jobs and less for low-paying jobs. The Phillips curve ensures inflation-slack coincidence: inflation is on target when the amount of slack is efficient. The Phillips curve is also convex in unemployment: inflation responds more strongly to unemployment when the economy is inefficiently tight than when the economy is inefficiently slack.

14.1. Overview of the model

The model follows the structure of the slackish business cycle model of chapter 13. Households purchase and consume services produced by other households. All the trades on the market for services are mediated by a matching function, which produces slack.

However, here the market is divided into many submarkets in which each household recruits workers to provide services for them. In each submarket, a household posts help-wanted ads that offer a specific price per service. Job search is directed: job seekers choose which submarket to join, depending on the price per service offered and of course the number of job seekers and help-wanted ads in that market. From a job seeker's perspective, the most attractive submarkets offer high prices, have many help-wanted ads, and have few other job seekers competing for the available positions.

Because search is directed, households must decide not only how many help-wanted ads they post but also what price they offer to pay for services rendered. A high price is expensive but allows households to recruit workers faster; a low price is cheap but imposes a long recruiting time.

This price-setting decision generates a Phillips curve. As usual in monetary models, to ensure that the Phillips curve is nondegenerate, we assume that households face price-adjustment costs—reflecting the type of fairness constraints discussed in chapter 5.

14.2. Slackish markets for services

Services are sold through long-term worker-household relationships. Once a worker has matched with a household, she becomes a full-time employee of the household, where she provides k services per unit time. She remains so until they separate, which occurs at rate λ.

To recruit workers, each household $j \in [0, 1]$ posts V_j help-wanted ads. These ads specify the price P_j that the household plans to pay for the services purchased. Each help-wanted ad is serviced by κ recruiters per unit time. The recruiters are employees of household j devoted to recruiting other workers.

Workers of other households only work for or apply to one household. In that way, each household j constitutes a submarket. Each worker decides which submarket to join, based on the price P_j offered for services and the submarket tightness θ_j. The submarket tightness is the ratio of the number of help-wanted ads and job seekers: $\theta_j = V_j/U_j$, where U_j is the number of unemployed workers from other households vying for jobs in household j. The tightness matters because it determines how quickly workers find a job. So job seekers trade off how much they are paid on the job with how long it takes to find a job.

In each submarket j, a matching function $m(U_j, V_j)$ determines the flow of new matches based on the number of job seekers and help-wanted ads. The matching function m satisfies the assumptions from chapter 3. Tightness θ_j determines the job-finding and ad-filling rates, as usual.

14.3. Equilibrium aggregate supply

In each submarket j, there are l_{ij} employed workers from household i, and a total $l_j = \int_0^1 l_{ij}(t)\, di$ employed workers from all households. There are also U_{ij} unemployed workers from household i, and a total $U_j = \int_0^1 U_{ij}(t)\, di$ unemployed workers from all households. The total number of workers from household i who are attached to submarket j is $h_{ij} = l_{ij} + U_{ij}$, and the labor force attached to submarket j is $h_j = \int_0^1 h_{ij}(t)\, di$.

Because it takes time to find a job in any submarket j, not all h_j workers are employed. The unemployment rate in submarket j is

$$u_j = \frac{U_j}{h_j}.$$

We assume that labor flows in each individual submarket are balanced: the number of workers who find a job with household j at any point in time equals the number who leave their jobs. This assumption implies that the unemployment rate in submarket j is a function of submarket tightness: $u_j = u(\theta_j)$, where the function $u(\theta)$ is given by (8.8).

The equilibrium submarket supply is the amount of services provided at any point in time given the submarket tightness, and given that submarket flows are balanced:

$$y_j^s(\theta_j) = \left[1 - u(\theta_j)\right] kh_j. \tag{14.1}$$

The equilibrium aggregate supply is the amount of services provided at any point in time across the entire economy, given that all submarket flows are balanced: $y^s = \int_0^1 y_j^s(\theta_j)\, dj$.

14.4. Directed search and price-tightness tradeoff

Household j observes the prices and tightnesses in all submarkets, and picks the one that suits them best. Indeed, different households may offer different prices P_j for services rendered. They may also post a different number of help-wanted ads and attract a different number of workers, generating a different tightness in submarket j—making it easier or harder to find a job there. In this section, we analyze how directed search links tightnesses and prices in all submarkets. This link will be the basis for the Phillips curve.

14.4.1. Arbitrage condition

When workers are unemployed, they do not receive any income. When they are employed, they produce k services per unit time, each paid a price P_j. Furthermore, in the submarket, the unemployment rate is $u(\theta_j)$, so a fraction $u(\theta_j)$ of the workers sent to submarket j are in unemployment, and a fraction $1 - u(\theta_j)$ are in employment. Accordingly, household i's expected income when they send h_{ij} workers to submarket j is $kP_j[1 - u(\theta_j)]h_{ij}$. The expected income per worker is $kP_j[1 - u(\theta_j)]$.

Household i does not prefer one employer or the other, so they pick their employer based solely on the expected income in the associated submarket j. They arbitrage across submarkets and send their workers to the submarket offering the highest expected income per worker. In their role as employer, households are aware of this arbitrage, so all households offer to pay a price for services that allows them to compete with other households. Arbitrage across submarkets therefore imposes that

$$P_j[1 - u(\theta_j)]$$

is the same across all submarkets j. Sellers direct their search toward the most attractive buyers, which induces competition across all buyers. Through competition, buyers set prices so sellers are indifferent across all buyers. If a buyer set a lower price than the rest, so the expected income on their submarket is lower than the rest, they would simply not attract any applicants.

By arbitrage, there is a price level P such that for all j,

$$P_j \cdot [1 - u(\theta_j)] = P \cdot [1 - u(\theta)], \tag{14.2}$$

where the aggregate tightness is the ratio of the aggregate number of help-wanted ads to the aggregate number of job seekers, given by

$$\theta = \frac{\sum_j V_j}{\sum_j U_j}.$$

14.4.2. Effect of price on tightness

The price posted by household j determines the tightness θ_j in their submarket, and therefore the ease with which the household is able to recruit workers. For instance, if the household offers to pay a high price for services, it attracts many workers—who want to work for a generous employer—so it can tap into a large pool of job seekers. This means that the household faces a low tightness, and that it can easily recruit workers.

This matters to household j because when it is harder to recruit—when help-wanted ads take longer to fill—they must devote more services to recruiting, and cannot consume as much. Indeed, the l_j workers in household j's employ produce $y_j = kl_j$ services. But the V_j help-wanted ads require each κ recruiters, so a total of $k\kappa V_j$ services are allocated to recruiting instead of consumption. Household j's consumption is therefore only

$$c_j = y_j - k\kappa V_j = k\left[l_j - \kappa V_j\right] = kh_j\left[1 - u_j - \kappa v_j\right], \tag{14.3}$$

where $u_j = U_j/h_j = 1 - l_j/h_j$ is the unemployment rate in submarket j and $v_j = V_j/h_j$ is the ad rate in submarket j.

Because we assume that flows are balanced in submarket j, we can express the gap between output and consumption of services using a matching wedge:

$$y_j = \left[1 + \tau(\theta_j)\right] c_j, \tag{14.4}$$

where the matching wedge $\tau(\theta_j)$ is given by (8.13).

Then, the arbitrage condition (14.2) tells us that when the price posted in submarket j is P_j, so the relative price in submarket j is $p_j = P_j/P$, then tightness in submarket j is

$$\theta_j(p_j) = u^{-1}\left(1 - \frac{1 - u(\theta)}{p_j}\right). \tag{14.5}$$

The function u^{-1} is strictly decreasing, so the submarket tightness $\theta_j(p_j)$ is decreasing in the relative price p_j. This means that a high relative price leads to low tightness and thus a low matching wedge $\tau(\theta_j)$. In fact, using again the results from appendix B, we see that the elasticity of the function $\theta_j(p_j)$ with respect to p_j is:

$$\epsilon_p^\theta = \frac{1}{\epsilon_\theta^u} \cdot \frac{-[1 - u(\theta)]/p_j}{1 - [1 - u(\theta)]/p_j} \cdot (-1) = \frac{-1}{[1 - \eta(\theta_j)]u(\theta_j)}, \tag{14.6}$$

where the elasticity ϵ_θ^u of $u(\theta)$ with respect to θ comes from (13.13) and is evaluated at θ_j, and we used $[1 - u(\theta)]/p_j = 1 - u(\theta_j)$.

In that way, employers have monopsonistic power in this economy. By offering higher prices for services, they attract more workers to their submarket, which allows them to recruit more easily. Conversely, if they choose to pay lower prices, fewer job seekers will join the submarket, and it will be difficult to recruit.

Household j therefore faces a tradeoff between the price per service, P_j, and the submarket tightness θ_j, which itself determines the amount of services that are devoted to matching, $\tau(\theta_j)$. All workers in household j are paid a price P_j per service. The expenditure by household j on workers therefore is

$$P_j y_j = \left[1 + \tau(\theta_j)\right] P_j c_j.$$

The effective price of services is not just P_j but $[1 + \tau(\theta_j)]P_j$. The price involves the price per service as well as the matching cost. Household j would prefer to pay a low price P_j and face a low tightness θ_j. But that is not possible, as (14.5) shows—hence the tradeoff.

The aim of the household is to minimize the effective price for services, $[1+\tau(\theta_j)]P_j$, which includes the cost of services consumed as well as the cost of services required for matching per unit of consumption. A low price P_j economizes on the price per service, but it imposes a high tightness θ_j and thus a high wedge $\tau(\theta_j)$.

14.4.3. Efficiency without price-adjustment costs

For reference, we describe what would happen in a submarket without any price-adjustment cost. Household j is free to set any price P_j she wants to minimize the effective price of services, $[1+\tau(\theta_j)]P_j$.

That is, the household chooses P_j to minimize $[1+\tau(\theta_j)]P_j$ subject to the arbitrage condition (14.2). Because of the arbitrage condition, the effective service price can be written

$$[1+\tau(\theta_j)]P_j = P\cdot[1-u(\theta)]\cdot\frac{1+\tau(\theta_j)}{1-u(\theta_j)}.$$

Hence, by choosing a price P_j, household j sets submarket tightness θ_j to minimize $[1+\tau(\theta_j)]/[1-u(\theta_j)]$, which is equivalent to maximizing $[1-u(\theta_j)]/[1+\tau(\theta_j)]$.

This in turn is equivalent to choosing tightness θ_j to maximize $1-u_j-\kappa v_j$. Indeed, notice by combining (14.1) and (14.4) that consumption by household j is

$$c_j = \frac{1-u(\theta_j)}{1+\tau(\theta_j)}kh_j.$$

But consumption is also given by (14.3), which shows that

$$\frac{1-u(\theta_j)}{1+\tau(\theta_j)} = 1-u_j-\kappa v_j.$$

This is equivalent to choosing the unemployment rate u_j to minimize $u_j+\kappa v(u_j)$, where the unemployment and ad rates are related by the Beveridge curve (8.5), which prevails here as flows are assumed to be balanced in submarket j. This is exactly the social planner's problem studied in chapter 9. Thus, tightness θ_j and unemployment rate u_j are chosen efficiently, so that condition (9.10) holds. Here we have just recovered the efficiency result of Moen (1997): directed search produces efficient market outcomes.

The key intuition is that directed search lets households internalize the planner's problem when choosing posted prices. In the absence of price-adjustment costs, each household's private pricing problem reproduces the planner's condition, so submarket tightness is efficient.

14.4.4. Introducing price rigidity

Generally, the unemployment rate is not efficient (chapter 12). To introduce inefficiency in the model, prices must be somewhat rigid.

Given that search is directed instead of random, prices are determined by buyers to minimize the effective cost of services—prices are not determined by norms here. To introduce price rigidity, we therefore assume that changing prices is costly for buyers. As in Rotemberg (1982), household j incurs a quadratic price-adjustment cost when inflation in submarket j departs from normal inflation π^n. The inflation in submarket j is

$$\pi_j(t) = \frac{\dot{P}_j(t)}{P_j(t)}. \tag{14.7}$$

The flow disutility caused by actual inflation deviating from normal inflation is

$$\frac{\varpi}{2} \cdot \left(\pi_j - \pi^n\right)^2.$$

The parameter $\varpi > 0$ governs the price-adjustment cost. This quadratic cost appears in the household j's utility function.

What does this price-adjustment mean? What does it capture in reality? First of all, when prices fall, or increase less than normal, workers in household j feel shortchanged. Indeed, the price p_j directly determines their salary, $p_j k$. And we saw in chapter 6 that workers' morale dips when their wages do not grow as expected, and that their performance may dip as well. The quadratic cost reflects the cost of restoring workers' morale, or the cost caused by diminished performance, when price growth is below the normal growth π^n.[1]

When prices rise, or increase more than normal, household j might be unhappy to pay more for the same services. We saw in chapter 6 that higher-than-normal inflation upsets customers, who feel unfairly treated when they shop and have to pay more for the same good or service. Here we assume that this displeasure with higher prices takes the form of a quadratic cost when price growth is above the normal growth π^n.

[1] The typical Keynesian wage-floor idea is that workers do not tolerate a drop in the level of nominal wages. But as Okun (1981, p. 93) notes, the same idea might apply to the growth rate of nominal wages, as we assume here: "Workers are likely to develop some notion of how fast they expect their wages to rise.... Any subsequent slowdown below the norm must be justified and explained to experienced workers. If the employer stresses the norm, a Keynesian wage-floor phenomenon may apply to the rate of increase of wages, and not just to their level."

14.5. Saving and pricing by households

We now present and solve the optimization problem of households. Households have essentially two decisions to make: how much income to save, and how to set prices for the services they purchase. From the optimal saving decision we will derive the aggregate demand curve, and from the optimal pricing decision the Phillips curve. We will then use these curves together with the aggregate supply curve to solve the model.

14.5.1. Household's utility function

Just as in chapter 13, household j cares about their consumption of services and their social status, derived from their relative real wealth. Here too, the household's nominal wealth $B_j(t)$ is held in government bonds. The household's relative nominal wealth is $B_j(t) - B(t)$, where $B(t) = \int_0^1 B_j(t)\, dj$ is average nominal wealth in the economy. The household's relative real wealth is $[B_j(t) - B(t)]/P(t)$, where the price level $P(t)$ is given by (14.2). In addition, households incur disutility from changing the prices they pay for services.

Hence, household j maximizes the discounted sum of flow utilities,

$$\int_0^\infty \exp(-\delta t)\left[c_j(t)^{1-\alpha} + \frac{1}{a}\cdot\frac{B_j(t) - B(t)}{P(t)} - \frac{\varpi}{2a}\cdot\left(\pi_j(t) - \pi^n\right)^2\right] dt, \tag{14.8}$$

where the parameter $\delta > 0$ is the time discount rate; the parameter $a > 0$ indicates the taste for consumption relative to other components of the utility function—saving and changing prices; and the parameter ϖ governs the cost of changing prices.

14.5.2. Household's budget constraint

Household j's nominal budget constraint is similar to that in chapter 13 except that the income of the household now comes from employment in different households that might pay different prices for services. The budget constraint therefore is

$$\dot{B}_j(t) = i(t)\cdot B_j(t) + \int_0^1 P_i(t) y_{ij}(t)\, di - P_j(t) y_j(t) - T(t),$$

where $T(t)$ is a lump-sum tax levied by the government.

Given that the unemployment rate is $u(\theta_i)$ in each submarket i, and given the arbitrage condition (14.2), the household's income can be rewritten as follows:

$$\begin{aligned}\int_0^1 P_i(t) y_{ij}(t)\, di &= \int_0^1 P_i(t)[1 - u(\theta_i(t))]kh_{ij}(t)\, di \\ &= P(t)[1 - u(\theta(t))]k \int_0^1 h_{ij}(t)\, di \\ &= P(t)[1 - u(\theta(t))]kh_j.\end{aligned}$$

Given the matching wedge (14.4) and tightness-price link (14.5), the household's expenditure can be written as a function of consumption and price:

$$P_j(t) y_j(t) = P_j(t)\left[1 + \tau\left(\theta_j\left(\frac{P_j(t)}{P(t)}\right)\right)\right] c_j(t).$$

Accordingly, the nominal budget constraint can be written as

$$\dot{B}_j(t) = i(t) \cdot B_j(t) + P(t)[1 - u(\theta(t))]kh_j - P_j(t)\left[1 + \tau\left(\theta_j\left(\frac{P_j(t)}{P(t)}\right)\right)\right] c_j(t) - T(t).$$

Then, just as in chapter 13, we translate the nominal budget constraint into a real budget constraint:

$$\dot{b}_j(t) = r(t) \cdot b_j(t) + [1 - u(\theta(t))]kh_j - p_j(t)\left[1 + \tau\left(\theta_j\left(p_j(t)\right)\right)\right] c_j(t) - \frac{T(t)}{P(t)}, \tag{14.9}$$

where $b_j(t) = B_j(t)/P(t)$ is real bond holdings. The real interest rate is defined by $r(t) = i(t) - \pi(t)$, where the inflation rate is $\pi(t) = \dot{P}(t)/P(t)$.

14.5.3. Law of motion for household's price

Besides the budget constraint, household j faces another constraint: the law of motion for the price it pays for services. The law of motion follows from (14.7), and says that any price change requires some inflation, which itself is costly. The law of motion for the nominal price $P_j(t)$ is

$$\dot{P}_j(t) = \pi_j(t) P_j(t).$$

Hence, using a result of the sort of (13.5), we find that the law of motion for the relative price $p_j(t) = P_j(t)/P(t)$ is

$$\dot{p}_j(t) = \left[\pi_j(t) - \pi(t)\right] p_j(t). \tag{14.10}$$

14.5.4. Household's problem

We now solve household j's maximization problem, which is to maximize (14.8) subject to the laws of motion (14.9) and (14.10). We do so using a Hamiltonian (result A.23):

$$\begin{aligned}
&\mathcal{H}(t, c_j(t), \pi_j(t), b_j(t), p_j(t)) = c_j(t)^{1-\alpha} + \frac{1}{a}\left[b_j(t) - b(t)\right] - \frac{\varpi}{2a}\left[\pi_j(t) - \pi^n\right]^2 \\
&+ \psi_j^b(t)\left\{r(t)b_j(t) + [1 - u(\theta(t))]kh_j - \left[1 + \tau\Big(\theta_j\big(p_j(t)\big)\Big)\right]p_j(t)c_j(t) - \frac{T(t)}{P(t)}\right\} \\
&+ \psi_j^p(t)\left[\pi_j(t) - \pi(t)\right]p_j(t).
\end{aligned}$$

The control variables are consumption $c_j(t)$ and inflation $\pi_j(t)$. The state variables are real bond holdings $b_j(t)$ and relative price $p_j(t)$. The costate variables are $\psi_j^b(t)$, which applies to the law of motion of real bond holdings (14.9), and $\psi_j^p(t)$, which applies to the law of motion of the relative price (14.10).

We now study the first-order conditions that characterize the household's optimal behavior (result A.24). We begin with the first-order condition with respect to consumption: $d\mathcal{H}/dc_j = 0$. It gives

$$(1 - \alpha)c_j(t)^{-\alpha} = \psi_j^b(t)\left[1 + \tau(\theta_j(t))\right]p_j(t). \tag{14.11}$$

Next we turn to the first-order condition with respect to inflation: $d\mathcal{H}/d\pi_j = 0$. It yields

$$\frac{\varpi}{a}\left[\pi_j(t) - \pi^n\right] = \psi_j^p(t)p_j(t). \tag{14.12}$$

The next first-order condition is that with respect to real bond holdings: $d\mathcal{H}/db_j = \delta\psi_j^b(t) - \dot{\psi}_j^b(t)$. It gives the same condition as in chapter 13:

$$\dot{\psi}_j^b(t) = [\delta - r(t)]\,\psi_j^b(t) - \frac{1}{a}. \tag{14.13}$$

The final first-order condition is that with respect to the relative price: $d\mathcal{H}/dp_j = \delta\psi_j^p(t) - \dot{\psi}_j^p(t)$. This condition becomes

$$\begin{aligned}
\dot{\psi}_j^p(t) - \delta\psi_j^p(t) &= \psi_j^p(t)[\pi(t) - \pi_j(t)] \\
&+ \psi_j^b(t)c_j(t)[1 + \tau(\theta_j(p_j(t)))]\left[1 + p_j(t)\frac{\tau'(\theta_j(t))}{1 + \tau(\theta_j(p_j(t)))}\theta_j'(p_j(t))\right].
\end{aligned}$$

We can rewrite the following term:

$$p_j(t)\frac{\tau'(\theta_j(t))}{1+\tau(\theta_j(p_j(t)))}\theta_j'(p_j(t)) = \frac{\theta_j(t)}{1+\tau(\theta_j(p_j(t)))}\cdot\tau'(\theta_j(t))\cdot\frac{p_j(t)}{\theta_j(t)}\cdot\theta_j'(p_j(t)) = \epsilon_\theta^{1+\tau}\cdot\epsilon_p^\theta,$$

where the elasticity $\epsilon_\theta^{1+\tau}$ is given by (6.12) and the elasticity ϵ_p^θ is given by (14.6), both evaluated at $\theta_j(t)$. Hence, the term boils down to

$$p_j(t)\frac{\tau'(\theta_j(t))}{1+\tau(\theta_j(p_j(t)))}\theta_j'(p_j(t)) = -\frac{\eta(\theta_j(t))}{1-\eta(\theta_j(t))}\cdot\frac{\tau(\theta_j(t))}{u(\theta_j(t))}.$$

Noting lastly that $[1+\tau(\theta_j(t))]c_j(t) = y_j(t)$, we simplify the first-order condition to

$$\dot{\psi}_j^p(t) = [\delta+\pi(t)-\pi_j(t)]\psi_j^p(t) + \psi_j^b(t)y_j(t)\left[1-\frac{\eta(\theta_j(t))}{1-\eta(\theta_j(t))}\cdot\frac{\tau(\theta_j(t))}{u(\theta_j(t))}\right]. \tag{14.14}$$

14.5.5. Symmetric solution

We focus on a symmetric solution to the model, in which all households behave the same, and all submarkets are therefore the same. In this symmetric situation, all submarket tightnesses θ_j are the same and equal the aggregate tightness θ. All submarket prices P_j are the same and equal the price level P, so all relative prices p_j equal 1, and submarket inflation π_j is the same as aggregate inflation π.

We can drop the index j on all individual variables.

The first-order conditions are necessary for an interior solution to the household's problem. So the symmetric solution must satisfy them. Under symmetry, the conditions (14.11), (14.12), (14.13), and (14.14) simplify as follows:

$$(1-\alpha)c(t)^{-\alpha} = \psi^b(t)\left[1+\tau(\theta(t))\right] \tag{14.15}$$

$$\frac{\varpi}{a}\left[\pi(t)-\pi^n\right] = \psi^p(t) \tag{14.16}$$

$$\dot{\psi}^b(t) = [\delta - r(t)]\,\psi^b(t) - \frac{1}{a} \tag{14.17}$$

$$\dot{\psi}^p(t) = \delta\psi^p(t) + \psi^b(t)y(t)\left[1-\frac{\eta(\theta(t))}{1-\eta(\theta(t))}\cdot\frac{\tau(\theta(t))}{u(\theta(t))}\right]. \tag{14.18}$$

14.6. Monetary policy

Before solving the model, we must specify monetary policy. Monetary policy has two dimensions in the model: setting interest rates and communicating around the inflation that can be expected in the future.

First, the central bank maintains a fixed real interest rate r. This is achieved by setting the nominal interest rate to $i(t) = r + \pi(t)$ at any point. There is no Taylor rule here: the central bank adjusts the nominal interest rate one-for-one with inflation to keep the real interest rate fixed. The central bank focuses on the real interest rate because this is the relevant price in the model: it is the relevant interest rate for aggregate demand.

Second, we assume that through effective communication and management of expectations, the Federal Reserve anchors people's inflation expectations to the inflation target, which is 2% in practice (Board of Governors of the Federal Reserve System 2026). Hence, we assume that normal inflation is just the Federal Reserve's inflation target: $\pi^n = \pi^*$. That is, the Federal Reserve manages to convince people that prices will grow at their target inflation rate. This assumption plays a central role in producing the inflation-slack coincidence, but it seems quite realistic. In the United States, since 1995, the long-run forecast of inflation elicited in the Survey of Professional Forecasters has been exceedingly stable, ever so slightly above 2% (Hazell et al. 2022, figure 2).

14.7. Equilibrium aggregate demand

We now derive the aggregate demand in the model. Just as in chapter 13, the aggregate demand is determined by two first-order conditions: (14.15) and (14.17).

These equations are the same equations as in chapter 13, so the aggregate-demand block is not affected by directed search or endogenous inflation. The reason is that by setting a real interest rate, monetary policy isolates aggregate demand from inflation. Inflation matters for aggregate demand only insofar as it affects the real interest rate. With the typical Taylor rule assumed in the New Keynesian model, inflation and real rate are linked. But that assumption appears there for convenience—to ensure the uniqueness of the solution of the model (Cochrane 2011). Here the solution is unique without it, so we simplify the analysis by making the reasonable assumption that the central bank fixes directly the price that matters: the real interest rate.

As in chapter 13, to respect the transversality condition and no-Ponzi condition, the costate variable ψ^b cannot diverge to ∞, and because the differential equation (14.17) is a source, the costate variable must jump to its equilibrium value at time 0, which is again given by equation (13.9): $\psi^b = 1/[(\delta - r)a]$.

Then, again as in chapter 13, the household's consumption is a function of the real interest rate and aggregate tightness. The function is obtained by plugging the value of the costate variable in (14.15). This gives again equation (13.10), from which we derive the same aggregate demand as in equation (13.11):

$$y^d(\theta, r) = \frac{[(1-\alpha)(\delta - r)a]^{1/\alpha}}{[1+\tau(\theta)]^{1/\alpha - 1}}. \tag{14.19}$$

14.8. Phillips curve

Finally, we derive the Phillips curve in the model. The Phillips curve comes from the two first-order conditions that we have not used so far: (14.16) and (14.18).

14.8.1. Expression of the Phillips curve

Equation (14.16) simply says that inflation is a linear function of the costate variable associated with the relative price: $\pi(t) = \pi^* + a\psi^p(t)/\varpi$.

The costate variable $\psi^p(t)$ is given by (14.18), which is a linear first-order differential equation. The equation is a source since $\delta > 0$. So if the costate variable does not jump to the equilibrium point of the differential equation, it exhibits explosive dynamics: going to ∞ as $t \to \infty$. Such dynamics imply that inflation would also diverge to $\pm\infty$, which cannot be optimal as that generates infinitely negative utility. So this costate variable must jump to the equilibrium point, just like the costate variable associated with real wealth:

$$\psi^p = \frac{y(\theta)}{\delta(\delta - r)a} \cdot \left[\frac{\eta(\theta)}{1 - \eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)} - 1\right].$$

Accordingly, inflation jumps at time 0 to the following value:

$$\pi = \pi^* + \frac{y(\theta)}{\varpi\delta(\delta - r)} \cdot \left[\frac{\eta(\theta)}{1 - \eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)} - 1\right]. \tag{14.20}$$

Equation (14.20) is the Phillips curve of the model. It links inflation to aggregate tightness, which is itself determined from the aggregate demand and supply curves. It has a few striking properties, which we review now.

14.8.2. Properties of the Phillips curve

The first property of the Phillips curve is that the inflation-slack coincidence holds: inflation is on target ($\pi = \pi^*$) whenever the economy operates efficiently ($u = u^*$). This can be seen because, when the economy operates efficiently, the term in square brackets in the Phillips curve is 0, as we established in equation (9.11).

In fact the term in square brackets in (14.20) measures the productive inefficiency of the economy. When the economy is inefficiently tight, the term is positive, so inflation is above target. When the economy is inefficiently slack, the term is negative, so inflation is below target. Furthermore, the response of inflation to departures from efficiency is governed by the price-adjustment cost ϖ. When prices are very costly to change, ϖ is large and inflation does not respond much to changes in tightness. By contrast, when prices are easy to change, ϖ is small and inflation responds much more forcefully to changes in tightness.

Thanks to the inflation-slack coincidence, the central bank does not face a tradeoff between inflation and unemployment: by bringing the unemployment rate to the FERU, it mechanically brings inflation to its target. Of course, the coincidence arises because we assumed that people take the inflation target as normal inflation. If that's not the case, the coincidence breaks down, as we discuss below. But when it's the case, the government automatically achieves its price mandate when it achieves its employment mandate.

We saw that inflation is above target when tightness is inefficiently high and below target when tightness is inefficiently low. But, more generally, inflation is strictly increasing in tightness around full employment and above. This can be seen because output $y(\theta)$, matching elasticity $\eta(\theta)$, matching wedge $\tau(\theta)$ are increasing in tightness, while the unemployment rate $u(\theta)$ is strictly decreasing in tightness. So above full employment ($\theta > \theta^*$), the term in square brackets is positive and growing with tightness and output is positive and growing with tightness so inflation is growing with tightness. Around full employment, the term in square brackets is 0, so the slope of the Phillips curve is determined by the derivative of the term in square brackets, which is strictly positive. Once again, inflation is increasing in tightness.

The logic actually continues to hold even below full employment ($\theta < \theta^*$) as long as output is large enough. Below full employment there are two opposing forces: output is increasing with tightness; the term in square brackets is increasing with tightness, but that term is negative and becoming less negative. Hence the product of the two terms is negative and could be increasing or decreasing. When tightness is close to full employment and the term in square brackets is close to 0, the change in the term dominates, so that inflation is increasing in tightness. When tightness is much lower and the term in square brackets is very negative, the negative change in output dominates, and inflation might decrease with tightness.

To illustrate the properties of the Phillips curve, we plot it in the calibrated model. Just as in chapter 13, we use the calibration from table 10.2, with in addition labor productivity normalized to $k = 1$, the discount rate set to $\delta = 91\%$, and the real interest rate set to $r = 0.5\%$. Under this calibration, the efficient tightness is $\theta = 1.03$ and the efficient unemployment rate is $u^* = 4.5\%$.

Here there is an important additional parameter to calibrate: the price-adjustment cost ϖ, which determines the slope of the Phillips curve. We calibrate it from microevidence collected by Zbaracki et al. (2004). Using time-and-motion methods, they study the pricing process of a large industrial firm. They find that the physical, managerial, and customer costs of changing prices amount to 1.22% of the firm's revenue in a given year. That year, the firm changed the price of 25% of its products, and most prices changed by about 4%. Based on this evidence, we calibrate ϖ so that if inflation is 4% on 25% of the services purchased by a household, the cost amounts to 1.22% of the household's purchases (which

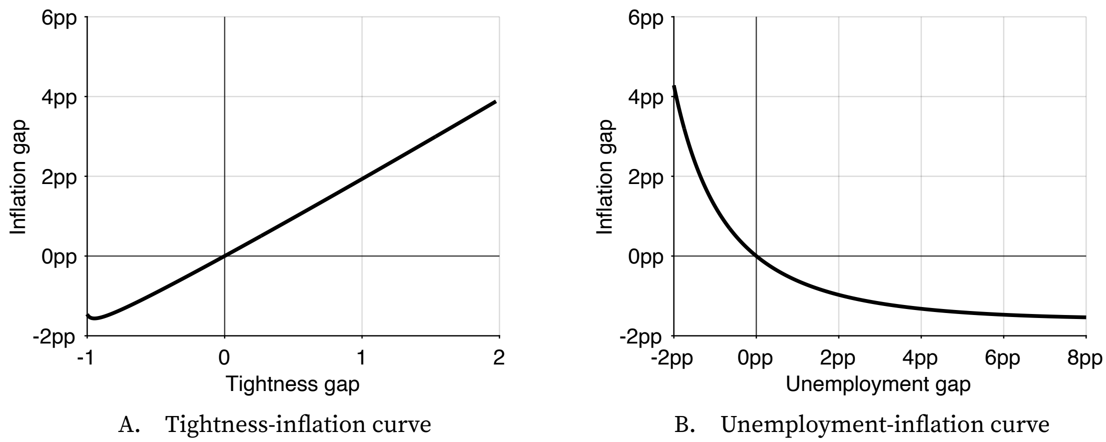


FIGURE 14.1. Phillips curves in the calibrated model

The Phillips curve is given by equation (14.20). Tightness and unemployment are related by equation (8.8). The model is calibrated as in table 10.2, with in addition labor productivity normalized to $k = 1$, the discount rate set to $\delta = 91\%$, the real interest rate set to $r = 0.5\%$, and the price-adjustment costs set to $\varpi = 58$.

correspond to the revenue of the household's workers) in utility terms. So we set the parameter so that

$$\frac{\varpi}{2a} \times 0.04^2 \times 0.25 = 0.0122 \times y,$$

where a is the taste for consumption and y is the real expenditure or income in the model. At the average tightness of 0.65, $y = 0.943$ and $a = 1$, so $\varpi = 58$.

Under this calibration, the Phillips curve is plotted in figure 14.1A. The non-monotonicity of the Phillips curve is observable but not relevant in practice. Inflation is increasing in tightness over the relevant range of tightness: for any tightness above 0.08. This means that generally, as expected, inflation rises as the economy becomes tighter. We also see that the Phillips curve grows broadly linear in tightness over a wide range: for tightness between 0.08 and 3.

14.8.3. The unemployment-inflation Phillips curve

The tightness-inflation Phillips curve appears almost linear under the calibration considered (figure 14.1A). However, once we represent the Phillips curve in a more traditional unemployment-inflation plane, the curvature of the Phillips curve changes drastically (figure 14.1B). The unemployment rate is a strictly decreasing function of tightness, given by (8.8), so tightness is a strictly decreasing function of the unemployment rate, $\theta(u)$. The figure plots $\pi(\theta(u))$, where $\pi(\theta)$ is given by (14.20).

As expected, the inflation rate is decreasing in the unemployment rate, but interestingly, inflation is now a convex function of unemployment. In the calibration, the point of inflation-slack coincidence occurs when inflation is $\pi^* = 2\%$ and unemployment is

$u^* = 4.5\%$. The Phillips curve goes through it, and it is much steeper when the unemployment rate is inefficiently low than when it is inefficiently high. When the unemployment gap goes up to 8pp, the inflation gap falls by just 1.5pp—so inflation falls to just 0.5%. But when the unemployment gap drops to -2pp, the inflation gap increases to 4.2pp, so inflation climbs to 6.2%.[2]

Although inflation varies smoothly with unemployment, the model produces relatively large fluctuations in inflation when the economy is inefficiently tight and relatively small fluctuations in inflation when the economy is inefficiently slack. Conversely, the model produces relatively small fluctuations in unemployment when the economy is inefficiently tight and relatively large fluctuations in unemployment when the economy is inefficiently slack. Here we obtain the result numerically; in appendix H, we establish analytically that the Phillips curve $\pi(u)$ is strictly convex under the common assumption that the matching function is Cobb-Douglas and symmetric: $m(u, v) = \mu\sqrt{uv}$.

14.8.4. Nonlinearity of the model

An interesting numerical property of the Phillips curve is that it is almost linear in tightness under this calibration (figure 14.1A). But that does not mean that the model is linear, because the aggregate supply curve is sharply nonlinear (figure 8.1B). For instance, aggregate demand shocks have small effects on tightness in a slack economy—they affect mostly output—but large effects on tightness in a tight economy. As inflation is linear in tightness, it behaves in a similar fashion: under aggregate demand shocks, it responds little when the economy is slack and much more when the economy is tight.

Relatedly, one wonders how the Phillips curve can have such different curvature depending on how it is represented: in a tightness-inflation plane or an unemployment-inflation plane. In fact, the near-linearity in the tightness-inflation representation is consistent with the strong curvature in the unemployment-inflation representation because unemployment is a nonlinear function of tightness through the Beveridge mapping $u(\theta)$, so a nearly linear $\pi(\theta)$ can generate a convex $\pi(u)$.

To conclude this discussion of nonlinearity, notice that in an output-unemployment plane, the aggregate supply curve is linear, given by $y^s(u) = (1 - u)kh$. So unemployment responds in a similar fashion to aggregate demand shocks whether the economy is slack or tight. However, when the economy is tight, inflation responds sharply to unemployment because of the convex Phillips curve, so it also responds sharply to aggregate demand shocks. By contrast, when the economy is slack, inflation barely responds to unemployment, so it responds little to aggregate demand shocks.

Of course, whether we analyze the model through tightness or unemployment, the

[2]The shape of the Phillips curve simulated in figure 14.1B bears an uncanny similarity with the shape of the original curve estimated by Phillips (1958, figure 1) from UK data for the years 1861–1913.

properties remain the same, although in the former case the nonlinearity comes from the aggregate supply curve while in the latter case it comes from the Phillips curve.

14.8.5. Intuition behind the Phillips curve

We have seen that price dynamics are driven by the amount of slack in the economy. When the economy is inefficiently slack, buyers are able to lower the prices they pay below what is normal. Conversely, when the economy is inefficiently tight, buyers are pushed to accept higher prices than what is normal. What is the intuition for this behavior?

When inflation is above normal, a buyer can reduce its price-adjustment cost by lowering its rate of inflation. Since pricing is optimal, however, there cannot exist any profitable deviation from the current situation. This means that the buyer must incur a commensurate cost when it lowers its rate of inflation. With lower inflation, the price offered by the buyer drops relative to the prices of other buyers. The absence of profitable deviation imposes that the price reduction is costly, so the price must already be below the effective-price-minimizing level. And since submarket tightness is inversely related to the buyer's price, submarket tightness must be above the effective-price-minimizing tightness, which is just the efficient tightness (section 14.4.3). In other words, when inflation is above normal, tightness must be inefficiently high—otherwise profitable deviations would exist for price-setting households.

The same logic holds when inflation is below normal. Then a buyer can reduce its price-adjustment cost by raising its rate of inflation. With higher inflation, the price offered by the buyer rises relative to the prices of other buyers. The absence of profitable deviation imposes that the price increase is costly, so the price must already be above the effective-price-minimizing level. And since submarket tightness is inversely related to the buyer's price, submarket tightness must be below the effective-price-minimizing level, which is the efficient tightness. So when inflation is below normal, tightness must be inefficiently low.

In this chapter, we have assumed that buyers set prices: they raise them when few workers are available to provide services and lower them when many workers are available. However, the results remain exactly the same if it is sellers and not buyers who set prices, as Michaillat and Saez (2024a) show. In that case, workers are able to raise prices for their services when they are in high demand, and are forced to lower their prices when demand slumps.

14.9. Solution of the model in normal times

We now solve the slackish business cycle model. We concentrate on normal times, when the nominal interest rate is positive. After that we will turn to the zero lower bound, when the nominal interest rate is at zero.

By symmetry, the aggregate supply is simply:

$$y^s(\theta) = [1 - u(\theta)]\, kh. \tag{14.21}$$

The aggregate demand is given by (14.19). The tightness that solves the model equalizes aggregate supply and demand:

$$y^d(\theta, r) = y^s(\theta). \tag{14.22}$$

Aggregate demand, aggregate supply, and solution are illustrated in figure 13.1B.

Then the inflation rate is given by the Phillips curve (14.20) and the unemployment rate by (8.8), and so on.

14.10. Implications for the conduct of monetary policy

This section now reviews a few implications of the slackish Phillips curve for the conduct of monetary policy.

14.10.1. Satisfying the dual mandate

In the United States, the Federal Reserve has a dual mandate: maintaining the economy at full employment while keeping prices stable. In the model, both mandates can be achieved through monetary policy. By selecting the real interest rate r, monetary policy moves the aggregate demand curve (14.19) along the aggregate supply curve. To reach the dual mandate, policy must bring the aggregate demand curve to the point where tightness is efficient ($\theta = \theta^*$). At this point, it is guaranteed through the Beveridge curve that the economy is at full employment ($u = u^*$) and along the Phillips curve that prices are stable ($\pi = \pi^*$).

We refer to the real interest rate that brings the economy to full employment as the full-employment rate of interest (FERI), and denote it by r^*. In the model, because of the inflation-slack coincidence, the FERI also ensures that prices are stable. The FERI is the real interest rate that guarantees that the aggregate demand and supply curves intersect at the efficient tightness:

$$y^d(\theta^*, r^*) = y^s(\theta^*).$$

From the expressions (14.19) and (14.21) for the aggregate demand and supply curves, we obtain an expression for the FERI:

$$r^* = \delta - \frac{1}{a} \cdot \frac{1 + \tau(\theta^*)}{(1 - \alpha)c(\theta^*)^{-\alpha}}. \tag{14.23}$$

The corresponding nominal interest rate is just $i^* = r^* + \pi^*$.

The Fed can achieve its dual mandate only if $i^* \geq 0$. If $i^* < 0$, then the dual-mandate solution would violate the zero lower bound constraint that $i \geq 0$. In that case the central bank would resort to setting $i = 0$. The zero lower bound becomes binding when $r^* < -\pi^*$. We see from (14.23) that this situation arises when the taste for consumption a is low enough, so that aggregate demand is depressed enough.

At the zero lower bound, monetary policy can no longer pin down the real interest rate independently of inflation: because the nominal interest rate is zero, the real interest rate is just negative inflation. So the normal-times decomposition of the model between the aggregate-demand-aggregate-supply block and the Phillips-curve block no longer applies. Below we tackle the analysis of the zero lower bound.

14.10.2. Choice of target variables

In the model, since the full-employment and price-stability mandates of the Fed coincide, aiming for the inflation target or the efficient unemployment rate is completely equivalent. In practice, however, the Fed should target the variable that is the most volatile—to observe more clearly departures from the dual mandate.

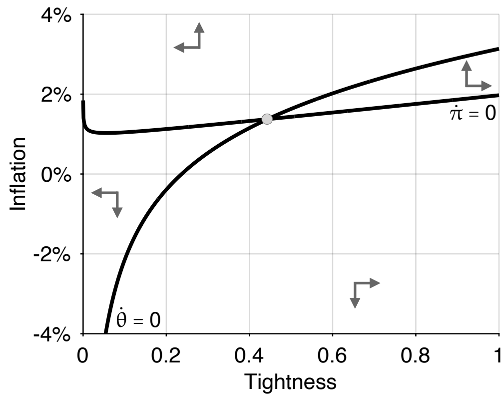
The slope of the Phillips curve in turn determines which of inflation and unemployment is the most volatile. If the Phillips curve is steep, aggregate-demand shocks will mostly generate movements in inflation, and targeting inflation will be easier. By contrast, if the Phillips curve is flat, aggregate-demand shocks will mostly generate movements in unemployment, and targeting unemployment will be easier.

In the model, the Phillips curve appears convex: flat when unemployment is inefficiently high, and steep when unemployment is inefficiently low. The implication is that the Fed should target the FERU u^* when the economy is inefficiently slack; and it should target an inflation rate of 2% when the economy is inefficiently tight.

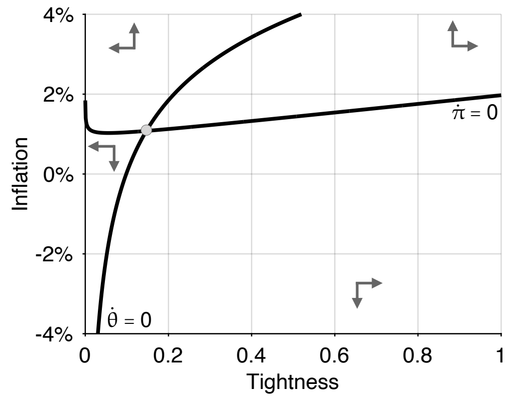
We saw that the inflation-tightness Phillips curve was broadly linear, which means that inflation and tightness vary commensurately. Hence, there is a corollary of the results for situations in which it is easier to measure tightness than inflation. In those cases, the Fed should target the FERU u^* when the economy is inefficiently slack and the efficient tightness θ^* when the economy is inefficiently tight. Shifting between unemployment and tightness in slumps and booms would give the Fed the maximum amount of information and precision to conduct policy.

14.11. Ineffective monetary communication

If monetary communication is ineffective, the inflation-slack coincidence breaks down. What we mean by ineffective monetary communication is that the inflation that people



A. Source at the zero lower bound



B. Source with lower aggregate demand

FIGURE 14.2. Phase diagrams at the zero lower bound in the calibrated model

The $\dot{\theta} = 0$ nullcline is given by equation (14.27). The $\dot{\pi} = 0$ nullcline is given by equation (14.28). The model is calibrated as in table 10.2, with in addition labor productivity normalized to $k = 1$, the discount rate set to $\delta = 91\%$, the real interest rate set to $r = 0.5\%$, and the price-adjustment costs set to $\varpi = 58$. In panel A, the taste for consumption is $a = 1.66$. In panel B, the taste for consumption is lower, set to $a = 1.62$.

consider normal, π^n, is not aligned with the inflation target, π^*.

With ineffective monetary communication, the derivations are the same, except that π^* is replaced by π^n in the Phillips curve (14.20). It is as if a shifter $\pi^n - \pi^*$ was added to the Phillips curve:

$$\pi = (\pi^n - \pi^*) + \pi^* + \frac{y(\theta)}{\varpi\delta(\delta - r)} \cdot \left[\frac{\eta(\theta)}{1 - \eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)} - 1\right]. \tag{14.24}$$

Because of the shifter $\pi^n - \pi^* \neq 0$, when the economy is at full employment (and the term in square brackets is 0), inflation is at π^n instead of π^*. So the inflation-slack coincidence breaks down: being at full employment does not guarantee price stability.

The two mandates of the central bank are not aligned here, and cannot be achieved at the same time. For instance if people believe that inflation will normally be above target ($\pi^n > \pi^*$), then inflation is too high when the economy is at full employment. Equivalently, when inflation is on target, tightness is too low, and the unemployment rate is inefficiently high.

Hence, in this model, the coincidence is not so much "divine" as the result of careful and effective messaging by the central bank.

14.12. Solution of the model at the zero lower bound

In normal times, the central bank sets a real interest rate, which separates the dynamics of inflation and tightness. Tightness can therefore be determined as in the fixed-inflation model of chapter 13, and inflation can be determined from the Phillips curve. At the zero lower bound, this separation disappears as the central bank loses control of the real interest rate. The nominal interest rate is stuck at 0 so the real interest rate is just negative inflation: $r = i - \pi = -\pi$. Tightness and inflation are therefore jointly determined by a two-dimensional dynamical system.

What might cause the economy to reach the zero lower bound? Typically, a large drop in aggregate demand, for instance a large increase in the marginal utility of wealth $1/a$. Aggregate demand must be depressed enough that a positive nominal interest rate is insufficient to maintain the economy at the divine equilibrium, where $\theta = \theta^*$ and $\pi = \pi^*$. The economy enters the zero lower bound only when at a zero nominal interest rate, aggregate demand is below the efficient level of output: $y^d(\theta^*, -\pi^*) < y^s(\theta^*)$. Hence, the marginal utility of wealth is high enough to push the economy to the zero lower bound when

$$\frac{1}{a} > \frac{(\delta + \pi^*)(1-\alpha)(c^*)^{-\alpha}}{[1+\tau^*]},$$

where c^* and τ^* are the efficient values of consumption and the matching wedge.

The tightness-inflation dynamical system describing the model at the zero lower bound is derived and analyzed in appendix I. We report the main results here. The system comprises two differential equations:

$$\frac{\dot{\theta}}{\theta} \cdot [\alpha [1-\eta(\theta)] u(\theta) + (1-\alpha)\eta(\theta)\tau(\theta)] = \frac{1}{a(1-\alpha)} \cdot [1+\tau(\theta)]\, c(\theta)^{\alpha} - (\delta + \pi) \tag{14.25}$$

$$\dot{\pi} = \delta\left[\pi - \pi^*\right] + \frac{a(1-\alpha)}{\varpi} \cdot c(\theta)^{1-\alpha}\left[1 - \frac{\eta(\theta)}{1-\eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)}\right]. \tag{14.26}$$

The $\dot{\theta}$ equation arises from equation (14.17), which describes households' consumption and saving behavior. The $\dot{\pi}$ equation arises from equation (14.18), which describes households' pricing behavior.

We construct the phase diagram of the nonlinear dynamical system to illustrate its properties, following the procedure presented in appendix A.6.2. The phase diagram is displayed in figure 14.2, under two possible calibrations.

The $\dot{\theta} = 0$ nullcline is a combination of the aggregate demand and aggregate supply curves from figure 13.1B. As inflation rises, the real interest rate falls, so the aggregate demand curve moves up along the aggregate supply curve, leading to a higher tightness. The nullcline admits the following expression:

$$\pi = \frac{1}{a(1-\alpha)} \cdot [1+\tau(\theta)]\, c(\theta)^{\alpha} - \delta. \tag{14.27}$$

The $\dot{\pi} = 0$ nullcline is mostly based on the Phillips curve from figure 14.1A, and it mostly describes inflation as an increasing function of tightness. The nullcline is given by

$$\pi = \pi^* + \frac{a(1-\alpha)}{\delta\varpi} \cdot c(\theta)^{1-\alpha} \left[\frac{\eta(\theta)}{1-\eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)} - 1 \right]. \tag{14.28}$$

Critically, the nullcline remains below the inflation target π^* on $(0, \theta^*)$, and is closer to it when prices are more rigid (higher ϖ).[3]

The system always admits an equilibrium, and when price rigidity ϖ is large enough, the equilibrium is guaranteed to be unique—as in figure 14.2. The large price rigidity keeps the nullcline $\dot{\pi} = 0$ flat enough, so close enough to $\pi = \pi^*$.

In that case, as we can see in figure 14.2, the dynamical system is a source around the equilibrium. Indeed, in all four quadrants of the phase diagram, the trajectories move away from the equilibrium.

This implies that the solution of the model is unique at the zero lower bound—although monetary policy is passive. The only solution is for tightness and inflation to jump to the equilibrium and stay there. If tightness and inflation jumped somewhere else, they would diverge away from the equilibrium.

An advantage of the slackish model is that we can use it to study permanent ZLB episodes. Unlike in the New Keynesian model, indeterminacy is never a risk at the zero lower bound. As we had already seen in chapter 13, the central bank can follow an interest-rate peg without creating indeterminacy. This is true even in this variant of the model with endogenous inflation.

So we can just assume that aggregate demand is depressed and the central bank keeps the interest rate at zero forever. The model admits a unique solution, where the economy is in a slump: inflation is below target and tightness is below its efficient level. The solution at the ZLB is the intersection of the two nullclines: (14.27) and (14.28).

When an unexpected and permanent shock occurs, the economy jumps to a new ZLB solution. We can use the phase diagram to study such jumps. For instance, negative aggregate-demand shocks continue to operate in a similar fashion at the zero lower bound. A negative aggregate demand shock—such as a decrease in the taste for consumption a—leads to an inward shift of the $\dot{\theta} = 0$ nullcline (from figure 14.2A to figure 14.2B). In response to the negative shock, tightness is lower and the unemployment rate is higher. If the solution is on the upward-sloping part of the $\dot{\pi} = 0$ nullcline, then inflation is lower.

14.13. Empirical evidence on the Phillips curve

In this section, we review recent US evidence supporting the slackish Phillips curve. In the modern US economy, the inflation-slack coincidence appears to hold, the infla-

[3] Recall that when tightness is inefficiently low, $\theta < \theta^*$, then $\tau(\theta)/u(\theta) < [1 - \eta(\theta)]/\eta(\theta)$, as equation (9.11) implies.

tion gap moves negatively with the unemployment gap, and the Phillips curve in the unemployment-inflation plane appears much steeper below the FERU than above it.

14.13.1. Divine coincidence

Figure 14.3A illustrates the parallel trajectories followed by the inflation gap and unemployment gap in the United States during the pandemic and after (2020–2024). Inflation is measured as annual change in the price index for personal consumption expenditures, which is itself produced by the BEA (2026b). The inflation gap is the inflation rate minus an inflation target of 2%. These are the inflation measure and inflation target used by the Federal Open Market Committee (Board of Governors of the Federal Reserve System 2026). The unemployment gap is computed in chapter 12 (figure 12.3).

The main observation is that unemployment gap and inflation gap evolve in tandem between 2020 and 2024. Both gaps are close to 0 at the beginning of 2020: the unemployment gap is just −0.2pp while the inflation gap is −0.4pp. So the economy is close to full employment and price stability—close to the point of inflation-slack coincidence.

Then, the following quarter, the unemployment gap rises very sharply and the inflation gap drops very sharply. In 2020:Q2, the unemployment gap reaches +6.4pp while the inflation gap bottoms at −1.4pp.

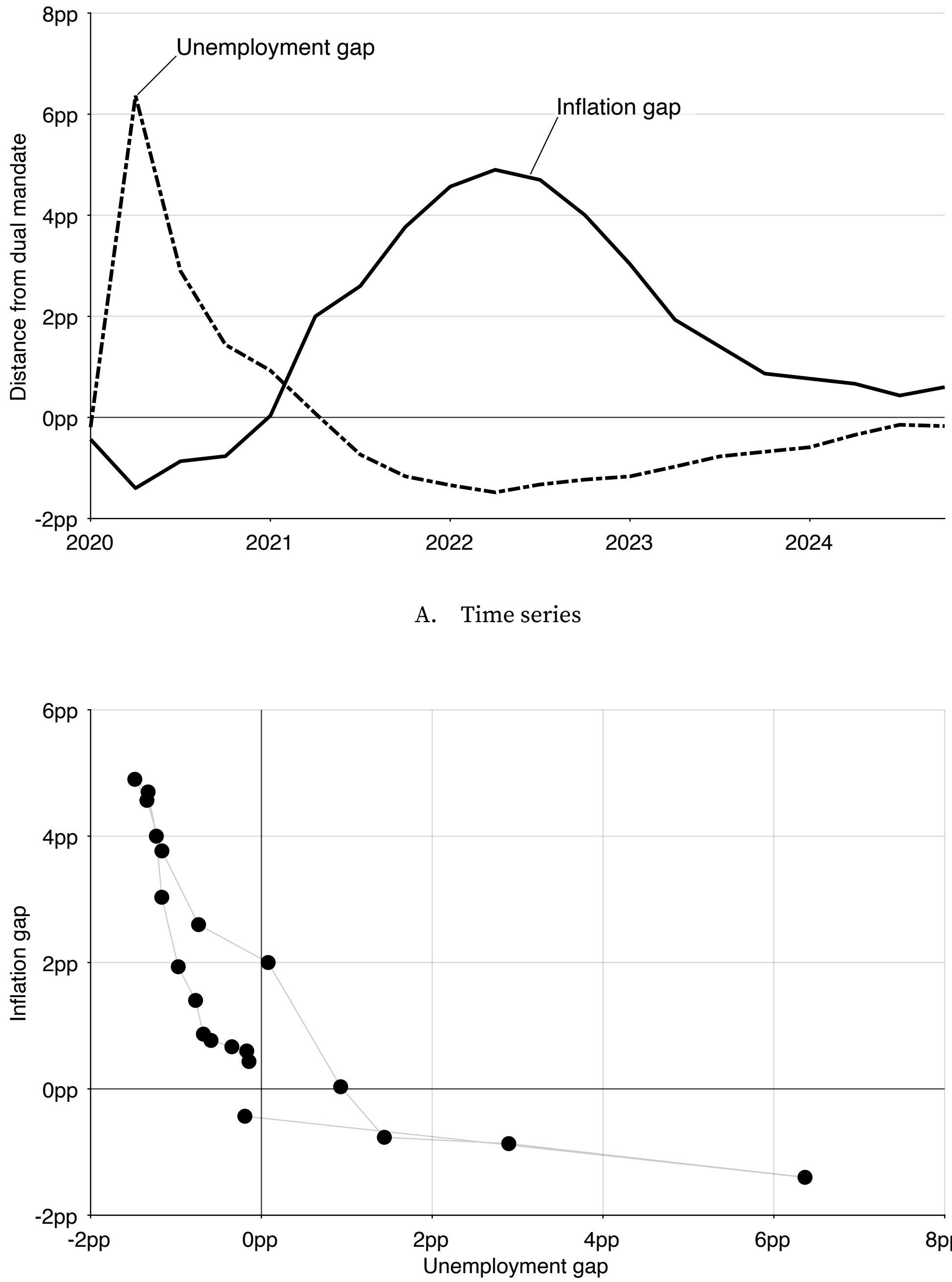
After that both gaps recover. They first return to 0 in 2021:Q1 for inflation and the following quarter for unemployment. Then, the inflation gap keeps rising and the unemployment gap keeps falling. In 2022:Q2, the inflation gap peaks at +4.9pp while the unemployment gap bottoms at −1.5pp.

After reaching these extreme values, both gaps normalize until the end of 2024. The inflation gap shrinks to reach +0.4pp in 2024:Q3. The unemployment gap shrinks to reach −0.1pp in 2024:Q3. So the economy is essentially back at full employment and stable prices at the end of 2024.

This parallel pattern suggests that the inflation-slack coincidence prevails in the modern US economy. The inflation gap is 0 whenever the unemployment gap is 0; or the inflation rate is on target whenever the unemployment rate is at the FERU; or prices are stable whenever the economy is at full employment. This coincidence occurred at the beginning of 2020, the beginning of 2021, and the end of 2024.

Moreover, inflation gap and unemployment gap are negatively related: inflation rises above target whenever the unemployment rate is below the FERU; and inflation falls below target whenever the unemployment rate is above the FERU.

The inflation-slack coincidence and negative relationship is also clearly visible when the inflation and unemployment gaps are displayed in a scatter plot (figure 14.3B). Interestingly, the empirical Phillips curve drawn by the US economy after the pandemic is quite close to the Phillips curve simulated from the slackish business cycle model (figure 14.1B).



A. Time series

B. Scatter plot

FIGURE 14.3. Phillips curve in the United States, 2020–2024

The unemployment gap comes from figure 12.3. The inflation rate is the quarterly average of the percentage change from a year ago of the monthly seasonally adjusted personal consumption expenditure price index, which is produced by the BEA (2026b). The inflation gap is the inflation rate minus the inflation target of 2%.

For the same fluctuations in unemployment gap, the inflation gap varies by comparable amounts in the data and simulations.

The basic evidence of inflation-slack coincidence presented in figure 14.3 is supported by other work. Benigno and Eggertsson (2023) use aggregate data for inflation and labor market tightness in the United States. They find clear evidence of coincidence in the 2008–2022 period: when the labor market tightness is efficient (at a level of 1), inflation is on target (at 2%) (Benigno and Eggertsson 2023, figure 4).

In the 1960s, the Council of Economic Advisers believed in the inflation-slack coincidence since they thought that at full employment the economy was not subject to inflationary pressures (Bernanke 2022, p. 18). And over the years the Fed through its FOMC statements often indicated that it believed in the coincidence, because it believed that price stability necessarily guaranteed full employment, so that its two mandates were equivalent (Thornton 2012, pp. 119–121, p. 130).

14.13.2. Convexity

Finally, it also seems that in the United States, the unemployment-inflation Phillips curve is convex.

The convexity of the Phillips curve appears clearly in figure 14.3B. In early 2020, the drop in inflation gap (down to −1.4pp) is much more subdued than the coinciding spike in unemployment gap (up to +6.4pp). So the economy was inefficiently slack and most of the movement occurred with the unemployment gap. By contrast, when the economy became inefficiently tight after 2021, most of the movement occurred with the inflation gap. In 2022, the spike in inflation gap (up to +4.9pp) was much larger than the drop in unemployment gap (down to −1.5pp). This indicates that the unemployment-inflation Phillips curve is much flatter when the economy is inefficiently slack than when it is inefficiently tight.

More generally, Babb and Detmeister (2017, table 4) and Smith, Timmermann, and Wright (2025, table 5) find that inflation is much more responsive to unemployment along the US Phillips curve when the unemployment rate is below 4.2% than when the unemployment rate is above 4.2%. In the postwar United States, the efficient unemployment rate is just 4.2% (figure 12.2), so the kink is detected at the efficient unemployment rate. It is true that these studies describe the pattern as a kink, whereas the slackish Phillips curve is smooth, but a convex Phillips curve can look kinked in finite samples.[4]

[4] Benigno and Eggertsson (2023, figure 4) even find that inflation is more responsive to labor market tightness along the US Phillips curve when tightness is inefficiently high (above 1) than when tightness is inefficiently low (below 1). In the numerical illustration of figure 14.1A, the slackish Phillips curve does not display this property: it is essentially linear in tightness. One way to generate the nonlinearity observed by Benigno and Eggertsson is to introduce an asymmetric price-adjustment cost, as Michaillat and Saez (2024a) do: when wage cuts are more costly than price hikes, the tightness-inflation Phillips curve is steeper when the economy is inefficiently tight than when it is inefficiently slack.

14.14. Soft landing after the coronavirus pandemic

After the coronavirus pandemic, inflation rose well above target—the inflation gap peaked at +4.9pp in 2022—while the US labor market became inefficiently tight (figure 14.3). Why policy allowed that configuration to emerge is a question we take up in chapter 16. Here we use the slackish business cycle model to study a more mechanical issue that dominated the policy debate once inflation was already high: whether the Federal Reserve could achieve a soft landing—bring inflation back down without a material rise in unemployment.

Inflation is governed by tightness through the Phillips curve, so cooling inflation requires reducing tightness. Monetary policy operates through the aggregate demand curve: to reduce tightness, it must raise the real interest rate and depress aggregate demand, which reduces tightness along the aggregate supply curve (figure 13.2A). As unemployment then moves with tightness along the aggregate supply curve, we see that a strict soft landing is therefore impossible in the model—the unemployment rate always increases after a decrease in tightness.

But because the aggregate supply curve is convex in slackish models, as we established in chapter 8, the rise in unemployment needed to restore efficient tightness can be modest when the labor market starts from a very tight position. When tightness is high and unemployment is low, the aggregate supply curve is steep, so a reduction in tightness raises unemployment only a little. In the flatter region of the curve, by contrast, the same reduction in tightness raises unemployment a lot. A "loose" soft landing is therefore possible: unemployment need not rise by much.

We can in fact compute the elasticity of the unemployment rate to tightness along the aggregate supply curve—and relate it to the elasticity of the Beveridge curve, which we have measured for the United States (chapter 2). By definition, $\theta = v(u)/u$, where $v(u)$ is the Beveridge curve—which is isomorphic to the aggregate supply curve. Implicitly differentiating in elasticity, we get

$$1 = -\beta \cdot \epsilon_{\theta}^{u} - \epsilon_{\theta}^{u},$$

where $\beta = -\epsilon_{u}^{v}$ is the elasticity of the Beveridge curve. Hence, the elasticity of the unemployment rate to tightness is directly determined by the Beveridge elasticity:

$$\epsilon_{\theta}^{u} = \frac{-1}{1+\beta}.$$

In the United States, $\beta = 1$, so $\epsilon_{\theta}^{u} = -1/2$.

Therefore, if the Fed wants to bring labor market tightness down from 2, its value at the peak of the post-pandemic boom (2022:Q2) to 1 (the efficient level), the unemployment rate would increase by $50\%/2 = 25\%$, since labor market tightness falls by 50%. Given that the unemployment rate was 3.5% at the peak of the post-pandemic boom (2022:Q3), it would increase to $3.5\% \times 1.25 = 4.4\%$. Thus, the effect on unemployment wouldn't be extreme, but it would not be zero either. In fact, the increase predicted by the Beveridge curve matches almost exactly the increase in unemployment observed after the pandemic: in July 2025, labor market tightness reached 0.99, and the unemployment rate was at 4.3% (Michaillat and Saez 2026).

14.15. Summary

This chapter develops a slackish model of the Phillips curve that links inflation to economic slack. Price dynamics are driven by the amount of slack in the economy. When the economy is slack, buyers can easily recruit workers and push the prices they pay below their normal level. Conversely, when the economy is tight, workers are scarce and buyers must post higher prices to attract workers. Without price rigidity, the model would be efficient, as in Moen (1997)'s model. But with price rigidity, the model generates a Phillips curve that describes inflation as a decreasing function of the unemployment gap.

The slackish Phillips curve ensures inflation-slack coincidence: inflation is on target if and only if the unemployment rate is efficient. In the unemployment-inflation plane, the Phillips curve is strictly convex in unemployment (see also appendix H). Inflation is therefore more sensitive to the unemployment gap when the labor market is inefficiently tight than when it is inefficiently slack.

Recent US evidence supports the model's predictions. The inflation-slack coincidence holds: inflation rose above target in 2021 as the unemployment rate fell below the FERU, and inflation has declined since its 2022 peak as unemployment has moved back toward the FERU. The unemployment-inflation Phillips curve also appears steeper when the labor market is inefficiently tight than when it is inefficiently slack.

A natural extension of this empirical analysis would be to move from a labor-market measure of slack to a broader measure of aggregate slack. The FERU and unemployment gap used in the chapter are labor-market objects, which is appropriate when the economy is modeled as a market for labor services. In a richer accounting of slack, slack would also incorporate idle labor inside firms, unsold goods, unused capacity, along the lines of the evidence in chapter 2. Constructing such an aggregate slack rate and slack gap is not yet feasible with existing US data, but it might yield new and interesting insights for assessing aggregate efficiency and understanding inflation fluctuations. Chapter 20 discusses the data that would make that measurement possible.

CHAPTER 15.

From unsold goods to unemployed workers

Chapter 12 showed that the US unemployment gap is generally positive and sharply countercyclical. The aim of part IV is to explain which aggregate shocks can generate these fluctuations, and to clarify how monetary and fiscal policy can undo such shocks so as to stabilize unemployment fluctuations.

In chapter 13, we developed the simplest possible slackish business cycle model to answer these questions. The main limitation of that chapter, however, is that it does not feature firms. All workers are self-employed and sell their production directly to other households. This is not very realistic as in the modern US economy, most production is mediated by firms, and only about 10% of the labor force is self-employed (Katz and Krueger 2019, table 1). To improve the realism of the slackish business cycle model, this chapter extends it by introducing firms that hire their employees on a labor market and sell their production on a product market.

This chapter's model therefore features two markets and two types of slack: a labor market with unemployed job seekers and a product market with unsold services and therefore idle workers. Both labor and product markets are organized around a matching function and are dynamic, as in the model of chapter 13. Just as workers and firms are in long-term employment relationships, consumers and firms are in long-term customer relationships. In that way, product and labor markets are perfectly symmetric.[1]

[1]Okun (1981, p. 142) also thought that the labor and product markets should be treated symmetrically. He argued that "In [many respects], customer markets share the characteristics of career labor markets. Both feature search costs, information costs, and bilateral-monopoly surpluses associated with established relations."

Critically, product market slack and labor market slack interact with each other, thus providing a richer description of labor demand. For instance, after an increase in aggregate demand, firms find more customers. This reduces the idle time of firms' employees and thus increases firms' labor demand. This in turn reduces unemployment. This richer labor demand allows us to answer many new questions that we could not address in the labor market model of chapter 10. We examine how aggregate demand shocks influence labor demand, how they can be distinguished from technology shocks that also influence labor demand, and how monetary and fiscal policy stimulate labor demand by boosting aggregate demand.

15.1. Overview of the model

This chapter brings together several model blocks developed in the previous chapters, and uses them to build a complete business cycle model with households, firms, a product market, and a labor market.

These markets are both organized around a matching function. Instead of having one big market where services are directly sold and all households are self-employed, we have two different markets and introduce firms in the model that transform labor into services. We keep the same notation in the product market and labor market, using only a subscript to designate the market (table 15.1).

First, households participate in the labor force, which generates a supply curve in the labor market, as in chapter 10.

Second, firms decide how many workers to hire, based on how easy it is to sell in the product market and how easy it is to recruit in the labor market. Firms' decisions generate the demand curve in the labor market, which is an extension of the labor demand in chapter 10. The labor demand depends on both product market tightness and labor market tightness; it therefore connects the two markets together.

Third, firms use labor to produce services that they sell to households in the product market. Firms' production capacity determines the aggregate supply, which constitutes the supply curve in the product market. The aggregate supply is an extension of the aggregate supply in chapter 13.

Fourth, households' decisions to spend or save determine the aggregate demand, which constitutes the demand curve in the product market. The aggregate demand takes the same form as in chapter 13.

Finally, the model features 4 equilibrating variables—2 per market. In the product market, the equilibrating variables are the product market tightness, which determines how easy or hard it is to buy and sell products, and the real interest rate, which determines the price of products relative to government bonds. In the labor market, the equilibrating variables are the labor market tightness, which determines how easy or hard it is to find jobs and employees, and the real wage, which determines the price of labor relative to products.

For simplicity, and following the assumption made in chapters 10 and 13, we assume that the real wage w is fixed, determined by a wage norm, and price inflation π is fixed, determined by an inflation norm. The central bank determines the real interest rate r, which with fixed inflation is tantamount to determining the nominal interest rate.

15.2. Building blocks of the model

In this section we collect the different building blocks of the model. Because all firms and households are the same, and because by now the distinction between individual and aggregate variables is hopefully clear, we directly use a representative household and representative firm to describe and solve the model.

15.2.1. Equilibrium labor supply

The equilibrium labor supply is the same as in chapter 10. The labor supply is an equilibrium object as it assumes that flows are balanced in the labor market. The equilibrium labor supply is the employment level given the labor force participation and labor market flows:

$$l^s(\theta_l) = [1 - u_l(\theta_l)]\, h, \tag{15.1}$$

where $h > 0$ is the size of the labor force, θ_l is the labor market tightness, and $u_l(\theta_l)$ is the equilibrium slack rate in the labor market—the equilibrium unemployment rate—given by

$$u_l(\theta_l) = \frac{\lambda_l}{\lambda_l + f_l(\theta_l)}. \tag{15.2}$$

In the expression for the unemployment rate, λ_l is the job separation rate, and $f_l(\theta_l) = m_l(1, \theta_l)$ is the job finding rate, itself determined by the labor market matching function $m_l(U_l, V_l)$.

15.2.2. Equilibrium labor demand

The equilibrium labor demand is almost the same as in chapter 10, but it must be extended to capture the fact that not all services offered for sale in the product market by the firm are actually sold. The labor demand is an equilibrium object as it assumes that flows are balanced on the labor and product markets.

TABLE 15.1. Notation in the slackish business cycle model with two markets

Product market			Labor market		
Symbol	Name	Definition	Symbol	Name	Definition
k	Aggregate capacity	–	h	Labor force	–
V_p	Shop visits	–	V_l	Job vacancies	–
U_p	Unsold services	–	U_l	Unemployed workers	–
v_p	Visit rate	$v_p = V_p/k$	v_l	Vacancy rate	$v_l = V_l/h$
u_p	Idleness rate	$u_p = U_p/k$	u_l	Unemployment rate	$u_l = U_l/h$
$m_p(u_p, v_p)$	Matching function	–	$m_l(u_l, v_l)$	Matching function	–
θ_p	Product market tightness	$\theta_p = v_p/u_p$	θ_l	Labor market tightness	$\theta_l = v_l/u_l$
f_p	Customer-finding rate	(3.4)	f_l	Job-finding rate	(3.4)
q_p	Buying rate	(3.5)	q_l	Job-filling rate	(3.5)
κ_p	Matching cost	–	κ_l	Recruiting cost	–
τ_p	Matching wedge	(8.13)	τ_l	Recruiting wedge	(10.4)
y	Output	–	l	Employment	–
y^s	Aggregate supply	–	l^s	Labor supply	–
y^d	Aggregate demand	–	l^d	Labor demand	–
c	Consumption	–	n	Producers	–
$\mathcal{U}$	Utility function	(13.2)	$\mathcal{Y}$	Production function	(10.5)
a_p	Taste for consumption relative to saving	–	a_l	Labor productivity	–
α_p	Diminishing marginal utility of consumption	–	α_l	Diminishing marginal productivity of labor	–
λ_p	Customer-producer separation rate	–	λ_l	Employer-employee separation rate	–
p	Price	–	w	Real wage	–

Because the product market is slackish, only a share $1 - u_p(\theta_p)$ of services are sold at any point in time, where θ_p is the product market tightness and u_p is the equilibrium slack rate in the product market—the equilibrium idleness rate—given by

$$u_p(\theta_p) = \frac{\lambda_p}{\lambda_p + f_p(\theta_p)}. \tag{15.3}$$

In the expression for the idleness rate, λ_p is the customer-separation rate, and $f_p(\theta_p) = m_p(1, \theta_p)$ is the customer-finding rate, itself determined by the product market matching function $m_p(U_p, V_p)$.

As a result, although the firm production function remains concave and given by

$$k(n) = a_l \cdot n^{1-\alpha_l}, \tag{15.4}$$

the flow of real profits is modified to

$$\mathcal{P} = \left[1 - u_p(\theta_p)\right] k(n) - w\left[1 + \tau_l(\theta_l)\right] n, \tag{15.5}$$

where w is the real wage, n is the number of producers in the firm, $k(n)$ is the firm's production capacity, θ_l is the labor market tightness, and τ_l is the equilibrium recruiting wedge, given by

$$\tau_l(\theta_l) = \frac{\kappa_l \lambda_l}{q_l(\theta_l) - \kappa_l \lambda_l}. \tag{15.6}$$

In the expression for the recruiting wedge, λ_l is the job separation rate, and $q_l(\theta_l) = m_l(1/\theta_l, 1)$ is the job filling rate, itself determined by the labor market matching function $m_l(U_l, V_l)$. In a slackish model, the production function does not tell us how much is sold (the output), but instead tells us how much could be sold (the capacity).

The labor demand in chapter 10 can therefore be used here, but with an additional term that captures slack in the product market. The labor demand gives us the number of workers that firms want to hire, with the goal of maximizing profits, for a given product market tightness, labor market tightness, and real wage. In chapter 10, because we only focus on the labor market, all of the firm's production is assumed to be sold. Here, by contrast, only a share $1 - u_p(\theta_p)$ of the firm's production capacity $k(n)$ is sold—because there is slack in the product market—so the labor demand becomes

$$l^d(\theta_p, \theta_l, w) = \left[\frac{(1-\alpha_l)\left[1 - u_p(\theta_p)\right] a_l}{w}\right]^{1/\alpha_l} \cdot \frac{1}{\left[1 + \tau_l(\theta_l)\right]^{1/\alpha_l - 1}}. \tag{15.7}$$

The main difference with the labor demand in chapter 10 is that now the labor demand curve depends on the slack rate in the product market and thus the product market tightness. So the labor demand is much more realistic than in the labor market model: instead of just depending on labor productivity, it also now depends on the demand for the firm's product. So both changes in technology and changes in demand affect labor demand. However, while the labor demand shifters might be different, the shape of the labor demand curve—how it depends on labor market tightness—remains the same.

15.2.3. Equilibrium aggregate supply

The equilibrium aggregate supply is almost the same as in chapter 13, but it must be extended to capture the fact that aggregate capacity is not exogenous but is determined by firms' production function and employment decisions. The aggregate supply is an equilibrium object as it assumes that flows are balanced in the product market.

Because the representative firm only sells a share $1 - u_p(\theta_p)$ of its productive capacity $k(n)$, the equilibrium aggregate supply is

$$y^s(\theta_p, n) = \left[1 - u_p(\theta_p)\right] k(n), \tag{15.8}$$

where $u_p(\theta_p)$ is the equilibrium idleness rate, given by (15.3), and $k(n)$ is aggregate capacity, given by (15.4).

The aggregate supply gives the output in the product market. Output is always less than the aggregate capacity because the product market is slackish, so workers are idle some of the time: $u_p(\theta_p) > 0$.

In (15.8) we see that the product market and labor market interact through the aggregate supply curve as well. In the same way that the labor demand curve (15.7) depends both on product market tightness and labor market tightness, the aggregate supply curve depends on product market tightness and employment, which itself depends on labor market tightness. Thus, we see that the aggregate supply also acts as a bridge between the labor market and product market.

In fact, we know that the number of producers in the labor market is

$$n = \frac{1 - u_l(\theta_l)}{1 + \tau_l(\theta_l)} \cdot h,$$

as the producers are people who are in the labor force but are neither unemployed nor recruiters. With this result, we rewrite the aggregate supply curve as a function of the two tightnesses:

$$y^s(\theta_p, \theta_l) = \left[1 - u_p(\theta_p)\right] k\left(\frac{1 - u_l(\theta_l)}{1 + \tau_l(\theta_l)} \cdot h\right). \tag{15.9}$$

15.2.4. Equilibrium aggregate demand

The households in our two-market model behave very similarly to the households in the basic model of chapter 13, so there is no need to spend time on solving the household problem—everything is essentially the same as in chapter 13.

The household's real budget constraint is slightly modified, because households now receive a wage from firms instead of receiving income from self-employment. The real budget constraint is now:

$$\dot{b}(t) = rb(t) + w\left[1 - u_l(\theta_l(t))\right]h - \left[1 + \tau_p(\theta_p(t))\right]c(t) + \mathcal{P}(t) - \frac{T(t)}{p(t)}. \tag{15.10}$$

There are two differences from the real budget constraint in chapter 13. The first is that the income of the household comes from the wages received from firms. The real wage is w, and the number of workers in firms is just the share $1 - u_l$ of the labor force h that is employed. The second is that households receive the profits $\mathcal{P}$ from firms, which are rebated to them. The households receive the profits because they naturally own the firms, since there is no one else in the economy who could own them.

However, the modifications to the budget constraint do not affect the household's problem. The household chooses consumption $c(t)$ and real bonds $b(t)$ to maximize utility, and as (15.10) shows, the terms with $c(t)$ and $b(t)$ are unaffected, so the solution to the household's problem is the same.

From the optimal amount of consumption, we compute the amount of output that households wish to purchase: $y = \left[1 + \tau_p(\theta_p)\right]c$. That amount of output constitutes the equilibrium aggregate demand, and it has exactly the same expression as in chapter 13:

$$y^d(\theta_p, r) = \frac{\left[(1-\alpha_p)(\delta - r)a_p\right]^{1/\alpha_p}}{\left[1 + \tau_p(\theta_p)\right]^{1/\alpha_p - 1}}, \tag{15.11}$$

where a_p and α_p are parameters from the household's utility function, which takes the same form as in chapter 13:

$$\int_0^\infty \exp(-\delta t)\left[c_j(t)^{1-\alpha_p} + \frac{1}{a_p}\cdot\left[b_j(t) - b(t)\right]\right]dt, \tag{15.12}$$

where j is the index of the individual household.

15.3. Solution of the model

We are now ready to solve the model. We first prove the existence and uniqueness of the solution analytically, and we then represent the solution graphically—which is useful to understand the mechanics of the model.

15.3.1. Existence and uniqueness of the solution

The structure of the solution to this chapter's business cycle model is similar to the structure of the solution of the business cycle model of chapter 13, except that the number of variables is about double, since this model features not one but two slackish markets. In particular, the key step now is to determine both the product market tightness and the labor market tightness. From the values of tightnesses, we then infer the values of all the other variables in the model.

Since we have two markets that interact with each other, solving the two-market model requires a little more work. We need to find the product market tightness θ_p and labor market tightness θ_l such that the aggregate demand equals aggregate supply in the product market, and labor demand equals labor supply in the labor market:

$$y^d(\theta_p, r) = y^s(\theta_p, \theta_l) \tag{15.13}$$

$$l^d(\theta_p, \theta_l, w) = l^s(\theta_l), \tag{15.14}$$

where the supply and demand curves are introduced above.

We first need to transform our two-equation system into a one-equation system to establish that the model always admits a solution and the solution is unique. That one equation will also be helpful to study the properties of the solution in the rest of the chapter.

Let us first consider the product market equation, given by (15.13). Separating the terms in θ_p and θ_l, and using the production function (15.4), it gives:

$$\frac{\left[(1-\alpha_p)(\delta-r)a_p\right]^{1/\alpha_p}}{a_l h^{1-\alpha_l}}\left[\frac{1+\tau_l(\theta_l)}{1-u_l(\theta_l)}\right]^{1-\alpha_l} = \left[1-u_p(\theta_p)\right]\left[1+\tau_p(\theta_p)\right]^{1/\alpha_p-1}. \tag{15.15}$$

Now, let us look at the labor market equation, given by (15.14). Separating the terms in θ_p and θ_l, and raising both sides to the power of α_l, it gives:

$$\frac{wh^{\alpha_l}}{(1-\alpha_l)a_l}\left[1-u_l(\theta_l)\right]^{\alpha_l}\left[1+\tau_l(\theta_l)\right]^{1-\alpha_l} = 1-u_p(\theta_p). \tag{15.16}$$

The main challenge here is that the term $[1+\tau_l(\theta_l)]/[1-u_l(\theta_l)]$ is not monotonic in θ_l in equation (15.15). So we need to eliminate that term to make progress. We do that by dividing (15.15) by (15.16) and rearranging terms:

$$\left[1+\tau_p(\theta_p)\right]^{1-1/\alpha_p} = \frac{wh\left[1-u_l(\theta_l)\right]}{(1-\alpha_l)\left[(1-\alpha_p)(\delta-r)a_p\right]^{1/\alpha_p}}. \tag{15.17}$$

We now rewrite both equations in a similar form: as formulas giving product market tightness as a function of labor market tightness. Equation (15.16) then says that product market tightness is a function of labor market tightness: $\theta_p = \theta_p^l(\theta_l)$, where

$$\theta_p^l(\theta_l, w) = u_p^{-1}\left(1 - \frac{wh^{\alpha_l}\left[1 - u_l(\theta_l)\right]^{\alpha_l}\left[1 + \tau_l(\theta_l)\right]^{1-\alpha_l}}{(1-\alpha_l)a_l}\right). \tag{15.18}$$

Here u_p^{-1} is the inverse of $\theta_p \mapsto u_p(\theta_p)$, which is well defined since u_p is strictly decreasing on $[0, \bar{\theta}_p] \to [u_p(\bar{\theta}_p), 1]$ and hence invertible. We infer the properties of the function θ_p^l from the properties of the functions u_p^{-1}, u_l, and τ_l. We see that θ_p^l is strictly increasing in θ_l, and that $\theta_p^l(0) = 0$. There is also a $\theta_l^l < \bar{\theta}_l$ such that $\theta_p^l(\theta_l^l) = \bar{\theta}_p$.

To get our second relationship, we look at equation (15.17). It says that product market tightness is another function of labor market tightness: $\theta_p = \theta_p^p(\theta_l)$, where

$$\theta_p^p(\theta_l, r, w) = \tau_p^{-1}\left(\frac{(1-\alpha_l)^{\alpha_p/(1-\alpha_p)}\left[(1-\alpha_p)(\delta - r)a_p\right]^{1/(1-\alpha_p)}}{(wh[1-u_l(\theta_l)])^{\alpha_p/(1-\alpha_p)}} - 1\right). \tag{15.19}$$

Here τ_p^{-1} is the inverse of $\theta_p \mapsto \tau_p(\theta_p)$, which is well defined since τ_p is strictly increasing on $[0, \bar{\theta}_p) \to [\tau_p(0), \infty)$ and hence invertible. We see that the function θ_p^p is strictly decreasing in θ_l. We also see that $\theta_p^p(0) = \bar{\theta}_p$. Finally, we see that there is a labor market tightness θ_l^p such that the function θ_p^p is well defined on $[0, \theta_l^p]$. If

$$\frac{(1-\alpha_l)^{\alpha_p/(1-\alpha_p)}\left[(1-\alpha_p)(\delta - r)a_p\right]^{1/(1-\alpha_p)}}{\left(wh[1-u_l(\theta_l^l)]\right)^{\alpha_p/(1-\alpha_p)}} \geq 1 + \tau_p(0),$$

then θ_p^p is well defined on $[0, \theta_l^l]$ and $\theta_p^p(\theta_l^l) \geq 0$, so we set $\theta_l^p = \theta_l^l$. If the inequality above does not hold, then there exists $\theta_l^p < \theta_l^l$ such that

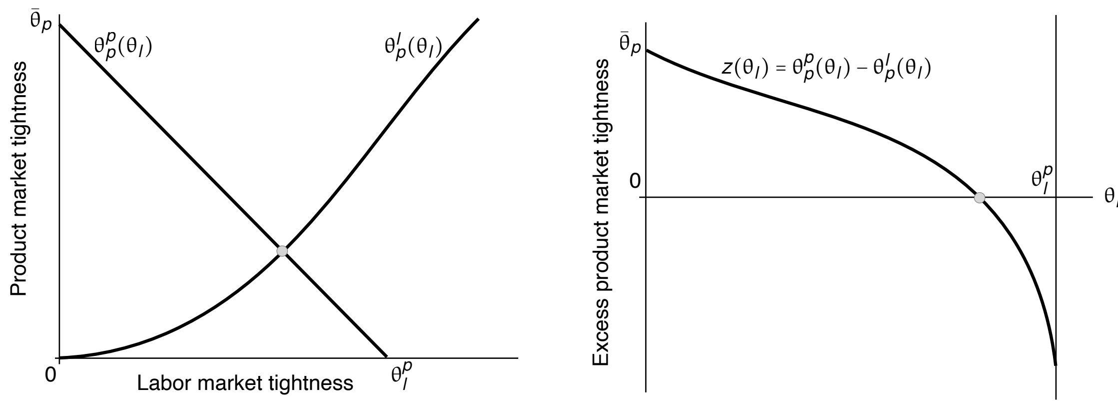
$$\frac{(1-\alpha_l)^{\alpha_p/(1-\alpha_p)}\left[(1-\alpha_p)(\delta - r)a_p\right]^{1/(1-\alpha_p)}}{\left(wh[1-u_l(\theta_l^p)]\right)^{\alpha_p/(1-\alpha_p)}} = 1 + \tau_p(0),$$

and then θ_p^p is well defined on $[0, \theta_l^p]$ and $\theta_p^p(\theta_l^p) = 0$.

We are now able to see that our two-market model has a unique solution. Indeed, solving the model is equivalent to finding the labor market tightness θ_l that solves the equation

$$\theta_p^l(\theta_l, w) = \theta_p^p(\theta_l, r, w). \tag{15.20}$$

We establish the existence and uniqueness of the solution by introducing the excess



A. Product market tightness curves B. Existence and uniqueness of the solution

FIGURE 15.1. One-diagram representation of the model solution

The $\theta_p^l(\theta_l)$ curve is given by equation (15.18). The $\theta_p^p(\theta_l)$ curve is given by equation (15.19).

tightness function:

$$z(\theta_l, r, w) = \theta_p^p(\theta_l, r, w) - \theta_p^l(\theta_l, w). \tag{15.21}$$

With the excess tightness function, the solution of the model simply is

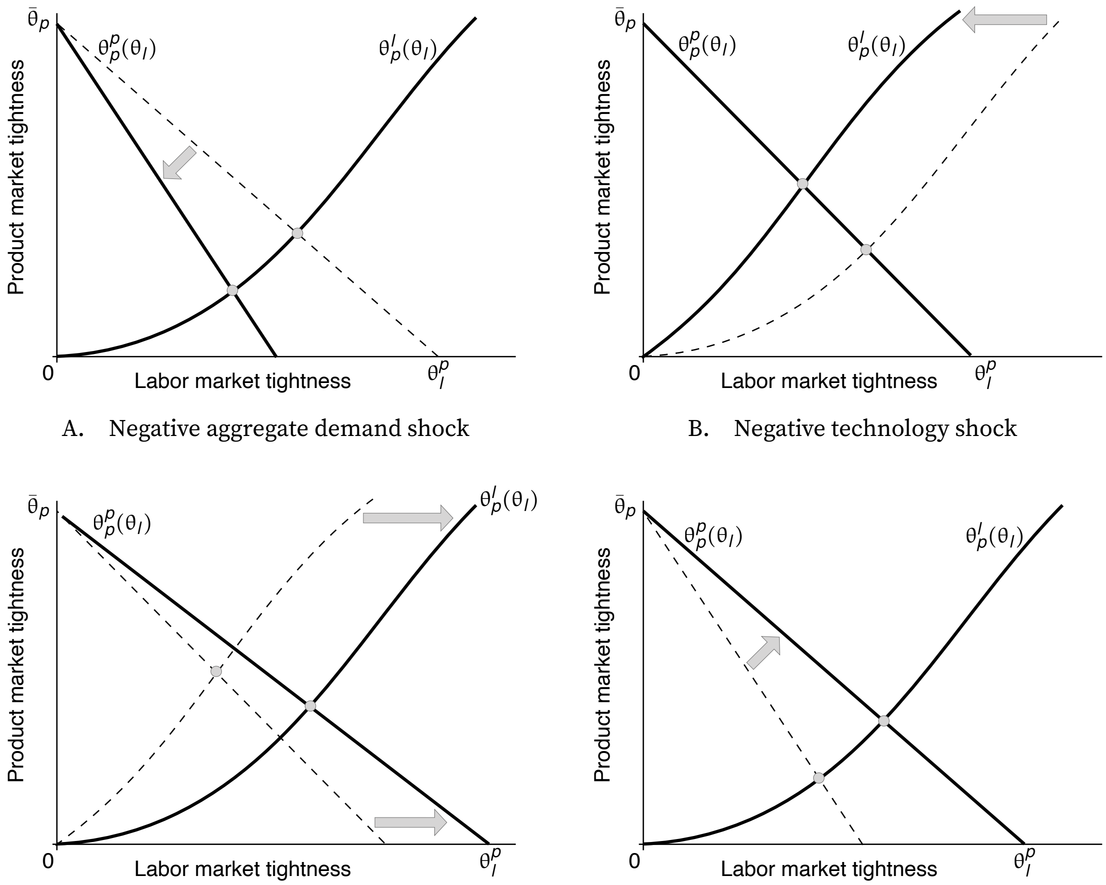
$$z(\theta_l, r, w) = 0.$$

We know that $\theta_p^p(\theta_l, r, w)$ is strictly decreasing in θ_l on $[0, \theta_l^p]$, starting from $\theta_p^p(0, r, w) = \bar{\theta}_p$ when $\theta_l = 0$, and reaching $\theta_p^p(\theta_l^p, r, w) = 0$ when $\theta_l = \theta_l^p$ if $\theta_l^p < \theta_l^l$ or $\theta_p^p(\theta_l^p, r, w) \geq 0$ when $\theta_l = \theta_l^p$ if $\theta_l^p = \theta_l^l$. At the same time, $\theta_p^l(\theta_l, w)$ is strictly increasing in θ_l on $[0, \theta_l^l]$, starting from $\theta_p^l(0, w) = 0$ when $\theta_l = 0$ and reaching $\theta_p^l(\theta_l^l, w) = \bar{\theta}_p$ when $\theta_l = \theta_l^l$. Moreover, both θ_p^p and θ_p^l are continuous in θ_l. The two curves are illustrated in figure 15.1A.

Thus, we know that the excess tightness function z is continuous in θ_l, and it declines from $z(0, r, w) = \bar{\theta}_p > 0$ to $z(\theta_l^p, r, w) < 0$ when θ_l increases from 0 to θ_l^p. Thus, by Bolzano's theorem, the equation $z(\theta_l, r, w) = 0$ has a solution on $(0, \theta_l^p)$ (result A.1). Moreover, since the excess tightness function is strictly decreasing, the solution is unique (figure 15.1B).

15.3.2. Graphical representation of the solution

We have seen that the two-market model reduces to a system of two supply-equals-demand equations: (15.13) and (15.14). To visualize how the model operates and how labor market and product market interact with each other, we represent the solution of the model graphically in two diagrams: one diagram plots labor demand and labor supply in an employment-labor market tightness plane, and the other diagram plots aggregate de-



A. Negative aggregate demand shock

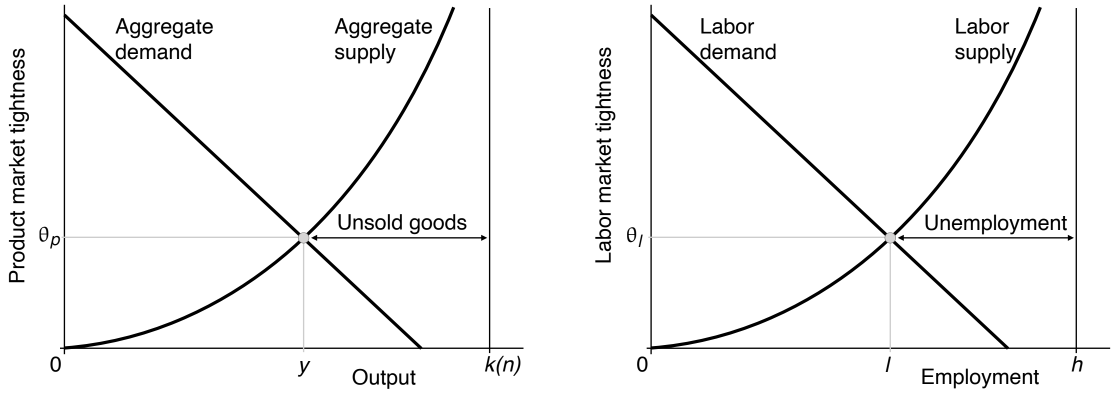
B. Negative technology shock

C. Negative labor supply shock

D. Expansionary monetary policy shock

FIGURE 15.2. Aggregate shocks in the one-diagram solution representation

The $\theta_p^l(\theta_l)$ curve is given by equation (15.18). The $\theta_p^p(\theta_l)$ curve is given by equation (15.19).



A. Product market

B. Labor market

FIGURE 15.3. Two-diagram representation of the model solution

The aggregate supply curve is given by equation (15.8). The aggregate demand curve is given by equation (15.11). The labor supply curve is given by equation (15.1). The labor demand curve is given by equation (15.7).

mand and aggregate supply in an output-product market tightness plane. This graphical representation of the solution is presented in figure 15.3.

Once we have the product market tightness, we deduce output by the aggregate demand curve since it does not depend on labor market tightness. Similarly, once we have the labor market tightness, we deduce employment by the labor supply curve since it does not depend on product market tightness. Then, both tightnesses must equate aggregate demand and aggregate supply as well as labor demand and labor supply. Thus, the interaction between the two markets occurs through the labor demand and aggregate supply curves.

15.3.3. Product-labor coincidence

In chapter 14 we discussed the concept of inflation-slack coincidence: such a coincidence emerges when the Phillips curve goes through the point where unemployment is efficient and inflation is on target. If the point of inflation-slack coincidence belongs to the Phillips curve, then the central bank can achieve its dual mandate: by modulating aggregate demand, it can bring the economy to full employment while maintaining inflation on target.

In this two-market model, an additional coincidence concept emerges: the product-labor coincidence, which occurs if product and labor markets can simultaneously be operating efficiently—if product-market slack and labor-market slack can simultaneously be efficient.

Thus, the product-labor coincidence emerges when curve $\theta_p^l(\theta_l)$ goes through the efficiency point $[\theta_l^*, \theta_p^*]$. That is, the coincidence emerges if $\theta_p^l(\theta_l^*) = \theta_p^*$. Looking at equation (15.18), we see that the product-labor coincidence appears if the real wage is just at the appropriate level:

$$w^* = \frac{(1-\alpha_l)a_l[1-u_p(\theta_p^*)]}{[1-u_l(\theta_l^*)]^{\alpha_l}[1+\tau_l(\theta_l^*)]^{1-\alpha_l}h^{\alpha_l}}. \tag{15.22}$$

If the real wage is at the level w^*, then the central bank maintains the product and labor markets at efficiency simultaneously by setting the real interest rate to

$$r^* = \delta - \frac{(w^*)^{\alpha_p}h^{\alpha_p}\left[1-u_l(\theta_l^*)\right]^{\alpha_p}\left[1+\tau_p(\theta_p^*)\right]^{1-\alpha_p}}{(1-\alpha_l)^{\alpha_p}(1-\alpha_p)a_p}. \tag{15.23}$$

At that real rate, $[\theta_l^*, \theta_p^*]$ is the solution of the model. The economy is therefore operating efficiently, and the labor market specifically is at full employment. The real rate r^* is therefore the full-employment rate of interest (FERI).

15.4. Keynesian, classical, and frictional unemployment

In chapter 10, we showed that the slackish labor market model offers a unified theory of unemployment, including both job rationing and frictional unemployment. One limitation of that model, however, is that job rationing is entirely due to labor productivity that is insufficient compared to real wages. Because the labor market model does not feature a separate product market, it cannot capture the idea that not all job rationing is due to excessive wages—some of it is due to insufficient product demand.

This chapter's model connects the slackish labor market to a slackish product market, which provides us with a complete theory of unemployment. In the model, unemployment can be decomposed into three components: a Keynesian component, a classical component, and a frictional component.[2] Unemployment has a Keynesian component because it depends on how easy or difficult it is for firms to sell their products; it has a classical component because it depends on how high or low the real wage is; and it has a frictional component because it depends on how easy or difficult it is for firms to recruit workers.

Formally, the labor demand curve is given by (15.7). We can see the different elements that drive unemployment. The real wage, w, captures the classical component of unemployment. When it is high, the labor demand decreases and unemployment goes up as a result. We also clearly see the Keynesian component of unemployment, which shows up in the $u_p(\theta_p)$ term, which is determined by aggregate demand. When the aggregate demand is small, product market slack increases, which reduces labor demand. This in turn increases unemployment. Lastly, we also see frictional unemployment in the

[2]This typology comes from Malinvaud (1977, figure 3).

$\tau_l(\theta_l)$ term, which is determined by recruiting costs. When recruiting costs increase, the recruiting wedge increases, which means labor demand is lower. As a result, there is an increase in unemployment.

15.5. Disentangling labor demand shocks

In chapter 10, we argued that labor demand shocks are the main source of unemployment fluctuations in the United States. We reached this conclusion by observing that employment level and labor market tightness are positively correlated. This chapter's model allows us to go one step further: it allows us to determine whether these labor demand shocks are caused by aggregate demand shocks or technology shocks.

15.5.1. Aggregate demand shocks

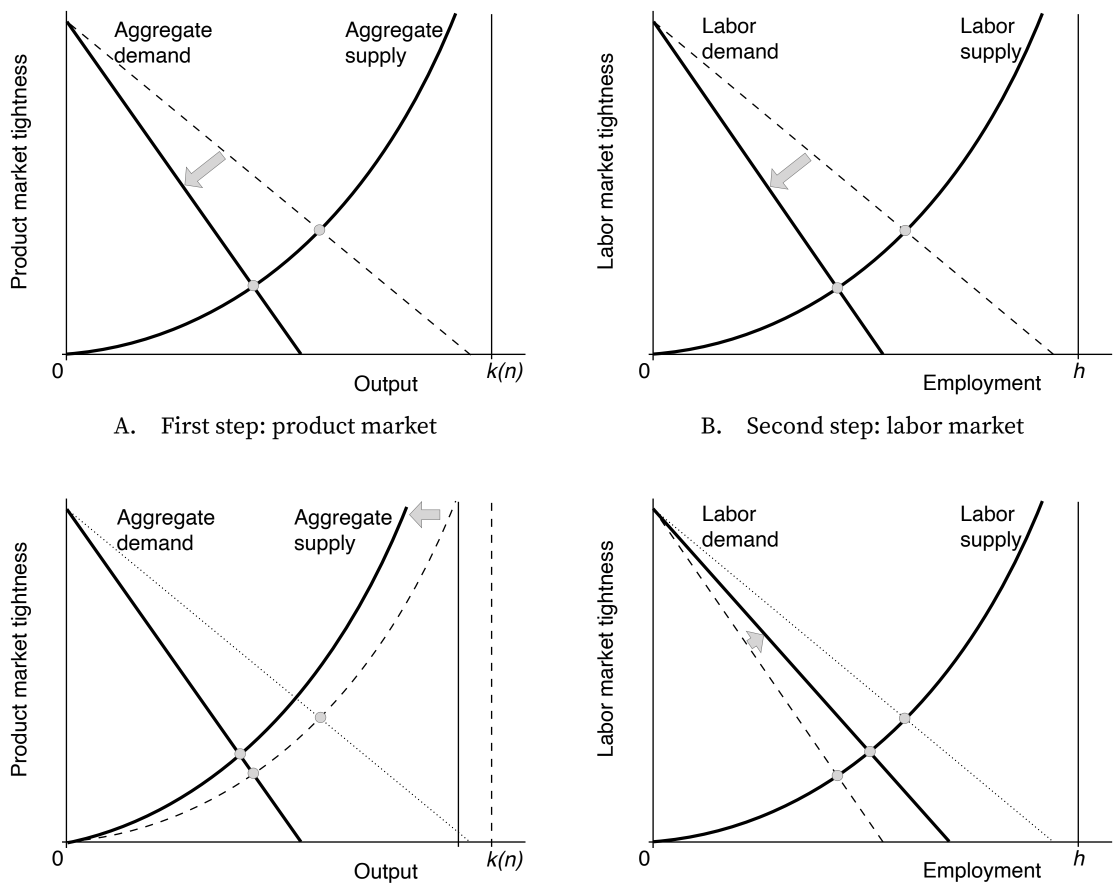
The most natural source of a negative aggregate demand shock is a decrease in the taste for consumption relative to saving, a_p. Another possible negative aggregate demand shock is a decrease in the time discount rate, δ. After such a shock, households become more thrifty: they desire to save more and consume less, which depresses aggregate demand (equation (15.11)).

Graphically, the shock triggers an inward rotation of the aggregate demand curve (figure 15.4A). This leads to a reduction in product market tightness θ_p and an increase in product market slack $u_p(\theta_p)$. The increased product market slack in turn depresses the labor demand curve, which rotates inward (figure 15.4B). This reduction in labor demand depresses the labor market tightness θ_l, and raises the unemployment rate $u_l(\theta_l)$.

Because labor market tightness drops, the number of producers $n(\theta_l)$ might change, so the aggregate capacity $k(n(\theta_l))$ might change, and there might be either a dampening or an accentuation of the initial shock in the product market. If the labor market is operating efficiently, $n(\theta_l)$ is maximized so does not respond to a small change in labor market tightness θ_l. If the labor market is inefficiently tight, $n(\theta_l)$ and $k(n(\theta_l))$ increase as θ_l drops. If the labor market is inefficiently slack, $n(\theta_l)$ and $k(n(\theta_l))$ decrease as θ_l drops.

The changes in aggregate capacity themselves lead to shifts in the aggregate supply curve. If the aggregate supply curve shifts inward, product market tightness moves back up; if the aggregate supply curve shifts outward, product market tightness moves further down. Of course, these changes in product market tightness further affect labor market tightness. The effect of the aggregate demand shock and linkage across markets is illustrated in figure 15.4C in the case of an inefficiently slack labor market, such that aggregate capacity drops after labor market tightness drops. That increase in product market tightness feeds back to the labor market, which pushes the labor demand slightly back up (figure 15.4D).

Given the feedback from the labor market to the product market, it is not entirely



A. First step: product market

B. Second step: labor market

C. Third step: back to product market

D. Fourth step: back to labor market

FIGURE 15.4. Negative aggregate demand shock in the slackish business cycle model

The aggregate supply curve is given by equation (15.8). The aggregate demand curve is given by equation (15.11). The labor supply curve is given by equation (15.1). The labor demand curve is given by equation (15.7).

clear from equations (15.13) and (15.14) what eventually happens to product and labor market tightnesses. But using the functions $\theta_p^p(\theta_l)$ and $\theta_p^l(\theta_l)$, we can establish how the tightnesses respond.

After a negative aggregate demand shock, whether a decrease in the taste for consumption a_p or discount rate δ, we see that $\theta_p^p(\theta_l)$, given by (15.19), shifts down, while $\theta_p^l(\theta_l)$, given by (15.18), stays the same. Thus, given equation (15.20), we infer that both tightnesses, θ_p and θ_l, decrease (figure 15.2A). Thus, maybe as expected, the response of the tightnesses is determined by the first-round response to the negative shock.

From the tightness results we infer a few other results: the unemployment rate, $u_l(\theta_l)$, increases and the idleness rate, $u_p(\theta_p)$, also increases: there is more slack on both markets. Because the unemployment rate increases, the employment level l decreases.

It is difficult, however, to determine how output changes just by looking at tightnesses. As the aggregate demand curve shifts inward, output tends to be lower. But as the product market tightness drops, the economy moves down along the aggregate demand curve, which raises output. To determine what happens overall, we need to use the labor-share result from chapter 10. In equation (10.9) we found using the firms' first-order condition that firms' wage bill is a share $1 - \alpha_l$ of their sales. The logic carries over here, even if not all productive capacity is sold, so that

$$(1 - \alpha_l)\, y = wl. \tag{15.24}$$

From this we clearly see what happens to output: since employment decreases but the real wage is unaffected, the wage bill drops, and therefore output y drops.

This completes our comparative statics for aggregate demand shocks, which are summarized in table 15.2.

15.5.2. Technology shocks

We now look at the model's response to technology shocks. A negative technology shock corresponds to a decrease in the technology parameter in the production function (15.4), a_l.

The negative technology shock simultaneously triggers an inward shift of the aggregate supply curve (figure 15.5A) and an inward rotation of the labor demand curve (figure 15.5B). These movements lead to an increase in product market tightness and a decrease in labor market tightness. This means in turn that the idleness rate decreases while the unemployment rate increases.

These tightness changes feed back to the product and labor markets. The increased product market tightness, and reduced product market slack, boosts labor demand, so the labor demand curve rotates back outward (figure 15.5D). Just as with the aggregate demand shock, the initial drop in labor market tightness might affect the number of producers and aggregate capacity, and there might be either a dampening or an accentuation of the initial aggregate supply shock in the product market. Figure 15.5C illustrates what occurs in the product market if the labor market is inefficiently slack, such that aggregate capacity drops after labor market tightness drops. Then the aggregate supply curve shifts inward further, and product market tightness moves up further. Of course, the change in product market tightness further affects labor market tightness.

Once again, we use the functions $\theta_p^p(\theta_l)$ and $\theta_p^l(\theta_l)$ to establish how the tightnesses respond in the presence of linkage across markets. We see that when technology a_l decreases, the function $\theta_p^p(\theta_l)$, given by (15.19), is unaffected, while the function $\theta_p^l(\theta_l)$, given by (15.18), shifts up. Thus, given equation (15.20), we infer that labor market tightness, θ_l, decreases while product market tightness, θ_p, increases (figure 15.2B). Once again, maybe as expected, the response of the tightnesses is determined by the first-round response to the negative shock.

From the tightness results we infer a few other results: the unemployment rate, $u_l(\theta_l)$, increases but the idleness rate, $u_p(\theta_p)$, decreases: there is more slack in the labor market but less in the product market. Because the unemployment rate increases, the employment level l decreases. As product market tightness increases, the economy is moving up along the fixed aggregate demand curve, so the output level y decreases too.

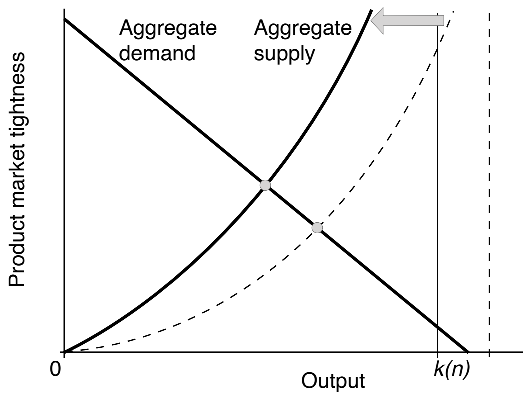
The comparative statics for technology shocks are summarized in table 15.2.

15.5.3. Labor supply shocks

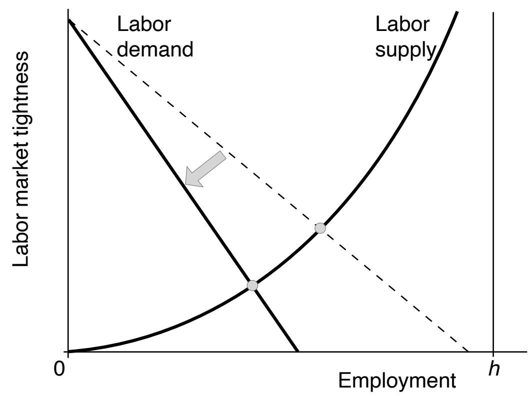
For completeness, and to contrast labor demand shocks, we finally consider the effects of labor supply shocks in the model. The most natural source of a negative labor supply shock is a decrease in the size of the labor force, h.

The negative labor supply shock triggers an inward shift of the labor supply curve (figure 15.6A). The shift leads to an increase in labor market tightness. This means in turn that the unemployment rate decreases and the recruiting wedge increases.

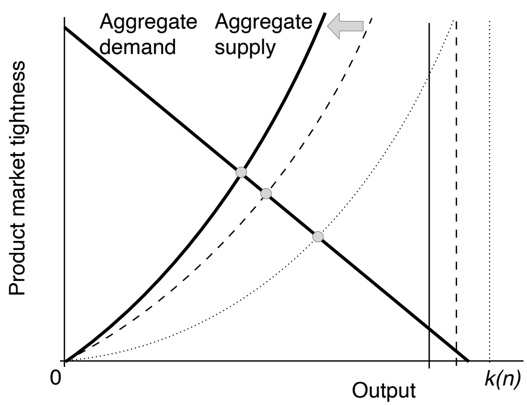
The labor market change feeds back to the product market. The labor market moves up along the labor demand curve, so employment decreases. Since the recruiting wedge increases due to the tighter labor market, the number of producers clearly drops. Aggregate capacity therefore declines, which means that the aggregate supply curve shifts inward (figure 15.6B). As the aggregate supply curve shifts inward, product market tightness increases. Of course, the change in product market tightness further affects labor market tightness: a higher product market tightness means a lower idleness rate, which stimulates the labor demand curve, and pushes labor market tightness further up (figure 15.6C). That change in labor market tightness might further change the number of producers and aggregate supply—for instance boost the aggregate supply curve if the labor market is



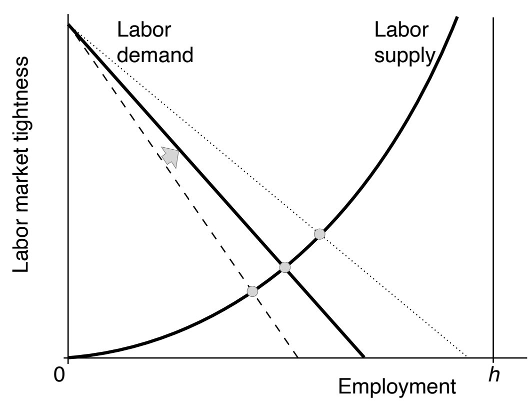
A. First step: product market



B. First step: labor market



C. Second step: back to product market



D. Second step: back to labor market

FIGURE 15.5. Negative technology shock in the slackish business cycle model

The aggregate supply curve is given by equation (15.8). The aggregate demand curve is given by equation (15.11). The labor supply curve is given by equation (15.1). The labor demand curve is given by equation (15.7).

inefficiently slack (figure 15.6D).

Once again, we use the functions $\theta_p^p(\theta_l)$ and $\theta_p^l(\theta_l)$ to establish how the tightnesses respond, once the cross-market linkages are accounted for. We see that when the labor force h decreases, the function $\theta_p^p(\theta_l)$, given by (15.19), shifts up, while the function $\theta_p^l(\theta_l)$, given by (15.18), shifts down. Thus, we infer that labor market tightness, θ_l, unambiguously increases (figure 15.2C).

What happens to product market tightness, θ_p, is not entirely clear from the shifts in the two functions $\theta_p^p(\theta_l)$ and $\theta_p^l(\theta_l)$ (figure 15.2C again). But we can use a proof by contradiction to show that θ_p eventually increases. Let us assume that θ_p decreases. As the product market moves down the aggregate demand curve, output y increases. Now, let us look at the labor demand equation, (15.7). Under our assumption that θ_p decreases, the idleness rate $u_p(\theta_p)$ increases, which depresses the labor demand. In addition, the labor market moves up the labor demand curve as labor market tightness increases. Both changes imply that employment decreases. But here we reach a contradiction, because output and employment must move in the same direction. Indeed, the labor share is simply $1 - \alpha_l$, as shown by (15.24). An increase in output means that employment increases too, since real wages are fixed. From this contradiction we learn that the product market tightness must increase. Once again, we find that the response of the tightnesses is determined by the first-round response to the shock.

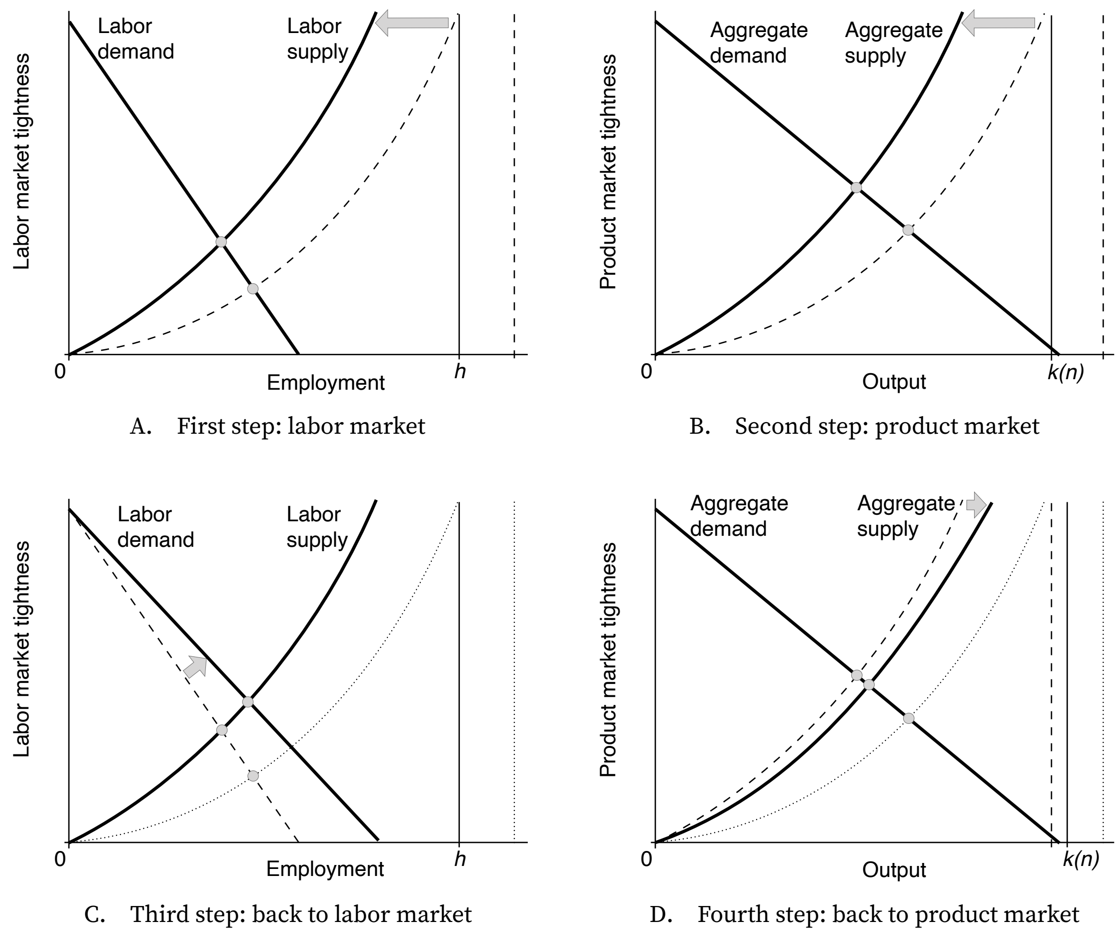
From the tightness results we infer a few other results: the unemployment rate, $u_l(\theta_l)$, decreases and the idleness rate, $u_p(\theta_p)$, also decreases: there is less slack on both markets after the negative labor supply shock. As the product market moves up the aggregate demand curve, output decreases. Using the labor share logic, we infer that employment decreases with output.

The comparative statics for labor supply shocks are summarized in table 15.2 as well.

15.5.4. Source of unemployment fluctuations

The comovements of quantities and prices, or in a macroeconomic model, of output and inflation, are commonly used to identify the sources of market or business cycle fluctuations. In a slackish model, prices and inflation are determined by norms, so the approach is not available. Instead, as we already saw in chapter 10, the sources of fluctuations in slackish models can be identified from the comovements of quantities and tightnesses. In our business cycle model, in particular, Okun curves allow us to separate labor demand versus labor supply and technology versus aggregate demand shocks.

In chapter 10 we used labor market data to argue that unemployment fluctuations are driven by labor demand shocks. Here we can use the standard Okun curve of changes in output against changes in unemployment rate to reach the same conclusion. Labor demand shocks can either be driven by technology or aggregate demand, as both enter the



A. First step: labor market

B. Second step: product market

C. Third step: back to labor market

D. Fourth step: back to product market

FIGURE 15.6. Negative labor supply shock in the slackish business cycle model

The aggregate supply curve is given by equation (15.8). The aggregate demand curve is given by equation (15.11). The labor supply curve is given by equation (15.1). The labor demand curve is given by equation (15.7).

TABLE 15.2. Comparative statics in the two-market slackish business cycle model

Shock	Product market tightness θ_p	Idleness rate u_p	Output y	Labor market tightness θ_l	Unemployment rate u_l	Employment l
Decrease in aggregate demand a_p	–	+	–	–	+	–
Decrease in technology a_l	+	–	–	–	+	–
Decrease in labor supply h	+	–	–	+	–	–
Decrease in nominal interest rate i	+	–	+	+	–	+

The comparative statics are obtained from figures 15.2A, 15.2B, 15.2C, and 15.2D. A variable's increase is denoted by "+" and a decrease by "-".

labor demand. But thankfully, both aggregate demand and technology shocks generate negative comovements between output and unemployment rate, so a downward-sloping Okun curve (table 15.2). By contrast, labor supply shocks generate positive comovements between output and unemployment rate, so they would produce an upward-sloping Okun curve (table 15.2). We showed in figure 2.11 that the Okun curve is downward sloping, as Okun's law stipulates, and from this we confirm that labor demand shocks cause unemployment fluctuations.

But which sort of labor demand shocks are the cause of the fluctuations: technology or aggregate demand shocks? Aggregate demand shocks generate negative comovements between output and idleness rate whereas technology shocks generate positive comovements between output and idleness rate (table 15.2). Hence, to ascertain whether the labor demand shocks causing unemployment fluctuations are aggregate demand or technology shocks, we simply plot the product-market Okun curves, relating output and product market slack. Under technology shocks, the Okun curve is upward sloping; under aggregate demand shocks, it is downward sloping.

As it turns out, in the United States, the product-market Okun curve is downward sloping—just like the labor-market Okun curve. That is, fluctuations in output and in idleness rate are negatively correlated. We established this fact with figures 2.12 and 2.13. Such a negative slope indicates that business cycle fluctuations are mostly caused by aggregate demand shocks, not by technology shocks.

15.5.5. Origins of aggregate demand shocks

We have just seen how aggregate demand shocks, transmitted across the economy through slack, are the main source of US business cycles. But what generates these shocks in the first place? For a few episodes, evidence is available to explain what caused a collapse in aggregate demand.

Hausman, Rhode, and Wieland (2019) find that in 1933, the economy recovered steadily from the Great Depression once crop prices rose and boosted the income and consumption of farmers. In that case, aggregate demand was stimulated through redistribution to an indebted population with high marginal propensity to consume—the farmers. Crop prices rose because the United States left the Gold Standard in the spring of 1933, which allowed the dollar to depreciate and the domestic prices of farm products to rise.

Farmer (2012) finds that during the Great Recession, the colossal drop in households' wealth, due to the collapse of the housing and stock markets, coincided with the rise in unemployment. He argues further that this is because the drop in wealth led to a collapse in aggregate demand. Mian and Sufi (2014) provide microevidence supporting this idea: US counties with a larger decline in housing wealth experienced a larger decline in nontradable employment between 2007 and 2009.

But it still remains to understand why fluctuations in aggregate demand seem to endlessly reappear to trigger the next recession. One possibility is that some of the events unleash the "animal spirits" that lie dormant otherwise, as Keynes (1936, chapter 12) proposed:

> Even apart from the instability due to speculation, there is the instability due to the characteristic of human nature that a large proportion of our positive activities depend on spontaneous optimism rather than on a mathematical expectation, whether moral or hedonistic or economic. Most, probably, of our decisions to do something positive, the full consequences of which will be drawn out over many days to come, can only be taken as the result of animal spirits—a spontaneous urge to action rather than inaction, and not as the outcome of a weighted average of quantitative benefits multiplied by quantitative probabilities.

It might be that after periods of stability, the animal spirits always awaken and produce bubbles and then crashes. These crashes destroy wealth and stymie aggregate demand. This could explain the dot-com recession of 2001 and the Great Depression (stock market bubbles), the Great Recession (housing market bubble), and maybe the next recession (bubble around artificial intelligence and cryptocurrency). Speculative manias, followed by financial crashes, have happened time and time again throughout history, driven by irrational exuberance from investors, and often exacerbated by fraudulent behavior.[3]

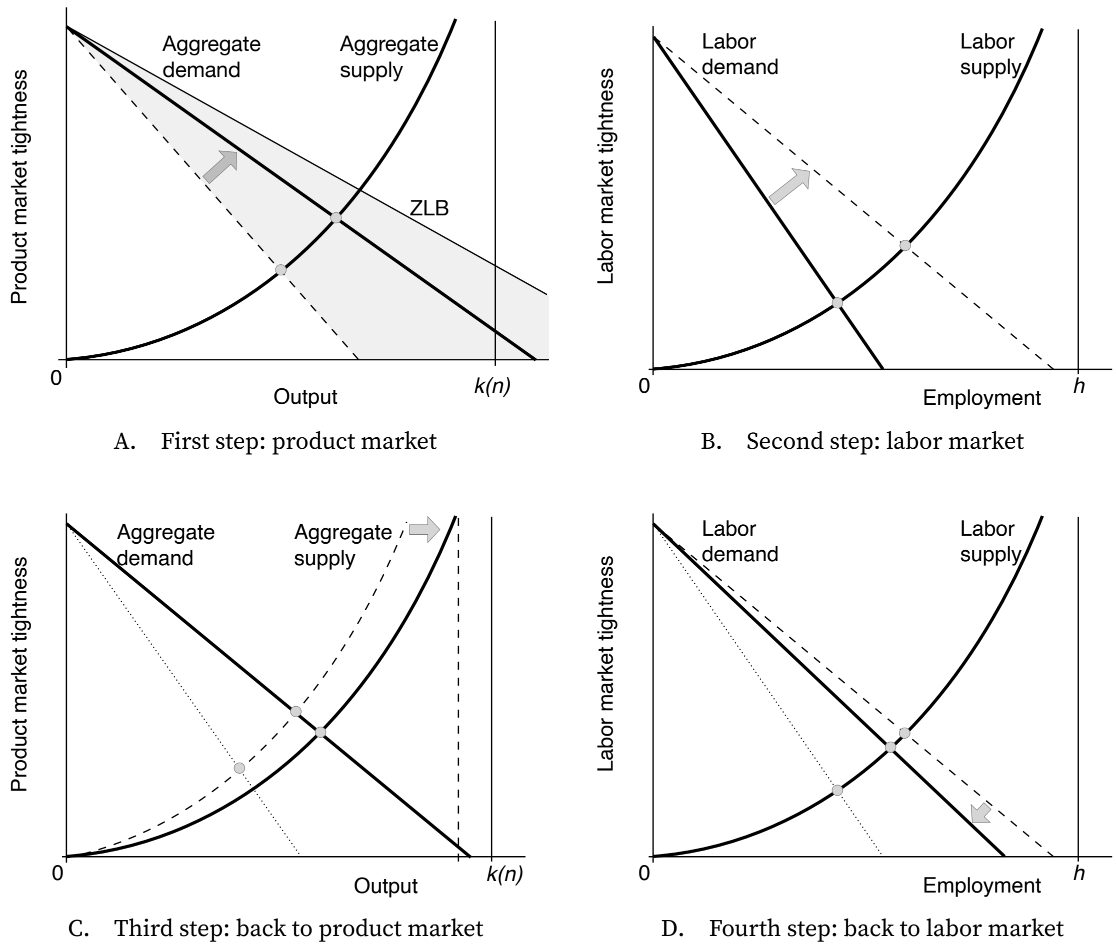
More generally, animal spirits might be an irrational fervor that drives economic decisions of crowds of people, possibly ignited by news of certain events, and assuredly exacerbated by media coverage.[4] When animal spirits are triggered, they may lead to brash, large-scale reactions—consumers suddenly cutting spending and firms suddenly cutting hiring. The aggregate demand shock in the model is a simple way to formalize these swings in collective optimism.

15.6. Revisiting the effect of monetary policy on unemployment

In chapter 13, we showed how monetary policy is able to stabilize unemployment in a one-market business cycle model: it does so by modulating the nominal interest rate, which steers aggregate demand and then stabilizes aggregate tightness and thus unemployment. But how does monetary policy operate in a more realistic, two-market business cycle model? The nominal interest rate affects the aggregate demand and then product market tightness, while unemployment is determined by the labor market tightness.

[3] For evidence, see Kindleberger and Alibert (2005), Shiller (2005), and Akerlof and Romer (1993).

[4] Akerlof and Shiller (2009), Beaudry and Portier (2014), and Nimark (2014) develop these ideas.



A. First step: product market

B. Second step: labor market

C. Third step: back to product market

D. Fourth step: back to labor market

FIGURE 15.7. Expansionary monetary policy in the two-market slackish business cycle model

The aggregate supply curve is given by equation (15.8). The aggregate demand curve is given by equation (15.11). The labor supply curve is given by equation (15.1). The labor demand curve is given by equation (15.7). In panel A, the gray cone indicates all the positions that the aggregate demand curve can reach when the nominal interest rate is reduced from its current level to the zero lower bound. The aggregate demand curve at the zero lower bound is the most outward position that the aggregate demand curve can reach; it is obtained from equation (15.11) when $i = 0$ and $r = -\pi$.

15.6.1. How does monetary policy operate?

We consider an expansionary monetary policy: the central bank lowers the nominal interest rate i, which reduces the real interest rate $r = i - \pi$, given that inflation is fixed in this model too (just as in chapter 13). Since the real interest rate is lower, the financial returns on wealth are lower, so households are keener to spend and consume instead of saving, which stimulates the aggregate demand (equation (15.11)).

Graphically, the policy triggers an outward rotation of the aggregate demand curve (figure 15.7A). This raises product market tightness θ_p and lowers product market slack $u_p(\theta_p)$.

The reduced product market slack means that firms sell a larger share of their services, which stimulates labor demand: the labor demand curve rotates outward, which boosts labor market tightness θ_l, and lowers the unemployment rate $u_l(\theta_l)$ (figure 15.7B).

Because labor market tightness rises, the number of producers employed by firms $n(\theta_l)$ might change, so the aggregate capacity $k(n(\theta_l))$ might change, which might affect the aggregate supply curve in the product market, just as we discussed in the case of aggregate demand shocks. For instance, if the labor market is inefficiently slack—the usual situation in which expansionary monetary policy is enacted—then the number of producers increases, so the aggregate supply curve shifts outward (figure 15.7C). In turn, output increases further but product market tightness moves back down somewhat.

Of course, these changes in product market tightness further affect labor market tightness. That decrease in product market tightness feeds back to the labor market: it pushes the labor demand curve slightly back down (figure 15.7D).

15.6.2. Overall effects of monetary policy on the product and labor markets

Given the connection between labor market and product market, it is not entirely clear from equations (15.13) and (15.14) what eventually happens to product and labor market tightnesses. But using the functions $\theta_p^p(\theta_l)$ and $\theta_p^l(\theta_l)$, we can establish again how the tightnesses respond overall.

After a decrease in the real interest rate r, the curve $\theta_p^p(\theta_l)$, given by (15.19), shifts up, while the curve $\theta_p^l(\theta_l)$, given by (15.18), stays the same. Thus, overall, both product market tightness, θ_p, and labor market tightness, θ_l, increase after the expansionary monetary policy (figure 15.2D). Once again, the response of the tightnesses is determined by the first-round response to the shock.

From the tightness results we infer a few other results. Importantly, the unemployment rate, $u_l(\theta_l)$, falls after the decrease in the nominal interest rate. This is because the idleness rate, $u_p(\theta_p)$, has fallen once aggregate demand has been stimulated, which has raised labor demand. Overall, there is less slack on both markets.

Because the labor market tightness increases, the employment level l increases, as the labor market moves up the labor supply curve. It is not easy to determine how output changes just by looking at tightnesses, because both aggregate supply and demand shift. But, using the labor-share result in equation (15.24), we conclude that output y rises because employment increases.

The comparative statics for a decrease in the nominal interest rate, which is an expansionary monetary policy shock, are summarized in table 15.2.

15.6.3. The monetary multiplier

In chapter 13 we computed the monetary multiplier du/di, which was given by equation (13.16). The transmission of monetary policy to unemployment is more complex in the two-market model, as the shift in aggregate demand caused by the change in interest rate has to travel first through product market tightness and then labor market tightness before reaching unemployment. Nevertheless, we can compute the monetary multiplier from equation (15.20). As appendix J shows, in this model the multiplier is given by:

$$\frac{du_l}{di} = \frac{1 - u_l(\theta_l)}{\delta - r} \cdot \frac{1}{\alpha_p + (1-\alpha_p)\left[\alpha_l + (1-\alpha_l)\frac{\eta_l(\theta_l)}{1-\eta_l(\theta_l)} \cdot \frac{\tau_l(\theta_l)}{u_l(\theta_l)}\right]\frac{\eta_p(\theta_p)}{1-\eta_p(\theta_p)} \cdot \frac{\tau_p(\theta_p)}{u_p(\theta_p)}}. \tag{15.25}$$

Many of the properties of the monetary multiplier remain the same in the two-market model as in the one-market model, with a few subtle differences.

First, the multiplier remains positive, reflecting our earlier finding that the unemployment rate rises when the nominal interest rate increases.

Second, when both product and labor markets are operating efficiently, we infer from (9.11) that

$$\frac{\eta_l(\theta_l^*)}{1-\eta_l(\theta_l^*)} \cdot \frac{\tau_l(\theta_l^*)}{u_l(\theta_l^*)} = \frac{\eta_p(\theta_p^*)}{1-\eta_p(\theta_p^*)} \cdot \frac{\tau_p(\theta_p^*)}{u_p(\theta_p^*)} = 1.$$

Hence, at efficiency, the monetary multiplier has the same value as in the one-market model:

$$\frac{du_l}{di} = \frac{1 - u_l(\theta_l^*)}{\delta - r}.$$

So, at efficiency, the monetary multiplier solely depends on the efficient unemployment rate, discount rate, and real interest rate.

Third, the multiplier continues to be slack dependent, but it now depends both on labor market slack, governed by tightness θ_l, and product market slack, governed by tightness θ_p. In particular, the monetary multiplier is strictly decreasing in product market tightness, so monetary policy reduces unemployment more effectively when the product market is slack than when it is tight. The same is true for the labor market, at least if we omit the movements in $1 - u_l(\theta_l) \approx 1$, as we did in the informal discussion of chapter 13. Then, the monetary multiplier is strictly decreasing in labor market tightness, so monetary policy reduces unemployment more effectively when the labor market is slack than when it is tight. Of course, if both labor and product markets are slack, the monetary multiplier is even higher, so monetary policy is a more potent tool to tackle unemployment.

The effect of monetary policy is symmetric so the same results hold if monetary policy wanted to raise unemployment by raising rates instead. Overall, monetary policy has a larger effect on unemployment in a slack economy than in a tight economy.

15.7. Revisiting other aggregate demand policies

In chapter 13, we discussed two policies besides conventional monetary policy that could be used to stabilize unemployment in a one-market business cycle model: fiscal policy in the form of public spending, and unconventional monetary policy in the form of forward guidance.

Both public spending and forward guidance operate by stimulating aggregate demand. Public spending does it mechanically. When the government spends in the product market, that immediately boosts aggregate demand, as equation (13.26) shows. Forward guidance boosts aggregate demand indirectly, by pushing down the costate variable on wealth in households' consumption-saving problems, which itself boosts aggregate demand, as equation (13.17) shows.

The two policies continue to stimulate aggregate demand in the exact same way in this chapter's two-market model. Their effects then percolate to the labor market just as in the case of conventional monetary policy, described in figure 15.7. Once the policy has stimulated aggregate demand, product market tightness increases, which reduces slack in the product market (figure 15.7A). Firms are able to sell more of their products, so it is profitable for them to hire more labor. This increase in labor demand boosts labor market tightness, which reduces unemployment as job seekers are able to find jobs more rapidly (figure 15.7B). The increase in employment feeds back into the product market, producing a secondary adjustment in product market tightness (figure 15.7C), which itself feeds back into the labor market to trigger a secondary adjustment in tightness there (figure 15.7D), and so on.

In fact, following the same approach as for the monetary multiplier, it is not difficult to compute the fiscal multiplier in the two-market model (appendix J). That multiplier gives the percentage-point reduction in the unemployment rate when public spending increases by 1% of private spending:

$$-\frac{du_l}{dg} = \frac{1 - u_l(\theta_l)}{1+g} \cdot \frac{1}{1 + \frac{1-\alpha_p}{\alpha_p}\left[\alpha_l + (1-\alpha_l)\frac{\eta_l(\theta_l)\tau_l(\theta_l)}{[1-\eta_l(\theta_l)]u_l(\theta_l)}\right]\frac{\eta_p(\theta_p)\tau_p(\theta_p)}{[1-\eta_p(\theta_p)]u_p(\theta_p)}}. \tag{15.26}$$

The fiscal multiplier of course shares many of the properties of the monetary multiplier since monetary policy and fiscal policy operate similarly through the aggregate demand curve.

In particular, when both product and labor markets operate efficiently, the multiplier reduces to

$$-\frac{du_l}{dg} = \alpha_p \cdot \frac{1 - u_l(\theta_l^*)}{1+g}.$$

So, at efficiency, the fiscal multiplier depends only on the efficient unemployment rate, curvature of the utility function, and public spending.

And just like the monetary multiplier, the fiscal multiplier is sharply slack dependent: it depends both on labor market slack, governed by tightness θ_l, and product market slack, governed by tightness θ_p. Like the monetary multiplier, the fiscal multiplier is strictly decreasing in product market tightness and strictly decreasing in labor market tightness (as long as we omit the movements in $1 - u_l(\theta_l) \approx 1$). Therefore, fiscal policy reduces unemployment more effectively when the product and labor markets are slacker.

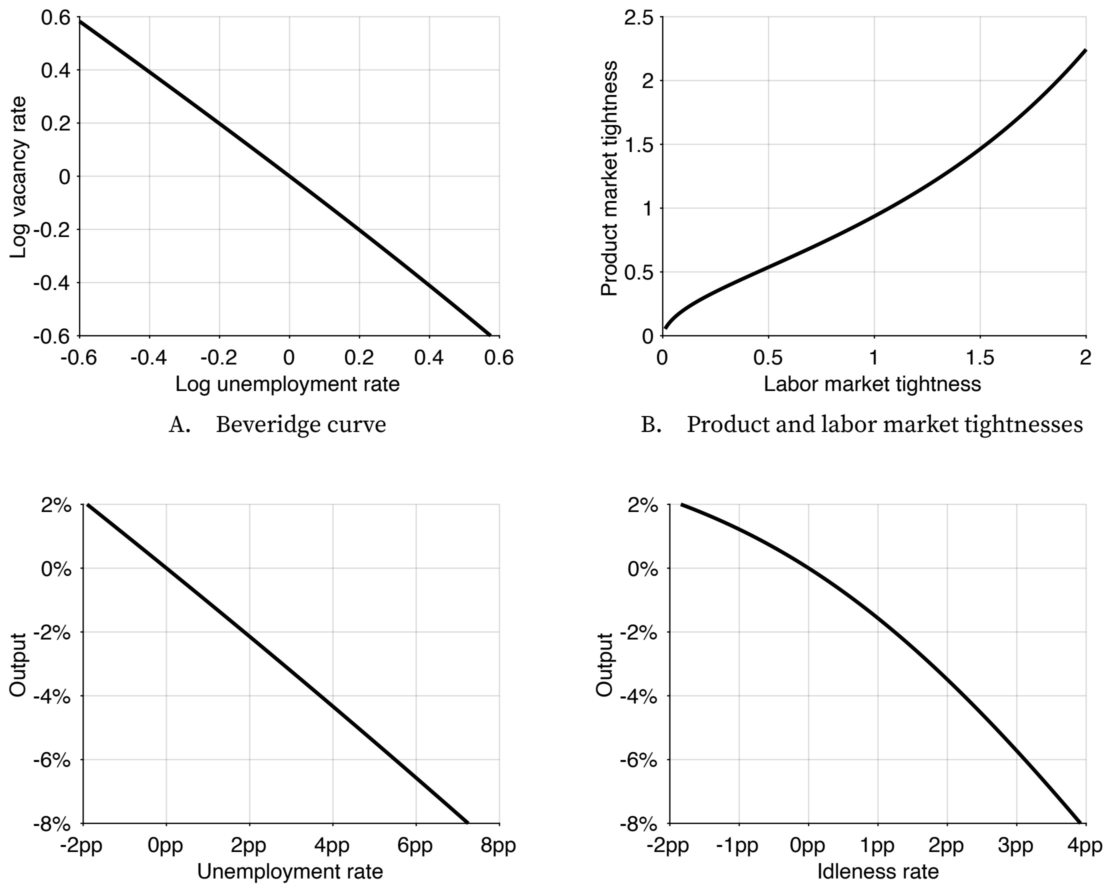
15.8. Simulating Beveridge and Okun curves

To conclude, we explore whether the most realistic slackish business cycle model, with firms and separate product and labor markets, can reproduce the key empirical patterns described in chapter 2: the Beveridge curve of figure 2.8, the standard, labor-market Okun curve of figure 2.11, and the product-market Okun curves of figures 2.13 and 2.12.

We simulate the model under aggregate demand shocks: variations in the taste for consumption, a_p. Hence, in the simulations, the economy's productive capacity, determined by the labor force h and labor productivity a_l, is entirely fixed. Nevertheless, we will see that the model generates substantial fluctuations in output: those are entirely produced by fluctuations in slack, both on the labor market and on the product market. There is not much evidence on slack on the product market, so we calibrate all the product market parameters just like the corresponding labor market parameters. This symmetric calibration is described by table 10.2.

First, we see that the model generates substantial fluctuations in labor market slack, along a Beveridge curve that is an almost perfect rectangular hyperbola, just like in the US labor market (figure 15.8A). The labor market matching function is calibrated to produce a Beveridge curve with an elasticity of -1 at the average unemployment rate; it turns out that the elasticity remains close to -1 over a large range of unemployment and vacancy rates.

Second, we confirm that under aggregate demand shocks, labor market tightness and product market tightness are positively correlated. In fact, the curve presented in figure 15.8B is just the curve $\theta_p^l(\theta_l)$ given by equation (15.18). The solution of the model



A. Beveridge curve

B. Product and labor market tightnesses

C. Labor-market Okun curve

D. Product-market Okun curve

FIGURE 15.8. Simulation of two-market slackish business cycle model under aggregate demand shocks

The model is simulated under a symmetric calibration: product market and labor market parameters are calibrated as indicated in table 10.2. The model is simulated under aggregate demand shocks: the taste for consumption a_p spans $[1.3, 1.6]$. All rates are reported as deviations from their average values; output is reported as log-deviation from its average value.

moves along $\theta^l_p(\theta_l)$ as the curve $\theta^p_p(\theta_l)$ given by equation (15.19) shifts with the aggregate demand parameter a_p. What we see is that as aggregate demand rises, the product market tightness rises, so firms sell a larger share of their capacity, which pushes them to recruit more workers, so labor market tightness rises as well.

Third, we see that fluctuations in unemployment rate and output are negatively correlated, so that a 1pp increase in unemployment rate coincides approximately with a 1% decrease in output (figure 15.8C). Thus, the model's labor-market Okun curve has a slope of about −1, just like the labor-market Okun curve in the US economy. When unemployment is higher, the effect on output is slightly larger: the Okun curve is slightly concave.

Why does the model's Okun curve match the US economy's Okun curve so well? This property follows directly from the labor-share result (15.24). Taking the logarithm of the labor-share result, we get

$$\ln(y) = \ln(w) + \ln(h) + \ln(1 - u_l) - \ln(1 - \alpha_l).$$

Hence, when aggregate demand shocks generate fluctuations in unemployment rate, the relative changes in output are:

$$\frac{d\ln(y)}{du_l} = \frac{-1}{1 - u_l} \approx -1,$$

where the approximation holds when the unemployment rate is small enough.[5] We also see why the labor-market Okun curve is slightly concave: when the unemployment rate rises, the slope $d\ln(y)/du_l$ of the curve becomes slightly more negative.

Last, we see that fluctuations in idleness rate and output are negatively correlated (figure 15.8D). When the product market is tight (idleness below average), a 1pp increase in idleness rate coincides approximately with a 1% decrease in output; when the product market is slack (idleness above average), a 1pp increase in idleness rate coincides approximately with a 2% decrease in output. Thus, the model's product-market Okun curve has a slope of about −1 on its tight portion and a slope of about −2 on its slack portion. The model's curve is clearly concave.

Here we see that the model does not match the data as well. The model's product-market Okun curve is downward sloping—just like the empirical curve. But the model's curve is significantly steeper than the empirical curve. We saw in figures 2.13 and 2.12 that a 1pp increase in idleness rate is associated with a decrease in output by 0.3%–0.5%. This is much less than in the model, where output decreases by 1%–2%.

[5]Since the real wage w and labor force h do not respond to aggregate demand, they drop out in the derivative. In the two-market model we have assumed that the real wage w is fixed. But the slope of the Okun curve would remain the same if the real wage responded to labor productivity (as in chapter 10) since we do not consider such shocks—we only consider aggregate demand shocks.

One possible explanation for this discrepancy between empirical and model Okun curves—the most flattering for the model—is that slack is not well measured on the product market, so the measured idleness rates are more volatile than the true idleness rates. Because the ISM is not a governmental agency, it is not required to be as transparent, so we do not know exactly how their surveys are conducted, how large their sample of firms is, and so on. We saw that the idleness rates are quite volatile—much more than the unemployment rate measured by the BLS—so this could be an explanation.

Another possibility is that the model structure or calibration does not describe the US product market well. We can rule out a calibration issue, because the link between output and idleness rate is fairly hard wired into the model. From the aggregate supply curve (15.8), we see that

$$\ln(y) = \ln(1 - u_p) + \ln(k(n(\theta_l))).$$

The key here is that around an efficient labor market, the number of producers $n(\theta_l)$ is maximized, so $\ln(k(n(\theta_l)))$ does not vary under aggregate demand shocks. In that case, we obtain as above:

$$\frac{d\ln(y)}{du_p} = \frac{-1}{1 - u_p} \approx -1.$$

So around an efficient labor market, irrespective of any calibration, the slope of the product-market Okun curve should be around −1. When the labor market is inefficiently slack and aggregate demand drops further, the idleness rate u_p goes up, and as labor market tightness falls, the number of producers $n(\theta_l)$ and productive capacity $k(n(\theta_l))$ drop. Accordingly, the slope of the product-market Okun curve becomes more negative, below −1. So at or below full employment, no calibration can make the model's product-market Okun curve flatter.

15.9. Summary

This chapter develops a slackish business cycle model with distinct labor and product markets. In the model, there are different types of unemployment: Keynesian unemployment, which is caused by a lack of aggregate demand; classical unemployment, which is caused by excessively high real wages; and frictional unemployment, which is caused by the cost of matching workers with firms. Most models do not capture these different types of unemployment—usually, a model focuses on one type of unemployment so different models are used to think about different types of unemployment. This is an issue because it does not provide a comprehensive view of what drives unemployment and unemployment fluctuations. One advantage of our slackish two-market model is that it is able to capture all these components, offering a completely unified theory of unemployment.

Previously we saw that labor demand shocks and not labor supply shocks cause fluctuations in US unemployment. In this chapter we show using product-market Okun curves that these labor demand shocks stem from aggregate demand shocks, not the technology shocks that are so often used in business cycle research.

The chapter shows how aggregate demand policies are able to stabilize unemployment. In the case of conventional monetary policy, for instance, a reduction in the interest rate stimulates aggregate demand, which raises product market tightness and thus reduces slack in the product market. Firms find it easier to sell their production, so they want to hire more labor, post more vacancies, which boosts labor market tightness and reduces unemployment. Fiscal policy works in a similar fashion by stimulating the aggregate demand curve through public spending.

Finally, numerical simulations of the model under aggregate demand shocks show that it can reproduce the main empirical patterns from chapter 2. With the labor force and technology held fixed, the model still generates substantial fluctuations in output; these arise entirely from fluctuations in slack on the labor market and on the product market driven by fluctuations in aggregate demand (modeled as fluctuations in the taste for consumption relative to saving). The simulations produce a hyperbolic Beveridge curve, positively correlated product and labor market tightnesses, and a labor-market Okun curve with a slope of about −1, closely matching US data. The product-market Okun curve is downward sloping as well, as in the data, though steeper than the empirical relationship.

An advantage of the two-market structure introduced in this chapter is that the model now displays the structure that is typical of modern business cycle models: a product market and a labor market, with firms hiring labor in one and selling production in the other, and households supplying labor in one and consuming products in the other. It is therefore easier to compare this slackish business cycle model to other business cycle models, such as the Real Business Cycle model or New Keynesian model, as we do in chapter 21.

PART V.

Policies to stabilize slackish cycles

CHAPTER 16.

Monetary policy

We have seen that the unemployment rate in the United States is always inefficiently high in slumps, and sometimes inefficiently low in booms (chapter 12). Therefore, the economy requires active and systematic stabilization policy to align unemployment with its efficient level. A natural policy to achieve such stabilization is monetary policy, which has scope to stabilize the unemployment rate. The link is the interest-rate channel: by shifting interest rates, monetary policy can move aggregate demand and thus unemployment toward the efficient rate—at least when conventional interest-rate policy is unconstrained (chapter 13).

16.1. Social planner's problem

To study optimal monetary policy, we follow the approach we used to study the unemployment gap in chapter 12: we specify a generic framework and solve the social planner's problem in that framework, which we then use to derive a sufficient-statistic formula for optimal monetary policy.

16.1.1. Economic slack

We assume that there is slack in the economy. This means that a share $u \in (0,1)$ of the aggregate productive capacity cannot be sold. Furthermore, we assume that slack determines social welfare, and so that social welfare is maximized when the slack rate is at its efficient level, u^*. Last, we assume that the efficient slack rate can be measured by sufficient statistics, as in chapter 12. The models of chapters 13 and 14 satisfy these assumptions.

The model of chapter 15 has two types of slack: product market slack and labor market slack. So social welfare generally depends on the two slack rates. But if the product-labor coincidence holds in that model—which requires that the real wage satisfies condition (15.22)—then product market slack and labor market slack are efficient at the same time. Thus, monetary policy can maximize social welfare by simply maximizing welfare on one of the markets—by targeting one of the slack rates. Hence, the model also satisfies the assumption.

16.1.2. Inflation

A corollary of the above assumption about slack is that inflation does not independently affect social welfare. This could be because inflation is fixed, as in the models of chapter 13 and 15, or because the inflation-slack coincidence holds, as in the model of chapter 14. This means that either inflation does not respond to monetary policy, or inflation is on target when the slack rate is efficient. This is distinct from saying that inflation has no welfare cost: it is saying that stabilizing slack at its efficient rate does not conflict with price stability.

16.1.3. Monetary policy

In the efficiency analysis of chapter 9, we simply assumed that the planner could pick the amount of slack in the market to maximize welfare. Here, we are studying optimal monetary policy, which means that our planner is more restricted: they can only affect slack indirectly, through monetary policy.

Specifically, we assume that monetary policy is non-neutral and conducted through the nominal interest rate. This means that the planner determines the nominal interest rate i, and the nominal interest rates can stabilize the economy. The slackish models in part IV are an example of this: in those models interest rates control aggregate demand and then output and slack. Formally, the nominal interest rate affects the slack rate through a generic function $u(i)$.

16.1.4. Planner's problem

We now formalize the social planner's problem. The planner uses the nominal interest rate i to maximize social welfare through the slack rate $u(i)$. Because social welfare is maximized at the efficient slack rate u^*, the optimal nominal interest rate i^* satisfies:

$$u(i^*) = u^*. \tag{16.1}$$

16.2. Sufficient-statistic formula for optimal monetary policy

We now derive our sufficient-statistic formula for optimal monetary policy.

16.2.1. Sufficient-statistic formula

Given the current slack rate, u, and current nominal interest rate, i, the sufficient-statistic formula gives the optimal nominal interest rate, i^*, that the central bank should set.

To derive the formula, we perform a first-order Taylor expansion of the function $u(i)$ around i:

$$u(i^*) = u(i) + \frac{du}{di} \cdot \left(i^* - i\right),$$

where the derivative du/di is evaluated at the current interest rate and slack rate. By definition of the function $u(i)$ and optimal interest rate i^*, we have $u(i) = u$ and $u(i^*) = u^*$, so the Taylor expansion gives

$$u^* = u + \frac{du}{di} \cdot \left(i^* - i\right).$$

Rewriting this to express the optimal interest rate as a function of the prevailing slack gap $u - u^*$, we get

$$i^* = i - \frac{u - u^*}{du/di}. \tag{16.2}$$

This is our sufficient-statistic formula. The statistic $u - u^*$ is the slack gap. The statistic du/di is the monetary multiplier: it gives the percentage-point change in slack when the nominal interest rate increases by 1 percentage point. It is positive if an exogenous increase in nominal interest rate raises slack, as it typically does in the real world.

16.2.2. Interpretation of the formula

The optimal monetary-policy formula (16.2) tells us that the central bank should adjust the nominal interest rate to bring the slack rate to its efficient level.

Consider the realistic case where the monetary multiplier is positive, so an increase in interest rates raises the slack rate. Then if the slack rate is inefficiently high, the central bank needs to cut interest rates. On the other hand, if the slack rate is inefficiently low, interest rates must be raised.

We can also see that how much the central bank should respond to the slack gap depends on the monetary multiplier. If the monetary multiplier is large, monetary policy has a strong effect on the slack gap so rates need to be changed by a smaller amount. If the multiplier is small, monetary policy only has a limited impact on the slack gap so rates need to be changed by a larger amount.

16.2.3. Calibration of the sufficient statistics

To calibrate formula (16.2), we need empirical measures of the slack rate, the efficient slack rate, and the monetary multiplier. Since aggregate slack is not readily measurable in the United States, we use the unemployment rate as our measure of slack and the FERU as the corresponding efficient slack rate. Accordingly, we measure the monetary multiplier by the response of unemployment to the nominal interest rate.

The nominal interest rate i in the United States is just the federal funds rate controlled by the Board of Governors of the Federal Reserve System (2025).

The US unemployment gap $u - u^*$ can be estimated directly from the unemployment and vacancy rates, as we saw in chapter 12.

A challenge in measuring the monetary multiplier du/di—and measuring the impact of policies in general—is that nominal interest rate changes are never exogenous: they are endogenous to the current situation. This makes it difficult to estimate the monetary multiplier properly. But several methods have been developed to isolate exogenous variations in the nominal interest rate in the United States and measure the effect of monetary policy on aggregate variables, particularly on the unemployment rate.[1]

Many papers use vector autoregressions (VARs) to identify exogenous variations in monetary policy and compute the monetary multiplier. For instance, using such an approach, Coibion (2012, p. 5) finds $du/di = 0.16$. This value accords with previous estimates of the monetary multiplier obtained by Christiano, Eichenbaum, and Evans (1996), Bernanke, Boivin, and Eliasz (2005), and others, with the exception of the larger estimate $du/di = 0.6$ obtained by Bernanke and Blinder (1992).

However, VAR does come with its issues—it is quite difficult to estimate the exogenous changes in monetary policy. An alternative approach is the narrative approach pioneered by Romer and Romer (1989, 2004). The idea is to carefully read accounts of policy decisions to separate endogenous and exogenous changes in the interest rate by isolating idiosyncratic decisions made by the Federal Reserve that were not driven by current economic conditions. Romer and Romer (1989, 2004) construct a series of monetary policy shocks using only these idiosyncratic decisions and, based on this series, measure the impact of monetary policy on the economy. Using the narrative approach, Coibion (2012, p. 7) obtains a much larger estimate of the monetary multiplier: $du/di = 0.93$.

This is significantly bigger than the value computed using VAR, and using this number would lead to very different outcomes in our analysis. Thus, it is important to try to get a narrower range of estimates for the monetary multiplier so that we can be more certain about how to conduct monetary policy. To reconcile the difference between the two

[1] See Christiano, Eichenbaum, and Evans (1999) and Ramey (2016) for surveys.

approaches, Coibion (2012, p. 8) proposes a hybrid approach that yields a medium estimate of the monetary multiplier: $du/di = 0.40$. The hybrid approach appears robust: across numerous specifications, it yields monetary multipliers between 0.4 and 0.6 (Coibion 2012, p. 12). We use the midpoint of this range, $du/di = 0.5$, to carry out our analysis and study optimal monetary policy.

16.3. Optimal response to unemployment fluctuations

Now that we have estimates for the sufficient statistics in our optimal interest-rate formula, we can put everything together to describe what the Fed's optimal behavior is. Combining the optimal interest-rate formula, given by (16.2), with the midrange estimate of the monetary multiplier, $du/di = 0.5$, we obtain the following formula:

$$i^* = i - 2 \times \left(u - u^*\right). \tag{16.3}$$

Therefore, for any unemployment gap, the optimal response of the nominal interest rate is twice the size of the gap. For example, if the unemployment gap is 1pp, the federal funds rate should drop by 2pp.

This is a very simple rule that should guide the Fed's response to unemployment fluctuations. Of course, it differs from the typical Taylor (1993) rule that appears in New Keynesian macroeconomics: rule (16.3) links monetary policy to the unemployment rate whereas the Taylor rule links it to the inflation rate. Moreover, the optimal response of monetary policy to the unemployment rate can be obtained empirically, by estimating the monetary multiplier. By contrast, the most appropriate response of monetary policy to inflation is typically determined by simulations in New Keynesian models.[2]

16.4. Optimal monetary policy without inflation-slack coincidence

The same logic applies to optimal monetary policy in models without inflation-slack coincidence. When the coincidence fails, monetary policy faces a tradeoff between closing the slack gap and bringing inflation to its target, so targeting the efficient slack rate is no longer optimal. In that case, the optimal interest rate is determined by weighting the slack gap against the inflation gap.

To see this, consider an extension of the social planner's framework in which social welfare depends not only on slack but also on inflation—for instance as in chapter 14. Let's denote the efficient inflation rate by π^*. The welfare loss around the efficient allocation

[2]See for instance the survey by Taylor and Williams (2010).

(u^*, π^*) admits the following quadratic approximation:

$$\mathcal{L}(u, \pi) = \left(\pi - \pi^*\right)^2 + \Lambda \left(u - u^*\right)^2, \tag{16.4}$$

where $\Lambda > 0$ measures the importance of slack relative to inflation in the social welfare function.

Additionally, the social planner faces a Phillips curve that relates inflation to slack. Around the efficient allocation, the Phillips curve admits the following linear approximation:

$$\pi - \pi^* = -\Gamma \left(u - u^*\right) + \Omega, \tag{16.5}$$

where $\Gamma > 0$ gives the slope of the downward-sloping Phillips curve, and $\Omega \neq 0$ is introduced to break the inflation-slack coincidence. Indeed, if $\Omega = 0$, then $\pi = \pi^*$ when $u = u^*$: inflation is on target when slack is efficient, so the inflation-slack coincidence holds. But if $\Omega > 0$, then $\pi > \pi^*$ when $u = u^*$: inflation is too high when slack is efficient, so slack must be inefficiently high to bring inflation to target, just like in the 1970s. Conversely, if $\Omega < 0$, then $\pi < \pi^*$ when $u = u^*$: inflation is too low when slack is efficient.

The Phillips curve developed in chapter 14 links the slack and inflation gaps. It guarantees that the inflation-slack coincidence holds if inflation expectations are anchored to the inflation target, so $\Omega = 0$ in (14.20). But the coincidence fails if expectations about normal inflation are not anchored to the inflation target. In that case, the Phillips curve features a shifter $\Omega \neq 0$, as seen in (14.24). The shifter is positive if the inflation that people consider normal is above target ($\pi^n > \pi^*$); the shifter is negative if the inflation that people consider normal is below target ($\pi^n < \pi^*$).

The social planner minimizes the welfare loss (16.4) subject to the Phillips curve (16.5). Inflation can be substituted out of the welfare loss using the Phillips curve (result A.22). Then, the planner's problem is to find $u \in [0, 1]$ to minimize

$$\left[-\Gamma \left(u - u^*\right) + \Omega\right]^2 + \Lambda \left[u - u^*\right]^2.$$

This objective function is strictly convex in u, so the first-order condition is sufficient to find its minimum (result A.19). We take the function's derivative with respect to u and set it to 0:

$$0 = -2\Gamma \left[-\Gamma \left(u - u^*\right) + \Omega\right] + 2\Lambda \left[u - u^*\right].$$

Rearranging terms, we express the optimal slack gap as a function of the parameters:

$$u - u^* = \frac{\Omega\Gamma}{\Gamma^2 + \Lambda}. \tag{16.6}$$

Plugging (16.6) in the Phillips curve (16.5), we express the optimal inflation gap as a function of the same parameters:

$$\pi - \pi^* = \frac{\Omega\Lambda}{\Gamma^2 + \Lambda}. \tag{16.7}$$

From these two expressions, we obtain a simple expression for the ratio between the optimal slack and inflation gaps:

$$\frac{u - u^*}{\pi - \pi^*} = \frac{\Gamma}{\Lambda} > 0. \tag{16.8}$$

The tradeoff between inflation and slack in the absence of inflation-slack coincidence clearly appears in these equations. Consider the case where inflation is too high when slack is efficient ($\Omega > 0$). Then (16.6) and (16.7) say that it is optimal to keep slack above its efficient level ($u > u^*$) and inflation above its target ($\pi > \pi^*$). Furthermore, (16.8) shows that the optimal gaps are determined by the welfare cost of slack relative to inflation (Λ) and the response of inflation to slack (Γ). If slack is particularly costly (high Λ) or if the Phillips curve is flat (low Γ), the optimal slack gap is small relative to the optimal inflation gap. In such cases, the optimal slack rate remains close to the efficient slack rate, either due to the high marginal cost of slack (high Λ) or the limited marginal benefit of slack (low Γ).

The equations provide two additional insights. First, if the inflation-slack coincidence holds ($\Omega = 0$), then it is optimal to keep slack at its efficient level ($u = u^*$ in (16.6)), which guarantees that inflation hits its target ($\pi = \pi^*$ in (16.7)). Second, it is never optimal for the slack and inflation gaps to have opposite signs ($(u - u^*)/(\pi - \pi^*) < 0$ in (16.8)). This is because welfare can be improved if the gaps take opposite signs. For example if the slack gap is negative and the inflation gap is positive (as in 2021–2024), then raising slack moves the slack gap toward 0 and, by cooling inflation, moves the inflation gap toward 0—thereby improving welfare.

16.5. Might the price mandate explain departures from full employment?

Despite the US government's full-employment mandate, the US labor market has consistently fallen short of full employment in the past century (chapter 12). In a world of fixed inflation or inflation-slack coincidence, there is no justification for falling short of full employment. But if the inflation-slack coincidence does not hold, one possible justification is that the dual mandate forced the Federal Reserve to keep unemployment inefficiently high so as to tame inflation, along the lines of equation (16.8). We explore this possibility here.

A central insight of equation (16.8) is that when monetary policy is conducted optimally, the unemployment gap and inflation gap should have the same sign. In other words, an unemployment rate above the FERU can only be justified if inflation is above target—as a way to cool inflation. Conversely, an unemployment rate below the FERU can only be justified if inflation is below target—as a way to boost inflation. In fact, optimality requires that the unemployment gap and inflation gap are linearly related through equation (16.8). A weaker but useful diagnostic is that they must at least have the same sign: if they do not, welfare can be improved by moving them toward the ratio in (16.8).

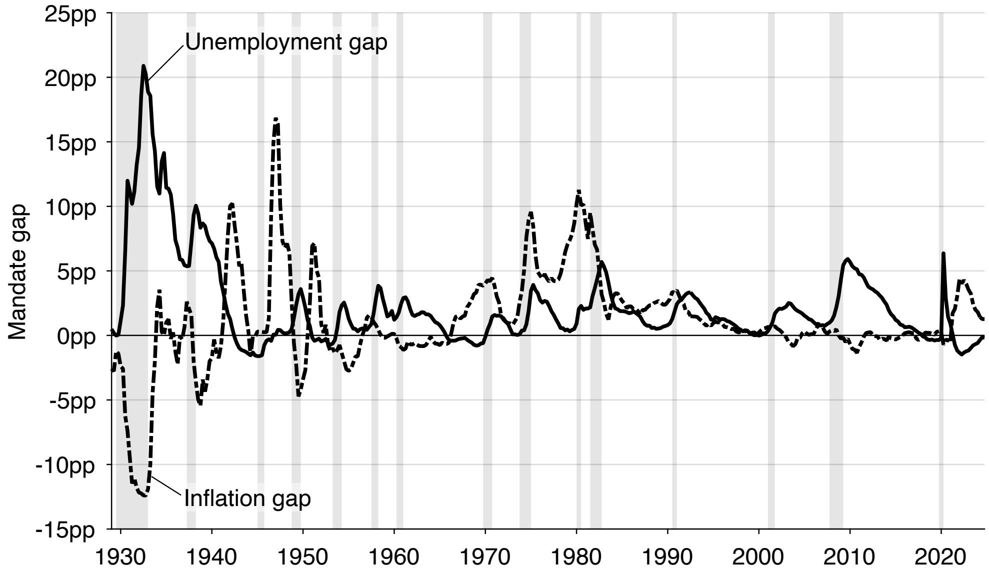
To assess whether this basic optimality requirement is satisfied, we first plot the unemployment gap (the distance between the unemployment rate and FERU) and inflation gap (the distance between the inflation rate and target of 2%) over time (figure 16.1A). Whenever the two gaps have opposite signs, monetary policy cannot be optimal. What we first see is that in many notable recessions, when the unemployment gap is positive, the inflation gap is negative, so clearly monetary policy was suboptimal. There was for instance notable deflation during the Great Depression—when the inflation gap was below −10pp. There were also notable negative inflation gaps, together with positive unemployment gaps, during the 1937–1938, 1948–1949, and 1953–1954 recessions.

To evaluate more systematically monetary policy, and assess whether the price-stability mandate could really explain departures from full employment, we produce scatter plots of the unemployment gap against the inflation gap (figure 16.1B). Under our baseline necessary condition for optimality, the unemployment and inflation gaps must have the same sign (equation (16.8)). Quarters that fall in the northwest or southeast quadrants therefore violate this condition: in those quadrants the gaps have opposite signs, so monetary policy cannot be optimal.

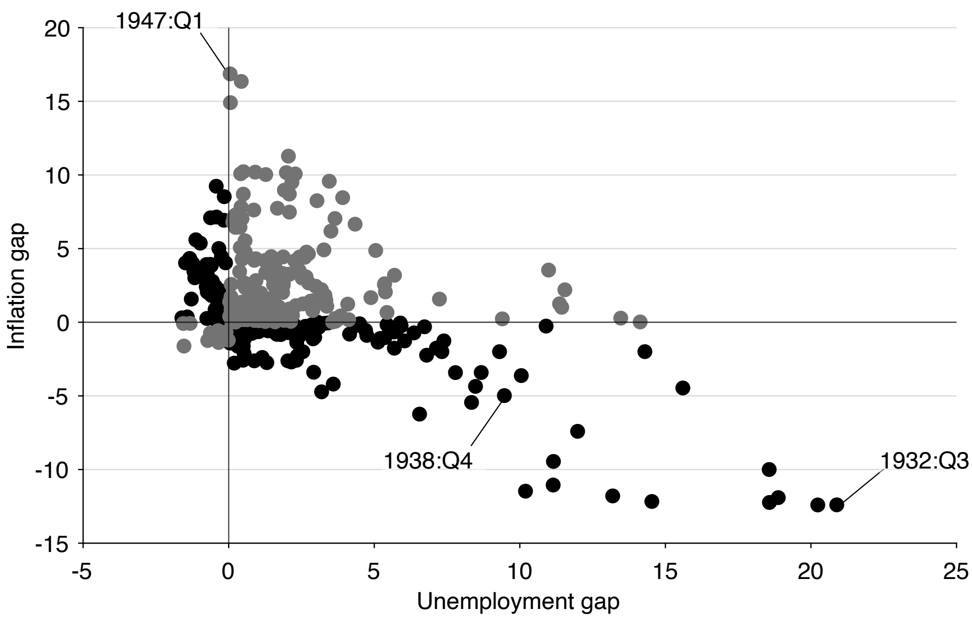
In the entire, 1929–2024 period, monetary policy is non-optimal at least half of the time: 47% of the 384 quarters cannot be optimal. This is not to say that monetary policy is optimal in 53% of the quarters: just that optimality cannot be rejected on the grounds that unemployment and inflation gaps have opposite signs. Indeed, when monetary policy is conducted optimally, the unemployment and inflation gaps should be linearly related (equation (16.8)). Such a linear relationship does not appear in the scatter plots.

One possible issue with the test of monetary policy in figure 16.1 is that the Federal Reserve did not officially adopt a 2% target for inflation until January 2012 (Board of Governors of the Federal Reserve System 2012). However, inflation targeting spread rapidly around the globe in the 1990s, with many countries adopting an inflation target of 2% (Svensson 2010, table 1). Even in the United States, Federal Reserve officials started openly discussing numerical targets for inflation in the 2000s, often around 2% (Shapiro and Wilson 2019).

So we now examine whether our test delivers different results in the modern era, 2000–2024, once the 2% inflation target was more or less in place. In figure 16.2A, we zoom



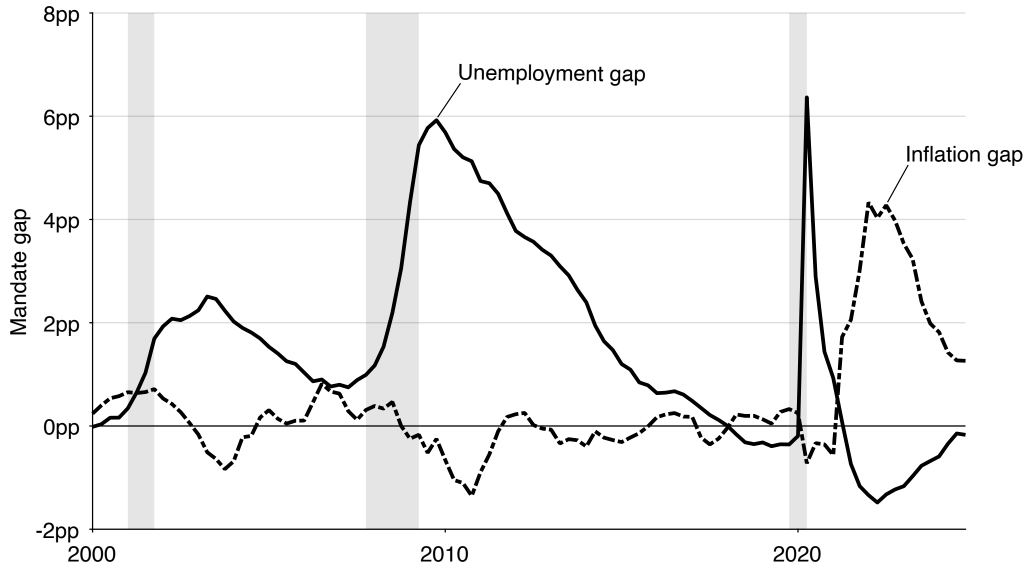
A. Time-series representation



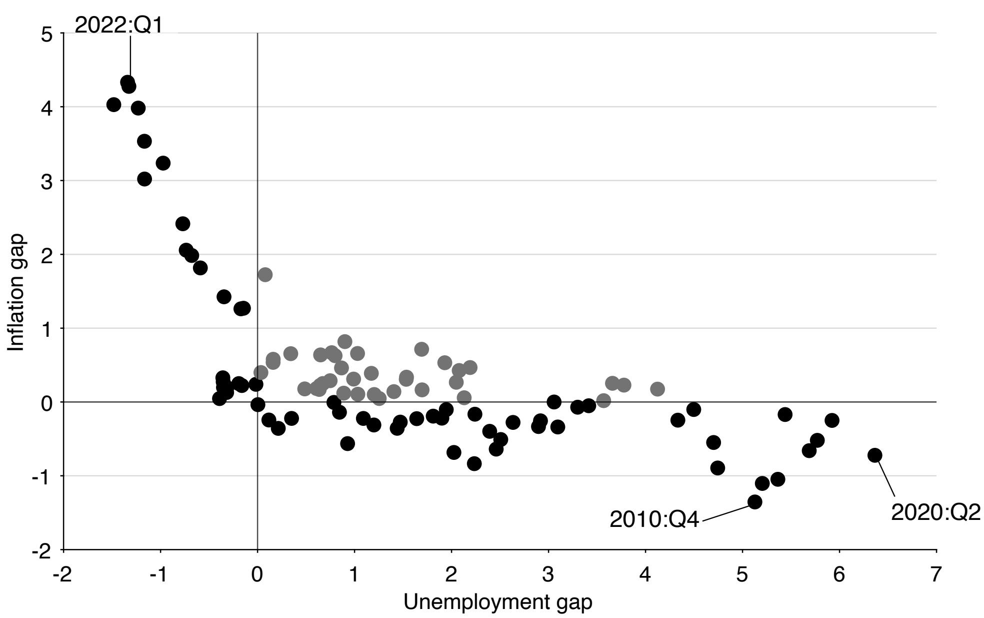
B. Scatter-plot representation

FIGURE 16.1. Unemployment gap and inflation gap in the United States, 1929–2024

The unemployment gap is $u - u^*$, where the unemployment rate u comes from figure 2.1 and the FERU u^* comes from figure 12.2. The inflation gap is $\pi - 2\%$, where π is the quarterly average of the monthly, seasonally adjusted, percentage change from a year ago of the consumer price index for all urban consumers which is produced by the BLS (2025d). After 1958, we use the core consumer price index produced by the BLS (2025e), which is typically preferred by the Federal Reserve because it does not include the prices of food and energy. Shaded areas indicate recessions dated by the NBER (2023).



A. Time-series representation



B. Scatter-plot representation

FIGURE 16.2. Unemployment gap and inflation gap in the United States, 2000–2024

The unemployment gap is $u - u^*$, where the unemployment rate u comes from figure 2.1 and the FERU u^* comes from figure 12.2. The inflation gap is $\pi - 2\%$, where π is the quarterly average of the monthly, seasonally adjusted, percentage change from a year ago of the consumer price index for all urban consumers, less food and energy, which is produced by the BLS (2025e). Shaded areas indicate recessions dated by the NBER (2023).

in on the 2000–2024 period. We see again that the large unemployment gaps observed after the dot-com recession (+2.5pp), and especially the Great Recession (+5.9pp) and pandemic recession (+6.3pp) cannot be explained by the price-stability mandate. When the unemployment gap peaked after these recessions, the inflation gap was always negative. This is especially true after the Great Recession, when the inflation gap fell as low as −1.4pp; the inflation gap bottomed at −0.8pp after the dot-com recession and −0.7pp after the pandemic recession.

Another striking period of suboptimal monetary policy is the post-pandemic period. The inflation gap was positive from 2021:Q2 until the end of 2024, peaking at +4.3pp in 2022:Q3, but that cannot be justified by a positive unemployment gap. To the contrary, unemployment was below the FERU from 2021:Q3 onward, and the unemployment gap fell as low as −1.5pp in 2022:Q2. So in 2021–2024, monetary policy was unambiguously too lax, leading to excessive inflation and labor market tightness.

Because the idea of targeting an inflation rate of 2% was not widespread until 2000, it is possible that monetary policy does better in the scatter-plot test in the 2000–2024 period. One might expect that the share of non-optimal quarters is lower after 2000, once the concept of price stability aligns with the 2% used in the test. That is not what we observe, however. Over the 2000–2024 period, we see that 63% of the 100 quarters display non-optimal monetary policy (figure 16.2B).

We finally zoom in on the even more recent period, 2008–2024, when the inflation target of 2% was generally accepted at the Federal Reserve (Shapiro and Wilson 2019, figure 1). We find that 81% of the 68 quarters display non-optimal monetary policy! That is, since the Great Recession, only 19% of the quarters feature same-sign unemployment and inflation gaps. Clearly, modern departures from full employment are not because of the price-stability mandate.

For the most part, therefore, we can reject the explanation that the US government has failed to maintain the labor market at full employment over the past century because of its price-stability mandate. The only period when large departures from full employment could be explained by the price-stability mandate is the 1970s and 1980s. During these two decades, unemployment was above the FERU and inflation was above target, so any reduction in unemployment might have been deemed undesirable as it might have pushed inflation even higher. Maybe surprisingly, the stagflation of the 1970s is a sign that the Fed was doing the best it could, given the constraints imposed by the Beveridge and Phillips curves. In that way, if the Fed had not let unemployment rise during the period, it would not have done enough to combat rampant inflation.

16.6. So why did the US economy fall short of full employment for a century?

The US government has consistently failed to maintain the labor market at full employment over the past century, and this failure cannot easily be explained by the price-stability mandate. Here we review other explanations as to why the US labor market has deviated from full employment in different periods.

16.6.1. Great Depression and its aftermath

During the Great Depression and its aftermath, the US economy was exceedingly slack. From the beginning of 1930, when our data begin, to the end of 1941, when the United States entered World War 2, the unemployment gap averaged +9.6pp. So the US economy was extremely far from full employment.

Three factors may explain this large amount of slack. The first is that the US government and the Federal Reserve did not have a full-employment mandate at the time. The mandate was introduced with the Employment Act of 1946, as a result of the Great Depression. A second factor is that the Federal Reserve was committed to the gold standard. The gold standard generated a deep deflation in the early 1930s, with dramatic consequences (Eichengreen and Temin 2000). A third factor is that the Fed failed to curb recurrent banking panics in the 1930s (Friedman and Schwartz 1963, chapter 7). Overall, as former Fed Chair Bernanke (2022, p. xvii) writes, "Blaming the Depression entirely on the Fed is an exaggeration, but the relatively new and unseasoned central bank did perform poorly."

16.6.2. World War 2, Korean War, Vietnam War

The US labor market was pulled out of its Great Depression slackness by World War 2 (figure 12.1). In fact, the labor market became inefficiently tight during the war, with tightness averaging 2.8 over the 1942–1945 period. The labor market was once again inefficiently tight during the Korean War, with tightness averaging 1.12 over the 1951–1953 period, and during the Vietnam War, with tightness averaging 1.22 over the 1966–1969 period.

Why was tightness so high during the wars? Part of the reason is that government spending was substantial during these three major wars (Ramey and Shapiro 1998). Such expenditure boosts aggregate demand, which increases tightness. Another part of the reason is that millions of potential labor-force participants are recruited by the military (Department of Veteran Affairs 2023). Such drastic reduction in civilian labor force reduces labor supply, which raises tightness and reduces the unemployment rate among civilian workers.

So why didn't the Fed tighten monetary policy to reduce tightness in wartime? Indeed, a high real interest rate curbs aggregate demand, which reduces tightness and raises unemployment. An appropriate increase in interest rates could have brought tightness back to its full-employment level of 1. In the case of World War 2, there is a simple answer. As Bernanke (2022, p. xviii) explains, during and shortly after World War 2, "at the Treasury's request, the Fed held interest rates at low levels to reduce the government's cost of financing the war."

The same happened at the beginning of the Korean War, when "facing new hostilities in Korea, President Truman pressed the Fed to keep rates low" (Bernanke 2022, p. xviii). The Fed did rebel and was allowed to phase out the low interest-rate peg that had been in place. But the phasing out came too late to cool down the labor market.

The situation during the Vietnam War was different (Bernanke 2022, pp. 20–22). The Fed raised interest rates by half a percentage point at the end of 1965, at the exact time when the economy had reached full employment. However, President Johnson was furious that the Fed tightened monetary policy. He needed low rates to help finance the war. Despite the pressure exerted by Johnson, the Fed continued increasing rates in 1966, which rapidly cooled the labor market. Worried about a possible recession, the Fed reversed its previous tightening. Under pressure from the White House, and facing a chaotic political situation, the Fed continued swinging between tightening and loosening until 1970. The lack of decisive tightening explains why the labor market remained so hot from 1966 to 1969.

16.6.3. Postwar period

In the postwar period, the US labor market was generally inefficiently slack. A manifestation of such pervasive slack is that the unemployment gap averaged 1.4pp between 1946 and 2019. Another manifestation is that the labor market did not achieve full employment once between 1970:Q1 and 2018:Q1—it was inefficiently slack for almost half a century.

A reason that might explain the slackness of the US labor market in the postwar period, especially after 1970, is that the Fed prioritized inflation at the expense of unemployment. Thornton (2011) reviews policy directives by the Federal Open Market Committee (FOMC) and finds that it made no reference to unemployment or full employment between 1979 and 2008—despite the dual mandate introduced in 1977. Instead, Thornton finds that the FOMC preferred "to state its objectives in terms of price stability and economic growth." This changed at the end of 2008, when the FOMC started mentioning its dual objective of "maximum employment and price stability" in policy directives and statements. Kaya et al. (2019) also detect this focus on inflation in FOMC transcripts. They find that from 1960 to 2010 FOMC discussions increasingly emphasized inflation relative to unemployment, and that this shift occurred during the Volcker era and continued even as inflation declined. They conclude that "the emphasis on inflation has become entrenched and disconnected from actual inflation."

The prioritization of inflation might be due to a change in the Fed's preferences or in macroeconomic theory. It might also come from Congress. Hess and Shelton (2016) examine legislative activity to determine when Congress pressures the Fed, and whether this pressure affects monetary policy. They find that by the late 1980s Congress shifted from threatening the Fed when unemployment was high to threatening when inflation was high. This finding is consistent with Weir (1987, p. 377)'s view that "By the mid-1980s full employment had been all but erased as a major political issue in the United States." In fact, Weir (1987, p. 395) argues that although the Kennedy CEA identified an unemployment rate of 4% as full employment, in the following decades "more conservative economists [offered] ever-increasing rates of unemployment as the 'true' definition of full employment."

16.6.4. Great Recession

The Great Recession saw the highest unemployment gap of the 1946–2019 period, at +5.9pp, and it presented new challenges to the Fed. Although the unemployment gap skyrocketed in 2008–2009, the Fed was unable to respond because it ran against the zero lower bound on nominal interest rates from the end of 2008 until 2015 (Board of Governors of the Federal Reserve System 2025). The Fed could not stimulate aggregate demand through lower interest rates because it was constrained by the zero lower bound, so it could not boost tightness and lower unemployment. Hence, unemployment remained inefficiently high until 2018.

Indeed, during the Great Recession, the unemployment gap shot up to around 6 percentage points. In this case, using our formula, the Fed should have lowered the federal funds rate by 12 percentage points. In reality, the federal funds rate at that time was 5%, so it would not have been possible for it to drop so much due to the zero-lower-bound (ZLB) on the nominal interest rate. The ZLB therefore became binding.

The Fed did resort to unconventional monetary policy, including forward guidance and quantitative easing, to reduce long-term interest rates in the aftermath of the Great Recession (Kuttner 2018). But the effectiveness of such policies is debatable (Greenlaw et al. 2018; Michaillat and Saez 2021b). Moreover, the Fed may not have used these policies aggressively enough because once again they targeted an unemployment rate that was too high. The Fed commonly uses the CBO's NRU to indicate full employment. During the Great Recession the CBO (2021) adjusted the NRU upward by 1pp because they believed that structural factors temporarily kept the unemployment rate high. As a result, in 2011:Q4, the short-term NRU reached 5.8%. We do find that the outward shift of the Beveridge curve after the Great Recession raised the FERU by 0.5pp, but the FERU only stood at 4.5% in 2011:Q4—1.3pp below the short-term NRU.

16.6.5. Coronavirus pandemic

The coronavirus pandemic sharply slowed down economic activity. In 2020, the US economy reached the largest unemployment gap since the Great Depression, at +6.3pp. As during the Great Recession, the Fed could not respond more aggressively to the slackness of the economy because of the zero lower bound (Board of Governors of the Federal Reserve System 2025).

Thanks to aggressive expansionary fiscal policy, however, the US economy recovered rapidly from the pandemic (Romer 2021). The US economy reached full employment in 2021:Q2, and the labor market continued tightening after that. In 2022:Q2, labor market tightness reached 1.98, a level it had not seen since the end of World War 2. It is only then, in the spring of 2022, that the Fed started tightening monetary policy.

It is unclear why the Fed did not start tightening monetary policy earlier. After the labor market reached full employment (2021:Q2), an entire year passed before the Fed increased rates (2022:Q2). This delay is all the more surprising since inflation was also above its target of 2% at the time. Core inflation was 3.7% in 2021:Q2 and rose to 6.3% in 2022:Q1 (figure 16.2). Combined with the two years required by monetary policy to be fully effective (Coibion 2012), this delay explains well why the labor market remained inefficiently tight until the end of 2024.

One possible explanation is that the Fed relied on the unemployment rate alone to judge whether the economy had recovered to full employment. It is true that the unemployment rate fell to the FERU in 2021:Q2, but the FERU was elevated at that time, at $u^* = 5.8\%$ (figure 12.2). This is because the Beveridge curve shifted far out during the pandemic, which raised the FERU (figure 2.7). So if the Fed was not aware of the increase in the FERU after the pandemic, they would see an unemployment rate that was much higher than before the pandemic, and would conclude that the economy had not reached full employment yet. Indeed, in the summer of 2021, Powell (2021, p. 4), the chairman of the Fed, offered the following assessment of the labor market:

> The unemployment rate has declined to 5.4%, a post-pandemic low, but is still much too high, and the reported rate understates the amount of labor market slack.

In fact, given that the FERU was well above its pre-pandemic level, the reported unemployment rate overstated the amount of labor market slack: the unemployment gap $u - u^*$ was much lower than it would have been before the pandemic with the same unemployment rate u.

The mistake was to not look at labor market tightness, which showed a labor market at full employment by mid-2021. Typically, the FERU is utterly stable, so observing unemployment or tightness yields the same insights. But the pandemic was different: the dramatic shift in the Beveridge curve affected the FERU unusually, so simply tracking the unemployment rate was misleading. The insight here is that the efficient tightness is unaffected by shifts in the Beveridge curve, so tightness is much more informative about the state of the labor market when the Beveridge curve is moving. Of course, tracking the unemployment gap (not the unemployment rate) provides the same insights as tracking labor market tightness.

A subtle secondary mistake was to take into account the reduced labor force participation at the time into the calculations. Powell wrote in the quote above that the unemployment rate "understated" the amount of slack. In fact, Powell (2021, chart 2) also displayed an unemployment rate adjusted "for diminished labor force participation induced by the pandemic (as estimated by Federal Reserve Board staff)." That unemployment rate was higher than the regular unemployment rate, prompting the Fed to argue that the unemployment gap was larger than one might think. But as we saw in chapter 9, this reasoning is not correct: even when labor force participation is endogenous, the efficient tightness and unemployment rate remain given by the same formula. So the FERU formula $u^* = \sqrt{uv}$ is also valid when people adjust their labor force participation due to the increased risk of working caused by the coronavirus or to the increased value of staying at home caused by school closures. The Fed should not have used an adjusted unemployment rate in their analysis.

Finally, in 2021:Q2, inflation was elevated, which was another reason to tighten monetary policy. But the Fed might have thought that the inflation spike was temporary, so it might have been reluctant to tighten monetary policy on account of inflation. Powell (2021, p. 5) wrote at the time that:

> Over the 12 months through July, measures of headline and core personal consumption expenditures inflation have run at 4.2% and 3.6%, respectively—well above our 2% longer-run objective.... Inflation at these levels is, of course, a cause for concern. But that concern is tempered by a number of factors that suggest that these elevated readings are likely to prove temporary.

16.7. Application to the slackish business cycle model

The aim of the sufficient-statistic approach is to cover a broad class of models. In this final section, we apply the sufficient-statistic to the slackish business cycle model of chapter 13.

Clearly, the slackish business cycle model with fixed inflation developed in chapter 13 falls into the class of models studied in this chapter. First, social welfare is solely determined by the unemployment rate, so social welfare is maximized when the rate of unemployment is efficient. Second, monetary policy determines the unemployment rate by modulating aggregate demand, without creating any distortions. Hence, the optimal monetary policy is to set the nominal interest rate so as to reach the efficient unemployment rate u^* and eliminate the unemployment gap. Formally, the optimal nominal interest rate i^* must be set so that $u(\theta(i^*)) = u^*$, where the unemployment rate is a function $u(\theta)$ of tightness given by (8.8), and tightness is a function of the interest rate given by (13.12).

Figure 16.3A illustrates the optimal response of monetary policy when the unemployment gap is small. Initially, aggregate demand is given by the dashed aggregate demand curve, and tightness is θ. Since $\theta < \theta^*$, the initial unemployment gap is positive. It is optimal for monetary policy to eliminate the unemployment gap, so the interest rate should drop to stimulate aggregate demand and bring tightness to its efficient level, θ^*.

Since the efficient allocation is in the gray cone in figure 16.3A, optimal monetary policy can be implemented. The gray cone represents the range of aggregate demand curves for any positive interest rate below the current interest rate; the outward boundary of the cone is the highest aggregate demand curve, obtained when the nominal interest rate is 0.

Due to the zero lower bound, the optimal policy is only feasible if $i^* \geq 0$. In figure 16.3B, aggregate demand is initially more depressed, and the efficient allocation falls outside of the gray cone. Hence, monetary policy cannot eliminate the unemployment gap before reaching the zero lower bound. The best that monetary policy can do is reduce the nominal interest rate to 0, and bring the economy to the zero lower bound. There tightness is $\theta^z < \theta^*$, so the unemployment gap remains positive.

Finally, we come back to the conventional situation illustrated in figure 16.3A. There the drop in interest rate to bring tightness up from θ to θ^*, and the unemployment rate down from u to u^*, is given by the sufficient-statistic formula (16.2).

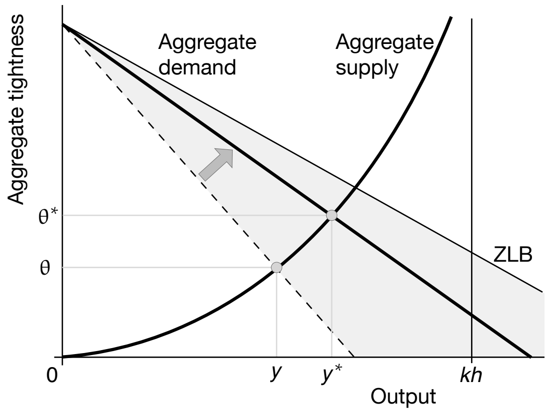
We saw in chapter 13 that the slackish business cycle model can produce the sort of fluctuations in the unemployment gap observed in the US labor market. One naturally wonders whether the slackish business cycle model is also able to match the values of the monetary multiplier estimated in US data—to rationalize empirical estimates of the effectiveness of monetary policy.

To assess whether the model produces a realistic monetary multiplier, we use formula (13.16), which gives the monetary multiplier in the model. At full employment, the multiplier equals

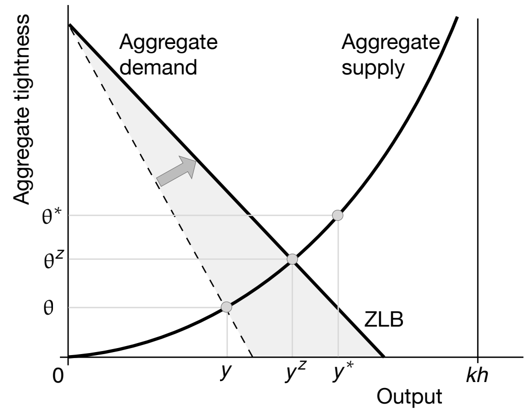
$$\frac{du}{di} = \frac{1 - u^*}{\delta - r^*}.$$

To assess the value of the monetary multiplier, we must calibrate u^*, δ, and r^*. For the FERU, we adopt the average US value between 1948 and 2019, which is $u^* = 4.2\%$ (chapter 12). For the natural real interest rate and time discount rate, we adopt the calibration from chapter 13: $r^* = 0.5\%$ and $\delta = 91\%$. From these estimates, we calibrate the monetary multiplier to be $du/di = (1 - 0.042)/(0.91 - 0.005) = 1.06$.

The calibrated value of the monetary multiplier (1.06) is higher than the estimate from



A. Small initial unemployment gap: optimal monetary policy restores efficient unemployment



B. Large initial unemployment gap: optimal monetary policy reaches the ZLB

FIGURE 16.3. Optimal monetary policy in the slackish business cycle model

The aggregate supply and aggregate demand curves are constructed in figure 13.1B. The ZLB curve is constructed in figure 13.4A. The tightness θ^* and output y^* constitute the efficient allocation.

Coibion (2012)'s hybrid approach (0.5). But it is quite close to the estimate from Coibion (2012)'s narrative approach (0.93). The takeaway is that the monetary multiplier in the basic slackish model is fairly consistent with the estimates coming from the narrative approach, but is much higher than the estimates coming from VAR approaches. If we used a lower value of the time discount factor in our calibration, closer to the median estimate, the calibrated value of the monetary multiplier would be higher, and further away from empirical estimates. So it seems fair to remark that the slackish business cycle model, in its basic form, might predict effects of monetary policy that are stronger than what we observe empirically.[3]

16.8. Summary

In this chapter, we have seen that if inflation is fixed or the inflation-slack coincidence holds, the optimal monetary policy is to adjust interest rates to eliminate the slack gap entirely. The central bank should lower rates in bad times, when slack is inefficiently high, and raise rates in good times, when slack is inefficiently low.

In fact, optimal monetary policy is given by a simple formula that relates the optimal interest rate to two sufficient statistics: the slack gap and the monetary multiplier (the effect of the federal funds rate on the slack rate). In the United States, the monetary

[3] A possible reason why changes in the federal funds rate are less potent in reality than in the model is that the changes do not affect households directly. Households respond to the interest rates on their credit cards, loans, and mortgages, not directly to the fed funds rate. If these interest rates respond sluggishly or partially to the fed funds rate, then of course the fed funds rate would have a limited effect on aggregate demand and the economy in reality (Gregor, Melecky, and Melecky 2021).

multiplier is about 0.5: the unemployment rate drops by 0.5pp when the nominal interest rate is cut by 1pp. The formula therefore indicates that the Fed should lower the federal funds rate by 2 percentage points for any 1 percentage point of unemployment gap.

When inflation responds to monetary policy and the inflation-slack coincidence fails, optimal monetary policy is more complex but still calls for reducing the slack gap. In this case, it is optimal for the Fed to tolerate inefficiently high slack when inflation is above target, and to tolerate inefficiently low slack when inflation is below target. The sizes of the deviations of slack and inflation from their targets are determined by the welfare cost of inflation relative to slack and by the slope of the Phillips curve.

Finally, we examine why the US labor market has generally been too slack in the past century. We find that the price-stability mandate is not the answer: most of the time, when unemployment was too high, inflation was too low. Other constraints on monetary policy, such as political pressure and the zero lower bound, as well as misestimates of full employment, are more plausible explanations.

CHAPTER 17.

Fiscal policy

In chapter 16 we saw how monetary policy should respond when the economy is not at full employment. In various circumstances, it is either not possible or not desirable for monetary policy to eliminate the unemployment gap entirely. In that case, it is natural to wonder: can monetary policy be complemented by fiscal policy? As we see in this chapter, the answer to this question is yes.

The first case when fiscal policy might be useful is when monetary policy is impotent. This occurs at the zero lower bound on the nominal interest rate. Once the zero lower bound is reached, interest rates can no longer be reduced, so monetary policy can no longer be used to prop up the economy.

The second case when monetary policy cannot be used is under monetary unions. In a monetary union, monetary policy is decided at a higher level and cannot be controlled by the local government. One example of a monetary union is the United States; here, monetary policy is controlled by the Federal Reserve and not by individual states. States may be subject to local recessions that are not synchronized with national recessions and that are therefore not tackled by the Federal Reserve (Owyang, Piger, and Wall 2005). But states are able to determine how much they want to spend, how much they want to tax, and they can even issue bonds. Hence states are in complete control of their fiscal policy, and can use it to tackle local business cycles.

17.1. Social planner's problem

To study optimal fiscal policy, we follow the same approach we used to study the unemployment gap in chapter 12 and optimal monetary policy in chapter 16. We specify a generic framework and solve the social planner's problem in that framework, which we will use to derive a sufficient-statistic formula for optimal fiscal policy.

17.1.1. Economic slack

We assume that there is slack in the economy. This means that a share $u \in (0, 1)$ of the aggregate productive capacity cannot be sold. As usual, we assume that slack is purely wasteful. This means that the unsold capacity does not contribute to welfare—just as in the models of part IV.

17.1.2. Fiscal policy

In the efficiency analysis of chapter 9, we simply assumed that the planner could pick the amount of slack in the economy to maximize welfare. Here, we are studying optimal fiscal policy, which means that our planner is more restricted: she can only affect slack indirectly, through fiscal policy.

Specifically, we assume that the planner determines the amount g of public goods and services provided to society. And we assume that, in addition to being valuable, public goods contribute to stabilization, just as in the models of part IV. That is, the amount of public goods provided affects the slack rate through a generic function $u(g)$.

17.1.3. Beveridge curve

Both the private sector and the government aim to purchase goods and services for private and public consumption. Purchasing goods and services requires posting help-wanted ads. In such a general framework, help-wanted ads represent many different things that households, firms, and the government do to purchase goods and services: they capture job vacancies posted to hire labor services, actual help-wanted ads posted online or in newspapers, requests for proposal or requests for tender posted to hire services or purchase goods, or any other actions or processes that buyers must follow to purchase goods or services.

With each help-wanted ad a buyer might purchase one good or service. The number of help-wanted ads advertised is v, and it is expressed as a share of the aggregate productive capacity. Moreover, a Beveridge curve $v(u)$ links the ad rate v to the slack rate u, just as in the models of chapters 13 and 14.

Why do help-wanted ads enter the planner's problem? Because ads require resources to be successful and translate into purchases. A matching cost $\kappa > 0$ measures the number of goods and services that have to be devoted to any one help-wanted ad per unit time. These goods and services devoted to matching are not consumed by households, so they do not contribute to social welfare, although they are required to create new trades.

17.1.4. Usage of aggregate productive capacity

Let's summarize how aggregate productive capacity is used. We take productive capacity as exogenous—and in particular as unaffected by fiscal policy.

A share u of aggregate productive capacity is slack and therefore unused. A share κv is used for matching. We therefore have a share $1 - (u + \kappa v)$ of aggregate productive capacity that constitutes goods and services that are consumed. There are two types of goods and services: private (purchased directly by households) and public (purchased and provided by the government). A share c of aggregate productive capacity constitutes private goods and services, while a share g constitutes public goods and services. Note that $c + g = 1 - (u + \kappa v)$ represents total consumption. Thus,

$$c = 1 - (u + \kappa v) - g, \tag{17.1}$$

where c is private consumption, g is public consumption, and $u + \kappa v$ is nonproductive use of capacity (matching and slack).

The government provides public goods that benefit everyone, and that can potentially reduce the amount of slack in the economy. However, there is a resource cost to providing public goods, since productive capacity used for public goods cannot be used for private goods: for instance, workers employed in the public sector cannot also be employed in the private sector. Productive capacity is finite, so the planner faces a tradeoff: it can divert more goods to the public sector, but then fewer goods are left for the private sector. So fiscal policy creates a distortion: it depletes resources from the private sector. This tradeoff is at the heart of the Samuelson (1954, 1955) framework.

17.1.5. Social welfare

Social welfare depends on both public and private goods and services. We denote by c the share of the economy's productive capacity that goes to private goods and services. As we saw earlier, g is the share of the economy's productive capacity that is allocated to public goods and services. Both private and public goods and services enter social welfare, $\mathcal{W}(c, g)$. We assume that $\mathcal{W}$ is strictly increasing in c and g, homogeneous of some positive degree, and strictly quasiconcave (result A.11). These assumptions are standard in welfare and utility theory.

17.1.6. Planner's problem

We now formalize the social planner's problem. The planner's goal is to maximize welfare $\mathcal{W}(c, g)$ subject to the resource constraint (17.1). Substituting private consumption from the welfare function, and inserting the Beveridge curve $v(u)$ and stabilization function $u(g)$, the planner's problem simplifies to

$$\max_{g \geq 0} \left(\mathcal{W}(1 - (u(g) + \kappa v(u(g))) - g, g) \right). \tag{17.2}$$

Here, we see the tradeoffs that the government faces. Increasing public goods raises welfare mechanically. But it also mechanically reduces private goods via the resource constraint. Moreover, public goods may contribute to stabilization because they affect the term $u(g) + \kappa v(u(g))$, which measures the nonproductive use of the economy's capacity, and which should ideally be as small as possible.

17.1.7. Solution to the planner's problem

We now start solving the social planner's problem, given by (17.2). We begin with the first-order condition of the problem.

We take the derivative of social welfare with respect to g to capture the different effects of public spending on welfare:

$$\frac{d\mathcal{W}}{dg} = \frac{\partial \mathcal{W}}{\partial g} - \frac{\partial \mathcal{W}}{\partial c} + \frac{\partial \mathcal{W}}{\partial c} \cdot \left[-u'(g) \right] \cdot \left[1 + \kappa v'(u) \right]. \tag{17.3}$$

Here, we see all the ways through which public spending affects welfare. The first term tells us that having more public goods increases welfare, and the second term tells us that if more public goods are provided, fewer private goods are available, which depresses welfare. The third term reflects stabilization by public spending—it shows how public goods affect unproductive use of resources.

We now determine the optimal amount of public spending g. Optimal public spending maximizes $\mathcal{W}(c(g), g)$. We assume that the planner's problem is a well behaved so that it admits a unique extremum which is an interior maximum. The first-order condition is therefore necessary and sufficient to find the problem's solution. The first order condition simply is $d\mathcal{W}/dg = 0$, where the total derivative is given by (17.3). Dividing the first-order condition by the marginal utility of private consumption, we get

$$1 = \frac{\partial \mathcal{W}/\partial g}{\partial \mathcal{W}/\partial c} + \left[-u'(g) \right] \cdot \left[1 + \kappa v'(u) \right]. \tag{17.4}$$

In the next section we start from this somewhat abstract condition and introduce several sufficient statistics to transform it into a usable formula for optimal fiscal policy.

17.2. Implicit formula for optimal fiscal policy

This section introduces several sufficient statistics that we will use to characterize optimal fiscal policy. We combine them into a first formula for optimal fiscal policy. This formula is only implicit. We use it as a stepping stone to then build an explicit formula for optimal fiscal policy.

17.2.1. Marginal rate of substitution between public and private goods

A first important statistic in the optimality condition (17.4) is the marginal rate of substitution between public and private services:

$$MRS(g/c) = \frac{\partial \mathcal{W}/\partial g}{\partial \mathcal{W}/\partial c} > 0. \tag{17.5}$$

Since the welfare function $\mathcal{W}$ is homogeneous of some positive degree and strictly quasi-concave, the marginal rate of substitution is well defined as a function of g/c alone, and is strictly decreasing in g/c (result A.3).

The marginal rate of substitution indicates how people value public goods relative to private goods. Because it is strictly decreasing in g/c, the relative value of a public good decreases when there are more public goods relative to private goods. We also assume that $MRS(g/c = 0) > 1$, so that when there are no public goods at all, a public good is more valuable than a private good. Under this assumption, it is desirable to provide some public goods, and so we conveniently have an interior solution to the planner's problem.

17.2.2. Fiscal multiplier

Another important statistic in the optimality condition (17.4) is the fiscal multiplier:

$$m = -u'(g). \tag{17.6}$$

The fiscal multiplier measures the decrease in the slack rate in percentage points when the share of aggregate capacity devoted to public production increases by 1 percentage point. In the special case $m = 0$, the slack rate does not respond to public spending. Although this is a possibility, empirical evidence from the United States suggests that the slack rate drops when public spending goes up: $m > 0$. Note that our framework also allows for $m < 0$, which would mean that the slack rate goes up when public spending increases. It's unclear how this would occur, but the sufficient-statistic approach allows for it nevertheless.

17.2.3. Inefficiency gauge

The third and final statistic to obtain an implicit formula for optimal fiscal policy is the inefficiency gauge:

$$IG(u) = 1 + \kappa v'(u). \tag{17.7}$$

The gauge describes how a change in the slack rate u affects the total amount unproductive resources in the economy, $u + \kappa v(u)$.

In effect, the sign of the inefficiency gauge tells us whether the economy is operating efficiently or is inefficiently slack or tight. Recall that when the economy operates efficiently, the slack rate u^* satisfies (9.3), which gives

$$IG(u^*) = 0.$$

So, when the economy operates efficiently, the inefficiency gauge is 0. Then, a change in the slack rate has no first-order effect on the amount of unproductive resources in the economy and thus social welfare.

In addition, the Beveridge curve is convex, so that $v'(u)$ is increasing in u. This implies that $u > u^*$ is equivalent to $v'(u) > v'(u^*)$, which itself is equivalent to $1+\kappa v'(u) > 1+\kappa \mathbf{v}'(u^*)$ or $IG(u) > 0$. By the same logic, $u < u^*$ is equivalent to $IG(u) < 0$.

Hence we have established that the sign of the inefficiency gauge $IG(u)$ determines the state of the economy: at $IG(u) = 0$ the economy is efficient; when $IG(u) < 0$ the economy is inefficiently tight, so slack is too low; and when $IG(u) > 0$ the economy is inefficiently slack, so slack is too high.

17.2.4. Implicit formula

We rewrite the abstract first-order condition (17.4) using the three statistics that we have just introduced. Inserting the statistics into the condition, we immediately obtain:

$$1 = MRS(g/c) + m \cdot IG(u). \tag{17.8}$$

This is our first formula for optimal fiscal policy. It says what the optimal amount of public spending, expressed as the ratio between public and private goods g/c, must satisfy in a world with slack.

The formula can be interpreted as the Samuelson rule plus a correction term. Indeed, the Samuelson rule says that public goods should be provided until the marginal utility from public goods equals the marginal utility from private goods—in other words, until the marginal rate of substitution between public and private goods equals 1. The Samuelson amount of public spending, denoted $(g/c)^*$, is therefore given by

$$1 = MRS((g/c)^*).$$

The correction term is $m \cdot IG(u)$. The fiscal multiplier m indicates how much public spending reduces slack. The inefficiency gauge $IG(u)$ indicates how much a reduction in slack improves welfare. So the correction term indicates how much public spending improves welfare through slack. The correction term is positive whenever public spending brings slack closer to its efficient level: either by lowering slack when it is too high ($m > 0$ and $IG > 0$), or by raising slack when it is too low ($m < 0$ and $IG < 0$).

The implicit formula (17.8) simply says that when the provision of public goods is optimal, the marginal cost and benefits from public goods are equalized. The marginal cost of public goods is 1 private good, which appears on the left-hand side of the formula. Indeed, for a given amount of production, 1 good or service provided by the public sector cannot be provided by the private sector. The marginal benefits of public goods appear on the right-hand side of the equation. One benefit is that people value public goods, which is captured by the marginal rate of substitution between public and private goods, MRS. Another benefit appears when public spending brings slack closer to its efficient level; in that case the benefit is measured by $m \cdot IG > 0$. If public spending pushes slack away from its efficient level, it imposes a cost, or negative benefit measured by $m \cdot IG < 0$.

17.2.5. Optimal deviation from Samuelson spending

Let's see what we can learn from our implicit formula (17.8). It turns out that while the formula is only implicit, it is useful to understand how optimal public spending deviates from public spending under the Samuelson rule.

We see from formula (17.8) that when the economy operates efficiently, the Samuelson rule remains valid. This can be seen because when slack is efficient, $IG(u) = 0$, so the formula boils down to $1 = MRS(g/c)$, which is just the Samuelson rule. Although the Samuelson rule was originally derived in a Walrasian model, it remains valid in a slackish model as long as the amount of slack is efficient.

Intuitively, why does the Samuelson rule continue to apply when slack is present but efficient? Government spending affects welfare in two ways: directly by providing public goods to people, and indirectly by affecting the amount of production in the economy. In the Walrasian model, the economy is efficient, so changes in production have no first-order effects on welfare. In the slackish model, the economy is generally inefficient, but when it is efficient, then changes in slack have no first-order effects on welfare. In both cases—Walrasian and efficient slackish models—the only first-order effect of public spending on welfare is through the provision of public goods, so the formulas are the same.

TABLE 17.1. Optimal deviation from Samuelson rule when slack is inefficient

	Fiscal multiplier		
Economic slack	$m < 0$	$m = 0$	$m > 0$
$u > u^*$	$g/c < (g/c)^*$	$g/c = (g/c)^*$	$g/c > (g/c)^*$
$u = u^*$	$g/c = (g/c)^*$	$g/c = (g/c)^*$	$g/c = (g/c)^*$
$u < u^*$	$g/c > (g/c)^*$	$g/c = (g/c)^*$	$g/c < (g/c)^*$

The optimal deviations from the Samuelson rule are derived from the formula (17.8).

Another case in which the Samuelson rule remains valid despite the presence of slack is when the fiscal multiplier is 0. Indeed, with $m = 0$, the formula boils down again to the Samuelson rule $1 = MRS(g/c)$. The reason is that when the fiscal multiplier is 0, public spending has no effect on slack, so it does not affect welfare through production, which brings us back to the Walrasian case.

There is an extra term in formula (17.8), however, which appears because we are no longer in a Walrasian model. We are considering a world with slack, and the amount of slack is generally inefficient, and public spending generally affects slack. When slack is indeed inefficient, and public spending affects it, an extra term appears in the formula ($m \cdot IG$). The term describes the effect of public spending on welfare through its contribution to stabilization.

The main lesson from formula (17.8) is that whenever public goods are more beneficial than envisioned by the Samuelson rule, we should provide more of them than stipulated by rule. Public goods are more beneficial when they bring slack closer to its efficient level. This appears with $m \cdot IG(u) > 0$ in the formula, which then requires $MRS(g/c) < 1$. As the marginal rate of substitution is decreasing in g/c, this indeed requires $g/c > (g/c)^*$.

Overall, formula (17.8) shows that it is optimal to raise public spending over Samuelson spending when public spending brings slack closer to its efficient level; conversely, it is optimal to lower public spending below Samuelson spending whenever public spending pushes slack away from its efficient level.

Thus, optimal public spending satisfies the Samuelson rule only if the fiscal multiplier is zero or if slack is at an efficient level. In all other situations, optimal public spending deviates from the Samuelson rule. Based on this logic, table 17.1 describes how optimal public spending deviates from Samuelson spending. It separates the different scenarios to consider, depending on the sign of the fiscal multiplier and the amount of slack in the economy.

Table 17.1 displays an important and interesting insight: it is not optimal for public spending to deviate from the Samuelson rule so as to completely fill the slack gap. We saw in chapter 16 that it was often optimal for monetary policy to completely fill the unemployment gap. That is because monetary policy often does not create any distortions. By contrast, fiscal policy always creates distortions: it distorts the consumption basket by providing more or less public consumption than what the Samuelson rule recommends. Because of the welfare cost of the distortion, it is not optimal to eliminate the slack gap. This result appears in the table, as when the slack gap is 0 public spending ought to be at the Samuelson level. The result also appears in equation (17.8). The equation must hold when fiscal policy is optimal. If optimal fiscal policy eliminates the slack gap, the equation reduces to $1 = MRS(g/c)$, which would not be possible if public spending deviated from Samuelson spending.

17.3. Sufficient-statistic formula for optimal fiscal policy

One limitation of formula (17.8) that appears immediately is that it is only implicit. The right-hand side of the formula, $MRS + m \cdot IG$, is determined by public spending g, so indeed the formula determines the optimal level of public spending. But public spending does not appear explicitly in the formula, so it's not possible to plug in some parameter values and compute the optimal amount of spending.

In this section, we introduce a few sufficient statistics that will help us understand how the marginal rate of substitution MRS depends on g, how the inefficiency gauge IG depends on the slack rate u and then how the slack rate itself depends on g. Once we have understood this dependence, we will construct a sufficient-statistic formula, from which we will be able to compute optimal public spending directly.

17.3.1. Elasticity of substitution between public and private goods

We are now in a position to introduce the first sufficient statistic that appears in the formula for optimal fiscal policy. The statistic is the elasticity of substitution between public and private services, σ, defined as the inverse of the positive elasticity of the marginal rate of substitution with respect to the ratio g/c:

$$\frac{1}{\sigma} = -\frac{(g/c)}{MRS(g/c)} \cdot \frac{dMRS}{dg/c}. \tag{17.9}$$

Since MRS is strictly decreasing in g/c, the definition (17.9) ensures that $\sigma > 0$. There are three different possibilities. First, when $\sigma < 1$, public and private services are complements. Second, when $\sigma = 1$, public and private services are independent. Last, when $\sigma > 1$, public and private services are substitutes.

The elasticity of substitution admits a few interesting special cases. When $\sigma \to 0$, public and private services are perfect complements. In this case, both c and g are required to generate welfare, and we have a Leontief welfare function:

$$\mathcal{W}(c, g) = \min(c, g).$$

Another special case is when $\sigma = 1$ so that public and private services are independent. Then, we would have a Cobb-Douglas welfare function:

$$\mathcal{W}(c, g) = c^{1-\alpha} \cdot g^{\alpha}.$$

The last special case is when $\sigma \to \infty$: then public and private goods are perfect substitutes. In this case, c and g indistinguishably generate welfare, so the welfare function is linear:

$$\mathcal{W}(c, g) = c + g.$$

The elasticity of substitution describes the shape of marginal rate of substitution, and how it varies with the ratio g/c. It is helpful here because it allows us to express how the marginal rate varies when g/c varies, which will then allow us to introduce g/c explicitly in the formula for optimal fiscal policy.

Indeed, we take a first-order approximation of $MRS(g/c)$ around Samuelson spending $(g/c)^*$, we get:

$$MRS = MRS((g/c)^*) + \frac{dMRS}{dg/c} \cdot \left[g/c - (g/c)^*\right].$$

As we have seen, $(g/c)^*$ is the amount of public spending implied by the Samuelson rule so $MRS((g/c)^*) = 1$. Given the definition of the elasticity of substitution (17.9), this approximation simplifies to:

$$1 - MRS = \frac{1}{\sigma} \cdot \frac{g/c - (g/c)^*}{(g/c)^*}, \tag{17.10}$$

where σ is the elasticity of substitution evaluated at $(g/c)^*$, so

$$\frac{1}{\sigma} = -\frac{(g/c)^*}{MRS((g/c)^*)} \cdot \frac{dMRS}{dg/c} = -(g/c)^* \cdot \frac{dMRS}{dg/c}.$$

A new but key term appearing in (17.10) is the relative deviation between optimal public spending and the Samuelson level of spending: $[g/c - (g/c)^*]/(g/c)^*$. It indicates how much optimal spending deviates from Samuelson spending, as a share of Samuelson spending. We refer to this relative deviation as the amount of stimulus spending. So for instance, a stimulus spending of 10% means that optimal public spending is 10% higher than Samuelson spending.

17.3.2. Beveridge curvature

The second sufficient statistic that appears in the formula for optimal fiscal policy is the positive elasticity of the slope of the Beveridge curve:

$$\beth = -\frac{u}{v'(u)} \cdot v''(u). \tag{17.11}$$

We refer to this elasticity as the Beveridge curvature: it describes how the slope of the Beveridge curve is changing with the slack rate; specifically, it describes how sharply the amplitude of the slope declines as the slack rate increases. It is helpful here because it allows us to express how the inefficiency gauge varies when the slack rate varies, which will then allow us to introduce the slack rate u explicitly in the formula for optimal fiscal policy.

If the Beveridge curve is isoelastic (as in the US labor market), the Beveridge curvature is directly related to the Beveridge elasticity:

$$\beth = 1 + \beta.$$

Indeed, when the Beveridge curve is isoelastic, $v(u) = A \cdot u^{-\beta}$, so $v'(u) = -\beta \cdot A \cdot u^{-(\beta+1)}$, and $v''(u) = -(\beta + 1) \cdot v'(u)/u$. So, clearly, $\beth = 1 + \beta$.

Using the Beveridge curvature, we take a first-order approximation of $IG(u)$ around the efficient slack rate u^*. We get:

$$IG(u) = IG(u^*) + IG'(u^*) \cdot (u - u^*).$$

Recall that $IG(u^*) = 0$. Moreover,

$$IG'(u) = \kappa v''(u) = -\kappa\beth \cdot \frac{v'(u)}{u}.$$

The efficiency condition (9.3) implies that $-v'(u^*) = 1/\kappa$, so at u^* the derivative of the efficiency gauge simplify to

$$IG'(u^*) = \frac{\beth}{u^*}.$$

From that, we see that the first-order approximation of $IG(u)$ around u^* simply is

$$IG(u) = \beth \cdot \frac{u - u^*}{u^*}. \tag{17.12}$$

A key term appearing in (17.12) is the relative deviation between optimal slack rate and the efficient slack rate: $[u - u^*]/u^*$. It indicates how much the optimal slack rate (the slack rate under optimal fiscal policy) deviates from the efficient slack rate (the slack rate minimizing the nonproductive use of capacity), as a share of the efficient slack rate. We refer to this relative deviation as the slack gap. So for instance, a slack gap of 10% means that the optimal slack rate is 10% higher than the efficient slack rate.

17.3.3. The fiscal multiplier again

At this point, by inserting the results given by (17.10) and (17.12) into the first-order condition (17.8), we obtain the following formula for optimal fiscal policy

$$\frac{g/c - (g/c)^*}{(g/c)^*} = \daleth\sigma m \cdot \frac{u - u^*}{u^*}. \tag{17.13}$$

In formula (17.13), optimal public spending g/c appears explicitly in the left-hand side, so it might appear that the issue of (17.8) has been addressed: optimal public spending is not implicit anymore. However, success is only illusory here: the slack rate u depends on g/c itself, so g/c appears in the right-hand side too, and the formula does not explicitly give optimal public spending.

However, thanks to the fiscal multiplier, we can make the dependence between slack and fiscal policy explicit, and thus rewrite the formula so that it actually explicitly quantifies optimal fiscal policy. Given that the fiscal multiplier m gives the effect of public spending g on the slack rate u, the main challenge is really to link changes in g/c to changes in g. Because the marginal rate of substitution between public and private goods depends on g/c, the ratio is in formula (17.13), but the multiplier (17.6) only involves the effect of g on u, so we must determine how changes in g/c and g are connected. This is mostly boring algebra, but which is greatly simplified by the fact that formula (17.13) is obtained around the efficient slack rate u^* and Samuelson spending $(g/c)^*$. The share of capacity devoted to private consumption is just the share of aggregate capacity that's productive and not devoted to public consumption:

$$c(g) = 1 - [u(g) + \kappa\nu(u(g))] - g.$$

Hence, the derivative of c with respect to g is

$$c'(g) = -u'(g) \cdot \left[1 + \kappa\nu'(u)\right] - 1.$$

But at the efficient slack rate, the nonproductive share of capacity is minimized, so as (9.3) shows, $1 + \kappa\nu'(u^*) = 0$. So private consumption simply moves one-for-one with public consumption: $c'(g) = -1$.

With this result, we easily compute $d(g/c)/dg$ around $[u^*, (g/c)^*]$:

$$\frac{d(g/c)}{dg} = \left(\frac{1}{c^*} - \frac{g^*}{(c^*)^2} \cdot c'(g^*)\right) = (g/c)^* \cdot \left(\frac{1}{g^*} + \frac{1}{c^*}\right), \tag{17.14}$$

where c^* and g^* are the share of aggregate capacity devoted to private and public consumption at $[u^*, (g/c)^*]$.

We can now link the slack rate u to fiscal policy g/c in (17.13). We assume that public spending is at the Samuelson level $(g/c)^*$ and slack is at an inefficient rate $u_0 \neq u^*$. We have in mind the following scenario. Initially everything is going well: slack is efficient, and public spending satisfies the Samuelson rule. Then a shock occurs, pushing slack to the inefficient level u_0. In response, public spending deviates from the Samuelson level to reach its optimal level. As it does so, it affects slack too, which does not stay at the initial rate u_0.

The goal here is to express the slack gap $[u - u^*]/u^*$ as a function of the initial slack gap $[u_0 - u^*]/u^*$ as well as stimulus spending $[g/c - (g/c)^*]/(g/c)^*$, and then make (17.13) completely explicit. We start by writing the first-order approximation of $u(g)$ and $(g/c)(g)$ around the initial $[u_0, g^*]$situation. We get:

$$u = u_0 + \frac{du}{dg} \cdot (g - g^*)$$

$$(g/c) = (g/c)^* + \frac{d(g/c)}{dg} \cdot (g - g^*).$$

Let's rewrite both first-order approximation to introduce the terms we are interested in:

$$u - u^* = u_0 - u^* - m \cdot (g - g^*)$$

$$g - g^* = \frac{1}{1/g^* + 1/c^*} \cdot \frac{(g/c) - (g/c)^*}{(g/c)^*}.$$

Here we could introduce the fiscal multiplier because $m = -du/dg$, and we used result (17.14) to replace $d(g/c)/dg$.

Combining these results we link the change in the slack rate u to the change in public spending g/c:

$$\frac{u - u^*}{u^*} = \frac{u_0 - u^*}{u^*} - \frac{m}{u^*} \cdot \frac{1}{1/g^* + 1/c^*} \cdot \frac{(g/c) - (g/c)^*}{(g/c)^*}. \tag{17.15}$$

We have now linked the current slack gap to the initial slack gap as well as stimulus spending and the fiscal multiplier. By plugging this result in the intermediate formula (17.13), we will obtain our final sufficient-statistic formula.

17.3.4. Sufficient-statistic formula

We have now expressed the slack gap using the initial slack gap, our sufficient statistics, and stimulus spending. We now plug expression (17.15) into the right-hand side of formula

(17.13) to make it explicit:

$$\frac{g/c-(g/c)^*}{(g/c)^*} = \beth\sigma m \cdot \frac{u_0-u^*}{u^*} - \frac{\beth\sigma m^2}{u^*}\cdot\frac{1}{1/g^*+1/c^*}\cdot\frac{(g/c)-(g/c)^*}{(g/c)^*}.$$

Collecting the stimulus-spending terms on the left-hand side, we obtain our explicit, sufficient-statistic formula for optimal fiscal policy:

$$\frac{g/c-(g/c)^*}{(g/c)^*} = \frac{\beth\sigma m}{1+z\beth\sigma m^2}\cdot\frac{u_0-u^*}{u^*}, \tag{17.16}$$

where the constant z is given by

$$z = \frac{1}{u^*}\cdot\frac{1}{\frac{1}{g^*}+\frac{1}{c^*}}.$$

Recall that m is the fiscal multiplier, defined by (17.6), σ is the elasticity of substitution between g and c, defined by (17.9), and $\beth$ is the Beveridge curvature, defined by (17.11). Hence, the optimal amount of fiscal policy, given by formula (17.16), depends on four sufficient statistics: initial slack gap u_0-u^*, elasticity of substitution between public and private goods σ, fiscal multiplier m, and Beveridge curvature $\beth$.

We saw in table 17.1 that it is not optimal to completely fill the slack gap with fiscal policy. The question, then, is: how much of the initial slack gap should we fill and how does that depend on our sufficient statistics? For an initial slack gap u_0-u^*, we showed that the resulting slack gap $u-u^*$ after public spending is given by (17.15). We have also just seen that optimal public spending given some initial slack gap u_0-u^* is given by (17.16). Combining optimal public spending (17.16) with the effect of public spending on slack as described by (17.15), we are able to compute how much of the initial slack gap is left once public spending is optimal:

$$\frac{u-u^*}{u^*} = \frac{u_0-u^*}{u^*} - \frac{z\beth\sigma m^2}{1+z\beth\sigma m^2}\cdot\frac{u_0-u^*}{u^*},$$

and which leads to

$$u-u^* = \frac{1}{1+z\beth\sigma m^2}\cdot(u_0-u^*). \tag{17.17}$$

This formula tells us that the optimal slack gap—the slack gap achieved through optimal fiscal policy—is a share $1/(1+z\beth\sigma m^2)\in(0,1)$ of the initial slack gap. As the share is less than 1, optimal fiscal policy reduces the original slack gap. But the share is strictly positive, so optimal fiscal policy never eliminates the slack gap entirely—it only reduces it.

17.3.5. Role of the slack gap

We first look at the role of the initial slack gap on the design of optimal fiscal policy. We consider a positive fiscal multiplier for concreteness ($m > 0$).

As we discussed above, if the economy operates efficiently and the slack rate is efficient, the optimal policy is to keep public spending at the Samuelson level.

On the other hand, if the slack gap is positive ($u_0 > u^*$), optimal stimulus spending is positive ($g/c > (g/c)^*$). Fiscal policy therefore reduces but does not completely fill the slack gap ($u_0 > u > u^*$).

This result is simple to understand. At the Samuelson rule, an increase in public spending has no first-order effect on welfare when we ignore its effect on slack. Now, when the fiscal multiplier is positive, an increase in public spending lowers slack; and when the slack gap is positive, slack is inefficiently high, so lowering slack raises welfare. Hence, overall, an increase in public spending generates a positive first-order effect on welfare. It is therefore optimal to increase public spending above the Samuelson rule.

If the slack gap is negative ($u_0 < u^*$), optimal stimulus spending is negative ($g/c < (g/c)^*$). Fiscal policy therefore reduces but does not completely eliminate the slack gap ($u_0 < u < u^*$).

We see in formula (17.16) that optimal stimulus spending is larger when the slack gap is larger. This makes sense because a larger slack gap means there is more inefficient slack, which must be tackled with more spending.

17.3.6. Role of the fiscal multiplier

Next, we look at the role of the fiscal multiplier. The relationship between the fiscal multiplier and stimulus spending is a little more interesting—the function is hump-shaped. Specifically, when the fiscal multiplier is 0, there is no stimulus spending: public spending should be at the Samuelson level. Stimulus spending then increases and reaches a peak value at:

$$m^* = \frac{1}{\sqrt{z\beth\sigma}}.$$

This is the value of m that maximizes stimulus spending, by maximizing the function

$$m \mapsto \frac{\beth\sigma m}{1 + z\beth\sigma m^2}$$

Optimal stimulus spending then decreases when $m > m^*$, and it reaches 0 when $m \to \infty$.

What is the intuition behind the hump shape? When public spending is optimal, the marginal social cost from consuming too many public services and too few private services equals the marginal social value from reducing slack. This marginal social value is determined by two factors: the current fiscal multiplier, which measures how much slack can be reduced by additional expenditure, and the current slack gap, which measures the social value from lower slack. For a given amount of stimulus spending and a given initial slack gap, a larger initial multiplier has conflicting effects on the two factors: it means a larger current multiplier (a higher marginal social value) but a smaller current slack gap (a lower marginal social value). The first effect advocates for more spending, the second for less spending.

It turns out that for small multipliers, the first effect dominates, so optimal stimulus spending is increasing in the multiplier; for large multipliers, the second effect dominates, so optimal stimulus spending is decreasing in the multiplier. In fact, for very large multipliers, it becomes optimal to nearly entirely fill the slack gap. Naturally, less spending is required to fill the gap when the multiplier is larger, so optimal stimulus spending is decreasing in the multiplier.

Hence, the bang-for-the-buck logic often used in policy discussions—the view that a larger multiplier entails a larger stimulus spending—is not generally valid. Stimulus skeptics usually believe in small multipliers and infer that stimulus spending should be small in slumps. Similarly, stimulus advocates usually believe in large multipliers and infer that stimulus spending should be large in slumps. We have just seen that this bang-for-the-buck logic holds for small multipliers but not for large ones: a large multiplier is not a justification for a large stimulus. Instead, since the relationship between multiplier and optimal stimulus spending is hump-shaped, optimal stimulus spending is similar for some small and large multipliers.

17.3.7. Role of the elasticity of substitution

The policy debate on fiscal policy often revolves around slack gaps and multipliers. Formula (17.16) confirms that optimal fiscal policy is indeed a function of the slack gap and the fiscal multiplier. Yet, these statistics are not sufficient to measure the effect of public expenditure on welfare because an increase in public expenditure also modifies the composition of households' consumption. Consequently, optimal fiscal policy also depends on the elasticity of substitution between public and private consumption; this statistic should probably play a more prominent role in the policy debate.

What exactly is the role of the elasticity of substitution? We see that stimulus spending is increasing in σ. When there is higher substitutability, the stimulus package is larger. At $\sigma = 0$, there is no stimulus spending, and at $\sigma \to \infty$, we have that

$$\frac{g/c - (g/c)^*}{(g/c)^*} = \frac{1}{zm} \cdot \frac{u_0 - u^*}{u^*}.$$

What is the intuition behind the case $\sigma \to 0$? In this case, additional public services have zero value: additional public workers dig and fill holes in the ground. Then, optimal stimulus spending is zero, irrespective of the slack rate and multiplier. Intuitively, above the Samuelson level, public consumption has no value, but it crowds out private consumption; therefore, it cannot be optimal to provide more public consumption than the Samuelson level.

But for any $\sigma > 0$, public services have some value, and optimal stimulus spending is nonzero as long as the economy is inefficient and the multiplier is nonzero. This result clarifies the link between usefulness of public expenditure and optimal stimulus spending. A concern of stimulus skeptics is that additional public expenditure could be wasteful. It is true that when the elasticity of substitution between public and private consumption is zero, public expenditure should remain at the Samuelson level. But in the more realistic case where the elasticity of substitution is positive, some stimulus spending remains desirable in slumps.

17.3.8. Stabilization from optimal fiscal policy

The general pattern is that optimal public expenditure deviates from the Samuelson rule to partially fill the initial slack gap. To understand these results, imagine that public expenditure satisfies the Samuelson rule, the fiscal multiplier is positive, and slack is inefficiently high. Keeping total consumption constant, increasing public consumption reduces private consumption one-for-one. Since the marginal utilities of public and private consumption are equal at the Samuelson rule, the increase in public consumption has no first-order effect on welfare so far. Now, since the fiscal multiplier is positive, increasing public consumption lowers slack; and since slack is inefficiently high, reducing slack raises total consumption. Hence, through its effect on slack, the increase in public consumption raises welfare. It is therefore optimal to increase public consumption above the Samuelson rule, and thus reduce the slack gap.

Why is it not optimal to completely fill the slack gap? If the government did that, we would reach a situation where increasing public consumption reduces private consumption one-for-one (since crowding out is one-for-one when the slack gap is zero), but extra public consumption is less valuable than extra private consumption (since public consumption is above the Samuelson level). The situation is clearly suboptimal: welfare can be improved by reducing public consumption.

Essentially, even though fiscal policy can be used for stabilization, it creates some distortion. When the government decides to increase spending by providing more public goods, it distorts the provision of goods in the economy away from private goods. While private goods can be substituted for public goods—public schools and hospitals can replace private schools and hospitals—they are generally not perfect substitutes. Due to these distortions, fiscal policy is not able to bring the economy all the way to the efficient allocation. Therefore, it is not optimal to stabilize the economy perfectly with fiscal policy alone: it is not optimal to close the slack gap entirely.

An interesting special case is that the final slack gap converges to 0 when the goods are indeed perfect substitutes: $\sigma \to \infty$. In this case, public consumption perfectly substitutes for private consumption, and optimal fiscal policy completely fills the slack gap. This result holds even if the multiplier is very small and public expenditure severely crowds out private consumption. Intuitively, public consumption and private consumption are interchangeable, so it is optimal to maximize total consumption, irrespective of its composition. This is achieved by completely filling the slack gap.

17.4. Stimulus spending during the Great Recession

We now complement our theoretical results with a numerical application to the Great Recession in the United States. We calibrate our sufficient-statistic formula and compute optimal fiscal policy at the onset of the recession. This exercise illustrates how much optimal public expenditure may deviate from the Samuelson rule.

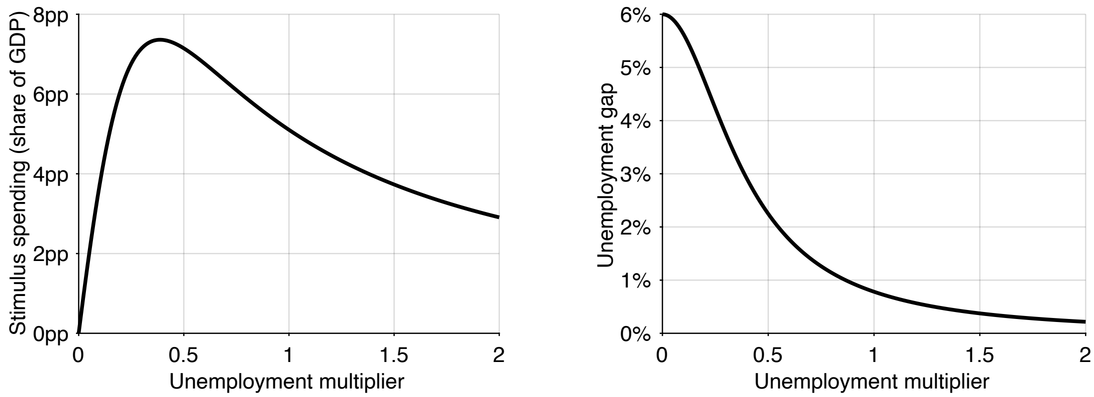
The motivation is that monetary policy hit the zero lower bound and became ineffective in 2009 (Board of Governors of the Federal Reserve System 2025), while the unemployment gap reached 5.9pp in the same year (figure 1.1). There was therefore a broad scope for fiscal policy in the form of public spending at the time. So according to our formula, how much should the US government have spent in 2009?

To calibrate the model, we need an empirical measure of slack. As discussed earlier, data to measure aggregate slack are not readily available, so we use the labor market's unemployment rate as our measure of slack u. Accordingly, we use the FERU as our measure of efficient slack u^*. For the fiscal multiplier, we use the effect of public spending on unemployment. For consistency, we use public and private employment as our measures of public and private consumption g and c.

17.4.1. Calibration of the sufficient statistics

We set the Samuelson level of spending to $(g/c)^* = 19.1\%$. To construct g/c, we set g to the seasonally adjusted employment level in the government industry and c to the seasonally adjusted employment level in the private industry. Both series are constructed by the BLS (2026a,b) from the Current Employment Statistics (CES) survey. Between 1939 and 2024, the average public expenditure satisfies $g/c = 19.1\%$, which we assume to reflect the Samuelson level.

To estimate the elasticity of substitution between public and private consumption, the empirical strategy is to isolate variations in the ratio of public-consumption price to private-consumption price and to assess their impact on the ratio of public consumption to private consumption. For example, if the consumption ratio stays constant in spite



A. Optimal stimulus spending

B. Optimal unemployment gap

FIGURE 17.1. Optimal fiscal policy for an initial unemployment gap of 6pp

Optimal fiscal policy is given by the explicit sufficient-statistic formula (17.16). The optimal unemployment gap is given by formula (17.17).

of secular variations in the price ratio, then the elasticity of substitution is about one. Applying this approach to US data, Amano and Wirjanto (1997, 1998) estimate elasticities of 0.9 and 1.56. Hence, we pick a midrange value for the elasticity of substitution: $\epsilon = 1$.

Next, we determine plausible values for the fiscal multiplier. The fiscal multiplier is estimated by measuring the percentage-point change in the unemployment rate when public expenditure increases by one percent of GDP. Monacelli, Perotti, and Trigari (2010, pp. 533–536) estimate a structural vector autoregression on US data and find fiscal multipliers between 0.2 and 0.6. Ramey (2013, pp. 40–42) also estimates structural vector autoregressions on US data with various identification schemes and sample periods. She finds fiscal multipliers between 0.2 and 0.5, except in one specification where the multiplier is 1. In sum, the average fiscal multiplier is estimated to be in the 0.2–1 range.

The multiplier entering our formula could be larger if multipliers are larger when slack is higher, as we find in slackish models (chapters 8, 13, and 15). To account for the uncertainty about the exact value of the multiplier in bad times, we compute optimal fiscal policy for a range of fiscal multipliers: $0 < m < 2$.

The last step before using the formulas is calibrating the constant z. We calibrate it just as above: we set $(g/c)^* = 19.1\%$, $(g/y)^* = 16.0\%$, $(c/y)^* = 1 - (g/y)^* = 84.0\%$, $u^* = 4\%$, which implies $z = 3.34$.

17.4.2. Calculation with sufficient-statistic formulas

Figures 17.1A and 17.1B display optimal fiscal policy as a deviation from Samuelson spending expressed as a share of GDP ($g/y - (g/y)^*$), and unemployment rate under optimal fiscal policy.[1] Several observations stand out.

First, even with small multipliers, optimal stimulus spending is significant. For example, take a tiny multiplier of 0.05: optimal stimulus spending is 2.0 points of GDP. For a small multiplier of 0.1, optimal stimulus spending is 3.7 points of GDP. And for a modest multiplier of 0.2, optimal stimulus spending is 6.1 points of GDP.

Second, the multiplier warranting the largest stimulus is fairly modest. The largest stimulus, which is 7.4 points of GDP, occurs at a multiplier of 0.39.

Third, optimal stimulus spending is the same for small and large multipliers. For instance, optimal stimulus spending is the same, at 3.7 points of GDP, for a small multiplier of 0.1 and a large multiplier of 1.5. Of course the resulting unemployment rates are very different.

Fourth, for small multipliers, the unemployment gap barely falls below the initial value of 6pp although optimal stimulus spending is large. For example, with a multiplier of 0.1, the unemployment gap only falls by 0.4pp to 5.6pp. This is because public spending has little effect on unemployment when the multiplier is small.

Fifth, with a multiplier above 1, optimal stimulus spending almost brings back unemployment to its efficient level, so the unemployment gap almost vanishes. Indeed, when the multiplier is above 1, the remaining unemployment gap is less than 0.8pp. And when the multiplier reaches 2, the remaining unemployment gap is just 0.2pp.

Finally, we calculate optimal fiscal policy at the onset of the Great Recession using midrange values for the fiscal multiplier of 0.5. Under this calibration, optimal stimulus spending is 7.1 points of GDP, and the unemployment gap is projected to fall by 3.8pp to 2.2pp. US GDP in 2009 is $14.5 trillion (BEA 2026a). Hence, optimal stimulus spending is $1.0 trillion. How does this optimal stimulus package compare to the actual stimulus package? The outlays from the American Recovery and Reinvestment Act, enacted into law in February 2009, are estimated to be $663 billion between 2009 and 2019 (CBO 2015, table 1). This is two-thirds of the optimal stimulus spending, but spread over a decade.

17.5. Robustness to distortionary taxation

In this chapter, we assume that taxation is nondistortionary: it does not affect workers' decisions to participate or not in the labor force. But in fact, the results are unaffected if taxation is distortionary, as Michaillat and Saez (2019) show.

Consider a model in which productive capacity is endogenous, as in chapter 7, and assume that public spending is financed by a linear income tax, which of course distorts participation to production and thus productive capacity. Following the argument by Michaillat and Saez (2019), we would find that while Samuelson spending is lower with the

[1]To obtain public spending as a share of GDP from (17.16), we use the identity $g/y = (g/c)/(1 + g/c)$.

linear income tax, the correction to the Samuelson rule remains the same. Accordingly, the sufficient-statistic formula developed in this chapter remains valid even with tax distortions.

Hence, the theory alleviates a common concern of stimulus skeptics. They worry that output is too low in slumps, and that increasing public spending would further reduce output through tax distortions. But our theory shows that if it is only because of tax distortions that public spending reduces output, then stimulus spending should be positive in slumps. Indeed, public spending affects output through two channels: the slack channel (public spending reduces slack) and the aggregate-supply channel (more public spending leads to higher taxes, which reduces productive capacity). Then, starting from the Samuelson rule, a small increase in public spending reduces slack, reduces aggregate supply, and increases public consumption, which are all good for welfare; but it reduces output and thus private consumption, which is bad for welfare. By construction, at the Samuelson rule, the cost of lower private consumption offsets the benefits of higher public consumption and lower aggregate supply; the only remaining effect on welfare is the positive effect from lower slack. Therefore, increasing public spending above the Samuelson rule is desirable. The key is that the Samuelson rule takes into account the negative welfare effect caused by the reduction in output stemming from lower aggregate supply, but not the positive welfare effect caused by the increase in output stemming from lower slack.

17.6. Summary

In this chapter we studied how fiscal policy should be adjusted when slack is inefficient. We found that optimal public spending deviates from the Samuelson rule to reduce, but not eliminate, the slack gap. The amplitude of the deviation—which corresponds to the amount of stimulus spending—depends on four sufficient statistics: slack gap, fiscal multiplier, elasticity of substitution between public and private goods, and Beveridge curvature. Since the slack gap is countercyclical, optimal public spending is also countercyclical. That is, the government should spend more in bad times and less in good times.

In the chapter we highlight three main results. The first is that when the fiscal multiplier and slack gap are positive, optimal stimulus spending is positive. This is because an increase in public spending reduces slack, which is inefficiently high, so it generates a positive effect on welfare that is not accounted for by the Samuelson rule and that warrants raising public spending above the Samuelson rule. The second result is that optimal stimulus spending is zero for a zero multiplier, increasing in the multiplier for small multipliers, maximized for a moderate multiplier, and decreasing in the multiplier for larger multipliers. The monotonicity breaks down because when the multiplier is large, it is optimal to fill the slack gap nearly entirely, and then less spending is required to fill the gap when the multiplier is larger. The third result is that optimal stimulus spending is increasing in the elasticity of substitution between public and private consumption. This result is natural: a higher elasticity of substitution means that extra public goods are more valuable, making stimulus spending more desirable.

Given that it is optimal to adjust public spending to the prevailing slack gap, it would be beneficial to have an institution that constantly monitors and adjusts public spending, just like there are institutions in place to constantly adjust monetary policy. In the United States, the Federal Reserve constantly monitors the economy so that they can respond quickly to fluctuations in the business cycle. On the other hand, fiscal policy is determined by the Congress and is heavily influenced by political factors. This makes it harder for the legislative branch to act quickly in stabilizing the economy—although fiscal policy, just like monetary policy, should respond to any deviation from full employment. It would be beneficial to develop an agency that adjusts fiscal policy around some average level determined by Congress, so as to stabilize the economy better. The average level of spending ensures that public goods are provided at an appropriate level. The cyclical adjustments allow public spending to pick up the slack in private spending in bad times, and to make room for private spending in good times, which stabilizes the amount of employment and consumption over the business cycle (even though the public/private mix of consumption varies slightly).

CHAPTER 18.

Recession detection

So far in this book, we have used vacancy and unemployment data to compute the tightness and unemployment gaps, which are central determinants of optimal stabilization policies. These optimal policies respond to the unemployment gap to try to keep the economy as close as possible to full employment.

However, recessions often lead to rapid changes in slack, and require rapid and sizable policy adjustments. It would therefore be useful to policymakers if recessions could be detected as early as possible. It turns out, as we explain in this chapter, that unemployment and vacancy data are a powerful tool to nowcast recessions. In that way, the vacancy-unemployment combination has not only normative power but some predictive power too: it can signal early that the economy is about to slump.

In this chapter we develop a threshold rule to detect recessions. The rule is inspired by the Michez rule proposed by Michaillat and Saez (2025), in that it relies on a recession indicator that is the minimum of an unemployment indicator and a vacancy indicator. However, the rule proposed here improves the Michez rule by choosing optimized ways to smooth the data and detect turning points. For convenience, this new rule is referred to as the Michez+ rule.

The Michez+ rule leverages two insights from chapter 15. First, recessions are mostly driven by drops in aggregate demand, which trigger drops in labor demand. Second, such shocks produce negative comovements between unemployment rate and vacancy rate as the economy moves along the Beveridge curve. Therefore, a typical recession features both a drop in vacancy rate and a rise in unemployment rate. By combining data on unemployment and job vacancies—two noisy but independent measures of aggregate demand—we obtain a clearer signal of latent aggregate demand than recession rules relying only on the unemployment rate, such as the Sahm rule.

18.1. Labor market data and their availability

To detect recessions in real time, the Michez+ rule relies on monthly data. Fortunately, the unemployment and vacancy data presented in chapter 2 are available at monthly frequency. Accordingly, we use these monthly data both for assessing the historical performance of the Michez+ rule and for detecting recessions in real time.

Moving forward, the unemployment and vacancy data required to apply the Michez+ rule in any given month are released in the first week of the following month, usually on a Tuesday for the JOLTS data and on a Friday for the CPS data (BLS 2024b).[1] So the rule can be applied in real time.

We should be conscious of the fact that the value of the recession indicator constructed in real time might not be its final value because the unemployment and vacancy data are revised after their initial release. The number of job openings released by the BLS (2025i) is preliminary and updated one month after its initial release, to incorporate additional survey responses received from businesses and government agencies and from the recalculation of seasonal factors (BLS 2024a). Additionally, the BLS revises the prior five years of CPS and JOLTS data each year at the beginning of January, to account for revisions to seasonal factors, population estimates, and employment estimates (BLS 2024a, 2025s). Yet, in practice, revisions to labor market data are generally minimal, especially compared to GDP revisions, so the information provided in real time is almost indistinguishable from the information provided in the final version (Crump, Giannone, and Lucca 2020).

18.2. Construction of the Michez+ rule

We start by constructing the recession indicator used by the Michez+ rule. We begin by combining unemployment and vacancy data into a single recession indicator. We then select a threshold that allows the rule to detect recessions in a timely manner and with accuracy.

18.2.1. Unemployment and vacancy indicators

First, we smooth the monthly unemployment and vacancy series to reduce their noisiness. To smooth them, we use an exponentially weighted moving average, which is defined

[1] Recall that we shift forward by one month the number of job openings reported in the JOLTS. Hence we have access to the vacancy and unemployment rates required to apply the Michez+ rule in the same week, as soon as the month is over.

recursively by:

$$\bar{u}(t) = 0.4 \times u(t) + 0.6 \times \bar{u}(t-1) \tag{18.1}$$

$$\bar{v}(t) = 0.4 \times v(t) + 0.6 \times \bar{v}(t-1), \tag{18.2}$$

where $u(t)$ and $v(t)$ are the raw monthly series, $\bar{u}(t)$ and $\bar{v}(t)$ are the smoothed series, and the initial conditions simply are $\bar{u}(0) = u(0)$ and $\bar{v}(0) = v(0)$. The amount of smoothing is governed by the scalar 0.4. (A value closer to 1 means less smoothing; a value closer to 0 means more smoothing.) The smooth series are displayed in figure 18.1A.

To detect turning points in the unemployment rate, we first take the 9-month minimum of the unemployment rate:

$$u^{\min}(t) = \min_{0 \le k \le 9} \bar{u}(t-k).$$

Then, the increase in unemployment rate from the turning point is computed as

$$\hat{u}(t) = \bar{u}(t) - u^{\min}(t). \tag{18.3}$$

We proceed analogously to determine turning points in the vacancy rate. We first take the 9-month maximum of the vacancy rate:

$$v^{\max}(t) = \max_{0 \le k \le 9} \bar{v}(t-k).$$

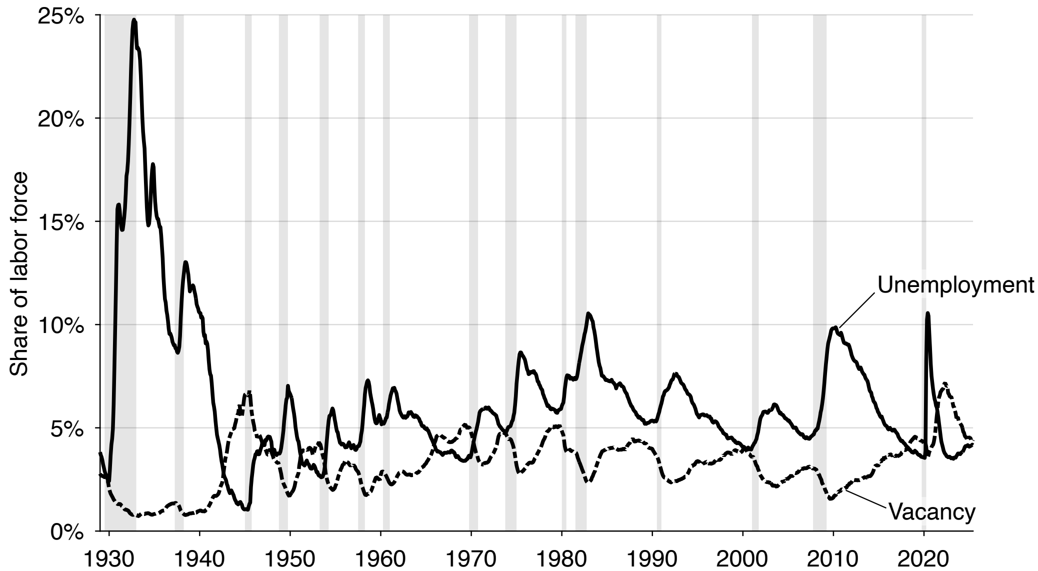
Then, the decrease in vacancy rate from the turning point is computed as

$$\hat{v}(t) = v^{\max}(t) - \bar{v}(t). \tag{18.4}$$

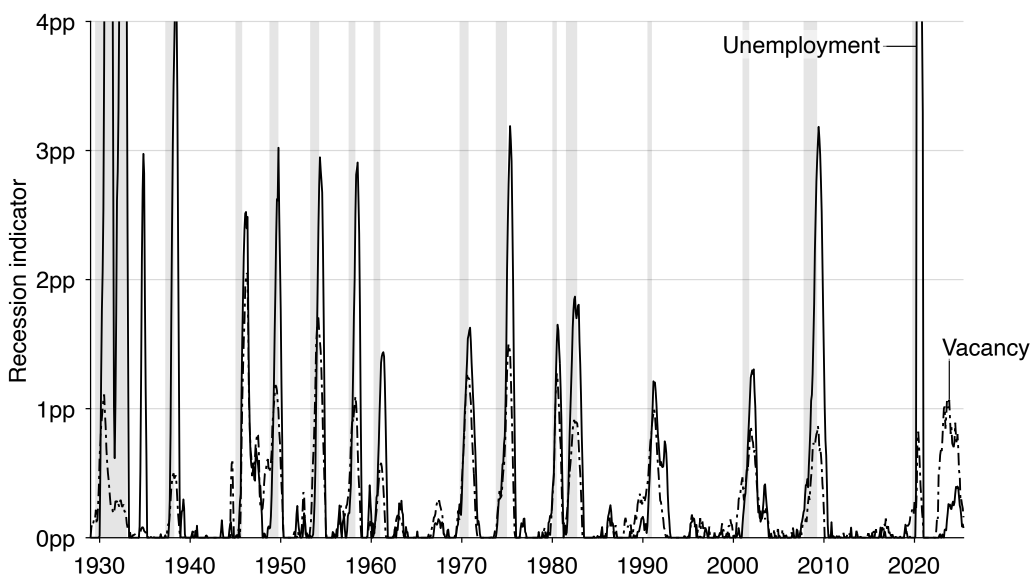
The unemployment and vacancy indicators $\hat{u}(t)$ and $\hat{v}(t)$ are displayed in figure 18.1B. The amount of smoothing and construction of the turning points differs from the approach followed in the Sahm and Michez rules: there the data are smoothed with a 3-month moving average and turning points are computed based on a 12-month window (Sahm 2019; Michaillat and Saez 2025). This departure might seem arbitrary, but processing the data in that specific way leads to one of the best possible recession detection rules (Michaillat 2025). We will verify the performance of the rule in the rest of the chapter.

18.2.2. Minimum indicator

The final step is to combine the unemployment and vacancy indicators that we have just constructed. To obtain a less noisy signal of recessions than either indicator, we take the minimum of these two indicators. Formally, the minimum indicator is constructed as



A. Smooth unemployment and vacancy rates



B. Unemployment and vacancy indicators

FIGURE 18.1. Construction of the Michez+ recession indicator in the United States, 1929–2024

The monthly unemployment rate comes from figure 2.1 while the monthly vacancy rate comes from figure 2.5. The unemployment rate is smoothed according to (18.1) and the vacancy rate is smoothed according to (18.2). The unemployment indicator is constructed from (18.3) and the vacancy indicator is constructed from (18.4).

follows:

$$m(t) = \min(\hat{u}(t), \hat{v}(t)). \tag{18.5}$$

The minimum indicator $m(t)$ is plotted in figure 18.2A. The indicator is zero when either the unemployment rate is trending down or the vacancy rate is trending up. It is positive when both the unemployment rate rises and vacancy rate declines.

Figure 18.1B shows the advantage of taking the minimum of the vacancy and unemployment indicators. Each indicator mostly spikes during recessions, but they also have some uninformative blips. Given that the blips of the unemployment and vacancy indicators do not occur at the same time, taking the minimum of the two indicators smooths out the blips and gives us a less noisy, more accurate recession indicator.

Let us look at a few examples. A striking situation occurred in 1934: the unemployment indicator peaked at 2.97pp, although no recession occurred then. The vacancy indicator is not subject to that blip: it remained close to zero. By construction, the minimum indicator is not subject to the blip either. This situation occurred again in 1959. The unemployment indicator reached 0.41pp, but there was no recession. Thankfully the vacancy indicator remained much lower during that episode, so the minimum indicator was only subject to a minor blip in 1959.

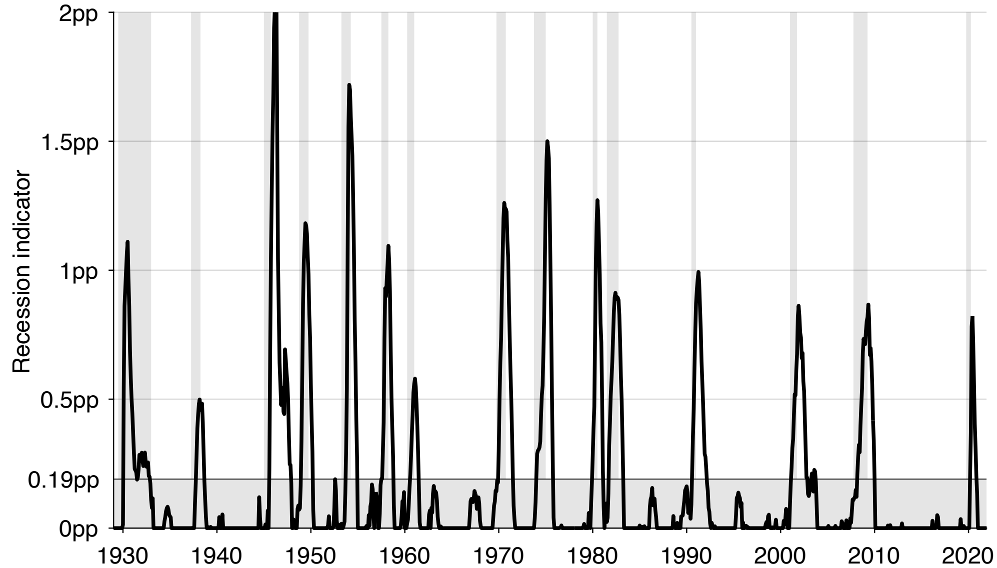
Sometimes, the situation is reversed: the vacancy indicator is subject to an uninformative blip that does not appear in the unemployment indicator. For instance, the vacancy indicator spiked at 0.59pp in 1944 while no recession happened then. Because the unemployment indicator did not rise much at the time, the minimum indicator did not rise much either. In 1967, the vacancy indicator spiked again while no recession was officially identified then—but the unemployment indicator remained low, keeping the minimum indicator low too.

18.2.3. Threshold and detection methodology

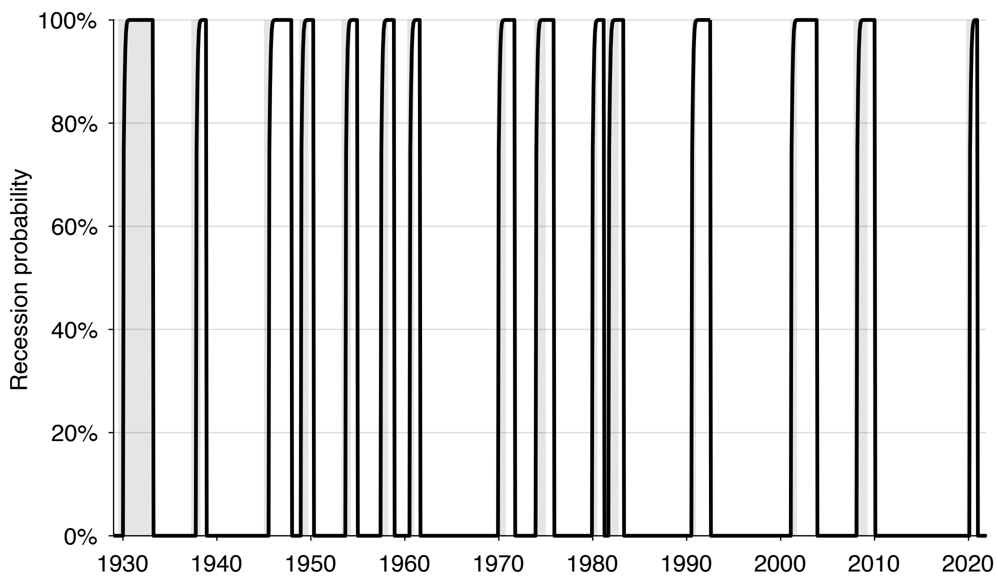
To complete the construction of the Michez+ rule, we must specify a recession threshold. The threshold is set to 0.19 pp. This threshold produces a detection rule that makes no mistake over the entire 1929–2021 period, and leads to early and accurate detection of recessions (Michaillat 2025).

Figure 18.2A illustrates why the threshold value works well. A lower threshold would create false positives, for instance in 1952. A higher threshold would just increase the detection delay. Hence, 0.19pp represents the optimal threshold for this indicator.

The Michez+ rule defines recessions as episodes when the indicator crosses the threshold from below, provided the economy was in expansion immediately beforehand. The economy remains in recession until it returns to expansion, which occurs once the indi-



A. Recession indicator and threshold



B. Recession probability

FIGURE 18.2. Michez+ recession rule in the United States, 1929–2021

The recession indicator is constructed from (18.5): it is the minimum of the unemployment and vacancy indicators displayed in figure 18.1B. The recession probability is computed from (18.6).

cator falls back to 0. If the indicator crosses the threshold several times after the initial crossing but before the return to expansion, only the first crossing is recorded as the recession's start.

A simpler alternative would be to label recessions as all periods when the indicator exceeds the 0.19 pp threshold—an approach used by the Sahm and Michez rules. This approach, however, misclassifies temporary rebounds at the end of recessions. As downturns fade, the unemployment rate rises more slowly and vacancies stop falling, causing the indicator to decline toward the threshold. During this adjustment, it may briefly dip below and then climb above the threshold before finally settling below it again. Such blips occurred after the Great Depression, after World War 2, and after the dot-com downturn (figure 18.2A). These transient fluctuations do not signal new recessions. To address such misclassifications, the Michez+ rule requires that the economy is in expansion before entering a new recession, and that a return to expansion is confirmed only once the indicator falls to 0. This slightly more complex procedure yields a markedly less noisy and more reliable recession rule.

18.3. Historical evaluation of the Michez+ rule (1929–2021)

We now evaluate the performance of the Michez+ rule between 1929 and 2021. We stop the evaluation at the end of 2021 because it is too early to say if and when a recession started after that. We focus on false positives and negatives as well as the timeliness of detection.

18.3.1. No false positives or negatives

The Michez+ rule has a perfect track record between January 1929 and December 2021 (figure 18.2A). First, the Michez+ rule does not produce false negatives: it does not fail to detect existing recessions. Indeed, the rule detects the 15 recessions that occurred between 1929 and 2021. This is because the recession indicator peaks well above the threshold of 0.19pp during each of the recessions.

Second, the Michez+ rule does not produce false positives: it does not detect nonexistent recessions. This is because the recession indicator always remains below the 0.19pp threshold outside of recessionary periods.

In sum, the Michez+ rule identifies all the recessions that occurred between 1929 and 2021, without any false alarms. This is of course due to how the recession rule is constructed: the threshold is chosen to be high enough to avoid false positives but low enough to avoid false negatives.

18.3.2. Timeliness of detection

The Michez+ rule detects the 15 recessions that occurred between 1929 and 2021 without any false alarms. This is a good first step, but it is not sufficient to be sure that we have a good rule to detect recessions—there are many, many recession rules that detect exactly 15 recessions over the period (Michaillat 2025).

However, the Michez+ rule also detects recessions systematically early and accurately. In fact, we picked the rule from the high-precision segment of the anticipation-precision frontier created in Michaillat (2025), so it is one of the most precise recession rules among all fast rules.

The dates at which the Michez+ rule detects US recessions are reported in table 18.1, together with the mean, standard deviation, minimum, and maximum of detection errors. How is the performance of the rule assessed? For each recession $j = 1, 2, \dots, 15$, we compute the detection error, which is the difference between the date when the recession officially started, $s(j)$, and the date when the rule first detected the recession, $d(j)$: $e(j) = d(j) - s(j)$. This is doable because the rule detects the correct number of recessions, so there are as many start dates as detection dates. If $e(j) > 0$, recession j is detected with some delay. If instead $e(j) < 0$, the recession is detected with some anticipation.

We then compute two performance measures over the training period. The first measure is the mean of the detection errors, given by

$$\mu = \frac{\sum_{j=1}^{15} e(j)}{15}.$$

The second measure is the standard deviation of the detection errors, given by

$$\sigma = \sqrt{\frac{\sum_{j=1}^{15} \left[e(j) - \mu\right]^2}{15}}.$$

We observe that the Michez+ rule always detects recession starts within 5 months of their actual occurrence. It takes 5 months to detect the first three recessions in the sample (1929, 1937, 1945). But after that, the rule performs better, and never takes that long to detect a recession. The next longest delay is 3 months, for the 1960 recession. After that the longest detection delay is only 2 months, for the 1981 and 2008 recessions.

The Michez+ rule generally detects recessions with some delay: the average detection delay over the entire period is 1.5 months—so about 6 weeks. The detection delay is cut in half over the 1960–2021 period: the average detection delay then is 0.7 months—about 3 weeks. It is not surprising that the Michez+ rule often detects recessions after their official start dates because the official dates are identified retrospectively (NBER 2021). Unlike what the Michez+ rule aims to do, the NBER identifies recessions with hindsight, not in real time.

However, a few recessions are detected 1 or 2 months before they have officially started. For instance, the 1957 recession was detected in July, although the NBER determined that it started in September. The 1980 and 2001 recessions are also detected 1 month before their official starts.

Furthermore, the Michez+ rule is able to detect recessions significantly before the official dates are announced by the NBER (2021). For instance, the NBER waited until December 2008 to announce that the previous business cycle peak had occurred in December 2007, and therefore that the Great Recession had started in January 2008. By contrast, the Michez+ rule detects the Great Recession in March 2008, so 9 months before the official NBER announcement. On average, between 1979 and 2021, the NBER announces recession starts 7.3 months after a recession's onset. Over the same period, the Michez+ rule detects recession starts with a delay of 0.3 months only. So on average the Michez+ rule detects recessions 7 months faster than the NBER.

We have just seen that the maximum detection delays occurred with the first 3 recessions in the sample, which were the hardest to detect for the Michez+ rule. It appears more generally that the rule does better in the recent part of the sample (1960–2021) than the old part (1929–1959). The mean detection delay is 2.7 months over 1929–1959 but only 0.7 months over 1960–2021. The standard deviation of the detection error is 2.7 months over 1929–1959 and only 2.2 months over 1960–2021. So the rule is able to detect recessions earlier and more precisely in the last two-thirds of the sample. A possible explanation is that the unemployment and vacancy data are noisier in the first third of the sample. This noisiness might be explained by the facts that the unemployment data come from a patchwork of sources before 1948 (when the BLS started collecting data via the CPS), and that the vacancy data were collected by private entities (MetLife and Conference Board) and not by the BLS until 2000.

18.3.3. Recession probability

We can also use the structure of detection errors to compute a recession probability from the Michez+ rule. If for instance the rule is on average exactly on time, the recession has started with 50% probability when the rule is triggered—assuming a symmetric detection error. If the rule is early on average, the probability is less than 50%. If the rule is late on average, the probability is more than 50%, and so on. In the months following detection, the probability converges to 1 along the detection error's cumulative distribution function.

More formally, each detection generates a detection date d and detection error e. Then, it is easy to compute the probability that a new recession's start time, s, truly occurred before time t_1, given that the rule detected a recession at time $d = t_0 \leq t_1$:

$$P(t_1) = \mathbb{P}(s < t_1 \mid d = t_0, \mu, \sigma) = \mathbb{P}(d - s > t_0 - t_1 \mid \mu, \sigma) = \mathbb{P}(e > t_0 - t_1 \mid \mu, \sigma).$$

For convenience, we assume that the detection error is normally distributed with mean μ and standard deviation σ. We therefore get

$$P(t_1) = \mathbb{P}\left(\frac{e - \mu}{\sigma} > \frac{t_0 - t_1 - \mu}{\sigma}\right) = 1 - \Phi\left(\frac{t_0 - t_1 - \mu}{\sigma}\right),$$

where Φ is the cumulative distribution function of the standard normal distribution. Hence, the probability that a recession has started at time t if a recession is detected at d:

$$P(t) = \Phi\left(\frac{t + \mu - d}{\sigma}\right), \tag{18.6}$$

as long as the economy remains in a recession state. When the rule signals that the economy is in expansion, we set the probability to $P(t) = 0$. The probability $P(t)$ is only positive when the rule signals that the economy is in recession. The value d used in the probability is then the most recent recession detection date.

Figure 18.2B shows the recession probability given by the Michez+ rule on the period 1929–2021, on which the rule was selected. In each of the 15 recessions, the recession probability rapidly rises at the onset of the recession and quickly reaches 1. The probability then starts declining right after or somewhat after the recession has ended.

18.4. Application of the Michez+ rule to the current situation (2022–2024)

Finally, we apply the Michez+ rule to contemporary data to assess the current risk of recession in the United States. Has the US economy entered a recession between January 2022 and December 2024?

The recession indicator crossed the 0.19pp threshold in October 2023, so the Michez+ rule detected a recession at that time (figure 18.3A). After crossing the threshold, the recession indicator continued climbing. In October 2024, the indicator attained 0.40pp. The indicator then fell somewhat, as the labor market stabilized at the end of 2024.

Finally, we apply formula (18.6) to current data to assess recession risk in real time (figure 18.3B). The recession probability turned positive in October 2023, when the recession was detected, and kept rising after that. Given that the standard deviation of detection errors is only 2.2 months, the probability that the detection error is more than 5 months is tiny (and indeed in the 15 past recessions the Michez+ rule has never been more than 5 months late). Accordingly, the recession probability reaches almost 1 (99.8%) in March 2024 and remained there afterwards.

TABLE 18.1. Detection of US recession starts by the Michez+ rule, 1929–2021

		Detection date					
Official start date		Michez+		Michez		Sahm	
Year	Month	Year	Month	Year	Month	Year	Month
1929	September	1930	February	1930	February		
1937	June	1937	November	1937	December		
1945	March	1945	August	1945	September		
1948	December	1949	January	1949	January		
1953	August	1953	October	1953	October		
1957	September	1957	July	1957	July		
1960	May	1960	August	1960	August	1960	October
1970	January	1970	January	1970	February	1970	March
1973	December	1974	January	1974	February	1974	July
1980	February	1980	January	1980	January	1980	February
1981	August	1981	October	1981	October	1981	November
1990	August	1990	August	1990	September	1990	October
2001	April	2001	March	2001	March	2001	July
2008	January	2008	March	2008	April	2008	February
2020	March	2020	March	2020	April	2020	April
Detection error for 1929–2021							
Mean:		1.5 months		1.9 months			
Minimum:		–2 months		–2 months			
Maximum:		5 months		6 months			
Standard deviation:		2.2 months		2.3 months			
Detection error for 1960–2021							
Mean:		0.7 months		1.2 months		2.7 months	
Minimum:		–1 month		–1 month		0 months	
Maximum:		3 months		3 months		7 months	
Standard deviation:		1.3 months		1.4 months		2.1 months	

Notes. Official recession start dates are provided by the NBER (2023). The Michez+ rule signals a recession when the indicator displayed in figure 18.2A crosses the threshold of 0.19pp. The detection dates for the Michez and Sahm rules are provided by Michaillat and Saez (2025, tables 1 and 2).

The Michez+ rule's detection of a recession starting in October 2023 has not yet been confirmed by the NBER. As of late 2025, available GDP figures and equity markets remain strong, perhaps buoyed by developments in artificial intelligence. However, the labor market indicators that form the basis of the rule tell a different story. The rule's historical track record suggests that labor market deterioration of this magnitude and persistence has always coincided with recessions. Whether the NBER will eventually backdate a recession to late 2023 or 2024, or whether this represents the rule's first false positive in its 95-year history, remains to be determined.

18.5. Comparison with the Sahm and Michez rules

To conclude this chapter, we briefly compare the performance of the Michez+ rule to that of the Michez rule, on which it is based, and that of the Sahm rule, which is the most famous slack-based recession detection rule.

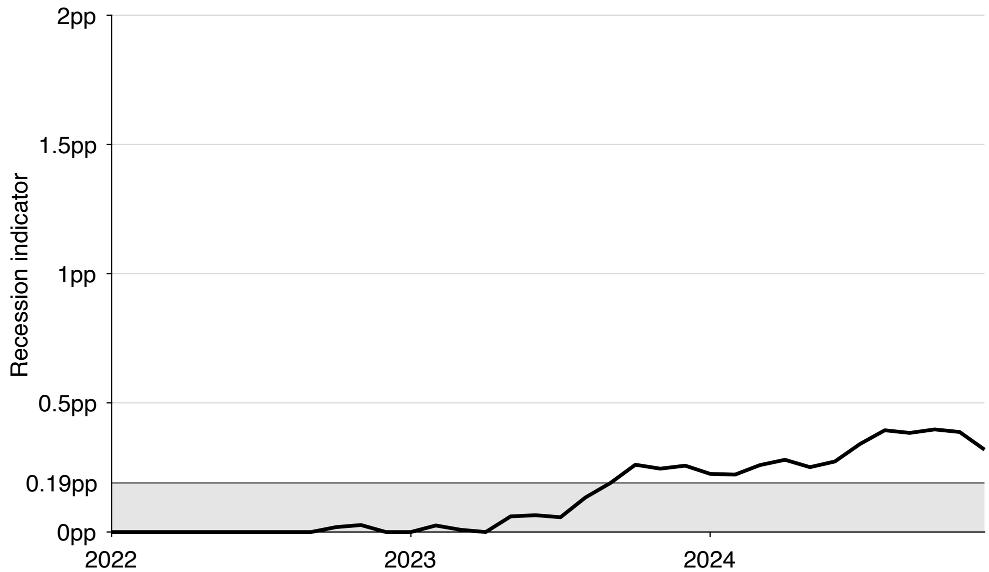
18.5.1. Comparison with the Michez rule over 1929–2021

The Michez+ and Michez rules both perfectly detect the 15 recessions that occurred between 1929 and 2021 (table 18.1). However, the Michez+ rule detects these recessions slightly faster than the Michez rule. On average, the Michez+ rule detects recessions 1.5 months after their official start dates, whereas the Michez rule takes on average 1.9 months to detect recessions. Furthermore, the Michez+ rule detects all 15 recessions within 5 months of their official starts, while the Michez rule sometimes takes 6 months to detect them. The Michez+ rule also outperforms the Michez rule on the recent part of the data (1960–2021): on the more recent segment, the mean detection error is 1.2 months for the Michez rule but only 0.7 months for the Michez+ rule.

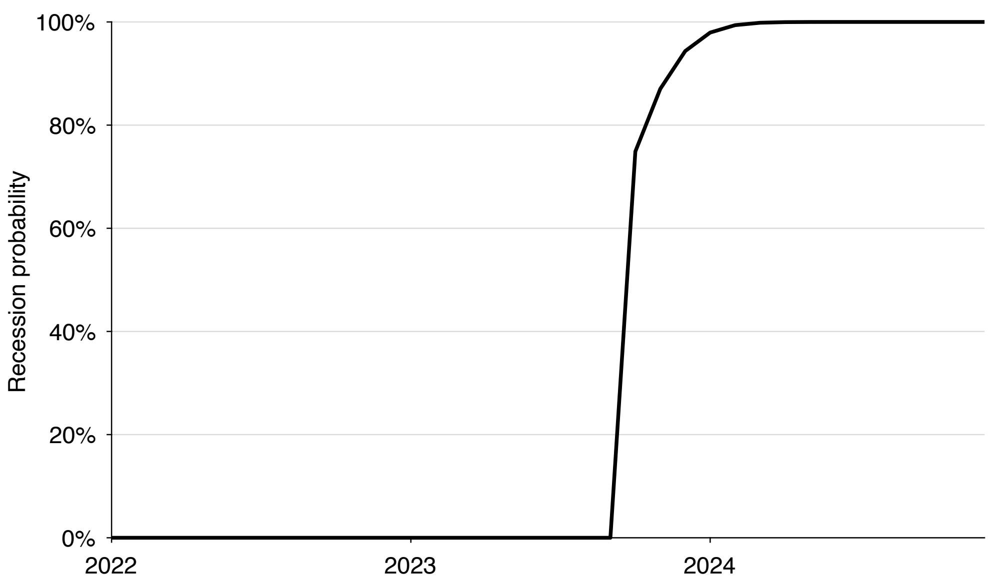
In fact, the Michez+ rule detects each and every recession as fast or faster than the Michez rule. For 8 recessions, the detection occurs in the same month; but for 7 recessions (1937, 1945, 1970, 1974, 1990, 2008, 2020), the Michez+ rule detects the recession start 1 month earlier than the Michez rule.

In addition to being a bit faster than the Michez rule, the new Michez+ rule is a bit more precise: the standard deviation of detection errors is 2.3 months for the Michez rule but only 2.2 months for the Michez+ rule.

It is not surprising that the Michez+ rule does better than the Michez rule, since it was constructed to reduce the mean and standard deviation of the detection error as much as possible. Compared to the Michez rule, the smoothing method and selection of turning points have been optimized.



A. Recession indicator and threshold



B. Recession probability

FIGURE 18.3. Michez+ recession rule in the United States, 2022–2024

The recession indicator is constructed from (18.5): it is the minimum of the unemployment and vacancy indicators displayed in figure 18.1B. The recession probability is computed from (18.6).

18.5.2. Comparison with the Sahm rule over 1960–2021

The Sahm rule only works between 1960 and 2021—it produces false positives before that (Michaillat and Saez 2025). Hence, we focus on that period for the comparison with the Michez+ rule.

The Michez+ rule detects recessions markedly faster than the Sahm rule (table 18.1). On average, the Michez+ rule detects recessions 0.7 months after their official start dates, whereas the Sahm rule takes on average 2 additional months, so a total of 2.7 months, to detect recessions. Furthermore, the Michez+ rule detects all 9 recessions within 3 months of their official starts, while the Sahm rule sometimes takes 7 months to detect them.

In fact, the Michez+ rule detects each recession at least 1 month faster than the Sahm rule, with the exception of the Great Recession, which it detects 1 month more slowly. In 2008, the Michez+ rule detected the recession in March, whereas the Sahm rule was able to detect it in February. The slight delay is because job vacancies took some time to drop at the onset of the Great Recession (figure 18.1).

In addition to being faster than the Sahm rule, the Michez+ rule is much more precise: the standard deviation of detection errors is 1.3 months for the Michez+ rule while it is 2.1 months for the Sahm rule.

It is not surprising that the Michez+ rule does better than the Sahm rule. The unemployment rate—which the Sahm rule uses—is only a noisy measure of the latent state of the economy. We know that recessions feature not only an increase in the unemployment rate but also a decline in the vacancy rate as the economy moves along the Beveridge curve. These joint movements are clearly visible on figure 18.1. Therefore, by combining data on unemployment and job vacancies, we obtain a recession indicator that is less noisy than unemployment-only indicators, such as the Sahm indicator. Thanks to the reduced noisiness, the detection threshold can be lowered once both unemployment and vacancy data are used, and recessions can be detected earlier and more precisely.

18.6. Summary

In this chapter, we develop and evaluate the Michez+ rule, a new tool for real-time recession detection based on unemployment and vacancy data. Building on the insights of previous chapters, the Michez+ rule exploits the negative comovement between unemployment and vacancies along the Beveridge curve—a defining feature of recessions driven by aggregate demand shocks. By combining both indicators into a single minimum-based measure, the Michez+ rule extracts a cleaner and timelier signal of economic downturns than rules relying on either variable alone. Its construction involves optimized smoothing and turning-point detection procedures, designed to detect recessions as early and accurately as possible.

Between 1929 and 2021, the Michez+ rule performs exceptionally well. It identifies all 15 recessions recognized by the NBER without any false positives and with an average delay of only 1.5 months—reduced to 0.7 months since 1960. Compared with the Sahm and Michez rules, the Michez+ rule is both faster and more precise, consistently detecting recessions earlier and with lower variance in detection errors.

The NBER Business Cycle Dating Committee has existed since 1979. During the Committee's existence, it has taken on average 7.3 months to announce recession starts. By contrast, over the 1979–2021 period, the Michez+ rule only takes 0.3 months to identify recessions after they have started. Hence, the rule detects recessions roughly 7 months earlier than the NBER's official announcements. The Michez+ rule therefore offers policymakers and analysts a practical, transparent, and near real-time instrument to identify recessions and adjust stabilization policies accordingly.

Applied to current data, the Michez+ rule indicates that the US economy entered a recession in late 2023, with a recession probability approaching 100% by early 2024. Regardless of the eventual NBER call, the labor market has been deteriorating at a recession-like pace since late 2023—which is what the rule is detecting.

PART VI.

Conclusion

CHAPTER 19.

Ten takeaways

Modern business cycle models focus on prices and quantities, overlooking a key feature of real economies: pervasive slack. Accounting for slack is essential to understanding and managing business cycles. This is what we endeavored to do in this book. In this chapter, we review what we have learned by isolating ten takeaways for business cycle modeling and policy.

19.1. Slack is everywhere

Our first takeaway is that slack is everywhere. In almost all markets, it takes time and effort to sell and buy goods, so some goods remain unsold at all times. The underlying cause for pervasive slack is that in reality there is no centralized marketplace and goods and tastes differ, so buyers must spend time and effort to find the goods they desire. A manifestation of these selling and buying difficulties is that buyers and sellers are always relieved when they are able to trade. Another manifestation is the prevalence of long-term trading relationships: once a buyer and seller have found each other, they aim to keep trading for as long as possible.

A corollary is that the Walrasian model—in which trading is seamless and slack does not exist—is inadequate to describe real-world markets.

19.2. Slack fluctuations are the business cycle

Our second takeaway is that fluctuations in slack account for the vast majority of business cycle fluctuations. Indeed, productive capacity is built from the existing capital stock, labor force, and existing technological knowledge, which are acyclical. So productive capacity moves very smoothly over time. GDP and consumption, on the other hand, are fluctuating in the short run. These fluctuations are largely explained by fluctuations in slack on the labor and product markets.

A corollary is that macroeconomic models without slack miss a central element of business cycles and will struggle to describe business cycle fluctuations realistically.

19.3. Markets equilibrate not only via prices but also via tightness

Our third takeaway is that markets equilibrate through both prices and tightness—two complementary adjustment margins. And if prices are particularly rigid, or fixed, markets operate entirely through tightness. It is therefore possible to identify demand and supply shocks by examining the correlation between quantity and tightness on any market: a positive correlation indicates a demand shock while a negative correlation indicates a supply shock.

A corollary is that there is much to gain by switching from supply-demand analysis in price-quantity planes to supply-demand analysis in tightness-quantity planes.

19.4. Demand matters mostly in slack markets and supply in tight markets

Our fourth takeaway is that demand matters mostly in slack markets and supply mostly in tight markets. This means that markets are state dependent: they operate differently when they are slack and tight. A policy that boosts demand effectively stimulates output in slack times but not in tight times; conversely a policy that boosts supply effectively stimulates output in tight times but not in slack times.

A corollary is that supply-side policies are generally ineffective in slack markets—for instance, policies that push unemployed workers to search harder for jobs in bad times do not lower unemployment much.

19.5. Markets are generally inefficient

Our fifth takeaway is that markets are generally inefficient—the efficient market allocation is unlikely to prevail in reality. The reason is that price norms do not generally guarantee efficient outcomes. Furthermore, the efficient tightness can be computed from a few sufficient statistics, so it is possible to assess in real time how far from efficiency a specific market is.

A corollary is that one cannot expect market economies to operate efficiently without some government intervention: inefficiency is a systematic outcome of market economies. This is a core departure from the Walrasian guarantee of efficient markets.

19.6. The US labor market has been excessively slack in the past century

Our sixth takeaway is that the US labor market has been excessively slack in the past century. We estimate that the efficient labor market tightness for the US economy is 1—corresponding to one vacant job per job seeker. In the past century the tightness has always been below 1, except in a few wartime episodes (World War 2, Korean War, Vietnam War) and around the pandemic. Moreover, tightness has fallen especially far below 1 in recessions. This finding implies that the US labor market has spent most of the past century at less than full employment, and that it has fallen especially far below full employment in recessions. The unemployment rate that corresponds to this full-employment tightness is the full-employment unemployment rate (FERU), given by $u^* = \sqrt{uv}$.

A corollary is that the Federal Reserve and federal government have fallen short of their full-employment mandate over the past century. The zero lower bound on nominal interest rates, pressure from the White House and Congress, and overly high unemployment targets might all have played a role.

19.7. US business cycles are mostly driven by aggregate demand shocks

Our seventh takeaway is that US business cycles are mostly driven by aggregate demand shocks. A clear manifestation of aggregate demand shocks is Okun's law: a negative relationship between detrended output and unemployment.

A corollary is that business cycle fluctuations can be stabilized with demand-side policies such as monetary and fiscal policy. For instance, reducing the federal funds rate lowers the real interest rate, which makes consumption more appealing than saving and boosts aggregate demand. A higher aggregate demand raises labor demand, which lowers unemployment. Fiscal policy in the form of public spending or public employment also stimulates labor demand and reduces unemployment.

19.8. Optimal monetary policy eliminates the slack gap

Our eighth takeaway is that monetary policy should set interest rates to eliminate the slack gap. This is true in a model in which inflation is fixed. But it is also true in models with endogenous inflation, as long as the inflation-slack coincidence holds—which seems realistic in the modern US economy. In such models, there is no trade-off between inflation and slack: lower slack leads to higher inflation, but when slack is efficient, inflation is on target. If the coincidence fails, it is not optimal to eliminate the slack gap, but the optimal policy still depends on the slack gap.

A corollary is that it is not enough for the Federal Reserve to follow a Taylor rule based on inflation. In fact the Federal Reserve does not have to adopt an interest-rate rule at all: an interest-rate peg is perfectly fine. The peg can be adjusted as new slack data become available.

19.9. Optimal fiscal policy departs from microeconomic principles to reduce the slack gap

Our ninth takeaway is that fiscal policy should depart from microeconomic principles to reduce the slack gap. For example, when public spending is not a perfect substitute for private spending, it is not optimal to use public spending to eliminate the slack gap. Instead, optimal public spending should be above the Samuelson rule in slack times to reduce—but not eliminate—the slack gap.

A corollary is that multipliers larger than 1 are not necessary to justify countercyclical public spending. Any positive multiplier justifies countercyclical public spending; the only justification for acyclical spending is a zero multiplier.

19.10. Recessions are detectable early from slack data

Our tenth and final takeaway is that recessions are detectable early using slack data—in particular unemployment and vacancy data. Fluctuations in slack are early indicators of recessions, so monitoring slack allows an early, proactive response to economic downturns.

A corollary is that it is difficult to detect recessions early by using only data on quantities (such as GDP or industrial production) and ignoring data on slack.

19.11. Some unconventional aspects of the theory of slack

The theory of slack developed in this book overturns several assumptions and implications of conventional macroeconomics. In reality, markets rarely clear: trading takes time, goods remain unsold, and job seekers remain unemployed for periods at a time. Slack is an inherent feature of market economies, and fluctuations in slack are the essence of business cycles. Because price rigidity is not a pathology but a ubiquitous feature of markets, markets equilibrate principally through tightness. Finally, free markets cannot be presumed efficient: efficiency requires the appropriate price norm, supporting an appropriate level of tightness. In fact, business cycle fluctuations are best understood as more or less severe departures from efficiency.

The theory of slack also modifies several principles of macroeconomic policy. Standard unemployment targets such as the NAIRU fail to capture full employment, and tend to be excessively high. Macroeconomic policies should be designed to bring slack closer to its efficient level. Monetary policy should target the slack gap rather than follow a Taylor rule, while fiscal policy should deviate from microeconomic principles to reduce the slack gap whenever monetary policy is constrained. Finally, because slack is strongly cyclical, unemployment and vacancy data provide valuable indicators of recessions.

Overall, the book replaces the price-centered, market-clearing paradigm with a slack-centered, matching-based view of the economy, where managing slack is the key to good policy.

CHAPTER 20.

Slack around the world

As we saw in chapter 2, by its free-market nature, the United States might be the economy least hospitable to slack, and yet slack is ubiquitous in the US economy. It is therefore natural to expect that other countries present at least as much slack as the United States—which implies that the book's theory of slack might be fruitful there too.

In this chapter we review worldwide evidence on slack. We find that slack is prevalent all over the world, in developed and developing countries. We then discuss how the theory of slack might help us better understand business cycles and development in other countries.

20.1. Prevalence of slack in other countries

In this section we review evidence that slack is present in labor and product markets all around the world.

20.1.1. Labor market slack in other countries

Unemployment is a universal problem, found in labor markets all around the world. For instance, across 73 developed and developing countries, in the years 2000 to 2003, the unemployment rate averages 8.7%, with a minimum value of 1.4% and a maxium value of 30.7% (Feldmann 2009, table 2). This is why it is so important to study unemployment, include it in our models, and account for it in policymaking.

Unemployment is present throughout the developed world (Elsby, Hobijn, and Sahin 2013). The Beveridge curve is more or less visible in all these countries (Elsby, Michaels, and Ratner 2015). Unemployment is also prevalent in the developing world (O'Higgins 2003, figure 11). Although developing countries do not seem to have more unemployment than developed countries, they experience a much higher share of self-employment. And self-employment appears to be a mild form of unemployment, as workers choose self-employment primarily because they face unemployment in the formal sector. Hence, it is likely that there is even more labor market slack in the developing world.[1]

The psychological cost of unemployment has also been confirmed in other developed countries. Just like in the United States, unemployed workers in the United Kingdom suffer from low mental well-being (Clark and Oswald 1994; Blanchflower and Oswald 2004). And they are much more likely to be anxious and depressed and to suffer from low self-esteem—even compared to workers in low-pay jobs (Theodossiou 1998). A typical concern about studies linking unemployment to poor health is that they might not be able to distinguish between causality (unemployment causes poor health) and selection (people in poor health are more likely to become unemployed). But panel data, following workers over time, are able to identify the causal effect of unemployment on health. For instance, using a large, representative panel dataset of German workers, Winkelmann and Winkelmann (1998) find that unemployment reduces life satisfaction and that the nonpecuniary cost of unemployment is much larger than the pecuniary cost. Using the same panel dataset, Lucas et al. (2004) find that people do not get used to becoming unemployed: workers who have already lost jobs in the past react as badly to a new spell of unemployment as workers who are becoming unemployed for the first time.

Unemployment also seems to have extremely high psychosocial costs in the developing world. A field experiment by Hussam et al. (2022) in Bangladesh illustrates just how large the psychosocial cost of unemployment is. The experiment shows that paid employment raises psychosocial well-being substantially more than the same amount of cash alone. In fact, two-thirds of the workers who were employed in the experiment would be willing to forgo cash payments and continue working for free.

[1] Poschke (2025, figure 1) documents the prevalence of self-employment in the developing world. (Robinson 1936) originally suggested that self-employment was a form of "disguised unemployment." Schoar (2010) argues that a substantial share of self-employed workers are "subsistence entrepreneurs," who are likely to come into self-employment from unemployment, to leave self-employment for better jobs whenever they can, and are primarily motivated by the need to support their family and lack of other options. Breza, Kaur, and Shamdasani (2021) confirm that in India, workers are pushed into self-employment when they cannot find jobs in the formal sector. Donovan, Lu, and Schoellman (2023) confirm this finding in a broad set of developing countries.

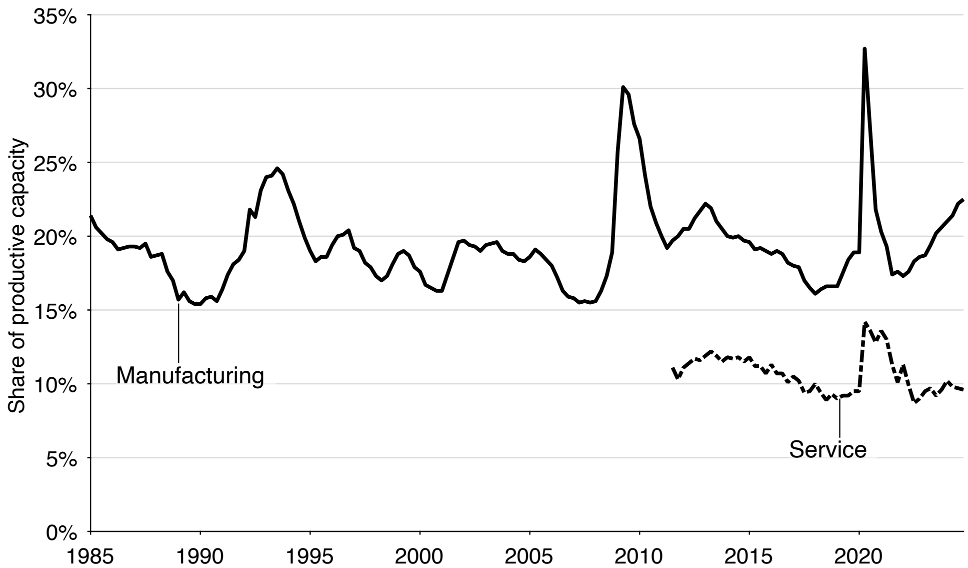


FIGURE 20.1. Idle capacity in the European Union, 1985–2024

The idleness rates are quarterly series constructed from the capacity utilization rates in the manufacturing and service sectors measured by European Commission (2025) from business surveys. For the service sector, the idleness rate is only available after 2011:Q3.

20.1.2. Product market slack in other countries

Slack is also present in product markets all over the world, and probably even more in developing countries than in developed countries, so the slackish framework might be especially useful there.

In the European Union, the European Commission (2025) measures slack in the product market through a large-scale firm survey. Just like the Institute for Supply Management in the United States, it asks firms the following question: "If the demand addressed to your firm expanded, could you increase your volume of activity with your present resources? Yes-No. If so, by how much?" Any firm with idle capacity will answer "yes" and say how much more they could sell (European Central Bank 2014). From the measured capacity utilization rate, we can create a rate of idleness, which is just 1 – utilization rate. In the manufacturing sector, between 1985:Q1 and 2024:Q4, the idleness rate averages 19.3%, and it fluctuates between 15.4% and 32.7%.[2] In the service sector, between 2011:Q3 and 2024:Q4, the idleness rate averages 10.7%, and it fluctuates between 8.7% and 14.2%. So just like in the United States, product markets display substantial slack in Europe (figure 20.1).

Product market slack is not limited to Europe, of course. Walker et al. (2024, figure 1)

[2]The question asked to manufacturing firms is slightly different from the question above, which is asked to service firms, but in practice it seems that firms answer this question (European Central Bank 2007).

show that slack prevails in firms across the globe, and that rates of slack are higher in low-income countries. They also provide survey evidence for urban and rural Kenya firms, which shows widespread product market slack.

A visible source of product market slack is wasted food. UN Environment Programme (2021) documents that food unsold by grocery stores and restaurants generates a significant amount of food waste across the globe. Brancoli (2021) provides a specific example: he explains why bread remains unsold at grocery stores in Sweden and documents how this generates food waste.

Another element suggesting that product market slack is prevalent globally is that sellers strive to form long-term relationships with buyers in all countries. For example, 70% of firms in the Euro area have long-term relationships with their customers (Fabiani et al. 2006, table B3). Firms in developing countries also appear to rely heavily on long-term relationships to operate. Macchiavello (2022, p. 340) reviews a wide range of evidence and concludes that in developing countries:

> Many—and perhaps most—transactions between firms occur in long-term relationships rather than in spot markets, as typically theorized in economic models.

Long-term customer relationships are governed by implicit contracts that alleviate matching problems, as I found in a survey of French bakers in the summer of 2007.[3] First, customer relationships alleviate the uncertainty associated with random demand. A baker told me that demand is difficult to predict and that having a large clientele of loyal customers who make it a habit to purchase bread in the shop was therefore important. In fact, "good" customers are expected to come every day to the bakery. Customer relationships also alleviate the uncertainty associated with random supply. Being a customer means having the assurance that your usual bread will be available, even on days when supply runs low. Of course, this is possible because bakers know exactly what customers order every day through their long association. A baker told me that it would be "unacceptable" to run out of bread for a customer, and that customers would probably "leave the bakery" if that happened.

20.1.3. International trade

Domestic product markets around the world appear slackish, in that it takes time to sell goods and services, it takes effort to buy them, and buyers and sellers alleviate these difficulties by forming long-term customer relationships. International product markets appear at least as slackish: the geographical, cultural, and legal distance makes it difficult to trade with buyers or sellers in other countries—even more so in other continents.

[3]The survey is described in Eyster, Madarasz, and Michaillat (2021).

It is because of these difficulties that sellers and buyers are keen to form long-lasting cross-country relationships (Egan and Mody 1992, table 3).

20.1.4. Price and wage rigidities in other countries

The price and wage rigidity used in the book's theory also appear in other countries, which makes a framework with rigid price and wage norms particularly applicable. Indeed, across the globe, prices and wages do not appear to be fully responsive to shocks. Administrative data and household surveys across 16 Western countries show that wages are fairly rigid (Dickens et al. 2007). Wage rigidity is also prevalent in developing countries (Kaur 2019). Large-scale firm surveys show that prices are quite rigid throughout the Euro area (Fabiani et al. 2006). Given such rigidities, we expect fluctuations in slack to be substantial—because adjustments in slack are required to equilibrate supply and demand when prices are rigid.

We argued that price and wage rigidities might be caused by fairness concerns, and we provided evidence from the United States. But there is substantial evidence that people in other countries also care about fairness: Frey and Pommerehne (1993) provides evidence from Switzerland and Germany, Gielissen, Dutilh, and Graafland (2008) from the Netherlands, and Boycko and Shiller (2016) from Russia. Firms across the globe also identify fairness as a major concern in price setting. Researchers have surveyed more than 12,000 firms across developed economies about their pricing practices; among the theories that the managers deem most important in explaining price rigidity, some version of "being fair to customers" invariably appears (Eyster, Madarasz, and Michaillat 2021, tables 1 and 2).

More generally, we assumed that social norms determine prices and wages in slackish markets. This assumption is consistent with Lewis (1954, p. 189)'s observation that in the developing world, many workers were available to work at a wage determined by a social norm: "a conventional view of the minimum required for subsistence." Ranis (2004, pp. 717–718) explains that such a wage is "an exogenous unskilled agricultural real wage," instead of a real wage "determined endogenously by the interaction of demand and supply," and he argues that such a wage is supported by historical and anthropological evidence.

In general, social norms determine how people think they should behave and others should behave (Akerlof and Kranton 2010, chapter 2). Hence, workers will refuse to work for an employer who violates the norm—who pays a wage that is too low—and they will be willing to punish other workers who accept to work at a norm-violating wage. Breza, Kaur, and Krishnaswamy (2025) find that all of this occurs in Indian labor markets, even when unemployment is high: Indian workers not only refuse to work at below-normal wages, but they are willing to pay to punish workers who accept such abnormally low wages.

20.2. Applying the theory of slack to other countries

In this book we built the theory of slack based on US evidence, calibrated the models to the US economy, and applied the theory to US policies. But, as unemployment and slack more generally exist everywhere, this book's theory should be helpful in other countries too: to understand how markets operate, how business cycles affect the economy, and how policies can be used to improve market outcomes.

20.2.1. Labor market policies

The slackish labor market model developed in this book has already been helpful to understand the effects of various labor market policies in several European countries: for example, in France (Crepon et al. 2013), in Austria (Lalive, Landais, and Zweimuller 2015), and in Sweden (Cheung et al. 2025). It might be helpful to study other policies in comparable labor markets.

The model might also be helpful to understand developing economies and design appropriate policies to combat poverty and foster development, and thus to provide more fulfilling lives to workers everywhere. Indeed, although developing countries see large flows of workers in the labor market (Donovan, Lu, and Schoellman 2023), job rationing in the formal sector is prevalent (Breza, Kaur, and Shamdasani 2021). The model implies for instance that policies that stimulate labor demand, such as public employment, might reduce unemployment and poverty effectively.[4] Public employment might be especially potent in slack labor markets, when it crowds out private employment less. The theory provides support for India's public work program, the National Rural Employment Guarantee Act, and suggests that the program might be especially effective in slack labor markets, just as Breza, Kaur, and Shamdasani (2021, section 8) find.

20.2.2. Self-employed economies

In chapter 13, we developed the simplest possible slackish model of an aggregate economy. In that model, all workers are self-employed and sell their production directly to other people. This is not very realistic to describe the modern US economy or other developed economies, where most production is mediated by firms. But it might be useful to describe developing economies because there a large share of the labor force is indeed self-employed. As Porta and Shleifer (2014, figure 4) show, the share of workers in the labor force who are self-employed decreases sharply with GDP per capita; the share is close to 100% in the world's poorest countries. The model of chapter 13 might be particularly

[4] A further advantage of public employment programs is that they allow policy to reach the rural poor (Ravallion 1991).

well adapted to describe the economy of such countries, and to explain why slack is so prevalent there.

20.2.3. Aggregate shocks

In the book we find that fluctuations in unemployment in the United States are driven by labor demand shocks. It would be interesting to assess if this conclusion holds in other countries. Because we reached that conclusion from Okun's law, a first step would be to ascertain whether Okun's law holds in other countries. Ball, Leigh, and Loungani (2017) show that it holds in 20 developed economies, which suggests that labor demand shocks also play a central role there. Ball et al. (2019) extend these results and show that Okun's law holds in many developing economies as well. The slackish business cycle model might therefore be useful in all these economies, as it features labor demand shocks that are nonneutral and produce meaningful unemployment fluctuations that respect Okun's law.

We went one step further and found that these labor demand shocks in fact arise from underlying aggregate demand shocks, not technology shocks. It would be good to know if this conclusion holds in other countries as well. We reached the conclusion by producing product-market Okun curves, which link short-run fluctuations in output and product market slack. It would be interesting to produce similar plots in other countries. The challenge would be to obtain good measures of product market slack there.

Finally, because agriculture is such an important component of developing economies, seasons generate fluctuations in labor demand and thus slack.[5] It might be fruitful to adapt the slackish model to agricultural economies with seasonal cycles, to understand better the fluctuations in slack that seasons produce and to design better policies to alleviate seasonal poverty.

20.2.4. Labor and total factor productivity

A task related to understanding the sources of slack fluctuations is to use slack data to obtain better measures of labor productivity and total factor productivity, purified from fluctuations in slack. Although labor and total factor productivity might not be an important driver of business cycles, productivity is of course a primary driver of economic growth, so measuring it well is important to understand economic development. As we discussed, measuring productivity accurately requires to measure slack accurately and to properly separate fluctuations in slack from fluctuations in productivity—as both enter fluctuations in measured productivity.

[5] See for instance (Boserup 1965) and Devereux, Vaitla, and Swan (2008) for general discussions and evidence. See Paxson (1993) for evidence from Thailand and Bryan, Chowdhury, and Mobarak (2014) for evidence from Bangladesh.

Another question related to labor productivity in the developing world stems from the work of Lewis (1954), who postulated that these economies were divided between a productive capitalist sector and a less-productive noncapitalist sector—including agriculture, domestic and other services, handicrafts, local retail, and so on. Lewis argued that moving a chunk of the labor force from the noncapitalist to the capitalist sector would boost the country's productivity without affecting much noncapitalist production—especially food production. This view has been challenged by modern development economists who observe that people do earn an income in the noncapitalist sector. Gollin (2014, pp. 79–80) proposed that the resolution between the two views might come from the market structure in developing countries:

> When one additional individual joins the queue of roadside sellers of popcorn or flyswatters in Kampala or Chittagong, the marginal product of that individual's labor may be positive for that individual and his or her family; but whether there is a positive social value is unclear. Arguably, this individual is simply taking business from other sellers, creating little or no additional social value.

The theory of slack immediately resolves this conundrum: as the noncapitalist market is typically very slack, a negative supply shock such as a reduction in the noncapitalist labor force would have a very small effect on output, just as Lewis (1954) argued. However, of course, workers do earn an income on the noncapitalist market, just as modern development economists observed. The resolution is that when workers move out of the market, market tightness goes up, so it becomes easier for remaining workers to sell their goods and services. Therefore, as Gollin (2014) hypothesized, output does not drop as much as what the micro evidence might have suggested, because the micro effects are counteracted by the response of tightness, which increases.

20.2.5. FERU and unemployment gap

As the slackish framework developed in the book is likely to be applicable in other countries, the policy tools introduced in the book should also be applicable beyond the United States. Many countries have a full-employment mandate, but no clear full-employment target. Computing the FERU in these countries would be helpful to guide policymakers towards full employment.

Unemployment rate, vacancy rate, and labor market tightness can be computed in other countries in which data on the number of job seekers, number of job vacancies, and number of labor force participants are available. From this the FERU formula $u^* = \sqrt{uv}$ can be applied, and from it the unemployment gap $u - u^*$ can be computed. For instance, Gokten, Heimberger, and Lichtenberger (2024) apply the formula to Germany, Sweden, Austria, Finland, and the United Kingdom, and analyze the evolution of the unemployment gap in these countries since the 1970s. Liargovas et al. (2025) apply the formula and compute the unemployment gap in Greece, 2009–2024. Moving forward, the real-time unemployment gaps could be used to adjust monetary and fiscal policy in response to shocks in these countries, as described in the book.

However, the FERU formula does require assumptions on the cost of unemployment, the cost of recruiting, and the shape of the Beveridge curve. In countries in which these statistics take different values than in the United States, the FERU must be computed from a more general formula. The shape of the Beveridge curve is particularly likely to differ across countries (Gaddnas and Keranen 2023, table 1). In that case, it is necessary to use the generalized formula for the efficient labor market tightness and adjust the calibration of the parameters. Gaddnas and Keranen (2023) take a step in that direction and compute the FERU in Finland, Sweden, Germany, and the Netherlands using country-specific Beveridge elasticities. Germain (2024) compute the FERU in Belgium's three regions—Brussels, Flanders, and Wallonia—using region-specific Beveridge elasticities.

20.2.6. Phillips curve

In the book we build a slackish Phillips curve that links slack to inflation. There is good evidence in favor of the slackish Phillips curve in the United States, and it would be fascinating to see whether the slackish Phillips curve holds in other countries too: Do we see that slack is a superior indicator of inflationary pressures? And do we see that when the economy is at full employment, inflation is on target? Benigno and Eggertsson (2024) have started to explore these questions. They find similar patterns in other countries, but more research to confirm and expand these findings would be valuable. A key empirical challenge here again is to find measures of slack that are of good quality and that cover a sufficiently long period.

One piece of evidence suggests that the slackish Phillips curve might describe well the behavior of inflation in the developing world. Egger et al. (2022) find that a large fiscal stimulus in Kenya led to a sizeable increase in consumption but a minimal change in price inflation. At the same time, Walker et al. (2024) showed that the Kenyan economy is exceedingly slack. This is consistent with the property of the slackish Phillips curve, which we found to be almost flat in slack economies; hence, even with endogenous inflation, we would expect inflation to increase very little in response to the fiscal stimulus.

20.2.7. Slack dependence of policy multipliers

The slackish model is sharply state dependent. This implies that the effectiveness of policies varies sharply with the state of the economy. For example, in the slackish labor market model, we saw that when the labor market is slacker, public employment reduces unemployment more effectively, in-migration raises unemployment more sharply, and unemployment insurance—although it continues to reduce job-search effort—raises unemployment less. In the slackish business cycle model, when the economy is slacker, fiscal and monetary policy reduce unemployment more effectively.

It would be interesting to investigate these predictions systematically in other countries. This investigation would help designing better policies—because it would help understanding when certain policies are particularly effective. The investigation would also be useful to evaluate the theory of slack.

Several papers offer promising starting evidence, as they found slack-dependent policy multipliers abroad. For instance, Auerbach and Gorodnichenko (2013) estimate state-dependent fiscal multipliers in numerous OECD countries; Holden and Sparrman (2016) estimate state-dependent public-spending multipliers in 20 OECD countries; and Kopiec and Walerych (2026) estimate a slack-dependent monetary multiplier in Poland. These papers find that public spending and monetary policy reduce the unemployment rate more effectively when the economy is slack than when it is tight, as the theory of slack would predict. And Gerard, Naritomi, and Silva (2026, table B3) find that cash transfers in Brazil increase formal employment by stimulating local demand; critically, cash transfers are much more effective in municipalities with above-average unemployment rate than in those with below-average unemployment rate, consistent with the theory's prediction that demand shocks have much stronger effects on employment in slacker labor markets than in tighter labor markets.

20.2.8. Recession detection

The Michez+ recession indicator can also be computed for any country in which unemployment and vacancy data are available, but the recession threshold of 0.19pp will need to be recalibrated. In other countries, the recession indicator takes different values, and recessions occur at different times, so the threshold must be recomputed to detect recessions as early and accurately as possible. It might also be possible to optimize how the unemployment and vacancy data are smoothed and combined to best detect recessions in other countries, using the method described in Michaillat (2025). For instance, Sikand and Zhang (2026) show that the approach performs well in Japan.

20.2.9. Out-of-equilibrium dynamics

When we considered dynamic markets, where trading relationships are formed and destroyed over time, we concentrated on the equilibrium—where market flows are balanced so slack is constant. In the US labor market, flows are large, so out-of-equilibrium dynamics are fast and unlikely to matter much. But such dynamics may matter more on other

US markets, or in other countries. For instance, labor market flows are much smaller in continental Europe (Elsby, Hobijn, and Sahin 2013, table 4). So there it would be important to study and incorporate out-of-equilibrium market dynamics.

20.3. Collecting better evidence on slack

Because slack is not a core concept in macroeconomic research, statistical agencies have not systematically collected data on it—beyond unemployment. Hence, many measures of slack reported in this book—for the United States and other countries—are imperfect in some ways. Collecting better slack data would help improve the design of slack-centered economic models and the conduct of slack-based economic policies. To conclude the chapter, this section discusses which type of data and evidence would be especially valuable.

20.3.1. Data on labor market slack in the United States

When we computed the full-employment rate of unemployment (FERU), we saw that a key statistic is the amount of man-hours devoted to recruiting by firms. There is little evidence on that, so we used the number of job vacancies reported by firms, together with an estimate of the amount of recruiting per vacancy, to infer the amount of recruiting at any point in time. This lack of evidence is not ideal to measure the FERU and unemployment gap in the past, but it could easily be remedied in the future. The BLS would only need to add one question to the Job Opening and Labor Turnover Survey. The recruiting cost could be measured every month by asking firms to report how many man-hours they devote to recruiting in addition to how many vacancies they are currently advertising. As part of the American Time Use Survey, the BLS already asks unemployed workers how many hours they spend every day searching for a job (Aguiar, Hurst, and Karabarbounis 2013). It should be easy to ask firms the same thing. Moving from job vacancies to recruiting hours would improve the measurement of the unemployment gap and thus the conduct of stabilization policies.

Another key statistic that could influence the FERU is the social cost of unemployment. We have seen substantial evidence from sociology, psychology, public health, and medical science that unemployment is detrimental to physical and especially mental health. But few studies have been able to quantify the social cost of unemployment. One such study, by Borgschulte and Martorell (2018), uses administrative military data to estimate the value of nonwork relative to the value of work. It would be valuable to measure the social value of nonwork for other workers and in other datasets to confirm or refine the findings and ascertain more confidently the social cost of unemployment. Using a wonderful field experiment, Hussam et al. (2022) find dramatic psychosocial costs of unemployment in

Bangladesh—to the point where many of the subjects were happier to work for some salary than remaining unemployed for the same amount of money. It would be interesting to see whether such findings also hold in the developed world.

20.3.2. Data on product market slack in the United States

Data on product market slack are especially scarce in the United States. Given the importance of the issue, product market slack is not something that is well-measured or that the government tracks at all—probably because it is not part of standard macroeconomic theory. It would be valuable to collect more data on product market slack. It would be good to know how many goods remain unsold across the economy at a monthly frequency, and how many employees remain idle each month across the economy—waiting for customers to purchase their labor services.

It would also be good to know how many buyers there are for goods and services at any point in time, which would then allow us to measure product market tightness. Unlike in the labor market, where job vacancies are measured systematically, the desired amount of purchases in the product market is not recorded in any centralized dataset.

And while we have some idea of the cost of filling a job vacancy, there is very little data on the cost of visiting shops and filling help-wanted ads. Collecting some data on the matching cost in the product market would be very valuable—for instance, to calibrate the product market in the two-market business cycle model and to compute the efficient product market tightness.

20.3.3. Data on slack in other countries

The most important form of slack is unemployment, so the most important statistic is the number of unemployed workers at any point in time. Developed countries now collect such data, but many developing countries still do not. It seems critical to start measuring unemployment in these countries to track and stabilize business cycle fluctuations.

One challenge is that many workers are self-employed in the developing world, and workers who are self-employed are neither as idle as unemployed workers nor as occupied as formal workers, so they only partially contribute to slack in the economy. To complicate matters, Maloney (2004, p. 1161) reports that in Latin America about 30% of the self-employed are in the occupation involuntarily, while 70% choose to be self-employed voluntarily, enjoying the flexibility and independence from self-employment. Presumably, the involuntarily self-employed could be more productive and work longer hours if they were employed by firms, so they contribute more significantly to aggregate slack. The voluntarily self-employed may be producing and earning as much as they can in their occupation, and they might not contribute to aggregate slack at all. Keeping track of slack among the self-employed might require measuring unemployed hours. The unemployment rate among the self-employed would be the share of the time that they are idle.

To assess the tightness of foreign labor markets, it would be key to collect monthly data on job vacancies. To assess the full-employment rate of unemployment, as well as the unemployment gap, it would be important to measure the amount of labor devoted to recruiting in these labor markets. At the minimum, vacancies can be translated into recruiters by assuming some constant number of recruiters per vacancy; but it would be even better to have direct, monthly measures of the number of recruiters.

Of course, in addition to data on slack in the labor market, it would be extremely instructive to measure slack in the product market systematically. This would require measuring the slack rate in firms, small and large, as the European Commission does in the European Union. These surveys ask firms how much more they could produce, given their current capital stock and workforce and organization, if only they had more demand for their product. In the slackish framework, this is the ideal information to compute the slack rate in the product market.

By processing the content of major newspapers—about 20,000 news articles per month—Caldara, Iacoviello, and Yu (2026, figure 2) construct an index of shortage in the United States from 1900 to 2024. The index includes shortages of food, of energy, in the industrial sector, and in the labor market. What is fascinating is that the episodes of labor market shortage that they measure perfectly overlap with the episodes during which labor market tightness is inefficiently high: World War 2, the Korean War, the Vietnam War, and the post-pandemic period. This suggests that it might be possible to obtain accurate measures of tightness in the labor market and possibly in the product market by textual analysis of newspapers: this might be a promising avenue to measure slack when government data are unavailable.

CHAPTER 21.

Scientific context of the slackish business cycle model

This chapter uses the Kuhnian perspective on science to understand how, over the past century, business cycle research evolved from the Old Keynesian model, built around nonclearing Walrasian markets, to the Real Business Cycle model, built around Walrasian markets, and finally to the New Keynesian model, built around monopolistic markets. We review how models improved over time along the three dimensions that Kuhn (1957, 1996) emphasizes: being descriptive, being economical, and being fruitful. Finally, we see how this book's slackish business cycle model fits in that context. We discuss the elements that it borrows from the previous traditions, and the novel elements that it introduces, in an attempt to be as descriptive, economical, and fruitful as possible.

Economists have been developing models of the business cycle since the Great Depression. From the outside, the literature looks fragmented and difficult to parse. It is quite difficult to identify the common thread. The models make strikingly different assumptions, and it is not always clear why. What is particularly frustrating is that although many assumptions seem fairly arbitrary, people adhere to them very firmly. The different branches also focus on different questions and problems, so it is hard to see what the debate is about. And of course the models have diametrically different policy implications. That is why this chapter uses a Kuhnian perspective: to clarify the strengths and limits of major business cycle models, and to understand the historical evolution from model to model.

21.1. The Kuhnian model of science

We begin by summarizing Kuhn's model of science, which was developed in two books. The first book is *The Copernican Revolution*, in which Kuhn (1957) explains the evolution of planetary astronomy from Antiquity to today. The second is *The Structure of Scientific Revolutions*, in which Kuhn (1996) generalizes the model of science developed in the context of astronomy and applies it to other sciences. That model of science will help us understand what has been happening in the world of macroeconomics, and how the slackish model fits in.

21.1.1. Organization of scientific knowledge in paradigms

The core building block of Kuhn's model of science is a paradigm, which is a way that scientists organize a whole body of knowledge. They then use that paradigm to interpret what they observe in nature. It is a lens through which scientists of a particular era look at the world. And at the core of a paradigm, there is a model that all the scientists that belong to that paradigm use.

But a paradigm is not only a model—there are a lot of other things that are attached to it. A paradigm also encompasses the facts that scientists are particularly interested in. Indeed, scientists who belong to different paradigms focus on different facts and look at different objects. Paradigms also include acceptable research practices and methodologies.

Overall, a paradigm is a model and shared norms and standards that are universally accepted by scientists in the field. Paradigms can typically be identified by the existence of a textbook or several textbooks that summarize everything that the paradigm stands for. These textbooks are used to educate students to pass along theory, facts, methods, and norms.

21.1.2. Cycling through paradigms

The question now is: How do we get to the current paradigm? Science is often described as a cumulative exercise where, over time, we learn and discover more and more. People think of scientific progress as continuous improvement—like adding bricks to a wall, making it higher and higher. However, this is quite an inaccurate picture. In practice, science does not evolve linearly: it is not really a cumulative exercise. Science is much more cyclical: periods of creation and improvement are followed by periods of destruction, when scientists abandon what they knew and start afresh.

What does this scientific cycle look like? The cycle starts with normal science. In a period of normal science, a paradigm is well established and accepted by everyone in the community. All the researchers look at the facts that the paradigm is interested in and use the research methods that the paradigm recommends. Often, at the center of the paradigm are several parameters or constants that need to be measured or estimated. A lot of time is spent trying to measure these things using the data. Sometimes, the paradigm makes predictions about how the world is supposed to work, so scientists spend time testing these predictions and assessing whether the world is in line with the predictions of the model. If the facts are not exactly consistent with the predictions, scientists go back and further articulate and refine the model to describe these new observations properly.

At some point during the normal-science process, as scientists keep looking at the world through the lens of the paradigm, they start to discover anomalies, which are empirical regularities or observations that cannot be easily reconciled with the model. Once there are too many anomalies that cannot be explained properly with the paradigm, some scientists start looking for different paradigms, or a different way to explain the world.

When we reach that stage, we move away from normal science and into revolutionary science. During these episodes, the existing paradigm has to compete with new paradigms that have been created to explain the anomalies that have been observed. Revolutionary science takes time: the battle between the paradigms is generally not resolved overnight.[1] Scientists come up with one or many alternative paradigms, and there is a big fight within the community to decide who is going to take over. After a while, one paradigm emerges as victor and replaces the old paradigm. In the case of astronomy, this period of revolutionary science occurred during the Renaissance. It involved Copernicus, Giordano Bruno, Tycho Brahe, Kepler, and Galileo, and was a battle between the old geocentric model and the new heliocentric model.

Once the new paradigm is adopted, there is a new period of normal science during which scientists refine the new paradigm and use it to look at new research questions and isolate new anomalies until it is time for a new period of revolutionary science where new paradigms are proposed and possibly take over.

21.1.3. A good paradigm is descriptive

Science evolves and improves by cycling through paradigms. To understand how new paradigms are created, and how a new paradigm might emerge victorious from the period of revolutionary science, we must understand what the qualities of a good paradigm and its underlying model are.

The first quality that Kuhn (1957, pp. 38–39) talks about is being descriptive. A model should be able to accurately describe and explain all the facts that have been collected by scientists. Indeed, the emphasis is usually only on this quality. When we are taught about science in school, we learn that models are fundamental truths that explain the world.

[1] Akerlof and Michaillat (2018) explain why better paradigms might take a long time to be accepted, or might not be accepted at all.

But, of course, no model is true—models are just an articulation of the knowledge that has been collected. A quality of a good model is that it can explain a lot of that accumulated knowledge. If a model is unable to describe the basic phenomena that are observed in the real world, it is not going to be useful. In the case of astronomy, these facts have to do with the movement of the Sun and the stars during the day and over the seasons in different locations on Earth.

21.1.4. A good paradigm is economical

The second quality of a good paradigm is to be economical. An economical model compresses knowledge into a compact form that scientists can use as a mental model: that they can manipulate in their mind to explain existing facts and analyze new problems.

An economical model provides assistance to a scientist's memory because by memorizing a few relationships that are easy to manipulate, the scientist can recover a long list of facts and patterns (Kuhn 1957, p. 37). An economical model is simple and logical enough to be cognitively usable, so scientists can keep it in their head and articulate mentally what they observe in the real world.

Being economical is a logical quality of the model, so it can hold even if a model is not descriptive. For instance, the geocentric view of the universe is not a good description of the universe. Nevertheless, that paradigm is very economical. In fact it is possible to recover all the facts that were known to astronomers from Antiquity to the Middle Ages from a two-sphere model—with the earth as a small stationary sphere whose center coincides with that of a much larger rotating stellar sphere (Kuhn 1957, figure 11). This is why, even today, the geocentric two-sphere model continues to be used in surveying and navigation (Kuhn 1957, p. 38). Even though the model is not as accurate as the heliocentric model, it can explain a number of things well enough and is especially simple to remember.

21.1.5. A good paradigm is fruitful

The third quality of a good paradigm is to be fruitful: the paradigm must be a good guide to the unknown (Kuhn 1957, pp. 39–41). It should make useful predictions about things that we have not seen yet. The paradigm is a tool for scientists exploring the unknown because it tells them where to look and what they may expect to find. A good paradigm must describe all the things that have been observed, but there are also many things that scientists have not looked at yet—either because they never thought about looking at certain things or because they did not have the right instruments to do so. A good paradigm makes predictions about what these unobserved, unknown things might look like and how they might behave.[2]

[2] By making predictions about the unknown that may be wrong, a model makes what Popper (1963) calls risky predictions—which he sees as necessary to validate or falsify a model. Economists are Popperian at

Planetary astronomy offers a striking example of this function of models as guides to empirical exploration. Before it was ever observed, Neptune was predicted to exist from Newtonian mechanics and the observation of disturbances in the orbit of Uranus. This prediction led astronomers to explore the space around Uranus with telescopes and ultimately to discover Neptune (Kuhn 1957, pp. 261–262).

21.2. Applying the Kuhnian model to business cycle research

We now apply the Kuhnian model of science to business cycle research. We first identify the paradigms of business cycle research: the Old Keynesian paradigm, the Real Business Cycle paradigm, and the New Keynesian paradigm. We describe the different elements that make up these paradigms: the core model used by the adherents to the paradigm, the facts that are considered relevant, the methodologies that are acceptable, and the research problems that are considered interesting. We then discuss why business cycle research evolved as it did, cycling from paradigm to paradigm.

21.2.1. The structure of business cycle models

Let us first briefly discuss the structure of business cycle models. Indeed, all the main business cycle models share a common structure.

All the models are composed of households—these are the people that make up the economy. The households do two things: they consume and produce. Household members are therefore both workers and consumers: the goods and services produced by one household are consumed by other households.

These models describe market economies: producers trade their goods and services and use the income to buy the goods and services that they want to consume. The models do not describe more primitive economies in which all production is done at home by self-sufficient households.

Typically, people are not self-employed but are instead employed by firms—although the presence of firms is not as important as it might seem. The production takes place in the firms through a production function, using labor that firms hire from households. The firms then sell the produced goods and services back to households. So these models have two markets: a labor market and a product market. Furthermore, the models are symmetric in that the product and labor markets have the same structure.

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heart: they enjoy examining the testable predictions of models, and use those to reject models. The Kuhnian view of models is more constructive: rather than simply offering up predictions to be shot down, models open new avenues for research, new domains of exploration. Of course, Kuhn does recognize that if too many risky predictions are wrong—if there are too many anomalies—models are eventually replaced. This is what occurs during scientific revolutions.

We now describe the different business cycle paradigms based on how they model the labor and product markets and how they model prices and wages. Hence we also discuss the other elements of each paradigm: norms, methods, parameters of interest, questions of interest, and so on.

21.2.2. The Old Keynesian model

We start with the oldest microfounded business-cycle model: the Old Keynesian model. With respect to the labor and product markets, the Old Keynesian model assumes non-clearing Walrasian markets. This means that firms and households are price takers, but prices and wages are fixed at levels that may not clear the markets. As a result, when shocks hit the economy, demand is no longer equal to supply, generating either excess demand or excess supply.

That paradigm's decisive paper is Barro and Grossman (1971), which developed a fully fledged Old Keynesian model of the business cycle. An early textbook that inspired the Old Keynesian paradigm is *Money, Interest, and Prices* by Patinkin (1965). The main textbooks of this paradigm are *Money, Employment and Inflation*, written by Barro and Grossman (1976), and *The Theory of Unemployment Reconsidered*, written by Malinvaud (1977). These books present the whole paradigm and articulate its implications.

We saw above that a paradigm is not just a model. It also comes with a shared set of norms: methods that scientists follow, research questions that are considered interesting, assumptions that are considered legitimate, concepts that are used and studied, and statistics that are measured in the data.

A concept specific to the Old Keynesian paradigm is the set of regimes that appear depending on whether the product and labor markets are in excess supply or demand. Malinvaud (1977, figure 1.2) refers to these regimes as Keynesian unemployment (excess supply on the labor and product markets), classical unemployment (excess supply in the labor market, excess demand in the product market), repressed inflation (excess demand on the labor and product markets), and underconsumption (excess demand in the labor market, excess supply in the product market).

Other concepts specific to the Old Keynesian model are the notional and effective demand and supply curves. The notional curves are derived by assuming that the market would clear. The effective curves are derived under the realization that, in fact, prices are fixed and so the market might not clear after all.

21.2.3. The Real Business Cycle model

The Real Business Cycle paradigm followed the Old Keynesian paradigm. It also assumes Walrasian labor and product markets, but unlike in the Old Keynesian model, it assumes that prices and wages are always at the market-clearing levels. Thus, the labor and product markets always clear, as prices and wages move flexibly.

The seminal paper for the Real Business Cycle paradigm was written by Kydland and Prescott (1982). The Real Business Cycle paradigm's textbook is *Frontiers of Business Cycle Research*, edited by Cooley (1995), with contributions from the main researchers in the field. The book summarizes everything that the Real Business Cycle paradigm stands for. It includes chapters about models and many chapters that describe the methodologies that are acceptable for the paradigm, including for instance recursive methods to compute equilibria.

An assumption specific to the Real Business Cycle paradigm is that technology shocks play a very important role: they are the main drivers of the business cycle. So researchers in the Real Business Cycle paradigm pay a lot of attention to technology and technology shocks. This is not true in all paradigms: in the Old Keynesian model, technology is not even mentioned.

There is a very clear procedure about how to do things in the Real Business Cycle paradigm: build a model; calibrate that model; simulate the dynamics of the model under stochastic shocks; and finally compare what happens in the simulated model to the real world—typically compare the moments of simulated variables to the same moments in the data.

Another methodology that is associated with the Real Business Cycle paradigm is to detrend the data with a Hodrick and Prescott (1997) filter, and to define business cycle fluctuations as the deviation from that trend. Under the typical calibration, the procedure produces small fluctuations, because the HP trend absorbs a good amount of the business cycle movements.[3] Hence the model is left with less to explain.

A last methodological norm attached to the Real Business Cycle paradigm is that it is utterly unacceptable to use diagrams and graphs. There is a saying that "a picture is worth a thousand words", but members of the Real Business Cycle paradigm say that "an equation is worth a thousand pictures." That may seem silly, since diagrams are representations of equations, and often allow us to uncover interesting and hidden properties of equations. Famously, the Feynman diagrams changed how postwar theorists did and thought about physics (Kaiser 2005). Nevertheless, this is the research methodology embraced by that paradigm.

At the core of the Real Business Cycle paradigm is a business cycle model with Walrasian markets, so the model is always efficient. Hence, policy questions regarding the stabilization of business cycles are not considered interesting in that paradigm. In fact, business cycles are considered virtually costless and therefore not worth worrying about (Lucas 2003).

[3]This is obvious by looking at labor market data detrended by HP filter (Shimer 2012, footnote 10).

And since prices are flexible, money is neutral in the Real Business Cycle paradigm, so there is no room for monetary policy. Any question related to the effectiveness of monetary policy, or the optimal conduct of monetary policy, is absent from that paradigm.

21.2.4. The New Keynesian model

After the Real Business Cycle paradigm came the New Keynesian paradigm. The New Keynesian model replaces the Walrasian markets of previous paradigms with monopolistic markets. Moreover, the paradigm assumes that prices and wages are rigid, in that they adjust slowly over time, in a staggered fashion: each period only a fraction of prices and wages are changed, as proposed by Calvo (1983).[4]

The foundational papers of the New Keynesian paradigm include Blanchard and Kiyotaki (1987) and Kimball (1995). Its main textbooks are *Interest and Prices* by Woodford (2003) and *Monetary Policy, Inflation, and the Business Cycle* by Gali (2008). These textbooks are used today in graduate programs all over the world.

In the New Keynesian paradigm, one statistic that researchers attempt to measure as well as possible is the frequency of price changes (Klenow and Malin 2010). That frequency is a preeminent parameter of interest in the New Keynesian model: it corresponds to the frequency at which the Calvo fairy operates, which in turn determines the shape of the Phillips curve and the effectiveness of monetary policy.

Another central statistic is the price markup, which New Keynesian economists have spent a lot of time measuring and estimating (Rotemberg and Woodford 1999). This is because of the monopolistic nature of competition in the model. Markups were not interesting in previous paradigms, but they attract a lot of attention in the New Keynesian paradigm.

Of course, because prices are rigid in the New Keynesian model, monetary policy is nonneutral. Therefore, researchers in the paradigm investigate what is the best way to conduct monetary policy (Woodford 2010). They also investigate the effectiveness of monetary policy in the real world, using a range of empirical strategies (Christiano, Eichenbaum, and Evans 1999).

Another concept that plays an important role in the New Keynesian paradigm is the Taylor (1993) rule for monetary policy. This is an important building block of the New Keynesian model because it ensures its determinacy. The Taylor rule had not appeared in previous paradigms, but it is a central object of research for New Keynesian researchers (Taylor and Williams 2010).

A last concept that is specific to the New Keynesian paradigm and generated a tremen-

[4]The simplest version of the New Keynesian model assumes rigid prices but flexible wages, but for quantitative applications, researchers typically assume both rigid prices and rigid wages, as in the model by Erceg, Henderson, and Levin (2000).

dous amount of research is inflation expectation. Gali (2008, p. 185) concludes his textbook by emphasizing that one of the two main implications of the New Keynesian model is the importance of expectations about future short-term interest rates and inflation. Indeed, these expectations play a key role in the New Keynesian model's Phillips curve (Coibion, Gorodnichenko, and Kamdar 2018). The importance of expectations in the model has pushed central banks toward more communication and transparency to anchor expectations at the appropriate level (Coibion, Gorodnichenko, and Weber 2022). Their importance has also led researchers to design specific surveys to measure inflation expectations of households and firms (Weber et al. 2022). It is interesting that New Keynesian researchers talk about expectations so much, while macroeconomists in the previous paradigms did not care about these expectations much. This shows that it is not just the models that are different across paradigms—the facts that scientists look at are different.

21.2.5. From the Old Keynesians to the New Keynesians

We have covered the three existing paradigms of business cycle research: Old Keynesian, Real Business Cycle, and New Keynesian. Now, let us place them in time and see how the macroeconomic community moved from one to the other.

The Great Depression prompted Keynes (1936) to write the *General Theory*. Because the *General Theory* was not mathematical, Hicks (1937) was prompted to formalize it—introducing the IS-LM framework. The IS-LM framework was developed before the mathematical revolution in economics (Debreu 1991). Although it is somewhat mathematical in that it is built on some equations and diagrams, it is not microfounded: it does not describe what workers and firms do, how markets operate, how prices and wages are set, and so on. As a result, for instance, it cannot be used for welfare analysis and optimal policy design.

These limitations of the IS-LM model prompted the appearance of the Old Keynesian paradigm in the 1960s. In *Money, Employment, and Inflation*, Barro and Grossman (1976) explain why they build the Old Keynesian model. They were worried about the weak foundations of the IS-LM framework. They wanted to provide a strong microfoundation to Keynesian ideas. For instance, they wanted to understand the interrelation between the behavior of individual economic agents and the realization of macroeconomic phenomena. As their book description explains:

> This is a textbook on macroeconomic theory that attempts to rework the theory of macroeconomic relations through a re-examination of their microeconomic foundations. In the tradition of Keynes's *General Theory of Employment, Interest and Money* (published in 1936), and Patinkin's *Money, Interest, and Prices,* published in 1956 and revised in 1965, this book represents a third generation of macroeconomic theory. This book presents a comprehensive choice-theoretic

> analysis of the determination of the level of employment and the rate of inflation. A central feature of the book is the recasting of macroeconomic analysis in terms of a theory of exchange under non-market-clearing conditions.

The Old Keynesian model did not aim to address empirical anomalies associated with the IS-LM model: it was touted as a methodological improvement. Thus, the community joined the Old Keynesian paradigm because it was providing a better methodology, with microfoundations, not because it described the world better. It managed to describe the ideas and mechanisms in the *General Theory* with a consistent mathematical framework consisting of firms and households maximizing utility and profit in a modern way.

The Old Keynesian paradigm was very active in the 1970s, but was rapidly displaced by the Real Business Cycle paradigm in the 1980s. Very quickly after Kydland and Prescott (1982) wrote their paper, the Real Business Cycle paradigm took over and was adopted by the macroeconomic community until the late 2000s.

The motivation for moving from the Old Keynesian paradigm to the Real Business Cycle paradigm is a little harder to see, but appears again to be solely methodological. A common view is that the US stagflation in the 1970s led to drastic changes in macroeconomics, as a result of which the Old Keynesian model perished. At the time, however, Real Business Cycle researchers sold their contribution differently. In *Frontiers of Business Cycle Research*, Cooley and Prescott (1995) provide a very simple explanation for what they were trying to do. They started with the view that growth and fluctuations are not distinct phenomena to be studied with separate data and analytical tools. They wanted to build a model that captures both short-term economic fluctuations, and long-term economic growth. In fact, the abstract of Kydland and Prescott (1982) starts with:

> The equilibrium growth model is modified and used to explain the cyclical variances of a set of economic time series, the covariances between real output and the other series, and the autocovariance of output.

So, the stated goal was to have a unified treatment of growth and fluctuations. This motivation explains where all the assumptions of the Real Business Cycle model come from: why it starts from a growth model, why prices are flexible (they ought to be in the long run), and why technology plays a crucial role (it is a key driver of economic growth).

A secondary factor that might have contributed to the demise of the Old Keynesian paradigm and accelerated the transition to the Real Business Cycle paradigm is an emerging reluctance to assume rigid prices and wages. One possible source of this reluctance is a famous argument by Barro (1977), in which he criticized the Old Keynesian assumption of wage rigidity on the grounds that firing workers due to wage rigidity violates bilateral efficiency. Firms and workers forego Pareto improvements: they could both be better off by renegotiating a lower wage and continuing to work together. Barro deemed such a level of inefficiency too implausible to be included in the models.

The Real Business Cycle literature was very active for two or three decades, after which the New Keynesian paradigm rose to prominence. The New Keynesian paradigm became very active in the late 1990s and is still dominant today. So, the 1990s and 2000s were a period of revolutionary science because there was a fierce competition between the two paradigms—Real Business Cycle and New Keynesian—and, eventually, the New Keynesian paradigm took over.

Again, it is interesting to understand the motivations for building the New Keynesian model. In this case, however, the motivation was very much empirical: the goal was to describe the world better than the Real Business Cycle model. In the preface of *Interest and Prices*, Woodford (2003) explains that the goal was to improve on two things. First, the model was designed to understand the inefficient fluctuations that result from rigid wages and prices. In fact, early on, Summers (1986, p. 26) criticized the Real Business Cycle model for being unable to account for the Great Depression—to account for market failure on a grand scale:

> A fourth fundamental objection to [the Real Business Cycle model] is that it ignores the fact that partial breakdowns in the exchange mechanism are almost surely dominant factors in cyclical fluctuations. Consider two examples. Between 1929 and 1933, the gross national product in the United States declined 50 percent, as employment fell sharply.... I submit that it defies credulity to account for movements on this scale by pointing to intertemporal substitution and productivity shocks.... If some other force is responsible for the largest fluctuations that we observe, it seems quixotic methodologically to assume that it plays no role at all in other smaller fluctuations. Whatever mechanisms may have had something to do with the depression of the 1930s in the United States ... presumably have at least some role in recent American cyclical fluctuations. What are those mechanisms? We do not yet know.

Second, Woodford (2003) argues that the New Keynesian framework was designed to help understand why monetary policy is effective, what central bankers do, how they could conduct monetary policy better. As Gali (2008, pp. 3–4) writes in the introduction of *Monetary Policy, Inflation, and the Business Cycle*:

> The attempts ... to introduce a monetary sector in an otherwise conventional Real Business Cycle model ... were not perceived as yielding a framework that was relevant for policy analysis.... The resulting framework ... generally predicts neutrality (or near neutrality) of monetary policy with respect to real variables. That finding is at odds with the widely held belief (certainly among central bankers) in the power of that policy to influence output and employment developments, at least in the short run. That belief is underpinned

> by a large body of empirical work, tracing back to the narrative evidence of Friedman and Schwartz (1963), up to the more recent work using time series techniques, as described in Christiano, Eichenbaum, and Evans (1999). In addition to the empirical challenges mentioned above, the normative implications of [these] models have also led many economists to call into question their relevance as a framework for policy evaluation. ... The conflict between theoretical predictions and evidence, and between normative implications and policy practice, can be viewed as a symptom that some elements that are important in actual economies may be missing in [Real Business Cycle] models.... Those shortcomings are the main motivation behind the introduction of some Keynesian assumptions, while maintaining the [Real Business Cycle] apparatus as an underlying structure.

Overall, the motivation for the New Keynesian model was understanding cyclical inefficiencies and thinking about monetary policy. Macroeconomists wanted to have a model that could explain market failures in recessions and that could guide monetary policy, which the Real Business Cycle model could not do. That is why the New Keynesian paradigm took over.

21.3. Connecting the slackish model to previous business cycle models

In this section, we draw connections between the slackish business cycle model and its predecessors. We explain the elements that the slackish model borrows from these previous models, and we discuss what the slackish model aimed to improve upon. We have seen in the previous section what are the main assets and liabilities of existing business cycle models. Given this assessment, the construction of the slackish model is quite natural: it aimed to preserve as many assets as possible and move away from the liabilities. Here we consider the two-market version of the slackish business cycle model, presented in chapter 15. That model has separate product and labor markets and firms, so it shares the same structure as the core models in the other paradigms.

21.3.1. Connections to the Old Keynesian model

The slackish business cycle model shares several similarities with the Old Keynesian model. A first similarity is that both models feature slack. In the Old Keynesian model, the model features slack in any situation of excess supply. For instance, there is slack in the product market in the Keynesian unemployment and underconsumption regimes, and slack in the labor market in the Keynesian unemployment and classical unemployment regimes. However, in the classical-unemployment and repressed-inflation regime, there is no slack in the product market: there is excess demand so firms can sell all their production. And in the repressed-inflation and underconsumption regimes, there is no unemployment in the labor market. This is all a bit unrealistic. In a slackish model, just as in the real world, there is always some slack and some unemployment—goods that cannot be sold and workers who cannot find jobs.

A second similarity is that both models are generally inefficient. The Old Keynesian model operates efficiently only when the fixed prices and wages happen to be at the market-clearing levels. The slackish model operates efficiently only when the price and wage norm coincide with the efficient price and wage levels.

The Old Keynesian model has several limitations, however, and the slackish model attempts to improve on them. A first limitation of the Old Keynesian model is that it is a little cumbersome to analyze: the model is described by four different systems of equations, one for each possible regime. These four systems are required to describe all the possible cases as either supply or demand can be rationed on each market. Studying the model is therefore difficult because each regime requires a different analysis. By contrast, the slackish model is described by a single system of smooth equations, irrespective of the slackness or tightness of the product and labor markets. A case-by-case analysis is not required.

A secondary, related limitation is that the Old Keynesian model is quite binary. Its behavior is determined either by the market supply (in excess demand) or by the market demand (in excess supply) on any market. Demand and supply never matter at the same time. The slackish market offers a more balanced perspective. Market demand and supply both matter at all times, although because the market is highly nonlinear, demand takes a more important role in slack times and supply in tight times. But the relative importance of supply and demand evolves continuously with the state of the market, instead of jumping from all demand to all supply when the nonclearing Walrasian market moves from excess supply to excess demand.

This is visible when we look at unemployment. The slackish model borrows the concepts of Keynesian and classical unemployment from the Old Keynesian model, but the advantage of the slackish model is that these unemployment components all occur at the same time and at all times with different relative importance. In the Old Keynesian model, by contrast, classical and Keynesian unemployment cannot coexist (Malinvaud 1977, figure 3).

In addition, the Old Keynesian model faces tricky theoretical challenges in dealing with the rationing that sellers or buyers face. Consider a situation of excess demand: the buyers expect to be able to buy anything they want at the market price, they plan accordingly, but when they get to the market they face rationing and are actually unable to buy what they planned to buy. The model must then explain how goods are rationed: by a lottery, by willingness to pay, or in some other way? The model must also explain how buyers might anticipate this situation in the first place, and how they may alter their behavior. A buyer who anticipates being rationed does not behave like someone who does not anticipate it.

In effect, the slackish model resolves this situation by introducing matching functions that mediate trades on the product and labor market, so there is always slack but no theoretical challenges. The matching function is the device that resolves the theoretical issues that plagued the Old Keynesian model. The matching function dictates what the trading probabilities are, and everyone takes them as given when they make decisions. So nobody is surprised at any point, and everyone anticipates the market conditions that are realized.

21.3.2. Connections to the Real Business Cycle model

The Real Business Cycle and slackish models are diametrically different. The Real Business Cycle model never has any slack or unemployment while the slackish model always has some slack and unemployment. The Real Business Cycle model generates business cycles from technology shocks while the slackish model generates business cycles from aggregate demand shocks. The Real Business Cycle model is generically efficient while the slackish model is generically inefficient. Prices are flexible and monetary policy is neutral in the Real Business Cycle while prices are rigid and monetary policy is nonneutral in the slackish model. In fact, prices are the main equilibrating variables in the Real Business Cycle model, while tightnesses are the main equilibrating variables in the slackish model. The Real Business Cycle paradigm emphasizes dynamic programming while we learn a lot about the slackish model through comparative-static analysis. The Real Business Cycle literature only uses stochastic models; we presented models that are deterministic and yet appear insightful. And we made almost all our arguments using graphs and diagrams, which would be heretical in the Real Business Cycle literature.

Nevertheless, the two models—Real Business Cycle and slackish—share some commonalities. Both feature price-taking households and firms. In both, all exchanges are always bilaterally efficient. Both aim to be as economical as possible, and favor theoretical cleanliness. The slackish model also shares with the Real Business Cycle model a desire for a unified framework, but over slack and tight states, or slumps and booms, rather than over economic growth and economic fluctuations.

21.3.3. Connections to the New Keynesian model

The slackish business cycle model also shares several commonalities with the New Keynesian model.

First, in both models, aggregate demand arises from the households' consumption and saving decisions, described by an Euler equation. However, we modify the Euler equation by introducing wealth in the utility function. In the New Keynesian model, the equilibrium aggregate demand is degenerate in that it does not involve any quantities—it just pins down a real interest rate. With wealth in the utility function, the equilibrium aggregate demand is nondegenerate: it describes output as a decreasing function of the real interest rate. As a result, the model behaves better, especially at the zero lower bound or under an interest-rate peg.

Second, monetary policy is nonneutral in the slackish model, just as in the New Keynesian model, and monetary policy continues to operate through the Euler equation. By adjusting the policy rate, the central bank influences households' consumption and saving decisions and, consequently, aggregate demand, just as in the New Keynesian model.

Third, just as in the New Keynesian model, monetary policy is nonneutral in the slackish model because prices are rigid. But pricing in the slackish model is very different from pricing in the New Keynesian model. The slackish model assumes that prices and wages are determined by price and wage norms instead of through Calvo pricing. Because the Calvo pricing assumption requires sellers to sell any quantity demanded at the current price—even if the price has not been reset in a very long time—a significant number of sellers are forced to engage in involuntary exchange (Huo and Rios-Rull 2020). In other words, many sellers are forced to sell at a loss: for monopolistic firms, it means selling goods at a price below marginal cost; for labor unions, it means selling labor at a wage below the marginal rate of substitution. By contrast, in the slackish model, all exchanges are voluntary: sellers and buyers who trade are always happy to do it.

The version of the slackish model presented in chapter 14 does feature a Phillips curve, and it uses the quadratic price-adjustment cost introduced by Rotemberg (1982) into the New Keynesian literature. However, the slackish Phillips curve is quite different from the New Keynesian Phillips curve. It links the unemployment gap to inflation instead of linking output gap to inflation. As such, it specifies how changes in slack translate into inflationary or disinflationary pressures, instead of changes in marginal costs (Gali and Gertler 1999).

Fourth, both the New Keynesian model and the slackish model are generally inefficient. In both cases, it is because prices and wages are rigid and can therefore not remain at their efficient level. And in both models, stabilization policies should reduce these inefficiencies as much as possible. However, in the New Keynesian model, measuring departures from efficiency requires measuring markups, which is notoriously difficult to do. In the slackish model, departures from efficiency are measured by tightness gaps, which can be computed in real time. Accordingly, the slackish perspective on inefficiency is more practical for policymaking.

A central difference between the two models is that the slackish business cycle model is simpler and more diagram-based than the New Keynesian apparatus. Krugman (2000, 2018) argues that policymakers still use the IS-LM model for their day-to-day thinking about policies because the New Keynesian model is just too complicated. His point is reinforced by the fact that popular undergraduate textbooks do not use the New Keynesian model to teach business cycles, but instead still rely on the IS-LM model (Abel, Bernanke, and Croushore 2017; Mankiw 2019).

Additionally, because the slackish product and labor markets are organized around a matching function, while the New Keynesian product and labor markets are monopolistically competitive, the two models operate very differently. Unlike in the slackish model, there is no slack in the New Keynesian model, as firms sell all their production and unions sell all their labor. Because firms and unions charge markups, they under-supply goods and labor, so the New Keynesian model is not well designed to think about tight, overheating economies. By contrast, in the slackish model, markets can be overly tight as easily as overly slack. Finally, the New Keynesian model is generally analyzed in a linear form, which features no state-dependence (Parker 2011). By contrast, because of the matching process, the slackish model is sharply nonlinear and hence sharply state-dependent: it behaves very differently in slack and tight times.

21.4. Kuhnian assessment of the slackish model

According to Kuhn, a good model has three qualities: it is descriptive, economical, and a good guide to the unknown. To conclude this book, we briefly assess the slackish model. This assessment is necessarily partial. Its purpose is to clarify the design decisions behind the slackish model, and to let readers judge for themselves how far it meets the Kuhnian qualities.

21.4.1. Is the slackish model descriptive?

The slackish business cycle model is designed to explain why slack always prevails and is sharply countercyclical. It aims to explain why buyers of goods and labor always coexist with sellers of goods and labor. It also aims to explain why price and wage norms prevail on markets, and why they are insulated from day-to-day market forces and shocks. The model also explains how slack fluctuations generate business cycle fluctuations in output and consumption even when the economy's productive capacity is acyclical. It also generates realistics Okun and Beveridge curves.

21.4.2. Is the slackish model economical?

One motivation throughout the book was to keep the framework as straightforward as possible so it could be represented and studied through simple diagrams: supply-and-demand diagrams in tightness-quantity planes to solve the model and perform comparative statics; Beveridge diagrams to study the efficiency properties of the model. The goal was to develop diagrammatic representations that are usable for teaching and policy work, and that are portable enough to keep in one's head and apply to new data and problems.

Besides diagrams, the book relies extensively on sufficient statistics when we design optimal monetary and fiscal policy. The aim here was to simplify the policy analysis, clarify the policy tradeoffs, and tighten the connection between policy theory and data.

21.4.3. Might the slackish model be fruitful?

A fruitful model guides empirical exploration, suggesting where to look and what might be there. It is too early to say how well the slackish model will do on that front. However, on a few occasions, the slackish model pointed to directions that led to interesting discoveries.

In chapter 12, we observed that US labor market tightness is systematically too high during major US wars: World War 2, the Korean War, and the Vietnam War. The only peacetime episode of an inefficiently tight labor market came in the aftermath of the pandemic. Motivated by this pattern, and by the high inflation observed during these episodes, Benigno and Eggertsson (2023) uncover a Phillips curve linking labor market tightness to inflation. Furthermore, the observed Phillips curve displays the inflation-slack coincidence that emerges in the slackish Phillips curve of chapter 14: inflation is on target when tightness is efficient while inefficiently high tightness generates excessively high inflation.

A prediction of the slackish labor market model of part III is that if a worker searches less for a job, the worker is less likely to find a job, but such reduced search will help other workers find jobs more rapidly. As in a rat race, as the worker moves down in the job queue, other workers move up ahead of them. This prediction prompted Lalive, Landais, and Zweimuller (2015) to look for spillovers in an Austrian natural experiment: a reform of the Austrian unemployment insurance system which extended the duration of benefits for selected workers in selected regions. And they did detect rat-race spillovers: workers who were not eligible for the higher benefits in treated regions had higher job-finding rates and lower unemployment durations than similar workers in nontreated regions.

Rat-race spillovers conversely imply that if a policy helps a worker find a job, this worker is more likely to find a job, but at the expense of other workers. In this case, the assisted worker moves ahead in the job queue and other workers move down. Moreover, the rat-race spillover is predicted to be weak in tight markets but strong in slack markets. Based on this prediction, Crepon et al. (2013, tables 6 and 10) looked for state-dependent rat-race spillovers in a large-scale double-randomization experiment that they had conducted in France, in which some jobseekers had received job-search assistance and some had not, and some local labor markets had been treated and some had not. They did find the predicted results: rat-race spillovers not only appeared, but they were stronger in local labor markets that were slacker.

Relatedly, in the slackish labor market model, unemployment insurance not only affects individual search effort but also labor market tightness. Tightness goes up when unemployment insurance increases, as the labor supply curve recedes along a downward-sloping labor demand curve. To grasp the complete effect of unemployment insurance, we should look not only at the response of search effort (which public economists had been doing for a long time), but also at the response of labor market tightness (which nobody had done). So, using a rich dataset with job vacancies and job applications from a large job board, Marinescu (2017) decided to estimate the impact of the unemployment benefits extensions enacted during the Great Recession on labor market tightness. As predicted, she found that labor market tightness went up when benefits were extended, and therefore that the macro effect of unemployment insurance was significantly less than the micro effect on which public economists had focused until then.

As a final example, the slackish labor market model says that public employment is more effective in bad times, when unemployment is high, than in good times. Hiring one public worker in bad times is more beneficial because public hiring crowds out private hiring less then. Holden and Sparrman (2016) used that prediction to examine the effect of government spending on employment in OECD countries and found that indeed government employment boosts employment more effectively when the labor market is slack than when it is tight.

Technical appendix

APPENDIX A.

Mathematical background

This appendix lists background mathematical results that we use repeatedly throughout the book.[1] These results are well known. They are presented, proved, generalized, and discussed in other texts, so we do not rederive them but simply collect them here for convenience. A complete treatment of the results is provided in the following textbooks: Apostol (1967) for calculus; Boyd and Vandenberghe (2004) for convexity of functions and convex optimization; Acemoglu (2009) for optimal control; and Hirsch, Smale, and Devaney (2013) for differential equations and dynamical systems.

A.1. Miscellaneous results

RESULT A.1 (Bolzano's theorem). *Consider a function f that is continuous on $[a, b]$. If $f(a)$ and $f(b)$ have opposite signs ($f(a)f(b) < 0$), then there exists $c \in (a, b)$ such that $f(c) = 0$. If, in addition, f is strictly monotone on $[a, b]$, then this zero is unique.*

RESULT A.2 (Euler's theorem for homogeneous functions). *Consider a differentiable function $f : (0, +\infty)^m \to (0, +\infty)$ that is homogeneous of degree $k > 0$, so that for all $x \in (0, +\infty)^m$ and $z > 0$:*

$$f(z \cdot x) = z^k \cdot f(x).$$

[1]The book assumes only basic familiarity with common functions (power, exponential, logarithmic, and so on), differential calculus (derivatives, linearization, and so on), and introductory probability and statistics (least-squares regressions, normal distribution, and so on). This appendix does not repeat these fundamentals; it collects results that are slightly more advanced.

Then the partial elasticities of the function add up to k:

$$\sum_{i=1}^{m} \epsilon_{x_i}^{f} = \sum_{i=1}^{m} \frac{x_i}{f(x)} \cdot \frac{\partial f}{\partial x_i} = k.$$

In the case of a function that has constant returns to scale (homogeneous of degree 1), the partial elasticities of f add up to 1:

$$\sum_{i=1}^{m} \epsilon_{x_i}^{f} = \sum_{i=1}^{m} \frac{x_i}{f(x)} \cdot \frac{\partial f}{\partial x_i} = 1.$$

RESULT A.3. *Consider a differentiable function $f : (0, +\infty)^2 \to (0, +\infty)$ that is homogeneous of degree $k > 0$, strictly increasing in each argument, and strictly quasiconcave. The ratio of partial derivatives*

$$\frac{\partial f / \partial y}{\partial f / \partial x}$$

depends on (x, y) only through the ratio y/x, and is a strictly decreasing function of y/x. The ratio of partial derivatives gives the slope of the level curves of f—the set of points such that $f(x, y) = c$ for some fixed $c > 0$. It is known as the marginal rate of substitution when f is a utility function and the level curve is an indifference curve, and as the marginal rate of technical substitution when f is a production function and the level curve is an isoquant.

RESULT A.4 (Poisson process). *A counting process $\{N(t), t \geq 0\}$ is a Poisson process with rate $\lambda > 0$ if $N(0) = 0$, and for any $t > 0$ and $h > 0$, the increments $N(t + h) - N(t)$ are independent over disjoint time intervals and they are Poisson distributed with parameter λh:*

$$\mathbb{P}(N(t + h) - N(t) = k) = \frac{\exp(-\lambda h) \cdot (\lambda h)^k}{k!},$$

for any $k = 0, 1, 2, \ldots$. This implies that over a short time interval dt:

$$\mathbb{P}(N(t + dt) - N(t) = 1) = \lambda\, dt + o(dt), \qquad \mathbb{P}(N(t + dt) - N(t) > 1) = o(dt).$$

Thus, the probability that one event occurs in the near future is always just $\lambda\, dt$. Furthermore, the waiting time to the next event, T, has an exponential distribution with parameter λ:

$$\mathbb{P}(T < t) = 1 - \exp(-\lambda t), \qquad \mathbb{P}(T > t) = \exp(-\lambda t).$$

This implies in particular that the waiting time to the next event is memoryless:

$$\mathbb{P}(T > s + t \mid T > s) = \mathbb{P}(T > t).$$

It also implies that the mean waiting time simply is:

$$\mathbb{E}(T) = \frac{1}{\lambda}.$$

RESULT A.5 (Spectral identities). *For any square real matrix, the trace is equal to the sum of its eigenvalues and the determinant is equal to the product of its eigenvalues, counting repeated eigenvalues. If the matrix is symmetric, all its eigenvalues are real. In general, a real matrix may have complex eigenvalues, but nonreal eigenvalues occur in conjugate pairs, so trace and determinant remain real.*

RESULT A.6 (Leibniz's integral rule). *Consider the integral*

$$I(z) = \int_{a(z)}^{b(z)} f(x,z)\,dx,$$

where x is the integration variable, z is a parameter, the functions a and b are differentiable, and the function f is continuously differentiable. Then the effect of a change in the parameter on the integral is given by:

$$\frac{dI}{dz} = \int_{a(z)}^{b(z)} \frac{\partial f}{\partial z}\,dx + \frac{db}{dz} f(b(z),z) - \frac{da}{dz} f(a(z),z).$$

RESULT A.7 (Implicit function theorem). *Consider a point (x_0, y_0) that satisfies $f(x_0, y_0) = 0$, where the function f is continuously differentiable in a neighborhood of (x_0, y_0) and*

$$\frac{\partial f}{\partial y}(x_0, y_0) \neq 0.$$

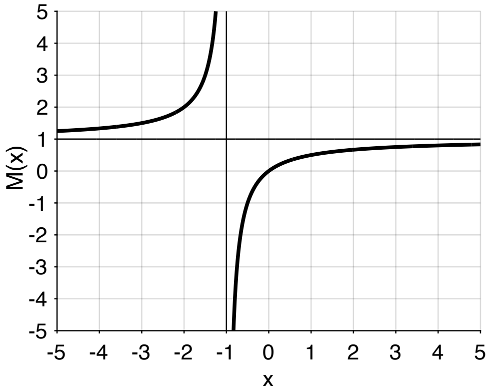
Then there exists a neighborhood of x_0 in which there is a unique differentiable function $y(x)$ such that $y(x_0) = y_0$ and $f(x, y(x)) = 0$. In that neighborhood, the derivative of y with respect to x satisfies

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dx} = 0.$$

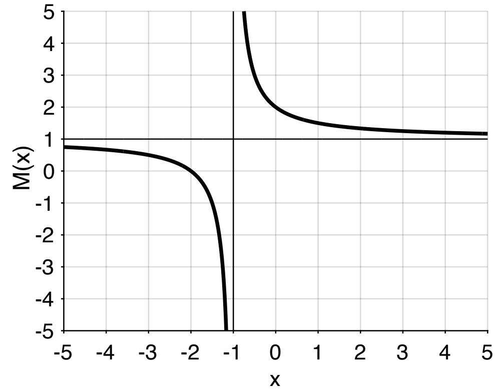
In other words, the derivative of y is determined by the partial derivatives of f:

$$\frac{dy}{dx} = -\frac{\partial f/\partial x}{\partial f/\partial y}.$$

RESULT A.8 (Maclaurin series of exponential). *The exponential function equals its Maclau-*



A. First type: $ad - bc > 0$



B. Second type: $ad - bc < 0$

FIGURE A.1. The two types of real Moebius transformations

Panel A plots $y = x/(x+1)$. Panel B plots $y = (x+2)/(x+1)$. For both, the pole is $x = -1$ and the asymptote is $y = 1$.

rin series (Taylor series at 0):

$$\exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

for all $x \in \mathbb{R}$. This is an exact identity, not a local approximation: because the exponential function is analytic on $\mathbb{R}$, this series converges to the function everywhere.

RESULT A.9 (Moebius transformation). *Consider the real Moebius transformation:*

$$M(x) = \frac{ax+b}{cx+d},$$

with $x \in \mathbb{R}$, $ad - bc \neq 0$ and $c \neq 0$.

- *The function M admits a single vertical asymptote at $x = -d/c$, where $M(x) \to \pm\infty$, and a single horizontal asymptote at $y = a/c$, when $x \to \pm\infty$.*
- *If $a \neq 0$, the function M admits a single root at $x = -b/a$, where $M(x) = 0$. If $a = 0$, the function has no roots and its horizontal asymptote is $y = 0$.*
- *If $ad - bc > 0$, the function M is strictly increasing and strictly convex on $(-\infty, -d/c)$ and strictly increasing and strictly concave on $(-d/c, +\infty)$ (as illustrated in figure A.1A).*
- *If $ad - bc < 0$, the function M is strictly decreasing and strictly concave on $(-\infty, -d/c)$ and strictly decreasing and strictly convex on $(-d/c, +\infty)$ (as illustrated in figure A.1B).*

A.2. Convexity of functions

The results below are adapted from Boyd and Vandenberghe (2004, chapter 3).

RESULT A.10. *Consider a function f defined on a convex set $\mathcal{X}$.*

- *The function f is convex if, for all $x, y \in \mathcal{X}$ and $\lambda \in [0,1]$,*

$$f(\lambda x + (1-\lambda)y) \le \lambda f(x) + (1-\lambda)f(y).$$

  Geometrically, the graph of f lies below the chord joining any two of its points. f is strictly convex if the inequality is strict whenever $x \neq y$ and $\lambda \in (0,1)$.

- *The function f is concave iff the function $-f$ is convex.*
- *The function f is strictly concave iff the function $-f$ is strictly convex.*

RESULT A.11. *Consider a function f defined on a convex set $\mathcal{X}$.*

- *The function f is quasiconcave if, for all $x, y \in \mathcal{X}$ and $\lambda \in [0,1]$,*

$$f(\lambda x + (1-\lambda)y) \ge \min(f(x), f(y)).$$

  Equivalently, every upper level set $\{x \in \mathcal{X} : f(x) \ge c\}$ is convex.

- *The function f is strictly quasiconcave if the inequality is strict whenever $x \neq y$ and $\lambda \in (0,1)$.*
- *The function f is quasiconvex iff the function $-f$ is quasiconcave.*
- *The function f is strictly quasiconvex iff the function $-f$ is strictly quasiconcave.*
- *Since a weighted average $\lambda f(x) + (1-\lambda)f(y)$ always lies above $\min(f(x), f(y))$, concavity implies quasiconcavity, and strict concavity implies strict quasiconcavity.*
- *Equivalently, convexity implies quasiconvexity, and strict convexity implies strict quasiconvexity.*

From result A.10, we can infer the concavity results below from the convexity results. Nevertheless, to improve user-friendliness, we state both convex and concave cases.

RESULT A.12. *For a twice-differentiable function of one variable, $f(x)$, defined on a convex set, concavity and convexity can be checked from its second derivative:*

- *The function is concave if $f''(x) \le 0$ on the set.*
- *The function is strictly concave if $f''(x) < 0$ on the set.*
- *The function is convex if $f''(x) \ge 0$ on the set.*
- *The function is strictly convex if $f''(x) > 0$ on the set.*

RESULT A.13. *The exponential function is strictly convex, so it is above all its tangents, and in* *particular above the tangent at $x = 0$:*

$$\exp(x) \geq 1 + x$$

for all $x \in \mathbb{R}$, with equality only at $x = 0$.

RESULT A.14. *For a twice-differentiable function of two variables, $f(x, y)$, defined on a convex set, concavity and convexity can be checked from its Hessian:*

$$\mathbf{H} = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}.$$

- *The function is concave if the Hessian is negative semidefinite on the set. This requires that the Hessian's two, real eigenvalues are negative, and in turn that its determinant is positive and its trace is negative.*
- *The function is strictly concave if the Hessian is negative definite. This requires that the Hessian's two, real eigenvalues are strictly negative, and in turn that its determinant is strictly positive and its trace is strictly negative.*
- *The function is convex if the Hessian is positive semidefinite on the set. This requires that the Hessian's two, real eigenvalues are positive, and in turn that its determinant is positive and its trace is positive.*
- *The function is strictly convex if the Hessian is positive definite. This requires that the Hessian's two, real eigenvalues are strictly positive, and in turn that its determinant is strictly positive and its trace is strictly positive.*

RESULT A.15. *Consider an affine function $g(x) = ax + b$ with $a \neq 0$.*

- *If f is convex, then $f \circ g$ is convex.*
- *If f is concave, then $f \circ g$ is concave.*
- *If f is strictly convex, then $f \circ g$ is strictly convex.*
- *If f is strictly concave, then $f \circ g$ is strictly concave.*
- *If f is convex and $a > 0$, then $g \circ f$ is convex.*
- *If f is concave and $a > 0$, then $g \circ f$ is concave.*
- *If f is strictly convex and $a > 0$, then $g \circ f$ is strictly convex.*
- *If f is strictly concave and $a > 0$, then $g \circ f$ is strictly concave.*

RESULT A.16. *The positive weighted sum of convex functions is convex. If any term in the sum* *is a strictly convex function with a strictly positive weight, the sum is strictly convex. Similarly, the positive weighted sum of concave functions is concave. If any term in the sum is a strictly concave function with a strictly positive weight, the sum is strictly concave.*

RESULT A.17. *Consider first two twice-differentiable functions $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$. The composition $f \circ g$ of the two functions has the following properties:*

- *If f is convex and increasing, and g is convex, then $f \circ g$ is convex.*
- *If f is convex and decreasing, and g is concave, then $f \circ g$ is convex.*
- *If f is concave and increasing, and g is concave, then $f \circ g$ is concave.*
- *If f is concave and decreasing, and g is convex, then $f \circ g$ is concave.*

If the two functions are defined not on $\mathbb{R}$ but on some convex subset of $\mathbb{R}$, the results continue to hold once we add a constraint on the domain of f:

- *If f is convex, f is increasing on an interval $(-\infty, a)$, and g is convex, then $f \circ g$ is convex.*
- *If f is convex, f is decreasing on an interval $(a, +\infty)$, and g is concave, then $f \circ g$ is convex.*
- *If f is concave, f is increasing on an interval $(a, +\infty)$, and g is concave, then $f \circ g$ is concave.*
- *If f is concave, f is decreasing on an interval $(-\infty, a)$, and g is convex, then $f \circ g$ is concave.*

RESULT A.18. *Consider a twice-differentiable bijection f. Then its inverse f^{-1} has the following properties:*

- *If f is strictly increasing and strictly convex, f^{-1} is strictly increasing and strictly concave.*
- *If f is strictly decreasing and strictly convex, f^{-1} is strictly decreasing and strictly convex.*
- *If f is strictly increasing and strictly concave, f^{-1} is strictly increasing and strictly convex.*
- *If f is strictly decreasing and strictly concave, f^{-1} is strictly decreasing and strictly concave.*

A.3. Convex optimization

Convex optimization is concerned with static optimization of convex (or concave) functions over convex sets. The results below are adapted from Boyd and Vandenberghe (2004, chapter 4).

RESULT A.19. *Consider the convex optimization problem:*

$$\min_{x \in \mathcal{X}} f(x) \tag{A.1}$$

If the objective function f is differentiable and convex and the feasible set $\mathcal{X} \subset \mathbb{R}^n$ is convex, then *any interior point satisfying the n first-order conditions*

$$\frac{\partial f}{\partial x_i} = 0, \quad i = 1, 2, \ldots, n,$$

is a global minimum. Furthermore, if the objective function is strictly convex, the global minimum is unique.

RESULT A.20. *Consider the convex optimization problem:*

$$\max_{x \in \mathcal{X}} f(x) \tag{A.2}$$

If the objective function f is differentiable and concave and the feasible set $\mathcal{X} \subset \mathbb{R}^n$ is convex, then any interior point satisfying the n first-order conditions

$$\frac{\partial f}{\partial x_i} = 0, \quad i = 1, 2, \ldots, n,$$

is a global maximum. Furthermore, if the objective function is strictly concave, the global maximum is unique.

RESULT A.21. *Consider the feasible set of an optimization problem described by m inequality constraints:*

$$f_i(x) \leq 0, \quad i = 1, 2, \ldots, m.$$

If the inequality-constraint functions f_i are convex, then the feasible set of the optimization problem is convex. For any interior point of the feasible set, the inequalities are not binding: $f_i(x) < 0$ for all i.

RESULT A.22. *If the optimization problem contains equality constraints, it is often convenient to eliminate them by substituting the appropriate number of variables out of the objective $f(x_1, x_2, \ldots, x_n)$. When this is possible, only inequality constraints are left after substitution, and the optimization problem can be written as in* (A.1) *or* (A.2). *The substitution approach generally works if the equality constraints are affine (of the form $\sum_{i=1}^{n} a_{i,k} x_i = b_k$) or take certain simple nonlinear forms. If the substitution approach is not available, or not convenient, one typically keeps the equality constraints and uses a Lagrangian approach.*

A.4. Optimal control

Optimal control is concerned with dynamic optimization in continuous time. The results in this section are adapted from Acemoglu (2009, chapter 7).

RESULT A.23. *Consider the optimal control problem with control variable $v_1(t), v_2(t), \ldots, v_n(t)$*

and state variable $w_1(t), w_2(t), \ldots, w_m(t)$:

$$\max_{v_1(t), v_2(t), \ldots, v_n(t), t \geq 0} \int_0^{\infty} e^{-\rho t} f(v_1(t), v_2(t), \ldots, w_1(t), w_2(t), \ldots, t)\, dt \tag{A.3}$$

subject to the m laws of motion

$$\dot{w}_i(t) = g_i(v_1(t), v_2(t), \ldots, w_1(t), w_2(t), \ldots, t), \quad i = 1, 2, \ldots, m,$$

and given initial conditions and admissibility constraints. The discount rate ρ is strictly positive, and the functions f and g_i are twice continuously differentiable in state and control arguments, and continuous in time. Then the current-value Hamiltonian is

$$\mathcal{H}(t) = f(v_1(t), \ldots, w_1(t), \ldots, t) + \sum_{i=1}^{m} q_i(t) \cdot g_i(v_1(t), \ldots, w_1(t), \ldots, t),$$

where $q_1(t), q_2(t), \ldots, q_m(t)$ are the costate variables for the state variables. Just like in result A.22, we assume here that all equality constraints have been substituted out of the optimization problem. If the substitution is not possible or convenient, extra Lagrangian terms must be added to the Hamiltonian.

RESULT A.24. *For the optimal control problem* (A.3)*, the necessary conditions for an interior maximum are the Pontryagin first-order conditions:*

$$\frac{\partial \mathcal{H}}{\partial v_j} = 0, \quad j = 1, 2, \ldots, n \tag{A.4}$$

$$\frac{\partial \mathcal{H}}{\partial w_i} = \rho q_i(t) - \dot{q}_i(t), \quad i = 1, 2, \ldots, m. \tag{A.5}$$

In addition, appropriate transversality conditions rule out explosive paths that violate optimality. A typical transversality condition is

$$\lim_{t \to \infty} \exp(-\rho t) q_i(t) w_i(t) = 0.$$

The transversality condition rules out asymptotically wasteful paths by requiring that the discounted shadow value of remaining state variables vanishes as $t \to \infty$.

RESULT A.25. *Consider the optimal control problem* (A.3)*, and assume that the laws of motion of the state variables are affine:*

$$\dot{\mathbf{w}}(t) = \mathbf{A}(t)\mathbf{w}(t) + \mathbf{B}(t)\mathbf{v}(t) + \mathbf{u}(t),$$

where $\mathbf{w}(t) = [w_1(t), \ldots, w_m(t)] \in \mathbb{R}^m$ is the vector of state variables, $\mathbf{v}(t) = [v_1(t), \ldots, v_n(t)] \in \mathbb{R}^n$ is the vector of control variables, and $\mathbf{u}(t) = [u_1(t), \ldots, u_m(t)] \in \mathbb{R}^m$ is a vector of forcing *terms in the laws of motion. Suppose that:*

- *The set of admissible controls and states is convex.*
- *For each t, the objective function $f(\mathbf{v}, \mathbf{w}, t)$ is concave in $(\mathbf{v}, \mathbf{w})$.*
- *Transversality conditions are satisfied.*

An admissible interior path is any path for states and controls that satisfies the initial conditions and laws of motion, stays strictly inside the feasible set at every date, and satisfies any additional admissibility conditions (such as no-Ponzi conditions). Then any admissible interior path $(\mathbf{v}(t), \mathbf{w}(t))$ that satisfies the Pontryagin first-order conditions is a global maximum. If the objective function is strictly concave, the optimum is unique.

RESULT A.26. *Consider the optimal control problem* (A.3)*. Suppose that:*

- *The set of admissible controls and states is convex.*
- *For each t, and along the relevant costate path, the Hamiltonian $\mathcal{H}(\mathbf{v}, \mathbf{w}, \mathbf{q}(t), t)$ is concave in $(\mathbf{v}, \mathbf{w})$.*
- *Transversality conditions are satisfied.*

Then any admissible interior path that satisfies the Pontryagin first-order conditions is a global maximum. If the Hamiltonian is strictly concave in $(\mathbf{v}, \mathbf{w})$, the optimum is unique.

A.5. Differential equations

Differential equations are functional equations in continuous time.

RESULT A.27. *The exponential function is defined as the only solution to the differential equation*

$$\dot{x}(t) - x(t) = 0,$$

with initial condition $x(0) = 1$. Hence, the only solution to the homogeneous autonomous linear first-order differential equation

$$\dot{x}(t) - \lambda x(t) = 0,$$

with initial condition $x(t_0) = x_0$, is

$$x(t) = x_0 \exp(\lambda (t - t_0)).$$

More generally, the only solution to the nonhomogeneous autonomous linear first-order differential equation

$$\dot{x}(t) - \lambda x(t) = \phi,$$

with initial condition $x(t_0) = x_0$, is

$$x(t) = x^* + \left(x_0 - x^*\right) \exp(\lambda\,(t - t_0)),$$

where $x^* = -\phi/\lambda$ is the equilibrium of the differential equation. The equilibrium x^* is a source when $\lambda > 0$ and a sink when $\lambda < 0$.

A.6. Dynamical systems

Dynamical systems are collections of differential equations. The results in this section are adapted from Hirsch, Smale, and Devaney (2013, chapters 1–6).

A.6.1. Linear dynamical system

We consider a two-dimensional homogeneous autonomous linear first-order dynamical system:

$$\begin{aligned}\dot{x}(t) &= a \cdot x(t) + b \cdot y(t)\\ \dot{y}(t) &= c \cdot x(t) + d \cdot y(t).\end{aligned}$$

The matrix governing the linear system is

$$\mathbf{M} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}. \tag{A.6}$$

We denote by $\lambda_1 \in \mathbb{C}$ and $\lambda_2 \in \mathbb{C}$ the two eigenvalues of $\mathbf{M}$, assumed to be nonzero and distinct. We denote by $\mathbf{v}_1 \in \mathbb{C}^2$ and $\mathbf{v}_2 \in \mathbb{C}^2$ the linearly independent eigenvectors associated with the eigenvalues λ_1 and λ_2.

RESULT A.28. *Assume that λ_1 and λ_2 are real. Without loss of generality, we assume $\lambda_1 < \lambda_2$. Then the solution to the linear system takes the form*

$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \alpha_1 \exp(\lambda_1 t)\mathbf{v}_1 + \alpha_2 \exp(\lambda_2 t)\mathbf{v}_2, \tag{A.7}$$

where $\alpha_1 \in \mathbb{R}$ and $\alpha_2 \in \mathbb{R}$ are constants determined by boundary conditions. The system exhibits nodal dynamics:

- *The system is a source when $\lambda_1 > 0$ and $\lambda_2 > 0$. The solutions are tangent to $\mathbf{v}_1$ when $t \to -\infty$ and are parallel to $\mathbf{v}_2$ when $t \to +\infty$.*
- *The system is a sink when $\lambda_1 < 0$ and $\lambda_2 < 0$. The solutions are tangent to $\mathbf{v}_1$ when $t \to -\infty$*

and are parallel to $\mathbf{v}_2$ *when* $t \to +\infty$.

- *The system is a saddle when* $\lambda_1 < 0$ *and* $\lambda_2 > 0$. *The vector* $\mathbf{v}_1$ *gives the direction of the stable line (saddle path) while the vector* $\mathbf{v}_2$ *gives the direction of the unstable line.*

In that way, in all cases, trajectories move from the $\mathbf{v}_1$*-direction to the* $\mathbf{v}_2$*-direction as time increases.*

RESULT A.29. *Assume that* λ_1 *and* λ_2 *are complex conjugates. We write the eigenvalues as* $\lambda_1 = \theta + i\vartheta$ *and* $\lambda_2 = \theta - i\vartheta$. *We also write the eigenvector associated with* λ_1 *as* $\mathbf{v}_1 + i\mathbf{v}_2$, *where the vectors* $\mathbf{v}_1 \in \mathbb{R}^2$ *and* $\mathbf{v}_2 \in \mathbb{R}^2$ *are linearly independent. Then the solution to the linear system takes the form*

$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \exp(\theta t)\,[\mathbf{v}_1, \mathbf{v}_2] \begin{bmatrix} \cos(\vartheta t) & \sin(\vartheta t) \\ -\sin(\vartheta t) & \cos(\vartheta t) \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix},$$

where $[\mathbf{v}_1, \mathbf{v}_2] \in \mathbb{R}^{2\times 2}$ *is a* 2×2 *matrix, and* $\alpha_1 \in \mathbb{R}$ *and* $\alpha_2 \in \mathbb{R}$ *are constants determined by boundary conditions. The system exhibits spiral dynamics:*

- *The system is a spiral source if* $\theta > 0$. *Then the solutions wind periodically around the origin, moving away from it.*
- *The system is a spiral sink if* $\theta < 0$. *Then the solutions wind periodically around the origin, moving toward it.*
- *The system is a center if* $\theta = 0$. *Then the solutions circle around the origin.*

RESULT A.30. *The discriminant of the matrix* $\mathbf{M}$,

$$\Delta = \mathrm{tr}(\mathbf{M})^2 - 4\det(\mathbf{M}), \tag{A.8}$$

distinguishes the cases with real and complex eigenvalues:

- *If* $\Delta > 0$, *the eigenvalues are real and the system exhibits nodal dynamics.*
- *If* $\Delta < 0$, *the eigenvalues are complex and the system exhibits spiral or center dynamics.*

The trace and determinant of the matrix $\mathbf{M}$ *further help classify the system dynamics:*

- *If* $\det(\mathbf{M}) < 0$, *eigenvalues are real and have opposite signs, so the system is a saddle.*
- *If* $\det(\mathbf{M}) > 0$ *and* $\mathrm{tr}(\mathbf{M}) < 0$, *both eigenvalues have negative real parts, so the system is a sink.*
- *If* $\det(\mathbf{M}) > 0$ *and* $\mathrm{tr}(\mathbf{M}) > 0$, *both eigenvalues have positive real parts, so the system is a source.*
- *If* $\det(\mathbf{M}) > 0$ *and* $\mathrm{tr}(\mathbf{M}) = 0$, *both eigenvalues are purely imaginary, so the system is a*

center.

The dynamics are degenerate in all other cases:

- *If* $\Delta = 0$*, the eigenvalues are real and repeated, so the system exhibits degenerate nodal dynamics.*
- *If* $\det(\mathbf{M}) = 0$*, at least one of the eigenvalues is 0, so the system exhibits degenerate dynamics.*

RESULT A.31. *A nonhomogeneous autonomous linear first-order dynamical system can be solved using the results on homogeneous systems. Consider the nonhomogeneous system* $\dot{\mathbf{v}}(t) = \mathbf{M}\mathbf{v}(t) - \mathbf{u}$*, where* $\mathbf{v}(t) \in \mathbb{R}^2$*,* $\mathbf{M} \in \mathbb{R}^{2\times 2}$ *with* $\det \mathbf{M} \neq 0$*, and* $\mathbf{u} \in \mathbb{R}^2$ *with* $\mathbf{u} \neq 0$*. The equilibrium point of the system is* $\mathbf{v}^* = \mathbf{M}^{-1}\mathbf{u}$*. Solving the nonhomogeneous system is equivalent to solving the homogeneous system* $\dot{\mathbf{w}}(t) = \mathbf{M}\mathbf{w}(t)$ *and then recovering the variable* $\mathbf{v}(t) = \mathbf{w}(t) + \mathbf{v}^*$*.*

A.6.2. Phase diagrams

Without solving for eigenvalues and eigenvectors explicitly, we can study the properties of a two-dimensional autonomous linear first-order dynamical system by drawing its phase diagram.

Here we construct the phase diagram for a nonhomogeneous dynamical system:

$$\dot{x}(t) = a \cdot x(t) + b \cdot y(t) + w \tag{A.9}$$

$$\dot{y}(t) = c \cdot x(t) + d \cdot y(t) + z, \tag{A.10}$$

Let $\mathbf{M}$ be the matrix associated with the system, given by (A.6). We assume that the eigenvalues of the system are nonzero, real, and distinct (so the discriminant of $\mathbf{M}$ is strictly positive and its determinant is nonzero). This implies that the dynamics of the system are nodal and nondegenerate.

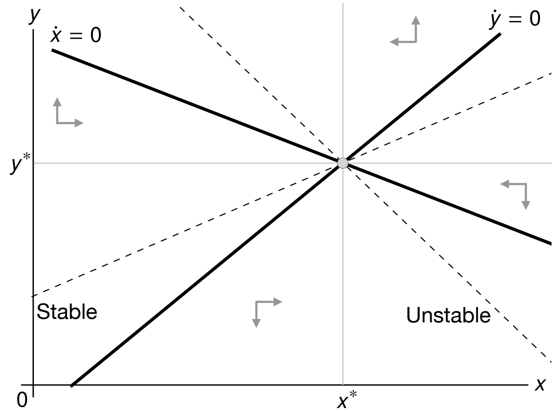
Drawing the phase diagram of a two-variable system is useful to understand the main features of the dynamical system without solving for $x(t)$ and $y(t)$ explicitly. Figure A.2 illustrates the construction of the phase diagram with six panels: A.2A–A.2B for the saddle case, A.2C–A.2D for the source case, and A.2E–A.2F for the sink case.

We first plot the nullclines, which are the loci $\dot{x} = 0$ and $\dot{y} = 0$. The locus for $\dot{x} = 0$ is a straight line given by

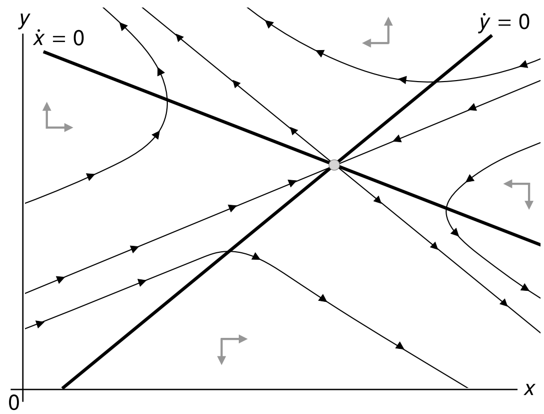
$$ax + by + w = 0. \tag{A.11}$$

The locus for $\dot{y} = 0$ is a straight line given by

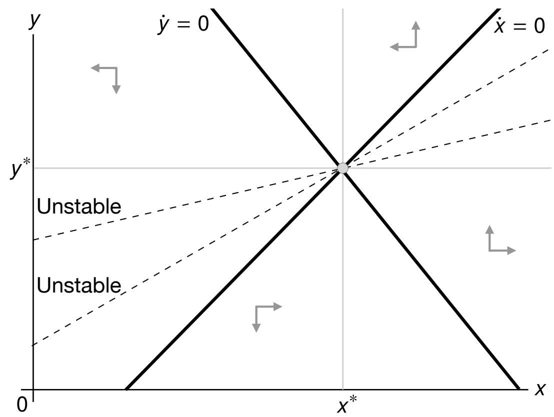
$$cx + dy + z = 0. \tag{A.12}$$



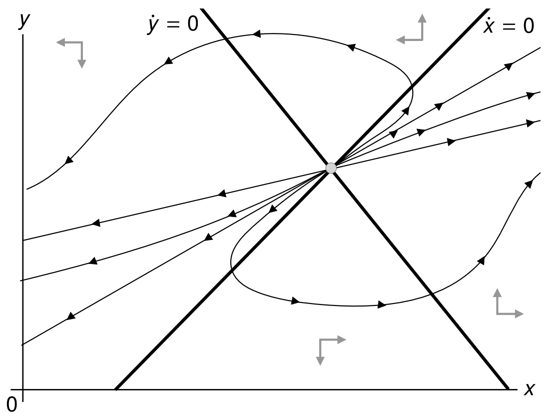
A. Saddle structure



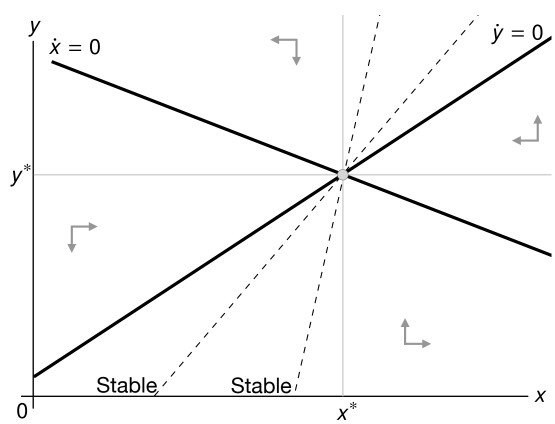
B. Saddle trajectories



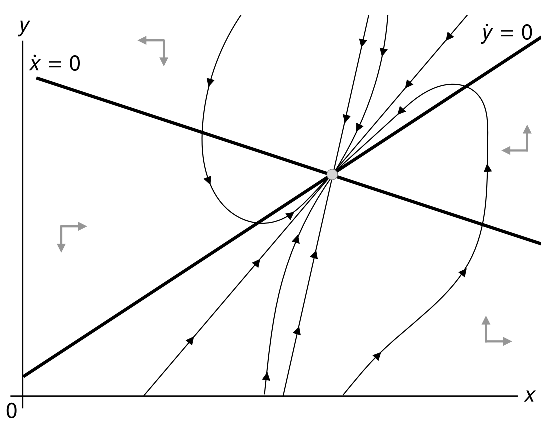
C. Source structure



D. Source trajectories



E. Sink structure



F. Sink trajectories

FIGURE A.2. Phase diagram for a two-dimensional linear dynamical system

In these illustrative phase diagrams, we assume that the eigenvalues of the system are real and distinct and nonzero, so the dynamics are nodal and nondegenerate.

Next we place the equilibrium point, (x^*, y^*). The equilibrium is given by the intersection of the two nullclines:

$$x^* = \frac{bz - dw}{ad - bc},$$
$$y^* = \frac{cw - az}{ad - bc}.$$

These two nullclines partition the plane into four regions where the signs of $\dot{x}$ and $\dot{y}$ are constant. In each quadrant, the solutions have specific directions: going up or down and going left or right. We describe the direction field with directional arrows in the diagram. These arrows determine the direction of the solution's trajectories over time anywhere on the phase diagram. For example, from (A.9), we see that any point above the $\dot{x} = 0$ line (given by (A.11)) has $\dot{x} > 0$ if $b > 0$ and $\dot{x} < 0$ if $b < 0$. Conversely, any point below the $\dot{x} = 0$ line has $\dot{x} < 0$ if $b > 0$ and $\dot{x} > 0$ if $b < 0$. We represent these properties by a horizontal arrow pointing east for any point with $\dot{x} > 0$ and a horizontal arrow pointing west for any point with $\dot{x} < 0$.

Similarly, from (A.10), we see that any point above the $\dot{y} = 0$ line (given by (A.12)) has $\dot{y} > 0$ if $d > 0$ and $\dot{y} < 0$ if $d < 0$. Conversely, any point below the $\dot{y} = 0$ line has $\dot{y} < 0$ if $d > 0$ and $\dot{y} > 0$ if $d < 0$. We represent these properties by a vertical arrow pointing north for any point with $\dot{y} > 0$ and a vertical arrow pointing south for any point with $\dot{y} < 0$.

We then draw the stable or unstable lines through the equilibrium point. These lines, corresponding to the eigenvectors of $\mathbf{M}$, organize all trajectories. Indeed, the stable and unstable lines act as separatrices that cannot be crossed: any solution remains in a region of the plane delimited by the lines.

Finally, using the directional arrows and stable and unstable lines, we can draw trajectories that satisfy the system of differential equations. These are solutions to the system. To select a specific solution among all possible solutions, we would need to specify an initial condition.

If the system is a saddle, the stable line is the set of initial conditions that converge to the equilibrium as $t \to +\infty$, while the unstable line is the set that converges to the equilibrium as $t \to -\infty$. Trajectories are tangent to the stable line as $t \to -\infty$ and tangent to the unstable line as $t \to +\infty$.

If the system is a source, trajectories move away from equilibrium in forward time. All the trajectories are tangent to one unstable direction as $t \to -\infty$ (the one associated with the smallest eigenvalue) and tangent to the other unstable direction as $t \to +\infty$ (the one associated with the largest eigenvalue).

If the system is a sink, trajectories move toward equilibrium in forward time. All the trajectories are tangent to one stable direction as $t \to -\infty$ (the one associated with the lowest eigenvalue) and tangent to the other stable direction as $t \to +\infty$ (the one associated with the highest eigenvalue).

A.6.3. Nonlinear dynamical system

The methodology that we developed for two-dimensional linear dynamical systems is also useful to study two-dimensional nonlinear dynamical systems.

Consider the autonomous nonlinear system

$$\dot{x}(t) = f(x(t), y(t))$$
$$\dot{y}(t) = g(x(t), y(t)).$$

An equilibrium (x^*, y^*) is defined by

$$f(x^*, y^*) = 0, \qquad g(x^*, y^*) = 0.$$

To study local dynamics around (x^*, y^*), define deviations

$$u(t) = x(t) - x^*, \qquad v(t) = y(t) - y^*.$$

A first-order linearization around the equilibrium gives

$$\begin{bmatrix} \dot{u}(t) \\ \dot{v}(t) \end{bmatrix} = J(x^*, y^*) \begin{bmatrix} u(t) \\ v(t) \end{bmatrix},$$

where the Jacobian matrix at (x^*, y^*) is defined by

$$J(x^*, y^*) = \begin{bmatrix} \frac{\partial f}{\partial x}(x^*, y^*) & \frac{\partial f}{\partial y}(x^*, y^*) \\ \frac{\partial g}{\partial x}(x^*, y^*) & \frac{\partial g}{\partial y}(x^*, y^*) \end{bmatrix}.$$

Therefore, the local behavior of the nonlinear system near the equilibrium (x^*, y^*) is determined by the Jacobian $J(x^*, y^*)$, as long as the equilibrium is hyperbolic (no eigenvalue with zero real part). In particular, results A.28 and A.29 describe the solutions $(u(t), v(t))$ of the linearized system based on the eigenvalues and eigenvectors of the Jacobian. The signs of the discriminant, trace, and determinant of the Jacobian classify the equilibrium locally as saddle, sink, or source, as described by result A.30. It is also possible to build a phase diagram using the Jacobian, just as in the linear case described in figure A.2. The phase diagram provides a local approximation of the nonlinear system around the equilibrium.

APPENDIX B.

Elasticities

In this appendix, we review the concept of elasticity, which we use repeatedly throughout the book, and we provide a set of results that are useful to handle elasticities. Throughout we consider a function $f(x)$ that is strictly positive and differentiable for $x > 0$.

B.1. Definition of elasticities

The elasticity of $f(x)$ with respect to x, denoted ϵ_x^f, is defined as

$$\epsilon_x^f = \frac{x}{f(x)} \cdot f'(x), \tag{B.1}$$

where $f'(x)$ is the standard derivative of f. All properties of elasticities follow from this definition.

B.2. Leibniz notation for differentials

To work with elasticities, it is convenient to introduce Leibniz's notation for differentials. In this notation, dx denotes an infinitesimal change in a variable x, and $df(x)$ denotes an infinitesimal change in a function $f(x)$. The differential $df(x)$ is related to the differential dx and the derivative $f'(x)$ of the function f by

$$df(x) = f'(x)\, dx.$$

From this definition, we see that the differential operator d obeys the same algebraic rules as derivatives.

The Leibniz notation is particularly convenient to use with the logarithm. Given that the derivative of $\ln(x)$ is $1/x$:

$$d\ln(x) = \frac{dx}{x}. \tag{B.2}$$

Hence, while dx measures the absolute change in x, $d\ln(x)$ measures a relative change in x.

Applying the logarithm to the function f, we obtain

$$d\ln f(x) = \frac{df(x)}{f(x)}. \tag{B.3}$$

So while $df(x)$ measures the absolute change in $f(x)$, $d\ln f(x)$ measures a relative change in $f(x)$.

These equations show that the log-differentials $d\ln(x)$ and $d\ln f(x)$ are simply a compact way of writing the relative change in the variable x or the function $f(x)$.

B.3. Interpretation of elasticities

What does an elasticity mean? To answer this, we rewrite the definition (B.1) using the Leibniz notation $f'(x) = df/dx$:

$$\epsilon_x^f = \frac{x}{f(x)} \cdot \frac{df(x)}{dx} = \frac{df(x)/f(x)}{dx/x}. \tag{B.4}$$

We see from this equation that the elasticity is the ratio of the relative change in f to the relative change in x.

This gives a simple interpretation. Consider a 1% change in x, which means its relative change is $dx/x = 0.01$. The resulting relative change in $f(x)$ is:

$$\frac{df(x)}{f(x)} = \epsilon_x^f \times \frac{dx}{x} = \epsilon_x^f \times 0.01.$$

Thus, a 1% change in x leads to an ϵ_x^f percentage change in $f(x)$: the elasticity measures the percentage response of a function to a 1% change in its input.

B.4. Connection between elasticities and log-differentials

Elasticities are so convenient because of their connection to natural logarithms. This connection comes from a basic result in calculus: the derivative of $\ln(x)$ is $1/x$. In Leibniz's notation, this means $d\ln(x) = dx/x$, as we saw in equation (B.2). Using this insight and expression (B.4), we can express the elasticity as the ratio of two log-differentials. This is not a new definition, but a convenient reformulation of (B.1):

$$\epsilon_x^f = \frac{d\ln f(x)}{d\ln(x)}. \tag{B.5}$$

This logarithmic form often helps compute the elasticity of complex functions.

Reshuffling (B.5), we also see that the elasticity relates the log-differential of the function to the log-differential of the argument:

$$d\ln f(x) = \epsilon_x^f \, d\ln(x).$$

B.5. Useful results on elasticities

We now introduce results that we use in the book to compute elasticities. Throughout, the functions $g(x)$ and $h(x)$ are assumed to be strictly positive and differentiable for $x > 0$.

RESULT B.1. *For any real a, the elasticity of the function $f(x) = x^a$ is $\epsilon_x^f = a$.*

PROOF. Since $\ln(f(x)) = a\ln(x)$, differentiating gives $d\ln(f(x)) = a\,d\ln(x)$. Dividing by $d\ln(x)$, we get $d\ln(f(x))/d\ln(x) = a$. We obtain the result by applying (B.5). □

RESULT B.2. *For any $a > 0$, the elasticity of the function $f(x) = a \cdot g(x)$ is $\epsilon_x^f = \epsilon_x^g$.*

PROOF. Since $\ln(f(x)) = \ln(a) + \ln(g(x))$, and $\ln(a)$ is just a constant, then differentiating gives $d\ln(f(x)) = d\ln(g(x))$, so $d\ln(f(x))/d\ln(x) = d\ln(g(x))/d\ln(x)$. Once again, we obtain the result by applying (B.5). □

RESULT B.3. *The elasticity of the function $f(x) = g(x) \cdot h(x)$ is $\epsilon_x^f = \epsilon_x^g + \epsilon_x^h$.*

PROOF. Since $\ln(f(x)) = \ln(g(x)) + \ln(h(x))$, then $d\ln(f(x)) = d\ln(g(x)) + d\ln(h(x))$. We reach the result after dividing by $d\ln(x)$ and using (B.5). □

RESULT B.4. *The elasticity of the function $f(x) = g(x)/h(x)$ is $\epsilon_x^f = \epsilon_x^g - \epsilon_x^h$.*

PROOF. Since $\ln(f(x)) = \ln(g(x)) - \ln(h(x))$, then $d\ln(f(x)) = d\ln(g(x)) - d\ln(h(x))$. Again, we obtain the result after dividing by $d\ln(x)$ and using (B.5). □

RESULT B.5. *The elasticity of the function $f(x) = g(x) + h(x)$ is*

$$\epsilon_x^f = \frac{g(x)}{g(x) + h(x)} \cdot \epsilon_x^g + \frac{h(x)}{g(x) + h(x)} \cdot \epsilon_x^h.$$

PROOF. For sums of functions, the logarithmic definition of elasticities is not helpful, so we return to the original definition, given by (B.1). Since $f'(x) = g'(x) + h'(x)$, we get

$$\begin{aligned}
\epsilon_x^f &= \frac{x}{f(x)} \cdot f'(x) \\
&= \frac{x}{g(x) + h(x)} \left[g'(x) + h'(x) \right] \\
&= \frac{g(x)}{g(x) + h(x)} \left[\frac{x}{g(x)} \cdot g'(x) \right] + \frac{h(x)}{g(x) + h(x)} \left[\frac{x}{h(x)} \cdot h'(x) \right].
\end{aligned}$$

We obtain the result from here, as the definition of elasticities also gives $\epsilon_x^g = xg'(x)/g(x)$ and $\epsilon_x^h = xh'(x)/h(x)$. □

RESULT B.6. *The elasticity of the function $f(x) = g(x) - h(x)$, with $h(x) < g(x)$, is*

$$\epsilon_x^f = \frac{g(x)}{g(x) - h(x)} \cdot \epsilon_x^g - \frac{h(x)}{g(x) - h(x)} \cdot \epsilon_x^h.$$

PROOF. Here again, we apply the elasticity definition (B.1). Since $f'(x) = g'(x) - h'(x)$, we get

$$\epsilon_x^f = \frac{x}{g(x) - h(x)} \left[g'(x) - h'(x) \right] = \frac{g(x)}{g(x) - h(x)} \left[\frac{x}{g(x)} \cdot g'(x) \right] - \frac{h(x)}{g(x) - h(x)} \left[\frac{x}{h(x)} \cdot h'(x) \right],$$

which again yields the result. □

RESULT B.7. *The elasticity of the function $f(x) = g(h(x))$ is*

$$\epsilon_x^f = \epsilon_x^h \cdot \epsilon_h^g,$$

where ϵ_h^g is the elasticity of g evaluated at $h(x)$.

PROOF. By the chain rule, $f'(x) = h'(x) \cdot g'(h(x))$. Hence, using the elasticity definition (B.1), we get

$$\epsilon_x^f = \frac{x}{g(h(x))} \cdot h'(x) \cdot g'(h(x)) = \left[\frac{x}{h(x)} \cdot h'(x) \right] \left[\frac{h(x)}{g(h(x))} \cdot g'(h(x)) \right].$$

The result follows from here as $\epsilon_x^h = xh'(x)/h(x)$ and $\epsilon_h^g = h(x)g'(h(x))/g(h(x))$. □

RESULT B.8. *Assume that the function $g(y)$ is invertible with differentiable inverse. The elasticity of the function $f(x) = g^{-1}(x)$ is*

$$\epsilon_x^f = \frac{1}{\epsilon_y^g},$$

where ϵ_y^g is the elasticity of g evaluated at $y = g^{-1}(x)$.

PROOF. By the inverse function theorem, $f'(x) = 1/g'(g^{-1}(x))$. Hence, using the elasticity definition (B.1), we get

$$\epsilon_x^f = \frac{x}{g^{-1}(x)} \cdot \frac{1}{g'(g^{-1}(x))}.$$

The result follows because $\epsilon_y^g = yg'(y)/g(y)$ with $y = g^{-1}(x)$. □

Finally, we introduce a bivariate function $k(y,z)$ which is strictly positive and differentiable for $y > 0$ and $z > 0$. We also assume that the arguments y and z are themselves functions of x, and the functions $y(x)$ and $z(x)$ are strictly positive and differentiable for $x > 0$.

RESULT B.9. *The elasticity of the function $f(x) = k(y(x), z(x))$ is*

$$\epsilon_x^f = \epsilon_y^k \cdot \epsilon_x^y + \epsilon_z^k \cdot \epsilon_x^z,$$

where ϵ_y^k and ϵ_z^k are partial elasticities of the function k:

$$\epsilon_y^k = \frac{y}{k(y,z)} \cdot \frac{\partial k}{\partial y} \quad \text{and} \quad \epsilon_z^k = \frac{z}{k(y,z)} \cdot \frac{\partial k}{\partial z}.$$

PROOF. By the multivariate chain rule,

$$f'(x) = \frac{\partial k}{\partial y} \cdot y'(x) + \frac{\partial k}{\partial z} \cdot z'(x).$$

Hence, using the elasticity definition (B.1), we get

$$\epsilon_x^f = \frac{x}{k(y,z)} \left[\frac{\partial k}{\partial y} \cdot y'(x) + \frac{\partial k}{\partial z} \cdot z'(x)\right] = \left[\frac{y}{k(y,z)} \cdot \frac{\partial k}{\partial y}\right]\left[\frac{x}{y} \cdot y'(x)\right] + \left[\frac{z}{k(y,z)} \cdot \frac{\partial k}{\partial z}\right]\left[\frac{x}{z} \cdot z'(x)\right].$$

We obtain the result from this equation and the definitions of the standard and partial elasticities. □

B.6. Implicit differentiation with elasticities

Implicit differentiation is very helpful to compute derivatives of functions that are defined implicitly. It turns out that we can also use the same technique to compute elasticities of functions defined implicitly.

RESULT B.10. *Assume that the function $y(x) > 0$ is defined implicitly by the equation $f(x, y) =$*

$g(x,y)$, where the bivariate functions $f(x,y)$ and $g(x,y)$ are strictly positive and differentiable for $x > 0$ and $y > 0$. Taking elasticities on both sides of the equation, the elasticity of y with respect to x satisfies

$$\epsilon_x^f + \epsilon_y^f \cdot \epsilon_x^y = \epsilon_x^g + \epsilon_y^g \cdot \epsilon_x^y,$$

where ϵ_x^f, ϵ_y^f, ϵ_x^g, ϵ_y^g, are partial elasticities defined by

$$\begin{aligned} \epsilon_x^f &= \frac{x}{f(x,y)} \cdot \frac{\partial f}{\partial x}, & \epsilon_y^f &= \frac{y}{f(x,y)} \cdot \frac{\partial f}{\partial y}, \\ \epsilon_x^g &= \frac{x}{g(x,y)} \cdot \frac{\partial g}{\partial x}, & \epsilon_y^g &= \frac{y}{g(x,y)} \cdot \frac{\partial g}{\partial y}. \end{aligned}$$

Collecting terms, we can express the elasticity of y as a function of the partial elasticities of the functions f and g:

$$\epsilon_x^y = \frac{\epsilon_x^g - \epsilon_x^f}{\epsilon_y^f - \epsilon_y^g}.$$

PROOF. As result A.7 states, the implicit differentiation of the equation $f(x,y) = g(x,y)$ gives

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot y'(x) = \frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} \cdot y'(x).$$

We then divide the left-hand side by $f(x,y)$ and the right-hand side by $g(x,y)$, which we can do because $f(x,y) = g(x,y)$, and we multiply both sides by x. We get

$$\frac{x}{f(x,y)} \cdot \frac{\partial f}{\partial x} + \left[\frac{y}{f(x,y)} \cdot \frac{\partial f}{\partial y}\right]\left[\frac{x}{y} \cdot y'(x)\right] = \frac{x}{g(x,y)} \cdot \frac{\partial g}{\partial x} + \left[\frac{y}{g(x,y)} \cdot \frac{\partial g}{\partial y}\right]\left[\frac{x}{y} \cdot y'(x)\right].$$

We obtain the result from this equation and the definitions of the standard and partial elasticities. □

APPENDIX C.

Hodrick-Prescott filter

Several of the variables that we examine in the book grow over time, or are otherwise influenced by slow-moving factors. To isolate cyclical fluctuations of these variables, we first need to remove their trends. This appendix explains how we detrend these time series.

In the book, we detrend time series using the Hodrick-Prescott (HP) filter. We pick the HP filter because it works well for our purposes, it is convenient, and it is widely used and understood in business cycle research.

C.1. History of the HP filter

The HP filter was introduced in modern macroeconomics by Hodrick and Prescott (1997), but it was discovered by mathematician Whittaker (1923), and popularized among actuaries by Henderson (1924). The Whittaker-Henderson smoothing method was well-known to macroeconomists in the 1930s (Macaulay 1931, chapter 6).

C.2. Definition of the HP filter

The HP filter takes a time series $y(t)$ and extracts a trend $\tau(t)$ from it, so that the initial series can be decomposed as $y(t) = \tau(t) + \delta(t)$, where $\tau(t)$ moves smoothly over time and $\delta(t)$ is the deviation of the series from trend, or in the context of business cycles, the cyclical fluctuations of the series.

Formally, given T observations of variable $y(t)$, the trend $\tau(t)$ is calculated by solving

$$\min_{\{\tau(t)\}} \left\{ \sum_{t=1}^{T} [y(t) - \tau(t)]^2 + \lambda \sum_{t=2}^{T-1} [[\tau(t+1) - \tau(t)] - [\tau(t) - \tau(t-1)]]^2 \right\}.$$

The parameter λ governs the smoothness of the trend produced by the HP filter. If $\lambda = 0$, the trend is just the series: $\tau(t) = y(t)$. When $\lambda \to \infty$, the trend is linear: $\tau(t) = a + b \cdot t$. In between, the trend evolves smoothly over time.

For the deviation from trend, $\delta(t)$, to be meaningfully interpreted as cyclical fluctuation, it should be stationary. So the filter can be applied directly to a series that is not growing, such as the unemployment rate or vacancy rate. But if the series grow exponentially, such as consumption or output, then applying the HP filter directly produces cycles whose variance grows with the level of the series. That means that $\delta(t)$ is not stationary, and cannot be interpreted as a cyclical component. Hence, the HP filter must be applied to the logarithm of a series that is growing exponentially. Taking the logarithm of the series makes it grow linearly over time, which ensures that its cyclical component is stationary.

C.3. Smoothing parameter

Before using the HP filter, we must specify a smoothing parameter λ, which determines how smooth the trend extracted by the filter is. Ravn and Uhlig (2002) recommend a smoothing parameter of 1,600 with quarterly data; this is the standard value in business cycle research. However, as Shimer (2012, p. 132) notes in the context of US labor market data, setting $\lambda = 1{,}600$ "seems to remove much of the cyclical volatility in the variable of interest." To keep more cyclicality, Shimer (2005, 2012) sets $\lambda = 100{,}000$ instead. Here, we want to keep fluctuations of interest but also apply the same filter to labor market data and to other business cycle data. So we pick an intermediate point between 1,600 and 100,000: we set the smoothing parameter to $\lambda = 10{,}000$. This means that our trends will be smoother than with $\lambda = 1{,}600$ but more flexible than with $\lambda = 100{,}000$. We use that parameter throughout the book, whenever detrending is required.

C.4. Limitations of the HP filter

Like any filter, the HP filter has some limitations (Hamilton 2018). One important limitation is that the HP filter struggles at the end of data samples. Accordingly, it's good practice to compare visually the trend produced by the filter to the original series.

APPENDIX D.

Geometry of the matching function

In this appendix, we establish several properties about the geometry of the matching function. Throughout we consider a matching function $m(s, b)$, with $s > 0$ and $b > 0$, that satisfies the following assumptions: the matching function is twice continuously differentiable, strictly concave in at least one argument (s or b), and with constant returns to scale.

D.1. Second derivatives of the matching function

In this section we establish geometric properties of the matching function based on the properties of its second derivatives.

D.1.1. Preliminary results

Since the matching function exhibits constant returns to scale, we can express it as

(D.1) $$m(s, b) = s \cdot f(\theta),$$

where $f(\theta) = m(1, \theta)$ and $\theta = b/s$.

Using (D.1), we compute the first derivatives of the matching function. The partial derivative with respect to s is

(D.2) $$\frac{\partial m}{\partial s} = f(\theta) + s \cdot f'(\theta) \cdot \frac{\partial \theta}{\partial s} = f(\theta) - \theta f'(\theta),$$

as $\partial\theta/\partial s = -b/s^2 = -\theta/s$. Next, the partial derivative with respect to b is

$$\frac{\partial m}{\partial b} = s \cdot f'(\theta) \cdot \frac{\partial \theta}{\partial b} = f'(\theta), \tag{D.3}$$

as $\partial\theta/\partial b = 1/s$.

From here, we compute the second derivatives of the matching function. Starting from (D.2), we obtain

$$\frac{\partial^2 m}{\partial s^2} = f'(\theta) \cdot \frac{\partial \theta}{\partial s} - \frac{\partial \theta}{\partial s} \cdot f'(\theta) - \theta \cdot f''(\theta) \cdot \frac{\partial \theta}{\partial s} = \frac{\theta^2}{s} \cdot f''(\theta). \tag{D.4}$$

Then, starting from (D.3), we obtain

$$\frac{\partial^2 m}{\partial b^2} = f''(\theta) \cdot \frac{\partial \theta}{\partial b} = \frac{1}{s} \cdot f''(\theta). \tag{D.5}$$

Starting from (D.3), we also obtain the cross-partial derivative

$$\frac{\partial^2 m}{\partial b \partial s} = f''(\theta) \cdot \frac{\partial \theta}{\partial s} = -\frac{\theta}{s} \cdot f''(\theta). \tag{D.6}$$

D.1.2. Main results

Using the preliminary results, we obtain two key properties of the matching function.

RESULT D.1. *The matching function is strictly concave in each argument:* $\partial^2 m/\partial s^2 < 0$ *and* $\partial^2 m/\partial b^2 < 0$.

PROOF. The matching function is strictly concave in at least one argument, which means that either $\partial^2 m/\partial s^2 < 0$ or $\partial^2 m/\partial b^2 < 0$. However, equations (D.4) and (D.5) show us that either assumption is equivalent: namely, they both mean that $f''(\theta) < 0$.[1] Accordingly, if either one is true, the other is true too. We therefore have both $\partial^2 m/\partial s^2 < 0$ and $\partial^2 m/\partial b^2 < 0$. □

RESULT D.2. *The matching function's two arguments are Edgeworth-Pareto complements:* $\partial^2 m/\partial b \partial s > 0$. *The cross-partial derivative is related to the own-partial derivatives by*

$$\frac{\partial^2 m}{\partial b \partial s} = \sqrt{\frac{\partial^2 m}{\partial b^2} \cdot \frac{\partial^2 m}{\partial s^2}}.$$

PROOF. Once again, the assumption that the matching function is strictly concave in at least one argument implies $f''(\theta) < 0$. With (D.6), this in turn implies $\partial^2 m/\partial b \partial s > 0$,

[1] Recall that $s > 0$ and $b > 0$ so $\theta^2/s > 0$ and $1/s > 0$.

which indicates Edgeworth-Pareto complementarity (Hicks 1946, p. 42).

Next, we establish the link between the cross-partial and own-partial derivatives. Equations (D.4) and (D.5) imply that

$$\frac{\partial^2 m}{\partial s^2} \cdot \frac{\partial^2 m}{\partial b^2} = \left[-\frac{\theta}{s} \cdot f''(\theta)\right]^2.$$

Since $-(\theta/s)f''(\theta) > 0$, the square root of the above equation is just $-(\theta/s)f''(\theta) > 0$, which from equation (D.6) we know is $\partial^2 m/\partial b \partial s$. □

D.2. Hessian of the matching function

In this section we establish geometric properties of the matching function based on the properties of its Hessian. Using again equations (D.4), (D.5), and (D.6), we see that the Hessian is given by

$$H = \begin{bmatrix} \frac{\partial^2 m}{\partial s^2} & \frac{\partial^2 m}{\partial b \partial s} \\ \frac{\partial^2 m}{\partial b \partial s} & \frac{\partial^2 m}{\partial b^2} \end{bmatrix} = \frac{f''(\theta)}{s} \cdot \begin{bmatrix} \theta^2 & -\theta \\ -\theta & 1 \end{bmatrix}.$$

RESULT D.3. *The matching function is jointly concave in its two arguments, but not strictly concave.*

PROOF. To establish that the matching function is jointly concave, we show that its Hessian is negative semidefinite (result A.14). First, we see that the determinant of the Hessian is 0: $\det(H) \propto \theta^2 \times 1 - (-\theta) \times (-\theta) = \theta^2 - \theta^2 = 0$. So one eigenvalue is 0. The trace of the Hessian is $\operatorname{tr}(H) = (1 + \theta^2)f''(\theta)/s < 0$ since the assumption that the matching function is strictly concave in at least one argument implies $f''(\theta) < 0$. So the second eigenvalue is strictly negative (result A.5). Hence, the Hessian is negative semidefinite. However, because the Hessian is rank one, it is not negative definite, so the matching function is not strictly concave. □

APPENDIX E.

Alternative matching cost

We have seen in chapter 4 that assessing the matching cost in terms of goods makes sense for markets in which the goods are labor or services. It is less clear how the assumption applies to markets for actual goods. In this appendix, we introduce the matching cost in a different way: as the cost of a sample that the buyer procures to determine whether they want to purchase the good or not. We show that this alternative representation is equivalent to the representation used in the model, and that it applies to many goods markets.

E.1. Model with sampling

Consider a market as described in chapter 4. Each of the v visits brings a buyer to one of the k goods offered for sale.

The buyer does not know whether they would like the good or not, having never bought it before. Hence the buyer buys a sample of the good to determine whether they want to purchase the good or not. The sample has a small size κ and is sold at price κp.

Samples do not contribute to consumption: they are used only for tasting. Only goods purchased after tasting contribute to consumption.

The probability that the buyer is happy with the good and purchases the full unit is set to $q(\theta) - \kappa$, where $q(\theta) = m(1/\theta, 1)$, as usual. In this interpretation, $q(\theta)$ is no longer the probability of purchasing the full-unit good. Instead, it represents the expected total quantity of goods purchased per visit: sample plus possible full-unit purchase. Note that

$q(\theta) \le 1$, and $q(\theta) \ge \kappa$ because $\theta \le \bar{\theta}$, so $q(\theta) - \kappa$ is a proper probability: it is between 0 and 1. The probability that the buyer does not ultimately buy the good—because they do not like it—is $1 - [q(\theta) - \kappa]$. Because $q(\theta) - \kappa$ is a proper probability, so is $1 - [q(\theta) - \kappa]$.

We parameterize the probability of purchasing the full unit as $q(\theta) - \kappa$ so that, together with the sure purchase of a sample of size κ, the expected total quantity purchased per visit remains equal to $q(\theta)$, and the matching function continues to give the total quantity of goods traded—including samples and full-unit goods.

E.2. Matching wedge with sampling

The market with sampling behaves exactly like the baseline market described in chapter 4.

Buyers make v visits. Each visit costs κ goods. The total amount of goods purchased is

$$y = v\kappa + v[q(\theta) - \kappa] = vq(\theta) = m(k, v),$$

since $q(\theta) = m(k/v, 1)$ and the matching function has constant returns to scale. The total amount of goods consumed simply is

$$c = y - v\kappa.$$

The numbers of goods purchased and sold are equal, so $m(k, v)$ goods are sold in total. This means that seller i, who places a share k_i/k of all goods for sale, sells an amount of goods given by

$$m(k, v) \cdot \frac{k_i}{k} = m(1, \theta) \cdot k_i = f(\theta)k_i.$$

On average, it is as if each good is sold with probability $f(\theta)$.

These sales take two forms: some goods are sold in the form of samples, and some in the form of complete goods to be consumed. Seller i receives a number $v \cdot k_i/k = \theta k_i$ of the visits, so sells $\theta k_i \kappa$ samples, and in addition sells $\theta k_i[q(\theta) - \kappa]$ complete goods. The total amount of sales therefore is $\theta k_i q(\theta) = f(\theta)k_i$, since $\theta q(\theta) = f(\theta)$.

The wedge between consumption and sales is

$$\tau(\theta) = \frac{y - c}{c} = \frac{\kappa v}{m(k, v) - \kappa v} = \frac{\kappa}{m(k, v)/v - \kappa} = \frac{\kappa}{q(\theta) - \kappa},$$

using the fact that the matching function has constant returns to scale. This is just the expression for the matching wedge in the baseline model, given by (4.8).

All in all, the market operates exactly as in the baseline model.

E.3. Markets with sampling in the real world

Many goods markets can be described using this equivalent representation. The wine market is a perfect example: people go to wineries, buy glasses of wine to taste different wines, and purchase bottles of the wines they enjoy.

The online market for goods also fits the sample representation, once the price paid for a good is separated between shipping and restocking fees and the rest. For many goods bought online, there is no guarantee that the good fits the buyer's needs. Indeed it's very hard to tell without seeing the good and trying it: most goods are experience goods (Che 1996). Goods that buyers do not like can be returned, but not for free: the buyer is typically not reimbursed for shipping and is commonly assessed a restocking fee (Janssen and Williams 2024). These fees, the logistical effort to bring the good back and repackage it, and the associated depreciation of the good, correspond to the sample sold by the seller. Buyers who keep their online purchases are the ones who actually consume the goods.

Any good sold in brick-and-mortar stores charging a restocking fee could also be described as above.

As a final example, many goods are sold in stores that offer the good both for short-term rental and for sale. Buyers can rent the good to try it out, and then buy it if they like it. Many surf shops and ski shops operate that way. Renting the good to test it corresponds to buying a sample; purchasing the good corresponds to consuming it.

Even internships in the labor market fit here: hiring an intern at a reduced wage for a short time corresponds to "sampling the good," and employing a well-performing intern at full pay on a long-term contract corresponds to "purchasing the good."

APPENDIX F.

Analysis in terms of visits

In chapter 4, we focused on buyers' decisions about how much to consume and purchase. But in many models of search and matching, the key decision is framed differently: it's about the effort someone puts into searching. In our model, this effort is the number of visits to sellers.

Buyers' decisions on how many sellers to visit determine what they buy, what they spend, and what they save. So, it's useful to analyze the model from this more traditional perspective. In this appendix, we recast the model in terms of visits, solve it, and show that this approach in terms of visits produces identical results. This exercise is instructive for various reasons. In particular, it reveals why our original output-based approach is more convenient than a visit-based approach.

F.1. Recasting the model in terms of visits

The first step is to rephrase the buyer's choice as a decision about visits (v_j) instead of consumption or purchases. This is straightforward because a buyer's consumption is directly proportional to their visits:

$$c_j = y_j - \kappa v_j = [q(\theta) - \kappa]\, v_j.$$

Buyer j chooses the number of visits v_j to maximize utility subject to their budget constraint, taking as given tightness and price. Thus, buyer j's problem becomes:

$$\max_{v_j \geq 0} a\left[q(\theta)-\kappa\right]^{1-\alpha} v_j^{1-\alpha}+b_j,$$

subject to the budget constraint:

$$b_j+pq(\theta)v_j=B_j.$$

Using the budget constraint to substitute money balances b_j out of the utility function, we rewrite the buyer's problem as

$$\max_{v_j \geq 0} a\left[q(\theta)-\kappa\right]^{1-\alpha} v_j^{1-\alpha}+B_j-pq(\theta)v_j.$$

This is just a concave maximization problem, so it suffices to take the first-order condition with respect to the number of visits to get the global maximum (result A.20). The first-order condition gives:

$$(1-\alpha)a\left[q(\theta)-\kappa\right]^{1-\alpha} v_j^{-\alpha}=pq(\theta).$$

Solving for the optimal number of visits for buyer j yields:

$$v_j=\left[\frac{(1-\alpha)a}{pq(\theta)}\right]^{1/\alpha}\left[q(\theta)-\kappa\right]^{1/\alpha-1}.$$

Then, the market-level number of visits is $v=\int_0^1 v_j\, dj$. Since all buyers visit the same number of sellers, and there is a mass 1 of buyers, the market-level number of visits is the same as the individual number of visits:

$$v=\left[\frac{(1-\alpha)a}{pq(\theta)}\right]^{1/\alpha}\left[q(\theta)-\kappa\right]^{1/\alpha-1}. \tag{F.1}$$

Equation (F.1) is equivalent to the market demand that we derived in chapter 4. Thinking in terms of visits is therefore isomorphic to thinking in terms of output. Indeed, if we multiply both sides of (F.1) by $q(\theta)$, we get

$$q(\theta)v=\left[\frac{(1-\alpha)a}{p}\right]^{1/\alpha}\left[\frac{q(\theta)-\kappa}{q(\theta)}\right]^{1/\alpha-1}.$$

Then we know that output satisfies $y=q(\theta)v$, via the matching function (4.1). By definition of the matching wedge (4.8), we also have

$$1+\tau(\theta)=\frac{q(\theta)}{q(\theta)-\kappa}.$$

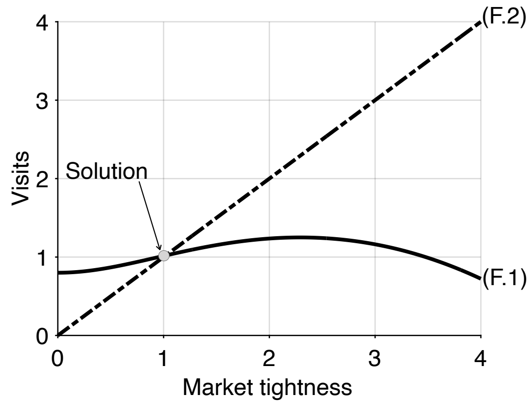


FIGURE F.1. Numerical illustration of the visit-based solution

Equation (F.1) gives the optimal number of visits for buyers as a function of tightness. Equation (F.2) gives the number of visits obtained from the definition of market tightness. The matching function is CES, given by (3.12). Parameters are set to $\sigma = 2$, $\kappa = 0.2$, $k = 1$, $\alpha = 0.5$, and $a = 2$.

Thus, the equation can be rewritten

$$y = \left[\frac{(1-\alpha)a}{p}\right]^{1/\alpha} [1+\tau(\theta)]^{1-1/\alpha}.$$

Here we have just recovered that $y = y^d(\theta, p)$ where the market demand curve is given by (4.15). Hence, focusing on the number of visits instead of consumption does not change the market demand.

F.2. Solving the model in terms of visits

Our next step is to solve the model by determining the aggregate number of visits and tightness. The solution is equivalent to what we obtained before; however, we learn some interesting things from this visit-based approach.

To find tightness θ and visits v, we use two key equations linking tightness θ to visits v. The first is the demand equation (F.1). The second is the definition of tightness, $\theta = v/k$, which gives

$$v = \theta k. \tag{F.2}$$

Note that this equation is equivalent to the market supply, given by (4.5). Multiplying (F.2) by $q(\theta)$ gives $vq(\theta) = \theta q(\theta)k$, which is just $y = f(\theta)k$, or $y = y^s(\theta)$. Hence, focusing on the number of visits instead of output does not change the market supply.

The solution of the model must satisfy the two equations (F.1) and (F.2). Since equations (F.1) and (F.2) are equivalent to equations (4.5) and (4.15), the visit-based and output-based solutions are equivalent. We can plot these on a graph to help us visualize everything better (figure F.1). The solution (θ, v) is at the intersection of the two curves. From there we could determine all the other variables in the model, just as in the standard representation of the model in chapter 4.

Nevertheless, figure F.1 shows why the visit-based solution is not as analytically convenient as the output-based solution. We see that the curve $v(\theta)$ defined by equation (F.1) is not monotonic, and it is neither concave nor convex. Hence, establishing that it crosses the curve defined by equation (F.2) once and only once would not be trivial. Performing comparative statics would also be more complicated. By contrast, the output supply and demand curves (y^s and y^d) are well-behaved, making the analysis of the solution and its properties much more straightforward.

APPENDIX G.

Solution concept

This appendix discusses the solution concept from the slackish market model of chapter 4.

G.1. How much rationality does the model assume?

Let us now briefly discuss the amount of rationality assumed in the model. Economic models are often criticized for assuming too much rationality from people. The rational-expectations model, for example, assumes that people have a complete understanding of the structure of the economy and are able to predict how aggregate variables are likely to behave in the future. This is often seen as a level of rationality that is so high that it is unlikely to be met by actual people. Although our model is static, it is worth going through the different layers of rationality that are assumed in order to convince ourselves that the amount of rationality in the model is reasonable.

The first assumption is that buyers maximize their utility subject to their budget constraint. This means that they try to make the best decisions possible given their wealth. While it may not be realistic to expect buyers to be able to solve this type of maximization problem exactly, it is reasonable to assume that they understand their budget and the trade-off between consumption and saving.

A second assumption is that buyers anticipate prices correctly. This seems realistic because buyers know what they have paid in the past for similar goods and therefore should be able to anticipate the prices they will pay in the future. This doesn't require too much rationality because prices follow price norms that are, by definition, understood and followed by everyone in the market.

Last, we assume that buyers and sellers anticipate market tightness correctly. This is more complicated because tightness depends on the behavior of all sellers and buyers. Anticipating market tightness correctly is therefore quite difficult, as it requires people to know, for instance, the total quantity of goods offered by all sellers. With utility functions that are not quasilinear, buyers of different wealth would consume and shop differently, requiring an individual to know all of these variables to predict tightness. In that case anticipating tightness might be an unrealistic assumption.

Anticipating tightness is very similar to anticipating traffic. It requires anticipating the behavior of a large number of people who are moved by similar motives, accounting for random disturbances. People are typically good at predicting traffic on their regular itineraries, so they should be good at predicting tightness on familiar markets. What might be more difficult is predicting tightness on markets that they are unfamiliar with—just like it's difficult to predict traffic on unfamiliar routes. And just like it's hard to predict traffic after big accidents, it might be hard to predict tightness after big shocks.

G.2. Forecasting tightness with a statistical agency

Assuming that buyers and sellers anticipate market tightness correctly might be unrealistic. An alternative assumption that might be more realistic—and that does not change how the market operates—is that a statistical agency announces a correct forecast of market tightness at the beginning of the period. Buyers and sellers take this announcement at face value and behave optimally based on it. The key assumption here is that the statistical agency is able to make an accurate forecast, so the announced tightness ends up being the realized tightness.

The agency's goal is to quote an accurate market tightness to buyers and sellers. Market participants take this quote as given and make decisions to maximize their utility based on it. The realized tightness is equal to the quoted tightness if the agency makes a correct forecast. Let us now discuss how a statistical agency can forecast the correct tightness.

Let's assume that the statistical agency quotes a market tightness θ^a which people take as given. Then, buyer j aims to buy $y_j(\theta^a, p) = y^d(\theta^a, p)$, since the market demand corresponds to the individual demand for goods in the model. The buyers therefore visit $v_j(\theta^a, p) = y_j(\theta^a, p)/q(\theta^a) = y^d(\theta^a, p)/q(\theta^a)$ sellers. As a result, the realized tightness θ is:

$$\theta = \frac{\int_0^1 v_j(\theta^a, p)\, dj}{\int_0^1 k_i\, di} = \frac{y^d(\theta^a, p)}{q(\theta^a)k} = \theta^a \cdot \frac{y^d(\theta^a, p)}{\theta^a q(\theta^a)k} = \theta^a \cdot \frac{y^d(\theta^a, p)}{y^s(\theta^a)}.$$

Since the statistical agency aims to make a correct forecast, they are looking for θ^a

such that $\theta^a = \theta$. Therefore, they announce θ^a such that

$$y^s(\theta^a) = y^d(\theta^a, p).$$

In other words, the statistical agency announces the tightness θ^a that equalizes market demand and supply! Since the realized tightness θ equals the announced tightness θ^a, then the realized tightness also equalizes market demand and supply: $y^s(\theta) = y^d(\theta, p)$.

This model is a little more appealing because it requires much less rationality from people. The only thing they have to do is listen to the statistical agency and do the best they can given the tightness that has been announced. If they do that, and the agency announces the tightness equalizing market supply and demand, then the prevailing tightness is the announced tightness. So the agency is correct, and the supply-equals-demand tightness prevails. This model does require the agency to solve the supply-equals-demand equation, but a statistical agency is better equipped than laypeople to understand market conditions and solve the equation.

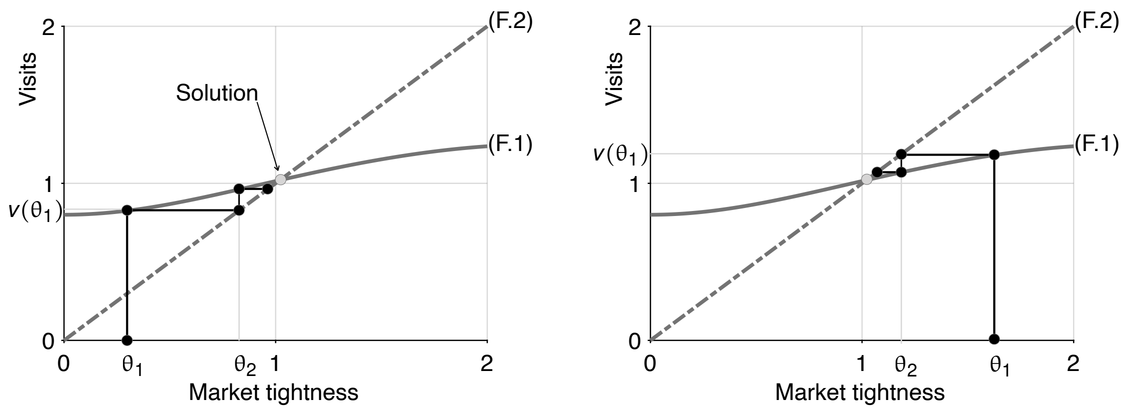
G.3. Convergence to the solution by tatonnement

Finally, what if the statistical agency cannot compute market demand and supply to forecast tightness? If computing this fixed point is difficult, the agency may attempt to learn it through an iterative process. Here we consider the tatonnement process through which the agency might converge to the correct forecast.

We imagine that the agency quotes a market tightness and asks buyers and sellers to report their plans under that tightness. Then the agency computes the tightness resulting from these plans. If the resulting tightness is not the same as the quoted tightness—which means that the agency would have misforecast tightness—the agency then re-quotes tightness, using the tightness that it has just computed from the plans they received. Then it asks buyers and sellers to report their new plans under the new tightness. The agency recomputes the tightness and checks whether it is the same as the quoted tightness. And so on, until the quoted and realized tightnesses converge.

The tatonnement process characterizes the evolution of tightness across iterations and shows whether the quoted tightness indeed converges to the model solution, which is given at the intersection of market supply and demand. Will the statistical agency eventually be able to quote the correct tightness, as a result of the tatonnement process?

Let's start from the first round of tatonnement. The statistical agency announces a tightness θ_1 and asks buyers and sellers to report their plans under that tightness. Sellers' plans do not depend on tightness, so they report their capacities k_i, which sum to $k = \int_0^1 k_i \, di$. We have seen in appendix F that if buyers believe that they will face a market tightness θ_1, the optimal number of visits for any buyer is $v(\theta_1)$, where $v(\theta)$ is defined by



A. Convergence to the solution from below B. Convergence to the solution from above

FIGURE G.1. Numerical illustration of convergence by tatonnement

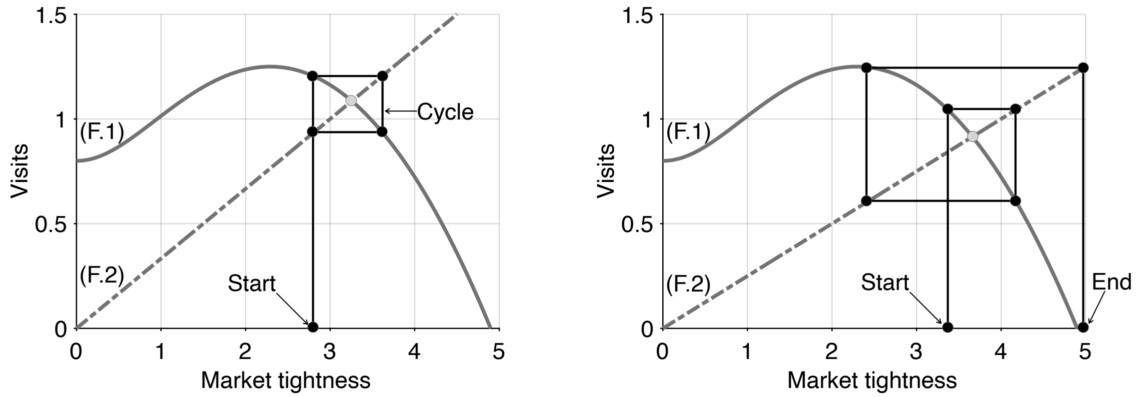
Equation (F.1) gives the optimal number of visits for buyers as a function of tightness. Equation (F.2) gives the number of visits obtained from the definition of market tightness. The matching function is CES, given by (3.12). Parameters are set to $\sigma = 2$, $\kappa = 0.2$, $k = 1$, $\alpha = 0.5$, and $a = 2$.

equation (F.1). Hence the aggregate number of visits is $v = \int_0^1 v(\theta_1)\, dj = v(\theta_1)$.

From the initial announcement θ_1, and the plans received from market participants, the agency then computes $\theta_2 = v(\theta_1)/k$. Are the tightnesses θ_2 and θ_1 the same? Well, it depends on the value of θ_1. Figure G.1A illustrates the tatonnement process when the agency announces a tightness that is too low, in the sense that it is below the solution θ of the model, which is at the intersection of the two curves. The agency announces θ_1, which we place on the x-axis. The optimal behavior by buyers if they believe that tightness will be θ_1 is to visit $v(\theta_1)$ sellers, as we established in appendix F. Receiving the plans of $v(\theta_1)$ visits, the agency computes a tightness of $\theta_2 = v(\theta_1)/k$, which can be obtained by projecting $v(\theta_1)$ on the line $v = \theta k$. We also place θ_2 on the x-axis and see that it is larger than θ_1, but still below the solution θ.

The next step for the agency is to announce θ_2. Following the same steps, buyers report a plan to visit $v(\theta_2)$ sellers, and the agency computes a new tightness $\theta_3 = v(\theta_2)/k$. Eventually, after many steps, the agency will approach the solution of the model, θ, which is at the intersection of the two curves. We can see this process visually in figure G.1, both when the agency starts from a tightness that is too low and when it starts with a tightness that is too high.

In the numerical case considered in figure G.1, the curve (F.1) is upward-sloping when it crosses the curve (F.2), so we can be sure that the tatonnement process converges to the model solution. But because it is difficult to characterize the properties of the curve (F.1) (as discussed in appendix F), it is difficult to guarantee that the tatonnement process always reaches the model solution. The curve (F.1) could be non-monotonic, and because of this the tatonnement process might not converge to the model solution.



A. Infinite cycle

B. Divergence to an infeasible tightness

FIGURE G.2. Failures of the tatonnement process to converge

Equation (F.1) gives the optimal number of visits for buyers as a function of tightness. Equation (F.2) gives the number of visits obtained from the definition of market tightness. The matching function is CES, given by (3.12). Parameters are set to $\sigma = 2$, $\kappa = 0.2$, $\alpha = 0.5$, and $a = 2$; in panel A, $k = 1/3$; in panel B, $k = 1/4$.

If the curve (F.1) is downward-sloping when it crosses (F.2), it is possible to construct examples where the tatonnement process does not converge. In figure G.2A, for instance, the tatonnement process cycles indefinitely, without ever converging. This is the type of pathology that Scarf (1960) had identified in the case of Walrasian models—showing that while the Walrasian model always has a unique solution, the tatonnement process might not converge to it. In figure G.2B, things are even worse: the tatonnement process rapidly leads the statistical agency to quote an infeasible tightness—a tightness so high that buyers do not want to enter the market at all (a θ such that (F.1) does not produce a positive v).

Hence, just like in Walrasian models, the tatonnement process is not guaranteed to converge to the solution in slackish models. An advantage of slackish models, however, is that the tatonnement process is completely natural. Once the statistical agency announces a tightness, market participants make decisions and report them, from which the agency can compute the realized tightness. This is quite different from the Walrasian tatonnement. There the auctioneer quotes a price, participants report the quantities that they would like to buy and sell, and if the quantities demanded and supplied are not the same, the auctioneer must quote a new price. Scarf (1960) assumes that the quoted price is adjusted by an amount commensurate to the excess market demand. But many other assumptions can be made, because the auctioneer may quote any price. In slackish models, the agency makes up a tightness only in the initial round. It then uses participants' plans to recompute tightnesses. This property might help determine formal conditions under which the tatonnement process is guaranteed to converge in slackish markets.

APPENDIX H.

Convexity of the Phillips curve

In this appendix, we establish that the Phillips curve $\pi(u)$ is strictly convex in u under the assumption that the matching function is Cobb-Douglas and symmetric:

$$m(u, v) = \mu\sqrt{uv}.$$

It is difficult to obtain more general analytical results, so for different matching functions, numerical analysis is required, as we do in chapter 14.

H.1. Symmetric Cobb-Douglas matching function

We assume in this appendix that the matching function is Cobb-Douglas with elasticity $\sigma = 1/2$. This means that the trading rates are

$$f(\theta) = \mu\sqrt{\theta} \quad \text{and} \quad q(\theta) = \frac{\mu}{\sqrt{\theta}}. \tag{H.1}$$

The matching elasticity is constant at $\eta(\theta) = \sigma = 1/2$, so

$$\frac{\eta(\theta)}{1 - \eta(\theta)} = 1. \tag{H.2}$$

The unemployment rate is given by (8.8), so

$$u(\theta) = \frac{\lambda}{\lambda + \mu\sqrt{\theta}}. \tag{H.3}$$

Inverting this strictly decreasing relationship, we write tightness as a strictly decreasing function of the unemployment rate:

$$\theta(u) = \left[\frac{\lambda(1-u)}{\mu u}\right]^2. \tag{H.4}$$

The highest possible value for tightness is $\theta \to \bar{\theta}$, when $\tau(\theta) \to \infty$. The upper bound $\bar{\theta}$ is defined such that $q(\bar{\theta}) = \kappa\lambda$, so here

$$\bar{\theta} = \left(\frac{\mu}{\kappa\lambda}\right)^2. \tag{H.5}$$

The lowest possible unemployment rate is achieved when $\theta \to \bar{\theta}$. We denote this minimum unemployment rate by $\bar{u} = u(\bar{\theta})$; it is given by

$$\bar{u} = \frac{\kappa\lambda^2}{\mu^2 + \kappa\lambda^2}. \tag{H.6}$$

H.2. Preliminary result

RESULT H.1. *The ratio between the matching wedge and unemployment rate admits the closed form*

$$\frac{\tau(u)}{u} = \frac{1-\bar{u}}{u-\bar{u}} - \frac{1}{u}. \tag{H.7}$$

The ratio is strictly decreasing and strictly convex in u on $(\bar{u}, 1)$.

The proof has three steps: derive the closed form (H.7), then sign the first and second derivatives.

H.2.1. Closed form for $\tau(u)/u$

Combining (H.1) and (H.4) allows us to express the ad-filling rate as a function of the unemployment rate:

$$q(u) = \frac{\mu^2}{\lambda} \cdot \frac{u}{1-u}.$$

Using expression (8.13), we can express the matching wedge as a function of the unemployment rate too:

$$\tau(u) = \frac{\kappa\lambda}{\dfrac{\mu^2}{\lambda}\dfrac{u}{1-u} - \kappa\lambda} = \frac{\kappa\lambda^2(1-u)}{\mu^2 u - \kappa\lambda^2(1-u)} = \frac{\kappa\lambda^2(1-u)}{(\mu^2+\kappa\lambda^2)u - \kappa\lambda^2} = \frac{\bar{u}(1-u)}{u-\bar{u}}.$$

Then, we find that the ratio between matching wedge and unemployment rate is a simple function of the unemployment rate:

$$\frac{\tau(u)}{u} = \frac{\bar{u}(1-u)}{u(u-\bar{u})} = \frac{\bar{u}-u+u-\bar{u}u}{u(u-\bar{u})} = \frac{1-\bar{u}}{u-\bar{u}} - \frac{1}{u},$$

which establishes (H.7).

H.2.2. First derivative of $\tau(u)/u$

Next, we compute the first derivative of the ratio τ/u. We know that τ is increasing in tightness, while the unemployment rate is decreasing in tightness, so τ/u is increasing in tightness, which means that it must be decreasing in the unemployment rate. So we know that the first derivative must be negative. But the expression for the first derivative will be useful to establish that the ratio is convex in the unemployment rate.

Differentiating (H.7) once with respect to the unemployment rate, we get

$$\frac{d\tau/u}{du} = \frac{1}{u^2} - \frac{1-\bar{u}}{(u-\bar{u})^2}. \tag{H.8}$$

We can quickly verify that the derivative is strictly negative. The sign of the derivative is the sign of

$$\begin{aligned}(u-\bar{u})^2 - (1-\bar{u})u^2 &= u^2 + \bar{u}^2 - 2u\bar{u} - u^2 + \bar{u}u^2 \\ &= \bar{u}\left[\bar{u} - 2u + u^2\right] \\ &= \bar{u}\left[(\bar{u}-u) + u(u-1)\right].\end{aligned}$$

Since $0 < \bar{u} < u < 1$, we have $u - 1 < 0$ and $\bar{u} - u < 0$, so the first derivative is strictly negative, which implies that τ/u is strictly decreasing in u.

H.2.3. Second derivative of $\tau(u)/u$

Differentiating (H.8) with respect to the unemployment rate, we now get the second derivative of the ratio:

$$\frac{d^2\tau/u}{du^2} = \frac{2(1-\bar{u})}{(u-\bar{u})^3} - \frac{2}{u^3}.$$

The sign of the second derivative is the sign of

$$\begin{aligned}(1-\bar{u})u^3 - (u-\bar{u})^3 &= u^3 - \bar{u}u^3 - u^3 + \bar{u}^3 + 3u^2\bar{u} - 3u\bar{u}^2 \\ &= \bar{u}\left[-u^3 + \bar{u}^2 + 3u^2 - 3u\bar{u}\right] \\ &= \bar{u}\left[u^2 - u^3 + \bar{u}^2 + u^2 - 2u\bar{u} + u^2 - u\bar{u}\right]\end{aligned}$$

$$= \bar{u}\left[u^2(1-u) + (\bar{u}-u)^2 + u(u-\bar{u})\right].$$

Since $0 < \bar{u} < u < 1$, all the terms in the square bracket are positive: $u^2(1-u) > 0$, $(\bar{u}-u)^2 > 0$, and $u(u-\bar{u}) > 0$. Hence the second derivative is strictly positive, which implies that $\tau(u)/u$ is strictly convex in u (result A.12).

H.3. Main result

We now pass from convexity of $\tau(u)/u$ to convexity of the equilibrium Phillips curve in an unemployment-inflation plane. The equilibrium Phillips curve $\pi(\theta)$ is given by (14.20), where here (H.2) holds. We compose it with the function $\theta(u)$ given by (H.4) to obtain a Phillips curve linking inflation to unemployment:

$$\pi(u) = \pi^* + \frac{1}{\varpi\delta(\delta-r)} \cdot y(u) \cdot \left[\frac{\tau(u)}{u} - 1\right].$$

The ratio $\tau(u)/u$ is given by (H.7) and output is given by $y(u) = (1-u)kh$. Given the linear expression for output, we can rewrite the Phillips curve as

$$\pi(u) = \pi^* + \frac{kh}{\varpi\delta(\delta-r)} \cdot (1-u) \cdot \left[\frac{\tau(u)}{u} - 1\right]. \tag{H.9}$$

RESULT H.2. *The Phillips curve $\pi(u)$ given by* (H.9) *is strictly decreasing in u around the efficient unemployment rate $u = u^*$ and strictly convex in u on $(\bar{u}, 1)$.*

To establish the result, we differentiate (H.9) twice. The first derivative is

$$\pi'(u) = \frac{kh}{\varpi\delta(\delta-r)} \cdot \left[(1-u) \cdot \frac{d\tau/u}{du} + 1 - \frac{\tau(u)}{u}\right].$$

One result that appears here is that $\pi'(u^*) < 0$. Indeed, we know from (9.11) and (H.2) that at efficiency, $\tau(u^*)/u^* = 1$. Hence, the sign of $\pi'(u^*)$ is the sign of $(1-u^*)d(\tau/u)/du$. But we know from result H.1 that $d(\tau/u)/du < 0$, so we see that $\pi'(u^*) < 0$. Hence, in the vicinity of efficiency, the Phillips curve is strictly decreasing in the unemployment rate.

Differentiating again, we obtain

$$\pi''(u) = \frac{kh}{\varpi\delta(\delta-r)} \cdot \left[(1-u) \cdot \frac{d^2\tau/u}{du^2} - 2 \cdot \frac{d\tau/u}{du}\right].$$

From result H.1, we know that $d(\tau/u)/du < 0$ and that $d^2(\tau/u)/du^2 > 0$. Both terms in the square brackets are strictly positive, so $\pi''(u) > 0$ on $(\bar{u}, 1)$. Hence, the Phillips curve is strictly convex in the unemployment rate.

APPENDIX I.

Inflation and tightness dynamics at the zero lower bound

In this appendix, we derive and study the dynamical system describing the behavior of tightness and inflation at the zero lower bound in the slackish business cycle model of chapter 14.

I.1. Relationships between inflation, tightness, and the costate variables

The first step in building the two-dimensional dynamical system is to express the costate variables ψ^b and ψ^p as a function of inflation π and tightness θ. Using these relationships, we will be able to transform the differential equations for the costate variables into differential equations for inflation and tightness.

This is easy to do for the costate variable ψ^p, as the relationship is directly given by equation (14.16):

$$\psi^p = \frac{\varpi}{a}\left[\pi - \pi^*\right] \tag{I.1}$$

$$\dot{\psi}^p = \frac{\varpi}{a}\dot{\pi}. \tag{I.2}$$

For the costate variable ψ^b, the relationship arises from equation (14.15). Indeed, in the model, aggregate consumption is directly determined by tightness:

$$c(\theta) = \frac{y(\theta)}{1+\tau(\theta)} = \frac{1-u(\theta)}{1+\tau(\theta)} \cdot kh. \tag{I.3}$$

Using the function $c(\theta)$, we can relate ψ^b to tightness using (14.15):

$$\psi^b = (1-\alpha) \cdot \frac{c(\theta)^{-\alpha}}{1+\tau(\theta)}. \tag{I.4}$$

Notice also, which will be useful later, that the product $\psi^b(\theta)y(\theta)$ is simply given by:

$$\psi^b(\theta)y(\theta) = \psi^b(\theta)c(\theta)[1+\tau(\theta)] = (1-\alpha)c(\theta)^{1-\alpha}. \tag{I.5}$$

Using a basic chain rule, the time derivative of the costate variable therefore is:

$$\dot{\psi}^b = \frac{\psi^b}{\theta} \cdot \epsilon_\theta^{\psi^b} \cdot \dot{\theta},$$

where $\epsilon_\theta^{\psi^b}$ is the elasticity of the costate ψ^b with respect to tightness θ.

To compute that elasticity (and others below), we use the result that the elasticity of $1+\tau(\theta)$ with respect to θ is given by (6.12):

$$\epsilon_\theta^{1+\tau} = \eta(\theta)\tau(\theta). \tag{I.6}$$

Another useful elasticity is the elasticity of $1-u(\theta)$ with respect to θ, which can be inferred from (13.13):

$$\epsilon_\theta^{1-u} = \frac{-u(\theta)}{1-u(\theta)} \cdot \epsilon_\theta^u = [1-\eta(\theta)]\, u(\theta). \tag{I.7}$$

From these results and (I.3), we infer that the elasticity of $c(\theta)$ with respect to θ is

$$\epsilon_\theta^c = [1-\eta(\theta)]\, u(\theta) - \eta(\theta)\tau(\theta).$$

Then, from these results and (I.4), we infer that the elasticity of ψ^b with respect to θ is

$$\epsilon_\theta^{\psi^b} = -\left[\alpha\, [1-\eta(\theta)]\, u(\theta) + (1-\alpha)\eta(\theta)\tau(\theta)\right].$$

And finally from this we relate the time derivative of the costate variable ψ^b to that of tightness θ:

$$\frac{\dot{\psi}^b}{\psi^b} = -\left[\alpha\, [1-\eta(\theta)]\, u(\theta) + (1-\alpha)\eta(\theta)\tau(\theta)\right] \cdot \frac{\dot{\theta}}{\theta}. \tag{I.8}$$

I.2. Differential equation for inflation

Using the results from the previous section, we now transform the differential equation for the costate variable ψ^p, given by (14.18), into a differential equation for inflation π. We simply replace ψ^p and $\dot{\psi}^p$ in the equation by the expressions (I.1) and (I.2), and $\psi^b y$ by the expression (I.5). We obtain:

$$\dot{\pi} = \delta\left[\pi - \pi^*\right] + \frac{(1-\alpha)a}{\varpi} \cdot c(\theta)^{1-\alpha}\left[1 - \frac{\eta(\theta)}{1-\eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)}\right]. \tag{I.9}$$

Accordingly, the nullcline $\dot{\pi} = 0$ is given by:

$$\pi = \pi^* + \frac{(1-\alpha)a}{\delta\varpi} \cdot c(\theta)^{1-\alpha}\left[\frac{\eta(\theta)}{1-\eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)} - 1\right]. \tag{I.10}$$

This expression is close but not the same as the Phillips curve in normal times, given by equation (14.20). This is because the costate variable ψ^b is fixed, determined by the real rate in normal times, whereas it is jointly determined by tightness and inflation at the zero lower bound.

I.3. Differential equation for tightness

Next, we transform the differential equation for the costate variable ψ^b, given by (14.17), into a differential equation for tightness θ. First, we divide the equation by ψ^b. We also set the real interest rate to $r = -\pi$. We obtain

$$\frac{\dot{\psi}^b}{\psi^b} = (\delta + \pi) - \frac{1}{a\psi^b}.$$

Next, we replace ψ^b and $\dot{\psi}^b$ in the equation by the expressions (I.4) and (I.8). This yields:

$$\left[\alpha\left[1-\eta(\theta)\right]u(\theta) + (1-\alpha)\eta(\theta)\tau(\theta)\right] \cdot \frac{\dot{\theta}}{\theta} = \frac{1}{a(1-\alpha)} \cdot \left[1+\tau(\theta)\right]c(\theta)^{\alpha} - (\delta+\pi). \tag{I.11}$$

From this equation, we see that the nullcline $\dot{\theta} = 0$ is given by:

$$\pi = \frac{1}{a(1-\alpha)} \cdot \left[1+\tau(\theta)\right]c(\theta)^{\alpha} - \delta. \tag{I.12}$$

I.4. Equilibrium of the dynamical system

We now determine the equilibrium of the dynamical system composed of differential equations (I.9) and (I.11). At the equilibrium, $\dot{\pi} = \dot{\theta} = 0$, so the equilibrium is determined by equations (I.10) and (I.12).

Let us introduce a few auxiliary functions to simplify the analysis:

$$
\begin{aligned}
P(\theta) &= \pi^* + \frac{(1-\alpha)a}{\delta\varpi} \cdot c(\theta)^{1-\alpha}\left[\frac{\eta(\theta)}{1-\eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)} - 1\right] \\
Q(\theta) &= \frac{1}{a(1-\alpha)} \cdot [1+\tau(\theta)]\, c(\theta)^{\alpha} - \delta \\
R(\theta) &= \frac{\theta}{\alpha\,[1-\eta(\theta)]\,u(\theta) + (1-\alpha)\eta(\theta)\tau(\theta)}.
\end{aligned}
$$

With these auxiliary functions, the nullcline (I.10) is given by $\pi = P(\theta)$ and the nullcline (I.12) is given by $\pi = Q(\theta)$. Let $S(\theta) = Q(\theta) - P(\theta)$. The tightness that solves the model is therefore the solution to $S(\theta) = 0$.

At $\theta = 0$, we have $u(0) = 1$ so $c(0) = 0$ and $S(0) = -(\delta + \pi^*) < 0$.

At $\theta = \theta^*$, we have $\tau/u = (1-\eta)/\eta$, from (9.11), so that

$$
S(\theta^*) = \frac{1}{a(1-\alpha)} \cdot \left[1+\tau(\theta^*)\right] c(\theta^*)^{\alpha} - (\delta + \pi^*).
$$

Because we are at the zero lower bound, we know that the marginal utility of wealth $1/a$ is quite large—otherwise a positive nominal interest rate would be sufficient to maintain the economy at the divine equilibrium, where $\theta = \theta^*$ and $\pi = \pi^*$. Studying the zero lower bound only makes sense when aggregate demand is depressed enough because of a high marginal utility of wealth. Hence, the economy enters the zero lower bound only when at a zero nominal interest rate, aggregate demand is below the efficient level of output: $y^d(\theta^*, -\pi^*) < y(\theta^*)$. Rearranging this inequality imposes that the marginal utility of wealth satisfies:

$$
\frac{1}{a} > \frac{(\delta + \pi^*)(1-\alpha)c(\theta^*)^{-\alpha}}{[1+\tau(\theta^*)]}.
$$

Hence, when the marginal utility of wealth is high enough to push the economy to the zero lower bound, then it is high enough so that $S(\theta^*) > 0$.

We have established that $S(0) < 0$ and $S(\theta^*) > 0$ and the function S is clearly continuous on $[0, \theta^*]$. According to Bolzano's theorem, the equation $S(\theta) = 0$ therefore admits a solution on $(0, \theta^*)$ (result A.1). This means that the dynamical system admits an equilibrium on $(0, \theta^*)$, which we denote (θ^z, π^z).

Is the equilibrium unique? It would be if S was strictly increasing on $(0, \theta^*)$. The function Q is strictly increasing on $(0, \theta^*)$, as $\tau(\theta)$ and $c(\theta)$ are increasing on the interval. But the function P is not monotone (recall that $P(0) = P(\theta^*) = \pi^*$), so we cannot say for sure that $S(\theta) = Q(\theta) - P(\theta)$ is strictly increasing over the entire interval.

However, when prices are rigid enough, so ϖ is large enough, the fluctuations in P can be made arbitrarily small. Indeed, at the limit where $\varpi \to \infty$, $P(\theta) \to \pi^*$ for any θ. So for

ϖ large enough, the fluctuations in P can be made small enough that S never returns to 0 once it has crossed 0 at θ^z—in which case the equilibrium of the system is unique. The equilibrium is for instance unique under the calibrations presented in figure 14.2. In that case, since $S(0) < 0$ and $S(\theta^*) > 0$, we know that S crosses 0 from below, so $S'(\theta^z) > 0$.

In the rest of the appendix we assume that price rigidity ϖ is sufficient to guarantee that the equilibrium is unique.

I.5. Classification of the dynamical system

In this section, we determine whether, around the zero-lower-bound equilibrium, the dynamical system is a source, sink, or saddle. To do that, we need to compute the Jacobian of the system (appendix A.6.3), which describes the local dynamics around the zero-lower-bound equilibrium.

With the above auxiliary functions, the differential equation (I.11) is given by

$$\dot{\theta} = R(\theta) \cdot [Q(\theta) - \pi]$$

and the differential equation (I.9) is given by

$$\dot{\pi} = \delta\left[\pi - P(\theta)\right].$$

With this notation the Jacobian of the system $[\pi(t), \theta(t)]$ is

$$\mathbf{J} = \begin{bmatrix} \delta & -\delta P'(\theta^z) \\ -R(\theta^z) & R(\theta^z)Q'(\theta^z) \end{bmatrix}.$$

(We simplified the Jacobian because $Q(\theta^z) - \pi^z = 0$ by definition of the equilibrium.)

To describe the nature of the dynamics of inflation and tightness around the zero-lower-bound equilibrium, we must compute the trace and determinant of the Jacobian. We find:

$$\begin{aligned} \mathrm{tr}(\mathbf{J}) &= \delta + R(\theta^z)Q'(\theta^z) \\ \det(\mathbf{J}) &= \delta R(\theta^z)\left[Q'(\theta^z) - P'(\theta^z)\right] = \delta R(\theta^z)S'(\theta^z). \end{aligned}$$

We know that $\delta > 0$, $R(\theta^z) > 0$, and $Q'(\theta^z) > 0$, so the trace of the Jacobian is strictly positive. We have also seen that $S'(\theta^z) > 0$, so the determinant of the Jacobian is strictly positive. Accordingly, the dynamical system describing inflation and tightness is a source around the zero-lower-bound equilibrium (result A.30).

APPENDIX J.

Policy multipliers in the two-market model

In this appendix, we compute policy multipliers in the two-market slackish business cycle model of chapter 15. We begin with the monetary multiplier, which describes the effect of a change in the nominal interest rate on the unemployment rate. We then turn to the fiscal multiplier, which gives the effect of a change in government spending on the unemployment rate.

J.1. Background

The key in computing the policy multipliers is to determine the response of labor market tightness to the policy change. From that it is simple to compute the response of other variables and eventually the multiplier. Given that labor market tightness is given by equation (15.20), we need to characterize the shapes of the curves $\theta_p^l(\theta_l, w)$ and $\theta_p^p(\theta_l, r, w)$. It is convenient to rewrite the definition of $\theta_p^l(\theta_l, w)$, given by equation (15.18), as follows:

$$1 - u_p(\theta_p^l) = \frac{wh^{\alpha_l}\left[1 - u_l(\theta_l)\right]^{\alpha_l}\left[1 + \tau_l(\theta_l)\right]^{1-\alpha_l}}{(1-\alpha_l)a_l}. \tag{J.1}$$

And it is convenient to rewrite the definition of $\theta_p^p(\theta_l, r, w)$, given by equation (15.19), as follows:

$$\left[1 + \tau_p(\theta_p^p)\right]^{1-\alpha_p} = \frac{(1-\alpha_l)^{\alpha_p}\left[(1-\alpha_p)(\delta - r)a_p\right]}{\left(wh\left[1 - u_l(\theta_l)\right]\right)^{\alpha_p}}. \tag{J.2}$$

To compute the multipliers, we will rely heavily on the elasticity results from appendix B. An elasticity that we will use repeatedly is the elasticity of $1+\tau(\theta)$ in any market, which is given by (6.12):

$$\epsilon_\theta^{1+\tau} = \eta(\theta)\tau(\theta). \tag{J.3}$$

Another useful elasticity is the elasticity of the slack rate $u(\theta)$ in any market, which is given by (13.13):

$$\epsilon_\theta^{u} = -\left[1-\eta(\theta)\right]\left[1-u(\theta)\right]. \tag{J.4}$$

A final useful elasticity is the elasticity of $1-u(\theta)$ in any market, which can be inferred from (13.13):

$$\epsilon_\theta^{1-u} = \frac{-u(\theta)}{1-u(\theta)} \cdot \epsilon_\theta^{u} = \left[1-\eta(\theta)\right]u(\theta). \tag{J.5}$$

With the appropriate matching wedge, matching elasticity, and slack rate, these results apply to both the product market and labor market.

J.2. Monetary multiplier

We are now ready to compute the monetary multiplier.

The solution of the two-market model is given by (15.20), which imposes $\theta_p^l(\theta_l, w) = \theta_p^p(\theta_l, r, w)$. We implicitly differentiate this equation in elasticity to obtain the elasticity of labor market tightness with respect to the real interest rate (result B.10). We get

$$\epsilon_r^{\theta_l} = \frac{\epsilon_r^{\theta_p^p}}{\epsilon_{\theta_l}^{\theta_p^l} - \epsilon_{\theta_l}^{\theta_p^p}}. \tag{J.6}$$

Hence, to obtain the response of labor market tightness to the real interest rate, we need to compute the three elasticities in the equation: $\epsilon_r^{\theta_p^p}$, $\epsilon_{\theta_l}^{\theta_p^l}$, and $\epsilon_{\theta_l}^{\theta_p^p}$.

First, we implicitly differentiate in elasticity the definition of θ_p^p given by (J.2) to obtain the elasticity of θ_p^p with respect to the real interest rate. We use (J.3) for the calculation. We get

$$\epsilon_r^{\theta_p^p} = \frac{-r/(\delta-r)}{(1-\alpha_p)\eta_p(\theta_p)\tau_p(\theta_p)}. \tag{J.7}$$

Second, we implicitly differentiate in elasticity the definition of θ_p^p given by (J.2) to obtain the elasticity of θ_p^p with respect to labor market tightness. We use both (J.3) and

(J.5) for the calculation. We get

$$\epsilon_{\theta_l}^{\theta_p^p} = \frac{-\alpha_p \left[1 - \eta_l(\theta_l)\right] u_l(\theta_l)}{(1-\alpha_p)\eta_p(\theta_p)\tau_p(\theta_p)}. \tag{J.8}$$

Third, we implicitly differentiate in elasticity the definition of θ_p^l given by (J.1) to obtain the elasticity of θ_p^l with respect to labor market tightness. We use both (J.3) and (J.5) for the calculation. We get

$$\epsilon_{\theta_l}^{\theta_p^l} = \frac{\alpha_l \left[1 - \eta_l(\theta_l)\right] u_l(\theta_l) + (1-\alpha_l)\eta_l(\theta_l)\tau_l(\theta_l)}{\left[1 - \eta_p(\theta_p)\right] u_p(\theta_p)}. \tag{J.9}$$

We now combine equations (J.6), (J.7), (J.8), and (J.9) to compute the elasticity of labor market tightness with respect to the real interest rate. The raw expression for the elasticity is:

$$\epsilon_r^{\theta_l} = \frac{\frac{-r}{\delta - r} \cdot \frac{1}{(1-\alpha_p)\eta_p(\theta_p)\tau_p(\theta_p)}}{\frac{\alpha_l[1-\eta_l(\theta_l)]u_l(\theta_l)+(1-\alpha_l)\eta_l(\theta_l)\tau_l(\theta_l)}{[1-\eta_p(\theta_p)]u_p(\theta_p)} + \frac{\alpha_p[1-\eta_l(\theta_l)]u_l(\theta_l)}{(1-\alpha_p)\eta_p(\theta_p)\tau_p(\theta_p)}}.$$

After some simplification, we get:

$$\epsilon_r^{\theta_l} = \frac{-r}{\delta - r} \cdot \frac{1}{[1-\eta_l(\theta_l)]u_l(\theta_l)} \cdot \frac{1}{(1-\alpha_p)\left[\alpha_l + (1-\alpha_l)\frac{\eta_l(\theta_l)\tau_l(\theta_l)}{[1-\eta_l(\theta_l)]u_l(\theta_l)}\right]\frac{\eta_p(\theta_p)\tau_p(\theta_p)}{[1-\eta_p(\theta_p)]u_p(\theta_p)} + \alpha_p}.$$

Then, given that the unemployment rate is given by $u_l(\theta_l)$, with an elasticity (J.4), the elasticity of the unemployment rate with respect to the real interest rate is given by $\epsilon_r^{u_l} = -[1-\eta_l(\theta_l)][1-u_l(\theta_l)]\epsilon_r^{\theta_l}$, which simplifies to:

$$\epsilon_r^{u_l} = \frac{r}{\delta - r} \cdot \frac{1-u_l(\theta_l)}{u_l(\theta_l)} \cdot \frac{1}{(1-\alpha_p)\left[\alpha_l + (1-\alpha_l)\frac{\eta_l(\theta_l)\tau_l(\theta_l)}{[1-\eta_l(\theta_l)]u_l(\theta_l)}\right]\frac{\eta_p(\theta_p)\tau_p(\theta_p)}{[1-\eta_p(\theta_p)]u_p(\theta_p)} + \alpha_p}. \tag{J.10}$$

Accordingly, the monetary multiplier $du_l/di = du_l/dr = (u_l(\theta_l)/r) \cdot \epsilon_r^{u_l}$ is given by:

$$\frac{du_l}{di} = \frac{1-u_l(\theta_l)}{\delta - r} \cdot \frac{1}{\alpha_p + (1-\alpha_p)\left[\alpha_l + (1-\alpha_l)\frac{\eta_l(\theta_l)\tau_l(\theta_l)}{[1-\eta_l(\theta_l)]u_l(\theta_l)}\right]\frac{\eta_p(\theta_p)\tau_p(\theta_p)}{[1-\eta_p(\theta_p)]u_p(\theta_p)}}. \tag{J.11}$$

J.3. Fiscal multiplier

Next we turn to the fiscal multiplier. We are interested in the effect of government spending on the unemployment rate. We express government spending as a share of private spending, so the aggregate demand with fiscal policy is

$$(1+g) \cdot y^d(\theta_p, r),$$

where $y^d(\theta_p, r)$ is the private demand given by (15.11). Expressing government spending in this fashion greatly simplifies the computation. It is also consistent with the way government spending appears in the analysis of optimal fiscal policy of chapter 17: what matters for welfare is the ratio between public and private consumption, which corresponds to g.

The term $1+g$ from the aggregate demand then appears in equation (J.2), which becomes:

$$\left[1+\tau_p(\theta_p^p)\right]^{1-\alpha_p} = \frac{(1+g)^{\alpha_p}(1-\alpha_l)^{\alpha_p}\left[(1-\alpha_p)(\delta-r)a_p\right]}{(wh\left[1-u_l(\theta_l)\right])^{\alpha_p}}. \tag{J.12}$$

Equation (J.1) on the other hand is unchanged.

We compute the fiscal multiplier just like the monetary multiplier, following the same steps. First, we implicitly differentiate equation (15.20), where $\theta_p^p(\theta_l, r, w, g)$ is now given by (J.12), in elasticity to obtain the elasticity of labor market tightness with respect to government spending. As above, we get

$$\epsilon_g^{\theta_l} = \frac{\epsilon_g^{\theta_p^p}}{\epsilon_{\theta_l}^{\theta_p^l} - \epsilon_{\theta_l}^{\theta_p^p}}. \tag{J.13}$$

The elasticities $\epsilon_{\theta_l}^{\theta_p^l}$ and $\epsilon_{\theta_l}^{\theta_p^p}$ remain the same as above: they are not affected by fiscal policy. The elasticity $\epsilon_g^{\theta_p^p}$ is simply obtained by implicitly differentiating in elasticity the definition of θ_p^p given by (J.12). We get

$$\epsilon_g^{\theta_p^p} = \frac{\alpha_p g/(1+g)}{(1-\alpha_p)\eta_p(\theta_p)\tau_p(\theta_p)}. \tag{J.14}$$

The elasticity is fairly similar to that in (J.7): the sole difference is that $-r/(\delta-r)$ is replaced by $\alpha_p g/(1+g)$.

Hence, we can compute the elasticities $\epsilon_g^{\theta_l}$ and $\epsilon_g^{u_l}$ just as in the case of monetary policy, but replacing $-r/(\delta-r)$ by $\alpha_p g/(1+g)$. Based on (J.10), this gives

$$\epsilon_g^{u_l} = \frac{-g}{1+g}\cdot\frac{1-u_l(\theta_l)}{u_l(\theta_l)}\cdot\frac{1}{1+\frac{1-\alpha_p}{\alpha_p}\left[\alpha_l+(1-\alpha_l)\frac{\eta_l(\theta_l)\tau_l(\theta_l)}{[1-\eta_l(\theta_l)]u_l(\theta_l)}\right]\frac{\eta_p(\theta_p)\tau_p(\theta_p)}{[1-\eta_p(\theta_p)]u_p(\theta_p)}}.$$

We normalize the fiscal multiplier to be positive, so we define the multiplier by $-du_l/dg = -[u_l(\theta_l)/g]\cdot\epsilon_g^{u_l}$. Given the expression above for the elasticity $\epsilon_g^{u_l}$, the fiscal multiplier is:

$$-\frac{du_l}{dg} = \frac{1-u_l(\theta_l)}{1+g}\cdot\frac{1}{1+\frac{1-\alpha_p}{\alpha_p}\left[\alpha_l+(1-\alpha_l)\frac{\eta_l(\theta_l)\tau_l(\theta_l)}{[1-\eta_l(\theta_l)]u_l(\theta_l)}\right]\frac{\eta_p(\theta_p)\tau_p(\theta_p)}{[1-\eta_p(\theta_p)]u_p(\theta_p)}}. \tag{J.15}$$

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